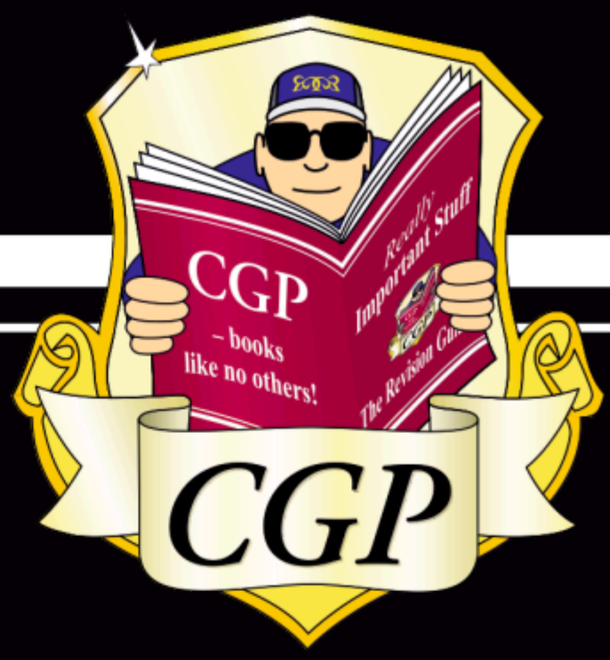
CGP



GCSE AQA

Physics

Complete

Revision & Practice

with new ***CGP RevisionHub***

New ***CGP RevisionHub*** includes:

- Quick Quizzes
- Summary Tests
- Videos
- And much more...

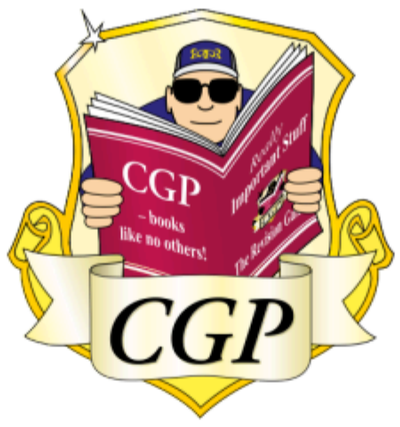
Ta-dah...

CGP RevisionHub

...is here!

- ✓ New online revision
- ✓ Quizzes, tests, videos and more
- ✓ All matched to your book

CGP RevisionHub included with this book



GCSE AQA

Physics

This CGP book is just what you need to ace GCSE Physics — it includes all the important facts, theory and practical skills for the exams.

There's also plenty of exam-style practice for every topic *and* a full set of practice papers. And we've put fully-worked answers at the back, so it's easy to mark your work.

We've also included lots of matching online revision in CGP RevisionHub — there are quick quizzes, summary tests, videos and more... It's too good to resist.

Complete

Revision & Practice

with new ***CGP RevisionHub***

Contents

How to Revise with CGP RevisionHub 1
What to Expect in the Exams 2

Paper One

Topic 1 — Energy

Topic 1a — Energy Stores and Transfers

Energy Stores and Systems 3
Kinetic and Potential Energy Stores 4
Specific Heat Capacity 5
Warm-Up & Exam Questions 6
Conservation of Energy and Power 7
Conduction and Convection 8
Thermal Insulation 9
Lubrication and Efficiency 10
Warm-Up & Exam Questions 11
Exam Questions 12
Revision Summary Test for Topic 1a 13

Topic 1b — Energy Resources

Energy Resources and Their Uses 14
Wind, Solar and Geothermal 15
Hydro-electricity, Waves and Tides 16
Bio-fuels and Non-renewables 17
Trends in Energy Resource Use 18
Warm-Up & Exam Questions 19
Revision Summary Test for Topic 1b 20

Topic 2 — Electricity

Topic 2a — Circuits

Current and Circuit Symbols 21
Resistance and $V = IR$ 22
Resistance and *I*-*V* Characteristics 23
Investigating *I*-*V* Characteristics 24
Warm-Up & Exam Questions 25
Circuit Devices 26
Series Circuits 27
Parallel Circuits 28
Investigating Resistance 29
Warm-Up & Exam Questions 30
Revision Summary Test for Topic 2a 31

Topic 2b — Everyday Electricity

Electricity in the Home 32
Power of Electrical Appliances 33
More on Energy Transfers and Power 34
The National Grid 35
Warm-Up & Exam Questions 36
Static Electricity 37
Electric Fields 38
Warm-Up & Exam Questions 39
Revision Summary Test for Topic 2b 40

Topic 3 — Particle Model of Matter

Density of Materials 41
Internal Energy and Changes of State 42
Specific Latent Heat 43
Particle Motion in Gases 44
Changes in Gas Pressure 45
Warm-Up & Exam Questions 46
Exam Questions 47
Revision Summary Test for Topic 3 48

Topic 4 — Atomic Structure

Developing the Model of the Atom 49
The Structure of the Atom 50
Isotopes and Nuclear Radiation 51
Nuclear Equations 52
Half-life 53
Warm-Up & Exam Questions 54
Exam Questions 55
Background Radiation and Contamination 56
Uses and Risk 57
Fission and Fusion 58
Warm-Up & Exam Questions 59
Revision Summary Test for Topic 4 60

Paper Two

Topic 5 — Forces

Topic 5a — Forces, Moments & Pressure

Contact and Non-Contact Forces 61
Weight, Mass and Gravity 62
Resultant Forces and Work Done 63
Calculating Forces 64
Warm-Up & Exam Questions 65
Forces and Elasticity 66
Investigating Springs 67
Moments 68
Warm-Up & Exam Questions 69
Fluid Pressure 70
Upthrust and Atmospheric Pressure 71
Warm-Up & Exam Questions 72
Revision Summary Test for Topic 5a 73

Topic 5b — Forces and Motion

Distance, Displacement, Speed and Velocity 74
Acceleration 75
Distance-Time and Velocity-Time Graphs 76
Terminal Velocity 77
Warm-Up & Exam Questions 78
Newton's First and Second Laws 79
Inertia and Newton's Third Law 80
Investigating Motion 81
Warm-Up & Exam Questions 82
Revision Summary Test for Topic 5b 83

Topic 5c — Stopping Distances & Momentum

Stopping Distances 84
More on Stopping Distances 85
Reaction Times 86
Warm-Up & Exam Questions 87
Momentum 88
Changes in Momentum 89
Warm-Up & Exam Questions 90
Revision Summary Test for Topic 5c 91

Topic 6 — Waves

Topic 6a — Wave Basics & EM Waves

Transverse and Longitudinal Waves 92
Experiments With Waves 93
Reflection 94
Refraction 95
Investigating Light 96
Warm-Up & Exam Questions 97
Exam Questions 98
Electromagnetic (EM) Waves 99
Radio Waves 100
Uses of EM Waves 101
More Uses of EM Waves 102
Dangers of EM Waves 103
Warm-Up & Exam Questions 104
Revision Summary Test for Topic 6a 105

Topic 6b — Light, Radiation & Other Waves

Lenses 106
Images and Convex Lenses 107
Concave Lenses and Magnification 108
Visible Light 109
Warm-Up & Exam Questions 110
Infrared Radiation and Temperature 111
Black Body Radiation 112
Sound Waves 113
Ultrasound 114
Seismic Waves 115
Warm-Up & Exam Questions 116
Exam Questions 117
Revision Summary Test for Topic 6b 118

Topic 7 — Magnetism & Electromagnetism

Permanent and Induced Magnets 119
Electromagnetism 120
Warm-Up & Exam Questions 121
The Motor Effect 122
Electric Motors and Loudspeakers 123
Warm-Up & Exam Questions 124
The Generator Effect 125
Generators and Microphones 126
Transformers 127
Warm-Up & Exam Questions 128
Revision Summary Test for Topic 7 129

Topic 8 — Space Physics

The Life Cycle of Stars 130
The Solar System and Orbits 131
Red-shift and the Big Bang 132
Warm-Up & Exam Questions 133
Revision Summary Test for Topic 8 134

Working Scientifically

The Scientific Method 135
Communication & Issues
Created by Science 136
Risk 137
Designing Investigations 138
Collecting Data 139
Processing Data 140
Presenting Data 141
Interpreting Graphs 142
Units 143
Maths Skills 144
Drawing Conclusions 145
Uncertainties and Evaluations 146

Practical Skills

Apparatus and Techniques 147
More Apparatus and Techniques 148
Working with Electronics 149

Practice Exams

Practice Paper 1 150
Practice Paper 2 171

Answers 191
Glossary 208
Index 213
Formulas 216

Throughout this book you'll see grade stamps like these:

These grade stamps help to show how difficult the questions are.
Remember — to get a top grade you need to be able to answer **all** the questions, not just the hardest ones.

In the real exams, some questions test how well you can write (as well as your scientific knowledge).
In this book, we've marked these questions with an asterisk (*).

Published by CGP
From original material by Richard Parsons.

Editors: Sammy El-Bahrawy, Josie Gilbert, Sean McParland, Charlotte Sheridan, Jack Simm, Dan Sreeves, Julie Wakeling and Maddie Wright.
Contributors: Paddy Gannon, Gemma Hallam, Jason Howell and Barbara Mascetti.

With thanks to Duncan Lindsay and Glenn Rogers for the proofreading.
With thanks to Jade Sim for the copyright research.

PAS49 - 0325 ~ 30573

Data used to construct stopping distance diagram on page 85, graph on page 163 and table on page 183: contains public sector information licensed under the Open Government Licence v3.0. https://www.nationalarchives.gov.uk/doc/open-government-licence/version/3/

Clipart from Corel®
Illustrations by: Sandy Gardner Artist, email sandy@sandygardner.co.uk

Text, design, layout and original illustrations © Coordination Group Publications Ltd. (CGP) 2025
All rights reserved.

0800 1712 712 • www.cgpbooks.co.uk

How to Revise with *CGP RevisionHub*

*There's more to **revision** than just reading this book — you need to **understand** and **remember** it too. Read on for how to get the most out of your **book** and **RevisionHub**...*

Here's a handy guide to using RevisionHub.

Video

1) You Need to **Learn Your Stuff**

This book covers everything you need to know for GCSE AQA Physics.

1) Read it — but not all at once... break it into manageable chunks.
2) Once you've read over a chunk, try making your own mind maps and flash cards — writing down and thinking about the content will help it to really stick.
3) We've also included some videos to help with the trickier bits — watch out for the QR codes.

2) Check That You Can **Remember** What You've Learnt

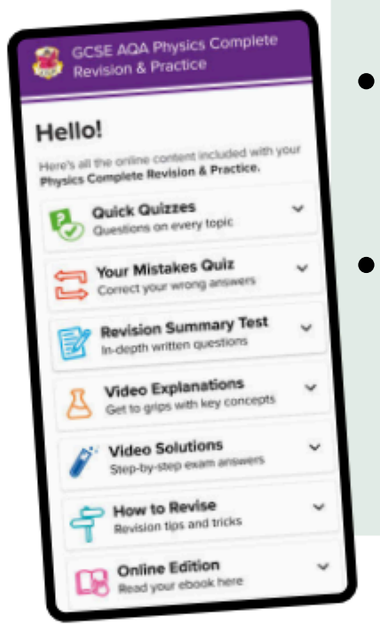
Ta-dah!

Revision Hub

If you don't test yourself, you won't know what you don't know. So try covering up a page and seeing if you can write it all out. Then try quizzing yourself using RevisionHub:

Quick Quizzes

- The QR code at the top of each main page takes you to a quick quiz of the stuff on that page.
- These quizzes are a quick check of the key facts and your maths skills.
- Revisit questions that you get wrong by doing the 'Your Mistakes' quiz, and there are Paper 1 and Paper 2 quizzes too.



Revision Summary Tests

- At the end of each topic in the book is a fiendish memory test on as much stuff as we could cram in.
- You can do these tests on paper or you can complete them on the RevisionHub — follow the QR code.
- Online, you can also find sample answers, assess your progress, and look back at your results for previous topics to see the areas you need to work on.

3) Make Sure That You're **Exam Ready**

To see if you're exam ready, try some exam-style questions. Luckily for you, there are loads in this book.

- You'll find one or more exam-style questions at the bottom of some pages.
- Every few pages, you'll find some warm-up and exam questions.
- There are also two practice papers at the back of the book.

Use these resources to check that you can apply what you've learnt to exams.

There are answers in the back of the book, and many questions at the bottom of the main pages also have a video solution for you to watch — just use the QR code next to the question.

You probably won't stick strictly to this order of steps — for example, you might want to try some exam-style questions as you go, and you'll need to look back at bits of this book if you get a question wrong or are finding a topic hard.

Revision — there's no better path to exam success...

Yep, revision can seem quite daunting. But it's a key part of preparing for your exams. If it's making you feel like you don't know where to start, then fear not — follow this QR code to see what we recommend.

Start Revising

What to Expect in the Exams

*Before you get cracking with your **revision** and **exam practice**, here's a **handy guide** to what you'll have to face in the exams. You're welcome.*

Topics are Covered in **Different Papers**

For GCSE AQA Physics, you'll sit two exam papers at the end of your course.

Paper	Time	No. of Marks	Topics Assessed	Maths Skills	Practical Skills
1	1 hr 45 mins	100	1, 2, 3 and 4	At least 30% of total marks	At least 15% of total marks
2	1 hr 45 mins	100	5, 6, 7 and 8		

You're expected to know the basic concepts of energy transfers in both papers.

There are **Different Question Types**

In each exam, you'll be expected to answer a mixture of multiple choice questions, structured questions, questions that have short, closed answers, as well as open response questions.

For some open response questions, you'll be marked on the overall quality of your answer, not just its scientific content. So...

Always make sure:

- You answer the question fully.
- You include detailed, relevant information.
- Your answer is clear and has a logical structure.

You'll be Tested on Your **Maths**...

At least 30% of the total marks for GCSE Physics will come from questions that test your maths skills. For these questions, always remember to:

We've indicated Maths Skills throughout this book with a handy stamp that looks like this:

- Show your working — you could get marks for this, even if your final answer's wrong.
- Check that the units of your answer are the same as the ones they asked for in the question.
- Make sure your answer is given to an appropriate number of significant figures.

...and on Your **Practical Skills**

Required practical activities are marked up with a stamp like this.

GCSE Physics contains 10 required practical activities that you'll do during the course. You can be asked about these, and the practical skills involved in them, in the exams.

E.g. you might have to comment on the experimental design (the apparatus and method), make predictions, analyse or interpret results... Pretty much anything to do with planning and carrying out investigations.

At least 15% of the total marks will be for questions that test your understanding of the practical activities and practical skills.

You'll Need to Know About **Working Scientifically**

Working Scientifically is all about how science is applied in the outside world by real scientists.

For example, you might be asked about ways that scientists communicate an idea to get their point across without being biased, or about the limitations of a scientific theory.

You need to think about the situation that you've been given and use all your scientific savvy. Always read the question and any data you've been given really carefully before you start writing.

Energy Stores and Systems

Quick Quiz

***Energy** is **never used up**. Instead it's just **transferred** between different **energy stores** and different objects...*

Energy can **Exist** in **Different Stores**

The energy stores you need to know are:

You may also see thermal energy stores called internal energy stores.

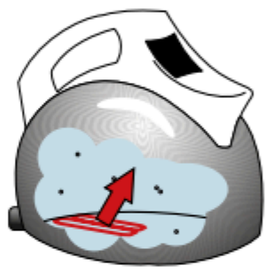
1) Thermal energy stores
2) Kinetic energy stores
3) Gravitational potential energy stores
4) Elastic potential energy stores
5) Chemical energy stores
6) Magnetic energy stores
7) Electrostatic energy stores
8) Nuclear energy stores

When a **System** Changes, Energy is **Transferred**

- A system can be a single object (e.g. the air in a piston) or a group of objects (e.g. colliding vehicles).
- When a system changes, energy is transferred. It can be transferred into or away from the system, between different objects in the system or between different types of energy stores.
- Closed systems are systems where matter and energy cannot enter or leave.
  The net change in the total energy of a closed system is always zero.

Energy can be **Transferred** in a Few Ways...

When a kettle boils water, energy is transferred electrically to the thermal energy store of the kettle's heating element. Energy is then transferred to the water's thermal energy store by heating.



Energy can also be transferred by radiation (e.g. by light).

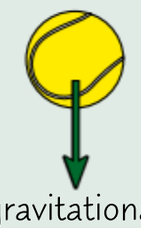
...Including by **Doing Work**

Work done is just another way of saying energy transferred. Work can be done when current flows (work is done against resistance in a circuit, see p.33) or by a force moving an object (see page 63).

You might hear energy transferred by a force described as energy transferred mechanically.

The initial force exerted by a person to throw a ball upwards does work. There is an energy transfer from the chemical energy store of the person's arm to the kinetic and gravitational potential energy stores of the ball and arm.

A ball dropped from a height is accelerated by gravity. The gravitational force does work. It causes energy to be transferred from the ball's gravitational potential energy store to its kinetic energy store.



The friction between a car's brakes and its wheels does work as it slows down. Energy is transferred from the wheels' kinetic energy stores to the thermal energy store of the surroundings.

In a collision between a car and a stationary object, the normal contact force between the car and the object does work. It causes energy to be transferred from the car's kinetic energy store to other energy stores, e.g. the elastic potential and thermal energy stores of the object and the car body.

No matter what store it's in, it's all energy...

In the exam, make sure you always refer to energy in terms of the store it's in. For example, if you're describing the energy in a hot object, you need to say that it 'has energy in its thermal energy store'.

Q1 Describe the energy transfers that occur when the wind causes a windmill to spin. [3 marks]

Q1 Video Solution

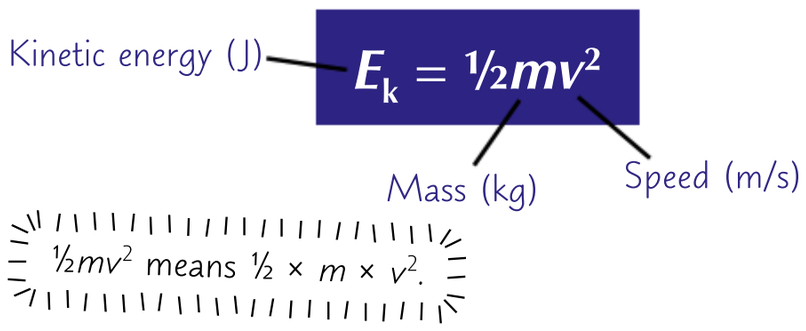
Quick Quiz

Kinetic and Potential Energy Stores

*Now you've got your head around **energy stores**, it's time to see how you can calculate the amount of energy in three of the most common ones — **kinetic**, **gravitational potential** and **elastic potential** energy stores.*

Movement Means There's Energy in an Object's **Kinetic Energy Store**

1) The energy in the kinetic energy store depends on the object's mass and speed.
The greater its mass and the faster it's going, the more energy there will be in its kinetic energy store.
2) The formula for kinetic energy is:



EXAMPLE

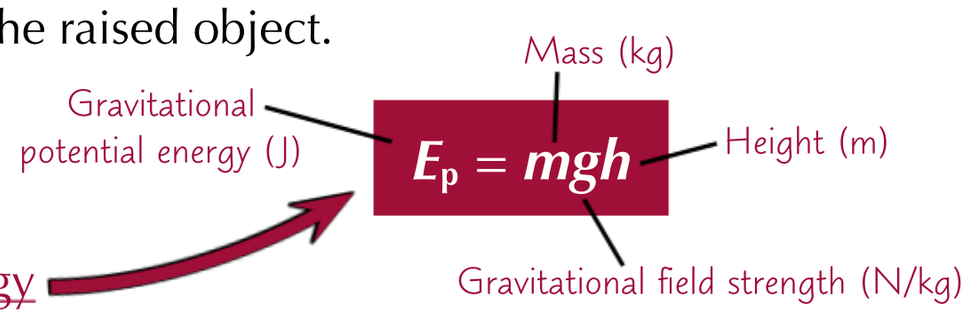
MATHS SKILLS

A car of mass 2500 kg is travelling at 20 m/s.
Calculate the energy in its kinetic energy store.

$E_k = \frac{1}{2} \times m \times v^2$
$= 0.5 \times 2500 \times 20^2 = 500\,000$ J

Raised Objects Store Energy in **Gravitational Potential** Energy Stores

1) Lifting an object in a gravitational field requires work. This causes a transfer of energy to the gravitational potential energy (g.p.e.) store of the raised object.
2) The amount of energy in a g.p.e. store depends on the object's mass, its height and the strength of the gravitational field the object is in (p.62).
3) You can use this equation to find the change in energy in an object's gravitational potential energy store for a change in height, h.



Falling Objects Also Transfer Energy

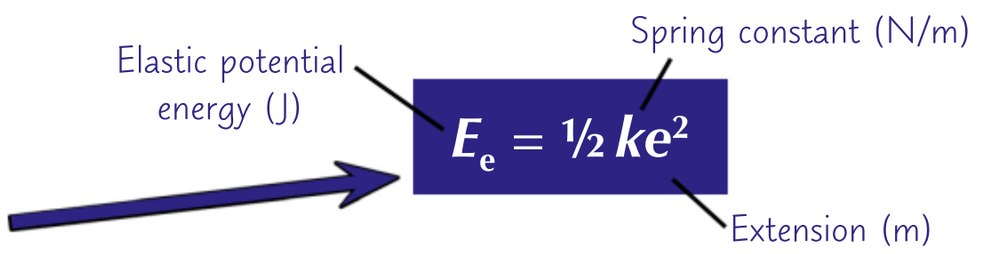
- When something falls, energy from its gravitational potential energy store is transferred to its kinetic energy store.
- For a falling object when there's no air resistance:

Energy lost from the g.p.e. store = Energy gained in the kinetic energy store

- In real life, air resistance (p.77) acts against all falling objects — it causes some energy to be transferred to other energy stores, e.g. the thermal energy stores of the object and surroundings.

Stretching can Transfer Energy to **Elastic Potential Energy Stores**

Stretching or squashing an object can transfer energy to its elastic potential energy store.
So long as the limit of proportionality has not been exceeded (p.66) energy in the elastic potential energy store of a stretched spring can be found using:



Greater height = more energy in gravitational potential stores...

Make sure you know what all the variables are in the equations above, and what units they're measured in.

Q1 A 2.0 kg object is dropped from a height of 10 m.
Calculate the speed of the object after it has fallen 5.0 m, assuming there is no air resistance.
Give your answer to 2 significant figures. g = 9.8 N/kg. [5 marks]

Q1 Video Solution

Specific Heat Capacity

Quick Quiz

Specific heat capacity *is really just a sciencey way of saying* ***how hard*** *it is to* ***heat*** *something up...*

Different Substances Have Different Specific Heat Capacities

1) More energy needs to be transferred to the thermal energy stores of some substances to increase their temperature than others.
2) Substances that need to gain lots of energy in their thermal energy stores to warm up also transfer loads of energy when they cool down again. They can 'store' a lot of energy.
3) Specific heat capacity is the amount of energy needed to raise the temperature of 1 kg of a substance by 1 °C.
4) Here's the equation that links energy transferred to specific heat capacity:

$$\Delta E = mc\Delta\theta$$

Change in thermal energy (J) — Mass (kg) — Specific heat capacity (J/kg°C) — Change in Temperature (°C)

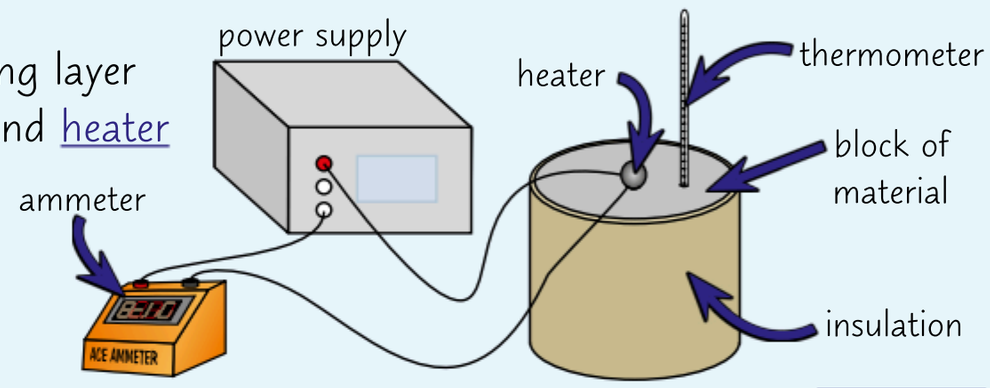
Δ just means "change in".

You Could Investigate Specific Heat Capacities Like This...

PRACTICAL

Method

1) Get a block of a solid material with two holes in it.
2) Measure the mass of the block, then wrap it in an insulating layer (e.g. a thick layer of newspaper). Insert the thermometer and heater and connect up the circuit as shown on the right.
3) Measure the initial temperature of the block and set the potential difference, V, of the power supply to be 10 V. Turn on the power supply and start a stopwatch.

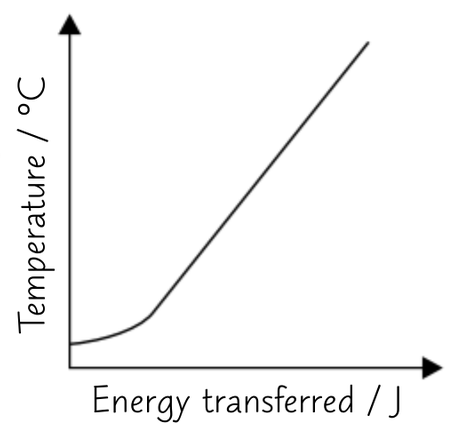


4) As the block heats up, take readings of the temperature and current, I, every minute for 10 minutes. You should find that the current through the circuit doesn't change as the block heats up.
5) When you've collected enough readings (10 should do it), turn off the power supply.

Video

- From this experiment, you can calculate the power supplied to the heater, using $P = VI$ (p.34). You can use this to calculate how much energy, E, has been transferred to the heater at the time of each temperature reading using the formula $E = Pt$, where t is the time in seconds since the experiment began.
- Assuming all the energy supplied to the heater has been transferred to the block, you can plot a graph of temperature against energy transferred to the thermal energy store of the block.
- Find the gradient of the straight part of the graph. This is equal to $\Delta\theta \div \Delta E$ (for more on gradients, see page 142).
- You know that $\Delta E = mc\Delta\theta$. If you divide both sides of the equation by ΔE, you will find that $1 = mc(\Delta\theta \div \Delta E) = mc \times \text{gradient}$.
- So the specific heat capacity, c, of the material of the block can be found using: $1 \div (\text{gradient} \times \text{the mass of the block})$.
- You can investigate the specific heat capacity of liquids too. This can be done by placing a heater and a thermometer in an insulated beaker filled with a known mass of the liquid you want to measure.

While the circuit is switched on, the current does work on the heater, transferring energy electrically from the power supply to the heater's thermal energy store. This energy is then transferred to the material's thermal energy store by heating, causing the material's temperature to increase.



Some substances heat up more easily than others...

Learn the definition of specific heat capacity and make sure you know how to use the formula for it.

Q1 Find the final temperature of 5 kg of water, at an initial temperature of 5 °C, after 50 kJ of energy has been transferred to it. The specific heat capacity of water is 4200 J/kg°C. [3 marks]

Q1 Video Solution

Warm-Up & Exam Questions

These questions give you chance to use your knowledge about energy transfers and specific heat capacity.

Warm-Up Questions

1) Give two methods of energy transfer.
2) How does the way energy is stored change when someone throws a ball upwards?
3) State the equation that links energy in an object's kinetic energy store with mass and speed.
4) Which has more energy in its kinetic energy store: a person walking at 3 miles per hour, or a lorry travelling at 60 miles per hour?

Exam Questions

For these questions, you can use the formula sheet on page 216.

1 A motor lifts a load of mass 20 kg.
The load gains 137.2 J of energy in its gravitational potential energy store. GRADE (4-6) GRADE

1.1 State the equation that links gravitational potential energy, mass, gravitational field strength and height.
Use this equation to calculate the height through which the motor lifts the load.
Assume the gravitational field strength = 9.8 N/kg

[4 marks]

1.2 The motor releases the load and the load falls.
Ignoring air resistance, describe the changes in the way energy is stored that take place as the load falls.

[2 marks]

1.3 Describe how your answer to **1.2** would differ if air resistance was not ignored.

[1 mark]

2 36 000 J of energy is transferred to heat a 0.5 kg concrete block from 20 °C to 100 °C. GRADE (6-7) GRADE

2.1 Calculate the specific heat capacity of the concrete block.

[4 marks]

2.2 Energy is transferred to the thermal energy store of an electric storage heater at night, and then transferred away to the thermal energy stores of the surroundings during the day.
Lead has a specific heat capacity of 126 J/kg°C.
Using your answer to **2.1**, explain why concrete blocks are used in storage heaters rather than lead blocks.

[2 marks]

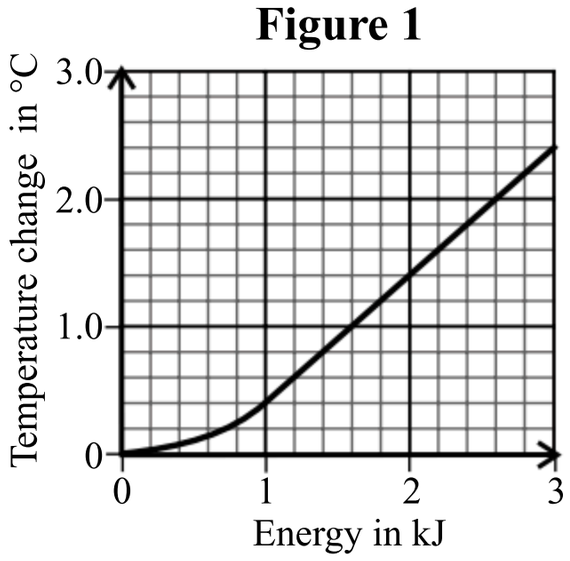
PRACTICAL

3 A student transfers energy steadily to a 1.0 kg aluminium block.
They produce a graph of the energy supplied against the increase in temperature of the block, shown in **Figure 1**. GRADE (6-7) GRADE

3.1 Use **Figure 1** to find a value for the specific heat capacity of aluminium in J/kg°C.

[4 marks]

Figure 1



3.2 Would you expect the true value for the specific heat capacity of aluminium to be higher or lower than the value found in this experiment? Explain your answer.

[3 marks]

Conservation of Energy and Power

Quick Quiz

*Repeat after me: **energy** is **NEVER** destroyed. Make sure you learn that fact, it's really important.*

You **Need** to Know the **Conservation of Energy Principle**

The conservation of energy principle is that energy is always conserved:

Energy can be transferred usefully, stored or dissipated, but can never be created or destroyed.

- Not all energy is transferred usefully when a transfer takes place — some energy is always dissipated.
- The dissipated energy can also be called 'wasted energy', and usually transfers to thermal energy stores.

In all systems, energy is wasted. Closed systems still waste energy, but the net change in energy is zero.

Mobile Phone

A mobile phone is a system. When you use the phone, energy is usefully transferred from the chemical energy store of the battery in the phone. But some of this energy is dissipated in this transfer to the thermal energy store of the phone, which is then partially transferred to the thermal energy store of the surroundings.

Sealed Flask

A cold spoon is dropped into an insulated flask of hot soup, which is then sealed. This is a closed system. Energy is transferred from the thermal energy store of the soup to the useless thermal energy store of the spoon, causing the soup to cool down slightly. Energy has been dissipated within the system, but no energy has left the system.

Power is the '**Rate of Doing Work**' — i.e. How Much **per Second**

1) Power is the rate of energy transfer, or the rate of doing work.
2) Power is measured in watts. One watt = 1 joule of energy transferred per second.
3) You can calculate power using these equations:

$$\text{Power (W)} = P = \frac{E}{t} \quad \begin{matrix}\text{Energy transferred (J)} \\ \text{Time (s)}\end{matrix}$$

$$\text{Power (W)} = P = \frac{W}{t} \quad \begin{matrix}\text{Work done (J)} \\ \text{Time (s)}\end{matrix}$$

A powerful machine is not necessarily one which can exert a strong force, but a machine which transfers a lot of energy in a short space of time.

Take two cars that are identical in every way apart from the power of their engines. Both cars race the same distance along a straight race track to a finish line. The car with the more powerful engine will reach the finish line faster than the other car — i.e. it will transfer more energy, as it reaches a higher speed.

EXAMPLE

It takes 8000 J of work to lift a stunt performer to the top of a building. Motor A can lift the stunt performer to the correct height in 50 s. Motor B would take 300 s to lift the performer to the same height. Which motor is most powerful? Calculate the power of this motor.

MATHS SKILLS

1) Both motors transfer the same amount of energy, but motor A would do it quicker than motor B. So, motor A is the more powerful motor.
2) Plug the time taken and work done for motor A into the equation $P = W \div t$ and find the power.
$P = W \div t = 8000 \div 50 = 160$ W

Remember: energy cannot be created or destroyed...

When energy is wasted, it's not destroyed — it still exists, it just isn't in a useful energy store anymore.

Q1 A motor transfers 4.8 kJ of energy in 2 minutes. Calculate its power output. [3 marks]

Q1 Video Solution

Conduction and Convection

Quick Quiz

*Unwanted energy transfers can be **reduced** by using **insulation**. But before you can tackle the topic of insulation, you need to understand some of the ways energy can be transferred by **heating**.*

Conduction Occurs Mainly in **Solids**

Conduction is the process where vibrating particles transfer energy to neighbouring particles.

Energy transferred to an object by heating is transferred to the thermal energy store of the object. On a particle level, this energy is shared across the kinetic energy stores of the particles in the object.

The particles in the part of the object being heated vibrate more and collide with neighbouring particles. These collisions cause energy to be transferred between particles' kinetic energy stores.

This process continues throughout the object until the energy is transferred to the other side. Energy is also transferred to the thermal energy store of the surroundings, or anything else touching the object, by conduction.

Thermal conductivity is a measure of how quickly energy is transferred through a material in this way. Materials with a high thermal conductivity transfer energy between their particles quickly.

Convection Occurs Only in **Liquids** and **Gases**

Convection is where energetic particles move away from hotter to cooler regions.

- Convection can happen in gases and liquids. Energy is transferred by heating to the thermal store of the liquid or gas. Again, this means energy is shared across the kinetic energy stores of the gas or liquid's particles.
- Particles in liquids and gases are able to move, so when you heat a region of a gas or liquid, the particles move faster and the space between individual particles increases. This causes the density (p.41) of the heated region to decrease.
- The warmer, less dense regions rise above denser, cooler regions.

Particles in liquids and gases are much more free to move around, which is why they usually transfer energy by convection instead of conduction.

If there is a constant heat source, a convection current can be created. Particles warmed by the heat source rise and get replaced by cooler particles.
The cooler particles are heated, whilst the warmer particles transfer energy to the thermal energy store of the surroundings, causing them to cool and sink again.
This cycle repeats, causing a convection current.

Some substances are better thermal conductors than others...

For example, denser materials (see page 41) are usually better conductors than less dense materials. This is because particles that are right next to each other are able to pass energy between their kinetic energy stores more effectively than particles that are further apart. This explains why water is a much better thermal conductor than air. Fascinating.

Thermal Insulation

Quick Quiz

*There are a few ways you can **reduce** the amount of energy scampering off to a **completely useless** store — **thermal insulation** is a biggie that you need to know about. Read on to find out more...*

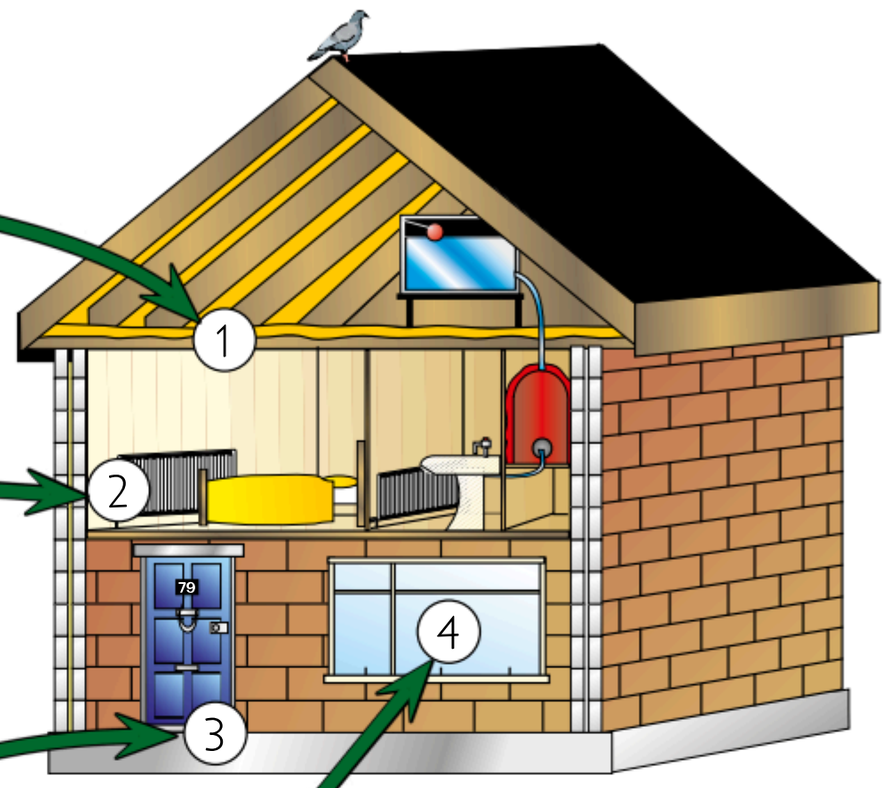
Insulation Reduces the Rate of **Energy Transfer** by **Heating**

The last thing you want when you've made your house nice and toasty is for that energy to escape outside. Thermal insulation is something you can use to prevent energy losses through heating.

Here are some examples of thermal insulation:

Loft insulation can reduce convection currents being created in lofts.

Some houses have cavity walls, made up of an inner and an outer wall with an air gap in the middle. The air gap reduces the amount of energy transferred by conduction through the walls. Cavity wall insulation, where the cavity wall air gap is filled with a foam, can also reduce energy transfer by convection in the wall cavity.



Draught excluders around doors and windows reduce energy transfers by convection.

Double-glazed windows work in the same way as cavity walls — they have an air gap between two sheets of glass to reduce energy transfer by conduction through the windows.

Having thick walls made from materials with a low thermal conductivity can also reduce energy loss. The thicker the walls and the lower their thermal conductivity, the slower the rate of energy transfer will be.

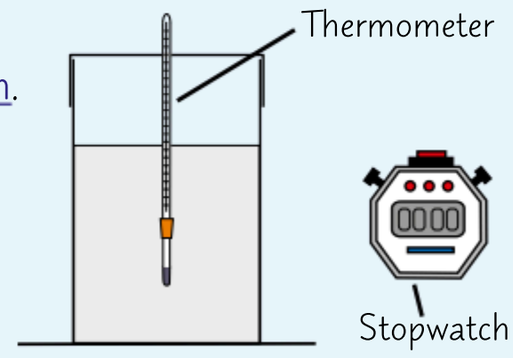
Investigating **Thermal Insulation**

PRACTICAL

Method

Here's one method for investigating the effectiveness of materials as thermal insulators:

1) Pour a fixed amount of hot water into a sealable container and measure the mass of water in the container (p.147).
2) Use a thermometer to measure the initial temperature of the water.
3) Seal the container and leave it for five minutes. Measure this time using a stopwatch.
4) Remove the lid and measure the final temperature of the water.
5) Pour away the water and allow the container to cool to room temperature.
6) Repeat this experiment, but wrap the container in a layer of material (*e.g.* foil, newspaper) once it has been sealed.
7) Try several different materials, using the same thickness of material each time.



The lower the temperature difference, the better that material is as a thermal insulator. This is because the smaller the temperature change of the water, the less energy is transferred. You should find materials like bubble wrap or cotton wool are good thermal insulators.

You could also investigate how the thickness of the material affects how good a thermal insulator it is. You should find that the thicker the insulating layer, the better the insulator.

Having a well-insulated house can reduce your heating bills...

Remember — when people talk of energy loss, it's not that the energy has disappeared. It still exists (see page 7), just not necessarily in the store we wanted the energy to be transferred to. For example, in a car, you want the energy to transfer to the kinetic energy store of the wheels, and not to the thermal energy stores of the moving components.

Quick Quiz

Lubrication and Efficiency

*As well as insulation, **lubrication** is another way to **reduce** energy transfers to stores that **aren't useful**. Insulation and lubrication can help increase **efficiency** — the **amount** of energy that is transferred **usefully**.*

Lubrication Reduces **Frictional Forces**

1) When something moves, there's usually at least one frictional force acting against it (p.77). This causes some energy in the system to be dissipated (p.7).
2) For objects that are being rubbed together, lubricants can be used to reduce the friction between the objects' surfaces when they move. Lubricants are usually liquids (like oil), so they can flow easily between objects and coat them.

Streamlining reduces air resistance too, see p.77.

Efficiency is the Proportion of **Energy Transferred Usefully**

1) The less input energy that is 'wasted' (transferred to an unwanted energy store) the more efficient the device is said to be.
2) You can improve the efficiency of energy transfers by insulating objects, lubricating them or making them more streamlined (see pages 9 and 77).
3) The efficiency of a device can be worked out using one of these equations:

You can give efficiency as a decimal or you can multiply your answer by 100 to get a percentage, i.e. 0.75 or 75%.

$$\text{Efficency} = \frac{\text{Useful output energy transfer}}{\text{Total input energy transfer}}$$

$$\text{Efficiency} = \frac{\text{Useful power output}}{\text{Total power output}}$$

EXAMPLE

A blender is 70% efficient. It has a total input power of 600 W. Calculate the useful output power.

MATHS SKILLS

1) Change the efficiency from a percentage to a decimal.
2) Rearrange the equation for useful power output.
3) Stick in the numbers you're given.

efficiency = 70% = 0.7

useful power output = efficiency × total power input

= 0.7 × 600 = 420 W

Useful Energy Output Isn't Usually Equal to **Total Energy Input**

- For any given example you can talk about the types of energy being input and output, but remember: NO device is 100% efficient and the wasted energy is usually transferred to useless thermal energy stores.
- Electric heaters are the exception to this. They're usually 100% efficient because all the energy is transferred electrically to "useful" thermal energy stores.

The less energy a device wastes, the higher its efficiency...

Q2 Video Solution

Make sure you can use and rearrange the equations for efficiency, then have a go at these questions.

Q1 A motor in a remote-controlled car transfers 300 J of energy into the car's energy stores. 225 J are transferred to the car's kinetic energy stores. Calculate the efficiency of the motor. [2 marks]

Q2 A machine has a useful power output of 900 W and a total power input of 1200 W. In a given time, 72 kJ of energy is transferred to the machine. Calculate the amount of energy usefully transferred by the machine in this time. [4 marks]

Warm-Up & Exam Questions

Don't let your energy dissipate. These questions will let you see how efficient your revision has been.

Warm-Up Questions

1) True or false? Energy can never be created or destroyed.
2) What are the units that power is measured in?
3) Name two mechanisms by which energy is transferred by heating.
4) For a given material, how does its thermal conductivity affect the rate of energy transfer through it?
5) Give one way you could reduce the frictional forces in the hinge of an automatic door?
6) Why is the efficiency of an appliance always less than 100%?

Exam Questions

For questions 1, 3 and 4, you can use the formula sheet on page 216.

1 The motor of an electric scooter moves the scooter 10 metres along a flat, horizontal course in 20 seconds. During this time the motor does 1000 J of work. GRADE 4-6 GRADE

1.1 Write down the equation that links power, work done and time.
Use this equation to calculate the power of the motor.

[3 marks]

1.2 The moving parts of the scooter are lubricated. The scooter then completes the course in 18 seconds.
Explain, in terms of energy transfer, why the scooter completes the course in a faster time.

[2 marks]

1.3 The scooter's motor is replaced with a more powerful, but otherwise identical, motor.
It moves along the same 10 m course.
Describe how its performance will differ from before. Explain your answer.

[2 marks]

PRACTICAL

2 A student wants to test whether the temperature difference between the inside and outside of a flask affects the rate of energy transfer from the flask. He sets up the experiment shown in **Figure 1** and records the temperature of the water in each flask every minute. GRADE 6-7 GRADE

Figure 1

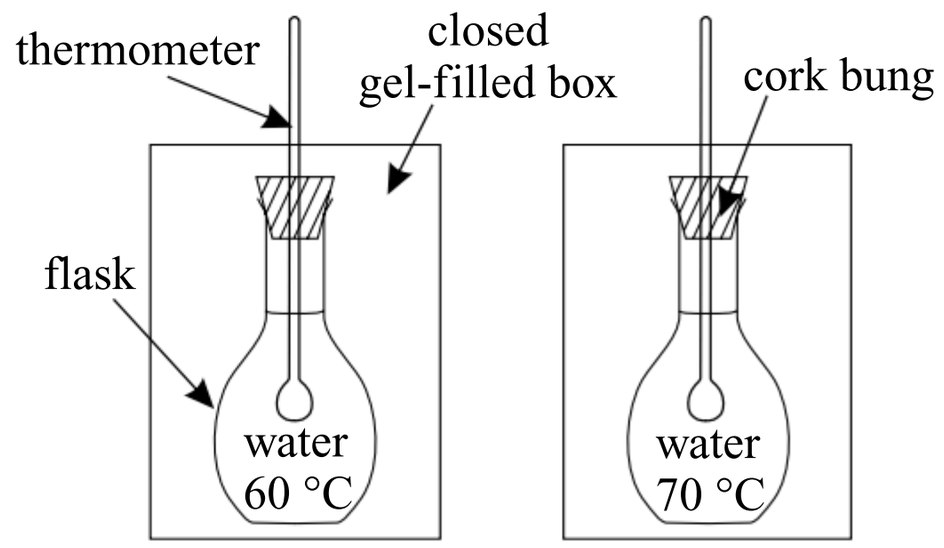
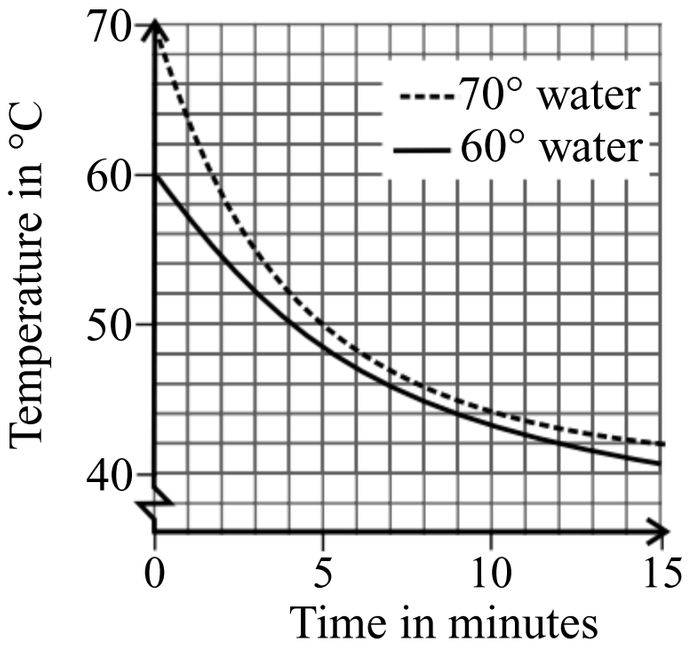


Figure 2



2.1 Give **two** variables that the student must control.

[2 marks]

2.2 The student produces the graph in **Figure 2** of his results.
Which temperature of water shows the greatest initial rate of energy transfer? Explain your answer using the graph.

[2 marks]

2.3 Calculate the rate of temperature change after 5 minutes for the 60 °C water. Give your answer in °C / minute.

[3 marks]

Exam Questions

3 Each second, a 1000 W hairdryer transfers 280 J of energy away through sound waves, 200 J of energy to kinetic energy stores, and the rest directly to the thermal energy store of the heating element.

GRADE 6-7 GRADE

3.1 Calculate how much energy the hairdryer transfers directly to thermal energy stores each second.

[1 mark]

3.2 Calculate the efficiency of the hairdryer.

[3 marks]

3.3 It takes 4 minutes for a particular person to dry their hair.
Calculate how much energy from electrical stores, in joules, is transferred in this time.

[3 marks]

4 Torch A transfers 1200 J of energy per minute.
480 J of this is transferred away usefully as light, 690 J is transferred to useless thermal energy stores and 30 J is transferred away as sound.

GRADE 6-7 GRADE

4.1 Write down the equation linking efficiency, useful output energy transfer and total input energy transfer.

[1 mark]

4.2 Calculate the efficiency of torch A.

[2 marks]

4.3 Torch B transfers 600 J of energy away usefully by light each minute.
Calculate the output power of torch B.

[2 marks]

4.4 Torch B has an efficiency of 0.55. Calculate input power of torch B.

[3 marks]

4.5 Each torch is powered by an identical battery. A student claims that the battery in torch B will go 'flat' quicker than in torch A because it transfers more energy away as light each minute.
Explain whether or not you agree with the student.

[2 marks]

PRACTICAL

5 A scientist is worried that her house is losing a lot of energy through its walls and windows.

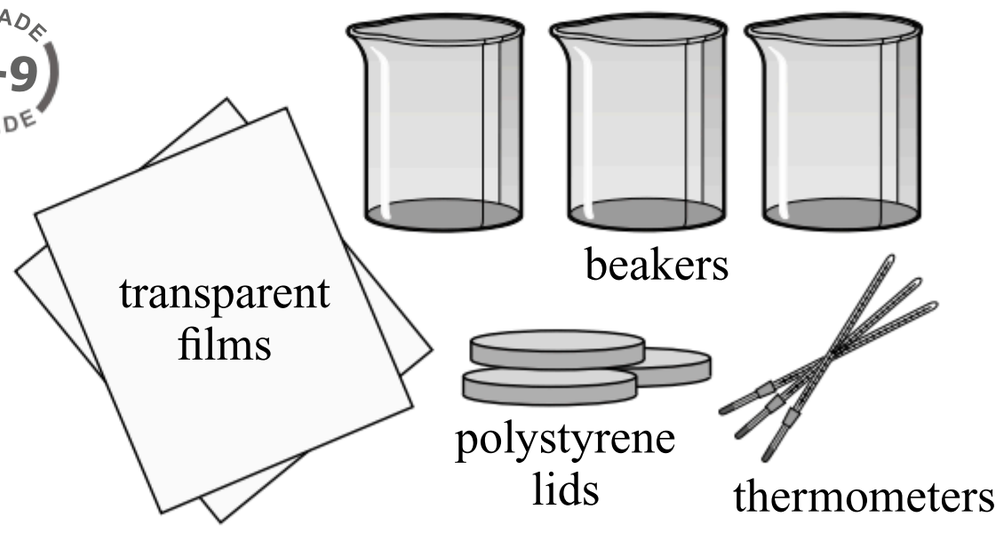
GRADE 7-9 GRADE

5.1 Suggest **one** way that the scientist could reduce the energy lost through the walls of her house.

[1 mark]

A shop sells two different types of transparent film that both claim to substantially 'reduce energy loss' when stuck over windows.

Figure 3



5.2* Describe a method the scientist could use, using the apparatus in **Figure 3**, to investigate whether the claims are true and which type of film is best.

[6 marks]

5.3 The scientist is concerned that her results are not reproducible.
Suggest **one** way she could check if her results are reproducible.

[1 mark]

Revision Summary Test for Topic 1a

I'm sure you've still got plenty of energy in your revision store to answer these summary questions:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

Energy Stores and Systems (p.3-4) ☑

1) List the eight types of energy store.
2) What is a system? What happens when a system changes?
3) What is a closed system? What is the net change in total energy for a closed system?
4) What energy transfers take place when water is heated in an electric kettle?
5) What is work done? Can you name two ways that work can be done?
6) Describe the energy transfers that occur when:
   a) a person throws a ball upwards, b) a ball is dropped from a height,
   c) a car slows down using its brakes, d) a car hits a stationary object.
7) What is the formula used to calculate the energy stored in:
   a) kinetic energy stores, b) gravitational energy stores, c) elastic potential energy stores?
8) Which energy stores is energy transferred between when an object falls from a height, if there is no air resistance?

Specific Heat Capacity (p.5) ☑

9) What is the specific heat capacity of a substance?
   And what is the formula for energy transferred that involves specific heat capacity?
10) Describe a method you can use to find the specific heat capacity of material.
11) Describe the energy transfers that take place between a power supply, a heater and a material when measuring a material's specific heat capacity.

Conservation of Energy and Power (p.7) ☑

12) Give the conservation of energy principle. Give an example of it for a closed system.
13) What is meant by dissipated energy? Give an example of energy being dissipated.
14) What is meant by power? Give two formulas for power involving time.
15) How can you tell which of two identical buses with different engines is more powerful?

Reducing Unwanted Energy Transfers (p.8-10) ☑

16) Describe how energy is transferred in conduction.
17) How are materials with a high thermal conductivity different to ones with a low thermal conductivity?
18) Describe what happens in the process of convection. What states of matter does it happen in?
19) How do convection currents form?
20) Describe how each of the following can help to thermally insulate a house:
    a) loft insulation b) cavity walls c) draught excluders d) double-glazed windows
21) Give two ways in which the walls can be designed to reduce energy loss from a house.
22) Give a method that could be used to investigate how effective a material is as a thermal insulator.
23) How do lubricants reduce the amount of energy being dissipated?
24) What is the efficiency of a device?
25) Give three ways that the efficiency of an energy transfer can be increased.
26) How can efficiency be calculated using energy transfers? How about with power input and output?

Quick Quiz

Energy Resources and Their Uses

*There are lots of **energy resources** available on Earth. They are either **renewable** or **non-renewable** resources.*

Non-Renewable Energy Resources Will Run Out One Day

Non-renewable energy resources are fossil fuels and nuclear fuel (uranium and plutonium). Fossil fuels are natural resources that form underground over millions of years. They are typically burnt to provide energy.

The three main fossil fuels are:

1) Coal 2) Oil 3) (Natural) Gas

PRO

- They are reliable sources of energy.

CONS

- They will all 'run out' one day.
- They do damage to the environment.

Renewable Energy Resources Will Never Run Out

Renewable energy resources are:

1) The Sun (Solar)
2) Wind
3) Water waves
4) Hydro-electricity
5) Bio-fuel
6) Tides
7) Geothermal

PROS

- They will never run out, since the resource can be 'renewed' as it's used.
- They are less harmful to the environment than fossil fuels.

CONS

- Most still cause some damage to the environment.
- Some are unreliable as they depend on the weather, so it's harder to produce as much energy on demand.

Energy Resources can be Used for Transport and Heating

TRANSPORT

- Petrol and diesel powered vehicles (including most cars) use fuel created from oil.
- Many vehicles can run on a mix of a bio-fuel and petrol or diesel, or even pure bio-fuels (p.17).
- Electricity can also be used to power vehicles (e.g. trains and some cars). It can be generated using renewable or non-renewable energy resources (p.15-17).

HEATING

- Natural gas is the most widely used fuel for heating homes in the UK. The gas is used to heat water, which is then pumped into radiators throughout the home.
- Coal can be burnt in fireplaces.
- The energy generated from burning bio-fuel can be used to heat homes.
- A geothermal heat pump uses geothermal energy resources (p.15) to heat buildings.
- Solar water heaters work by using the Sun to heat water which is pumped into radiators in the building.
- The electricity generated from renewable or non-renewable resources can also be used for heating, e.g. in electric heaters.

Learn these non-renewable and renewable resources...

Energy resources, both renewable and non-renewable, are mostly used to generate electricity, but there are two other major uses — transport and heating. While a lot of cars still use fossil fuels (petrol or diesel), people are gradually switching over to electric vehicles that can be run using electricity from renewable energy resources (see p.18).

Wind, Solar and Geothermal

Quick Quiz

***Renewable** energy resources, like **wind**, **solar** and **geothermal**, will not run out.*

Wind Power — Lots of Wind Turbines

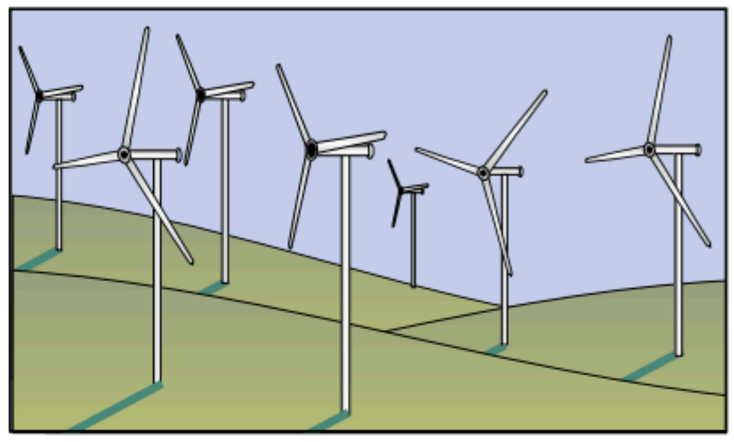
Wind turbines are put up in exposed places like on moors, around coasts or out at sea. Each turbine has a generator inside it — the rotating blades turn the generator and produce electricity (p.125).

PROS

- There's no pollution except during manufacture.
- There's no permanent damage to the landscape.
- There are no fuel costs and minimal running costs.

CONS

- The turbines can spoil the view — you need hundreds of wind turbines to replace one coal-fired power station.
- Their noise can annoy people living nearby.
- The initial costs of making them are quite high.
- The turbines will stop if there is no wind or if the wind is too strong.
- The supply cannot be increased with extra demand.



On average, wind turbines produce electricity 70-85% of the time.

Solar Cells — Electricity Generated Directly from Sunlight

1) Solar cells are usually used to generate electricity on a relatively small scale.
2) Solar power is often used in remote places where there's not much choice (e.g. the Australian outback) and to power electric road signs and satellites.
3) Solar cells are often the best source of energy to charge batteries in calculators and watches, which don't use much electricity.

PROS

- There's no pollution or environmental damage once the solar panels have been made.
- They are a very reliable source of energy in sunny countries. Can also be cost-effective in cloudy places like Britain.
- Once they're made, the energy is free and running costs are almost zero.

CONS

- They only produce energy in the daytime.
- Quite a lot of energy is used to manufacture the panels.
- The supply cannot be increased with extra demand.
- The initial costs of manufacture and set up are quite high.

Geothermal Power — Energy in Underground Thermal Energy Stores

Geothermal power can be used to generate electricity, or to heat buildings directly. It is possible in volcanic areas or where hot rocks lie quite near to the surface. The source of much of the energy is the slow decay of various radioactive elements, including uranium, deep inside the Earth.

PROS

- It's a reliable source of free energy, since it doesn't depend on the weather.
- It does little damage to the environment.

CONS

- There aren't many locations.
- The cost of building a plant is usually high compared to the amount of energy produced.

People love the idea of wind power — just not in their back yard...

It's easy to think that non-renewables are the answer to the world's energy problems. However, we definitely couldn't rely on them totally at present. Make sure you know the pros and cons for wind, solar and geothermal power.

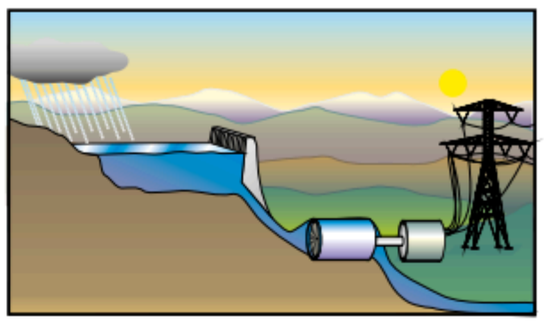
Hydro-electricity, Waves and Tides

Quick Quiz

*Good ol' **water**. Not only can we drink it, we can also use it to **generate electricity**.*

Hydro-electric Power Uses Falling Water

Hydro-electric power usually requires the flooding of a valley by building a big dam. Water is allowed out through turbines.



Hydroelectric power is reliable in areas with high rainfall. But it's not suitable in dry climates or in periods of drought.

PROS

- There is no pollution created directly from the generation of power once the stations are built.
- The flow of water can be controlled to meet the changing demands in electricity.
- There are no fuel costs and minimal running costs.

CONS

- The initial costs to build the power plants are high.
- The reservoirs can look very unsightly when they dry up.
- There's a big impact on the environment from flooding the valley, e.g. rotting vegetation releasing methane and CO_2 and possible loss of habitat for some species.

Putting hydroelectric power plants in remote valleys can reduce their impact on humans.

Wave Power — Lots of Little Wave-Powered Turbines

Like with wind power (p.15), the moving turbines are connected to a generator.
The turbines have to be in the sea, and it's usually easiest for them to be by the coast.

PROS

- There is no pollution.
- There are no fuel costs.
- It could be useful on islands, e.g. where there's a lot of coastline.

CONS

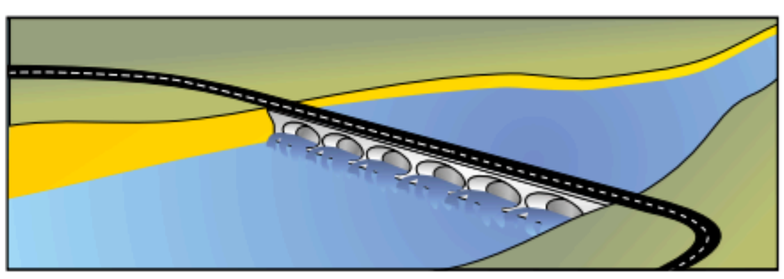
- The initial building and installation costs are quite high.
- The turbines are difficult and expensive to maintain.
- The turbines can disturb the habitats of marine animals, be a hazard to boats and spoil the view.
- The amount of energy they generate is fairly unreliable, as the waves tend to die out when the wind drops.

Tidal Barrages — Using the Sun and Moon's Gravity

Tidal barrages are big dams built across river estuaries, with turbines in them. As the tide comes in, it fills up the estuary. The water is then allowed out through turbines at a controlled speed.
The tides are produced by the gravitational pull of the Sun and Moon.

PROS

- There is no pollution once they're up and running.
- Tides are reliable — they always happen twice a day.
- There are no fuel costs and minimal running costs.
- It has the potential to generate a lot of energy.



CONS

- The dams can prevent free access to boats, alter the habitats of wildlife and spoil the view.
- The height of the tide is variable — lower tides don't provide as much energy.
- There aren't many suitable locations — you need an estuary where there's a big enough difference between high and low tides to power the turbines.
- They don't work when water levels are the same either side of the barrage (four times a day).

Wave and tidal — power from the motion of the ocean...

The first large-scale tidal barrages started being built in the 1960s, so tidal power isn't exactly a new development. Wave power is still pretty experimental though at the moment. Make sure you know the differences in how they work.

Bio-fuels and Non-renewables

Quick Quiz

*And the **energy resources** just keep on coming. It'll be over soon, I promise. Just a few more to go.*

Bio-fuels are Made from **Plants** and **Waste**

1) Bio-fuels are renewable energy resources created from either plant products or animal dung.
2) They can be solid, liquid or gas and can be burnt to produce electricity or run cars like fossil fuels can.
3) They are supposedly carbon neutral, although there is some debate about this as it's only really true if you keep growing plants at the rate that you're burning things.

PROS

- They are fairly reliable, as crops grow relatively quickly and can be grown all year round.
- Though they take time to grow, they can be continuously produced and stored for when needed, allowing them to respond to changes in energy demands.

CONS

- The cost to refine bio-fuels is very high.
- Some worry that growing crops specifically for bio-fuels will mean there isn't enough space or water to meet the demands for crops that are grown for food.
- Areas of forest may be cleared to make room to grow bio-fuels, destroying natural habitats.
- The decay and burning of vegetation increases CO_2 and methane emissions.

Non-Renewables are **Reliable**...

- There's enough fossil fuels and nuclear energy sources to meet current demand, and they are extracted from the Earth at a fast enough rate that power plants always have fuel in stock. Many of these power plants can respond quickly to changes in demand (p.35).
- There are fossil fuel power plants already set-up and running — replacing all of them with renewable power stations would be expensive.
- Nuclear fuel is relatively cheap, but the overall cost of nuclear power is high due to the cost of the power plant and final decommissioning.

Although fossil fuels are reliable, they are slowly running out. If no new resources are found, some fossil fuel stocks may run out within a hundred years.

...But Create **Environmental Problems**

Burning fossil fuels and using nuclear energy can lead to the following environmental problems:

	CAUSE	IMPACT
GLOBAL WARMING	Burning coal, oil and gas releases CO_2, adding to the greenhouse effect.	Loss of habitats and rising sea levels.
ACID RAIN	Burning coal and oil releases sulfur dioxide.	Harmful to trees, soil and ecosystems.
VISUAL POLLUTION	E.g. Coal mining, particularly open-cast mining, as well as power plants.	Makes a mess of the landscape.
OIL SPILLAGES	Accidents with tankers, pipelines or oil rigs.	Affects animals that live in and by the sea.
RADIATION EXPOSURE TO HUMANS AND ANIMALS	Radioactive nuclear waste is hard to dispose of safely, and nuclear catastrophes can release radiation.	Radiation can be very dangerous to all living things.

Some of these impacts can be minimised, e.g. removing sulfur from fuel before it's burned.

In theory, bio-fuels are carbon neutral...

When plants grow, they absorb CO_2 from the atmosphere. When you burn bio-fuels, you release CO_2. A bio-fuel is 'carbon neutral' if the amount of CO_2 released by burning it is equal to the amount absorbed by the plants as they grew.

Trends in Energy Resource Use

Quick Quiz

*Although we still rely on **non-renewables** for a lot of our energy needs at the moment, the balance is **shifting**.*

Currently we Still Depend on **Fossil Fuels**

See p.35 for more about the supply and demand of electricity.

1) Over the 20th century, the electricity use of the UK hugely increased as the population grew and people began to use electricity for more and more things.
2) Since the beginning of the 21st century, electricity use in the UK has been decreasing (slowly), as we get better at making appliances more efficient (p.10) and become more careful with energy use in our homes.
3) Some of our electricity is produced using fossil fuels and from nuclear power.
4) Generating electricity isn't the only reason we burn fossil fuels — oil (diesel and petrol) is used to fuel cars, and gas is used to heat homes and cook food.
5) However, we are trying to increase our use of renewable energy resources. This move towards renewable energy resources has been triggered by many things...

People Want to use **More Renewable** Energy Resources

- We now know that burning fossil fuels is very damaging to the environment (p.17). This makes many people want to use more renewable energy resources that affect the environment less.
- Because non-renewables will run out, some think it's better to get by without them before that happens.
- Pressure from other countries and the public causes governments to set targets for using renewable resources. This puts pressure on energy providers to build new power plants that use renewable resources to make sure they do not lose business and money.

Car companies have also been affected by the push for increased use of renewables — the popularity of electric cars and hybrids (cars powered by two fuels, e.g. petrol and electricity) is increasing.

The Use of **Renewables** is **Limited** by **Reliability**, **Costs** and **Politics**

There's a lot of scientific evidence supporting renewables, but although scientists can give advice, they don't have the power to make people, companies or governments change their behaviour (see p.136).

Reliability

Some renewables aren't as reliable as fossil fuels, or can't increase their output on demand. Switching completely to renewables would mean having to use a combination of different power plants or researching ways to improve reliability, which would be expensive.

Politics

Even if new power plants are built, there are arguments over where to put them. E.g. many people don't want to live next to a wind farm and there are arguments over whether it's fair to force them to.

Costs

- Building new renewable power plants costs money, so some energy providers are reluctant to do this, especially when continuing to use fossil fuels is so cost effective. These costs will have to be paid, either through energy bills or taxes. Some people don't want to or can't afford to pay, and there are arguments about whether it's ethical to make them.
- Research on improving the reliability and cost of renewables takes time and money.
- Making personal changes can also be expensive, e.g. hybrid and electric cars generally cost more than equivalent petrol cars. The cost is slowly going down, but they're still not an option for many people.

Switching to renewables isn't necessarily quick or easy...

With more people wanting to help the environment and non-renewables due to run out some day, the energy resources we use are changing. But high costs and reliability issues are slowing down the switch to renewables.

Warm-Up & Exam Questions

Here's a set of warm-up and exam questions for Topic 1b. They're not *too* horrendous, I promise.

Warm-Up Questions

1) What is meant by a 'renewable' energy resource?
2) Describe one way that renewable energy resources can be used to power vehicles.
3) Give three advantages of using solar power to generate electricity.
4) Give two ways in which using coal as an energy resource causes environmental problems.
5) Suggest two reasons why we can't just stop using fossil fuels immediately.

Exam Questions

1 The inhabitants of a remote island do not have the resources or expertise to build a nuclear power plant. They have no access to fossil fuels.

GRADE 4-6 GRADE

For question 3, you can use the formula sheet on page 216.

1.1 The islanders have considered using wind, solar and hydro-electric power to generate electricity. Suggest **two** other renewable energy resources they could use.

[2 marks]

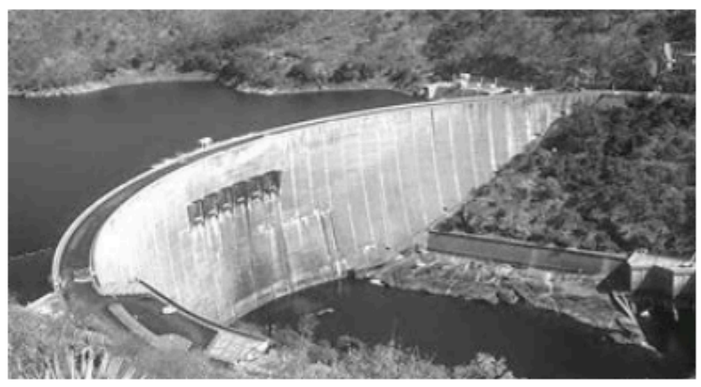
1.2 The islanders decide that hydro-electric power could reliably generate enough electricity for all their needs, but they are concerned about the environmental impact. Give **one** environmental impact of using hydro-electric power to generate electricity.

[1 mark]

2 In the hydro-electric power station in **Figure 1**, water is held back behind a dam before being allowed to flow out through turbines.

GRADE 4-6 GRADE

Figure 1



2.1 Describe the transfer between energy stores of the water which occurs during this process.

[2 marks]

2.2 The tides can also be used to generate electricity using tidal barrages. Give **two** environmental advantages of generating electricity using tidal barrages.

[2 marks]

3 A family want to install solar panels on their roof. They have 10 m^2 of space on their roof for the solar panels. They use 32 500 000 J of energy per day. A 1 m^2 solar panel has an output of 200 W in good sunlight.

GRADE 6-7 GRADE

3.1 Calculate the minimum number of 1 m^2 solar panels required to cover the family's daily energy use, assuming there are 5 hours of good sunlight in a day.

[5 marks]

3.2 Determine, using your answer from **3.1**, whether the family can install enough solar panels to provide all of the energy they use, assuming there are 5 hours of good sunlight every day.

[1 mark]

3.3 In reality, the number of hours of good sunlight in a day varies based on the weather and time of year. Discuss the reliability of energy from solar panels compared to from a local coal-fired power station.

[3 marks]

Revision Summary Test for Topic 1b

I hope your revision resources aren't running out just yet. To test what you've learnt, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

Energy Resources and Their Uses (p.14) ☑

1) Write down four examples of non-renewable energy resources.
   What are some pros and cons of non-renewable energy resources? ☑
2) List the seven types of renewable energy resources. ☑
3) Give two pros and two cons of using renewable energy resources. ☑
4) a) Describe three ways energy resources can be used for transport.
   b) Now describe six ways they can be used for heating. ☑

Renewable and Non-Renewable Energy Resources (p.15-17) ☑

5) Where are the most suitable places to put wind turbines? ☑
6) What are the advantages and disadvantages of using wind power? ☑
7) What kinds of devices are solar cells useful for? And what kinds of places is solar power useful in? ☑
8) What are four issues with using solar power? ☑
9) Where does the energy for geothermal power come from?
   Give two pros and two cons of using geothermal energy. ☑
10) How can hydroelectric power be used to meet changes in demand for electricity?
    What are two other advantages of using it? ☑
11) How does flooding a valley for hydroelectric power cause damage to the environment?
    Give two more disadvantages of using hydroelectric power. ☑
12) What are two advantages and four disadvantages of using wave power? ☑
13) What are tidal barrages and how do they work? ☑
14) What are four advantages of using tidal barrages? ☑
15) When do tidal barrages produce less energy? Give two other disadvantages of using them. ☑
16) What are bio-fuels made from? Why is there debate around whether bio-fuels are carbon neutral? ☑
17) Explain what makes bio-fuels reliable. What are four disadvantages of using bio-fuels? ☑
18) What makes using fossil and nuclear fuels relatively cost-effective? ☑
19) How does burning fossil fuels contribute to global warming?
    Give two negative impacts of global warming. ☑
20) Give the cause and impact of each of the following environmental problems:
    a) acid rain, b) visual pollution, c) oil spillages, d) radiation exposure to humans and animals. ☑

Trends in Energy Resource Use (p.18) ☑

21) a) What caused electricity use in the UK to increase in the 20th century?
    b) What has caused electricity use in the UK to decrease slowly at the start of the 21st century? ☑
22) Give two reasons that governments might introduce targets for using more renewable resources.
    How does this put pressure on energy providers? ☑
23) Suggest how the push for increased use of renewable energy resources has impacted car companies. ☑
24) How does the reliability of renewable energy resources limit their use? ☑
25) What are three ways in which costs limit the use of renewable energy resources? ☑

Current and Circuit Symbols

Quick Quiz

*Isn't **electricity** great? Mind you it's pretty bad news if the **words** don't mean anything to you...*

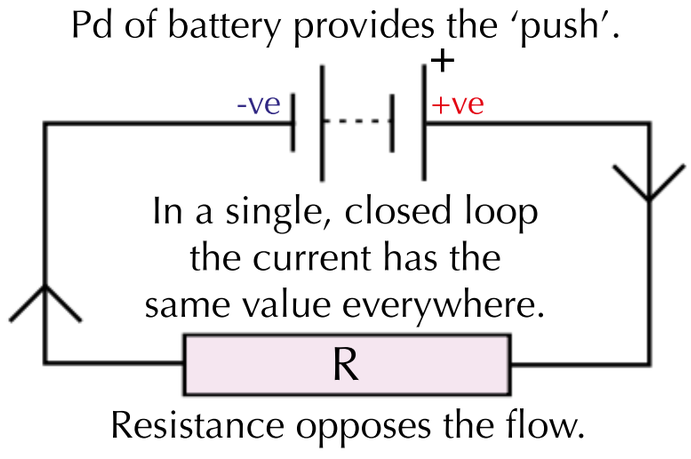
Current Depends on **Potential Difference** and **Resistance**

Term	Description	Units
Current	A flow of electrical charge.	Amperes, A
Potential difference	The driving force that pushes charge round.	Volts, V
Resistance	Anything that slows the flow of charge down.	Ohms, Ω

Potential difference is also called voltage or pd.

- There will only be a flow of electrical charge (a current) round a complete (closed) circuit if there is a source of potential difference.
- The current flowing through a component depends on the potential difference across it and the resistance of the component (p.22):

The greater the resistance, the smaller the current that flows (for a given potential difference).



Total Charge Through a Circuit Depends on **Current** and **Time**

The size of the current is the rate of flow of charge.
The charge that passes a point in a circuit in a given time is:

Charge flow (coulombs, C) — $Q = It$ — Current (A), Time (s)

EXAMPLE

A battery charger passes a current of 2.0 A through a cell over a period of 2.5 hours. How much charge passes through the cell?

1) Multiply to convert the time to seconds. t = 2.5 hours = (2.5 × 60 × 60) = 9000 s
2) Substitute the values into the formula. $Q = It = 2.0 \times 9000 = 18\,000$ C

MATHS SKILLS

Circuit Diagrams Use These **Symbols**

Cell	Battery	Switch open	Switch closed	Filament lamp (or bulb)	Fuse	LED
Resistor	Variable resistor	Ammeter	Voltmeter	Diode	LDR	Thermistor

- Along with these symbols, circuit diagrams use straight lines to represent wires.
- Circuits need to be closed for a current to flow, i.e. you can follow a wire from one end of the power supply, through any components, to the other end of the supply.

The size of an electric current equals the rate of flow of charge

Q1 A laptop charger passes a current of 8 A through a laptop battery. Calculate, in minutes, how long the charger needs to be connected to the battery for 28 800 C of charge to be transferred. [4 marks]

Q1 Video Solution

Quick Quiz

Resistance and $V = IR$

*Prepare yourself to meet one of the most **important equations** in electronics. It's all about **resistance**, **current** and **potential difference**... Now if that doesn't tempt you on to read this page, I don't know what will.*

There's a Formula Linking **Potential Difference** and **Current**

Potential difference (V) = Current (A) × Resistance (Ω)

$V = IR$

EXAMPLE

A 4.0 Ω resistor in a circuit has a potential difference of 6.0 V across it. What is the current through the resistor?

MATHS SKILLS

1) Rearrange $V = IR$ for current by dividing both sides by R. $V = IR \rightarrow I = V \div R$

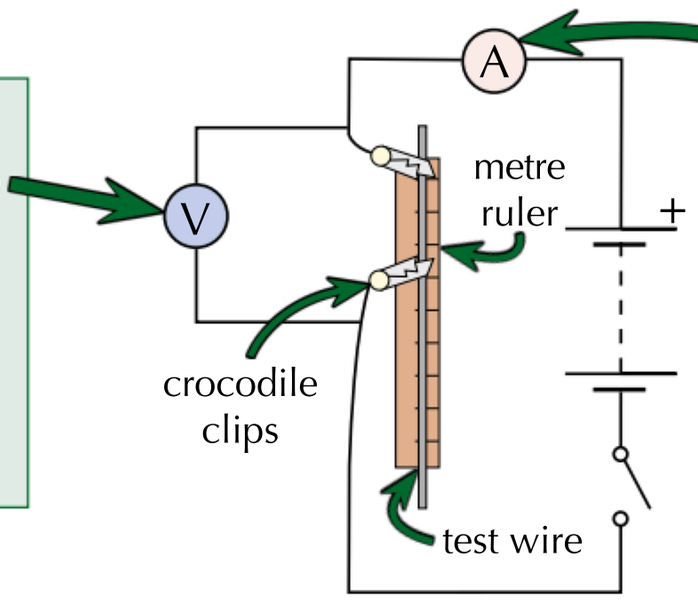
2) Substitute in the values you have. $I = V \div R = 6.0 \div 4.0 = 1.5$ A

The **Length of a Wire** in a Circuit Can Affect Its **Resistance**

PRACTICAL

You could use the circuit below to investigate the effect of wire length on resistance.

The VOLTMETER measures the potential difference (or pd) across the test wire (in volts). It must always be placed in parallel around whatever you're investigating — NOT any other bit of the circuit.



The AMMETER measures the current (in amps) flowing through the test wire. It must always be placed in series with whatever you're investigating.

See p.27-28 for more on series and parallel circuits, and p.29 to investigate how wiring in series and parallel can affect resistance.

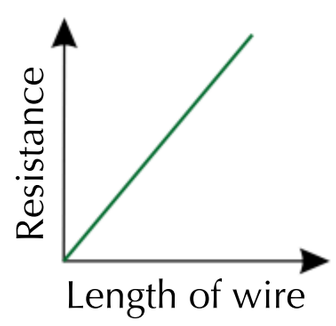
Method

1) Attach a crocodile clip to the wire level with 0 cm on the ruler.
2) Attach the second crocodile clip to the wire, *e.g.* 10 cm away from the first clip. Write down the length of the wire between the clips.
3) Close the switch, then record the current through the wire and the pd across it.
4) Open the switch, then move the second crocodile clip, *e.g.* another 10 cm, along the wire. Close the switch again, then record the new length, current and pd.
5) Repeat this for a number of different lengths of the test wire.
6) Use your measurements of current and pd to calculate the resistance for each length of wire, using $R = V \div I$ (from $V = IR$).
7) Plot a graph of resistance against wire length and draw a line of best fit.

A thin wire will give you the best results. Make sure it's as straight as possible so your length measurements are accurate.

The wire may heat up during the experiment, which will affect its resistance (p.23). Leave the switch open between readings to let the circuit cool down.

- Your graph should be a straight line through the origin, meaning resistance is directly proportional (p.144) to length — the longer the wire, the greater the resistance.
- If your graph doesn't go through the origin, it could be because the first clip isn't attached exactly at 0 cm, so all of your length readings are a bit out. This is a systematic error (p.139).



Resistance can be temperamental when temperature changes...

In general, resistance increases with temperature (though there are exceptions, like thermistors — see p.26).

Q1 An appliance is connected to a 230 V source. Calculate the resistance of the appliance if a current of 5.0 A is flowing through it. [3 marks]

Q1 Video Solution

Resistance and *I-V* Characteristics

Quick Quiz

*Time to look at just how **current** varies with **potential difference**. Then you can make some **sweet graphs**...*

Ohmic Conductors Have a Constant Resistance

- At a constant temperature, the resistance of an ohmic conductor (e.g. a wire or resistor) doesn't change with the current through the conductor.
- This means that the current flowing through an ohmic conductor at a constant temperature is directly proportional to the potential difference across it. (R is constant in $V = IR$.)

The resistance of some components DOES change with current, e.g. a diode or a filament lamp (see below).

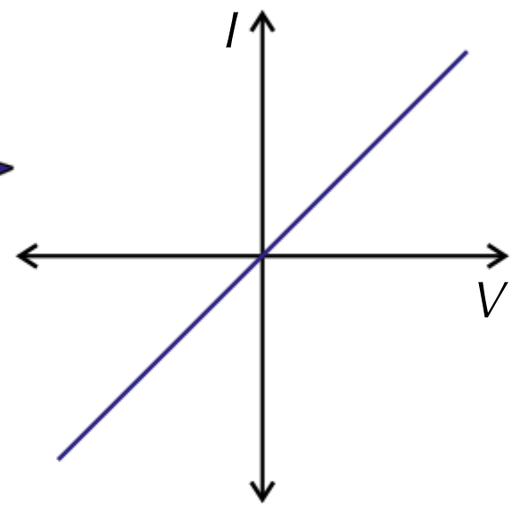
An *I-V* Characteristic is a Graph of Current Against Pd

The term '*I-V* characteristic' refers to a graph which shows how the current (I) through a component changes as the potential difference (V) across it increases.

Since $V = IR$, you can find the resistance at any point on an *I-V* characteristic by calculating $R = V \div I$.

Ohmic Conductors Have a Straight Characteristic

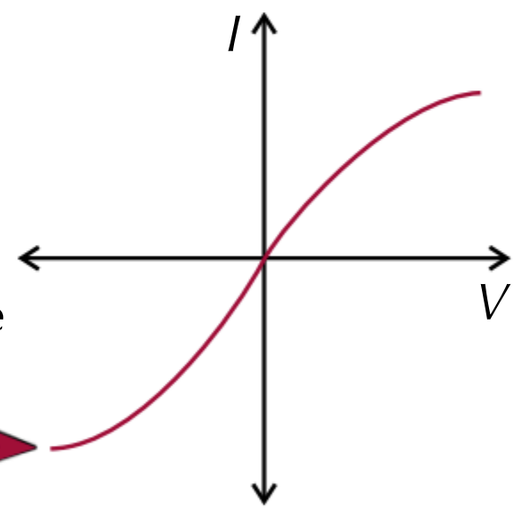
- The *I-V* characteristic of an ohmic conductor at a constant temperature is a straight line.
- This is because current and potential difference are directly proportional for ohmic conductors (see above).



Components with *I-V* characteristics that are straight lines are called linear components.

Filament Lamps Have a Funky Curve

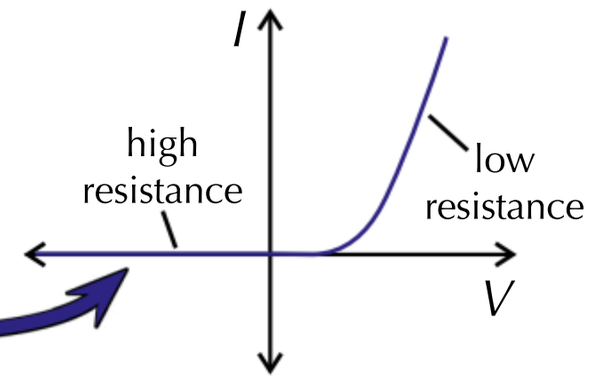
- When a current flows through a filament lamp, the moving electrical charges transfer some energy to the thermal energy store of the filament (p.3), which is designed to heat up.
- Increasing the current makes the filament heat up more and more. Because resistance increases with temperature, this means the resistance increases.
- This means less current can flow per unit pd, so the *I-V* characteristic gets shallower — the graph is a curve.



Components with curved *I-V* characteristics are called non-linear components.

Diodes Go From Flat to Steep Pretty Quickly

- For diodes, the resistance depends on the direction of the current.
- They will happily let current flow in one direction but have a very high resistance if the current is reversed.
- This means the *I-V* characteristic starts off flat, then curves up.



You may be asked to interpret an *I-V* characteristic...

Draw out those graphs until you're sketching them in your sleep — it's essential that you know them.

Q1 Explain the shape of the filament lamp *I-V* characteristic above, for the quadrant where I and V are positive. [3 marks]

Q1 Video Solution

Quick Quiz

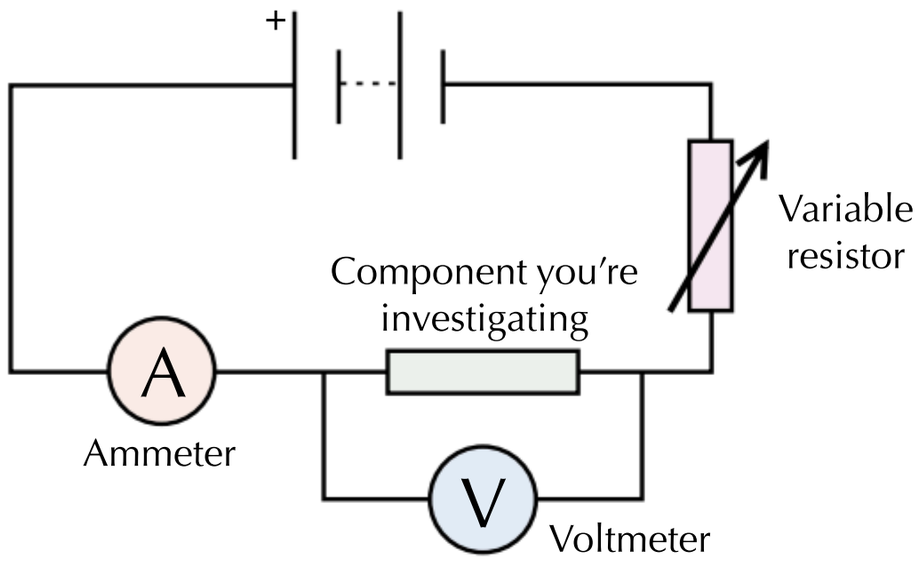
Investigating *I-V* Characteristics

PRACTICAL

*You've met **ohmic conductors** already on page 23, but most circuit components **aren't** ohmic.*
*You can see how circuit components behave by plotting an **I-V characteristic**.*

Use a Variable Resistor for **Filament Lamps** and **Ohmic Conductors**

One way to investigate the *I-V* characteristics of filament lamps or ohmic conductors is to use the test circuit shown below.



This alters the current flowing through the circuit and the potential difference across the component.

Method

1) Begin to vary the variable resistor.
2) Take several pairs of readings from the ammeter and voltmeter to see how the potential difference across the component varies as the current changes.
3) Swap over the wires connected to the battery, so the direction of the current is reversed. Use the same method as above to take more readings of pd and current.
4) Plot a graph of current against pd for the component (see p.141 for more on plotting graphs).

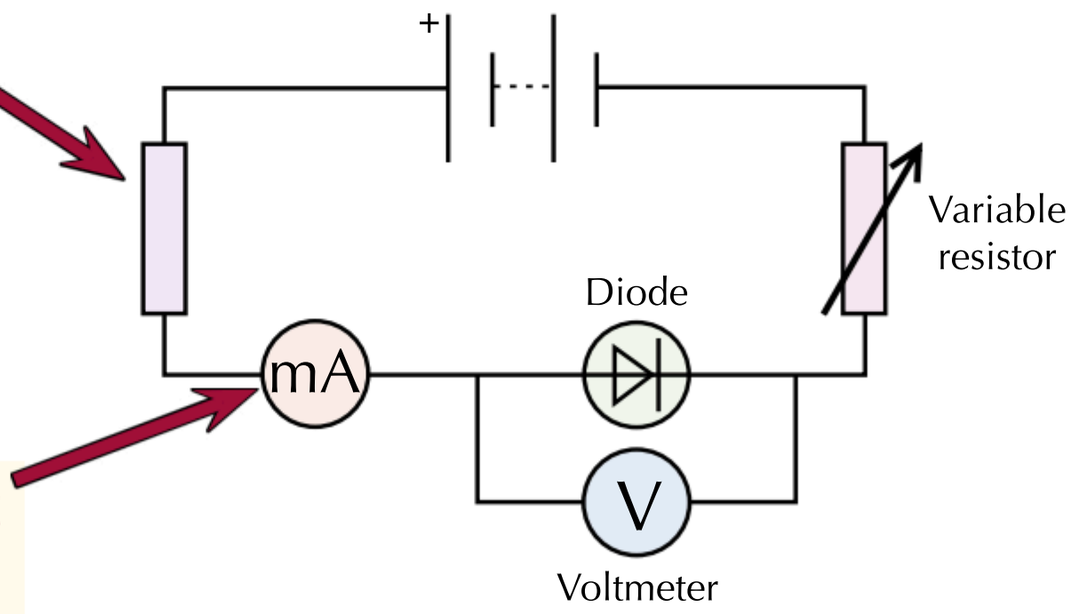
Your graphs should look like the ones on the previous page — a straight line for the ohmic conductor and a curve for the filament lamp.

You Need a **Protective Resistor** and **Milliammeter** for a **Diode**

You can do a very similar experiment to investigate the *I-V* characteristic of a diode.
The method is the same as for filament lamps and resistors, but you need to use a slightly different circuit:

You need to add a protective resistor.

- This is just a fixed resistor connected in series with the diode.
- It increases the overall resistance of the circuit, which protects the diode from too high a current.



You should use a milliammeter rather than a regular ammeter, since the current values you'll be measuring will be small.

The graph you plot should end up looking like the diode graph on the previous page.

A milliammeter has a higher resolution than a normal ammeter...

Diodes are greedy little so-and-sos when it comes to current — their resistance goes so low that above a certain potential difference, without a protective resistor, they'd draw as much current as the battery could supply. This would happen for a second or two before they burnt themselves out, which isn't very useful if you're investigating it...

Warm-Up & Exam Questions

Phew — circuits aren't the easiest thing in the world, are they? Make sure you've understood the last few pages by trying these questions. If you get stuck, just go back and re-read the relevant page.

Warm-Up Questions

1) What are the units of resistance?
2) How does current through a component vary with resistance for a fixed potential difference?
3) Draw the symbol for a light-emitting diode (LED).
4) How should a voltmeter be connected in a circuit to measure the pd across a component?
5) Give an example of an ohmic conductor.

Exam Questions

For these questions, you can use the formula sheet on page 216.

1 **Figure 1** shows is a circuit diagram for a standard test circuit. When the switch is closed, the ammeter reads 0.30 A and the voltmeter reads 1.5 V.

GRADE 6-7 GRADE

1.1 Calculate the resistance of the filament lamp.

[3 marks]

1.2 The switch is closed for 35 seconds.
Calculate the total charge that flows through the filament lamp.

[2 marks]

1.3 The variable resistor is used to increase the resistance in the circuit.
Describe how this will affect the current flowing through the circuit.

[1 mark]

Figure 1

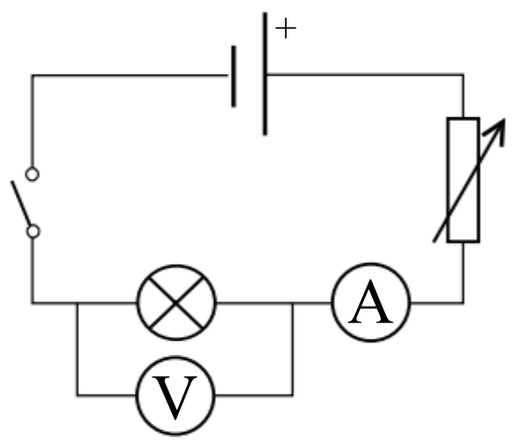
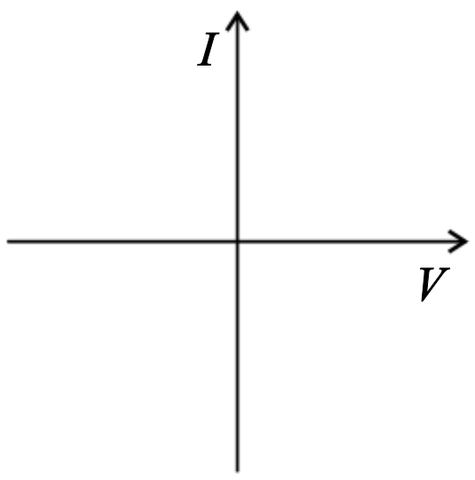


Figure 2



The resistance of a filament lamp changes with temperature.

1.4 On **Figure 2**, sketch the potential difference-current graph for a filament lamp.

[1 mark]

1.5 State what happens to the resistance of the filament lamp as the temperature of the filament increases.

[1 mark]

PRACTICAL

2 A student carried out an experiment using a standard test circuit where she varied the current and monitored what happened to the potential difference across a diode. **Figure 3** shows a graph of her results.

GRADE 6-7 GRADE

2.1 State the dependent variable in this experiment.

[1 mark]

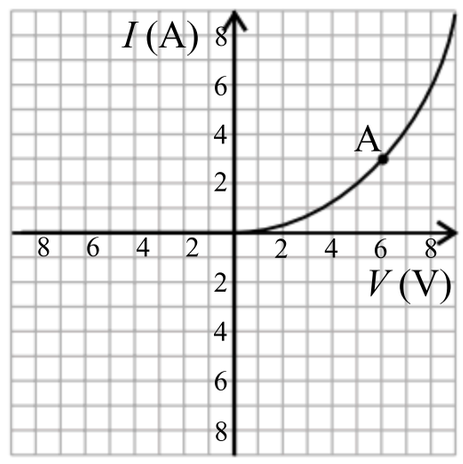
2.2 Explain why the graph in **Figure 3** shows zero current for negative pds.

[1 mark]

2.3 Calculate the resistance of the diode at the point marked A.

[4 marks]

Figure 3



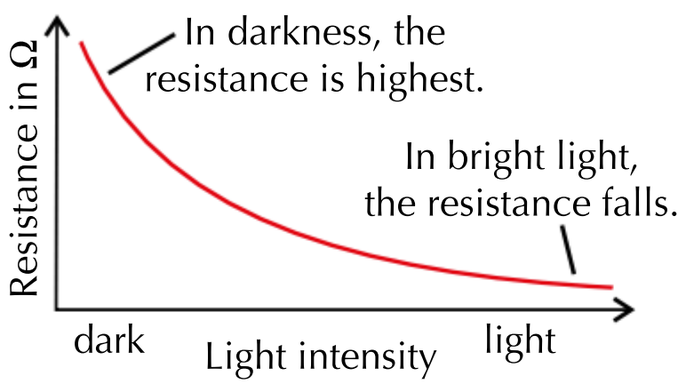
Circuit Devices

Quick Quiz

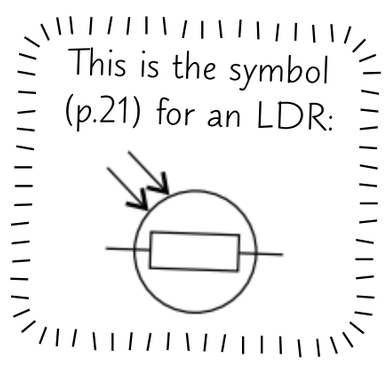
*It turns out that **resistance** changing with things like **temperature** (or even **light**) can be quite useful...*

LDR is Short for **Light Dependent Resistor**

An LDR is a resistor with a resistance that is dependent on the intensity of light. The clue's in the name.

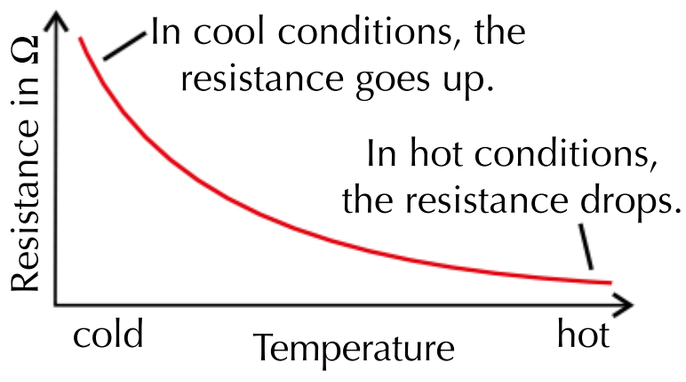


They have lots of applications including automatic night lights, outdoor lighting and burglar detectors.

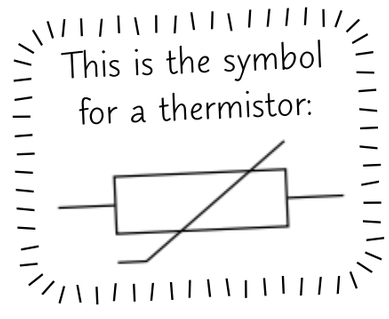


The Resistance of a **Thermistor** Depends on **Temperature**

A thermistor is a temperature-dependent resistor.



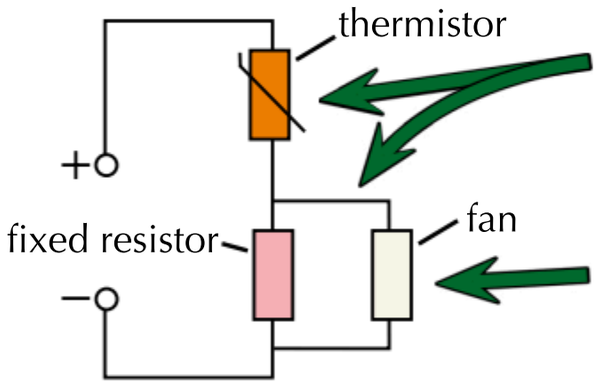
Thermistors make useful temperature detectors, e.g. car engine temperature sensors and electronic thermostats.



You Can Use LDRs and Thermistors in **Sensing Circuits**

Video

Sensing circuits can be used to turn on or increase the power to components depending on the conditions that they are in. The circuit below is a sensing circuit used to control a fan in a room.



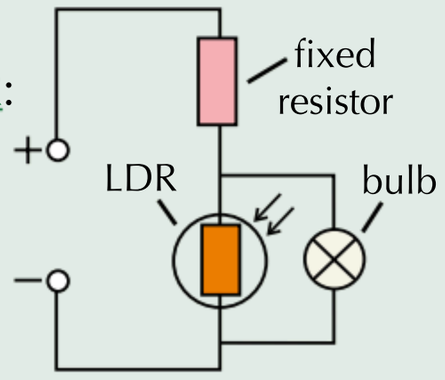
1) The pd of the power supply is shared out between the thermistor and the loop made up of the fixed resistor and the fan. It's split according to their resistances — the bigger the resistance, the more of the pd it takes.

2) The potential difference across the fan and the fixed resistor will always be equal (because they're connected in parallel — see p.28).

3) As the room gets hotter, the resistance of the thermistor decreases and it takes a smaller share of the pd from the power supply. So the pd across the fixed resistor and the fan rises.
The greater the pd across a component, the more energy it gets. This means the fan goes faster.

You can also connect the component you want to control across the LDR or thermistor instead. For example, if you connect a bulb in parallel with an LDR:

- The pd across both the LDR and the bulb will be high when it's dark and the LDR's resistance is high.
- So a bulb connected across an LDR would get brighter as the room got darker.



Thermistors and LDRs have many applications...

The uses of thermistors and LDRs aren't just limited to the examples on this page. Oh no. For example, LDRs are used in some cameras to control how long the shutter should stay open for. If the light level is low, changes in the resistance cause the shutter to stay open for longer than if the light level was higher. How interesting.

Series Circuits

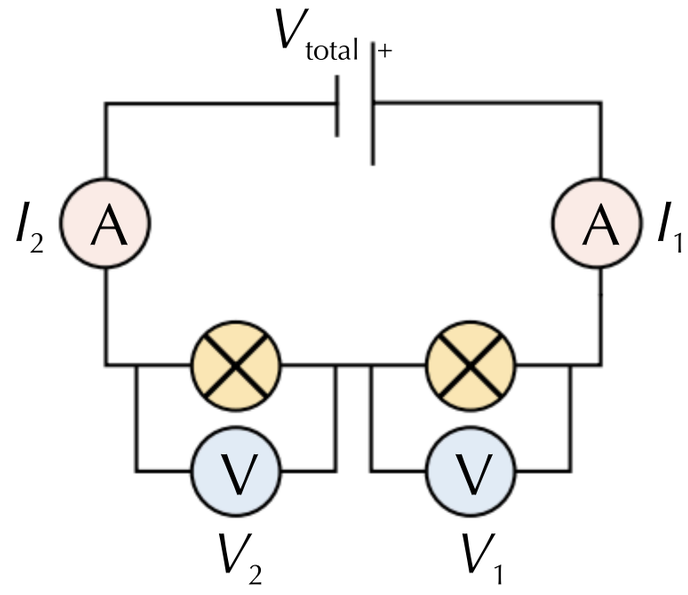
Quick Quiz

*You need to be able to tell if components are connected in series or parallel **just by looking at circuit diagrams**. You also need to know the **rules** about what happens with both types. Read on to find out more.*

Series Circuits Are Just Big Loops

1) In series circuits, the different components are connected in a line, end to end, between the +ve and –ve of the power supply.

Voltmeters are always connected in parallel, but they don't count as part of the circuit.



2) If you remove or disconnect one component, the circuit is broken and they all stop working. This is generally not very handy, and in practice very few things are connected in series.

3) Series circuits can be useful though, especially for measuring quantities and testing components (e.g. the test circuit on p.24).

4) You can use the following rules to analyse and design series circuits:

Potential difference	The total pd of the supply is shared between the components. The pds round the circuit add up to equal the source pd.	$V_{total} = V_1 + V_2 + ...$
Current	The same current flows through all components. The size of the current is determined by the total pd of the cells and the total resistance of the circuit: i.e. $I = V \div R$.	$I_1 = I_2 = ...$
Resistance	The total resistance of the circuit is the sum of the resistances of the components.	$R_{total} = R_1 + R_2 + ...$

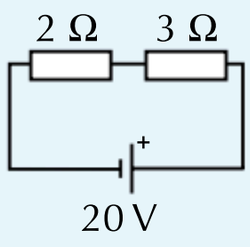
This is because when two resistors are connected in series, they have to share the total pd.

The potential difference across each resistor is lower, so the current through each resistor is also lower. In a series circuit, the current is the same everywhere, so the total current in the circuit is reduced when a resistor is added. This means the total resistance of the circuit has increased.

The bigger a component's resistance, the bigger its share of the total potential difference.

EXAMPLE

For the circuit diagram below, calculate the current passing through the circuit.



1) First find the total resistance by adding together the resistance of the two resistors. $R_{total} = 2 + 3 = 5\ \Omega$

2) Then rearrange $V = IR$ for I. $V = IR \rightarrow I = V \div R$

3) Substitute in the values you have. $I = V \div R = 20 \div 5 = 4$ A

MATHS SKILLS

Cell Potential Differences **Add Up** in Series Circuits

- There is a bigger pd when more cells are in series, if they're all connected the same way.
- For example, when two cells with a potential difference of 1.5 V are connected in series they supply 3 V between them.

Series circuits aren't used very much in the real world...

Once you think you've got your head around those rules, test yourself with the question below.

Q1 A battery is connected in series with a 4 Ω resistor, a 5 Ω resistor and a 6 Ω resistor. A current of 0.6 A flows through the circuit. Calculate the potential difference of the battery. [3 marks]

Q1 Video Solution

Quick Quiz

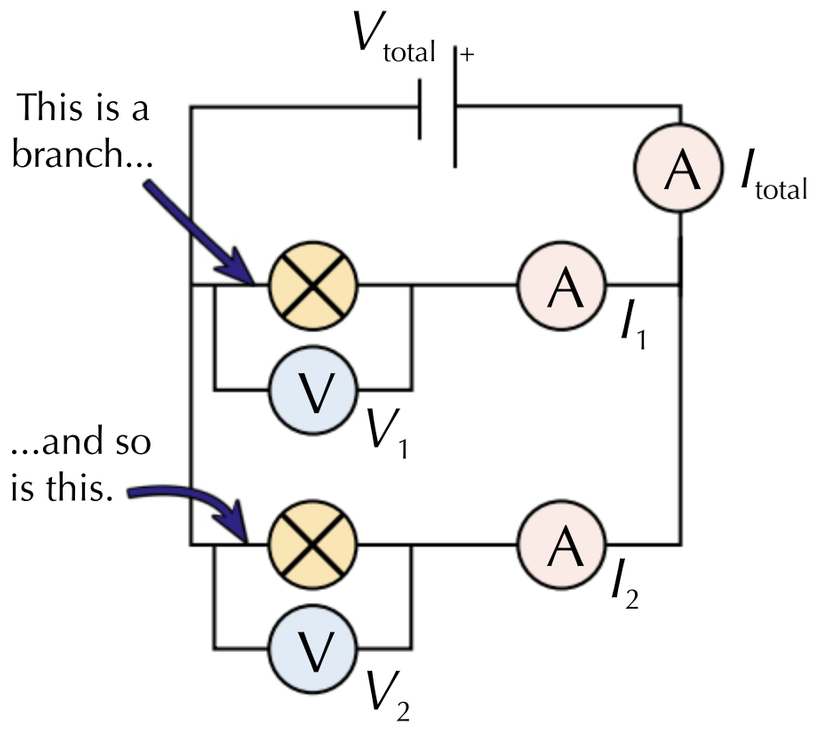
Parallel Circuits

Parallel circuits *are much more* ***sensible*** *than series circuits and so they're much more* ***common*** *in* ***real life****. For example, all the electrics in buildings, like your house, will probably be wired in parallel circuits.*

Parallel Circuits Are All About **Independence**

- In parallel circuits, each component is on its own loop (or branch) — each loop is separately connected to the supply.
- If you remove or disconnect one of them, it will hardly affect the others at all.
- This is obviously how most things must be connected, for example in cars and in household electrics — you have to be able to switch everything on and off separately.
- Some circuits have some components wired in series and some components wired in parallel.
- You can use these rules to analyse and design parallel circuits:

Ammeters are always connected in series, but they don't count as part of the circuit.



Potential difference	All components get the full source pd, so the pd is the same across all components.	$V_1 = V_2 = ...$
Current	The total current flowing around the circuit is equal to the total of all the currents through the separate branches. The total current going into a junction (where current splits or rejoins) has to equal the total current leaving.	$I_{total} = I_1 + I_2 + ...$
Resistance	The total resistance is less than the resistance of any of the individual resistors.	$R_{total} < R_1$ or R_2 or...

If two identical components are connected in parallel then the same current will flow through each component.

Adding a **Resistor** in Parallel **Reduces** the Total **Resistance**

1) In parallel, all resistors have the same potential difference across them as the source.
2) This means the 'pushing force' making the current flow is the same as the source pd for each resistor that you add.
3) But by adding another loop, the current has more than one direction to go in. This doesn't change the current through the existing branches — you're just adding on the current that flows through the new branch.
4) This means that total current that flows around the circuit increases.
5) Using $V = IR$, an increase in total current means a decrease in the total resistance of the circuit.

The current flowing into a junction equals the current out...

Q1 Video Solution

Parallel circuits are tricky, but they're very useful, so it's worth spending time learning how they work.

Q1 A circuit contains three resistors, each connected in parallel with a cell. Explain what happens to the total current and resistance in the circuit when one resistor is removed. [4 marks]

Q2 Draw a circuit diagram for two filament lamps connected in parallel to a battery. Both of the lamps can be switched on and off without affecting each other. [3 marks]

PRACTICAL

Investigating Resistance

Quick Quiz

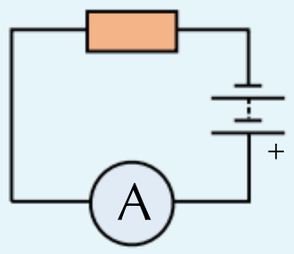
You saw on page 22 how the length of the wire used in a circuit affects its ***resistance****. Now it's time to do an* ***experiment*** *to see how placing* ***resistors*** *in series or in parallel can affect the resistance of the* ***whole circuit****.*

You Can **Investigate** Adding **Resistors in Series**...

This way of investigating resistance in series circuits calls for at least four identical resistors.

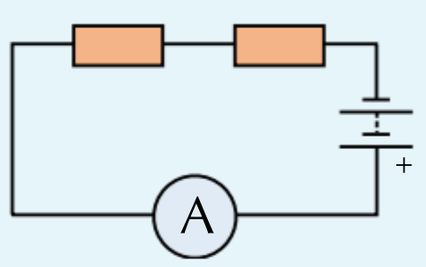
Method

1) Build the circuit shown on the right using one of the resistors. Make a note of the potential difference of the battery (V).



2) Measure the current through the circuit using the ammeter. Use this to calculate the resistance of the circuit using $R = V \div I$.

3) Add another resistor, in series with the first.



4) Again, measure the current through the circuit and use this and the potential difference of the battery to calculate the overall resistance of the circuit.

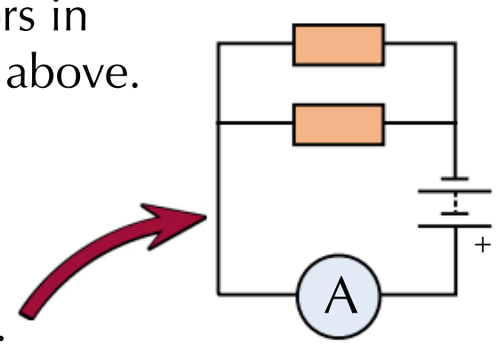
5) Repeat steps 3 and 4 until you've added all of your resistors.

6) Plot a graph of the total resistance of the circuit against the number of resistors (see below).

... or in **Parallel**

You can investigate the effect of adding resistors in parallel with a very similar method to the one above.

- Start with the same initial circuit.
- The only difference is that in step 3 (and every time you repeat it), you should add the resistor in PARALLEL with the first.

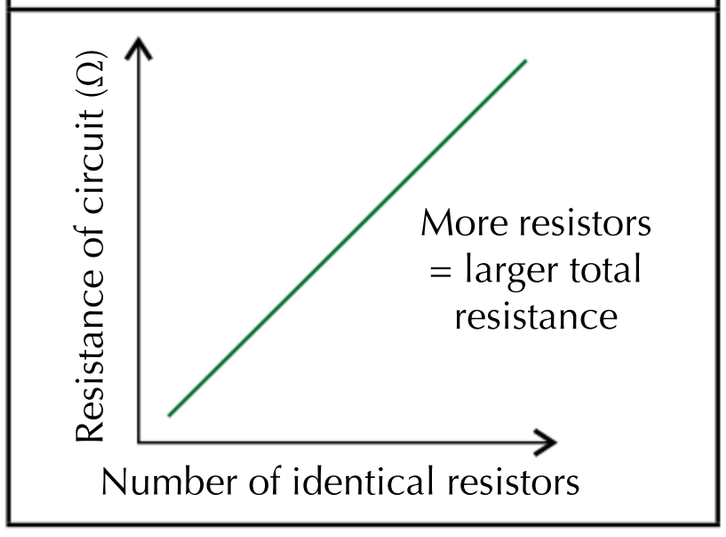


If you want to compare your results for series and parallel circuits, use the same equipment for each experiment. This ensures it's a fair test.

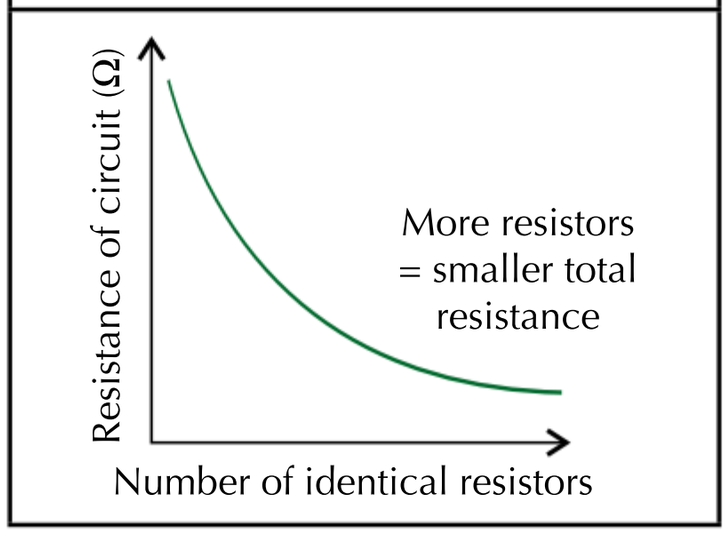
Your Results Should **Match** the **Resistance Rules**

You should find that your results from the two bits of the investigation show that:

Adding a resistor in SERIES...
...decreases the total current.
...increases the total resistance.



Adding a resistor in PARALLEL...
...increases the total current.
...decreases the total resistance.



These results agree with what you learnt about resistance in series and parallel circuits on pages 27 and 28.

Adding resistors in parallel will lower the resistance of the circuit...

Nothing too hard on this page, which makes a nice change from all of those rules about circuits. Make sure you remember that bit about how to make it a fair test — they love to ask about stuff like that in the exams...

Warm-Up & Exam Questions

Time to check and see what you can remember about those circuit devices, parallel and series circuits.

Warm-Up Questions

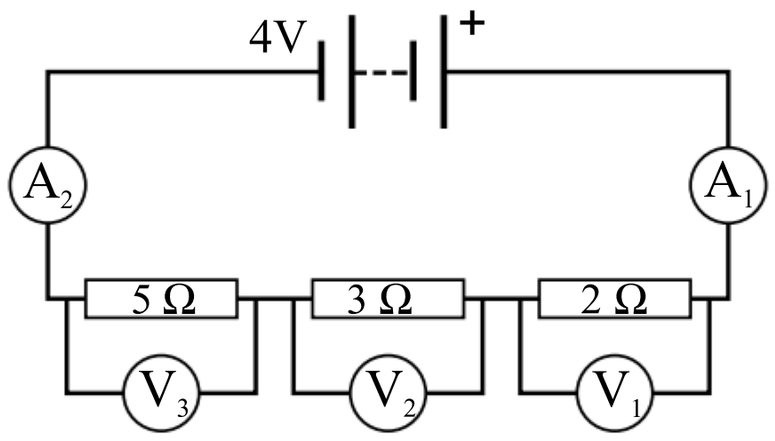
1) Give one use of a light-dependent resistor (LDR).
2) What happens to the resistance of a thermistor as its temperature increases?
3) Draw a circuit diagram of a sensing circuit where a bulb gets brighter with decreasing temperature.
4) Give one practical disadvantage of series circuits.
5) How do you work out the total resistance in a series circuit?
6) Which has the higher total resistance: two resistors in series, or the same two resistors in parallel?
7) Sketch a graph to show how the number of identical resistors connected together in parallel affects the total resistance in a circuit.

Exam Questions

For these questions, you can use the formula sheet on page 216.

1 **Figure 1** shows a series circuit. GRADE 4-6

Figure 1



1.1 Calculate the total resistance in the circuit.

[2 marks]

1.2 The current through A_1 is 0.4 A.
What is the current through A_2? Explain your answer.

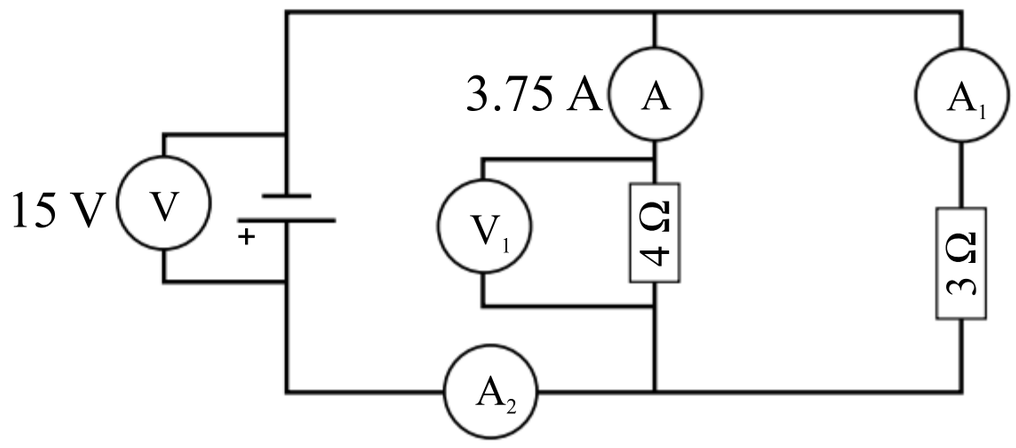
[2 marks]

1.3 V_1 reads 0.8 V and V_2 reads 1.2 V.
Calculate the reading on V_3.

[2 marks]

2 A parallel circuit is connected as shown in **Figure 2**. GRADE 7-9

Figure 2



2.1 Give the reading on voltmeter V_1.

[1 mark]

2.2 Calculate the reading on ammeter A_1.

[3 marks]

2.3 Calculate the reading on ammeter A_2.

[2 marks]

Revision Summary Test for Topic 2a

Your brain should now be charged with all that knowledge. To keep it flowing, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

Circuit Basics (p.21-22) ☑

1) What is meant by current, potential difference and resistance in a circuit? What are their units?
2) a) What causes current to flow around a closed loop?
   b) True or false? Current has the same value everywhere in a single, closed loop.
3) For a given potential difference, how does an increase in resistance affect the current?
4) How is the size of an electric current related to the rate of flow of charge?
5) What is the equation that links current, charge and time? What are the units for each term?
6) Draw the circuit symbols for a cell, battery, open switch, closed switch, filament lamp, fuse, LED, resistor, variable resistor, ammeter, voltmeter, diode, LDR and thermistor.
7) What is the equation that links potential difference, current and resistance?
8) What are ammeters and voltmeters used for? How should they each be connected in a circuit?
9) a) Explain how you would investigate how the length of a wire affects its resistance.
   b) What would you expect the results of this experiment to show?

I-V Characteristics and Resistance (p.23-26) ☑

10) What is an ohmic conductor? What two things are directly proportional for ohmic conductors (at a constant temperature)?
11) Why does the current through a filament lamp affect its resistance?
12) What is an *I-V* characteristic?
13) What is the difference between a linear component and a non-linear component?
14) Describe and explain the *I-V* characteristic of: a) a fixed resistor at a constant temperature, b) a filament lamp, c) a diode.
15) a) Describe how you could use a standard test circuit to find the *I-V* characteristic for a resistor.
    b) How would the circuit need to change for you to investigate the *I-V* characteristic of a diode?
16) How does the resistance of an LDR and a thermistor change with the conditions that they are in? Give a use for each.
17) a) Explain one way a thermistor could be used as part of a sensing circuit.
    b) Describe a sensing circuit that makes use of an LDR and a bulb.

Series and Parallel Circuits (p.27-29) ☑

18) What is a series circuit?
19) What is the rule relating the pds across the components of a series circuit to the source pd?
20) Describe how current behaves in a series circuit.
21) Explain how adding resistors in series affects a circuit's total resistance.
22) What is a parallel circuit?
23) What is the rule relating the pds across the components of a parallel circuit to the source pd?
24) Describe how current behaves in a parallel circuit.
25) Explain how adding resistors in parallel affects a circuit's total resistance.
26) Describe an experiment to investigate how the total resistance of a circuit changes when you add resistors: a) in series, b) in parallel.

Quick Quiz

Electricity in the Home

*Now you've learnt the basics of **electrical circuits**, it's time to see how **electricity** is used in **everyday life**.*

Mains Supply is **ac**, **Battery** Supply is **dc**

There are two types of electricity supplies — alternating current (ac) and direct current (dc).

Alternating current (ac)

- In ac supplies, the current is constantly changing direction. Alternating currents are produced by alternating voltages (p.126) in which the positive and negative ends keep alternating.
- The UK mains supply (the electricity in your home) is an ac supply at around 230 V. The frequency of the ac mains supply is 50 cycles per second or 50 Hz (hertz).

Direct current (dc)

- Cells and batteries supply direct current (dc).
- Direct current always flows in the same direction.
- It's created by a direct voltage.

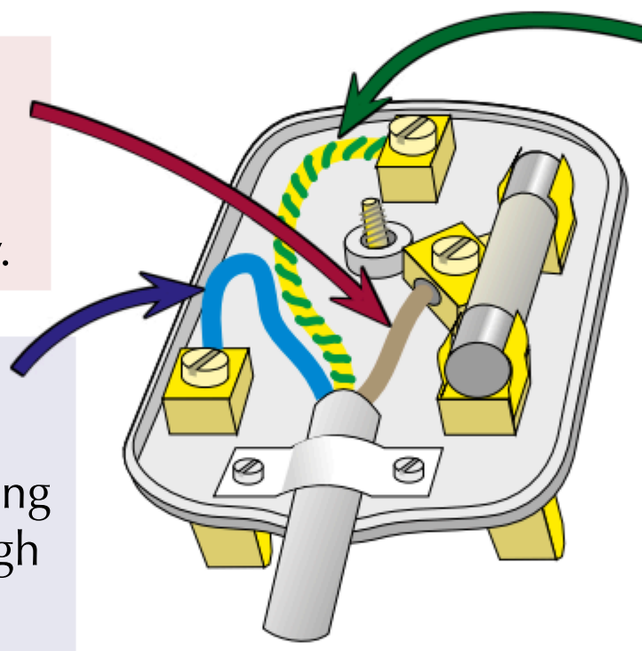
Most Cables Have **Three** Separate **Wires**

- Most electrical appliances are connected to the mains supply by three-core cables. This means that they have three wires inside them.
- The colour of the insulation on each wire shows its purpose.
- The colours are always the same for every appliance, so that it is easy to tell the different wires apart.

Each wire has a copper core and a coloured plastic coating.

1) LIVE WIRE — brown.
This provides the alternating potential difference (at about 230 V) from the mains supply.

2) NEUTRAL WIRE — blue.
This completes the circuit — when the appliance is operating normally, current flows through the live and neutral wires.
It is around 0 V.



3) EARTH WIRE — green and yellow.
It is for protecting the other wires, and for safety — it stops the appliance casing from becoming live. It doesn't usually carry a current — only when there's a fault. It's also at 0 V.

All the potential differences here are between the wires and the earth.

The **Live Wire** Can Give You an **Electric Shock**

1) If you touched the live wire when it's plugged in, your body would provide a link between the supply and the earth.
2) The earth is at 0 V, so this link means a large potential difference (pd) would be produced across your body, causing a current to flow through you.
3) This causes a large electric shock which could injure or even kill you.
4) Even if a plug socket or a light switch is turned off (i.e. the switch is open), there could still be a pd in the live wire. This means there would still be a large pd across your body if you touched the live wire, so you could still get an electric shock.

Any connection between live and earth can be dangerous. If the link creates a low resistance path to earth, a huge current will flow, which could result in a fire.

Plug sockets supply an alternating current...

You should know what each wire is for in a standard plug. Remember: the live wire provides the alternating potential difference, the neutral wire completes the circuit and the earth wire protects you if anything goes wrong. Good to know.

Power of Electrical Appliances

Quick Quiz

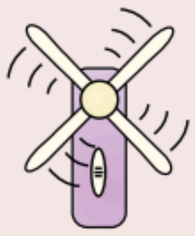
*You can think about **circuits** in terms of **energy transfer** — the charge carriers take energy around the circuit.*

Energy is Transferred from Cells and Other Sources

1) A moving charge transfers energy (see p.3) because it does work against the resistance of a circuit.
2) Electrical appliances are designed to transfer energy to components in the circuit when a current flows.

Kettles transfer energy electrically from the mains ac supply to the thermal energy store of the heating element inside the kettle.

Energy is transferred electrically from the battery of a handheld fan to the kinetic energy store of the fan's motor.



Remember, no appliance transfers all of its energy to useful stores (p.7).

Energy Transferred By an Appliance Depends on Power and Time

- The total energy transferred by an appliance depends on how long the appliance is on for and its power.
- The power of an appliance is the energy that it transfers per second.
  So the more energy it transfers in a given time, the higher its power.
- The amount of energy transferred by electrical work is given by:

Energy transferred (J) = Power (W) × Time (s) **$E = Pt$**

This equation should be familiar from page 7.

EXAMPLE

A 600 W microwave is used for 5 minutes. How long (in minutes) would it take a 750 W microwave to do the same amount of work?

MATHS SKILLS

1) Multiply to convert the time to seconds. — $t = 5 \text{ mins} = 5 \times 60 = 300 \text{ s}$
2) Calculate the energy transferred by the 600 W microwave in this time. — $E = Pt = 600 \times 300 = 180\,000 \text{ J}$
3) Rearrange $E = Pt$ for time by dividing both sides by P. — $E = Pt \rightarrow t = E \div P$
4) Calculate the time it takes for the 750 W microwave to transfer 180 000 J of energy. — $t = E \div P = 180\,000 \div 750 = 240 \text{ s}$
5) Divide to convert the time back to minutes. — $t = 240 \div 60 = 4 \text{ minutes}$

A Power Rating is the Maximum Safe Power an Appliance Can Run At

- Appliances are often labelled with a power rating. This is usually their maximum operating power.
- The power rating tells you the maximum amount of energy transferred between stores per second when the appliance is in use.
- This helps customers choose between models depending on what they want from an appliance:

The lower the power rating, the less electricity an appliance uses in a given time and so the cheaper it is to run.

The higher the power rating, the faster the energy transfer, so an appliance might not need to be on for as long.

A powerful, inefficient appliance might only transfer the same amount of energy (or less) to useful stores as a less powerful, more efficient model.

Power is the rate of energy transfer...

Q1 An appliance transfers 6000 J of energy in 30 seconds. Calculate its power. [2 marks]

Q2 Calculate the difference in the amount of energy transferred by a 250 W TV and a 375 W TV when they are both used for two hours. [4 marks]

Q2 Video Solution

Quick Quiz

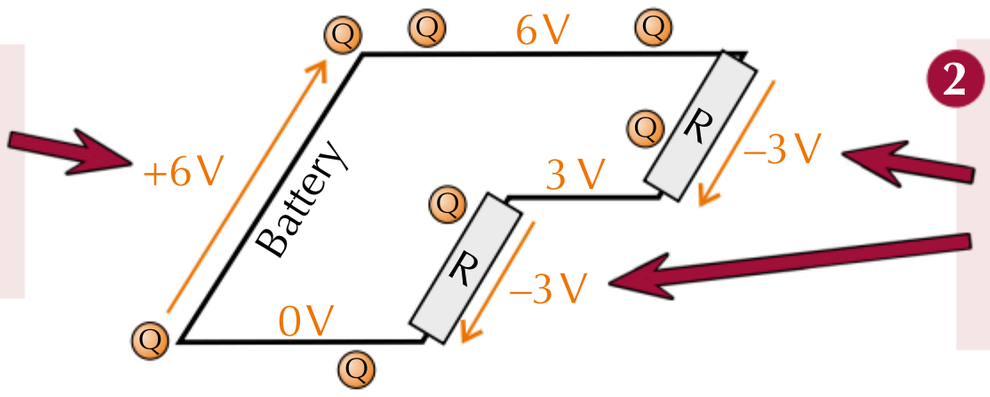
More on Energy Transfers and Power

*As you've seen, the **power** of a device tells you how much **energy** it transfers per second. In electrical systems, there are a load of useful formulas you can use to calculate energy and power.*

Potential Difference is **Energy Transferred** per **Charge Passed**

When an electrical charge goes through a change in electrical potential, then energy is transferred:

1 Energy is supplied to the charge at the power source to 'raise' it through a potential.



2 The charge gives up this energy when it 'falls' through any potential drop in components elsewhere in the circuit.

The formula is real simple:

Energy transferred (J) = Charge flow (C) × Potential difference (V) **$E = QV$**

That means that a battery with a bigger pd will supply more energy to the circuit for every coulomb of charge that flows round it, because the charge is raised up "higher" at the start.

EXAMPLE

The motor in an electric toothbrush is attached to a 3 V battery. 140 C of charge passes through the circuit as it is used. Calculate the energy transferred.

MATHS SKILLS

Just plug the numbers straight in. $E = QV = 140 \times 3 = 420$ J

This energy is transferred to the kinetic energy store of the motor, and to the thermal energy stores of the surroundings.

Power Also Depends on **Current** and **Potential Difference**

- As well as energy transferred in a given time, the power of an appliance can be found with:

Power (W) = Potential difference (V) × Current (A) **$P = VI$**

EXAMPLE

A 1.0 kW hair dryer is connected to a 230 V supply. Calculate the current through the hair dryer. Give your answer to two significant figures.

MATHS SKILLS

1) Rearrange the equation for current. $P = VI \rightarrow I = P \div V$
2) Make sure your units are correct. 1.0 kW = 1000 W
3) Then just stick in the numbers that you have. $I = 1000 \div 230 = 4.34... = 4.3$ A (to 2 s.f.)

- You can also find the power if you don't know the potential difference.

To do this, stick $V = IR$ (from page 22) into $P = VI$, which gives you:

Power (W) = Current^2 (A^2) × Resistance (Ω) **$P = I^2R$**

Power is measured in watts, W — one W is equal to one J/s...

Q3 Video Solution

Q1 Calculate the energy transferred from a 200 V source as 10 000 C of charge passes. [2 marks]

Q2 An appliance is connected to a 12 V source. A current of 4.0 A flows through it. Calculate the power of the appliance. [2 marks]

Q3 An appliance has a power of 2300 W and has a current of 10.0 A flowing through it. Calculate the resistance of the appliance. [3 marks]

The National Grid

Quick Quiz

*In case you didn't know — whoever you pay for your **electricity**, it's the **National Grid** that gets it to you.*

Electricity is Distributed via the **National Grid**

- The National Grid is a giant system of cables and transformers (p.127) that covers Great Britain.
- It transfers electrical power from power stations (the supply) to consumers (the demand — anyone who is using electricity) anywhere on the grid.

Electricity **Production** has to **Meet Demand**

1) Throughout the day, electricity usage (the demand) changes. Power stations have to produce enough electricity for everyone to have it when they need it.
2) Power stations often run at well below their maximum power output, so there's spare capacity to cope with a surge in demand, even if there's an unexpected shut-down of another station.
3) Lots of smaller power stations that can start up quickly are also kept on standby just in case.

Demand increases when people get up in the morning, come home from school or work and when it starts to get dark or cold outside.

The National Grid Uses a **High Pd** and a **Low Current**

To transmit the huge amount of power needed, you need either a high current or a high potential difference (as $P = VI$, from the previous page).

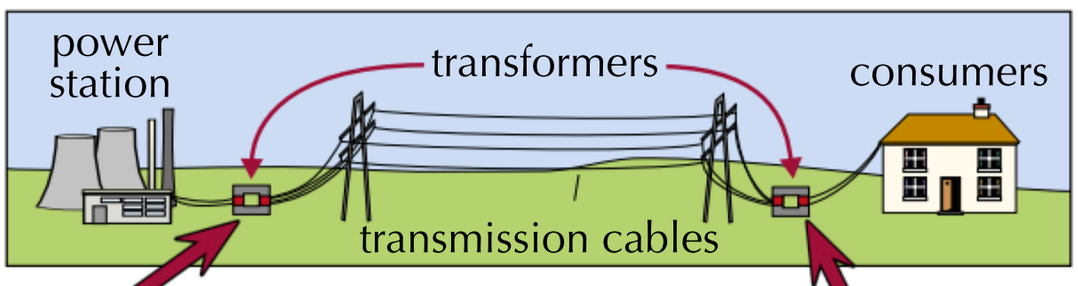
A high current means the wires heat up more, so lots of energy is lost to the thermal energy store of the surroundings.

- Instead, the National Grid transfers energy by boosting the pd up really high (400 000 V), which keeps the current as low as possible. This minimises the energy lost to heating the wires and surroundings.
- This is a much cheaper, more efficient way of transferring electricity.

A higher power means more energy transferred in a given time.

Potential Difference is Changed by a **Transformer**

To get the voltage up for efficient transmission, we use transformers (and big pylons with huge insulators).



For more on how transformers work, have a look at page 127.

1 • Potential difference (pd) is increased from the power station to the transmission cables using a step-up transformer.
• As pd is increased, current is decreased.

2 • The pd is then reduced again at the local consumer end using a step-down transformer. (This makes the pd safer for consumers.)
• As pd is decreased, current is increased.

Energy demands are ever increasing...

The National Grid has been working since the 1930s and has gone through many changes and updates to meet increasing energy demands. Using energy-efficient appliances and switching unneeded lights off can help to keep supply and demand in balance. It'll do wonders for your electricity bills too, as I'm sure your parents often remind you.

Warm-Up & Exam Questions

Who knew there was so much to learn about electricity in the home and across the country?
See if it's switched on a lightbulb in your brain by trying out these questions.

Warm-Up Questions

1) What are the three wires that make up a standard three-core cable?
2) What is the main energy transfer when electric current flows through an electric kettle?
3) What is the equation linking power, current and resistance?
4) What is the National Grid?

Exam Questions

For questions 2 and 4 you can use the formula sheet on page 216.

1 An electrical cable has become frayed so that the metal part of the live wire is exposed. Explain why you would get an electric shock if you touched it. GRADE 4-6 GRADE

[3 marks]

2 **Table 1** shows the power and potential difference ratings for two kettles. GRADE 6-7 GRADE

Table 1

	Power (kW)	Potential Difference (V)
Kettle A	2.8	230
Kettle B	3.0	230

2.1 State the equation linking power, potential difference and current.

[1 mark]

2.2 Calculate the current drawn from the mains supply by kettle A. State the correct unit.

[4 marks]

2.3 A student is deciding whether to buy kettle A or kettle B.
She wants to buy the kettle that boils water faster. Both kettles have an efficiency of 90%.
Suggest which kettle she should choose. Explain your answer.

[2 marks]

3 The National Grid transmits electricity from power stations to homes and businesses all over the country. GRADE 6-7 GRADE

Explain why the National Grid uses step-up transformers.

[3 marks]

4 A current of 0.5 A passes through a torch bulb. The torch is powered by a 3.0 V battery. GRADE 7-9 GRADE

4.1 The torch is on for half an hour.
Calculate the amount of energy transferred from the battery in this time.

[4 marks]

4.2 Calculate how much charge passes through the torch in half an hour.

[3 marks]

Static Electricity

Quick Quiz

*Static electricity is all about charges which are **not** free to move, e.g. in **insulating** materials.*

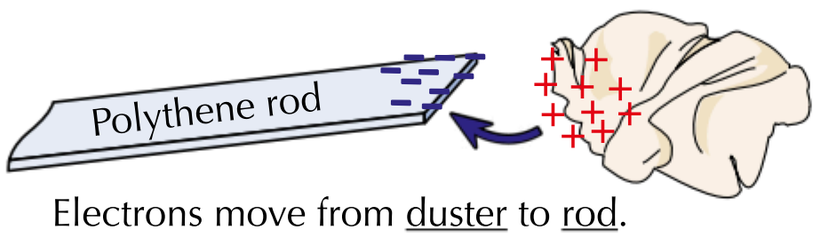
Build-up of **Static** Can Be Caused by Objects **Rubbing Together**

- When certain insulating materials are rubbed together, negatively charged electrons are transferred from one material to the other.
- This leaves both materials electrically charged, with a positive static charge on the one that loses electrons and an equal negative static charge on the one that gains electrons.
- Which way the electrons are transferred depends on the two materials involved.

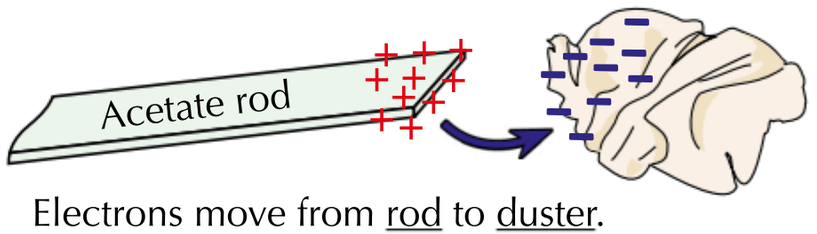
Both +ve and –ve electrostatic charges are only ever produced by the movement of ELECTRONS. The positive charges definitely DO NOT MOVE.

(The material that loses electrons loses some negative charge, so is left with a positive charge.)

The classic examples are polythene and acetate rods being rubbed with a cloth duster.



Electrons move from duster to rod.



Electrons move from rod to duster.

Video

Too Much Static Causes **Sparks**

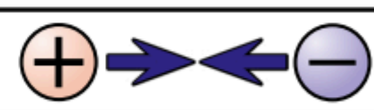
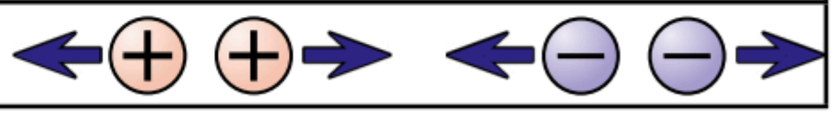
1) As electric charge builds on an object, the potential difference between the object and the earth (which is at 0 V) increases.
2) If the potential difference gets large enough, electrons can jump across the gap between the charged object and the earth — this is the spark.
3) This usually happens when the gap is fairly small. (But not always — lightning is just a big spark.)

For more on how sparks actually jump across gaps, see the next page.

Electrons can also jump to any nearby earthed conductor — which is why you can get static shocks getting out of a car. Charge builds up on the car's frame, and when you touch the car, the charge travels through you to earth.

Like Charges Repel, **Opposite** Charges Attract

1) When two electrically charged objects are brought close together they exert a force on one another.
2) This is electrostatic force. It's a non-contact force (the objects don't need to touch, p.61).

Objects with opposite types of charge...	...attract each other	
Objects with the same type of charge...	...repel each other	

3) The force on each object gets weaker the further apart the two things are.
4) These forces will cause the objects to move if they are able to do so. One way to see this is to suspend a rod with a known charge from a piece of string (so it is free to move):
   - Placing an object with the same charge nearby will cause the rod to move away from the object.
   - Using an oppositely charged object will cause the rod to move towards the object.

It shouldn't shock you to learn that static charges are everywhere...

You'll probably be pretty familiar with static charges from your day-to-day life. Whenever you're taking off a jumper or you brush your hair in a way that makes your hair stand on end — that's static electricity.

Electric Fields

Quick Quiz

As you've seen on the last page, ***electrical charges*** *exert a force on each other. You can explain these forces in terms of* ***electric fields****. They can be used to explain things like* ***sparks*** *as well.*

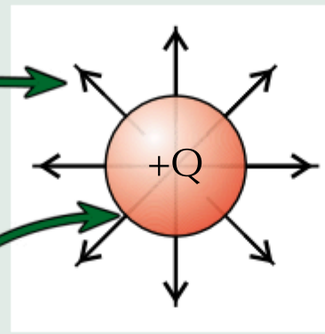
Electric **Charges** Create an **Electric Field**

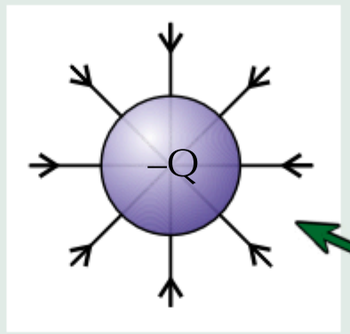
- An electric field is created around any electrically charged object.
- The closer to the object you get, the stronger the field is.
  (And the further you are from the object, the weaker it is.)
- You can show an electric field around an object using field lines.
  For example, you can draw the field lines for an isolated, charged sphere:

Isolated means it's not interacting with anything.

Electric field lines go from positive to negative — out of a positive sphere and into a negative one.

The lines are always at a right angle to the surface.





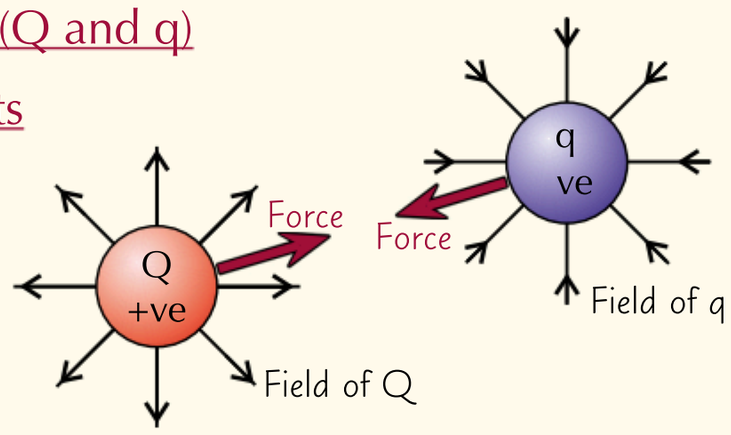
The closer together the lines are, the stronger the field is. The further from a charge you go, the further apart the lines are and so the weaker the field is.

Charged Objects in an Electric Field Feel a **Force**

- When a charged object is placed in the electric field of another object, it feels a force.
- The force is caused by the electric fields of each charged object interacting with each other.
- This force causes the attraction or repulsion you saw on the previous page. For example:

Two oppositely charged particles (Q and q)

1) The electric field of Q interacts with the electric field of q.
2) This causes forces to act on both Q and q.
3) These forces move Q and q closer together.



The interaction of the two fields shown in the diagram actually creates a more complicated field, but you don't need to worry about that at GCSE.

- The size of the force on an object is linked to the strength of the electric field it is in.
- As you increase the distance between the charged objects, the strength of the field decreases and the force between the objects gets smaller.

Sparking Can Be **Explained** By **Electric Fields**

1. A high potential difference between a charged object and an earthed object causes a strong electric field between the objects.
2. The strong electric field causes electrons in the air particles to be removed (known as ionisation).
3. Being ionised makes the air much more conductive, so a current can flow through it — this is the spark.

Anything that has a charge will have an electric field...

Electric fields work like gravitational fields (p.62) and magnetic fields (p.119) but for things that are charged — the closer a charged object is to something else that's creating an electric field, the stronger the force it will feel.

Warm-Up & Exam Questions

Have you been paying close attention to the last two pages? Only one way to tell really — have a go at these delightful warm-up and exam questions. If you get any wrong, go back and read the pages again.

Warm-Up Questions

1) Two objects are rubbed together and become charged.
What particles move to cause the objects to become charged?
2) What happens to an acetate rod when it is rubbed by a cloth?
3) What happens to the strength of an electric field as you move further away from a charged object?

Exam Questions

1 A student rubbed a plastic sphere with a cloth, and the sphere became negatively charged. The charge on the sphere is –1.6 nC.

GRADE 4-6 GRADE

1.1 State the charge on the cloth.

[1 mark]

1.2 Describe the movement of charge which caused the sphere to become charged.

[1 mark]

An image of the charged sphere is shown in **Figure 1**.

Figure 1

1.3 Give the position (A, B or C) where a test charge would feel the greatest force.

[1 mark]

1.4 Draw an arrow on **Figure 1** showing the force a negative test charge would feel at that point.

[1 mark]

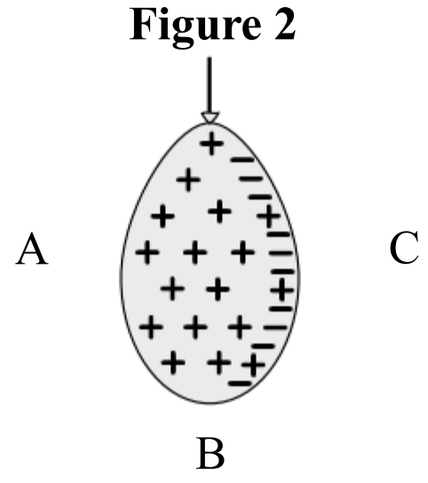
1.5 Describe how the presence of a strong electric field can cause a spark.

[2 marks]

2 Jaya hangs an uncharged balloon from a thread. She brings a negatively charged polythene rod towards the balloon. **Figure 2** shows how the positive and negative charges in the balloon rearrange themselves when she does this.

GRADE 7-9 GRADE

Figure 2



In which of the positions labelled A, B and C on the diagram did Jaya hold the polythene rod? Explain your answer.

[3 marks]

Revision Summary Test for Topic 2b

Well done for powering through that section. To keep the spark of knowledge alive, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

Electricity in the Home (p.32) ☑

1) What is the difference between ac and dc? What type of current does a battery supply? ☑
2) What is the potential difference and the frequency of the UK mains supply? ☑
3) What are the names and colours of the wires in a three-core cable? Why are they colour coded? ☑
4) Explain the function of each wire in a three-core cable. ☑
5) Give the potential difference between the wire and the earth for the three wires in a three-core mains cable. ☑
6) Explain why touching a live wire could be dangerous even if a plug socket is turned off. ☑

Power and the National Grid (p.33-35) ☑

7) Explain the energy transfers involved in using: a) a kettle, b) a handheld fan. ☑
8) Give three equations that can be used to calculate electrical power. ☑
9) What is the power rating of an appliance? How is it useful for consumers? ☑
10) What is the equation that relates energy transferred in a circuit to potential difference? ☑
11) What is the National Grid? What is it used for? ☑
12) Give two ways that the National Grid is set up to deal with surges in demand for electricity. ☑
13) Explain why electricity is transferred by the National Grid at a high pd but low current. ☑
14) What are the functions of step-up and step-down transformers? ☑

Static Electricity and Electric Fields (p.37-38) ☑

15) How does the rubbing of materials cause static electricity to build up? ☑
16) Which type of charge moves when static electricity is building up? ☑
17) Is electrostatic repulsion a contact or non-contact force? What does this mean? ☑
18) Do objects with the same type of charge attract or repel each other? How about objects with opposite types of charge? ☑
19) How do the forces between charged objects change when the objects move further apart? ☑
20) Explain how you can use a charged rod to demonstrate electrostatic attraction and repulsion. ☑
21) What type of object produces an electric field? ☑
22) How does the strength of an electric field change as you move closer to the object? ☑
23) In which direction do the arrows on electric field lines point? ☑
24) Describe the electric field lines around an isolated, positively charged sphere. How are they different for a negatively charged sphere? ☑
25) What happens when a charged object is placed in the electric field of another charged object? How does this change as the objects are moved further apart? ☑
26) Using the concept of electric fields, explain why a build-up of static electricity can cause a spark. ☑

Density of Materials

Quick Quiz

*The **particle model** of matter says that everything is made up of **lots** of **tiny particles**. It's really useful.*

The **Particle Model** can Explain **Density** and the **Three States of Matter**

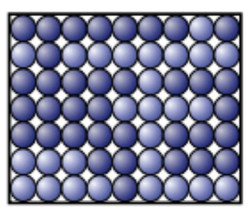
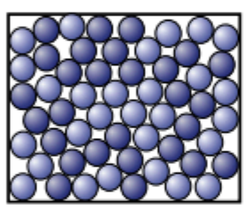
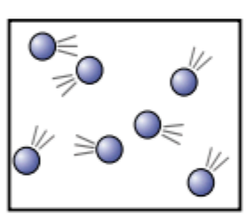
Density relates the mass of a substance to how much space it takes up (i.e. it's the mass per unit volume). It's usually given in kg/m^3 but can also be given in g/cm^3. 1 g/cm^3 = 1000 kg/m^3.

$$\text{Density (kg/m}^3\text{)} = \frac{\text{mass (kg)}}{\text{volume (m}^3\text{)}}$$

The symbol for density is a Greek letter rho (ρ).

$$\rho = \frac{m}{V}$$

- The density of an object depends on what it's made of and how its particles are arranged. The denser a material, the more tightly packed its particles are.
- Here's how the arrangement, energy and density of particles are different for each state of matter:

	Arrangement		Energy	Density
Solid		Strong forces hold the particles close together in a fixed, regular arrangement.	Not much energy — they vibrate around their fixed points.	Generally highest in this state.
Liquid		Weaker forces hold the particles close together, but they can move past each other, and form irregular arrangements.	More energy than solids — they move in random directions at low speeds.	Generally slightly less dense than solids.
Gas		There are almost no forces of attraction between the particles, so they are free to move.	More energy than liquids — they move in random directions at a range of higher speeds.	Generally much less dense than liquids.

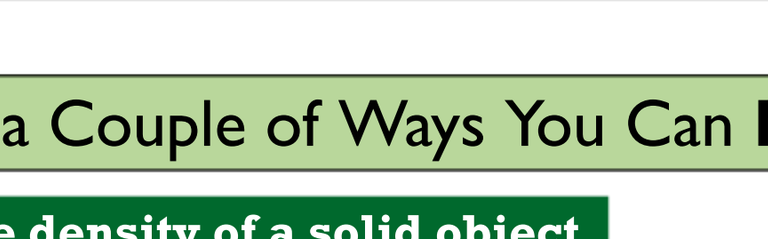
Video

Here are a Couple of Ways You Can **Measure Density**

PRACTICAL

To find the density of a solid object

1) Use a balance to measure its mass (see p.147).
2) For a regular solid, calculate its volume by measuring its dimensions (e.g. with a ruler) and using the relevant formula for that shape.
3) For an irregular solid, you can find its volume using a eureka can. The volume of water displaced by the object into the measuring cylinder is equal to the volume of the object.



4) Plug the object's mass and volume into the formula above to find its density.

To find the density of a liquid

1) Place a measuring cylinder on a balance and zero the balance.
2) Pour 10 ml of the liquid into the measuring cylinder (see p.148) and record the liquid's mass.
3) Pour another 10 ml into the measuring cylinder, repeating the process until the cylinder is full and recording the total volume and mass each time.
4) For each measurement, use the formula to find the density. (Remember that 1 ml = 1 cm^3.)
5) Finally, take an average of your calculated densities. This will reduce the effect of random errors on your results (see p.139).

Eureka cans help find the volume of irregularly shaped solids...

Remember — density is all about how tightly packed the particles in a substance are. Simple.

Q1 A 0.019 kg gemstone is placed into a full eureka can, causing 7.0 cm^3 of water to be pushed out the spout into a measuring cylinder. Calculate the density of the gemstone in g/cm^3. [3 marks]

Q1 Video Solution

Quick Quiz

Internal Energy and Changes of State

*This page is all about **heating** things up and **cooling** them down — and all the exciting things that happen then...*

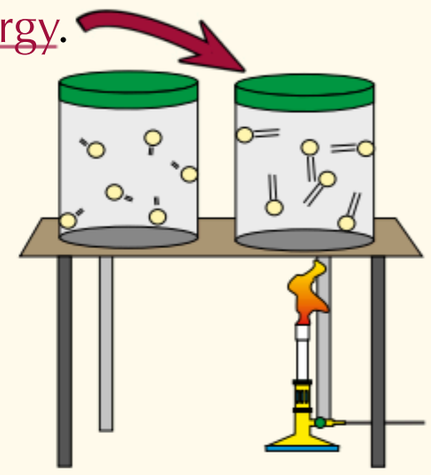
Internal Energy is Stored by the **Particles** That Make Up a **System**

The particles in a system (atoms and molecules) can store energy in their:

- Kinetic energy stores — energy due to movement or vibration of the particles.
- Potential energy stores — energy due to the positions of the particles.

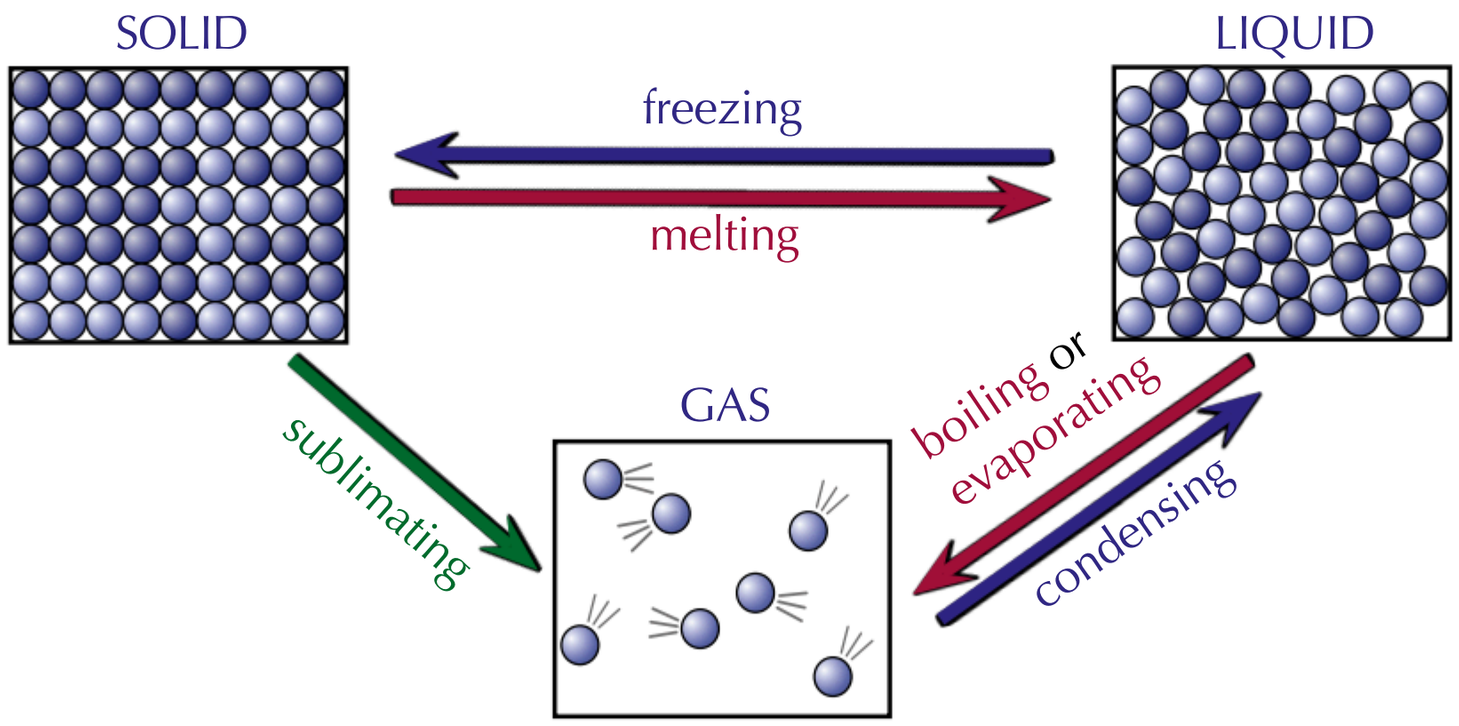
The internal energy of a system is the total energy that its particles have in their kinetic and potential energy stores.

1) Heating the system transfers energy to its particles, increasing the internal energy.
2) This leads to a change in temperature or a change in state.
   For a change in temperature, the particles gain energy in their kinetic energy stores and move faster. Make sure you can remember all of the stuff on specific heat capacity from p.5 — particularly how to use $\Delta E = mc\Delta\theta$.
3) A change in state occurs if the substance is heated enough — energy goes into breaking the bonds between the particles, increasing the energy in the potential energy stores of the particles.
4) Cooling the system reduces the energy in the kinetic energy stores of the particles. You get a change in state when bonds form between the particles, reducing the energy in their potential energy stores.



A **Change** of **State Conserves Mass**

When you heat a liquid, it boils and becomes a gas. When you cool a gas, it condenses and becomes a liquid. These are both changes of state. The changes of state are:



1) A change of state is a physical change (rather than a chemical change). This means you don't end up with a new substance — it's the same substance as you started with, just in a different form.
2) If you reverse a change of state (e.g. freeze a substance that has been melted), the substance will return to its original form and get back its original properties.
3) The number of particles doesn't change — they're just arranged differently. This means mass is conserved — none of it is lost when the substance changes state.

The hotter a system is, the more internal energy it has...

...and the cooler a system is, the less internal energy it has. You also tend to find that solids have the least internal energy, gases have the most and liquids lie somewhere in that Goldilocks zone in the middle. I guess I'm a solid...

Specific Latent Heat

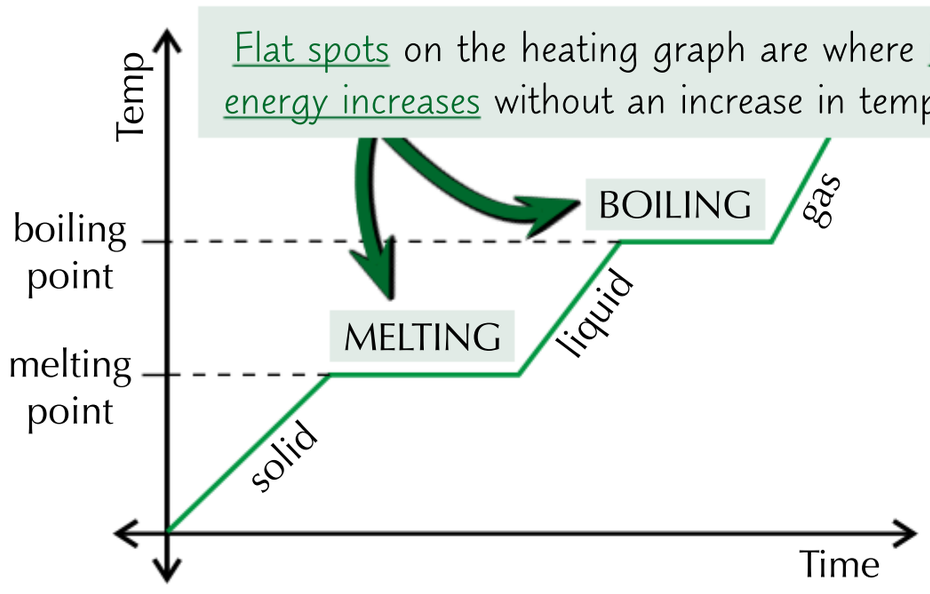
Quick Quiz

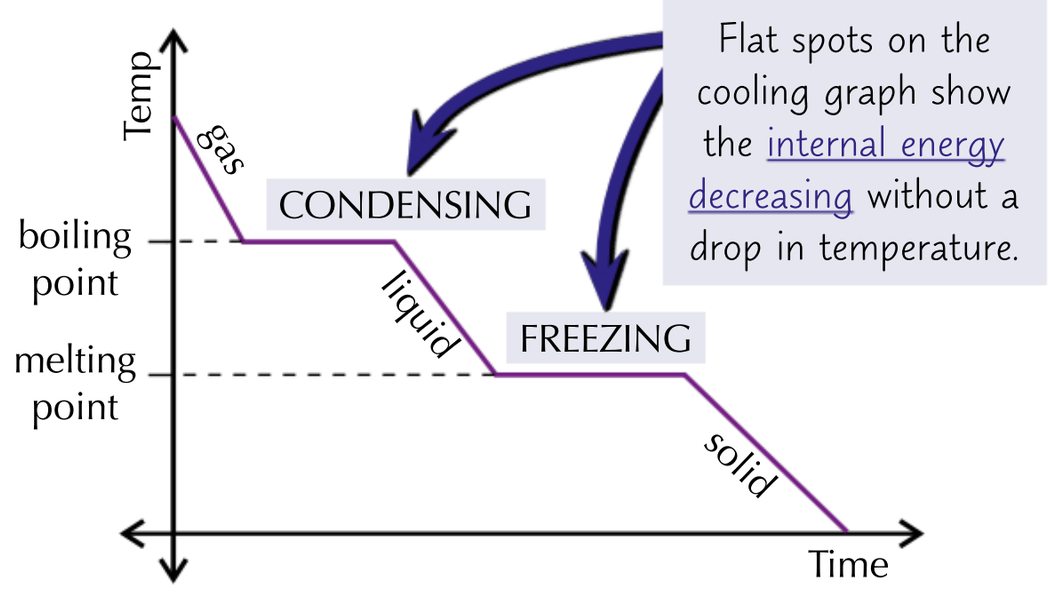
*If you heat up a pan of water on the stove, the water never gets any hotter than 100 °C. You can **carry on** heating it up, but the **temperature won't rise**. How come, you say? It's all to do with **latent heat**...*

Latent Heat is the Energy Transferred When There's a Change of State

1) When a substance is melting or boiling, energy is still transferred to the particles by heating, so internal energy increases. But this energy is used for breaking bonds between particles rather than raising the temperature.

2) When a substance is condensing or freezing, bonds form between particles, which releases energy. This decreases the internal energy, but the temperature doesn't go down until all the substance has changed state.





Specific Latent Heat is the Energy Needed to Change State

- The specific latent heat (SLH) of a substance is the amount of energy needed to change 1 kg of it from one state to another without changing its temperature.
- For cooling, specific latent heat is the energy released by a change in state.
- Specific latent heat is different for different materials, and for changing between different states.
- The specific latent heat for changing between a solid and a liquid (melting or freezing) is called the specific latent heat of fusion.
- The specific latent heat for changing between a liquid and a gas (evaporating, boiling or condensing) is called the specific latent heat of vaporisation.

There's a Formula for Specific Latent Heat

You can work out the energy needed (or released) when a substance of mass m changes state using this formula:

Energy (J) = Mass (kg) × Specific Latent Heat (J/kg) or: $E = mL$

Don't get confused with specific heat capacity (p.5), which relates to a temperature rise of 1 °C. Specific latent heat is about changes of state where there's no temperature change.

EXAMPLE

The specific latent heat of vaporisation for water (boiling) is 2 260 000 J/kg.
How much energy is needed to completely boil 1.50 kg of water at 100 °C?

1) Just plug the numbers into the formula.
2) The units are joules because it's energy.

$E = mL = 1.50 \times 2\,260\,000$
$= 3\,390\,000$ J

MATHS SKILLS

Temperature doesn't change during a change of state...

When it comes to the specific latent heat of vaporisation and fusion, the formula's the same, but the process is different. Make sure you understand which process you're actually looking at.

Q1 The SLH of fusion for a particular substance is 120 000 J/kg. How much energy is needed to melt 250 g of the substance when it is already at its melting temperature? [2 marks]

Q1 Video Solution

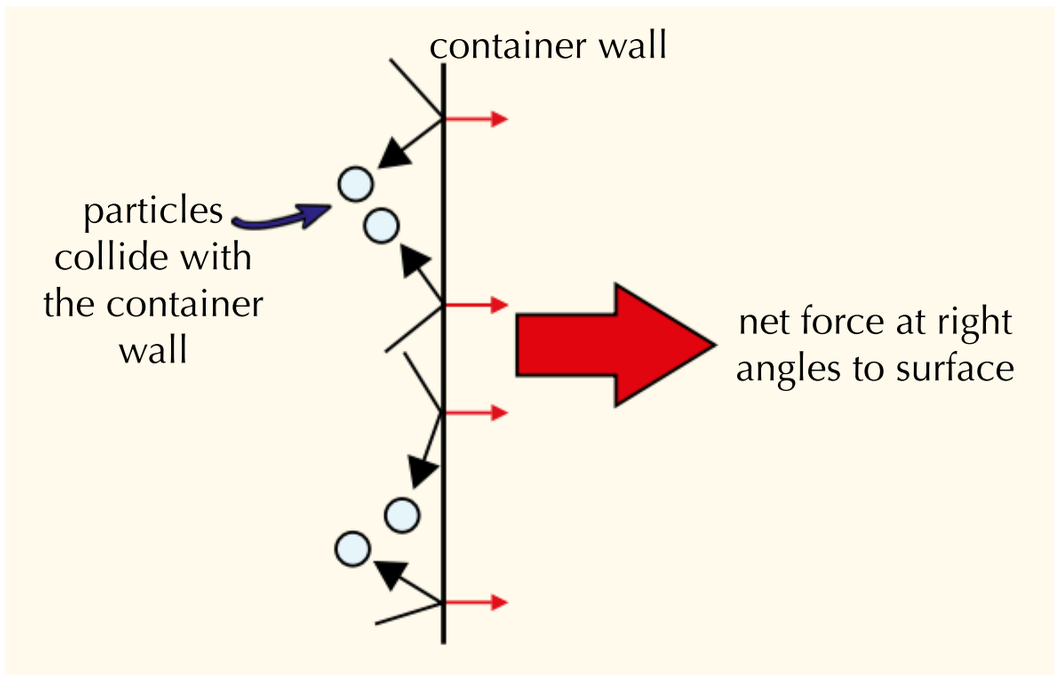
Particle Motion in Gases

Quick Quiz

*The **particle model** helps explain how **pressure**, **temperature** and the **energy** stored by particles are all related.*

Colliding Gas Particles Create **Pressure**

1) The particles in a gas are constantly moving in random directions at a range of speeds.
2) As the gas particles move about, they bang into each other and whatever else happens to get in the way.
3) When they collide with something, the particles exert a force and so a pressure on it — see page 70.
4) In a sealed container, the particles collide with the container walls and the net force is at right angles to the wall.
5) The outward gas pressure is the total force exerted by all of the particles in the gas on a unit area of the container walls. There's more about pressure in fluids on page 70.



Average Energy in **Kinetic Stores** is Related to **Temperature**

1) If you increase the temperature of a gas, you transfer energy into the kinetic energy stores of its particles (you saw this on p.42).
2) The temperature of a gas is related to the average energy in the kinetic energy stores of the particles in the gas:
3) So as you increase the temperature of a gas, the average speed of its particles increases. This is because the energy in the particles' kinetic energy stores is $\frac{1}{2}mv^2$ — p.4.

Higher temperature

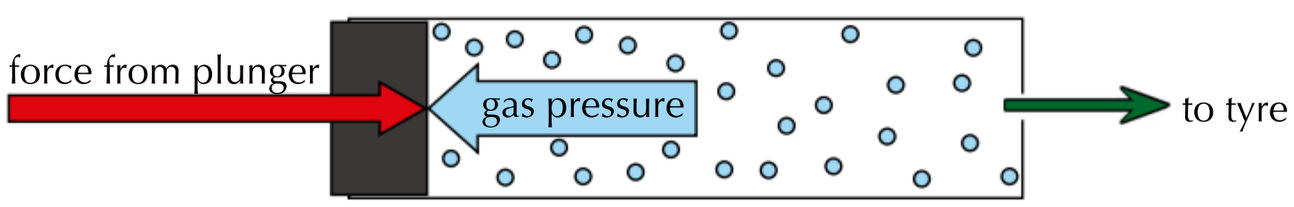
↓

Higher average energy

Doing Work on a Gas Can Increase its **Temperature**

There's more about doing work on p.63.

- If you transfer energy by applying a force, then you do work.
- Doing work on a gas increases its internal energy, which can increase its temperature.
- You can do work on a gas mechanically, e.g. with a bike pump. The gas inside the pump applies pressure to the plunger, and so exerts a force on it. Work has to be done against this force to push down the plunger.



- This transfers energy to the kinetic energy stores of the gas particles, increasing the temperature of the gas. If the pump is connected to a tyre, you should feel it getting warmer.

The more collisions in a given time, the higher the gas pressure...

That example of the bike pump is a great one — give it a try if you've got one kicking around. It doesn't just warm up a smidge — the metal bits on the pump can get really quite hot. Not that I've ever burnt myself... like a numpty...

Changes in Gas Pressure

Quick Quiz

*Since the **temperature**, **volume** and **pressure** of a gas are all related, changing one changes the others...*

Changing the **Temperature** or **Volume** of a Gas Changes its **Pressure**

Faster particles and more frequent collisions both lead to an increase in net force, and so gas pressure.

Change	Effect on particles	Effect on collisions	Effect on pressure
Higher temperature	Faster particles	Collisions are more frequent. Greater force from each collision.	Increases
Lower volume	Particles closer together	Collisions are more frequent.	Increases
Lower temperature	Slower particles	Collisions are less frequent. Smaller force from each collision.	Decreases
Higher volume	Particles more spread out	Collisions are less frequent.	Decreases

Pressure and **Volume** are **Inversely Proportional**

When volume goes up, pressure goes down (and when volume decreases, pressure increases).

For a gas of fixed mass at a constant temperature, the relationship is:

$$pV = \text{constant}$$

p = pressure, in pascals (Pa)
V = volume (m^3)

EXAMPLE

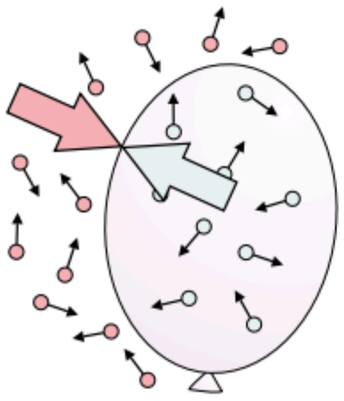
A bubble of gas starts with a volume of 0.4 m^3. The bubble then expands to a volume of 1.2 m^3 at a constant temperature. If the pressure of the gas at 1.2 m^3 is 100 000 Pa, what was the starting pressure?

MATHS SKILLS

1) Find the value of pV when $V = 1.2\ m^3$. $pV = 100\,000 \times 1.2 = 120\,000$
2) Rearrange pV = constant for p. $p = \text{constant} \div V$
3) Substitute in values for when $V = 0.4\ m^3$. $p = 120\,000 \div 0.4 = 300\,000$ Pa

A Change in **Pressure** can Cause a Change in **Volume**

1) The pressure of a gas causes a net outwards force at right angles to the surface of its container.
2) There is also a force on the outside of the container due to the pressure of the gas around it.
3) If a container can easily change its size (e.g. a balloon), then any change in these pressures will cause the container to compress or expand, due to the overall force.



If a helium balloon is released, it rises.

Atmospheric pressure decreases with height (p.71), so the pressure outside the balloon decreases.

This causes the balloon to expand and the pressure inside it to drop until the inward forces and outward forces balance.

Temperature, volume and pressure of a gas are all linked...

...that's why you get phrases like "at a constant temperature" in exam questions. You have to keep one of them constant to see how the other two are related — that way, a change in one is more clearly linked to a change in the other.

Warm-Up & Exam Questions

Once you think you've got to grips with everything in this topic, all the way through from the particle model to that stuff about gases, it's time to test yourself with these questions. Let's see how you get on.

Warm-Up Questions

1) What is density a measure of?
2) Describe the particles in a liquid in terms of their arrangement, energy and movement.
3) How does cooling a system affect its internal energy?
4) What is the specific latent heat of vaporisation?
5) What are the units of specific latent heat?

Exam Questions

For questions 3-6, you can use the formula sheet on page 216.

1 Substances can exist in different states of matter. GRADE 4-6

1.1 Describe the arrangement and movement of the particles in a solid.

[2 marks]

If a substance is heated to a certain temperature it can change from a solid to a liquid.

1.2 Give the name of this process.

[1 mark]

1.3 If a liquid is heated to a certain temperature it starts to boil and become a gas.
Name the other process that causes a liquid to start to become a gas.

[1 mark]

PRACTICAL

2 A student has a collection of metal toy soldiers of different sizes made from the same metal. GRADE 6-7

2.1 Which of the following statements about the toy soldiers is true? Tick **one** box.

- ☐ The masses and densities of each of the toy soldiers are the same.
- ☐ The masses of each of the toy soldiers are the same, but their densities may vary.
- ☐ The densities of each of the toy soldiers are the same, but their masses may vary.
- ☐ The densities and masses of each toy soldier may vary.

[1 mark]

The student wants to measure the density of one of the toy soldiers.
He has a eureka can, a measuring cylinder, a mass balance and some water.

2.2 State the **two** quantities the student must measure in order to calculate the density of the toy soldier.

[2 marks]

2.3* Describe the steps the student could take to find the density of the toy soldier using the equipment he has.

[6 marks]

Exam Questions

3 The following question is about specific latent heat. GRADE 6-7 GRADE

3.1 What is the name given to the specific latent heat of a substance when it's changing between a solid and a liquid?

[1 mark]

3.2 The energy required to convert 40.8 g of liquid methanol to gaseous methanol is 47 100 J.
Calculate the specific latent heat of vaporisation of methanol.
Give your answer in J/kg and to 3 significant figures.

[3 marks]

3.3 The energy used to change liquid methanol to gaseous methanol was supplied by heating the system.
How does the internal energy of the system change as the system is heated? Explain your answer.

[2 marks]

3.4 Explain, using the particle model, what happens when methanol is heated so that it changes from a liquid to a gas.

[2 marks]

4 The movement of particles in a gas can be described using the particle model. GRADE 6-7 GRADE

4.1 Describe the particles in a gas using the particle model.
In your answer, you should refer to the arrangement, energy and movement of the particles.

[3 marks]

A scientist is calculating the density of a gas in a sealed, rigid container.

4.2 Use the particle model to explain how gas particles create pressure in a sealed container.

[2 marks]

4.3 State the equation that links density with mass and volume.

[1 mark]

4.4 A certain gas had a mass of 8.2 g and a volume of 6.69 cm^3.
Calculate the density of the gas.
Give your answer to an appropriate number of significant figures.

[4 marks]

4.5 The scientist heats the container.
What happens to the pressure of the gas within the container?
Explain your answer using the particle model.

[3 marks]

5 A student is doing an investigation into the masses of different materials of different densities.
A cube has edges of length 1.5 cm and a density of 3500 kg/m^3.
Calculate the mass of the cube. GRADE 6-7 GRADE

[5 marks]

6 A sealed balloon contains 0.034 m^3 of gas at a pressure of 98 kPa.
The balloon is compressed to 0.031 m^3. The temperature of the air inside it remains constant. GRADE 7-9 GRADE
Calculate the air pressure inside the balloon after the compression.

[3 marks]

Revision Summary Test for Topic 3

I hope that section didn't leave you too hot under the collar. To make sure it sticks, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

Density of Materials (p.41) ☑

1) What is meant by the "density" of a material? What is the formula for density?
2) What are the three states of matter?
3) For each state of matter, describe: a) the arrangement of the particles, b) how the particles move.
4) How would you expect the energy and density of a material to be different in each of the three different states of matter?
5) Describe how you could find the volume of an irregular solid object.
6) Briefly describe an experiment to find the density of a liquid.

Internal Energy and Changes of State (p.42-43) ☑

7) What is the internal energy of a system?
8) What happens to the internal energy of a system when it is heated? What are the two possible effects this could have on the system?
9) Name all the possible changes of state between each of the three states of matter.
10) Is a change of state a physical change or a chemical change? What does that mean?
11) Why does the mass of a substance stay the same when it changes state?
12) Explain the cause of the flat sections on a graph of temperature against time for a substance being heated.
13) Define specific latent heat. Give a formula that relates energy transferred to specific latent heat.
14) What is the difference between the specific latent heat of vaporisation and the specific latent heat of fusion?
15) What is the difference between specific latent heat and specific heat capacity?

Particle Motion and Pressure in Gases (p.44-45) ☑

16) Explain how a gas in a sealed container exerts a pressure on the walls of the container.
17) Explain how the temperature of a gas and the average speed of its particles are related.
18) Explain why blowing up a football with a pump causes the ball to warm up.
19) A sealed container of gas is kept at a constant volume. The gas is heated. What happens to the pressure of the gas? Explain why.
20) True or false? For a gas at constant temperature, increasing the volume of the gas will also increase its pressure. Explain your answer.
21) For a fixed mass of gas at a constant temperature, what is the relationship between pressure and volume?
22) A balloon containing a fixed mass of helium gas is moved from an area of high atmospheric pressure to one of low atmospheric pressure. What will happen to the volume of helium in the balloon?

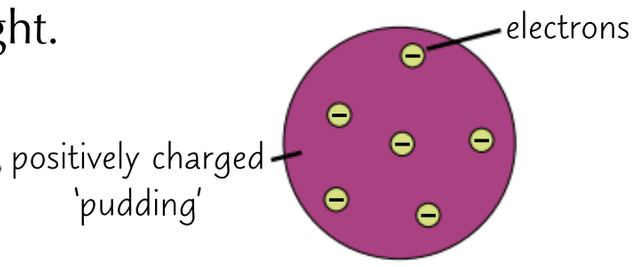
Developing the Model of the Atom

Quick Quiz

*All this started with a Greek fella called Democritus in the 5th Century BC. He thought that **all matter**, whatever it was, was made up of **identical** lumps called "atomos". And that's as far as it got until the 1800s...*

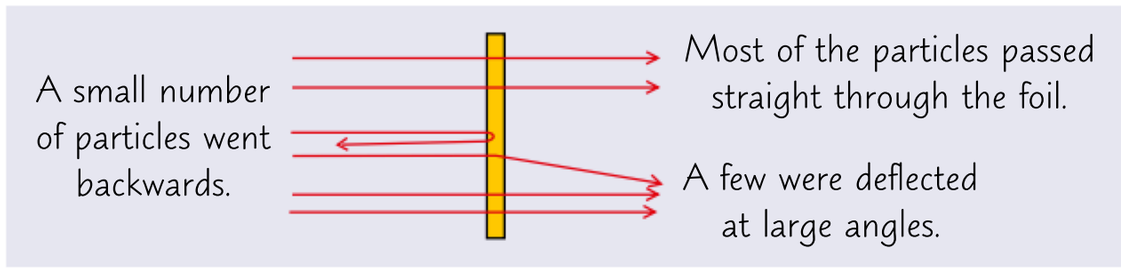
Thomson's **Plum Pudding Model** was the Accepted Model...

- In 1804 John Dalton agreed with Democritus that matter was made up of tiny spheres ("atoms") that couldn't be broken up, but he reckoned that each element was made up of a different type of "atom".
- Nearly 100 years later, J. J. Thomson discovered particles called electrons that could be removed from atoms, so Dalton's theory wasn't quite right.
- Thomson suggested atoms were spheres of positive charge with tiny negative electrons stuck in them like fruit in a plum pudding — the plum pudding model.



...Until Rutherford Conducted the **Alpha Scattering Experiment**...

1) In 1909, scientists in Rutherford's lab tried firing a beam of alpha particles (see p.51) at thin gold foil.
2) From the plum pudding model, they expected the particles to pass straight through the gold sheet, or only be slightly deflected. The actual results of this alpha scattering experiment were different:



3) The plum pudding model couldn't explain these observations, so the nuclear model was proposed.

These were the experimental observations and the key bits of the model that came from them:

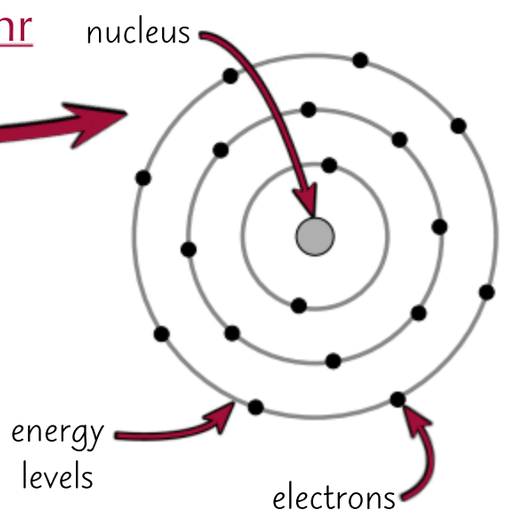
Most alpha particles passed straight through the foil. → Most of the atom is empty space.

A tiny number of particles were deflected backwards, suggesting they bounced off something very small. → Most of the mass is concentrated in a tiny nucleus at the centre of the atom.

Some of the positive alpha particles were deflected by a large angle, and like charges repel each other (p.37). → The nucleus is positively charged.

...Then Bohr Refined Rutherford's **Nuclear Model** of the Atom

- Rutherford's model had a cloud of electrons surrounding the nucleus. Niels Bohr tweaked this, saying that electrons orbit the nucleus at certain distances called energy levels. His theoretical calculations agreed with experimental data.
- Evidence from further experiments changed the model to have a nucleus made up of a group of particles (protons) which all had the same positive charge that added up to the overall charge of the nucleus.
- About 20 years after the idea of a nucleus was accepted, in 1932, James Chadwick proved the existence of the neutron, which explained the imbalance between the atomic and mass numbers (page 51).



Rutherford and Bohr helped to adapt the model of the atom...

This is the scientific method (p.135) in action, folks. Rutherford and his lot made a hypothesis, tested it out and found something they weren't expecting. They showed that the plum pudding model of the atom must be wrong, so it was changed. Bohr and Chadwick made a few more suggestions that were backed by the data... and so it goes on.

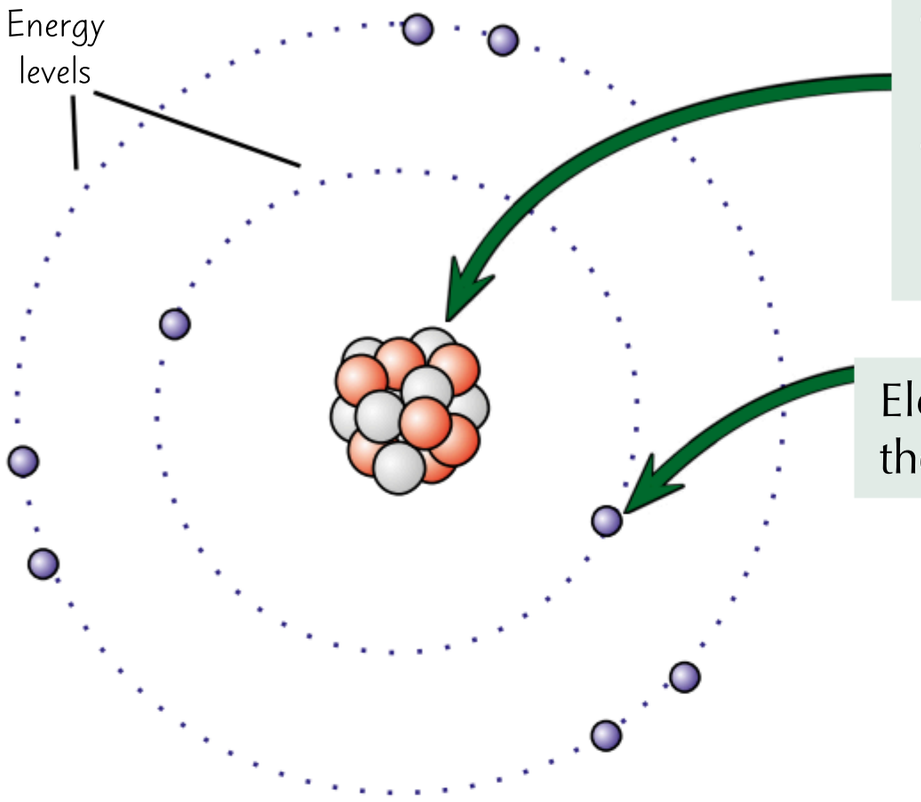
Quick Quiz

The Structure of the Atom

*The **nuclear model** has stood the test of time — you should know its **structure** as it's understood now.*

You Need to Know About this **Modern Model** of the Atom

There are a few different models of the atom used by scientists today — this is the one you need to know about. It's a nuclear model, meaning there's a nucleus in the centre surrounded by electrons:



- The nucleus is tiny, but it makes up most of the mass of the atom.
- The nucleus is made up of protons (which are positively charged) and neutrons (which are neutral). So the nucleus has an overall positive charge.

Electrons have a negative charge. They move around (orbit) the nucleus at different distances, known as energy levels.

Protons	positively charged	relative charge +1
Neutrons	neutral	relative charge 0
Electrons	negatively charged	relative charge –1

Atoms have no overall charge because protons and electrons have equal and opposite charges, and number of protons = number of electrons.

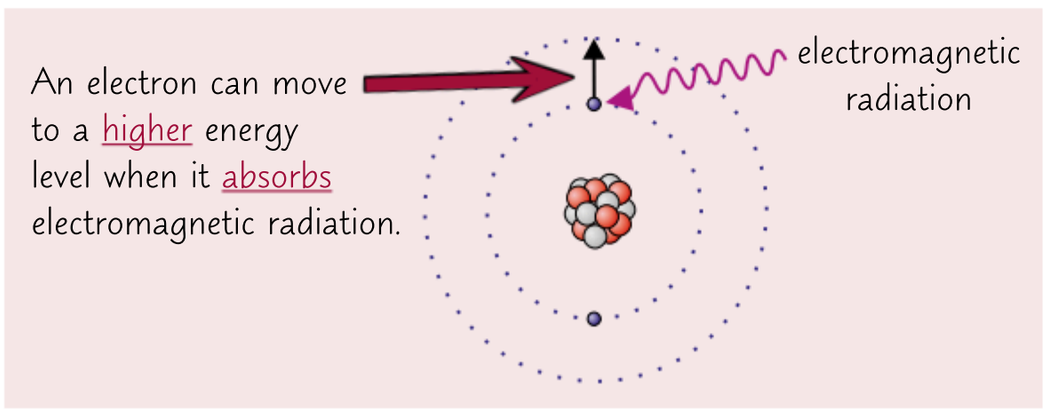
Atoms are very small. The radius of an atom is about 1×10^{-10} m (or 0.000 000 000 1 m — see page 144 for more on standard form).

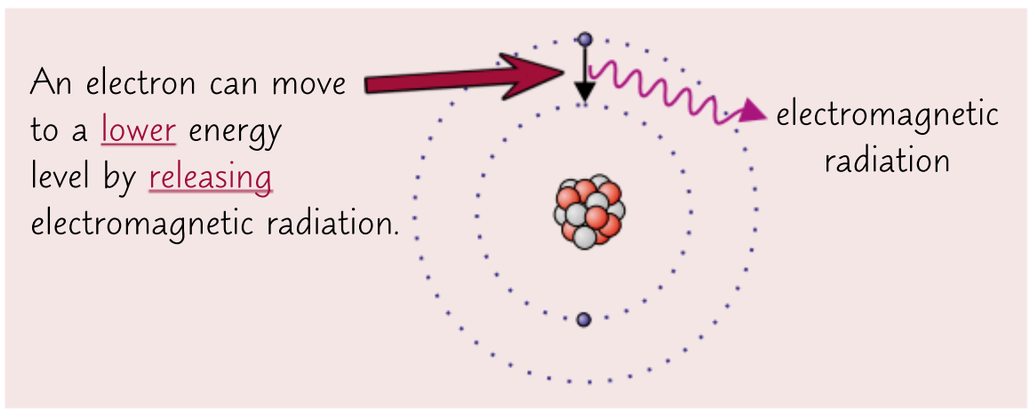
The radius of the nucleus is over 10 000 times smaller than the radius of the atom.

There's no saying our current models of the atom won't change. Just like for the plum pudding model, new experiments sometimes mean we have to change or get rid of current models (see page 135).

Electrons can **Move** Between **Energy Levels**

1) The further an energy level is from the nucleus, the more energy an electron in that energy level has.
2) Electrons can move between energy levels by absorbing or releasing electromagnetic radiation (p.99).





3) If an electron in an outer energy level absorbs electromagnetic radiation, it can leave the atom.
4) If an atom loses one or more of its outer electrons it turns into a positively charged ion.

This model of the atom has developed over time...

Protons, neutrons, electrons, energy levels — there's a whole lot going on inside those tiny little spheres that make up... well... everything. And this model is a great way to visualise what's going on in the tiny world of the atom.

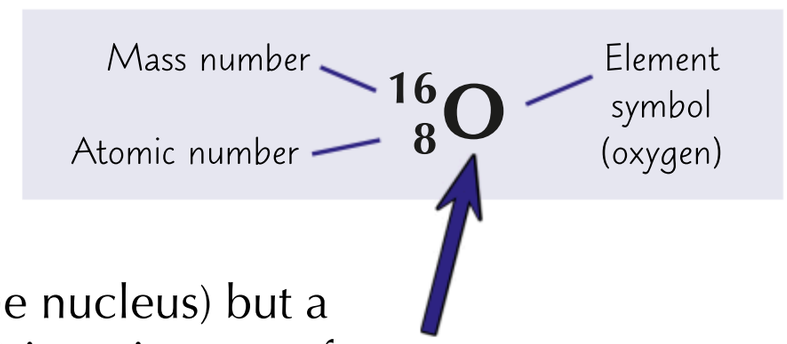
Isotopes and Nuclear Radiation

Quick Quiz

***Isotopes** of an element look pretty similar, but watch out — they have **different numbers of neutrons**.*

Isotopes are Different Forms of the Same Element

- All atoms of a given element have a set number of protons.
  The number of protons in an atom is its atomic number.
- The mass number of an atom is:
  the number of protons + the number of neutrons in its nucleus.
- Isotopes of an element are atoms with the same number of protons (the same atomic number, and so the same charge on the nucleus) but a different number of neutrons (a different mass number). E.g. $^{16}_{8}O$ is an isotope of oxygen.



Some Isotopes are Unstable

Unstable isotopes tend to decay into other elements and give out radiation from their nuclei as they try to become more stable. This process is called radioactive decay.

Radioactive isotopes can emit one or more types of nuclear radiation:

Alpha particles (α)

2 protons and 2 neutrons, like a helium nucleus

Beta particles (β)

Fast-moving electrons formed from a neutron changing into a proton

Gamma rays (γ)

Electromagnetic radiation (p.99) released by the nucleus

Neutrons (n) can also be released when atoms decay, to rebalance their atomic and mass numbers.

Each Form of Ionising Nuclear Radiation has Different Properties

Ionising radiation is radiation that knocks electrons off atoms, creating positive ions. The ionising power of a radiation source is how easily it can do this. Each kind of ionising nuclear radiation has different properties:

Type of radiation	Ionising power	Range in air	Stopped by
alpha particles	strong	a few centimetres	a sheet of paper
beta particles	moderate	a few metres	a sheet of aluminium
gamma rays	weak	a long distance	thick sheets of lead or metres of concrete

Gamma rays tend to pass through materials rather than colliding with their atoms. This is why they are weakly ionising and have a long range in air.

The properties of each kind of ionising radiation make them suitable for different practical uses.

- E.g. alpha radiation is used in smoke alarms — it ionises air particles, causing current to flow. If there is smoke around, it binds to the ionised air and causes the current to stop, setting off the alarm.
- E.g. beta radiation can be used to test the thickness of metal sheets, as beta particles would penetrate further than alpha particles, but would not penetrate as far as gamma rays.

Alpha particles are more ionising than beta particles...

...and beta particles are more ionising than gamma rays. Make sure you've got that memorised.

Q1 In order to sterilise medical equipment, radiation is directed at the equipment while it is sealed in packaging. Explain whether alpha radiation would be suitable for this use. [2 marks]

Q1 Video Solution

Quick Quiz

Nuclear Equations

Nuclear equations *show* ***radioactive decay*** *and once you get the hang of them they're* ***dead easy****. Get going.*

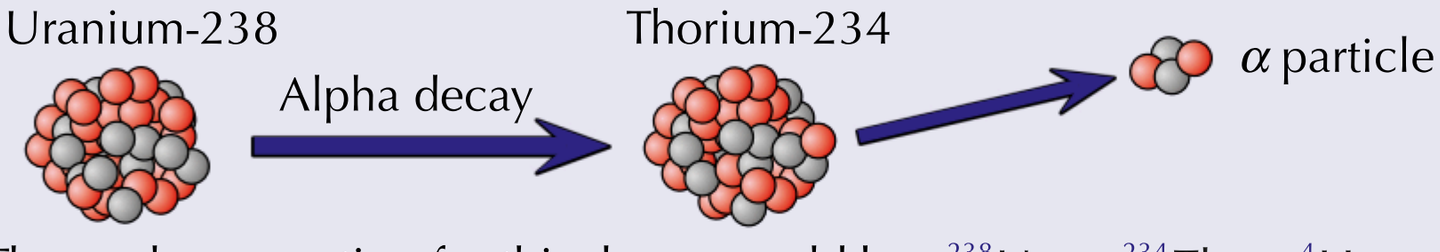
Mass Numbers and Atomic Numbers Have to Balance

1) Nuclear equations are a way of showing radioactive decay by using element symbols (p.51).
2) They're normally written like this: nucleus before decay → nucleus after decay + radiation emitted.
3) There is one golden rule to remember:

The total atomic and mass numbers on both sides of the arrow must be equal.

Alpha Decay Decreases the Charge and Mass of the Nucleus

- When a nucleus emits an alpha particle, it loses two protons and two neutrons. So its atomic number decreases by 2 and its mass number decreases by 4.
- A proton is positively charged and a neutron is neutral, so the charge of the nucleus decreases.
- In nuclear equations, an alpha particle can be written as a helium nucleus: $^{4}_{2}\text{He}$.



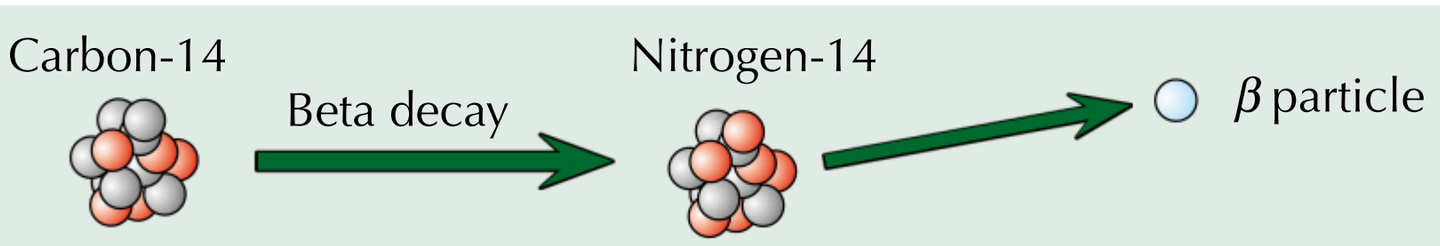
The nuclear equation for this decay would be: $^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + {}^{4}_{2}\text{He}$

$238 \rightarrow 234 + 4$
$92 \rightarrow 90 + 2$

Gamma rays are sometimes also released when a nucleus decays by alpha or beta decay.

Beta Decay Increases the Charge of the Nucleus

- When beta decay occurs, a neutron in the nucleus turns into a proton and releases a fast-moving electron (the beta particle).
- The number of protons in the nucleus increases by 1 and the number of neutrons decreases by 1. This increases the positive charge of the nucleus, and increases its atomic number by 1.
- However, the mass of the nucleus doesn't change (because protons and neutrons have the same mass).
- A beta particle is written as $^{0}_{-1}\text{e}$ in nuclear equations.



The nuclear equation for this decay would be: $^{14}_{6}\text{C} \rightarrow {}^{14}_{7}\text{N} + {}^{0}_{-1}\text{e}$

$14 \rightarrow 14 + 0$
$6 \rightarrow 7 - 1$

In both alpha and beta emissions, a new element will be formed, as the number of protons (atomic number) changes.

Gamma Rays Don't Change the Charge or Mass of the Nucleus

1) Gamma rays are a way of getting rid of excess energy from a nucleus.
2) This means that there is no change to the mass number or atomic number of the atom.

Mass and atomic numbers must balance in nuclear equations

Nuclear equations are simple, but that doesn't mean you shouldn't practise them. Give these questions a go.

Q1 What type of radiation is given off in this decay? $^{8}_{3}\text{Li} \rightarrow {}^{8}_{4}\text{Be}$ + radiation. [1 mark]

Q2 Write the nuclear equation for $^{219}_{86}\text{Rn}$ decaying to polonium (Po) by emitting an alpha particle. [3 marks]

Q2 Video Solution

Half-life

Quick Quiz

__How quickly__ unstable nuclei decay is measured using __activity__ and __half-life__ — two very __important__ terms.

__Radioactivity__ is Something That You Can __Measure__

1) Radioactive substances give out radiation from the nuclei of their atoms.
2) It's important to know the difference between the following ways of measuring radioactivity:

COUNT-RATE	The number of radiation counts measured by a detector (e.g. by a Geiger-Muller tube and counter) per second.
ACTIVITY	The rate at which a source decays. It is measured in becquerels (Bq), where 1 becquerel is equal to 1 decay per second.

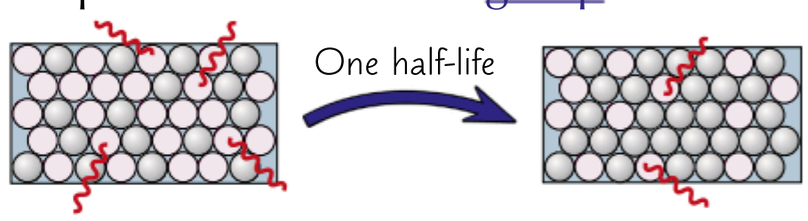
The count-rate can be measured in counts per second (cps).

The __Radioactivity__ of a Source __Decreases Over Time__

Video

- Each time a radioactive nucleus decays into a stable nucleus, the activity as a whole will decrease (older sources emit less radiation because there are fewer radioactive nuclei left).
- For some isotopes it takes just a few hours before nearly all the unstable nuclei have decayed, whilst others last for millions of years.
- Radioactive decay is always random. So you can't predict exactly which nucleus in a sample will decay next, or when any one of them will decay. But you can make predictions about a group of nuclei:

The half-life is the time taken for the number of radioactive nuclei in an isotope to halve.



- The half-life is also the time taken for activity (and count-rate) to halve.

EXAMPLE

The initial activity of a sample is 640 Bq. After 20 minutes, the activity falls to 160 Bq. How many half-lives has the sample undergone in 20 minutes?

MATHS SKILLS

1) Keep halving the initial activity until you get the final activity. 640 ÷ 2 = 320, 320 ÷ 2 = 160
2) The amount of half-lives it has undergone is how many times you halved the initial activity. Divided by 2 twice, so it has undergone 2 half-lives.

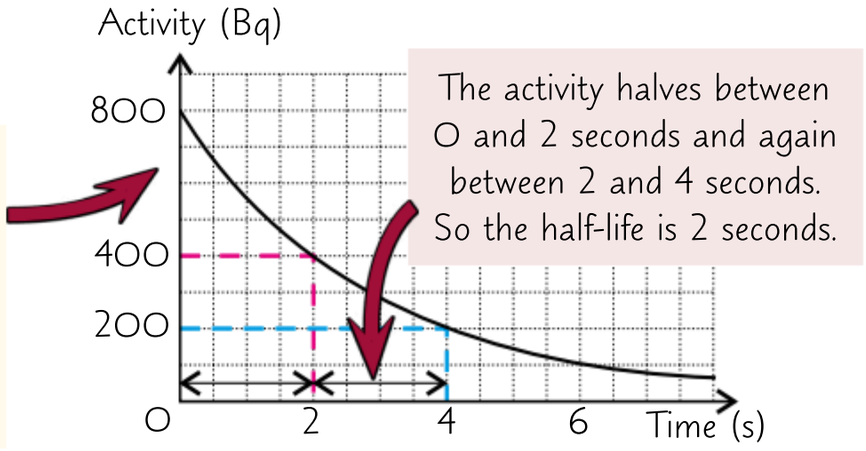
The kind of danger posed by a radioactive substance depends on the length of its half-life.

SHORT HALF-LIFE → Activity falls quickly → High amount of radiation at the start, but quickly becomes safe

LONG HALF-LIFE → Activity falls slowly → Smaller amounts of radiation, but for a long time — nearby areas constantly exposed

You Can __Measure Half-Life__ Using a Graph

- A graph of activity against time (with background radiation removed, p.56) will always be shaped like this.
- The half-life is found from the graph by finding the time interval on the bottom axis corresponding to a halving of the activity on the vertical axis. Easy.



__Different substances have different half-lives...__

Half-life — the time for the number of radioactive nuclei, the activity or the count-rate to halve.

Q1 The initial count-rate of a sample is 40 cps. Show that the ratio of its final count rate to its initial count rate is 1:8 after three half-lives. [3 marks]

Q1 Video Solution

Warm-Up & Exam Questions

Atoms may be tiny, but you could bag some big marks in your exams if you know them inside out. Here are some questions to check just how great your understanding of atoms and radiation really is...

Warm-Up Questions

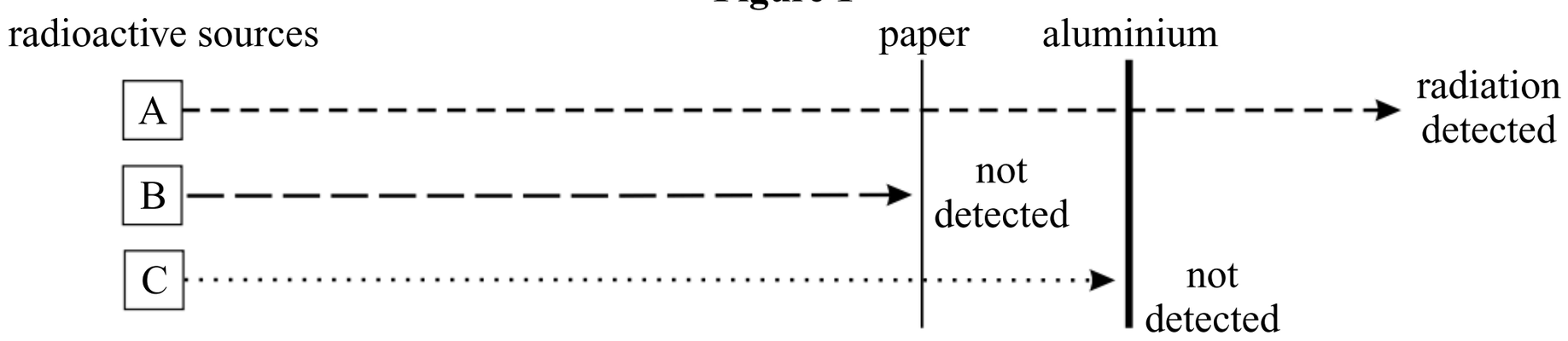
1) What component of an atom exists in energy levels?
2) Give the definition of the term 'isotope'.
3) Which type of radioactive decay releases fast-moving electrons?
4) Name the type of nuclear radiation that is an electromagnetic wave.
5) Which are the most ionising — alpha particles or gamma rays?
6) Outline why beta emitters, rather than alpha or gamma emitters, are used to test the thickness of sheets of metal.
7) Radioactive substances with short half-lives can be initially very dangerous. Explain why.

Exam Questions

1 Alpha, beta and gamma radiation sources were used to direct radiation at thin sheets of paper and aluminium. A detector was used to measure where radiation had passed through the sheets. The results are shown in **Figure 1**.

GRADE 4-6 GRADE

Figure 1



1.1 Name the type of radiation that source C emits. Explain your answer.

[2 marks]

1.2 Give **one** example of a detector that could have been used to detect the radiation.

[1 mark]

2 A sample of a highly ionising radioactive gas has a half-life of two minutes.

GRADE 6-7 GRADE

2.1 Define what is meant by the term 'half-life'.

[1 mark]

The sample contains a number of unstable nuclei.

2.2 Calculate the fraction of these nuclei that will be present after four minutes.

[2 marks]

Exam Questions

3 A radioactive isotope sample has a half-life of 40 seconds. The initial activity of the sample is 8000 Bq.

GRADE 6-7 GRADE

3.1 Calculate the activity after 2 minutes. Give your answer in becquerels.

[2 marks]

3.2 After how many half-lives will the activity have fallen to 250 Bq?

[2 marks]

3.3 The radioactive source is left until its activity falls to 100 Bq. Calculate the final activity as a percentage of the initial activity.

[2 marks]

4 **Table 1** contains information about three atoms.

GRADE 6-7 GRADE

Table 1

	Mass number	Atomic number
Atom A	32	17
Atom B	33	17
Atom C	32	16

4.1 Name the two types of particle that the nucleus of an atom contains.

[2 marks]

4.2 Define the term 'mass number' in the context of atoms.

[1 mark]

4.3 Which of the two atoms in **Table 1** are isotopes of the same element? Explain your answer.

[2 marks]

Alpha and beta particles are deflected in electric fields.

4.4 Suggest why alpha and beta particles are deflected in opposite directions.

[1 mark]

5 Nuclear equations show what is produced when unstable nuclei decay.

GRADE 7-9 GRADE

5.1 Draw a symbol that can be used to represent a beta particle in a nuclear equation.

[1 mark]

5.2 Describe what happens to the atomic number and the mass number of an atom when it undergoes beta decay.

[2 marks]

5.3 Describe what happens to the atomic number and the mass number of an atom when it undergoes gamma decay.

[2 marks]

5.4 Complete the nuclear equation, shown in **Figure 2**, which shows a polonium isotope decaying by alpha emission.

Figure 2

$$\text{.....}^{\text{.....}}_{84}\text{Po} \longrightarrow \text{.....}^{205}_{\text{.....}}\text{Pb} + \text{.....}^{\text{.....}}_{\text{.....}}\text{He}$$

[3 marks]

Background Radiation and Contamination

Quick Quiz

*Forget love — **radiation** is **all around**. Don't panic too much though — it's usually a pretty **small amount**.*

Background Radiation Comes from Many Sources

Background radiation is the low-level radiation that's around us all the time. It comes from:

- Naturally occurring unstable isotopes (e.g. in air, food and building materials)
- Rocks underneath our feet
- Radiation from space (cosmic rays)
- Radiation from human activity (e.g. fallout from nuclear explosions or nuclear waste)

The radiation dose tells you how harmful radiation can be. It's measured in sieverts (Sv) (p.103). The background radiation dose is small, so millisieverts are often used (1 Sv = 1000 mSv). Your radiation dose can depend on where you live or if you have a job that involves radiation.

You should always measure and subtract the background radiation when you're investigating the activity of a source.

Exposure to Radiation is called Irradiation

- Objects near a radioactive source are irradiated by it. This simply means they're exposed to it (we're always being irradiated by background radiation sources).
- Irradiating something does not make it radioactive (and won't turn you into a superhero).
- Keeping sources in lead-lined boxes, standing behind barriers or being in a different room and using remote-controlled arms are all ways of reducing the effects of irradiation.

Contamination is Radioactive Material Getting onto Objects

- If unwanted radioactive atoms get onto or into an object, the object is said to be contaminated.
- These atoms might then decay, releasing radiation which could cause you harm.
- Contamination is especially dangerous because radioactive material could get inside your body.
- Gloves and tongs should be used when handling sources, to avoid radioactive material getting stuck to your skin or under your nails. Some industrial workers wear protective suits to stop them breathing in radioactive dust and gas.

The Seriousness of Irradiation and Contamination Depends on the Source

Irradiation or contamination can cause different amounts of harm (p.57), based on the radiation type.

	Alpha sources	Beta sources	Gamma sources
IRRADIATION	Least dangerous of the sources as they can't penetrate the skin and are stopped by a small air gap.	Dangerous as both sources can penetrate the body and damage delicate organs.	
CONTAMINATION (inside of the body)	Most dangerous of the sources as they have a high ionising power and do all their damage in a very localised area.	Less damaging as radiation is absorbed over a wider area.	Least dangerous, as they mostly pass straight out of the body.

High irradiation levels from all sources are dangerous.

Outside of the body, alpha sources are the least dangerous to be contaminated by, as they can't get through the skin.

It's important that research about radiation is published. The data is peer-reviewed (see p.135) and can then become accepted, leading to improvements in how we protect ourselves when using radioactive sources.

Alpha sources are the most dangerous inside the body...

Alpha sources are the most ionising, so if they get into the body, they can wreak havoc. Beta and gamma sources are the most dangerous outside the body because they're more penetrating than alpha sources, so they can get through skin.

Uses and Risk

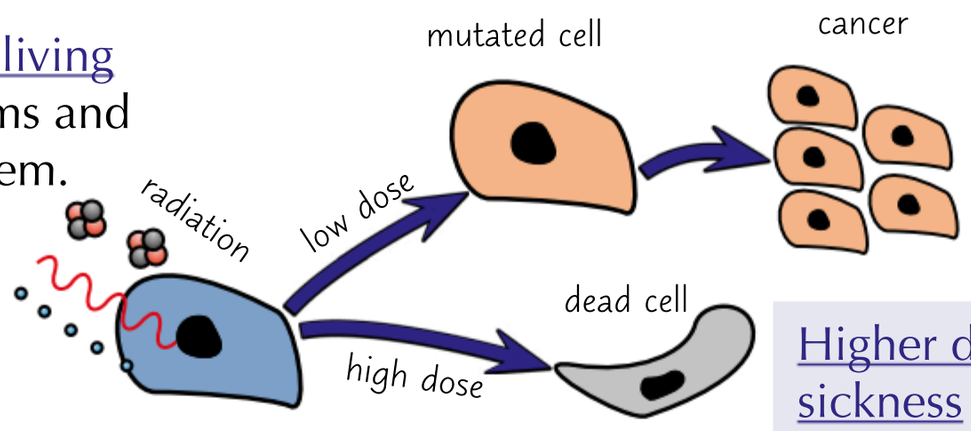
Quick Quiz

Radiation *can be pretty useful. Using it is all about* ***reducing the risks*** *whilst still keeping the* ***benefits****.*

There are **Risks** to Using Radiation

Radiation can enter living cells and ionise atoms and molecules within them.

This can lead to tissue damage.



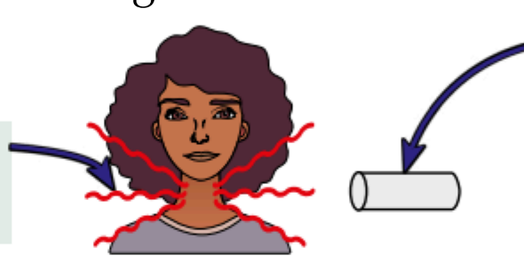
Lower doses tend to cause minor damage without killing the cells. This can lead to mutant cells which divide uncontrollably. This is cancer.

Higher doses tend to kill cells, causing radiation sickness if a lot of cells all get killed at once.

Medical Tracers Often Use **Gamma** Sources

1) Certain radioactive isotopes can be injected into people (or they can just swallow them) and their progress around the body can be followed using an external detector. This can then reveal where the strongest reading is coming from.
2) One example is the use of iodine-123:

The thyroid gland in the neck absorbs normal iodine-127 and the tracer iodine-123, which gives off gamma rays.



The amount of radiation detected from the thyroid gland shows whether it is taking in iodine as it should.

3) Isotopes which are taken into the body like this are usually gamma, which means that the radiation passes out of the body without causing much ionisation. The isotope should have a short half-life so the radioactivity inside the patient quickly disappears.

Radiotherapy — Treating **Cancer** with Radiation

- Since high doses of ionising radiation will kill all living cells, it can be used to treat cancers.
- Gamma rays are directed carefully and at just the right dosage to kill the cancer cells without damaging too many normal cells — but it still damages some normal cells, making patients feel ill after treatment.
- Radiation-emitting implants (usually beta emitters) can also be put next to or inside tumours.

You Have to **Weigh Up** the **Risks** and **Benefits**

There are risks and benefits to using tracers and radiotherapy which people must consider:

	Tracers	Radiotherapy
Risk	There is a very small risk of getting cancer.	Makes patients feel very ill and can pose some future risks, e.g. slightly increasing the chance of getting another type of cancer.
Benefit	Can be used to diagnose life-threatening conditions.	If it is successful, it will get rid of all the cancerous cells.

People decide whether to use radiation based on perceived risk (how risky they think something is) as opposed to actual risk. The perceived risk can vary from person to person (see p.137 for more on risks).

People use radiation if they think the benefits outweigh the risks...

The risks of using radiotherapy are quite high, and using radioactive sources for medical purposes will almost always mutate and/or kill ordinary cells. But radiotherapy is often worth the risk, as it can get rid of a patient's cancer.

Quick Quiz

Fission and Fusion

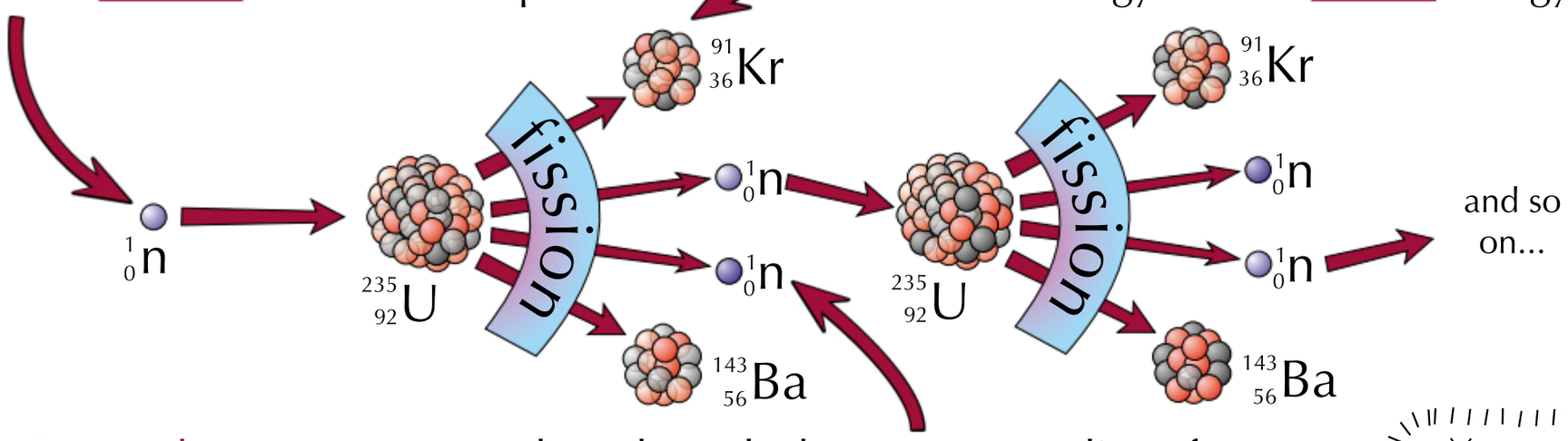
Splitting up *or* ***squishing together*** *nuclei releases lots of useful* ***energy****, but it can have* ***explosive*** *consequences.*

Nuclear **Fission** — **Splitting** a **Large**, **Unstable** Nucleus

Nuclear fission is a type of nuclear reaction that is used to release energy from large and unstable nuclei (e.g. uranium or plutonium) by splitting them into smaller nuclei.

1) Spontaneous (unforced) fission rarely happens. Usually, the nucleus has to absorb a neutron before it will split.

2) When the nucleus splits it forms two lighter elements that are roughly the same size (and that have some energy in their kinetic energy stores).



3) Two or three neutrons are also released when an atom splits. If any of these neutrons are moving slow enough to be absorbed by another nucleus, they can cause more fission to occur. This is a chain reaction.

You may have to draw or complete a diagram to show a chain reaction in the exam.

- The energy not transferred to the kinetic energy stores of the products is carried away by gamma rays.
- The energy carried away by the gamma rays (and in the kinetic energy stores of all the products) can be used to heat water, making steam that will turn turbines and generators (p.126).

The amount of energy produced by fission in a nuclear reactor is controlled by changing how quickly the chain reaction occurs.

Uncontrolled chain reactions quickly lead to lots of energy being released as an explosion — this is how nuclear weapons work.

Control rods are used to slow down the chain reaction. When they are lowered into the reactor, the rods absorb neutrons. This slows the chain reaction, reducing the energy released.

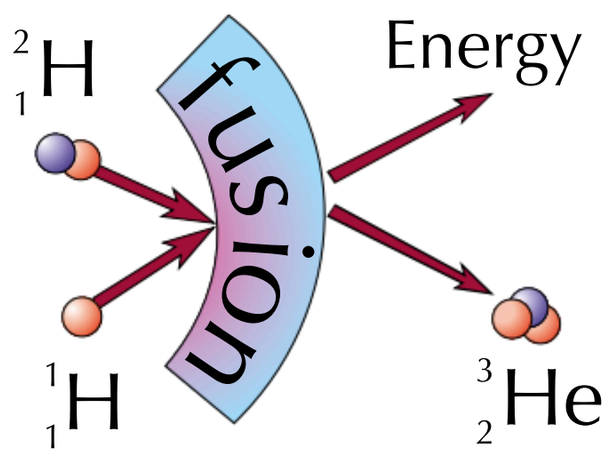
Nuclear **Fusion** — **Joining Small Nuclei**

Video

Nuclear fusion is the opposite of nuclear fission.

1) In nuclear fusion, two light nuclei collide at high speeds and join (fuse) to create a larger, heavier nucleus. For example, hydrogen nuclei can fuse to produce a helium nucleus.

2) The heavier nucleus produced by fusion does not have as much mass as the two separate, light nuclei did. Some of the mass of the lighter nuclei is converted to energy (don't panic, you don't need to know how) and released.



3) Fusion releases a lot of energy (more than fission for a given mass of fuel).

So far, scientists haven't found a way of using fusion to generate energy for us to use. Fusion reactors are expensive and hard to build because the temperatures and pressures needed for fusion are so high.

Nuclear fusion releases a lot more energy than nuclear fission...

Thankfully you don't need to know the complicated processes behind fission and fusion — you just need an idea of the steps involved. It's important to remember the differences between fission and fusion. For example, fission starts with big nuclei splitting into smaller nuclei, while fusion starts with small nuclei joining together to make bigger nuclei.

Warm-Up & Exam Questions

Radiation has pros and cons, and one of its downsides is that you need to know about it for your exam. So here's a dose of practice questions to see how well you know your stuff.

Warm-Up Questions

1) What is the difference between radioactive contamination and irradiation?
2) Describe two precautions that should be taken when working with radioactive sources in a lab.
3) Why is nuclear radiation dangerous?
4) Why is contamination by an alpha source more dangerous to humans than irradiation by an alpha source?

Exam Questions

1 Nuclear radiation can have harmful effects on the human body. (GRADE 4-6)

1.1 What is likely to happen to cells if they receive a high dose of nuclear radiation?

[1 mark]

1.2 Scientists have carried out research into the dangers of nuclear radiation.
Suggest **one** reason why it is important that the data from research is peer-reviewed.

[1 mark]

2 Nuclear reactors often use uranium-235. (GRADE 6-7)

2.1 Describe how a chain reaction is set up in a nuclear reactor.

[3 marks]

2.2 In a nuclear reactor, control rods are used to control the reaction.
What could happen if the nuclear reaction is not controlled?

[1 mark]

3 Iodine-123 is a radioactive isotope, commonly used as a tracer in medicine. (GRADE 7-9)

3.1 Describe how iodine-123 can be used to detect whether the thyroid gland is absorbing iodine as it normally should do.

[3 marks]

Table 1 shows the properties of three other radioisotopes.

Table 1

Radioisotope	Half-life	Type of emission
technetium-99m	6 hours	gamma
phosphorus-32	14 days	beta
cobalt-60	5 years	beta/gamma

3.2 State which of these would be best to use as a medical tracer. Explain your answer.

[3 marks]

Revision Summary Test for Topic 4

Hopefully you're positively radiating with knowledge now. To put it to the test, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

The Atomic Model (p.49-50) ☑

1) a) Describe Rutherford's alpha scattering experiment.
   b) How does Rutherford's nuclear model differ from Thomson's plum pudding model?
2) How was Bohr's model of the atom different from Rutherford's?
3) What kind of particles make up the nucleus? What is the overall charge of the nucleus?
4) Roughly how big is an atom? How does this compare to the size of its nucleus?
5) What happens to an electron in an atom when it: a) absorbs EM radiation, b) emits EM radiation?
6) What is the overall charge of an atom? What happens if it loses one or more of its electrons?

Nuclear Decay and Half-life (p.51-53) ☑

7) What is the atomic number of an atom? What is the mass number of an atom?
   How do they both compare between different isotopes of the same element?
8) What is radioactive decay? Describe four things that may be emitted during decay.
9) What are the ionising powers and ranges in air for alpha, beta and gamma radiation?
10) Give an example of a material that stops each of alpha, beta and gamma radiation.
11) What kind of radiation is used in smoke alarms? How is the radiation used?
12) What two things must be the same on both sides of a nuclear equation?
13) How do the mass numbers and atomic numbers of atoms change during:
    a) alpha decay, b) beta decay, c) gamma decay.
14) How would you represent an alpha particle and a beta particle in a nuclear equation?
15) What is the difference between count-rate and activity?
16) True or false? You can't predict when an individual nucleus will decay, but the half-life of an isotope can be used to make general predictions about radioactive decays.
17) Define half-life. How does the danger posed by a radiation source depend on its half-life?
18) How can you calculate half-life from a graph of activity against time?

Dangers and Uses of Radiation (p.56-58) ☑

19) Give four different sources of background radiation.
20) What does a radiation dose tell you? What can affect a person's radiation dose?
21) Define irradiation and contamination. Give an example of how you can protect against each of them.
22) Compare the hazards of alpha, beta and gamma sources when someone is:
    a) irradiated by them, b) contaminated by them.
23) What are two effects that radiation can have on cells?
24) How do medical tracers work? Give one example of a medical tracer.
25) What are the pros and cons of using radiation to treat cancer?
26) What happens in nuclear fission? How are gamma rays involved?
27) Draw a diagram showing a uranium-235 nucleus undergoing nuclear fission.
28) How can nuclear fission lead to a chain reaction? What can an uncontrolled chain reaction lead to?
29) Describe what happens in nuclear fusion. What can happen to some of the mass in this process?

Contact and Non-Contact Forces

Quick Quiz

*When you're talking about the **forces** acting on an object, it's not enough to just talk about the **size** of each force. You need to give its **direction** too — force is a **vector**, with a size and a direction.*

Physical Quantities Can be Vector or Scalar

	Definition	Examples
Vector quantity	Has magnitude and direction.	Force, velocity, displacement, acceleration, momentum.
Scalar quantity	Only has magnitude.	Speed, distance, mass, temperature, time.

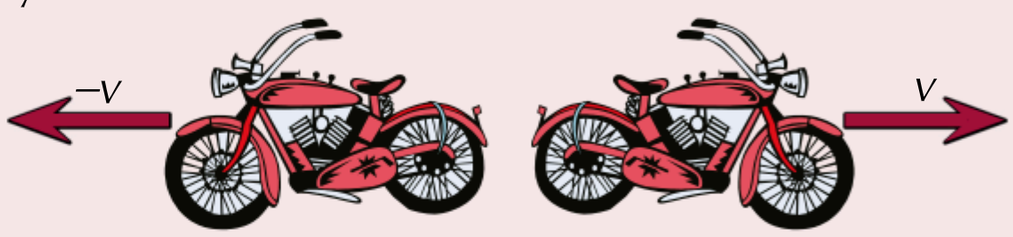
Vectors are usually represented by an arrow — the length of the arrow shows the magnitude, and the direction of the arrow shows the direction of the quantity.

Magnitude just means size.

Velocity is a vector, but speed is a scalar quantity.

Both bikes are travelling at the same speed, v (the length of each arrow is the same).

They have different velocities because they are travelling in different directions.



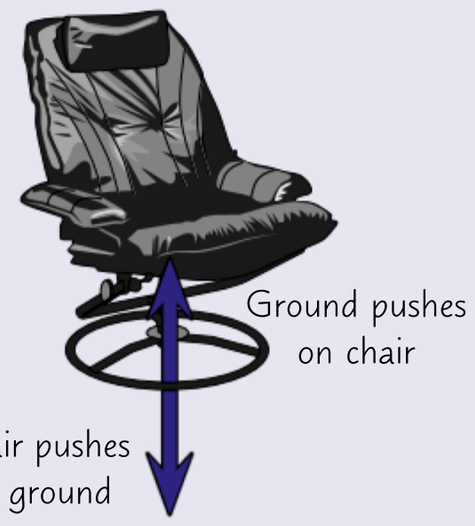
Forces Can be Contact or Non-Contact

- A force is a push or a pull on an object that is caused by it interacting with something.
- All forces are measured in newtons (N) and are either contact or non-contact forces:

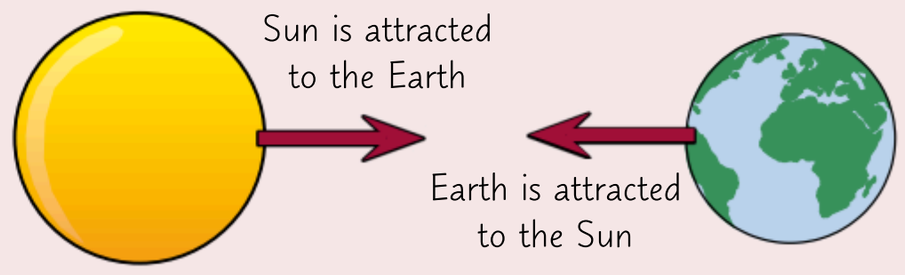
	Definition	Examples
Contact force	A force that acts when two objects touch.	Friction, air resistance, tension in ropes, normal contact force.
Non-contact force	A force that can act without objects touching.	Magnetic force, gravitational force, electrostatic force.

- When two objects interact, there is a force produced on both objects. An interaction pair is a pair of forces that are equal and opposite and act on two interacting objects. (This is just Newton's Third Law — see p.80.)

A chair exerts a force on the ground, whilst the ground pushes back on the chair with the same sized force (the normal contact force). Equal but opposite forces are felt by both the chair and the ground.



The Sun and the Earth are attracted to each other by the gravitational force. This is a non-contact force. An equal but opposite force of attraction is felt by both the Sun and the Earth.



Vectors have magnitude and direction, scalars just have magnitude...

This all seems pretty basic, but make sure you've 100% got it all before moving on through the rest of this topic. Don't get tripped up by thinking 'distance' when an exam question asks for 'displacement'. It's always good to have a few examples of vectors and scalars prepped in your brain (and the same goes for contact and non-contact forces).

Weight, Mass and Gravity

Quick Quiz

*Now for something a bit more **attractive** — the force of **gravity**. Bit of a **weighty** one, but we should manage...*

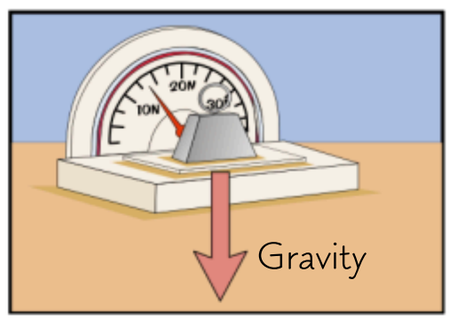
Gravitational Force is the Force of Attraction Between Masses

Gravity attracts all masses, but you only notice it when one of the masses is really really big, e.g. a planet. Anything near a planet or star is attracted to it very strongly.

This has two important effects:

1) Near the surface of a planet, it makes all things fall towards the ground.

2) It gives everything a weight.



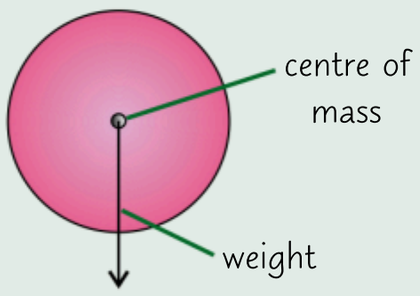
Weight and Mass are Not the Same

WEIGHT...	MASS...
...is the force acting on an object due to gravity.	...is the amount of stuff in an object.
...is measured in newtons (N).	...is measured in kilograms (kg).
...is measured using a calibrated spring-balance (or newtonmeter).	...is measured with a mass balance.
...depends on the object's location and mass.	...is the same for an object whatever its location.

Weight changes with location because of gravitational field strength — the gravitational field is stronger the closer you are to the mass causing the field. If the mass causing the gravitational field is larger, then the gravitational field strength will be larger too.

The weight of a 1 kg mass will be less on the surface of the Moon (about 1.6 N) than it is on Earth (about 9.8 N) because the gravitational field strength on the Moon is less.

- You can think of the force of weight as acting on a single point on the object. This point is called the centre of mass (it's where you assume the whole mass is concentrated).
- For a uniform object (one that's the same density throughout and is a regular shape), the centre of mass is at the centre of the object.



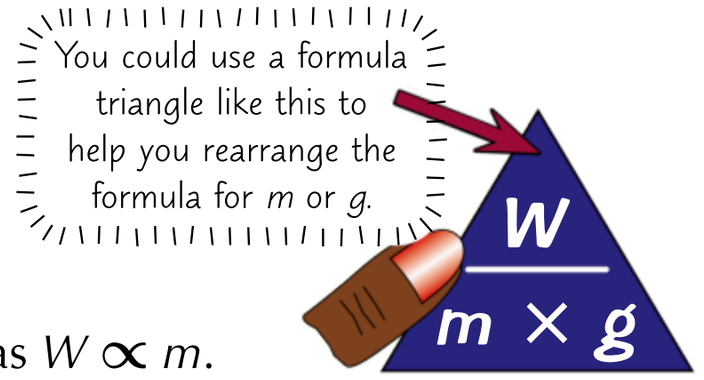
Mass and Weight are Directly Proportional

- You can calculate the weight of an object using the formula:

Weight (N) = Mass (kg) × Gravitational Field Strength (N/kg)

W = mg

- For Earth, $g \approx 9.8$ N/kg and for the Moon it's around 1.6 N/kg. Don't worry, you'll always be given a value of g to use in the exam.
- Increasing the mass of an object increases its weight. For a constant gravitational field strength, if you double the mass, the weight doubles too — you can say that weight and mass are directly proportional.
- You can write this using the direct proportionality symbol (see p.144) as $W \propto m$.



An object's weight depends on the gravitational field it's in...

Q1 Video Solution

Remember, weight is a force due to gravity that acts on an object's centre of mass. It changes depending on the strength of the gravitational field the object is in, and is directly proportional to the object's mass.

Q1 Calculate the weight in newtons of a 5 kg mass:
a) on Earth ($g \approx 9.8$ N/kg) b) on the Moon ($g \approx 1.6$ N/kg) [4 marks]

Resultant Forces and Work Done

Quick Quiz

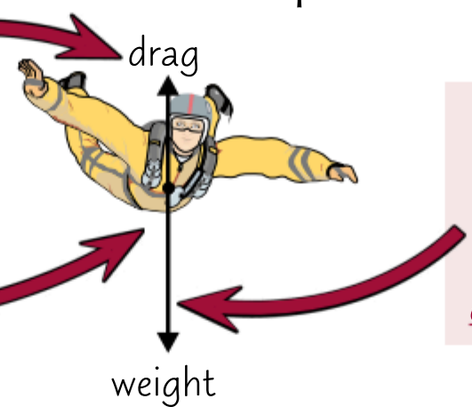
*When **multiple forces** act on an object, it's often helpful to **combine** their effects into a **single overall force**...*

Free Body Diagrams Show All the **Forces** Acting on an Object

Free body diagrams don't show any of the forces that the system or object exerts on the rest of the world, only the forces acting on the system or object. Here's an example of a skydiver falling towards the ground:

There are two forces acting on the skydiver: his weight and drag (air resistance).

The sizes of the arrows show the relative magnitudes of the forces. The arrow here is longer because the weight is bigger than the drag.



The arrows point in the direction that the forces act. Weight acts downwards (as it always does), pulling the skydiver towards the ground. Drag acts in the opposite direction to the weight, opposing the motion.

A **Resultant Force** is the **Overall Force** on a Point or Object

- In most real situations there are at least two forces acting on an object.
- If you have a number of forces acting at a single point, you can replace them with a single force that has the same effect as all the original forces together. This single force is called the resultant force.
- If the forces all act along the same line (they're all parallel), the overall effect is found by adding those going in the same direction and subtracting any going in the opposite direction.

There's a downward resultant force acting on the skydiver above.

EXAMPLE

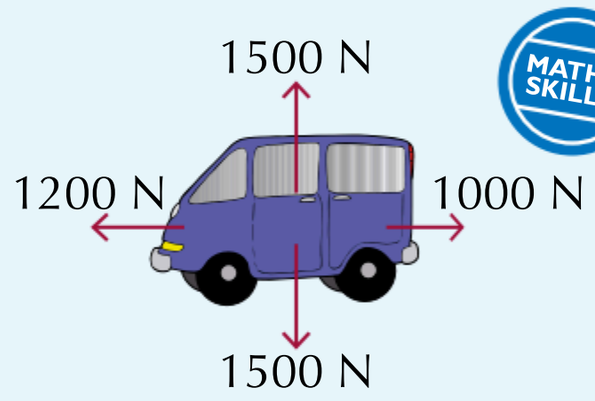
For the free body diagram shown on the right, calculate the resultant force acting on the van.

1) Consider the horizontal and vertical directions separately.
2) State the size and direction of the resultant force.

Vertical: 1500 – 1500 = 0 N

Horizontal: 1200 – 1000 = 200 N

The resultant force is 200 N to the left.



MATHS SKILLS

If a **Resultant** Force **Moves** an Object, **Work is Done**

When a force moves an object through a distance, energy is transferred and work is done on the object.

1) To make something move (or keep it moving if there are frictional forces), a force must be applied.
2) The thing applying the force needs a source of energy (like fuel or food).
3) The force does work to move the object and energy is transferred from one store to another (p.3).
4) Whether energy is transferred 'usefully' (e.g. lifting a load) or is 'wasted' (p.7) you can still say that 'work is done'.
5) You can find out how much work has been done using:
6) One joule of work is done when a force of one newton causes an object to move a distance of one metre.

When you push an object along a rough surface (like a carpet) you do work against frictional forces. Energy is transferred to the kinetic energy store of the object as it moves, but some is also transferred to thermal energy stores due to the friction. This causes the overall temperature of the object to increase.

Work done (J) = Force (N) × Distance (moved along the line of action of the force) (m)

$$W = Fs$$

$$1\ J = 1\ Nm$$

The resultant force — one force with the same result as many...

Free body diagrams make most force questions easier, so start by sketching one. Then get to work.

Q1 A force of 20 N pushes an object 20 cm. Calculate the work done on the object. [3 marks]

Q1 Video Solution

Calculating Forces

Quick Quiz

Scale drawings *are useful things — they can help you* ***resolve*** *forces or* ***work out*** *the* ***resultant force****.*

Use **Scale Drawings** to Find **Resultant Forces**

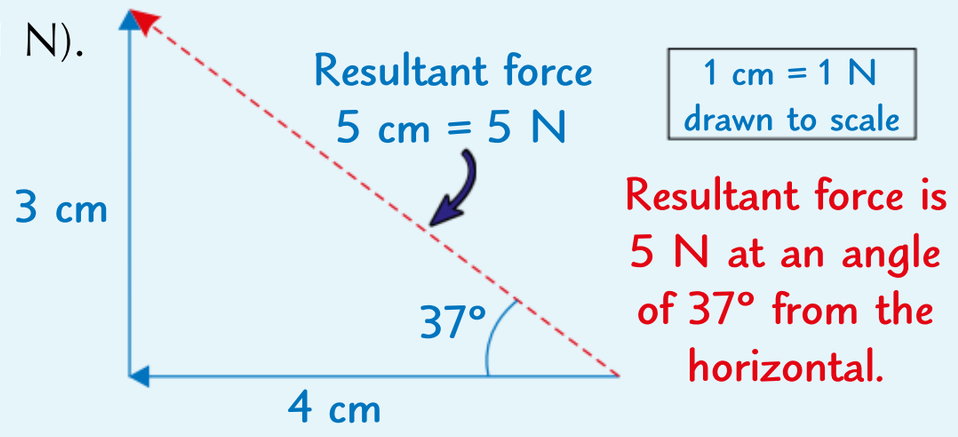
1) Draw all the forces acting on an object, to scale, 'tip-to-tail'.
2) Draw a straight line from the start of the first force to the end of the last force — this is the resultant force.
3) Using a ruler, measure the length of the resultant force on the diagram to find the magnitude
4) Using a protractor, measure the angle of the resultant force on the diagram to find its direction.

EXAMPLE

An object experiences a horizontal force of 4 N to the left and an upwards vertical force of 3 N. Find the magnitude and direction of the resultant force.

MATHS SKILLS

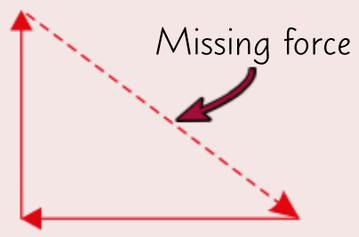
1) Start by drawing a scale drawing of the forces acting.
2) Make sure you choose a sensible scale (e.g. 1 cm = 1 N).
3) Draw the resultant from the tail of the first arrow to the tip of the last arrow.
4) Measure the length of the resultant with a ruler and use the scale to find the force in N.
5) Use a protractor to measure the angle of the resultant force from the horizontal.



An Object is in **Equilibrium** if the **Forces** on it are **Balanced**

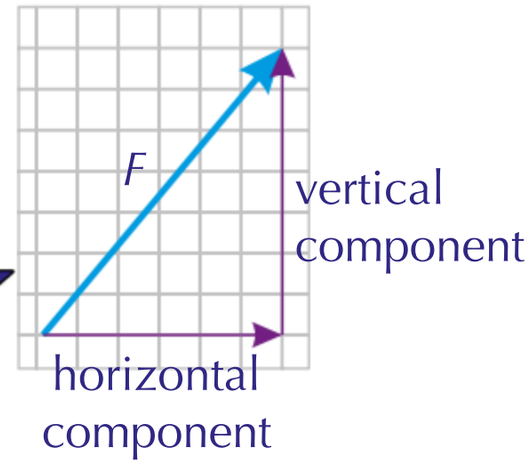
- If all the forces acting on an object add up to give a resultant force of zero, the object is in equilibrium.
- On a scale diagram, this means that the tip of the last force you draw should end where the tail of the first force you drew begins. E.g. for three forces, the scale diagram will form a triangle like below:

You might be given forces acting on an object and told to find a missing force, given that the object is in equilibrium. To do this, draw out the forces you do know (to scale and tip-to-tail), then join the end of the last force to the start of the first force. This line is the missing force so you can measure its size and direction.



You Can **Split** a Force into **Components**

- Not all forces act horizontally or vertically — some act at awkward angles.
- To make them easier to deal with, you can resolve (split) these forces into two components at right angles to each other (usually horizontal and vertical).
- Acting together, these components have the same effect as the single force.
- You can resolve a force by drawing it on a square grid. Draw the force to scale, and then add the horizontal and vertical components along the grid lines. Then you can just measure them.



With scale diagrams, it's important to keep things in proportion...

A pencil, ruler and protractor — the tools of the trade for scale diagram work, so keep them primed and ready. And make sure to choose a sensible scale before you start drawing. A teeny tiny diagram is no good for measuring lengths and angles accurately, and don't make life harder for yourself with something awkward like 1 cm = 0.892 N....

Warm-Up & Exam Questions

Now you've learnt the basics of forces, it's time to act on your new knowledge.
Give these questions a whack and test how well you've forced those facts into your brain.

Warm-Up Questions

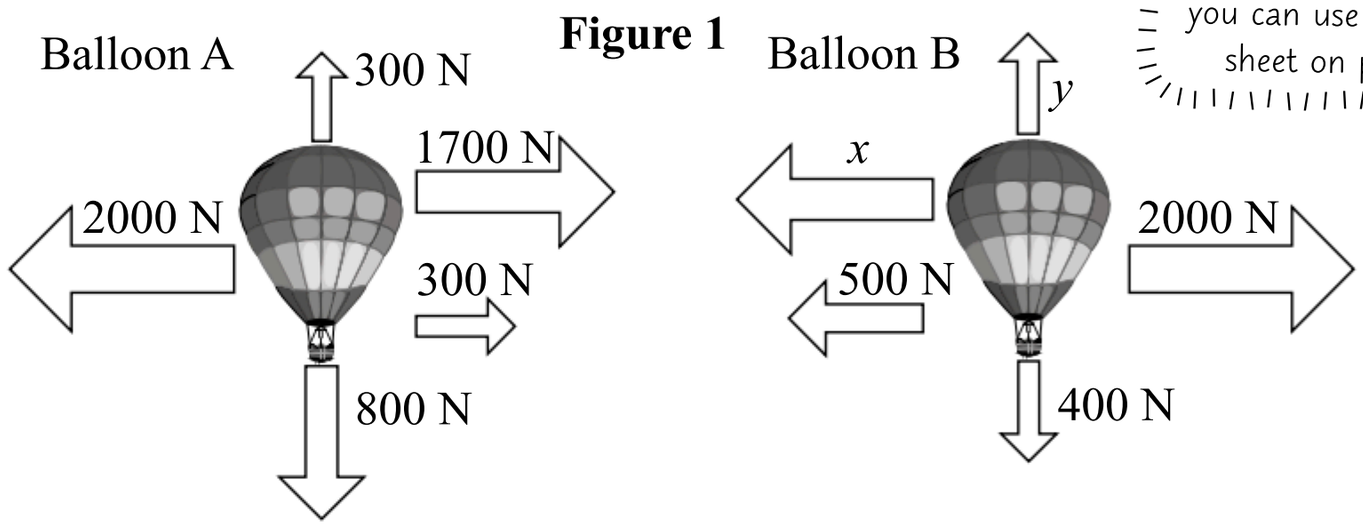
1) Give an example of a vector quantity and a scalar quantity.
2) A tennis ball is dropped from a height.
   Name one contact force and one non-contact force that act on the ball as it falls.
3) State the units of: a) gravitational field strength, b) mass, c) weight.
4) How can you tell if a set of forces are balanced using a scale diagram?

Exam Questions

For questions 2 and 3, you can use the formula sheet on page 216.

1 **Figure 1** shows two hot air balloons, labelled with the forces acting on them. GRADE 4-6 GRADE

Figure 1



1.1 Calculate the size of the resultant force acting on Balloon A and give its direction.

[3 marks]

1.2 The resultant force acting on Balloon B is zero. Calculate the size of forces x and y.

[2 marks]

2 A train moves 700 m in a straight line along a flat track.
The resultant force acting on the train is 42 000 N forwards along the track. GRADE 6-7 GRADE

2.1 Calculate the work done by the resultant force as the train moves 700 m. Give your answer in kJ.

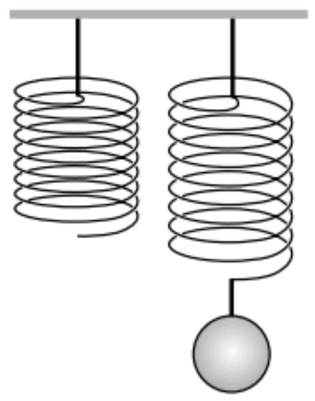
[3 marks]

2.2 The weight of the train is 200 000 N. The resistive force on the train is 15 000 N.
Draw a free-body diagram of the train. Model the train as a rectangle.

[4 marks]

3 A spring increases in length when masses are suspended from it, as shown in **Figure 2**. When a metal ball with a mass of 0.10 kg is suspended from the spring, the spring stretches by 3 cm. If the experiment was repeated on Mars, the spring would only be stretched by 1.1 cm. GRADE 7-9 GRADE

Figure 2



3.1 Suggest why the spring would stretch less on Mars than on Earth.

[3 marks]

3.2 The weight of the ball on Mars is 0.37 N. Calculate the value of g on Mars.

[3 marks]

Forces and Elasticity

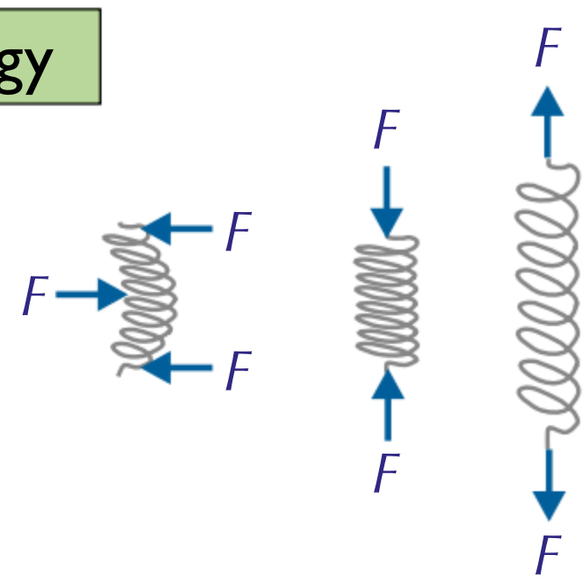
Quick Quiz

***Forces** don't just make objects **move**, they can also make them **change shape**. Whether they change shape **temporarily** or **permanently** depends on **the object** and the forces applied.*

Stretching, Compressing or Bending Transfers Energy

1) When you apply forces to an object you may change its shape by stretching, compressing or bending it.
2) To change the shape of an object (deform it) like this, you need more than one force acting on the object (otherwise the object would move in the direction of the applied force).
3) There are two types of deformation:

Elastic deformation	The object returns to its original shape and length after the forces have been removed.
Inelastic deformation	The object doesn't return to its original shape and length after the forces have been removed.



Objects that can be elastically deformed are called elastic objects (e.g. a spring).

4) Work is done when a force stretches, compresses or bends an object and causes energy to be transferred to the elastic potential energy store of the object. If it is elastically deformed, ALL this energy is transferred to the object's elastic potential energy store (see p.4).

Extension is Directly Proportional to Force...

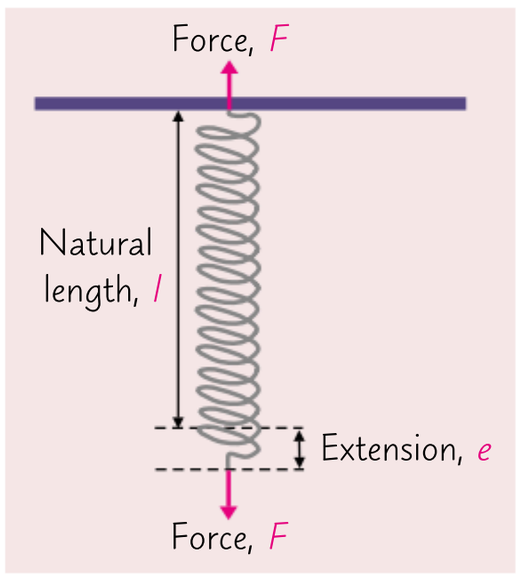
If a spring is supported at the top and has a weight attached to the bottom, it stretches.

- The extension of a stretched spring (or other elastic object) is directly proportional to the load or force applied — so $F \propto e$.
- The formula that shows this relationship is:

$$F = ke$$

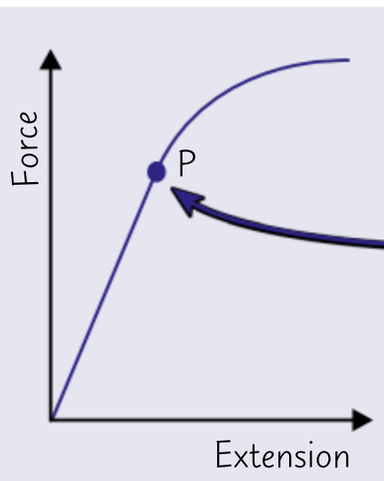
F = Force (N), k = Spring constant (N/m), e = Extension (m)

- The spring constant depends on the material being deformed — a stiffer spring has a greater spring constant.
- The equation also works for compression (where e is just the difference between the natural and compressed lengths).



...but this Stops Working when the Force is Too Great

If the force is increased beyond a certain limit, the extension is no longer proportional to it.



- The graph shows force against extension for an elastic object.
- There is a maximum force above which the graph curves. This means that extension is no longer proportional to force. This is known as the limit of proportionality (P).
- You might see graphs with these axes the other way around. The graph still starts with a straight part, but starts to curve upwards once you go past the limit of proportionality, instead of downwards.

Elastic objects are only elastic up to a certain point...

That equation is pretty simple, but that doesn't mean you can skip over it. Have a go at this question.

Q1 A spring is fixed at one end and a force of 1 N is applied to the other end, causing it to stretch. The spring extends by 2 cm. Calculate the spring constant of the spring. [4 marks]

Q1 Video Solution

Investigating Springs

Quick Quiz

*Oh look, here's another one of those **Required Practicals**... This one looks pretty straightforward, but there are a few ways this experiment can **stretch** you. **Read on**, so you won't be past your limits in the exam.*

You Can Investigate the Link Between Force and Extension

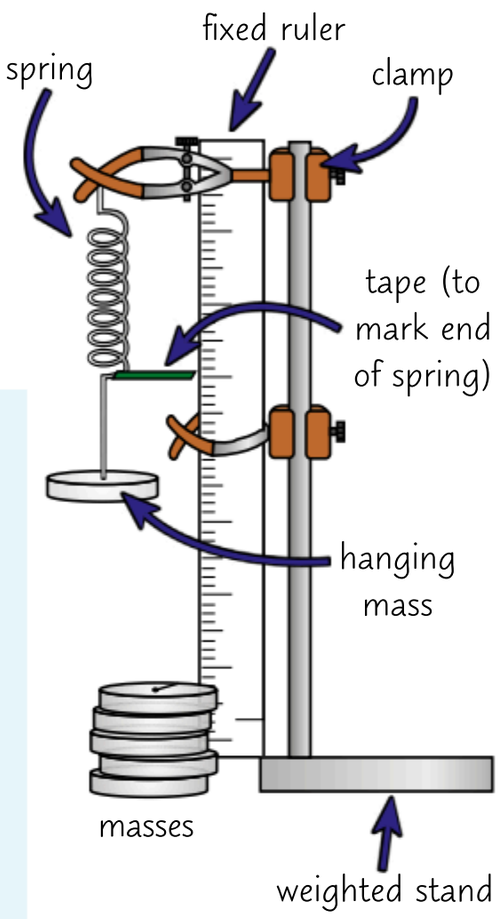
PRACTICAL

Video

One way to test the link between force and extension is to add masses to the end of a spring and record its extension.

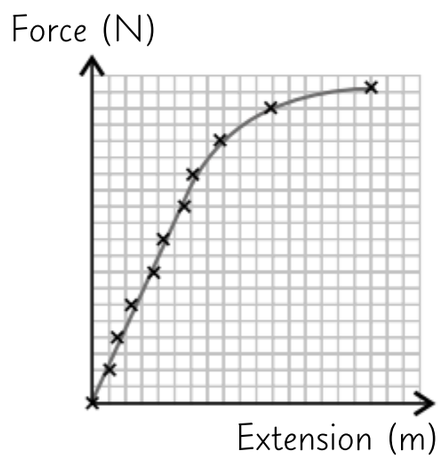
It's a good idea to do a quick pilot experiment first to check your masses are a good size:

- Using an identical spring to the one you'll be testing, load it with masses one at a time up to a total of five. Measure the extension each time you add a mass.
- Work out the increase in the extension of the spring for each mass you add. If any mass causes a bigger increase than the previous masses, you've gone past the spring's limit of proportionality. If this happens, you'll need to use smaller masses, or else you won't get enough measurements for your graph.



Method

1) Measure the exact mass of each of the masses (with a mass balance) and calculate its weight (the force it will apply) using $W = mg$ (p.62).
2) Measure the natural length of the spring (with no masses) with a millimetre ruler clamped to the stand. Make sure you take the reading at eye-level — you can add a marker (e.g. a thin strip of tape) to the bottom of the spring to make the reading more accurate.
3) Add a mass to the end of the spring and allow it to come to rest. Record the total mass and measure the new length of the spring. The extension is the change in length.
4) Repeat until you have enough measurements (no fewer than 6).

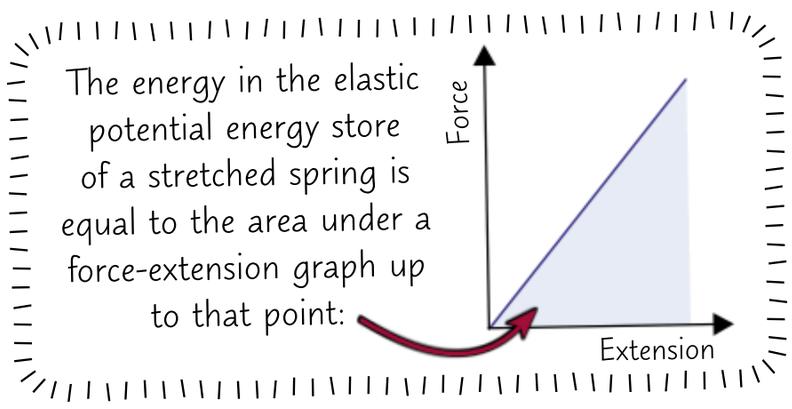


- Plot a force-extension graph of your results. It will only start to curve if you exceed the limit of proportionality.
- When the line of best fit is a straight line it means there is a linear relationship between force and extension (they're directly proportional). $F = ke$, so the gradient (see page 142) of the straight line is equal to k, the spring constant.
- When the line begins to bend, the relationship is now non-linear between force and extension — the spring stretches more for each unit increase in force.

You Can Work Out Energy Stored for Linear Relationships

1) As long as a spring is not stretched past its limit of proportionality, the work done in stretching (or compressing) a spring can be found using:

Elastic potential energy (J) — $E_e = \frac{1}{2} ke^2$ — Spring constant (N/m); Extension (m)



2) For elastic deformation, this formula can be used to calculate the energy stored in a spring's elastic potential energy store. It's also the energy transferred to the spring's elastic potential energy store as it's deformed (or transferred by the spring as it returns to its original shape).

The force-extension graph curves at the limit of proportionality...

You may not see the curved part in your force-extension graph for this experiment, depending on how many masses you add. It doesn't matter either way — you only need the gradient of your straight line to calculate the spring constant.

Moments

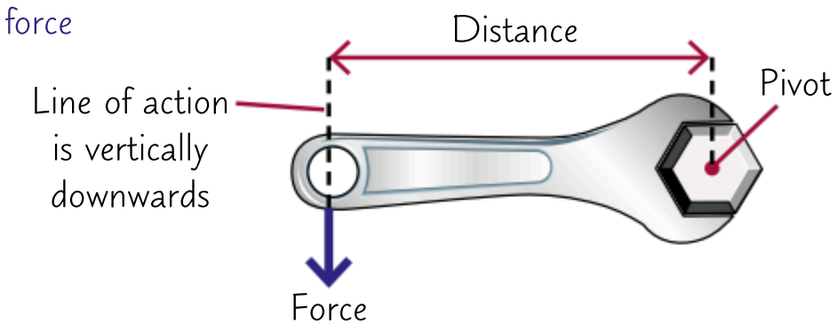
Quick Quiz

Moments *play a part in a lot of mechanical devices, from clocks to wheelbarrows. They can help you get the most out of a* ***small force****, and allow you to* ***transmit*** *forces across a distance.*

A **Moment** is the **Turning Effect** of a Force

A force can produce a turning effect (moment) on an object. The size of the moment of a force is given by:

Moment of a force (Nm) — $M = Fd$ — Force (N); Distance (m) — perpendicular distance from the pivot to the line of action of the force



- The force on the spanner causes a turning effect or moment on the nut (which acts as pivot). A larger force or a longer distance (spanner) would mean a larger moment.
- The line of action is just the direction the force acts.
  So in the diagram above, and on the exam, the distance you use for calculating a moment is just the distance between the point where the force acts and the pivot. This is the perpendicular distance.

If the total anticlockwise moment equals the total clockwise moment about a pivot, the object is balanced and won't turn. You can use the equation to find a missing force or distance in these situations.

EXAMPLE

A 6 m long steel girder weighing 1000 N rests horizontally on a pole 1 m from one end. What is the tension in a supporting cable attached vertically to the other end?

MATHS SKILLS

1) The total anticlockwise moment (caused by the weight) equals the total clockwise moment (caused by the tension).

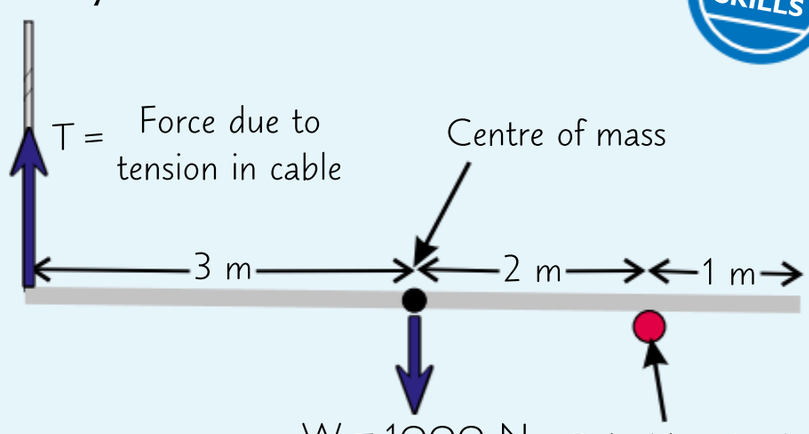
$$M = Fd \rightarrow F_1d_1 = F_2d_2 \rightarrow T \times d_1 = W \times d_2$$

2) Rearrange for T and stick in the values you know.

d_1 = total distance from pole to cable = 2 + 3 = 5 m

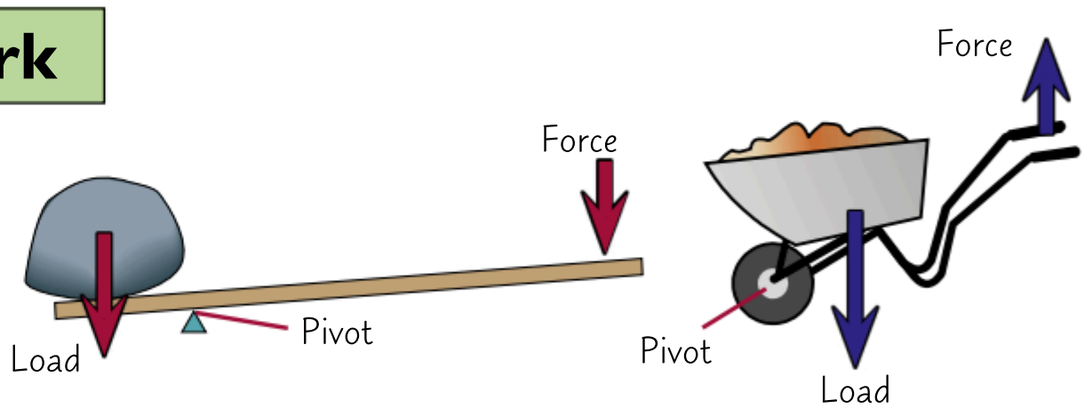
d_2 = distance from pole to centre of mass = 2 m

$T = (W \times d_2) \div d_1 = (1000 \times 2) \div 5 = 400$ N



Levers Make it **Easier** for us to **Do Work**

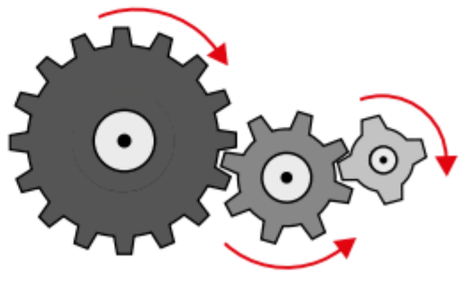
Levers increase the distance from the pivot at which the force is applied. Since $M = Fd$, this means less force is needed to get the same moment, making it easier to do work, e.g. lift a load or turn a nut.



Gears Transmit **Rotational Effects**

1) Gears are circular discs with 'teeth' around their edges.
2) Their teeth interlock so that turning one causes another to turn, in the opposite direction.
3) They are used to transmit the rotational effect (moment) of a force from one place to another.
4) Different sized gears affect the size of the moment of the force.
   If you apply a given force to turn a larger gear, the moment will be greater, as the distance to the pivot (centre of the gear) is greater.
5) When this moment is transmitted to a smaller gear, the smaller gear will turn faster than the larger gear.

Video



Moment of a force = force × distance...

Moments can be used in lots of different situations, so get your head around them sooner rather than later.

Q1 Your brother weighs 300 N and sits 2 m from the pivot of a seesaw. If you weigh 600 N, what distance from the pivot, on the other side of the seesaw, should you sit to balance it? [3 marks]

Q1 Video Solution

Warm-Up & Exam Questions

I can hear those gears whirring away in your brain from all that new information. So it's the perfect time to test what you know. Work through these questions then check to see if you got them right...

Warm-Up Questions

1) State the formula linking force, extension and spring constant.
2) What is meant by the limit of proportionality of a spring?
3) Give the equation used to calculate the moment of a force (around a pivot).
4) What are the units of the moment of a force?

Exam Questions

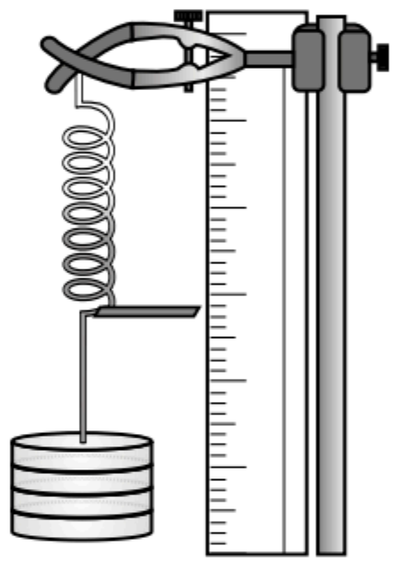
For these questions, you can use the formula sheet on page 216.

PRACTICAL

1 The teacher shows her students an experiment to show how a spring extends when masses are hung from it. She hangs a number of 90 g masses from a 50 g hook attached to the base of the spring. She records the extension of the spring and the total weight of the masses and hook each time she adds a mass to the bottom of the spring.

GRADE 6-7 GRADE

Figure 1



1.1 Give the independent variable in this experiment

[1 mark]

1.2 Give **one** control variable in this experiment.

[1 mark]

When a force of 4 N is applied to the spring, the spring extends elastically by 2.5 cm.

1.3 Calculate the spring constant of the spring.

[4 marks]

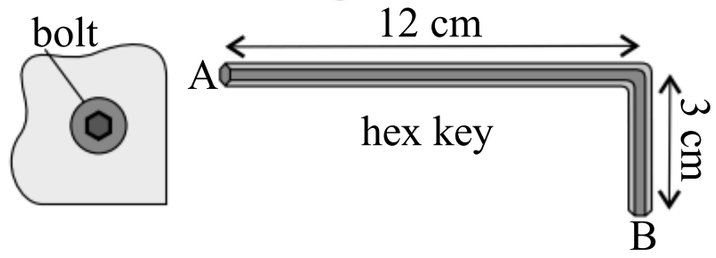
1.4 The teacher applies a 15 N force to the spring. When she removes the force, the spring is 7 cm long. The original length of the spring was 5 cm. Describe what has happened to the spring.

[1 mark]

2 Bolts are often secured using a hex key, as shown in **Figure 2**. One end of the hex key is put into the bolt and the other is turned to tighten the bolt.

GRADE 6-7 GRADE

Figure 2



2.1 Calculate the moment on the bolt when end A of the hex key is put into the bolt and a force of 15 N is applied to end B.

[3 marks]

2.2 Calculate the moment on the bolt when end B of the hex key is put into the bolt and a force of 15 N is applied to end A.

[3 marks]

2.3 Which end of the hex key (A or B) should be put into the bolt to make it easier to tighten? Explain your answer.

[2 marks]

Fluid Pressure

Quick Quiz

*I'm sure you're familiar with **pressure**, what with exams coming up, but it's an important concept in physics too.*

Pressure is the Force per Unit Area

- Fluids (gases or liquids) are substances that can 'flow' because their particles can move around. As these particles move around, they collide with surfaces and other particles, and exert a force on them.
- Pressure is force per unit area, so this means the particles exert a pressure.
- The pressure of a fluid means a force is exerted normal (at right angles) to any surface in contact with the fluid.
- You can calculate the pressure at the surface of a fluid by using:

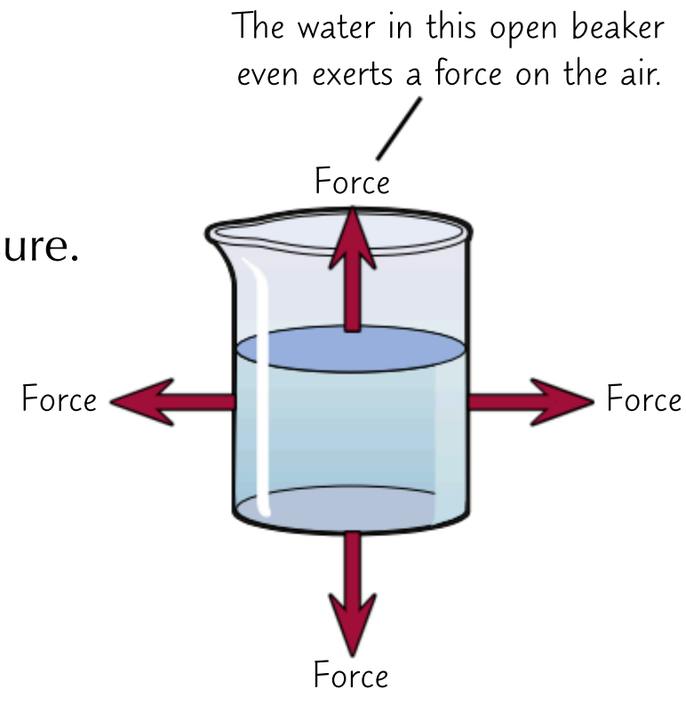
$$p = \frac{F}{A}$$

Pressure in pascals (Pa) — p

Force normal to a surface (N) — F

Area of that surface (m^2) — A

The water in this open beaker even exerts a force on the air.



Pressure in a Liquid Depends on Density and Depth

1) Density is a measure of how close together the particles in a substance are (p.41). The density of any given liquid is uniform and doesn't vary with shape or size.

Unlike liquids, the density of a gas can vary (see next page).

2) A more dense liquid has more particles in a certain space. So the number of particles in the column of liquid above a certain point is higher for a more dense liquid. The weight of these particles adds to the pressure at that point, so liquid pressure increases with density.
3) Going to a deeper point in a liquid also increases the number of particles above the point. Again, the weight of these particles adds to the pressure. So liquid pressure also increases with depth.
4) You can calculate the pressure at a certain depth due to the column of liquid above using:

$$p = h\rho g$$

Pressure (Pa) — p

Height of the column of liquid (m) — h

Gravitational field strength (N/kg) — g

Density of the liquid (kg/m^3) (the symbol is the Greek letter 'rho') — ρ

On Earth, g = 9.8 N/kg.

EXAMPLE

The density of water is 1000 kg/m³. Calculate the change in pressure between a point 25 m below the surface of water and a point 45 m below the surface.

MATHS SKILLS

1) Calculate the pressure caused by the water at a depth of 25 m.
2) Do the same for a depth of 45 m.
3) Take away the pressure at 25 m from the pressure at 45 m.

$p = h\rho g = 25 \times 1000 \times 9.8$
$= 245\,000$ Pa

$p = h\rho g = 45 \times 1000 \times 9.8$
$= 441\,000$ Pa

$441\,000 - 245\,000 = 196\,000$ Pa (or 196 kPa)

Check your answer makes sense (you can't get negative pressure).

The greater the depth, the higher the pressure...

Pressure = force ÷ area doesn't just apply to fluids — it's true for any situation where a force acts on an area.

Q1 Calculate the force exerted on a 10 m^2 area by a pressure of 200 kPa. [3 marks]

Q2 At a point 5.0 cm below the surface of a jug of olive oil, the pressure due to the oil is 450 Pa. Calculate the density of olive oil. The gravitational field strength of Earth is 9.8 N/kg. [3 marks]

Q2 Video Solution

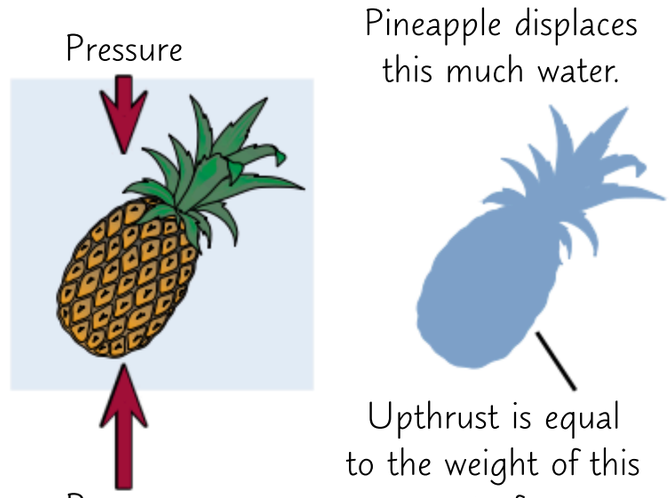
Upthrust and Atmospheric Pressure

Quick Quiz

*This page is about **upthrust** — a force that determines whether an object will **sink** or **float**.*
*So take some time to read up on it now, so you're not **submerged** in revision at the last minute.*

Objects in Fluids Experience **Upthrust**

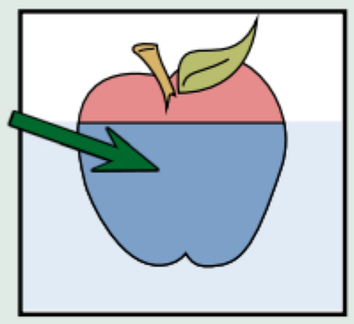
1) When an object is submerged in a fluid (either partially or completely), the pressure of the fluid exerts a force on it from every direction.
2) Pressure increases with depth, so the force exerted on the bottom of the object is larger than the force acting on the top of the object.
3) This causes a resultant force (p.63) upwards, known as upthrust.
4) The upthrust is equal to the weight of fluid that has been displaced (pushed out of the way) by the object.



The **Density** of an Object Determines if it **Sinks** or **Floats**

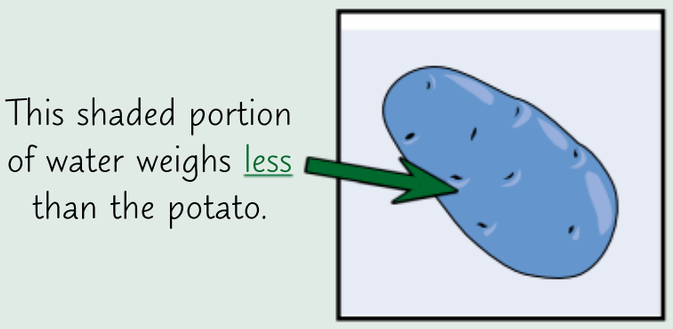
Floating

When weight = upthrust, the forces balance and the object floats. An object that is less dense than the fluid it's placed in weighs less than the same volume of fluid. The apple here is less dense than water:



Sinking

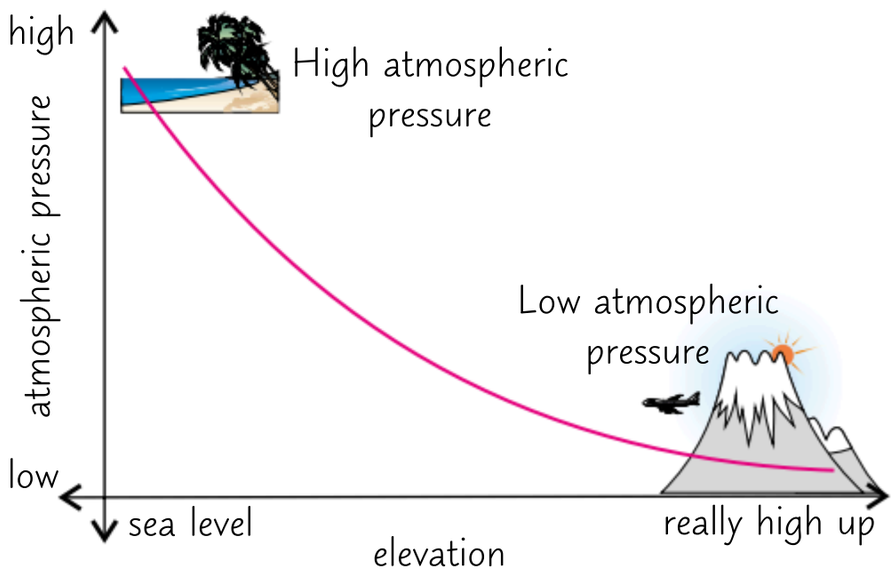
When weight is greater than upthrust, the object sinks. An object that is denser than the fluid it is placed in can't displace enough fluid to equal its weight. The potato here is more dense than water:



Submarines make use of upthrust. To sink, large tanks are filled with water to increase the weight of the submarine so that it is more than the upthrust. To rise to the surface, the water is emptied into the ocean and the tanks are filled with compressed air — this reduces the weight so that it's less than the upthrust.

Atmospheric Pressure Decreases with **Height**

1) The atmosphere is a layer of air that surrounds Earth. It is thin compared to the size of the Earth.
2) Atmospheric pressure is created on a surface by air molecules colliding with the surface.
3) As the altitude (height above sea level) increases, the atmosphere gets less dense, so there are fewer molecules around to collide with a surface at that height. This means atmospheric pressure decreases.
4) There are also fewer air molecules above the surface as its height increases. This means that the weight of the air above a surface, which contributes to atmospheric pressure, decreases with altitude.



Atmospheric pressure decreases as height above ground increases...

...because the density of air decreases as height increases. This is why mountain climbers carry tanks of oxygen. The lower density of air means there are fewer air particles, and so less oxygen, which makes it harder to breathe.

Warm-Up & Exam Questions

Well, now you've gotten through that section, have a go at these questions, and see how much you've learnt. If you can master these, it should ease the pressure a bit for the exam.

Warm-Up Questions

1) Define the term pressure.
2) Other than density, name two variables the pressure due to a column of liquid depends on.
3) An object with a weight of 3.5 N is floating in a bucket of water.
   Give the size of the upthrust acting on the object.
4) How does atmospheric pressure vary with altitude?

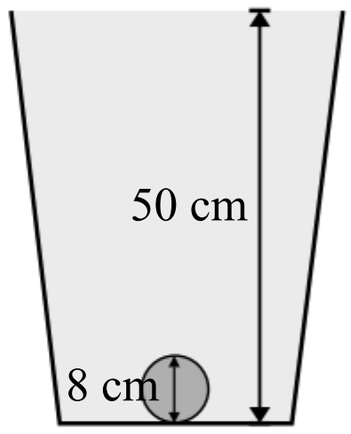
Exam Questions

For these questions, you can use the formula sheet on page 216.

1 A student places a ball into a bucket of water and it sinks to the bottom, as shown in **Figure 1**.
The water is 50 cm deep, and the ball has a diameter of 8 cm.

GRADE 6-7

Figure 1



1.1 What conclusion can be made about the density of the ball?
Tick **one** box.

- ☐ It is more dense than water.
- ☐ It is less dense than water.
- ☐ It has the same density as water.

[1 mark]

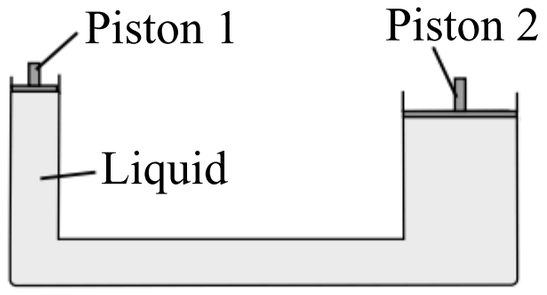
1.2 Water has a density of 1000 kg/m^3. The gravitational field strength is 9.8 N/kg.
Calculate the pressure due to the column of water above the ball.
Give your answer to 2 significant figures.

[4 marks]

2 **Figure 2** shows a simple hydraulic system containing a liquid. In a hydraulic system, pressure is transmitted from one piston to another through the liquid. The liquid cannot be compressed.

GRADE 7-9

Figure 2



2.1 A force of 175 N is applied to piston 1, which has a cross-sectional area of 0.25 m^2.
Calculate the pressure created at the first piston.

[2 marks]

2.2 Piston 2 has a cross-sectional area of 1.3 m^2.
Calculate the force acting on piston 2.

[3 marks]

Revision Summary Test for Topic 5a

There's not a moment to waste — if you want to become a force to be reckoned with, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

Contact and Non-Contact Forces (p.61-62) ☑

1) What is the difference between a vector and a scalar quantity? Give five examples of each. ☑
2) What do the length and direction of an arrow used to represent a vector tell you? ☑
3) What is a force? ☑
4) What is the difference between a contact and a non-contact force? Give three examples of each. ☑
5) Describe the forces acting between a stationary chair and the ground it's standing on. ☑
6) Give two effects that the gravity of a really big mass has on objects in its gravitational field. ☑
7) List four differences between weight and mass. ☑
8) What happens to Earth's gravitational field strength as you move away from the surface? ☑
9) Define the centre of mass of an object. Where is it in a uniform object? ☑
10) What's the formula that relates weight and mass? ☑

Resultant Forces, Work Done and Equilibrium (p.63-64) ☑

11) What does a free body diagram show? ☑
12) What is a resultant force? How is it calculated for two forces acting on an object along the same line? ☑
13) Why does the temperature of an object increase when it is pushed along a rough surface? ☑
14) What is the formula for the work done when a force moves an object? Give the definition of a joule. ☑
15) Describe how you can use a scale drawing to calculate the resultant force on an object. ☑
16) What does it mean if the forces acting on an object are in equilibrium?
    How could you tell whether they are in equilibrium from a scale drawing? ☑
17) Describe how you can resolve a force using a square grid. ☑

Elasticity and Moments (p.66-68) ☑

18) What's the difference between elastic and inelastic deformation? ☑
19) What does each letter in the formula $F = ke$ mean? How is k affected by the stiffness of a spring? ☑
20) What is the limit of proportionality? Where is it on a graph of force against extension? ☑
21) Describe a method that can be used to investigate the link between force and extension for a spring. ☑
22) What formula is used to find the elastic potential energy stored in an elastically deformed spring? ☑
23) What is a moment? Give two variables you can change to increase the moment of a force. ☑
24) What can you say about the moments acting on a balanced object? Describe the motion of the object. ☑
25) Describe how each of the following increase the moment of a force: a) levers, b) gears. ☑

Fluid Pressure and Upthrust (p.70-71) ☑

26) What is the formula for the pressure exerted at the surface of a fluid? Give the units of pressure. ☑
27) Explain how the pressure of a fluid depends on its density and depth. ☑
28) Explain what causes upthrust. How do submarines make use of upthrust? ☑
29) Explain how you can predict whether a particular object will sink or float when placed in a certain fluid. ☑
30) What causes atmospheric pressure? Why does it decrease with altitude on Earth? ☑

Quick Quiz

Distance, Displacement, Speed and Velocity

*You could walk a **distance** of a mile, but still have a **displacement** of zero afterwards. Intrigued? Read on...*

Distance and **Displacement** are Both **How Far You've Travelled**

- Distance is just how far an object has moved. It's a scalar quantity (p.61) so it doesn't involve direction.
- Displacement is a vector quantity. It measures the distance and direction in a straight line from an object's starting point to its finishing point — e.g. 5 metres north. The direction could be relative to a point (e.g. towards the school) or an angle (e.g. 35° from the horizontal).

If you move 5 m north, then 5 m south, your displacement is 0 m but the distance travelled is 10 m.

Speed is a **Scalar**, **Velocity** is a **Vector**

SPEED is just how fast you're going (e.g. 30 mph or 20 m/s).
VELOCITY is speed in a given direction, e.g. 30 mph north or 20 m/s at 60° from the vertical.

1) So objects can travel at a constant speed but with a changing velocity.
This happens when the object is changing direction.
2) An object moving in a circle has a constantly changing velocity,
as the direction is always changing (e.g. a car going around a roundabout).
3) To measure the speed of an object that's moving with a constant speed,
you can time how long it takes the object to travel a certain distance.
You can then calculate the object's speed from your measurements using this formula:

$$s = vt$$

Distance travelled (m) = Speed (m/s) × Time (s)

Objects rarely travel at a constant speed. E.g. when you walk, run or travel in a car, your speed tends to fluctuate. For these cases, the formula above gives the average (mean) speed during the total time.

You Need to Know Some **Typical** Everyday **Speeds**

Lots of things affect the speed something travels at, but you need to remember these typical speeds:

	Walking	Running	Cycling	Car	Train	Plane
Typical speed	1.5 m/s	3 m/s	6 m/s	25 m/s	30 m/s	250 m/s

- The speed at which a person actually walks, runs or cycles depends on their fitness, their age, the distance travelled and the terrain (the type of land they're moving over, e.g. roads, fields) as well as many other factors.
- It's not only the speed of objects that varies. The speed of sound (330 m/s in air) changes depending on what the sound waves are travelling through.
- Wind speed is affected by things like temperature, atmospheric pressure and if there are any large buildings or structures nearby (e.g. forests reduce the speed of the air travelling through them).

Speed equals distance over time...

Remember those typical speeds of objects — you might need to use them to make estimates.

Q1 A sprinter runs 200 m in 25 s. Calculate his speed. [3 marks]

Q1 Video Solution

Q2 Marie walks her dog after school. She takes a route of 1500 m that starts at and returns to her house. State: a) the distance she travels b) her displacement [2 marks]

Acceleration

Quick Quiz

Acceleration *is the* ***rate of change*** *of* ***velocity****. For cases of* ***constant acceleration****, there's a really useful* ***equation*** *you can use to calculate all sorts of* ***variables*** *of* ***motion****.*

Acceleration is How **Quickly** You're **Speeding Up**

1) Acceleration is the change in velocity in a certain amount of time.
2) You can find the average acceleration of an object using this formula:
3) Deceleration is just negative acceleration (if something slows down, the change in velocity is negative).

$$a = \frac{\Delta v}{t}$$

a = Acceleration (m/s²), Δv = Change in velocity (m/s), t = Time (s)

EXAMPLE

A cat accelerates at 2.5 m/s² from 2.0 m/s to 6.0 m/s. Find the time it takes to do this.

MATHS SKILLS

1) Rearrange $a = \Delta v \div t$ for t. $t = \Delta v \div a$
2) Plug in the values from the question. $t = (6.0 - 2.0) \div 2.5 = 1.6$ s

You Need to be Able to **Estimate Accelerations**

To do this, you need the typical speeds from the previous page:

EXAMPLE

A car is travelling along a road, when the driver sees a hazard ahead and applies the brakes. It takes 2.5 s for the car to come to a complete stop. Estimate the deceleration of the car.

1) First, give a sensible speed for the car to be travelling at.
2) Put these numbers into the acceleration equation.
3) The question asked for the deceleration, so you can lose the minus sign (which shows the car is slowing down):

The typical speed of a car is ~25 m/s.

The car comes to a stop in 2.5 s.

MATHS SKILLS

$a = \Delta v \div t$
$= (-25) \div 2.5$
$= -10$ m/s²

So an estimate of the deceleration is ~10 m/s²

The ~ symbol just means it's an approximate value (or answer).

Uniform Acceleration Means a **Constant Acceleration**

- Acceleration due to gravity (g) is uniform for objects in free fall. It's roughly equal to 9.8 m/s² near the Earth's surface and has the same value as gravitational field strength (p.62).
- You can use this equation for uniform acceleration:

$$v^2 - u^2 = 2as$$

v = Final velocity (m/s), u = Initial velocity (m/s), a = Acceleration (m/s²), s = Distance (m)

Initial velocity is just the starting velocity of the object.

EXAMPLE

A van travelling at 23 m/s starts decelerating uniformly at 2.0 m/s² as it heads towards a built-up area 112 m away. What will its speed be when it reaches the built-up area?

1) First, rearrange the equation so v^2 is on one side. $v^2 - u^2 = 2as \rightarrow v^2 = u^2 + 2as$
2) Now put the numbers in — remember, a is negative because it's a deceleration. $v^2 = 23^2 + (2 \times -2.0 \times 112)$ $= 529 - 448 = 81$
3) Finally, take the square root of the whole thing. $v = \sqrt{81} = 9$ m/s

MATHS SKILLS

An object is accelerating if its velocity is changing...

Don't forget — you can only use that second equation when the acceleration is uniform.

Q1 A ball is dropped from a height, h, above the ground. The speed of the ball just before it hits the ground is 7.0 m/s. Calculate the height the ball is dropped from. (acceleration due to gravity ≈ 9.8 m/s²) [3 marks]

Q1 Video Solution

Distance-Time and Velocity-Time Graphs

Quick Quiz

*You need to be able to **draw** and **interpret distance-time** and **velocity-time graphs**.*

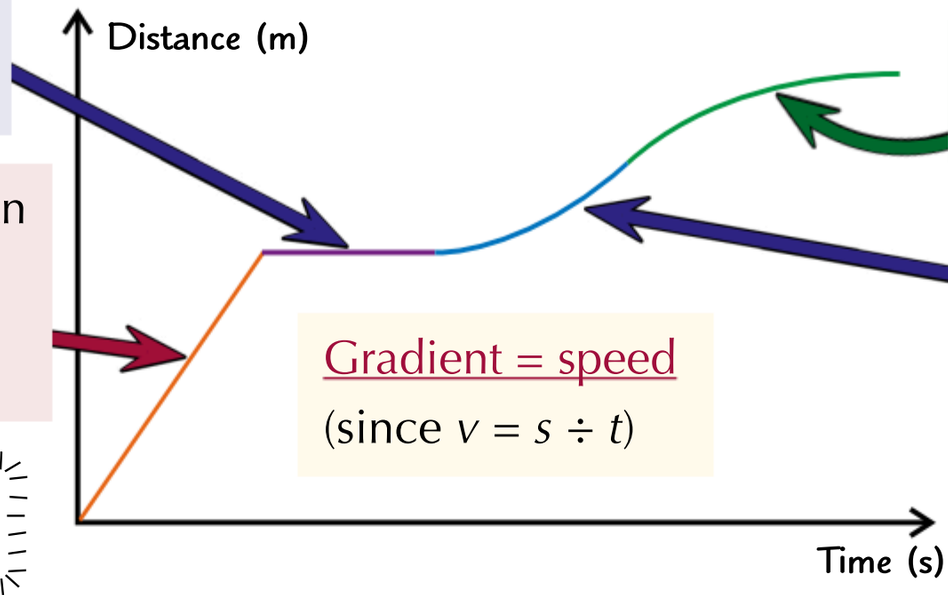
You Can **Show Journeys** on **Distance-Time Graphs**

If an object moves in a straight line, its distance travelled can be plotted on a distance-time graph:

Flat sections are where it's stationary — it's stopped.

Straight uphill sections mean it is travelling at a steady speed — the steeper the graph the faster it's going.

Gradient = change in vertical axis ÷ change in horizontal axis (see p.142)



Gradient = speed
(since $v = s \div t$)

A levelling off curve means it's slowing down — decelerating.

A steepening curve means it's accelerating.

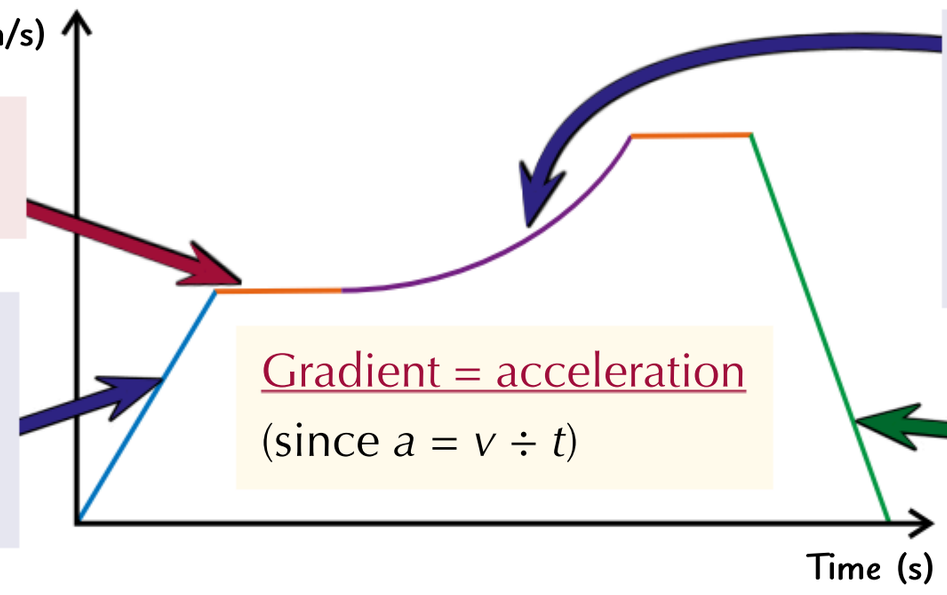
If the graph is curved, you can find the speed at a point by drawing a tangent to the curve, then speed = gradient of tangent.

You Can Also Show them on a **Velocity-Time Graph**

How an object's velocity changes as it travels can be plotted on a velocity-time graph:

Flat sections represent travelling at a steady speed.

Straight uphill sections show constant acceleration. A steeper graph means a greater acceleration.



Gradient = acceleration
(since $a = v \div t$)

A curve upwards means increasing acceleration.

If the graph is curved, you can find the acceleration at a point by calculating the gradient of a tangent to the curve.

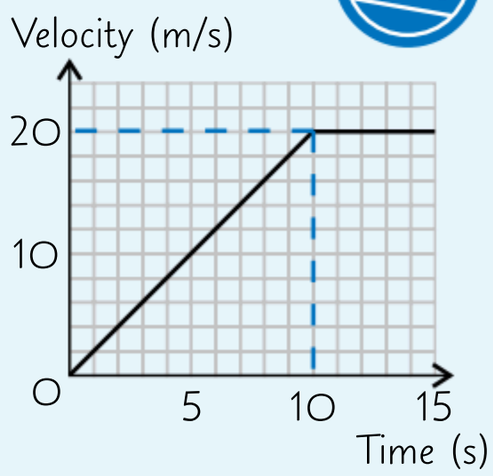
Downhill sections show deceleration.

The AREA under any section of the graph (or all of it) is equal to the distance travelled in that time interval. If the section under the graph is irregular, it's easier to find the area by counting the squares under the line and multiplying the number by the value of one square.

EXAMPLE

The velocity-time graph of a car's journey is plotted.
a) Calculate the acceleration of the car over the first 10 s.
b) How far does the car travel in the first 15 s of the journey?

MATHS SKILLS



1) This is just the gradient of the line:
$a = \Delta v \div t = 20 \div 10 = 2$ m/s²

2) Split the area into a triangle and a rectangle, then add together their areas.
Area = (½ × 10 × 20) + (5 × 20)
= 200 m

3) Or find the value of one square, count the total number of squares under the line, then multiply these two values together.
1 square = 2 m/s × 1 s = 2 m
Area = 100 squares
= 100 × 2 = 200 m

Read the axes of any graph you get given carefully...

Q2 Video Solution

Make sure you don't get confused between distance-time graphs and velocity-time graphs.

Q1 Sketch the distance-time graph for an object that accelerates before travelling at a steady speed. [2 marks]

Q2 A stationary car starts accelerating increasingly for 10 s until it reaches a speed of 20 m/s. It travels at this speed for 20 s until the driver sees a hazard and brakes. He decelerates uniformly, coming to a stop 4 s after braking. Draw the velocity-time graph for this journey. [3 marks]

Terminal Velocity

Quick Quiz

*If an object **falls** for long enough, it will reach its **terminal velocity**. It's all about **balance** between **weight** and **air resistance**. **Parachutes** work by **decreasing** your terminal velocity.*

Friction is Always There to **Slow Things Down**

- If an object has no force propelling it along it will always slow down and stop because of friction (unless you're in space where there's nothing to rub against).
- Friction always acts in the opposite direction to movement.
- To travel at a steady speed, the driving force needs to balance the frictional forces (see page 79).
- You get friction between two surfaces in contact, or when an object passes through a fluid (drag).

Drag **Increases** as **Speed Increases**

1) Drag is the resistance you get in a fluid (a gas or a liquid). Air resistance is a type of drag.
2) Frictional forces from fluids increase with speed. A car has much more drag to work against at 70 mph than at 30 mph. So at 70 mph the engine has to work much harder to maintain a steady speed.
3) The most important factor in reducing drag is using a streamlined shape. Parachutes work in the opposite way — they want to maximise drag.

A streamlined shape allows fluid to flow easily across it, reducing drag.

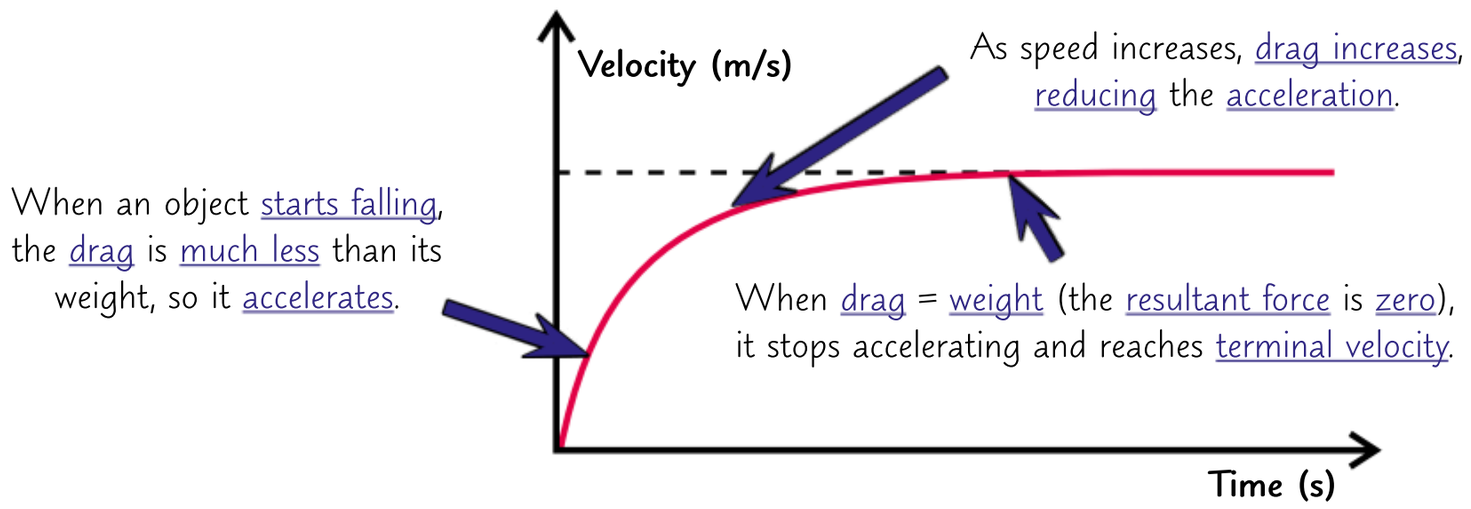
Objects **Falling** Through **Fluids** Reach a **Terminal Velocity**

The accelerating force acting on all falling objects is their weight (due to gravity). It would make all objects accelerate at the same constant rate until they hit the ground if it wasn't for drag.

An object falling through air (or another fluid) doesn't accelerate at a constant rate. It reaches a maximum speed (terminal velocity) that depends on drag and its weight:

Video

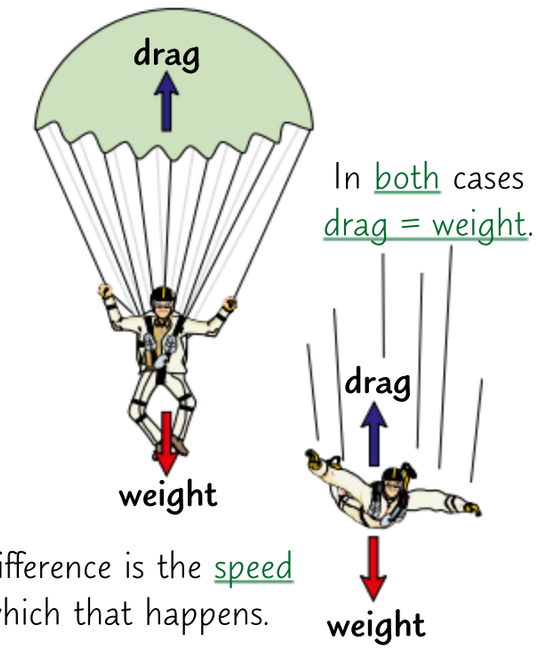
1) Weight acts downwards and doesn't change as the object falls.
2) Drag acts upwards and increases with speed.
3) The object reaches terminal velocity when drag = weight.



At any given speed, drag depends on the size and shape of the object — larger area = more drag.

For a skydiver falling through the air:

- With the parachute closed he has quite a small area, so the air resistance is low. He needs to be going pretty fast for the drag to equal his weight — a terminal velocity of about 120 mph.
- With the parachute open, there's much more air resistance, so drag = weight at a much lower speed. You get a terminal velocity of about 15 mph — a safe speed to hit the ground at.



In both cases drag = weight.

The difference is the speed at which that happens.

Terminal velocity is when drag = weight...

Make sure you're clear on what terminal velocity is and why it happens. If you're given the velocity-time graph of a falling object, you should be able to explain the object's motion in terms of the forces acting at any given time.

Warm-Up & Exam Questions

Slow down — it's not time to move on to the next section just yet. First, it's time to check that all the stuff you've just read is still running around your brain. Dive into these questions.

Warm-Up Questions

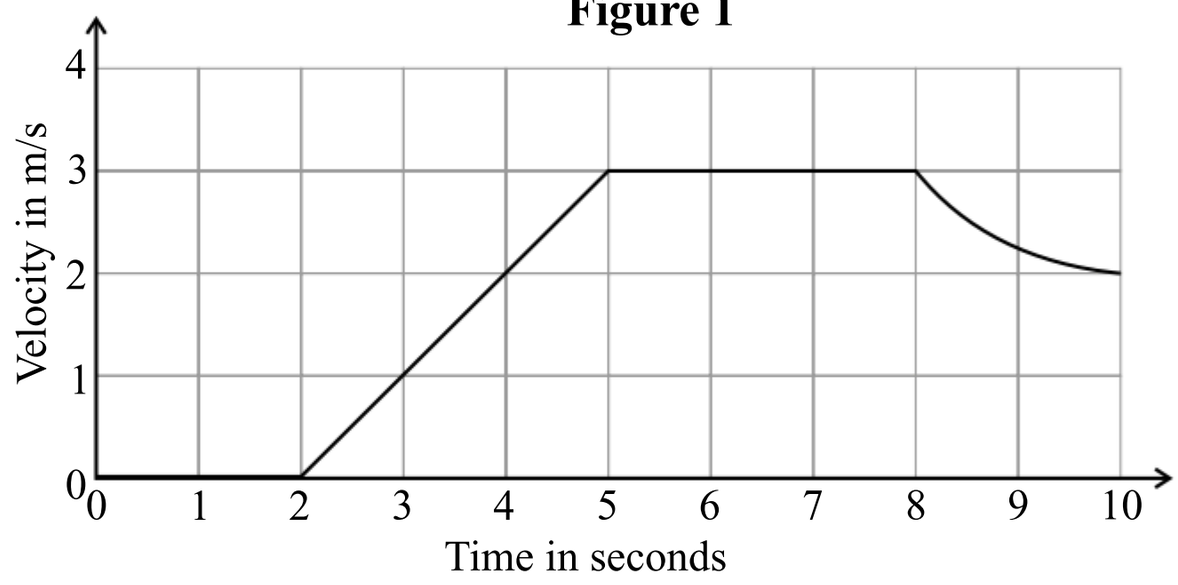
1) What is the difference between speed and velocity?
2) Suggest the typical speeds of: a) a person running, b) a train, c) a plane.
3) How is acceleration shown on a distance-time graph?
4) Describe the shape of the line on a velocity-time graph for an object travelling at a steady speed.
5) In general, how does the air resistance acting on a car change as its speed increases?
6) A skydiver is falling at terminal velocity when she opens her parachute. After a while, she reaches a new terminal velocity. How does her new terminal velocity compare to her original terminal velocity?

Exam Questions

For question 1, you can use the formula sheet on page 216.

1 **Figure 1** shows the velocity-time graph of a cyclist. GRADE 4-6

Figure 1



1.1 Describe the motion of the cyclist between 5 and 10 seconds.

[2 marks]

1.2 Calculate how far the cyclist travelled between 2 and 5 seconds.

[2 marks]

1.3 Calculate the acceleration of the cyclist between 2 and 5 seconds.

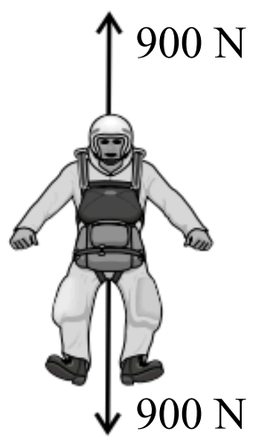
[2 marks]

1.4 Calculate the average deceleration of the cyclist between 8 and 10 seconds.

[3 marks]

2 Two parachutists, A and B, are members of the same club. **Figure 2** shows the forces acting on parachutist A. GRADE 6-7

Figure 2



2.1 Give the resultant force acting on parachutist A.

[1 mark]

2.2 Describe the velocity of parachutist A.

[1 mark]

Parachutist B is in free fall.
The total weight of parachutist B and her equipment is 700 N.

2.3 Determine the force of air resistance on parachutist B when she reaches terminal velocity. Explain your answer.

[2 marks]

2.4 Which parachutist, A or B, would you expect to have a higher terminal velocity? Explain your answer.

[4 marks]

2.5 Explain why a parachutist slows down when they open their parachute.

[2 marks]

Newton's First and Second Laws

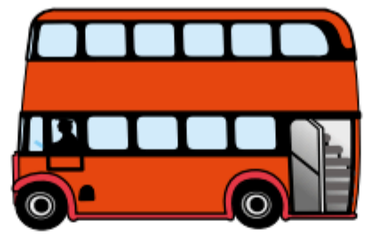
Quick Quiz

*In the 1660s, a chap called **Isaac Newton** worked out some useful **Laws of Motion**. Here are the first **two**.*

Newton's **First** Law — a **Resultant Force** is Needed to **Change Motion**

If the resultant force on a stationary object is zero, the object will remain stationary.
If the resultant force on a moving object is zero, it'll just carry on moving at the same velocity.

So, when a bus or train (or anything else) is stationary or moving at a constant velocity (same speed and direction), the resistive and driving forces on it must be balanced.



Stationary

Constant velocity

Have a look at page 63 for a reminder about resultant forces.

The velocity will only change if there's a non-zero resultant force acting on the object.

1) A non-zero resultant force produces an acceleration (or deceleration) in the direction of the force.
2) This "acceleration" can take five different forms: starting, stopping, speeding up, slowing down and changing direction.
3) On a free body diagram, the arrows will be unequal.

Accelerating

Newton's **Second** Law — **Acceleration** is **Proportional** to **Resultant Force**

1) The larger the resultant force acting on an object, the more the object accelerates — the force and the acceleration are directly proportional. You can write this as $F \propto a$.
2) Acceleration is also inversely proportional to the mass of the object — so an object with a larger mass will accelerate less than one with a smaller mass (for a fixed resultant force).

Which gives this incredibly useful formula:

$$F = ma$$

Resultant force (N) — Acceleration (m/s^2) — Mass (kg)

EXAMPLE

A van of mass of 2080 kg has an engine that provides a driving force of 5200 N. At 70 mph the drag force acting on the van is 5148 N. Find its acceleration at 70 mph.

1) Work out the resultant force on the van. (Drawing a free body diagram may help, see p.63.)
2) Rearrange $F = ma$ and stick in the values you know.

Resultant force = 5200 − 5148 = 52 N
$a = F \div m$
$= 52 \div 2080 = 0.025\ m/s^2$

MATHS SKILLS

You can use Newton's Second Law to get an idea of the forces involved in everyday transport. Large forces are needed to produce large accelerations:

EXAMPLE

Estimate the resultant force on a car as it accelerates from rest to a typical speed.

1) Estimate the acceleration of the car, using typical speeds from page 74.
2) Estimate the mass of the car.
3) Put these numbers into Newton's 2nd Law.

A car will reach its typical speed of ~25 m/s in ~10 s.
So $a = \Delta v \div t = 25 \div 10 = 2.5\ m/s^2$
Mass of a car is ~1000 kg.
So using $F = ma = 1000 \times 2.5 = 2500$ N
So an estimate of the resultant force is ~2500 N

MATHS SKILLS

F = ma — there's no acceleration without a resultant force...

Short and sweet, just how I like my equations. Sadly you can't get away with just learning those symbols, so make sure you've got your head around both of those laws before moving on to the next one — Newton's aptly named Third Law.

Inertia and Newton's Third Law

Quick Quiz

Inertia *and* ***Newton's Third Law*** *might seem simple on the surface, but they can quickly get confusing. Make sure you really understand what's going on with them — especially if an object is in* ***equilibrium****.*

Inertia is the Tendency for **Motion** to **Remain Unchanged**

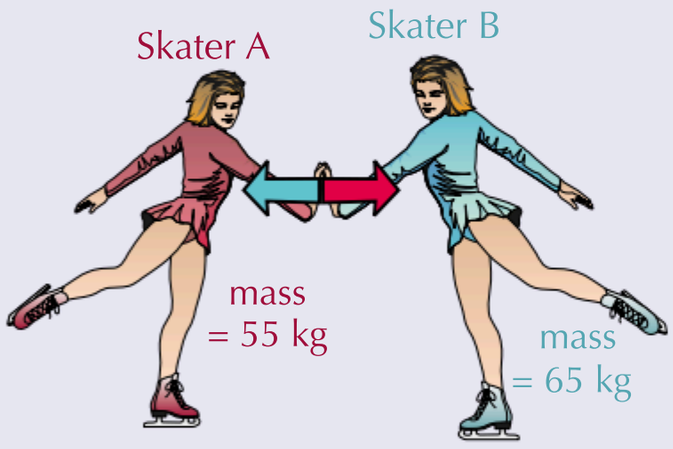
- Unless a resultant force acts on them, objects at rest stay at rest and objects moving at a steady speed keep moving at that speed (Newton's First Law). This tendency to continue in the same state of motion is called inertia.
- An object's inertial mass is a measure of how difficult it is to change the velocity of the object.
- Inertial mass can be found using Newton's Second Law — $F = ma$ (previous page). Rearranging this gives $m = F \div a$, so inertial mass is just force over acceleration.

Newton's **Third** Law — Objects Exert **Equal** and **Opposite Forces**

When two objects interact, the forces they exert on each other are equal and opposite.

1) If you push something, say a shopping trolley, the trolley will push back against you, just as hard.
2) And as soon as you stop pushing, so does the trolley. Kinda clever really.
3) So far so good. The slightly tricky thing to get your head round is this — if the forces are always equal, how does anything ever go anywhere? The important thing to remember is that the two forces are acting on different objects.

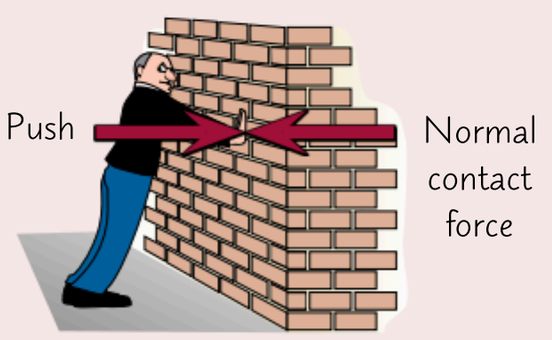
Video



When skater A pushes on skater B, she feels an equal and opposite force from skater B's hand (the 'normal contact' force). Both skaters feel the same sized force, in opposite directions, and so accelerate away from each other.

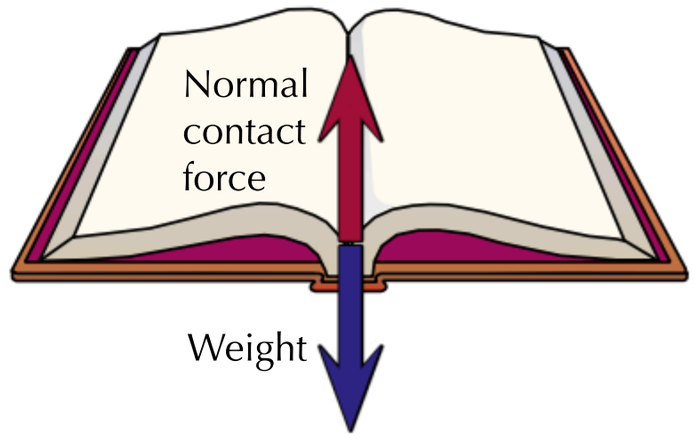
Skater A will be accelerated more than skater B, though, because she has a smaller mass — remember $a = F \div m$.

An example of Newton's Third Law in an equilibrium situation is a man pushing against a wall. As the man pushes the wall, there is a normal contact force acting back on him. These two forces are the same size. As the man applies a force and pushes the wall, the wall 'pushes back' on him with an equal force.



Be careful with Newton's Third Law when an object is in equilibrium. A book resting on the ground is in equilibrium. The weight of the book is equal to the normal contact force.

But this is NOT Newton's Third Law because the two forces are different types, and both acting on the book.



Interacting objects exert equal and opposite forces on each other

Q1 Video Solution

Newton's 3rd Law trips people up, so make sure you understand exactly what the forces are acting on and how that results in movement (or lack of it). Then have a go at this question to practise what you know.

Q1 A car moves at a constant velocity along a road, so that it is in equilibrium. Give an example of a pair of forces that demonstrate Newton's Third Law in this situation. [1 mark]

PRACTICAL

Investigating Motion

Quick Quiz

*Sure, you can learn the different **laws of motion**, but doing an experiment for yourself can really help you to understand what's going on. Read on for some snazzy ways to test how **mass** and **force affect motion**.*

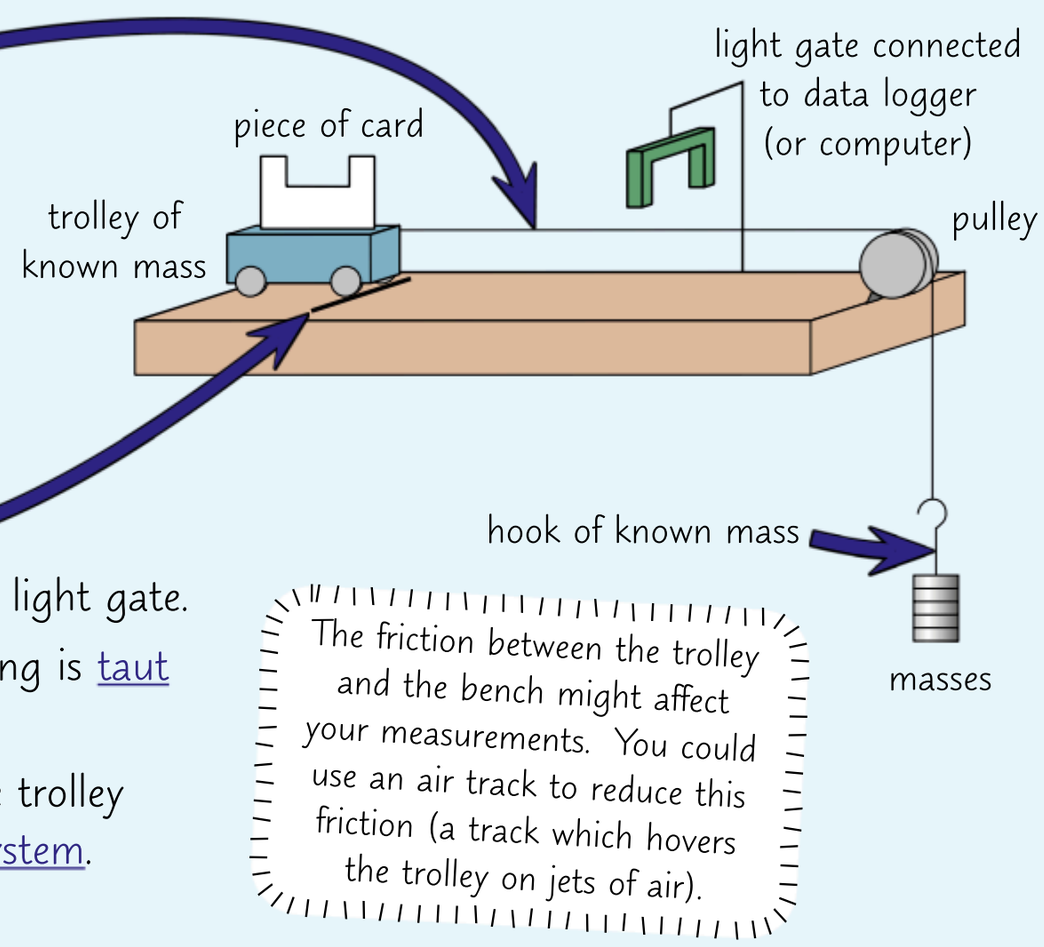
You can **Investigate** how **Mass** and **Force** Affect Acceleration

The set-up below shows an experiment to test Newton's 2nd law, $F = ma$ (p.79).
(This one uses a light gate and data logger, but you can do a similar experiment with a stopwatch and chalk lines marked on the table.)

The trolley holds a piece of card with a gap in the middle, so it will interrupt the signal on the light gate twice. If you input the length of each bit of card into the software, the light gate can measure the velocity of each bit as it passes. It'll use this to work out the acceleration of the trolley.

Method

1) Connect the trolley to a piece of string that goes over a pulley and is connected on the other side to a hook (of known mass that you can add masses to).
2) The weight of the hook and any masses attached to it provide the accelerating force ($W = mg$). This accelerates both the trolley and the masses, so you're investigating the acceleration of the whole system (the trolley, hook and masses together).
3) Mark a starting line on the table the trolley is on, so that the trolley always travels the same distance to the light gate.
4) Place the trolley on the starting line, holding it so the string is taut (not hanging loose), and release it.
5) Record the acceleration measured by the light gate as the trolley passes through it. This is the acceleration of the whole system.
6) Repeat this twice more to get an average acceleration.



The friction between the trolley and the bench might affect your measurements. You could use an air track to reduce this friction (a track which hovers the trolley on jets of air).

Add Masses to Change the **Mass**, **Transfer** Masses to Change the **Force**

Investigating Mass

- To investigate the effect of mass on acceleration, add masses to the trolley one at a time to increase the mass of the system.
- Don't add masses to the hook, or you'll change the force.
- Record the average acceleration for each mass.
- By adding masses to the trolley, the mass of the whole system increases, but the force applied to the system stays the same.
- You should find that the acceleration decreases as the mass increases ($a = F \div m$).

Investigating Force

- To investigate the effect of force, you need to keep the total mass of the system the same, but change the mass on the hook.
- So start with all the masses on the trolley, and transfer them to the hook one at a time to increase the accelerating force.
- Record the average acceleration for each force.
- By transferring the masses from one part of the system (the trolley) to another (the hook), you are increasing the accelerating force without changing the mass of the whole system.
- You should find that the acceleration increases as the force increases ($a = F \div m$).

These experiments have lots of steps — don't speed through them...

Make sure the string is the right length and there's enough space for the hanging masses to fall. The masses shouldn't hit the floor before the trolley has passed through the light gate fully. If they do, the force won't be applied the whole way through the trolley's journey, so you won't get an accurate measurement for the speed.

Warm-Up & Exam Questions

Now you've got yourself on the right side of the law(s of motion), it's time to put your knowledge on trial. Have a go at cross-examining these questions.

Warm-Up Questions

1) What is the resultant force on an object moving at a constant velocity?
2) Boulders A and B are accelerated from 0 m/s to 5 m/s in 10 s. Boulder A required a force of 70 N, and Boulder B required a force of 95 N. Which boulder has the greater inertial mass?
3) True or false? Two interacting objects exert equal and opposite forces on each other.
4) In a trolley-and-pulley system, as in the practical on page 81, where should you put masses to increase the mass of the system without increasing the force on the trolley?

Exam Questions

For question 2, you can use the formula sheet on page 216.

1 Dahlia's cricket bat has a mass of 1.2 kg.
She uses it to hit a ball with a mass of 160 g forwards with a force of 500 N. GRADE 4-6

1.1 State the force that the ball exerts on the bat. Explain your answer.

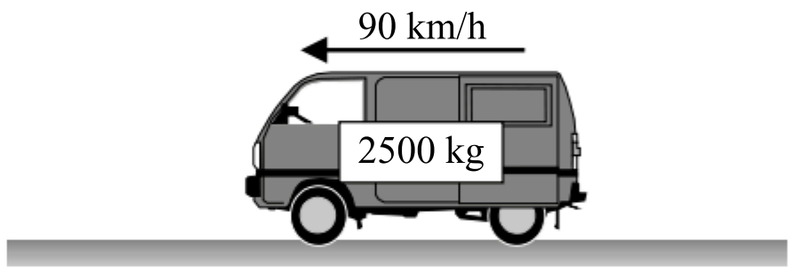
[2 marks]

1.2 Which is greater — the acceleration of the bat or the ball? Explain your answer.

[2 marks]

2 A camper van has a mass of 2500 kg.
It is driven along a straight, level road at a constant speed of 90.0 kilometres per hour. GRADE 6-7

Figure 1



2.1 A headwind begins blowing with a force of 200 N, causing the van to slow down.
Calculate the van's deceleration.

[3 marks]

The van begins travelling at a constant speed before colliding with a stationary 10.0 kg traffic cone.
The traffic cone accelerates in the direction of the van's motion with an acceleration of 29.0 m/s^2.

2.2 Calculate the force applied to the traffic cone by the van.

[2 marks]

2.3 Calculate the deceleration of the van during the collision.
Assume all of the force applied by the cone to the van causes the deceleration.

[3 marks]

PRACTICAL

3* Kaleb is investigating how acceleration varies with force.
He has a 1 kg trolley, attached by a pulley to a 0.5 kg hanging hook.
He also has eight 100 g masses. When the hook is released, the trolley rolls along a table, and passes through a light gate which calculates its acceleration. GRADE 7-9

Describe an experiment that Kaleb can perform using this equipment to investigate the relationship between force and acceleration.

[6 marks]

Revision Summary Test for Topic 5b

Nearly at the finish line — don't let all those lovely facts race away from you, now. Go on and:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

Displacement, Velocity and Acceleration (p.74-75) ☐

1) What does 'displacement' mean? How is it different from distance travelled? ☐
2) Write a definition of velocity. ☐
3) Explain why an object moving in a circle at constant speed has a changing velocity. ☐
4) What is the formula for working out speed in terms of distance travelled?
   If the object's speed isn't constant, what does this formula give you? ☐
5) What are typical speeds in m/s for:
   a) a person walking, b) a person running, c) a person cycling, d) a car, e) a train, f) a plane? ☐
6) Why might a person's walking speed be different from the typical walking speed? ☐
7) Define acceleration in terms of velocity and time. Write down the formula. ☐
8) What does the term 'uniform acceleration' mean? ☐

Motion Graphs and Terminal Velocity (p.76-77) ☐

9) What does the gradient represent for: a) a distance-time graph, b) a velocity-time graph? ☐
10) A sprinter sets off from stationary, accelerates with constant acceleration up to his maximum speed, then runs at a constant speed until he reaches the finish line.
    For his run, describe the shape of: a) the distance-time graph, b) the velocity-time graph. ☐
11) Describe how to find the distance travelled from a velocity-time graph. ☐
12) What is friction? Describe the two cases where you get friction. ☐
13) What is terminal velocity? ☐
14) Explain why an object reaches terminal velocity as it falls through air. ☐
15) Explain why the terminal velocity of a skydiver is lower with their parachute open than it is when the parachute is closed. ☐

Newton's Laws and Investigating Motion (p.79-81) ☐

16) What is Newton's First Law of motion? ☐
17) Use Newton's First Law to explain whether or not there's a resultant force acting when a car is:
    a) travelling in a straight line at a steady speed, b) stopped at traffic lights,
    c) travelling in a circle at a steady speed. ☐
18) Write down Newton's Second Law of motion as a formula. ☐
19) Two balls of different masses are fired from a slingshot. If the same force is applied each time, explain why the two balls will accelerate at different rates. ☐
20) What is inertia? Explain why it's harder to change the velocity of something with a larger mass. ☐
21) What is Newton's Third Law? ☐
22) True or false? "When a book is sitting on a table, the weight of the book and the normal reaction force of the table on the book are a Newton's Third Law pair." Explain your answer. ☐
23) Describe an experimental set-up to test Newton's Second Law using a light gate and data logger. ☐
24) How would you use the set-up you just described to investigate:
    a) the effect of mass on the acceleration of the system,
    b) the effect of accelerating force on the acceleration of the system? ☐

Quick Quiz

Stopping Distances

*Knowing what affects **stopping distances** is especially useful for everyday life, as well as the exam.*

Many Factors Affect Your Total Stopping Distance

Video

In an emergency, a driver may need to perform an emergency stop. This is where maximum force is applied by the brakes to stop the car in the shortest possible distance. The longer it takes to perform an emergency stop, the higher the risk of an accident.

The distance it takes to stop a car in an emergency (its stopping distance) is found by:

Stopping Distance = Thinking Distance + Braking Distance

The factors in the table increase stopping distance, so they affect safety — especially in an emergency.

	Definition	Is increased by...
Thinking Distance	How far the car travels during the driver's reaction time.	• Higher Speeds — the faster a car travels, the further it goes in the time taken to react. • Longer Reaction Times — the longer it takes for the brakes to be applied, the further the car travels.
Braking Distance	The distance taken to stop under the braking force (once the brakes are applied).	• Higher Speeds — faster vehicles take a longer distance to stop under a given braking force. • Poor Weather and Road Conditions — wet/icy roads or oil spills reduce grip between tyres and the road (reducing friction), increasing risk of skidding. • Worn Tyres — if tyres don't have any tread left, they can't get rid of water on the road, increasing risk of skidding in wet conditions. • Worn or Faulty Brakes — can't apply as much braking force.

Reaction time is the time between the driver seeing a hazard and applying the brakes. Lots of factors can affect a person's reaction time — see p.86.

Braking Relies on Friction Between the Brakes and Wheels

1) When the brake pedal is pushed, the brake pads are pressed onto the wheels. This contact causes friction, so work is done between the brakes and the wheels. The work done transfers energy from the kinetic energy stores of the wheels to the thermal energy stores of the brakes — so the brakes increase in temperature.
2) The faster a car travels, the more energy it has in its kinetic energy stores, so the more work needs to be done to stop it. This means a greater braking force (and larger deceleration) is needed to make it stop in a certain distance.
3) You can estimate the forces involved in accelerations and decelerations of vehicles using typical values:

Very large decelerations can be dangerous because they may cause brakes to overheat (so they don't work as well) or could cause the vehicle to skid. These both increase the risk of an accident.

EXAMPLE

A car travelling at a typical speed makes an emergency stop to avoid hitting a hazard 50 m ahead. Estimate the braking force needed to produce this deceleration.

MATHS SKILLS

1) Assuming deceleration is uniform, use $v^2 - u^2 = 2as$. Rearrange for a by dividing both sides by $2s$.

$(v^2 - u^2) \div 2s = 2as \div 2s$

$\rightarrow a = (v^2 - u^2) \div 2s$

2) Using a typical speed of a car in m/s, calculate the deceleration of the car.

u = ~25 m/s, v = 0 m/s and s = 50 m

$a = (v^2 - u^2) \div 2s = (0^2 - 25^2) \div (2 \times 50) = -6.25$ m/s^2

3) Calculate the force using $F = ma$. Use a typical value for the mass of the car.

m = ~1500 kg

F = 1500 × 6.25 = 9375 N, so F is ~9000 N

This is vital real-life knowledge for drivers...

Make sure you understand the difference between thinking distance and braking distance, and the factors that affect them.

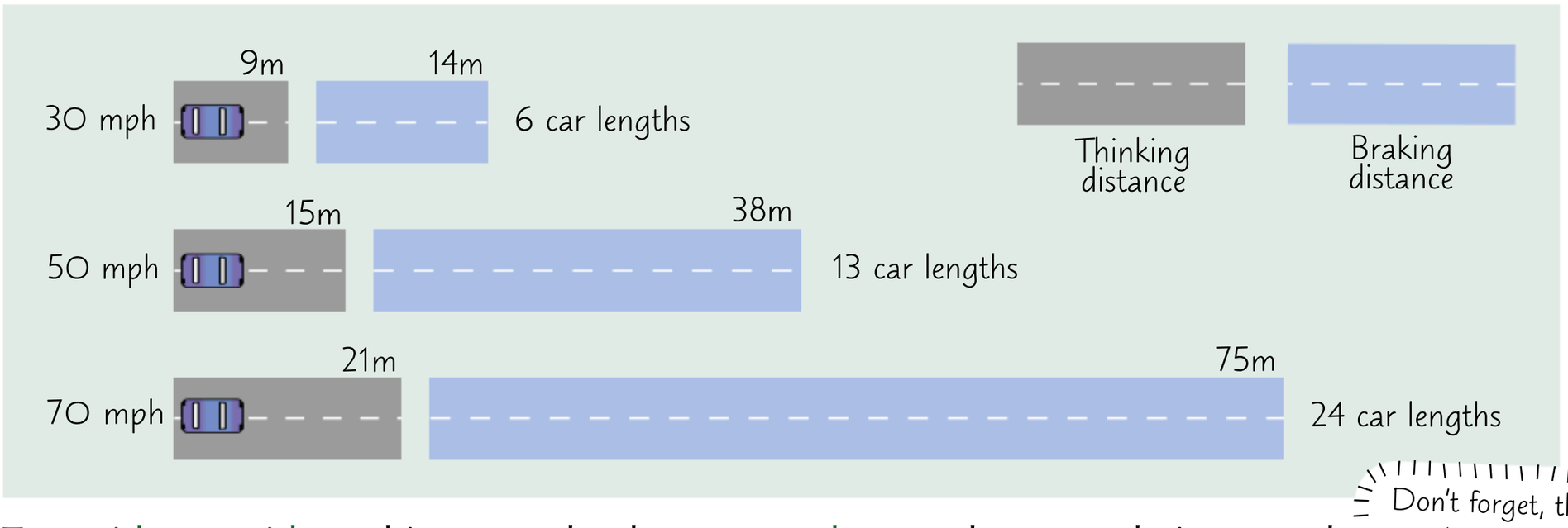
More on Stopping Distances

Quick Quiz

*So now you know what affects **stopping distances**, let's have a look at the **facts** and **figures**.*

Leave Enough **Space** to **Stop**

- The figures below for typical stopping distances are from the Highway Code.



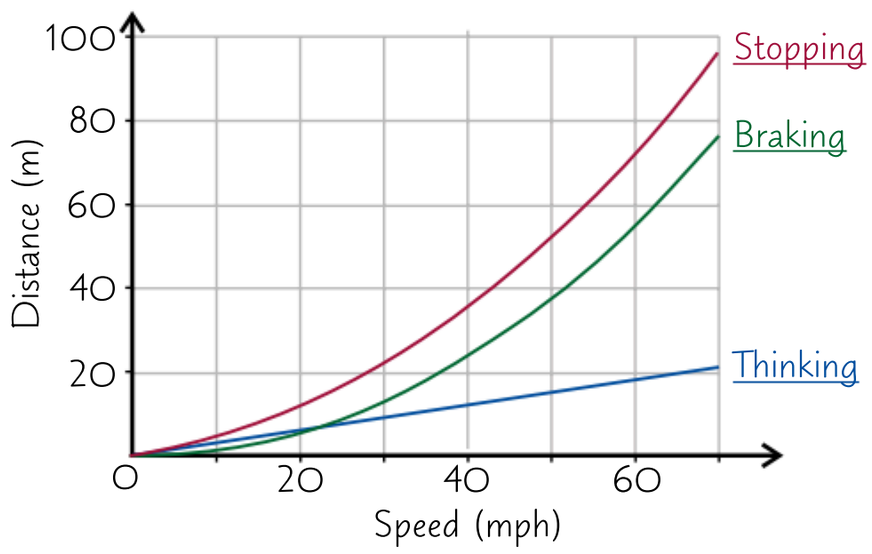
- To avoid an accident, drivers need to leave enough space between their car and the one in front so that if they had to stop suddenly, they would have time and space to do so safely. The longer the stopping distance, the more space needed.
- Speed limits are really important because speed affects the stopping distance so much.

Don't forget, things like bad weather and road conditions will make stopping distances even longer (see p.84).

Speed Affects Braking Distance **More Than** Thinking Distance

Braking distance increases faster the more you speed up. The work done to stop the car is equal to the energy in the car's kinetic energy store ($\frac{1}{2}mv^2$).

As the speed doubles, the kinetic energy increases 4-fold (2^2), and so does the work done to stop the car. Since $W = Fs$ (p.63) and the braking force is constant, braking distance increases 4-fold.



Thinking distance increases at the same rate as speed, so the graph is linear (a straight line). This is because the reaction time stays pretty constant — but the higher the speed, the more distance you cover in that same time.

Stopping distance is a combination of these two distances (p.84) so the graph of speed against stopping distance is a combination of the braking distance and thinking distance graphs.

You need to be able to interpret stopping distance graphs for a range of vehicles — they're all a similar shape.

EXAMPLE

Below is a graph of stopping distance against speed for two vehicles, A and B. Compare the stopping distance for both vehicles at a speed of 40 mph.

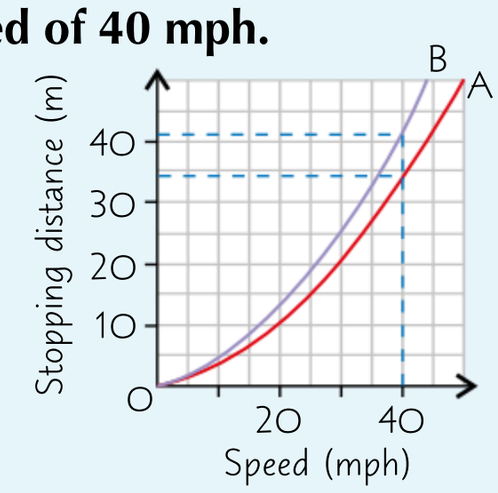
MATHS SKILLS

1) Read off the graph to find the stopping distance for each vehicle at 40 mph.

Vehicle A stopping distance = 34 m
Vehicle B stopping distance = 41 m

2) Find the difference between these two values.

41 – 34 = 7 m
So the stopping distance for vehicle B is 7 m longer than for vehicle A.



The higher your speed, the more space you need to stop...

Practise finding information from graphs of stopping distance against speed, so you're comfortable with it.

Q1 A driver performs an emergency stop. His thinking distance and braking distance are both 6 m. Estimate his total stopping distance if he had been travelling three times as quickly. [4 marks]

Q1 Video Solution

Reaction Times

Quick Quiz

Reaction times *are an* ***important factor*** *in* ***thinking distances****. They're also super easy to* ***test*** *for yourself.*

Reaction Time is **How Long** a Person Takes to **React** to an **Event**

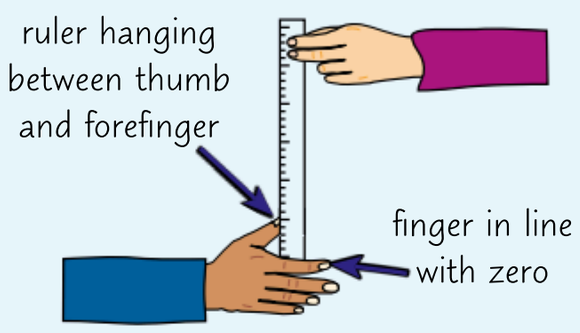
Everyone's reaction time is different, but a typical reaction time is between 0.2 and 0.9 s.
This can be affected by tiredness, drugs or alcohol. Distractions can also affect your ability to react.

Reaction Times are **Hard** to **Measure Accurately**

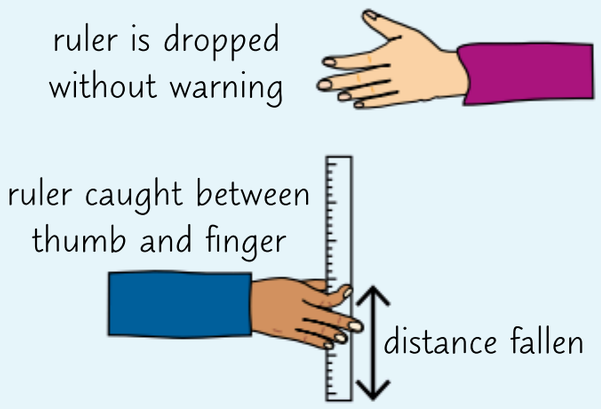
Reaction times are so short, you've got no chance measuring one with a stopwatch. One way to do it is to use a computer-based test (e.g. clicking a mouse when the screen changes colour). Another is the ruler drop test:

Method — Ruler Drop Test

1) Sit with your arm resting on the edge of a table (this should stop you moving your arm up or down during the test). Get someone else to hold a ruler so it hangs between your thumb and forefinger, lined up with zero.



You may need a third person to be at eye level with the ruler to check it's lined up.



2) Without giving any warning, the person holding the ruler should drop it. Close your thumb and finger to try to catch the ruler as quickly as possible.
3) The measurement on the ruler at the point where it is caught is how far the ruler dropped in the time it takes you to react. The longer the distance, the longer the reaction time.

You can calculate how long the ruler falls for (the reaction time) since acceleration due to gravity is constant:

EXAMPLE

A student carried out the Ruler Drop Test and caught the ruler at 20 cm. Calculate the student's reaction time.

MATHS SKILLS

1) Rearrange the formula $v^2 - u^2 = 2as$ (p.75) for v.
2) Substitute in your values to find v.
3) Rearrange $a = \Delta v \div t$ for t, then plug in your values.

$v^2 - u^2 = 2as \rightarrow v^2 = 2as + u^2 \rightarrow v = \sqrt{2as + u^2}$

$u = 0$ m/s, $a = 9.8$ m/s^2 (due to gravity), $s = 20$ cm $= 0.2$ m

$v = \sqrt{2 \times 9.8 \times 0.2 + 0} = 1.97...$ m/s

$t = \Delta v \div a = 1.97... \div 9.8 = 0.202... = 0.2$ s (to 1 s.f.)

It's hard to do this experiment accurately, but you can improve the quality of your results by:

- Repeating the experiment lots of times — so you can calculate a mean reaction time.
- Adding a small blob of clay to the bottom of the ruler — so it falls down in a straight line.
- Making it a fair test — e.g. use the same ruler for each repeat.

You could also investigate some of the factors affecting reaction time, e.g. you could introduce distractions by having some music playing. You can then compare results for reaction times with and without distractions.

The further the ruler falls, the longer the reaction time...

Q1 Video Solution

Q1 Mark's reaction time is tested using the ruler drop test. He is tested in the early afternoon and at night. In the afternoon, he catches the ruler after it has fallen a distance of 16.2 cm. At night, he catches the ruler after it has fallen 18.5 cm.

a) Calculate Mark's reaction time in the afternoon. Give your answer to 2 significant figures. [5 marks]

b) Explain why Mark's thinking distance might be longer when driving in the evening. [2 marks]

Warm-Up & Exam Questions

Time to apply the brakes for a second and put your brain through an MOT. Try out these questions. If you can handle these, your exam should be clear of hazards.

Warm-Up Questions

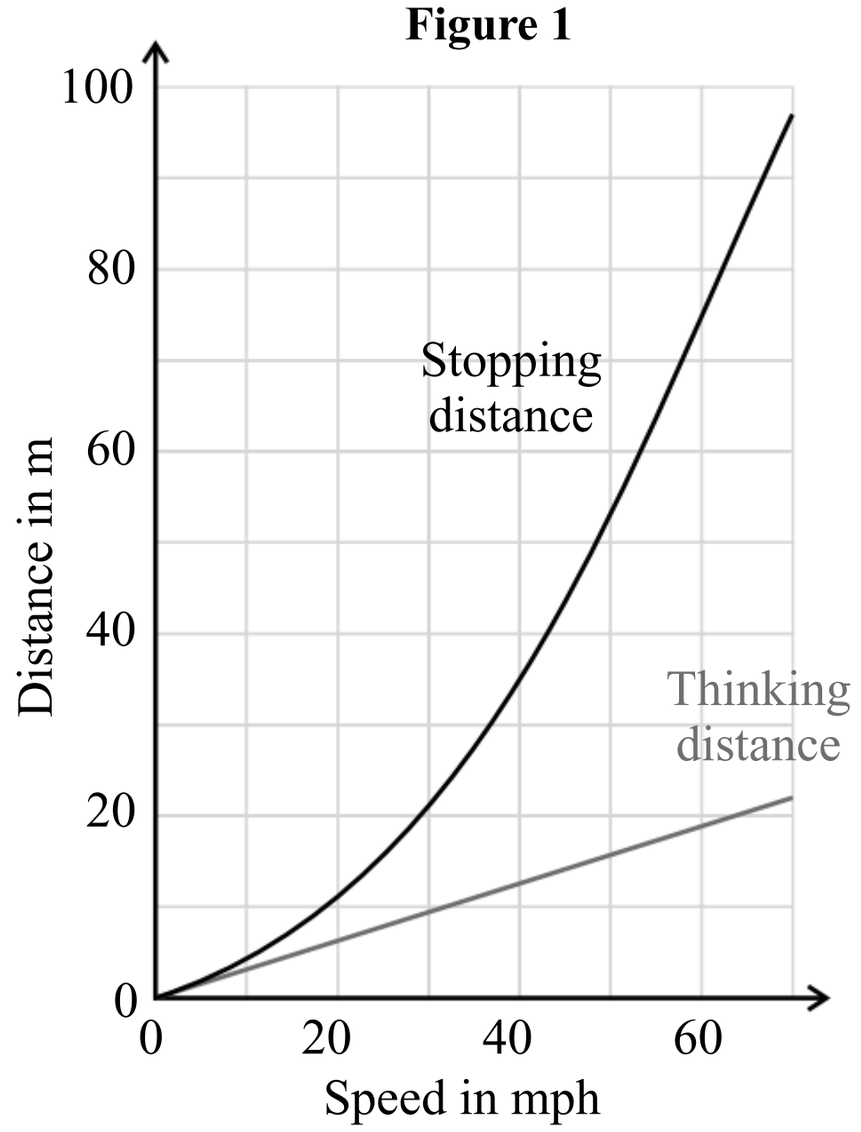
1) What must be added to the thinking distance to find the total stopping distance of a car?
2) What is meant by 'thinking distance'?
3) Give an example of how poor weather can affect your ability to stop a car before hitting a hazard.
4) What is the main energy transfer that occurs when a car brakes?
5) Describe an experiment you could carry out, using a ruler, to measure the reaction time of an individual.

Exam Question

1 **Figure 1** shows how thinking distance and stopping distance vary with speed for a car travelling on a clear day on a dry road.

GRADE 4-6 GRADE

Figure 1



1.1 Use **Figure 1** to determine the braking distance for a car travelling at 40 mph.

[4 marks]

1.2 Which is greater at 50 miles per hour, thinking distance or braking distance?

[1 mark]

1.3 Using **Figure 1**, determine whether stopping distance is directly proportional to speed. Explain your answer.

[2 marks]

1.4 Describe how the shape of each of the graphs on **Figure 1** would change if the data was taken for a driver travelling on an icy road.

[2 marks]

Momentum

Quick Quiz

*Momentum is mainly about how much '**oomph**' an object has. **Faster**, **heavier** objects have more oomph...*

All **Moving Objects** Have Momentum

1) The greater the mass of an object, or the greater its velocity, the more momentum the object has.
2) Momentum is a vector quantity — it has size and direction.
3) You can work out the momentum of an object using:

$p = mv$ **momentum (kg m/s) = mass (kg) × velocity (m/s)**

EXAMPLES

MATHS SKILLS

1) A 50 kg cheetah is running at 60 m/s. Calculate its momentum.
Just plug the values straight into the formula: $p = mv = 50 \times 60 = 3000$ kg m/s

2) A boy has a mass of 30 kg and a momentum of 75 kg m/s. Calculate his velocity.
Rearrange $p = mv$ for v, by dividing both sides by m. $v = p \div m = 75 \div 30 = 2.5$ m/s

Momentum **Before** = Momentum **After**

A closed system is just a fancy way of saying that no external forces act.

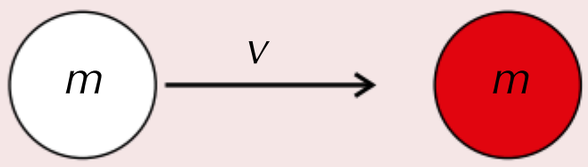
In a closed system, the total momentum before an event (e.g. a collision) is the same as after the event. This is called conservation of momentum. Here are some examples:

Example 1 — Snooker Balls

A white snooker ball of mass m collides with a stationary red ball with the same size and mass.

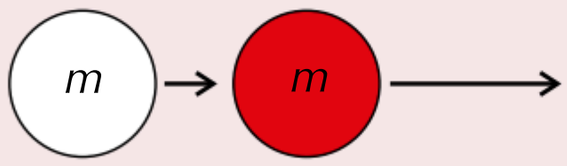
Before:

1) The white ball has momentum $p = mv$.
2) The red ball is stationary, so has zero momentum.
3) Total momentum before collision $0 + mv = mv$.



After:

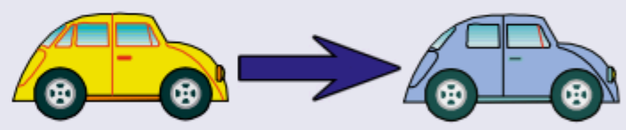
1) The collision causes the red ball to move, so it now has momentum.
2) The white ball continues moving, but at a much smaller velocity than before (and so a much smaller momentum).
3) Combined momentum of balls after = total momentum before = mv.



Example 2 — Car Collision

A moving car hits into the back of a parked car. The crash causes the two cars to lock together, and they continue moving in the direction that the original moving car was travelling, but at a lower velocity.

Before:



Total momentum = mass of moving car × its velocity

After:

The mass of the single moving object has increased, but its momentum is equal to the total momentum before the collision. So an increase in mass causes a decrease in velocity.

If the momentum before an event is zero, then the momentum after will also be zero.
E.g. in an explosion, the momentum before is zero. After the explosion, the pieces fly off in different directions so that the total momentum cancels out to zero (remember, momentum is a vector).

Don't forget — momentum is always conserved...

Conservation of momentum is incredibly handy — there's more on using it over on the next page.

Q1 Video Solution

Q1 Calculate the momentum of a 60 kg woman running at 3 m/s. [2 marks]

Q2 Describe how momentum is conserved by a gun recoiling (moving backwards) as it shoots a bullet. [4 marks]

Changes in Momentum

Quick Quiz

*A **force** causes the **momentum** of an object to **change**. A **bigger force** makes it change **faster**.*

Use **Conservation of Momentum** to **Calculate Velocities** or **Masses**

EXAMPLE

Misha fires a paintball gun. A 3.0 g paintball is fired at a velocity of 90 m/s. Calculate the velocity at which the paintball gun recoils if it has a mass of 1.5 kg. Momentum is conserved.

MATHS SKILLS

The word recoil means to move backwards.

1) Calculate the momentum of the pellet using $p = mv$. Remember to convert the mass to kg.

mass in kg = 3 ÷ 1000 = 0.003 kg
p = 0.003 × 90 = 0.27 kg m/s

2) The momentum before the gun is fired is zero. This is equal to the total momentum after the collision.

p before = p after
0 = p of paintball after + p of gun after

3) The momentum of the gun is $1.5 \times v$.

0 = 0.27 + (1.5 × v)

4) Rearrange the equation for v. First take 0.27 from both sides, then divide both sides by 1.5.

$-0.27 = 1.5 \times v \rightarrow -(0.27 \div 1.5) = v$

5) Calculate the velocity of the gun. The minus sign shows the gun is travelling in the opposite direction to the bullet.

$v = -(0.27 \div 1.5)$
$= -0.18$ m/s

Forces Cause a **Change in Momentum**

1) You know that when a non-zero resultant force acts on a moving object (or an object that can move), it causes its velocity to change (p.79). This means that there's a change in momentum.

2) You also know that $F = ma$ and that a = change in velocity ÷ change in time.

3) So $F = m \times \frac{v - u}{t}$, which can also be written as:

$$F = \frac{m\Delta v}{\Delta t}$$

Force (N) — Change in momentum (kg m/s) — Change in time (s)

This formula tells you that the force causing the change is equal to the rate of change of momentum.

4) The larger the force, the faster the change of momentum. This means that if someone's momentum changes very quickly (like in a car crash), the forces on their body will be very large, and more likely to cause injury.

5) This is why cars are designed to slow people down over a longer time when they have a crash — increasing the time it takes for a certain change in momentum reduces the force exerted. Smaller forces mean any injuries are likely to be less severe. Cars have a range of safety features:

Car Safety Feature	How it reduces the force exerted in a collision
Crumple Zones	Crumples on impact, increasing time taken for car to stop.
Seat Belts	Stretches slightly, increasing time taken for wearer to stop.
Air Bags	Inflates during a crash. Hitting a bag of compressible air slows you down more gradually than if you were to hit the hard dashboard.

6) Other pieces of safety equipment use the same principle, e.g. bike helmets, crash mats and playground surfaces contain soft, compressible (squishable) materials to increase the time taken to stop.

Momentum questions could ask you to analyse a scenario...

Q1 A 10 kg object is travelling at 6 m/s. It hits a stationary 20 kg object and the two objects join together and keep moving in the same direction. Calculate the velocity of the combined object, assuming that momentum is conserved. [3 marks]

Q1 Video Solution

Warm-Up & Exam Questions

Here it is, the final hurdle before the end of the topic. Keep up your momentum, and throw yourself into these questions. You'll be over the finish line before you know it.

Warm-Up Questions

1) Calculate the momentum of a 2.5 kg rabbit running through a garden at 10 m/s.
2) What is the total momentum before and after an explosion?
3) How does increasing the time over which an object changes momentum change the force on it?

Exam Questions

For these questions, you can use the formula sheet on page 216.

1 A 60 kg gymnast lands on a crash mat. When they hit the crash mat, they are moving at 5.0 m/s and come to a stop in a period of 1.2 seconds (after which their momentum is zero). GRADE 6-7

1.1 State the equation linking momentum, mass and velocity.

[1 mark]

1.2 Calculate the momentum of the gymnast immediately before they hit the crash mat.

[2 marks]

1.3 Calculate the size of the average force acting on the gymnast as they land on the crash mat.

[2 marks]

2 In a demolition derby, cars drive around an arena and crash into each other. GRADE 6-7

2.1 One car has a mass of 650 kg and a velocity of 15.0 m/s.
Calculate the momentum of the car.

[2 marks]

2.2 The car collides head-first with another car with a mass of 750 kg. The two cars stick together.
Calculate the combined velocity of the two cars immediately after the collision if the other car had a velocity of 10.0 m/s before the collision. Give your answer to 3 significant figures.

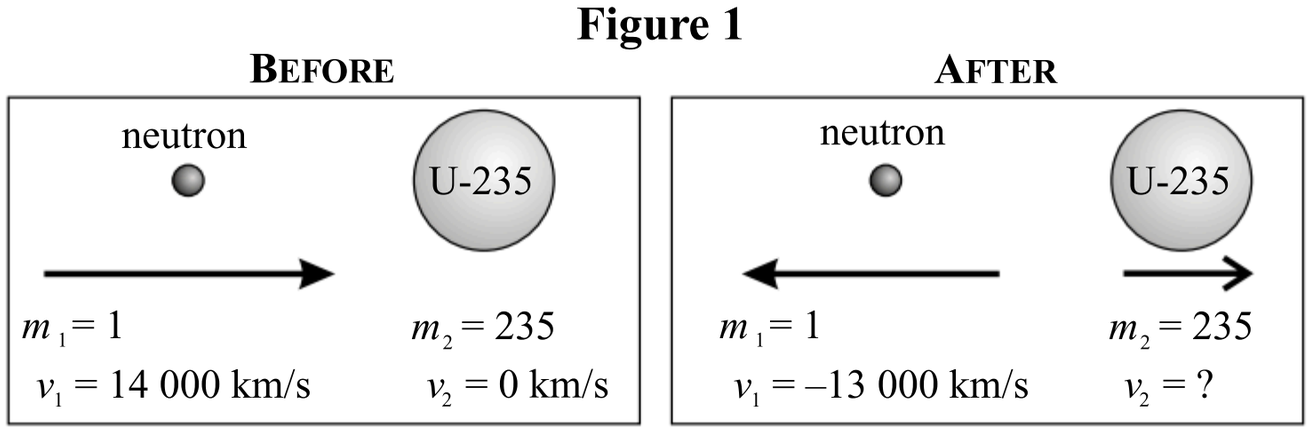
[5 marks]

2.3 The cars have crumple zones at the front of the car that crumple on impact.
Explain how a crumple zone reduces the forces acting on a driver during a collision.

[2 marks]

3 A fast-moving neutron collides with a uranium-235 atom and bounces off. **Figure 1** shows the particles before and after the collision. GRADE 7-9

Figure 1



Calculate the velocity of the U-235 atom after the collision. Give your answer to 3 significant figures

[5 marks]

Revision Summary Test for Topic 5c

Stop! Before you speed off into the next section, you should:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

Stopping Distances (p.84-85) ☑

1) What is an emergency stop? Why is it important to minimise the stopping distance of a car?
2) Define thinking distance and braking distance.
3) How can you calculate stopping distance from thinking distance and braking distance?
4) Explain how thinking distance is affected by: a) higher speeds, b) a longer reaction time.
5) What are four factors that could increase the braking distance of a car?
6) Explain why, when a driver uses the brakes of a car, the brakes increase in temperature.
7) What are two dangers of large decelerations whilst driving?
8) Why is it important that drivers leave enough space between them and the car in front? Explain why this space changes depending on the speed of the car.
9) Use formulas for kinetic energy and work done to explain how the braking distance of a car changes as its speed halves.
10) What does a graph of thinking distance against speed look like? Explain your answer.
11) For a car travelling at 60 mph, what has the greatest effect on its stopping distance — thinking distance or braking distance? Explain your answer.

Reaction Times (p.86) ☑

12) What are four things that can affect a person's reaction time or ability to react?
13) a) Briefly describe the method for the Ruler Drop Test to investigate reaction time.
    b) How could you calculate reaction time from the results of this experiment?
    c) List two things you could do to improve the accuracy of results from this experiment.
    d) How could you make sure this experiment is a fair test?

Momentum (p.88-89) ☑

14) What is the formula for calculating the momentum of an object? Include the units of all the variables.
15) Is momentum a scalar or a vector quantity?
16) What is the momentum of a stationary object?
17) What is meant by the conservation of momentum?
18) Trolley A of mass m travelling at speed v, collides with an identical trolley, trolley B, that is stationary. After the collision, the trolleys stick together and continue moving in the same direction as trolley A before the collision. Explain the changes in momentum of the trolleys due to the collision.
19) A 10 kg watermelon rolls along the floor at a velocity of 3 m/s. It collides with a stationary honeydew melon. After the collision, the velocity of the watermelon is 2 m/s, and the honeydew melon starts rolling in the same direction, at a velocity of 5 m/s. Calculate the mass of the honeydew melon.
20) What is the equation relating the resultant force acting on a moving object to the resulting change in momentum? Describe what this equation tells you.
21) List three car safety features that are designed to reduce the force exerted during a crash. How does each safety feature work?
22) What are three other pieces of safety equipment that are designed to reduce the forces involved during an impact?

Quick Quiz

Transverse and Longitudinal Waves

*Waves **transfer energy** from one place to another **without** transferring any **matter** (stuff).*

Waves Transfer **Energy** in the **Direction** they are **Travelling**

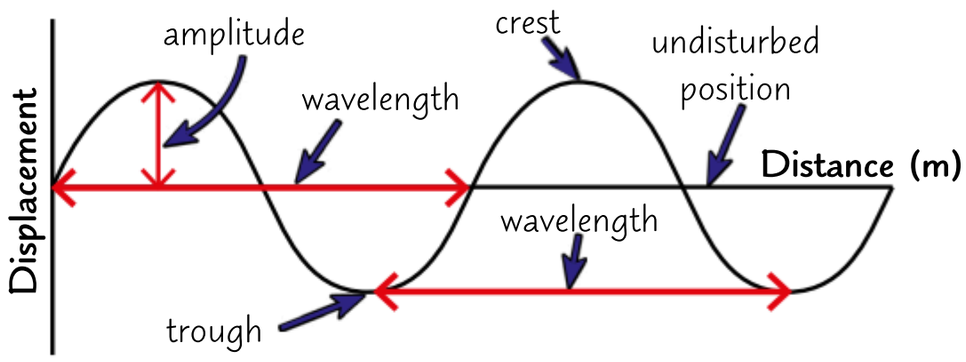
When waves travel through a medium, the particles of the medium oscillate and transfer energy between each other (see p.3). BUT overall, the particles stay in the same place — only energy is transferred.

If you drop a twig into a calm pool of water, ripples form on the water's surface. The ripples don't carry the water (or the twig) away with them though.

Similarly, if you strum a guitar string and create sound waves, the sound waves don't carry the air away from the guitar (i.e. a vacuum isn't created).

Waves have important properties you need to know about:

AMPLITUDE	Maximum displacement of a point on the wave from its undisturbed position.
WAVELENGTH	Distance between the same point on two adjacent waves (i.e. between two troughs).
FREQUENCY	Number of complete waves passing a certain point per second, measured in hertz (Hz). 1 Hz is 1 wave per second.
PERIOD	Time taken for one full cycle of a wave.



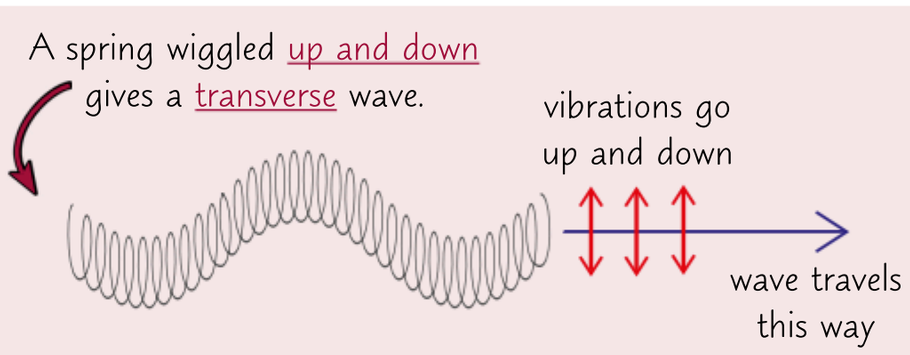
Period (s) — $T = \frac{1}{f}$ — Frequency (Hz)

All Waves are Either **Transverse** or **Longitudinal**

Transverse waves

The oscillations (vibrations) are perpendicular (at 90°) to the direction of energy transfer. Examples are:

1) All electromagnetic waves, e.g. light (p.99).
2) Ripples on the surface of water (see p.93).
3) A wave on a string.

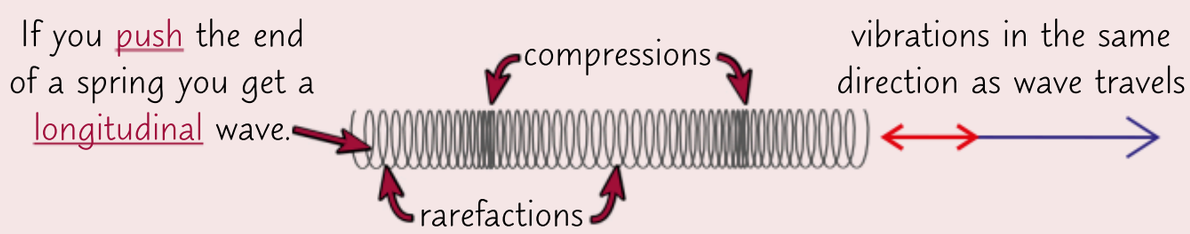


Video

Longitudinal waves

The oscillations are parallel to the direction of energy transfer. Examples are:

1) Sound waves in air, ultrasound.
2) Shock waves, e.g. some seismic waves.



Wave Speed = Frequency × Wavelength

The wave speed is the speed at which energy is being transferred (or the speed the wave is moving). The wave equation applies to all waves:

Wave speed (m/s) — $v = f\lambda$ — Wavelength (m); Frequency (Hz)

EXAMPLE

Find the wavelength of a radio wave with frequency 12.0×10^6 Hz travelling at a speed of 3.0×10^8 m/s.

MATHS SKILLS

Rearrange $v = f\lambda$ and plug the numbers in.

$\lambda = v \div f = (3.0 \times 10^8) \div (12.0 \times 10^6) = 25$ m

The only thing a wave transfers is energy...

Make sure this is all clear in your head, otherwise the rest of the topic will just be a wavey blur of nonsense.

Q1 A wave has a speed of 0.15 m/s and a wavelength of 7.5 cm. Calculate its frequency. [4 marks]

Q1 Video Solution

Experiments With Waves

Quick Quiz

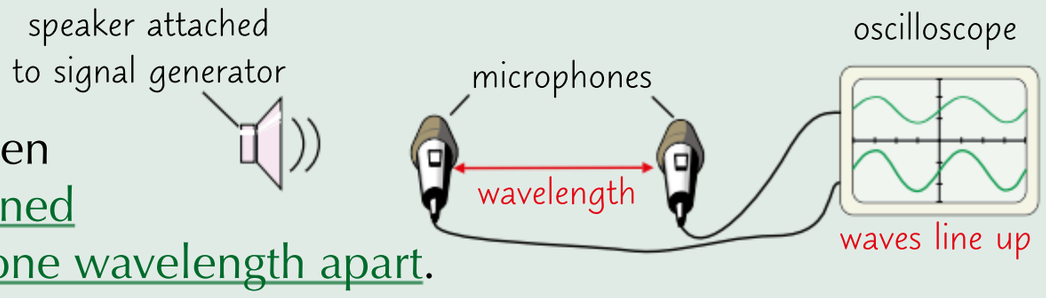
*Time to **experiment**. Make sure to choose **suitable equipment** to measure the **speed of different waves**.*

Use an **Oscilloscope** to Measure the **Speed** of **Sound**

Video

By attaching a signal generator to a speaker you can generate sounds with a specific frequency. You can use two microphones and an oscilloscope to find the wavelength of the sound waves.

1) Set up the oscilloscope so the sounds detected at each microphone are shown as separate waves.
2) Start with both microphones next to the speaker, then slowly move one away until the two waves are aligned on the display. The microphones are now exactly one wavelength apart.
3) Measure the distance between the microphones to find one wavelength (λ).
4) You can then use the formula $v = f\lambda$ (p.92) to find the speed (v) of the sound waves passing through the air — the frequency (f) is whatever you set the signal generator to (around 1 kHz is sensible).



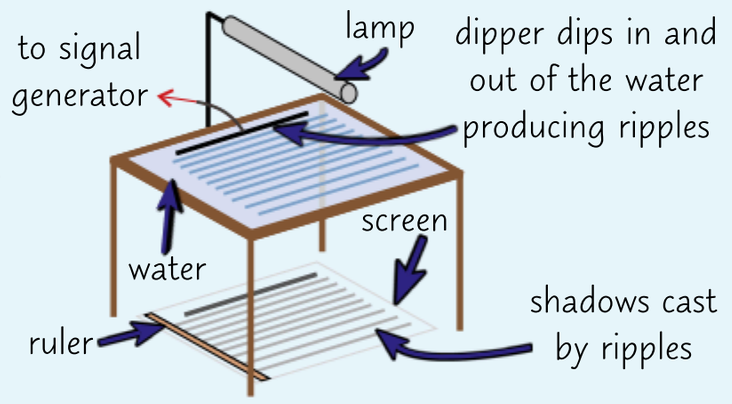
The speed of sound in air is around 330 m/s, so check your results roughly agree with this.

You can Measure the **Speed** of **Water Ripples** Using a **Lamp**

PRACTICAL

Method

1) To investigate wave speed, you can use a signal generator attached to the dipper of a ripple tank to create water waves at a set frequency.
2) Dim the lights and use a lamp to see wave crests on a screen below the tank.
3) The distance between each shadow line is equal to one wavelength. Measure the distance between shadow lines that are 10 wavelengths apart, then divide this distance by 10 to find the average wavelength.



- Use $v = f\lambda$ to calculate the wave speed of the waves.
- This set-up is suitable for investigating waves, because it allows you to measure the wavelength without disturbing the waves.

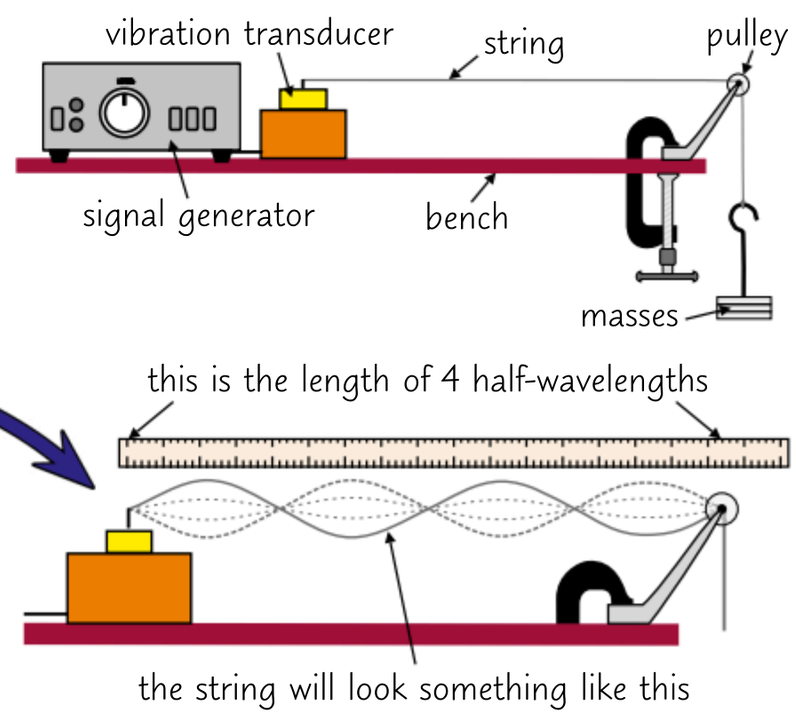
When setting up this experiment, you should position the lamp so the shadows cast are as close to the size of the water waves as possible.

You can Measure Waves on a **Vibrating String**

PRACTICAL

Method

1) Set up the equipment shown on the right, then turn on the signal generator and vibration transducer. The string will start to vibrate.
2) Adjust the frequency of the signal generator until there's a clear wave on the string.
3) Measure over as many half-wavelengths as you can, then divide by the number of half-wavelengths. This gives the mean half-wavelength (p.140). Double this mean to get a full wavelength.



You can find the speed of the wave using $v = f\lambda$. The frequency of the wave is whatever the signal generator is set to.

Learn the methods for all these practicals...

These experiments might seem complicated, but they all follow the same basic process — set the frequency on the signal generator, measure the length of the resulting wave, then calculate the wave speed with $v = f\lambda$. Not too bad.

Reflection

Quick Quiz

*You should be pretty familiar with the idea of **reflection**. Of course, there's a bit more to it than you'd think.*

All Waves Can be **Absorbed**, **Transmitted** or **Reflected**

When waves arrive at a boundary between two different materials, three things can happen:

Absorption	Transmission	Reflection
Energy is transferred to the material's energy stores (like in a microwave oven, page 101).	The waves carry on travelling through the new material. This often leads to refraction.	The waves bounce back off the boundary — more on this below.

What actually happens depends on the wavelength of the wave and the properties of the materials involved.

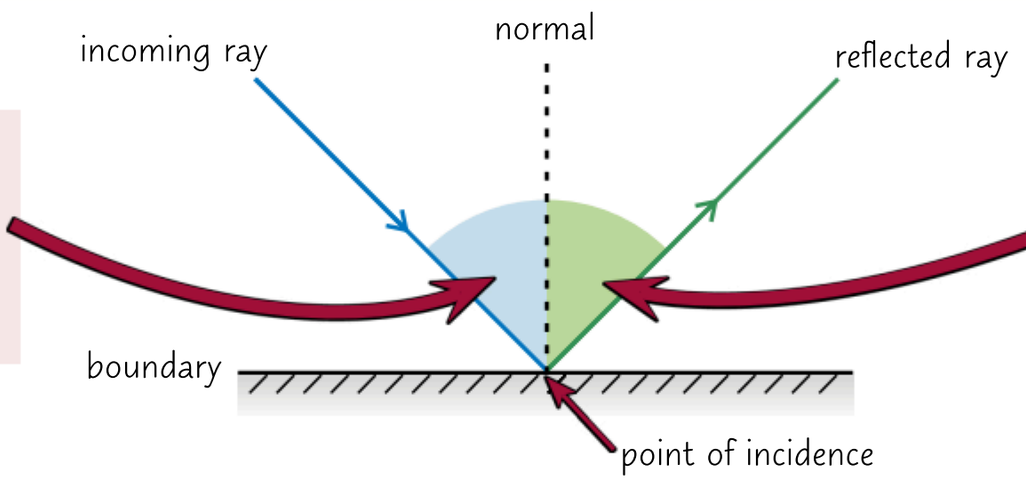
You can Draw a Simple **Ray Diagram** for **Reflection**

For all reflected waves:

Angle of incidence = Angle of reflection

A ray is a line showing the path a wave travels along. It's perpendicular to a wave's wave fronts (next page). Rays are always drawn as straight lines.

The normal is an imaginary line that's perpendicular (at right angles) to the surface at the point of incidence (the point where the wave hits the boundary).



The angle of incidence is the angle between the incoming ray (wave) and the normal.

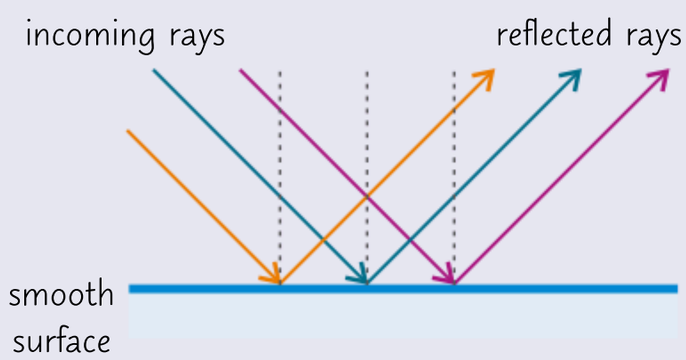
The angle of reflection is the angle between the reflected ray (wave) and the normal.

Reflection can be **Specular** or **Diffuse**

Waves are reflected at different boundaries in different ways (there's an investigation on this — see p.96).

Specular reflection

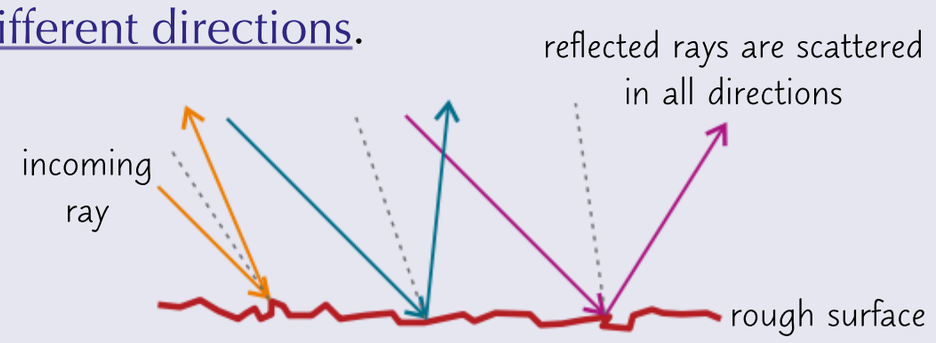
This happens when a wave is reflected in a single direction by a smooth surface.



For example, when light rays reflect off a mirror you get a nice clear reflection.

Diffuse reflection

This happens when a wave is reflected by a rough surface (e.g. a piece of paper) and the reflected rays are scattered in lots of different directions.



- This happens because the normal is different for each incoming ray, which means that the angle of incidence is different for each ray. The angle of incidence still equals the angle of reflection.
- When light is reflected by a rough surface, the surface appears matte (not shiny) and you don't get a clear reflection of objects.

The angle of incidence always equals the angle of reflection...

... for both types of reflection, specular and diffuse. Speaking of which, here's a way of remembering the difference — waves are reflected in a **s**ingle direction for **s**pecular reflection, and **d**ifferent directions for **d**iffuse reflection.

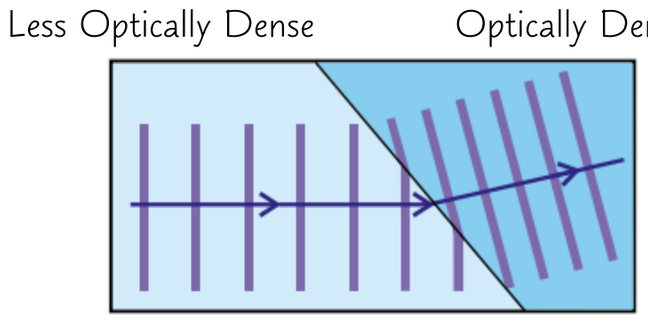
Refraction

Quick Quiz

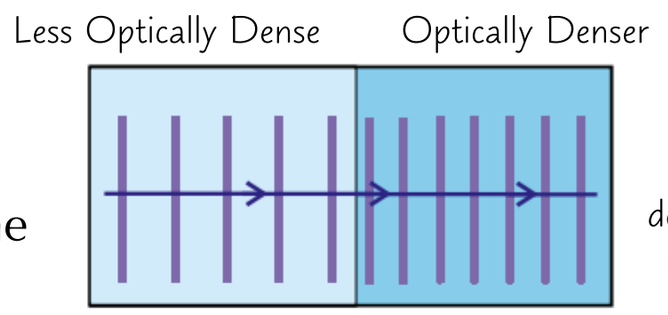
*Refraction is when light waves **bend** when they enter a **new media** (which is a posh word for material).*

Refraction — Waves Changing Direction at a Boundary

- When a wave crosses a boundary between materials at an angle it changes direction — it's refracted. The amount it's refracted by depends on how much the wave speeds up or slows down.
- If a wave crosses a boundary and slows down it will bend towards the normal. If it crosses into a material and speeds up it will bend away from the normal.
- The wavelength of a wave changes when it is refracted, but the frequency stays the same.
- If the wave is travelling along the normal it will change speed, but it's NOT refracted.
- The optical density of a material is a measure of how quickly light travels through it — the higher the optical density, the slower light travels through it.



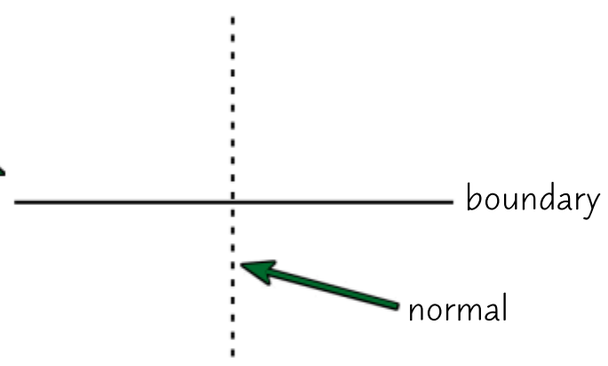
The wave hits a different medium at an angle, so the wave changes direction.

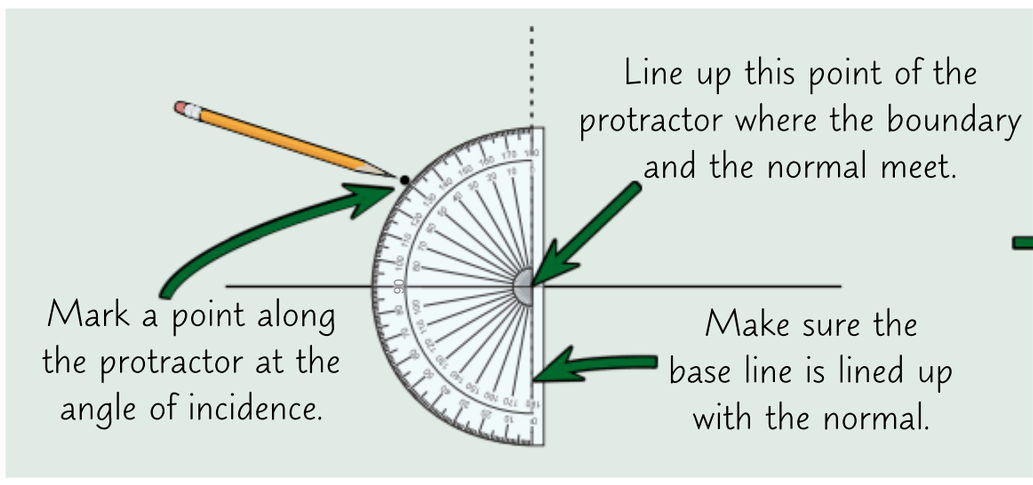


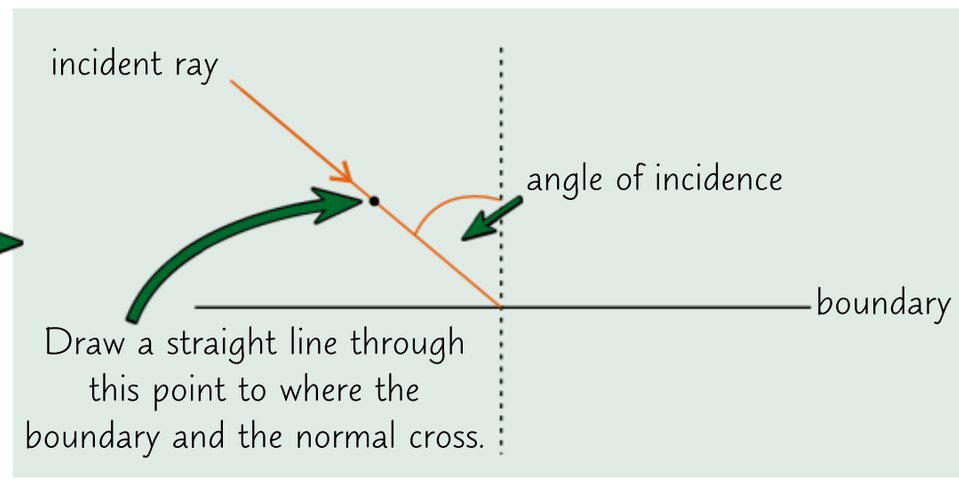
The wave fronts being closer together shows a decrease in wavelength (and so a decrease in velocity).

A Ray Diagram can Show the Path of a Refracted Ray

1) To construct a ray diagram for refraction, start by drawing the boundary between your two materials.
2) Then draw a normal at right angles to the boundary.
3) Next draw the incident ray. This is the incoming ray that meets the boundary at the normal. The angle between the incident ray and the normal is called the angle of incidence. Use a protractor and a ruler to draw it accurately:

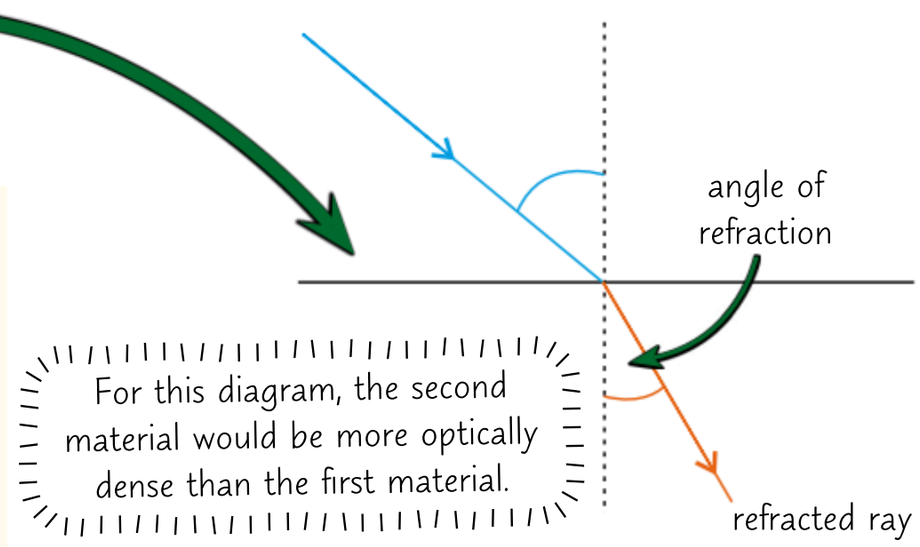






4) Now draw the refracted ray on the other side of the boundary using the same method. The angle of refraction is the angle between the refracted ray and the normal.

When light travels across a boundary, if the second material is optically denser than the first, the refracted ray bends towards the normal (like on the right). The angle of refraction is smaller than the angle of incidence. If the second material is less optically dense, the angle of refraction is larger than the angle of incidence.



Hitting a boundary at an angle leads to refraction...

Refraction appears a few times in GCSE Physics, so make sure you really understand it before moving on.

Q1 Draw a ray diagram for light entering a less optically dense medium, 40° to the normal. [3 marks]

Q1 Video Solution

Quick Quiz

Investigating Light

PRACTICAL

*Grab a **ruler**, **protractor** and a **nice sharp pencil** — it's time to get practical.*

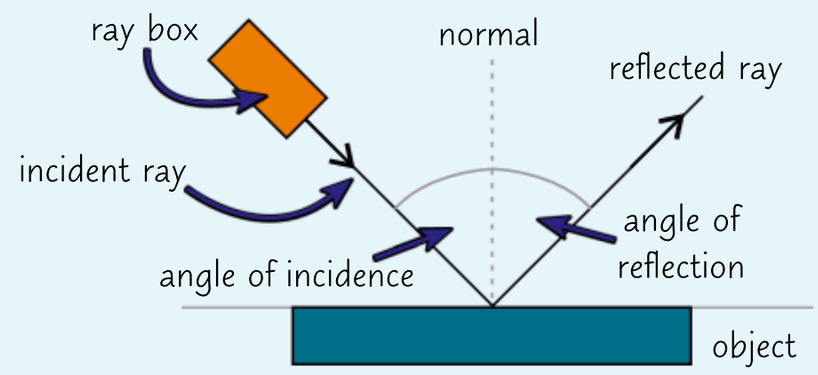
You Need to Do Both of These **Experiments** in a **Dim Room**

Both experiments use rays of light, so it's best to do them in a dim room so you can clearly see the light rays. They also both use either a ray box or a laser to produce thin rays of light. This is so you can easily see the middle of the ray when tracing it and measuring angles from it.

Different Materials **Reflect Light** Differently

Method

1) Take a piece of paper and draw a straight line across it. Place an object so one of its sides lines up with this line.
2) Use a ray box to shine a ray of light at the object's surface and trace the incoming and reflected light beams.
3) Draw the normal at the point where the ray hits the object. Use a protractor to measure the angle of incidence and the angle of reflection. Also, make a note of the width and brightness of the reflected light ray.
4) Repeat this experiment for a range of objects.



For a refresher on reflection, go back to page 94.

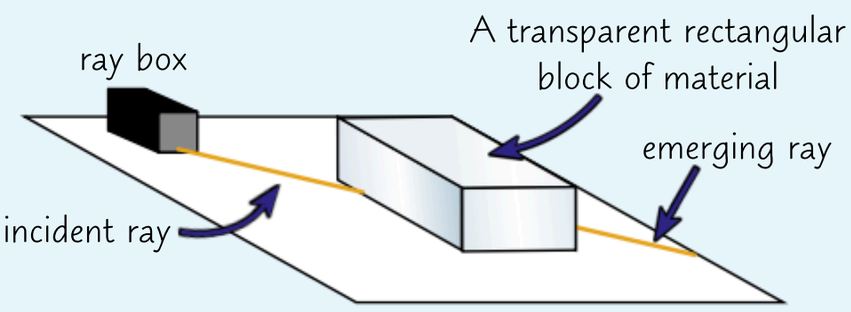
- You should see that smooth surfaces like mirrors give clear reflections (the reflected ray is as thin and bright as the incident ray). Rougher surfaces like paper cause diffuse reflection (p.94) — the reflected beam will be wider and dimmer (or not observable at all).
- You should also find that the angle of incidence ALWAYS equals the angle of reflection.

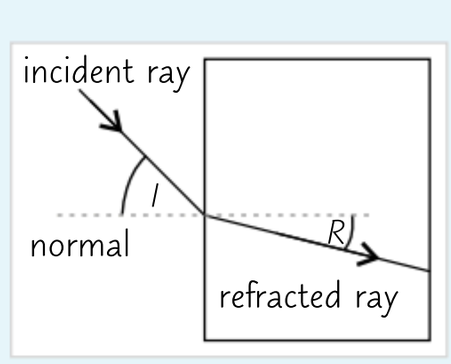
You Can Use **Transparent Materials** to Investigate **Refraction**

The boundaries between different substances refract light by different amounts. Below is a method you could use to investigate how much light is refracted when it passes from air into different materials:

Method

1) Place a transparent rectangular block on a piece of paper and trace around it. Use a ray box or a laser to shine a ray of light at the middle of one side of the block.
2) Trace the incident ray and mark where the light ray emerges on the other side of the block. Remove the block and join up the incident ray and the emerging point to show the path of the refracted ray.
3) Draw the normal through the point where the light ray entered the block. Use a protractor to measure the angle of incidence (I) and the angle of refraction (R).
4) Repeat this experiment using rectangular blocks made from different materials, keeping the incident angle the same throughout.





- You should find that the angle of refraction changes for different materials.
- The difference is due to the different optical densities of the materials (see page 95).

It's always a good idea to check your results are repeatable...

When you're doing one of these investigations, it's a good idea to repeat the test a few times just in case anything weird happened. You could have set up the experiment wrong, or you might just have read the wrong number off the protractor — it's very easily done. Having repeat results will make these odd results easy to spot.

Warm-Up & Exam Questions

Now to check what's actually stuck in your mind over the last five pages...

Warm-Up Questions

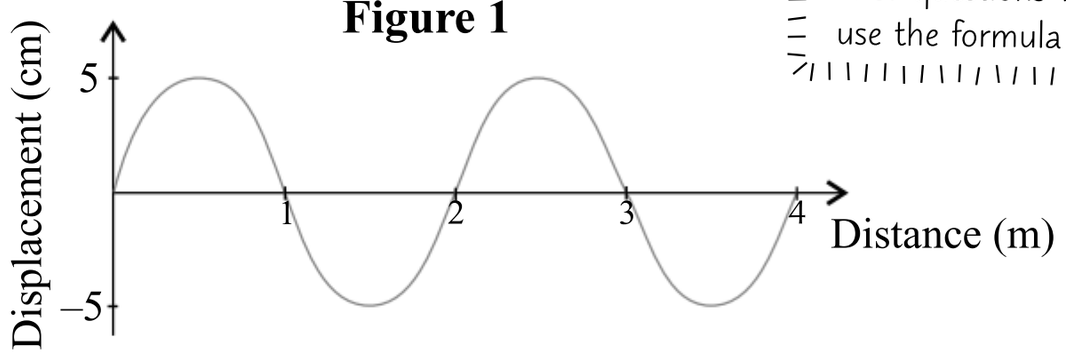
1) Describe the direction of vibrations in a longitudinal wave.
2) Give the formula for calculating the speed of a wave.
3) Outline a method you could use to measure the speed of water waves in a ripple tank.
4) True or false? A wave entering a new medium along the path of the normal won't be refracted.
5) How could you compare the refraction of light through two transparent blocks?

Exam Questions

For questions 1, 3 and 4, you can use the formula sheet on page 216.

1 **Figure 1** shows a graph of a water wave. GRADE 4-6 GRADE

Figure 1



1.1 State whether water waves are transverse or longitudinal.

[1 mark]

1.2 Give the amplitude of this wave.

[1 mark]

1.3 Find the wavelength of this wave.

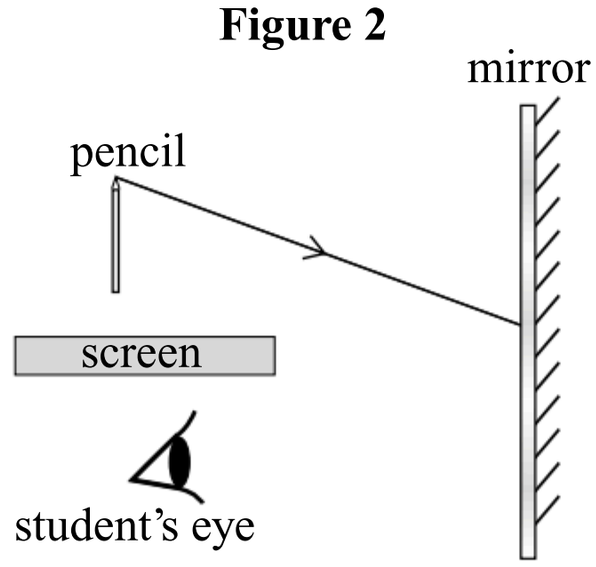
[1 mark]

1.4 If the frequency of the wave doubles but its speed stays the same, describe what will happen to its wavelength.

[1 mark]

2 **Figure 2** shows a student looking at a pencil from behind a screen using a plane mirror. GRADE 4-6 GRADE

Figure 2



2.1 State the type of reflection that occurs in **Figure 2** as light is reflected from the pencil tip to the student's eye.

[1 mark]

2.2 Complete the ray diagram in **Figure 2** to show how the light reflected by the tip of the pencil travels to the student's eye. Mark the angle of incidence and the angle of reflection.

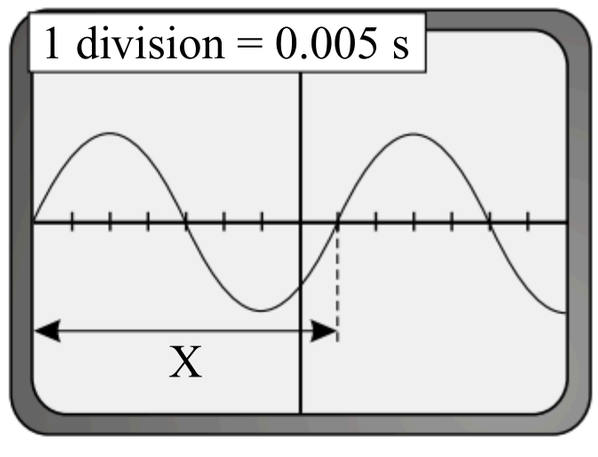
[3 marks]

Exam Questions

3 **Figure 3** shows a trace representing a sound wave on an oscilloscope screen. It is a displacement-time graph, which shows the displacement of a single particle as time passes.

GRADE 6-7 GRADE

Figure 3



3.1 The particle shown by the wave on **Figure 3** is at its undisturbed (rest) position at T = 0 seconds.
At what time will it next cross the undisturbed position?

[2 marks]

3.2 What quantity does the distance marked X represent?
Tick **one** box.

amplitude ☐ frequency ☐ period ☐

[1 mark]

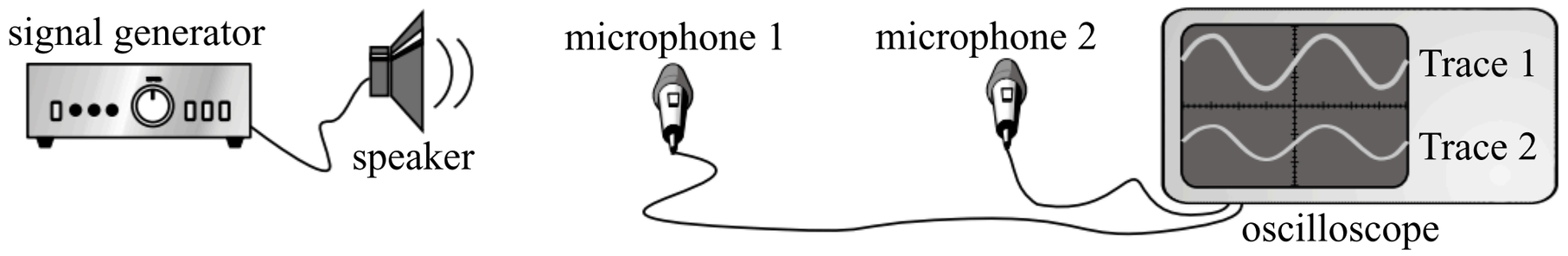
3.3 Calculate the frequency of the wave.

[4 marks]

4 **Figure 4** shows how an oscilloscope can be used to display sound waves by connecting microphones to it. Trace 1 shows the sound waves detected by microphone 1 and trace 2 shows the sound waves detected by microphone 2.

GRADE 6-7 GRADE

Figure 4



Both microphones begin at equal distances from the speaker, and the signal generator is set to a fixed frequency. A student gradually moves microphone 2 away from the speaker, which causes trace 2 to move. The student stops moving microphone 2 when both traces line up again as shown in **Figure 4**. The distance between the microphones is measured.

4.1 Explain how the measurement could be used to work out the speed of sound in air.

[2 marks]

4.2 With the signal generator set to 50 Hz, the distance between the microphones was measured as 6.8 m. Calculate the speed of sound in air. Give the correct unit.

[3 marks]

PRACTICAL

5 **Figure 5** shows a ray of red light entering a glass prism.

GRADE 7-9 GRADE

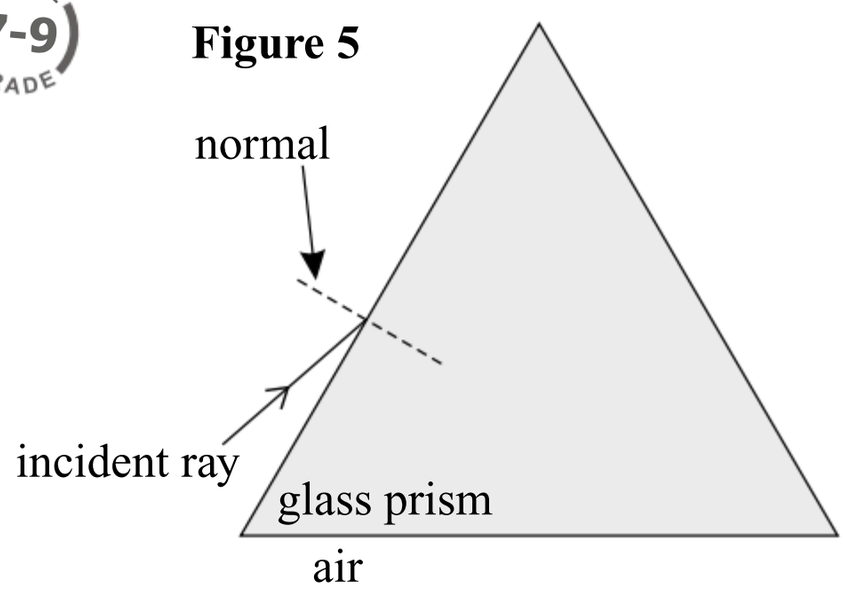
5.1 Complete the diagram to show the ray passing through the prism and emerging from the other side. Label the angles of incidence, *I*, and refraction, *R*, for both boundaries.

[3 marks]

5.2 Describe an experiment that could be carried out to measure *I* and *R* at both boundaries.

[4 marks]

Figure 5



Electromagnetic (EM) Waves

Quick Quiz

*The **light waves** our eyes detect are a **small part** of a spectrum of **EM waves** — there's lots more we can't see...*

Electromagnetic Waves Transfer Energy

1) Electromagnetic (EM) waves are transverse waves that transfer energy from a source to an absorber:

A camp fire transfers energy to its surroundings by emitting infrared radiation. These infrared waves are absorbed by objects in the surroundings and transfer energy to the thermal energy stores of the objects, causing them to warm up.

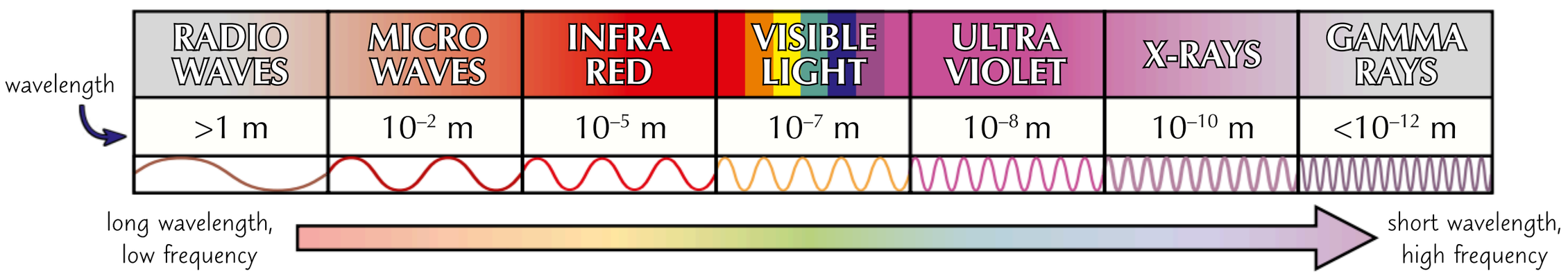
Radio waves transfer energy to the kinetic energy stores of electrons in radio receivers, which generates an electric current (see page 100).

2) Electromagnetic waves aren't vibrations of particles, they're vibrations of electric and magnetic fields (p.119). This means they can travel through a vacuum (e.g. space). All EM waves travel at the same speed through air or a vacuum.

3) They travel at different speeds in different materials (which can lead to refraction — p.95).

There's a Continuous Spectrum of EM Waves

Electromagnetic waves form a continuous spectrum over a range of frequencies. They are grouped into seven basic types, based on their wavelength and frequency.



- Our eyes can only detect a small part of this spectrum — visible light (see page 109).
- Because of their different properties, different EM waves are used for different purposes.
- The different wavelengths of EM waves are reflected, absorbed and transmitted differently when they meet a boundary.

Different Types of EM Waves are Produced in Different Ways

1) A wide range of frequencies and wavelengths of EM radiation can be produced by changes in atoms and their nuclei, for example:

- Electrons rearranging themselves inside atoms (see p.50) can create waves with frequencies from infrared up to X-rays.
- Changes in the nucleus of an atom can create gamma rays (see p.51).

2) Radio waves are produced outside the atom by oscillating charged particles, e.g. electrons in a circuit (see next page).

3) The same processes that produce EM waves happen in reverse when EM waves are absorbed.

EM waves are ordered by frequency and wavelength...

The next few pages dive deeper into the different types of EM waves — get confident with the basics before moving on. Here's a handy mnemonic for the EM wave spectrum: 'Rock Music Is Very Useful for eXperiments with Goats'.

Quick Quiz

Radio Waves

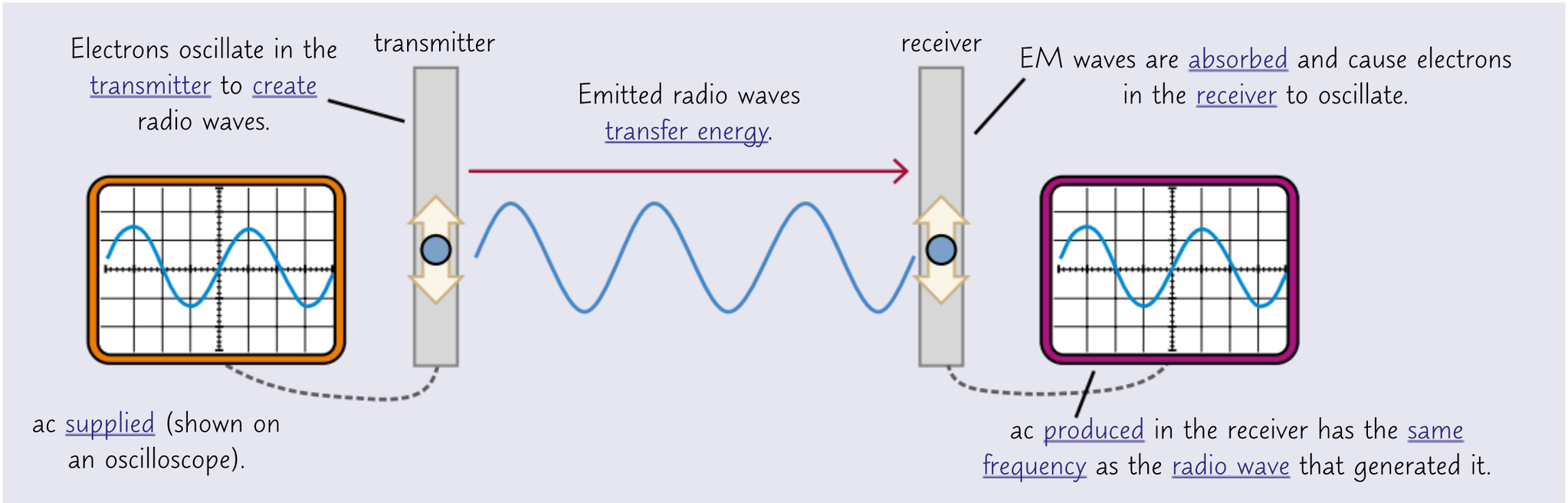
*EM waves have all sorts of uses, and **radio waves** might be the most entertaining. They're used to **transfer energy** to your **TV**, your **mobile phone** and your **wireless headphones** — what would you do without them?*

Radio Waves are Made by **Oscillating Charges**

- EM waves are made up of oscillating electric and magnetic fields.
- Alternating currents (ac) (p.32) are made up of oscillating charges. As the charges oscillate, they produce oscillating electric and magnetic fields, i.e. electromagnetic waves.

Head on over to pages 21-38 for more on electricity.

- You can produce radio waves using an alternating current in an electrical circuit as shown below:

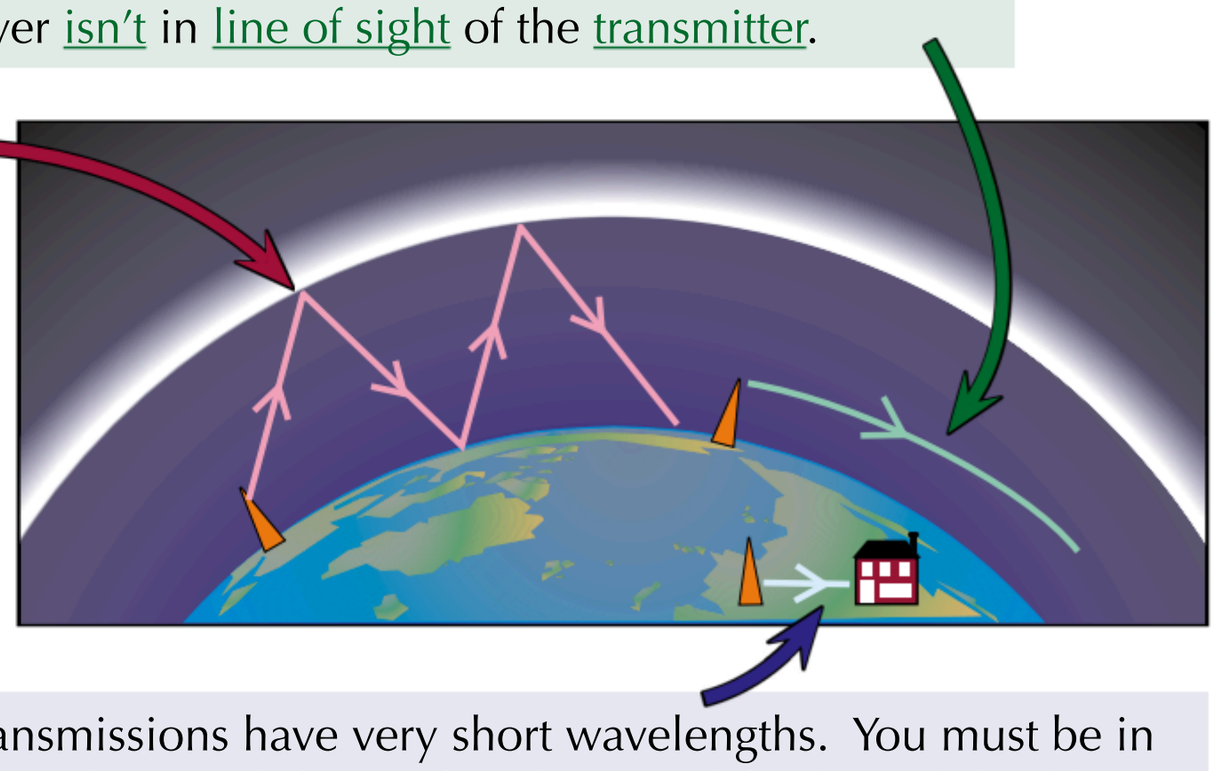


Radio Waves are Used **Mainly** for **Communication**

Radio waves have wavelengths longer than about 10 cm.
Different sized radio waves have different purposes:

Long-wave radio (wavelengths of 1 – 10 km) can be transmitted from halfway round the world because long wavelengths diffract (bend) around the curved surface of the Earth. Long-wave radio wavelengths can also diffract around hills, into tunnels and all sorts. This makes it possible for radio signals to be received even if the receiver isn't in line of sight of the transmitter.

- Short-wave radio signals (wavelengths of about 10 m – 100 m) are reflected from the ionosphere — an electrically charged layer in the Earth's upper atmosphere. They can also can be received at long distances from the transmitter.
- Medium-wave signals can also reflect off the ionosphere, depending on atmospheric conditions and the time of day.



- The radio waves used for TV and FM radio transmissions have very short wavelengths. You must be in direct sight of the transmitter to get reception — the signal doesn't bend or travel far through buildings.
- Bluetooth® uses even shorter wavelengths than TV and FM radio (about 10 cm) to send data over short distances between devices without wires (e.g. wireless headphones).

Frequency of the radio wave = frequency of the alternating current...

So an alternating current in the transmitter makes radio waves. These radio waves get absorbed by the receiver, which makes the electrons in the receiver oscillate. And hey presto — you have an alternating current again.

Uses of EM Waves

Quick Quiz

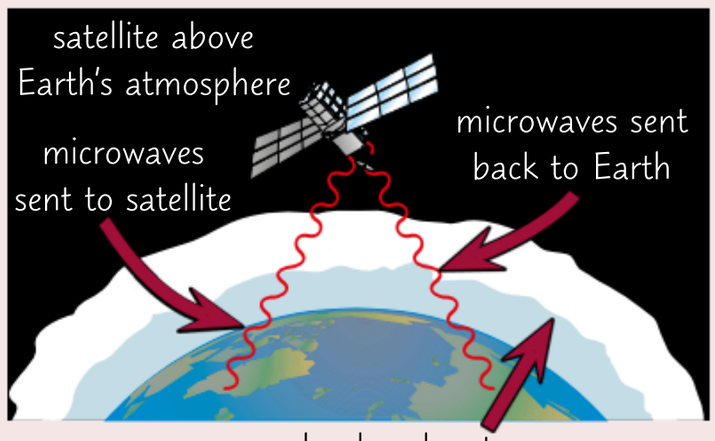
Radio waves *aren't the only waves used for* ***communication*** *— other EM waves come in pretty handy too. The most important thing is to think about how the* ***properties*** *of a wave relate to its* ***uses****.*

Microwaves are Used by Satellites

For satellite communication, it's best to use microwaves that can pass easily through the Earth's atmosphere.

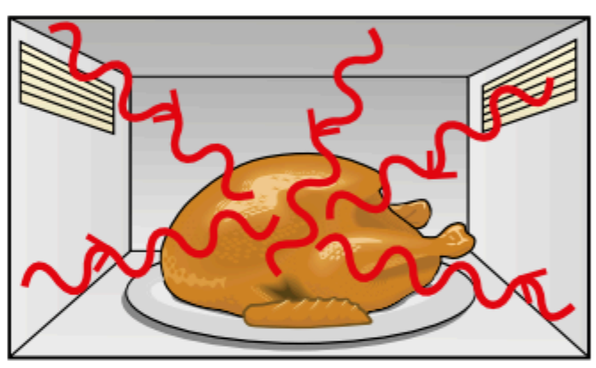
For satellite TV, the signal from a transmitter is transmitted into space...

- ... where it's picked up by the satellite receiver dish orbiting thousands of kilometres above the Earth. The satellite transmits the signal back to Earth in a different direction...
- ... where it's received by a satellite dish on the ground. There is a slight time delay between the signal being sent and received because of the long distance the signal travels.



Microwave Ovens Also Use Microwaves

1) In microwave ovens, shorter wavelength microwaves than the ones that are used for satellites are absorbed by water molecules in food.
2) The microwaves penetrate up to a few centimetres into the food before being absorbed and transferring the energy they carry to the thermal energy stores of the water molecules, causing the water to heat up.
3) The water molecules then transfer this energy to the rest of the molecules in the food by heating — which quickly cooks the food.



Infrared Radiation Can be Used to Increase or Monitor Temperature

Infrared (IR) radiation is given out by all objects — and the hotter the object, the more IR radiation it gives out.

Infrared cameras

They can be used to detect infrared radiation and monitor temperature. The camera detects the IR radiation and turns it into an electrical signal, which is displayed on a screen as a picture. The hotter an object is, the brighter it appears. E.g. infrared radiation emitted from different parts of a house can be detected using infrared cameras.

Different colours represent different amounts of IR radiation being detected. Here, the redder the colour, the more IR radiation is detected.

Heating

Absorbing IR radiation causes objects to get hotter.

- Food can be cooked using IR radiation — the temperature of the food increases when it absorbs IR radiation, e.g. from a toaster's heating element.
- Electric heaters heat a room in the same way. They contain a long piece of wire that heats up when a current flows through it. This wire then emits infrared radiation (and a little visible light — the wire glows). Energy is transferred by the IR waves to the thermal energy stores of the surroundings, causing the temperature to increase.

The uses of EM waves depend on their properties...

Differences in wavelength, frequency and energy between types of EM wave give them different properties. For example, some types of EM wave are very harmful (check out page 103). Luckily, microwaves with long wavelengths are considered safe to beam round the world. Infrared is fairly safe, although too much of it will burn you.

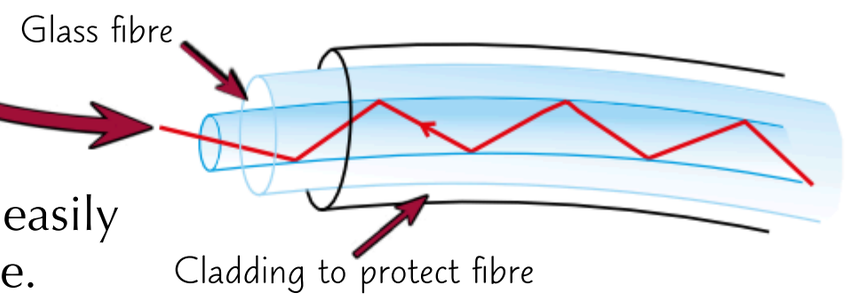
More Uses of EM Waves

Quick Quiz

*Here are just a few more uses of EM waves — complete with the all-important **reasons** why they're used.*

Fibre Optic Cables Use Visible Light to Transmit Data

1) Optical fibres are thin glass or plastic fibres that can carry data (e.g. from telephones or computers) over long distances as pulses of visible light.
2) They work because of reflection (p.94). The light rays are bounced back and forth until they reach the end of the fibre.
3) Visible light is suitable to use because its not easily absorbed or scattered as it travels along a fibre.



Ultraviolet (UV) Radiation — Fluorescent Lights and Suntans

- Fluorescent lights generate UV radiation, which is absorbed and re-emitted as visible light by a layer of phosphor on the inside of the bulb. They're energy-efficient (p.10) so they're good to use when light is needed for long periods.
- Security pens can be used to mark property with your name. Under UV light the ink will glow (fluoresce), but it's invisible otherwise. This can help the police identify stolen property.

Fluorescence is a property of certain chemicals, where ultra-violet radiation is absorbed and then visible light is emitted. That's why fluorescent colours look so bright — they actually emit light.

- Ultraviolet radiation is produced by the Sun, and exposure to it is what gives people a suntan.
- When it's not sunny, some people go to tanning salons where UV lamps are used to give them an artificial suntan. However, overexposure to UV radiation can be dangerous (fluorescent lights emit very little UV — they're totally safe).

There's more on the dangers of UV on p.103.

X-rays and Gamma Rays are Used in Medicine

Radiographers in hospitals take X-ray 'photographs' of people to see if they have any broken bones. X-rays pass easily through flesh but not so easily through denser material like bones or metal. So it's the amount of radiation that's absorbed that gives you an X-ray image.

This is a negative image — the plate starts off all white. The brighter bits are where fewer X-rays get through.

X-rays and gamma rays can be used to treat people with cancer (radiotherapy). This is because high doses of these rays kill all living cells — so they are carefully directed towards cancer cells, to avoid killing too many normal, healthy cells.

Gamma radiation can also be used as a medical tracer (p.57) — this is where a gamma-emitting source is injected into the patient, and its progress is followed around the body. Gamma radiation is well suited to this because it can pass out through the body to be detected.

There's more on gamma rays on p.51.

So many different uses — don't get them mixed up...

For each type of EM wave, you need to know what it's used for and why it's suitable. Examiners might ask you why different EM waves **can't** be used for something, too — like why you shouldn't try and get a suntan using gamma rays...

Dangers of EM Waves

Quick Quiz

*Okay, so you know how **useful** electromagnetic radiation can be — well, it can also be pretty **dangerous**.*

Some **EM Radiation** Can be **Harmful** to **People**

The effects of each type of radiation on living tissue are based on how much energy the wave transfers:

1) Low frequency waves, like radio waves, don't transfer much energy to the atoms in your body — they mostly pass through soft tissue without being absorbed.
2) High frequency waves like UV, X-rays and gamma rays all transfer lots of energy and so can cause lots of damage:
   - UV radiation damages surface cells, which can lead to sunburn and cause skin to age prematurely. Some more serious effects are blindness and an increased risk of skin cancer.
   - X-rays and gamma rays are types of ionising radiation (p.51). Ionisation can cause gene mutation or cell destruction, and cancer.

You Can **Measure Risk** Using the **Radiation Dose** in **Sieverts**

1) Whilst UV radiation, X-rays and gamma rays can all be harmful, they are also very useful (see page 102). Before any of these types of EM radiation are used, people look at whether the benefits outweigh the health risks.

   For example, the risk of a person involved in a car accident developing cancer from having an X-ray photograph taken is much smaller than the potential health risk of not finding and treating their injuries.

2) Radiation dose (measured in sieverts) is a measure of the risk of harm from the body being exposed to radiation. It is not a measure of the total amount of radiation that has been absorbed.
3) The risk depends on the total amount of radiation absorbed and how harmful the type of radiation is.
4) A sievert is pretty big, so you'll often see doses in millisieverts (mSv), where 1000 mSv = 1 Sv.

Radiation doses can be calculated for all types of radiation, not just UV, X-rays and gamma rays (see p.56).

Risk can be **Different** for **Different Parts** of the **Body**

A CT scan uses X-rays and a computer to build up a picture of the inside of a patient's body. The table shows the radiation dose received by two different parts of a patient's body when having CT scans.

	Radiation dose (mSv)
Head	2.0
Chest	8.0

If a patient has a CT scan on their chest, they are four times more likely to suffer damage to their genes (and their added risk of harm is four times higher) than if they had a head scan.

The risks and benefits of radiation exposure must be balanced...

Ionising radiation can be dangerous, but sometimes the risk is worth taking — and sometimes it's not. From 1920-1970, X-ray machines were installed in shoe shops for use in shoe fittings. But when people realised radiation was harmful, they were phased out. The risks far outweighed the benefits of using X-rays rather than tape measures...

Warm-Up & Exam Questions

There are quite a few different sorts of electromagnetic waves — and you never know which ones might come up in the exams... So check which you're still a bit hazy on with these questions.

Warm-Up Questions

1) Which type of EM wave has the highest frequency?
2) Briefly describe how radio waves produce an electrical signal in a receiver.
3) Explain how microwaves heat food.
4) What are optical fibres used for in phone lines?
5) Why is ionising radiation dangerous?

Exam Questions

1 EM radiation can be harmful but also useful. (GRADE 4-6)

1.1 Give **one** practical use of ultraviolet radiation.

[1 mark]

1.2 Give **one** hazard associated with ultraviolet radiation.

[1 mark]

1.3 Gamma rays can be used in medical tracers to check that your body is working correctly.
Explain how medical tracers work.

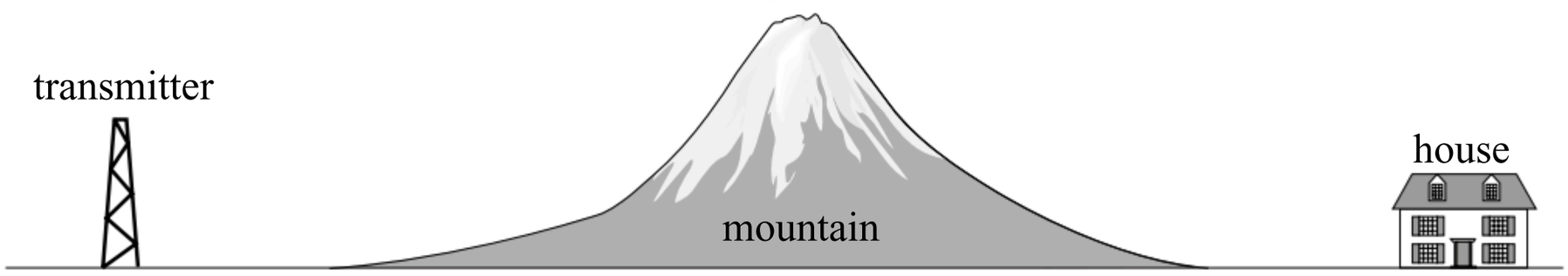
[3 marks]

1.4 Explain why gamma rays are suitable for use in medical tracers.

[1 mark]

2 **Figure 1** shows a transmitter which is transmitting a communications signal. There is a receiver for the signal inside the house.
There is a mountain between the transmitter and the house. (GRADE 6-7)

Figure 1



2.1 Explain why radio waves would be more suitable than light to send the transmission.

[2 marks]

2.2 The home owner decides to get satellite TV installed.
State what type of electromagnetic radiation is used to send signals to satellites and explain why it is suitable.

[2 marks]

2.3 Describe how satellite TV signals are transmitted from a transmitter on the ground to the house.

[2 marks]

Revision Summary Test for Topic 6a

Wave goodbye to those revision woes — I've got some questions here to help you master topic 6a:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

Wave Properties (p.92-93) ☑

1) Do waves cause matter to travel with them? Give an example to explain your answer.
2) What is the amplitude, wavelength, frequency and time period of a wave?
3) What's the difference between transverse and longitudinal waves? Give an example of each of them.
4) What formula is used to calculate wave speed? What does each symbol in the formula mean?
5) Describe an experiment you could do to measure:
   a) the speed of sound, b) the speed of ripples in water, c) the speed of a wave on a string.

Reflection and Refraction (p.94-96) ☑

6) What three things can happen when waves arrive at a boundary between two different materials?
7) What rule is true for all reflected waves?
8) Describe what the 'normal' is when drawing ray diagrams.
9) What is the difference between specular reflection and diffuse reflection?
10) What is refraction? What two properties of a wave change as it is refracted?
11) Draw a ray diagram for a light ray entering a more optically dense material.
    Label the angle of incidence (I) and the angle of refraction (R).
12) Give a method that could be used to investigate how each of the following are affected by different materials: a) reflection, b) refraction.

Electromagnetic Waves (p.99) ☑

13) Give an example of how EM waves transfer energy from a source to an absorber.
14) True or false? Different EM waves travel at different speeds through a vacuum.
15) List the seven basic types of EM wave in order from longest to shortest wavelength.
16) Give two ways that an atom can produce EM waves.

Uses and Dangers of EM Waves (p.100-103) ☑

17) Briefly describe how an alternating current and a transmitter are used to produce radio waves.
18) Give one reason why long-wave radio waves are suitable for long-distance communication.
19) Why are microwaves used by satellites? Explain how microwaves are used for satellite TV.
20) Explain how microwave ovens quickly heat food.
21) How is infrared radiation used in infrared cameras?
22) Give two things that use IR radiation for heating.
23) What are optical fibres? Why is visible light suitable to use in them?
24) What type of EM wave do fluorescent lights use? What is a benefit of using fluorescent lights?
25) Explain why radiographers use X-rays to take images of bones? Give one other use of X-rays.
26) Why are gamma rays good medical tracers?
27) Are high frequency EM waves or low frequency EM waves more dangerous? Why do you think that?
28) What damage can UV radiation cause to the body? What about X-rays and gamma rays?
29) What is radiation dose and what unit is it measured in? What are two things that it depends on?

Quick Quiz

Lenses

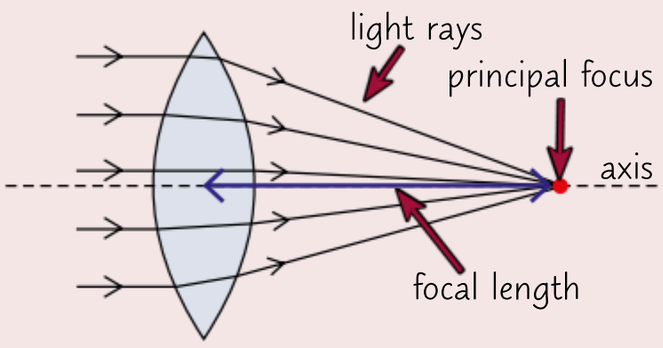
*These next few pages are about how **light** acts when it hits a **lens**. Be ready for lots of **diagrams**.*

Different Lenses Produce Different Kinds of Images

Lenses form images by refracting light (p.95) and changing its direction. There are two main types of lens — convex and concave. They have different shapes and have opposite effects on light rays.

Convex (or Converging) Lens

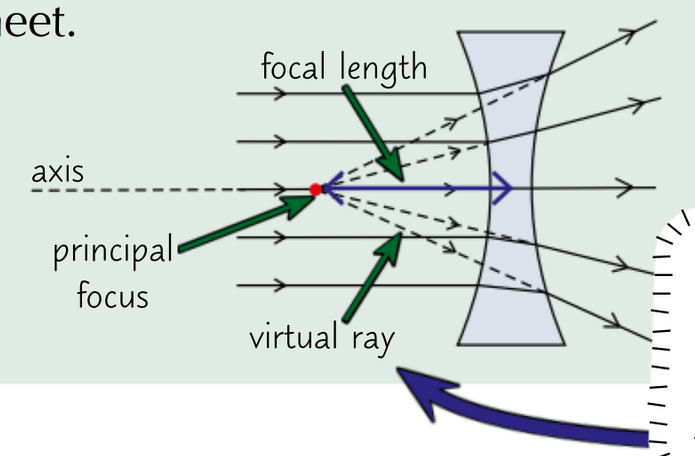
- Bulges outwards.
- Causes rays of light to be brought together (converge).
- Rays hitting the lens parallel to the axis meet at the principal focus.



Light can pass through a lens in either direction, so there is a principal focus on each side of the lens.

Concave (or Diverging) Lens

- Caves inwards.
- Causes rays of light to spread out (diverge).
- Rays hitting the lens parallel to the axis appear to come from the principal focus — you can trace them back to this point, where they all appear to meet.



The difference between real and virtual is explained on the next page.

Make sure you know what these key terms mean:

Axis	An imaginary line passing through the middle of a lens, perpendicular to the lens.
Principal Focus	The point where rays hitting a lens parallel to its axis get converged to (for a convex lens), or appear to have come from (for a concave lens).
Focal Length	The distance from the centre of the lens to the principal focus.

There are Three Rules for Refraction in a Convex Lens...

1) An incident ray parallel to the axis refracts through the lens and passes through the principal focus on the other side.
2) An incident ray passing through the principal focus refracts through the lens and travels parallel to the axis.
3) An incident ray passing through the centre of the lens carries on in the same direction.

An example ray diagram for a convex lens is on page 107.

... And Three Rules for Refraction in a Concave Lens

1) An incident ray parallel to the axis refracts through the lens and travels in line with the principal focus (so it appears to have come from the principal focus on the incident side of the lens).
2) An incident ray passing through the lens towards the principal focus on the other side refracts through the lens and travels parallel to the axis.
3) An incident ray passing through the centre of the lens carries on in the same direction.

The neat thing about these rules is that they allow you to draw ray diagrams without bending the rays twice — once as they go into the lens and once as they leave the lens. You can draw the diagrams as if each ray only changes direction once, in the middle of the lens (see next page).

Convex lenses bulge outwards, concave lenses cave inwards...

There's a lot of new stuff here, so use the old 'cover the page and see what you've remembered' method. Sketch out a convex and a concave lens, label all the key bits, then draw some rays of light passing through from different directions.

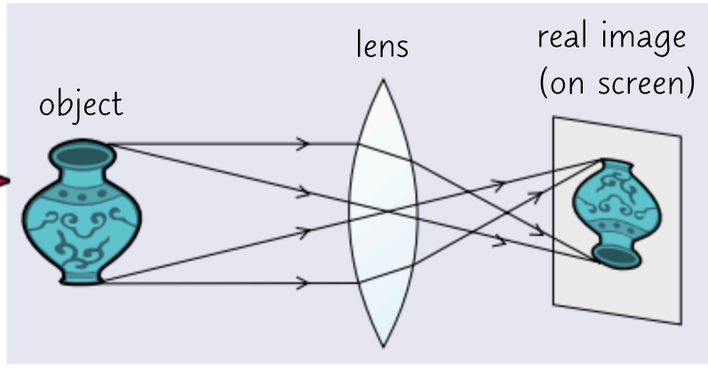
Images and Convex Lenses

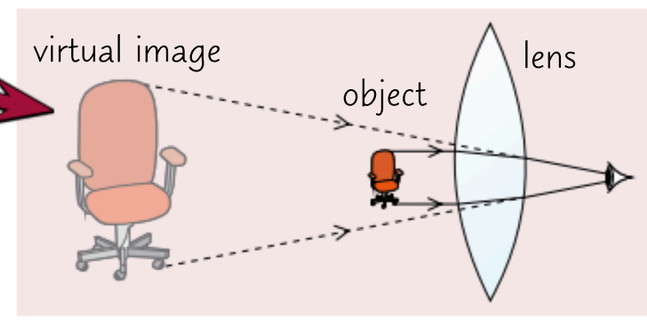
Quick Quiz

*Prepare yourself to learn about **virtual** and **real images** (don't worry, they're not as weird as they sound).*

Lenses Can Produce **Real** and **Virtual** Images

- A real image is where the light from an object comes together to form an image on a 'screen' — like the image formed on an eye's retina (the 'screen' at the back of an eye).
- A virtual image is when the rays are diverging, so the light from the object appears to be coming from a completely different place.
- When you look in a mirror you see a virtual image of your face because the object (your face) appears to be behind the mirror.
- You get a virtual image when you look at an object through a magnifying lens (see next page) — the virtual image appears bigger than the object.



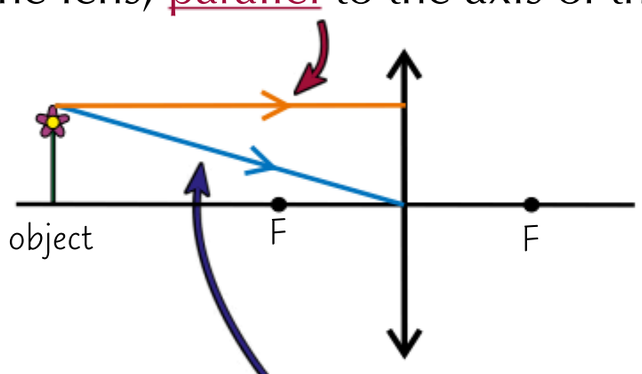


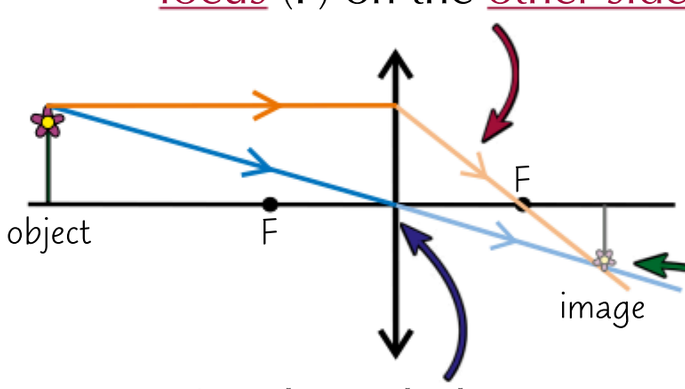
To describe an image properly, you need to say 3 things about it:

1) How big it is compared to the object.
2) Whether it's upright or inverted (upside down).
3) Whether it's real or virtual.

You Can Draw a **Ray Diagram** for an **Image** Through a **Convex Lens**

1. Draw a ray going from the top of the object to the lens, parallel to the axis of the lens.
2. This ray is refracted through the principal focus (F) on the other side of the lens.



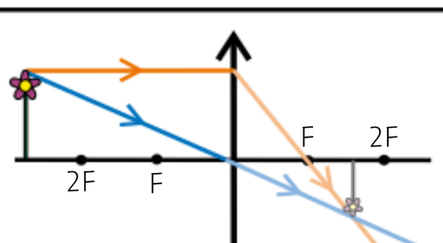
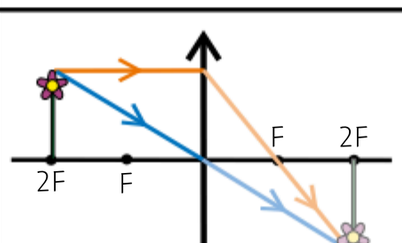
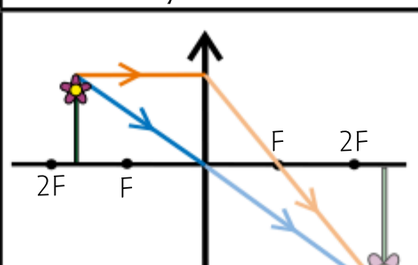
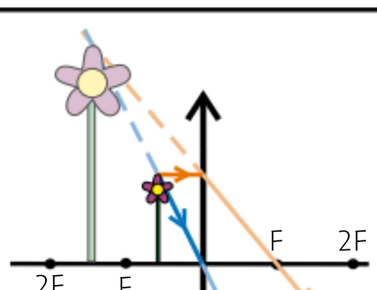


In ray diagrams, ↕ represents a convex lens.

3. Draw a ray going straight through the middle of the lens.
4. The ray passing through the middle of the lens doesn't bend.
5. Where the rays meet is the top of the image.
6. Repeat the process for a point on the bottom of the object. When the bottom of the object is on the axis, the bottom of the image is also on the axis.

Distance from the Lens Affects the **Image**

Video

Object's Position	Further than 2F	At 2F	Between F and 2F	Nearer than F
Image's Size	Smaller	Same size	Bigger	Bigger
Inverted?	Yes	Yes	Yes	No
Real or Virtual?	Real	Real	Real	Virtual
Image's Position	Between F and 2F	At 2F	Beyond 2F	Same side of the lens
Diagram				

Converging rays = real image, diverging rays = virtual image...

Make sure you know the differences between real and virtual images — they can be pretty tough.

Q1 Draw a ray diagram for an object at a distance of 2F in front of a convex lens. [3 marks]

Q1 Video Solution

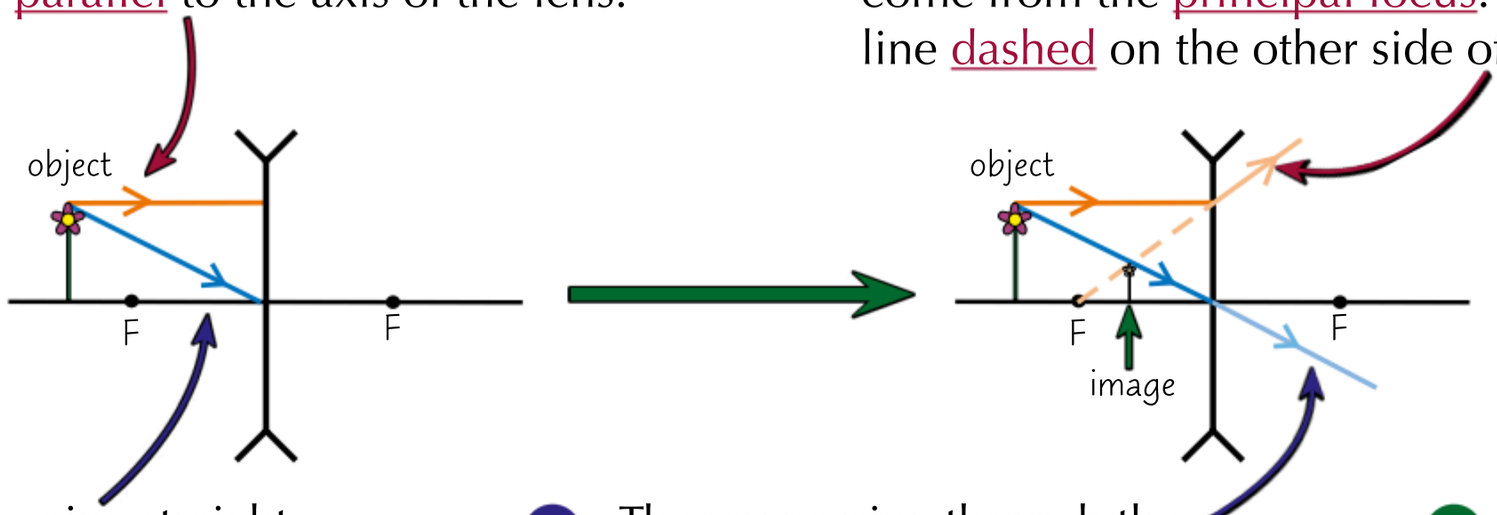
Quick Quiz

Concave Lenses and Magnification

*Now for **concave lenses** and getting familiar with a very handy equation — the **magnification formula.***

You Can Draw a **Ray Diagram** for an **Image** Through a **Concave Lens**

1. Draw a ray going from the top of the object to the lens, parallel to the axis of the lens.
2. This ray is refracted so it appears to have come from the principal focus. Make the line dashed on the other side of the lens.



In ray diagrams, I represents a concave lens.

3. Draw a ray going straight through the middle of the lens.
4. The ray passing through the middle of the lens doesn't bend.
5. Where the rays meet is the top of the image.
6. Repeat the process for a point on the bottom of the object. When the bottom of the object is on the axis, the bottom of the image is also on the axis.

Concave Lenses Always Create **Virtual Images**

Unlike a convex lens, a concave lens always produces a virtual image. The image is the right way up, smaller than the object and on the same side of the lens as the object — no matter where the object is.

The further an object is from the concave lens, the smaller the image produced. You can work this out by drawing a ray diagram.

Magnification Compares **Object Size** and **Image Size**

1) You can use the magnification formula to work out the magnification produced by a lens at a given distance:

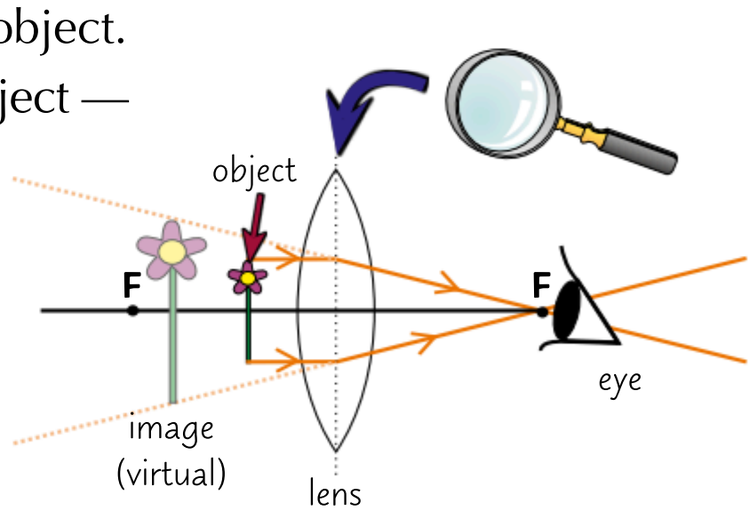
$$\textbf{magnification} = \frac{\textbf{image height}}{\textbf{object height}}$$

Magnification is a ratio, so it doesn't have any units. As long as the units for both heights are the same, you can measure them in whatever units you like (e.g. mm or cm).

2) So a magnification less than 1 means the image is smaller than the object.

3) A magnification greater than 1 means the image is bigger than the object — this is how magnifying glasses work:

- Magnifying glasses use a convex lens.
- The object must be closer to the lens than the focal length.
- The image produced is a virtual image, so the light rays don't actually come from the place where the image appears to be.



EXAMPLE

A coin with diameter 14 mm is placed behind a magnifying lens. The virtual image produced has a diameter of 35 mm. What is the magnification of the lens at this distance?

magnification = image height ÷ object height = 35 ÷ 14 = 2.5

You can use a magnifying glass to form a virtual image...

... so the image *appears* to be at the point where the diverging rays *seem* to have come from.

Q1 Find the magnification of a lens if a 12 cm tall object produces a 6 cm high image. [2 marks]

Q2 Calculate the height of an image if the object is 10 cm tall and the magnification is 2.5. [3 marks]

Q2 Video Solution

Visible Light

Quick Quiz

*We see **light** all the time. But it's a bit more **complicated** than you might have thought.*

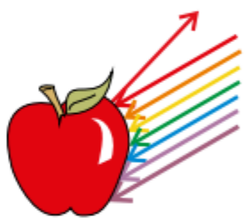
Visible Light is Made Up of a Range of Colours

- As you saw on page 99, EM waves cover a very large spectrum. We can only see a tiny part of this — the visible light spectrum. This is the range of wavelengths that we perceive as different colours.
- Each colour has its own narrow range of wavelengths (and frequencies) ranging from violets down at 400 nm up to reds at 700 nm.
- Colours can also mix together to make other colours. The only colours you can't make by mixing are the primary colours: pure red, green and blue.
- When all of these different colours are put together, it creates white light.

These are the primary colours of light. Don't confuse green with yellow — yellow is a primary colour of paint.

Colour and Transparency Depend on Absorbed Wavelengths

1) Different objects absorb, transmit and reflect different wavelengths of light in different ways (p.94).
2) Opaque objects are objects that do not transmit light. When visible light waves hit them, they absorb some wavelengths of light and reflect others.
3) The colour of an opaque object depends on which wavelengths of light are most strongly reflected:

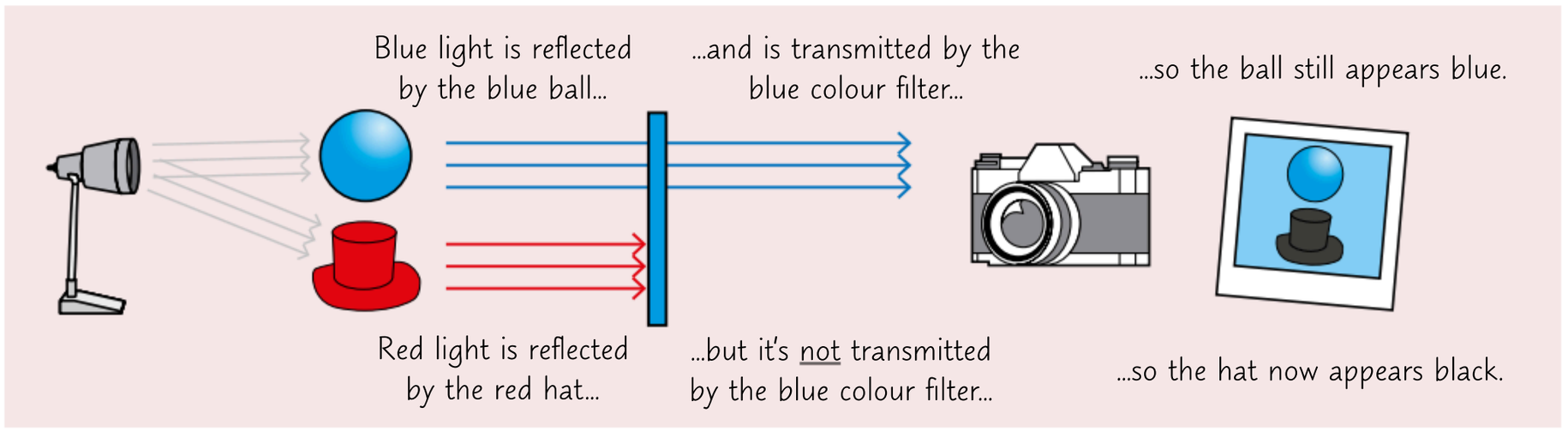


Colour of object	Black	White	Primary colour	Non-primary colour
Reflects...	None	All (equally)	That colour	That colour, and/or the primary colours that mix to make that colour
Absorbs...	All	None	Everything else	Everything else

4) Transparent (see-through) and translucent (partially see-through) objects transmit light, i.e. not all light that hits the surface of the object is absorbed or reflected — some can pass through.
5) Some wavelengths of light may be absorbed or reflected by transparent and translucent objects. The colour of the object is related to the wavelengths of light transmitted and reflected by it.

Colour Filters Only Let Through Particular Wavelengths

Colour filters are used to filter out different wavelengths of light, so that only certain colours (wavelengths) are transmitted — the rest are absorbed. A primary colour filter only transmits that colour, e.g. if white light is shone at a blue colour filter, only blue light will be let through. The rest would be absorbed.



Filters that aren't for primary colours let through both the wavelengths of light for that colour AND/OR the wavelengths of the primary colours that can be added together to make that colour.

We see the colour of light that objects reflect into our eyes...

So the only thing that makes red stuff red is the fact that it reflects the 'red' wavelengths of light and absorbs the rest. So I wonder... what colour is a red object in the dark when there's no light around for it to reflect? The mind boggles.

Warm-Up & Exam Questions

It's best to make sure this stuff's crystal clear now, or you'll regret it when you get to the exam... So here's lots of questions on lenses, as well as on colour and filters. Go on, answer them, you know you want to.

Warm-Up Questions

1) Name the two main types of lens.
2) Of the two types of lens, which always creates a virtual image and which can create both real and virtual images?
3) Explain why a red object appears red.
4) Why does a green object viewed through a red filter appear black?

Exam Questions

For question 1, you can use the formula sheet on page 216.

1 This question is about different types of lens. GRADE 6-7

1.1 Sunita is trying to start a campfire by focussing sunlight through her spectacle lens onto the firewood. The lens is concave. Explain why she cannot focus the sunlight onto the wood using this lens.

[2 marks]

1.2 Sunita finds a slug and uses a magnifying glass to look at it. Complete **Figure 1** to show where the image of the slug is formed.

Figure 1

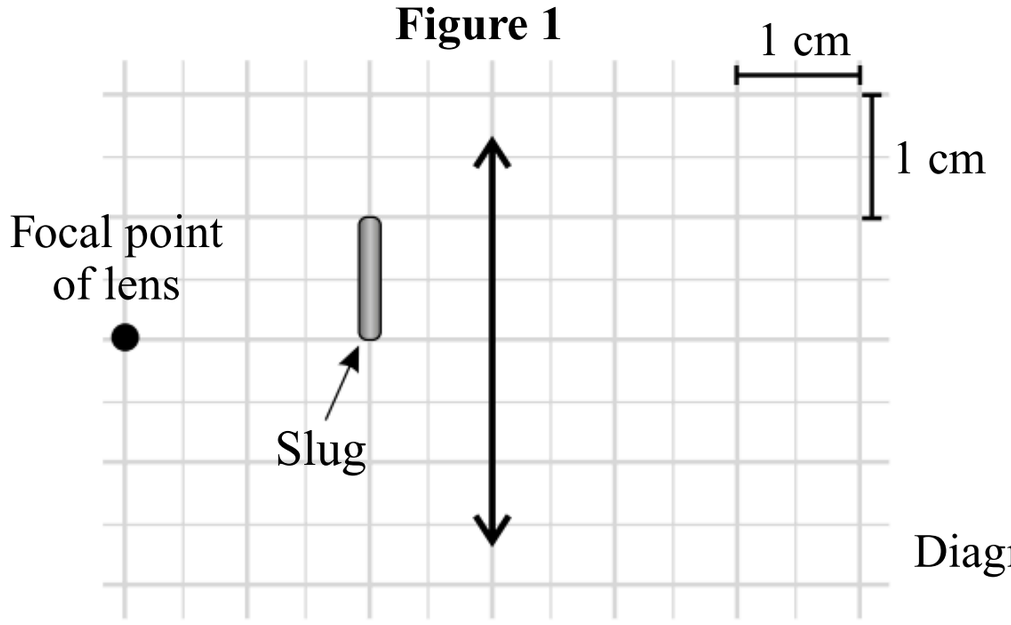


Diagram to scale.

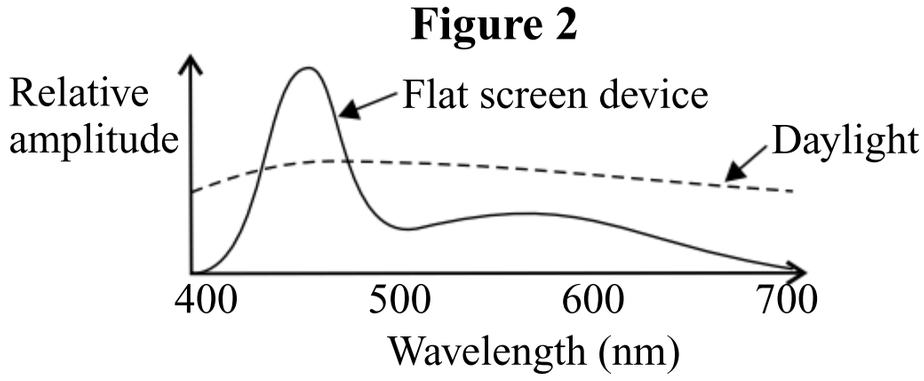
[3 marks]

1.3 Calculate the magnification of the lens for the slug at this distance.

[2 marks]

2 **Figure 2** shows the distribution of wavelengths of visible light emitted by a particular flat screen, compared to that of normal daylight. GRADE 7-9

Figure 2



2.1 Describe how light from the flat screen device differs from daylight.

[3 marks]

2.2 Exposure to blue light, which has wavelengths below 500 nm, has been linked to eye problems. Suggest how people could protect themselves from these problems while using the device. Explain your answer.

[2 marks]

Infrared Radiation and Temperature

Quick Quiz

*Another **Required Practical** coming up — in this one, you'll meet a fun new piece of kit called a **Leslie Cube**. Read on to find out about how you can use this equipment to investigate **infrared radiation emissions**.*

Every Object **Absorbs** and **Emits Infrared Radiation**

1) All objects are continually emitting and absorbing infrared (IR) radiation. IR is emitted from the surface of an object. This affects the temperature of the object. For example, an object that...

...emits less IR than it absorbs	...emits IR at the same rate that it absorbs it	...emits more IR than it absorbs
Warms up	Stays the same temperature	Cools down
E.g. a glass of ice water	E.g. a table	E.g. a mug of tea

The mug gives out more IR radiation than it absorbs, which cools it down.

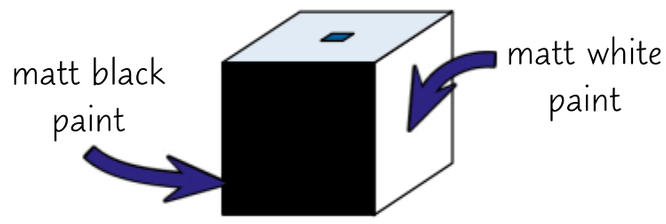
2) The hotter an object is, the more infrared radiation it radiates in a given time.

3) Some colours and surfaces absorb and emit radiation better than others. For example, a black surface is better at absorbing and emitting radiation than a white one, and a matt surface is better at absorbing and emitting radiation than a shiny one.

You Can Investigate **Emission** With a **Leslie Cube**

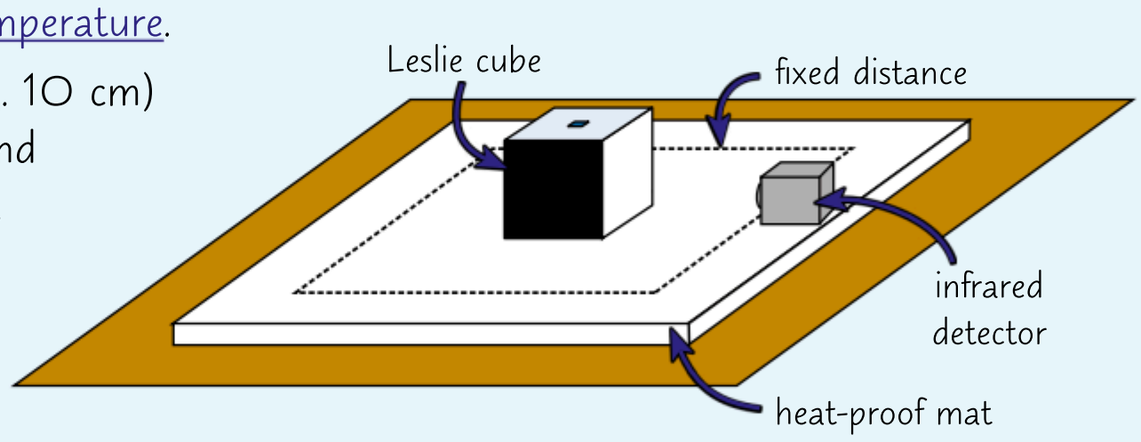
PRACTICAL

A Leslie cube is a hollow, watertight, metal cube made of e.g. aluminium, whose four vertical faces have different surfaces (for example, matt black paint, matt white paint, shiny metal and dull metal). You can use a Leslie cube to investigate IR emission by different surfaces:



Method

1) Place an empty Leslie cube on a heat-proof mat.
2) Boil water in a kettle and fill the Leslie cube with boiling water.
3) Wait a while for the cube to warm up, then hold a thermometer against each of its four vertical faces. You should find that they're all the same temperature.
4) Hold an infrared detector a set distance (e.g. 10 cm) away from one of the cube's vertical faces, and record the amount of IR radiation it detects.
5) Repeat this measurement for each of the cube's vertical faces. Make sure you position the detector at the same distance from the cube each time.



- You should find that you detect more IR radiation from the black surface than the white one, and more from the matt surfaces than the shiny ones.
- As always, you should do the experiment more than once, to make sure your results are repeatable (p.138).
- Handling boiling water is a hazard (p.137) in this experiment — don't try to move the cube just after you fill it or you might burn your hands. And be careful carrying a full kettle — your mate won't thank you if you spill boiling water down their back.

You can also investigate how absorption depends on surface. One way is to use wax to stick ball bearings to the back of two metal plates with different surfaces (e.g. black and white surfaces). You can then place the surfaces equal distances from a Bunsen burner and see which ball bearing falls off first. The wax will melt faster on the plate that is better at absorbing IR radiation.

Carry out your practicals carefully...

Watch out when dealing with potential hazards. Take care when you're pouring or carrying boiling water, and make sure any water or equipment has cooled down enough before you start handling it after your experiment is done.

Quick Quiz

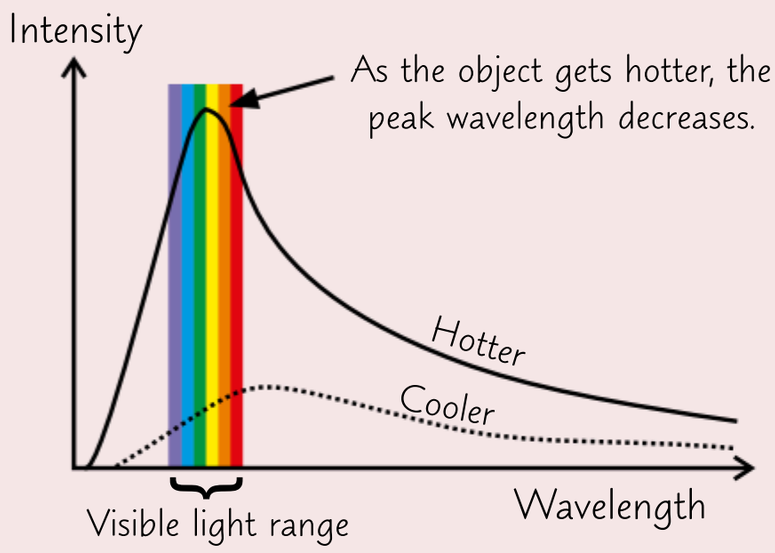
Black Body Radiation

Black body radiation *sounds complicated, and I'm afraid it is a bit — but just take your time and you'll get it.*

Black Bodies are the Ultimate Emitters

A perfect black body is an object that absorbs all of the radiation that hits it. No radiation is reflected or transmitted.

- As well as being perfect absorbers, black bodies are the best possible emitters of radiation.
- Black bodies emit electromagnetic radiation due to the energy in their thermal energy stores. This radiation isn't just in the infrared part of the spectrum — it covers a range of wavelengths.
- The intensity and distribution of the wavelengths emitted by a black body depend on its temperature. Intensity is the power per unit area, i.e. how much energy is transferred to a given area in a certain amount of time. The curves below show this in action:

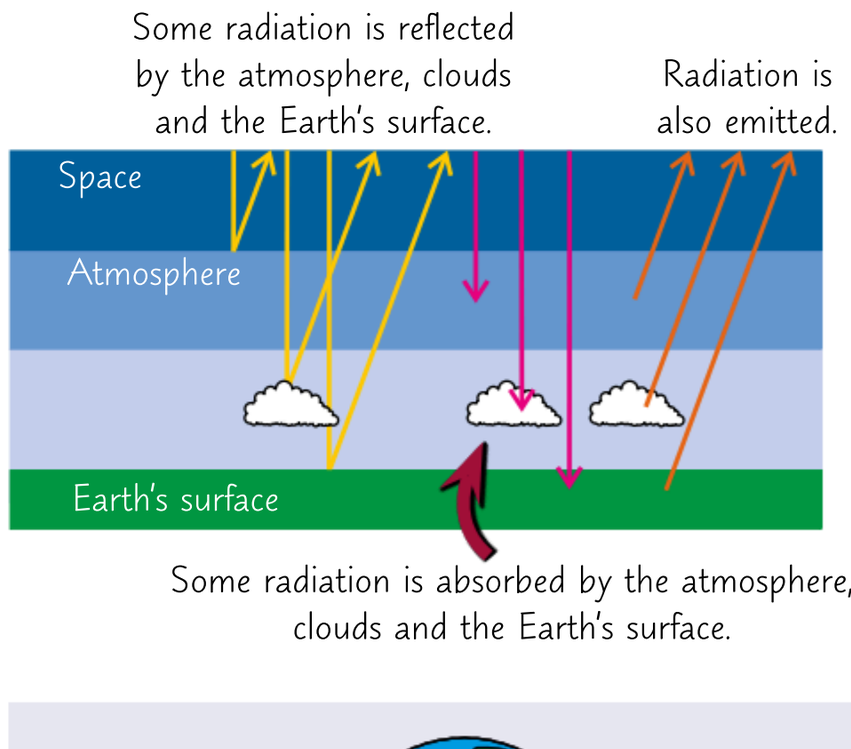


1) As the temperature of an object increases, the intensity of every emitted wavelength increases.
2) However, the intensity increases more rapidly for shorter wavelengths than longer wavelengths.
3) This causes the peak wavelength (the wavelength with the highest intensity) to decrease.

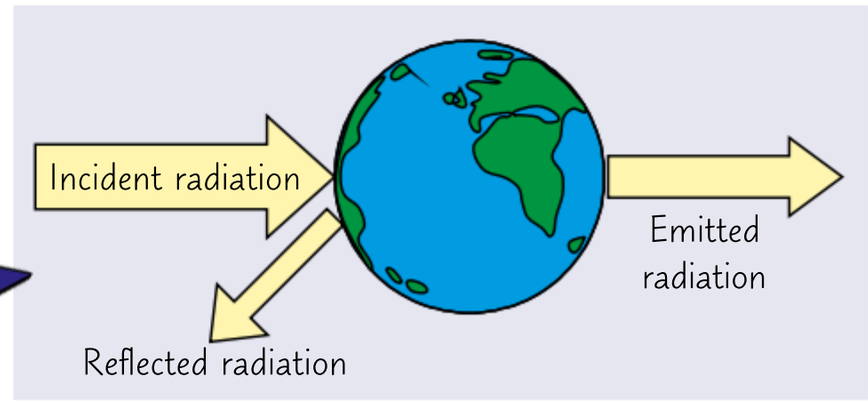
Real objects absorb and emit EM radiation less perfectly than black bodies, but their intensity and wavelength distribution follows a similar pattern, and is affected by temperature in the same way.

Radiation Affects the Earth's Temperature

1) The overall temperature of the Earth depends on the amount of radiation it reflects, absorbs and emits.
2) During the day, lots of radiation (like light) from the Sun gets absorbed. This causes an increase in local temperature.
3) At night, less radiation gets absorbed than gets emitted, causing a decrease in the local temperature.
4) Overall, the temperature of the Earth stays fairly constant. However, changes to the atmosphere can cause a change to the Earth's overall temperature.
5) If the atmosphere starts to absorb more radiation without emitting the same amount, the overall temperature will rise until absorption and emission are equal again.



You can show the Earth's balance of radiation on a diagram — thicker arrows mean more radiation. If the amounts reflected and emitted add up to less than the incident amount, then there is a net absorption of radiation and the Earth's temperature is increasing.



Absorbed radiation means a rise in temperature...

Q1 The peak wavelength of light from the Sun is about 500 nm. The peak wavelength of light from a second star is at about 850 nm. Which star is cooler? Explain your answer. [2 marks]

Q1 Video Solution

Sound Waves

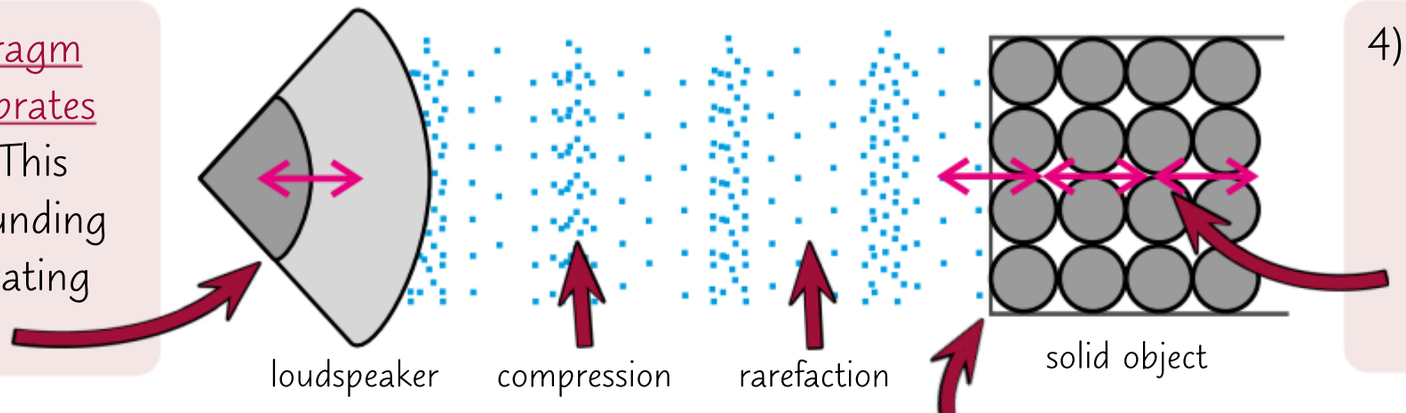
Quick Quiz

*Time to learn how we **hear** things. Don't panic — you won't be tested on each individual part of the **ear**.*

Sound Travels as a **Wave**

- Sound waves are caused by vibrating objects. These vibrations are passed through the surrounding medium as a series of compressions and rarefactions (sound is a type of longitudinal wave — page 92).
- Sound generally travels faster in solids than in liquids, and faster in liquids than in gases.
- When a sound wave travels through a solid it does so by causing the particles in the solid to vibrate.

1) The paper diaphragm in the speaker vibrates back and forth. This causes the surrounding air to vibrate, creating a sound wave.



4) These particles hit the next particles in line and so on — so they end up passing the sound wave through the object as a series of vibrations.

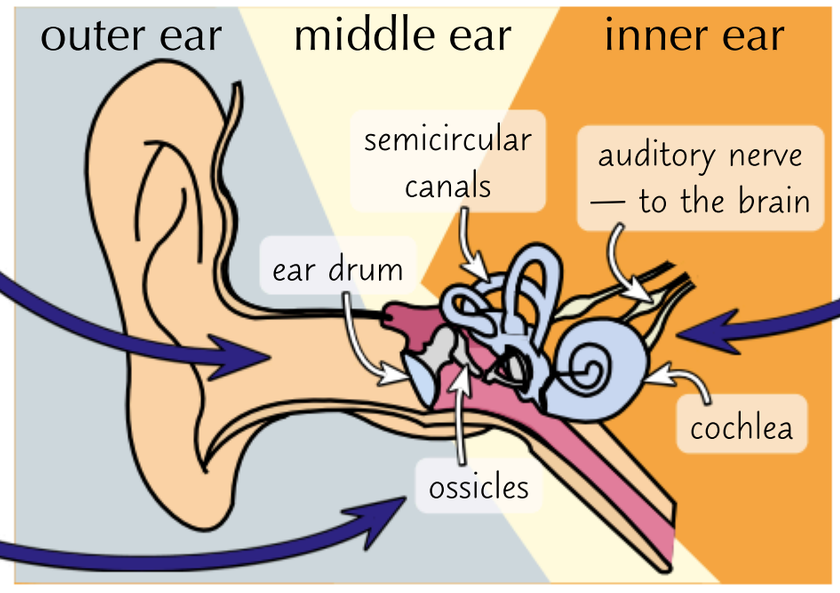
2) The sound wave travels through the air as a series of compressions and rarefactions.

3) The sound wave hits a solid object. The air particles hitting the object (the pressure — p.44) cause the particles in the solid to vibrate back and forth.

Sound can't travel in space because it's mostly a vacuum (there are no particles to move or vibrate).

You **Hear Sound** When Your **Ear Drum Vibrates**

1. Sound waves that reach your ear drum can cause it to vibrate.
2. These vibrations are passed on to tiny bones in your ear called ossicles, through the semicircular canals and to the cochlea.



3. The cochlea turns these vibrations into electrical signals which get sent to your brain and allow you to sense (i.e. hear) the sound.

Microphones work in a similar way. Sound waves cause a diaphragm to vibrate and this movement is transferred into an electrical signal.

- Different materials can convert different frequencies of sound waves into vibrations. For example, humans can hear sound in the range of 20 Hz – 20 kHz. Microphones can pick up sound waves outside of this range, but if you tried to listen to this sound, you probably wouldn't hear anything.
- Human hearing is limited by the size and shape of our ear drum, as well as the structure of all the parts within the ear that vibrate to transfer the energy from the sound wave.

Sound Waves Can **Reflect** and **Refract**

1) Sound waves will be reflected by hard, flat surfaces. Echoes are just reflected sound waves.
2) Sound waves will also refract as they enter different media. As they enter denser material, they speed up. This is because when a wave travels into a different medium, its wavelength changes but its frequency remains the same, so its speed must also change (p.92).
(However, since sound waves are always spreading out so much, the change in direction is hard to spot under normal circumstances.)

Sound waves are all about vibrations...

Vibrating objects cause sound waves, and sound waves make objects vibrate... and don't forget that your ear drum vibrating is what lets you hear stuff. Got that? Great — now have a break until your brain stops vibrating in your skull.

Quick Quiz

Ultrasound

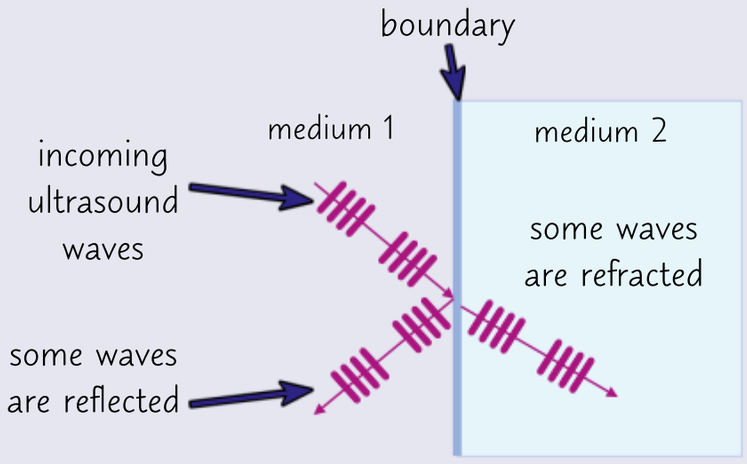
*Can you hear **that**? If not, '**that**' could be **ultrasound** — a handy wave used for **seeing hidden objects**.*

Ultrasound is Sound **Beyond** The Range of **Human Hearing**

Ultrasound waves are sound waves with frequencies above 20 000 Hz.
The way these waves reflect can help you measure things you can't see inside materials.

When an ultrasound wave passes from one medium into another, some of the wave is reflected off the boundary between the two media, and some is transmitted (and refracted).

This is partial reflection.



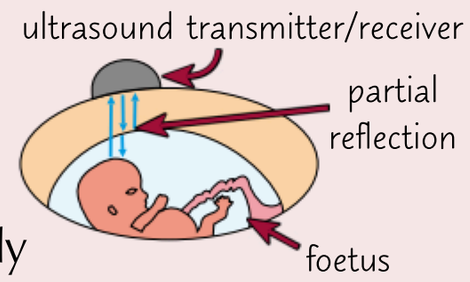
So wherever there are boundaries between substances, some of the waves get reflected back.

The time taken for the reflections to reach a detector can be used to measure how far away the boundary is.

Ultrasound is **Useful** in Lots of **Different Ways**

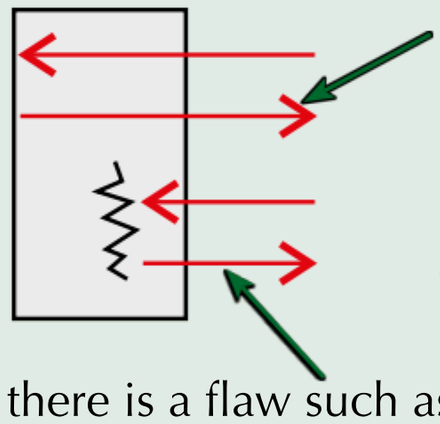
MEDICAL IMAGING

- Ultrasound waves can pass through the body, but whenever they reach a boundary between two different media (like fluid in a womb and the skin of a foetus) some of the wave is reflected back and detected.
- The exact timing and distribution of these echoes are processed by a computer to produce a video image of the body's internal structure.
- Ultrasound is a safer way to make images of inside the body than using X-rays (see p.102-103).



INDUSTRIAL IMAGING

Ultrasound can also be used to find flaws in objects such as pipes or materials such as wood or metal.



Ultrasound waves entering a material will usually be reflected by the far side of the material.

If there is a flaw such as a crack inside the object, the wave will be reflected sooner.

ECHO SOUNDING

High frequency sound waves (including ultrasound) can be used by boats and submarines to find out the depth of the water they are in, or to locate objects in deep water.

EXAMPLE

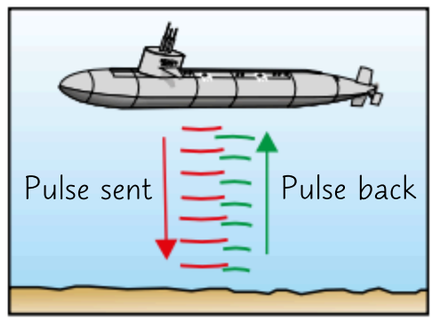
A pulse of ultrasound takes 4.5 seconds to travel from a submarine to the seabed and back again. If the speed of sound in seawater is 1520 m/s, how far away is the submarine from the seabed?

MATHS SKILLS

1) You already know the formula: distance = speed × time.
2) But the 4.5 s is for there and back, so remember to halve the distance.

$s = vt = 1520 \times 4.5 = 6840$

$6840 \div 2 = 3420$ m



Learn how ultrasound is used in medicine and industry...

It's all about measuring the time taken for ultrasound to be reflected. You can measure distances, scan bits of the body or get data about the internal structure of different objects. Useful stuff.

Q1 Calculate how long it takes for an ultrasound pulse to return to a submarine, if the speed of sound in seawater is 1520 m/s and the submarine is 2500 m above the seabed. [3 marks]

Q1 Video Solution

Seismic Waves

Quick Quiz

__Ultrasound__ waves __aren't__ the __only__ waves that can be used to __explore structures__ that we can't __see__.
You can also use __seismic waves__ produced by __earthquakes__ to investigate the Earth's __inner structure__. Fancy.

__Earthquakes__ and __Explosions__ Cause __Seismic Waves__

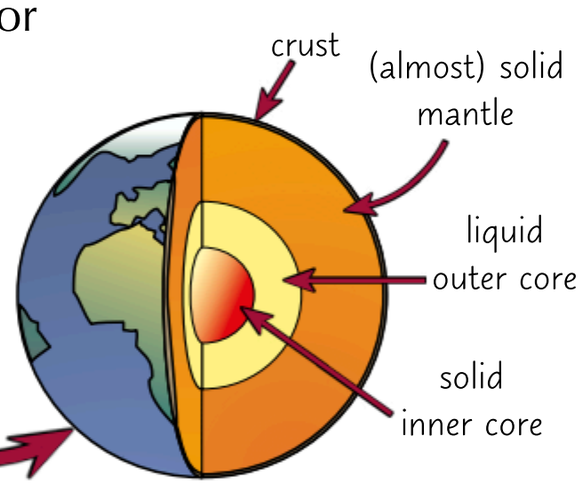
- When there's an earthquake somewhere, it produces seismic waves which travel out through the Earth.
- When seismic waves reach a boundary between different layers of material (which all have different properties, like density) inside the Earth, some waves will be absorbed and some will be refracted.
- Most of the time, if the waves are refracted they change speed gradually, resulting in a curved path. But when the properties change suddenly, the wave speed changes abruptly, and the path has a kink.

There are two types of seismic waves you need to know:

P-WAVES	S-WAVES
Longitudinal	Transverse
Can travel through solids and liquids	Can't travel through liquids
Faster than S-waves	Slower than P-waves

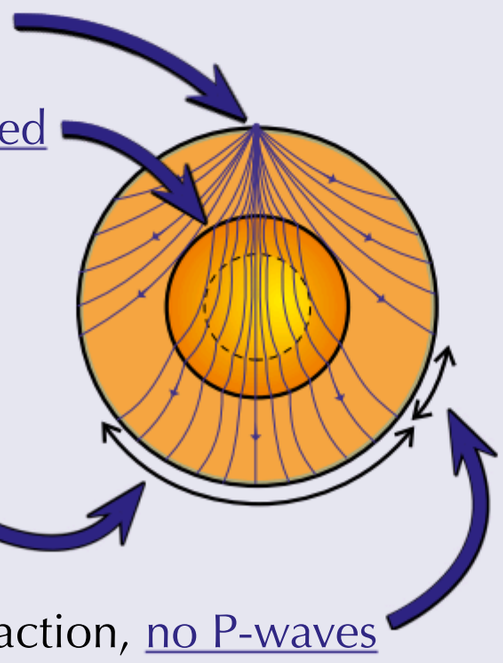
__Seismic Waves__ Can Be Used to Investigate __Earth's Internal Structure__

1) Seismic waves can be detected at the surface of the Earth using seismometers.
2) When an earthquake happens, seismologists work out the time it takes for the shock waves to reach seismometers in various locations. They also note which parts of the Earth don't receive the shock waves at all.
3) By observing how the different types of seismic waves are absorbed and refracted (see below), scientists have been able to work out where the properties of the Earth change dramatically.
4) Our current understanding of the internal structure of the Earth and the size of the Earth's core is based on these observations.



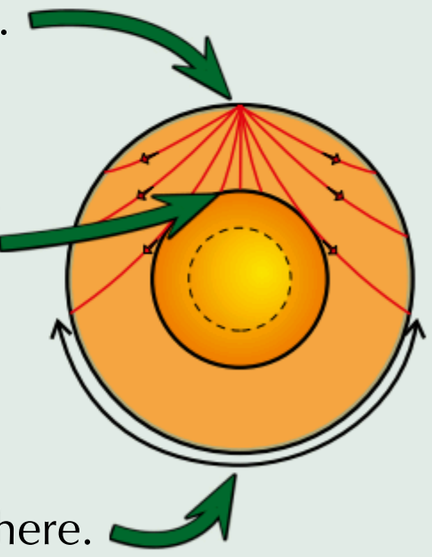
P-waves inside the Earth

1) Earthquake occurs.
2) P-waves are refracted when they enter different parts of the Earth's core.
3) P-waves that pass through the core are detected here.
4) Because of the refraction, no P-waves reach here. Measuring the size and position of these areas can tell you about the structure of the core.



S-waves inside the Earth

1) Earthquake occurs.
2) S-waves can't pass through the liquid outer core.
3) Because of this, no S-waves reach here. Measuring the size of this area can tell you about the size of the core.



Learn the differences between S-waves and P-waves...

... it might not save your life in an earthquake, but it'll save you marks in an exam. Write a couple of bullet point lists and see how many properties you can remember. Remember that S-waves can't travel through liquids, and no S-waves reach the opposite side of the Earth when there's an earthquake — so at least some of the Earth's core must be liquid.

Warm-Up & Exam Questions

This is the last set of questions in the waves topic. Phew, I hear you say. There's some tricky stuff on these pages, such as black body radiation. But plough through them carefully and you'll get there.

Warm-Up Questions

1) What specialised piece of apparatus do you fill with hot water in an experiment to investigate IR radiation emissions from different surfaces?
2) In terms of radiation absorbed and emitted, explain why local temperatures drop at night.
3) What sort of wave is a sound wave — transverse or longitudinal?
4) What does it mean if a wave experiences partial reflection at a boundary between two media?
5) Name a natural event that produces seismic waves.
6) Which type of seismic wave is longitudinal?
7) Why are S-waves not detected on the opposite side of Earth from which they are generated?

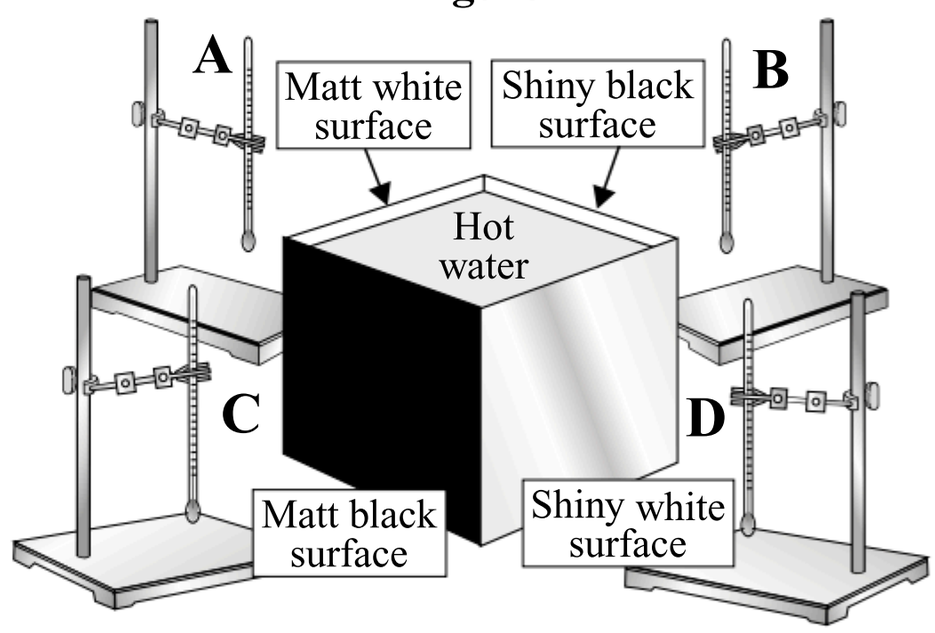
Exam Questions

PRACTICAL

GRADE 6-7

1 A student is investigating the infrared radiation emitted by different surfaces using a Leslie Cube, as shown in **Figure 1**. The student records how long it takes the temperature on each thermometer to increase by 5 °C.

Figure 1



1.1 Suggest **one** thing the student should do to make the experiment a fair test.

[1 mark]

Of the faces of the Leslie Cube, the face at thermometer C behaves most like a perfect black body, and face at thermometer D behaves least like a perfect black body.

1.2 In which of the following orders (fastest to slowest) would you expect the thermometers to increase by 5 °C?
Tick **one** box.

☐ A, B, C, D ☐ C, A, B, D ☐ D, C, B, A ☐ B, A, D, C

[1 mark]

1.3 The water used in the experiment was initially at boiling point, 100 °C.
The experiment is repeated using water at 60 °C.
Predict how this would affect the results. Explain your prediction.

[2 marks]

1.4 Another student suggests that using digital thermometers connected to data loggers to measure the temperatures would improve the investigation.
The digital thermometers measure temperature in °C to two decimal places.
Give **two** reasons why this student is correct.

[2 marks]

Exam Questions

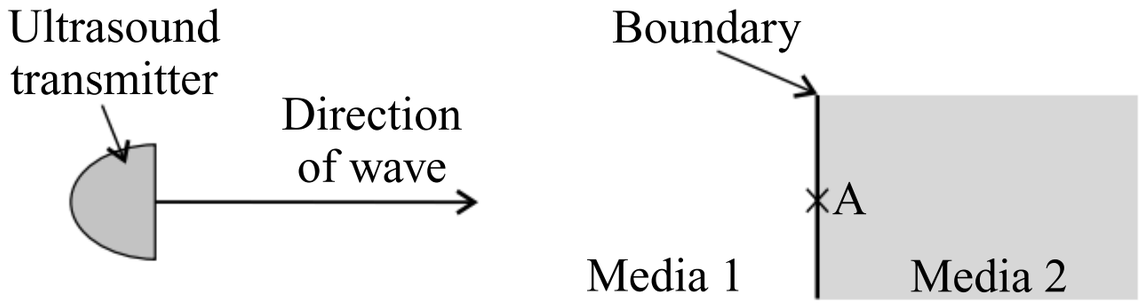
2 Ultrasound waves have frequencies above the normal range of human hearing. GRADE 6-7 GRADE

2.1 What is the minimum frequency of an ultrasound wave?

[1 mark]

2.2 An ultrasound transmitter uses mechanical vibrations to produce ultrasound waves.
On **Figure 2**, draw on the direction of the mechanical vibrations.

Figure 2



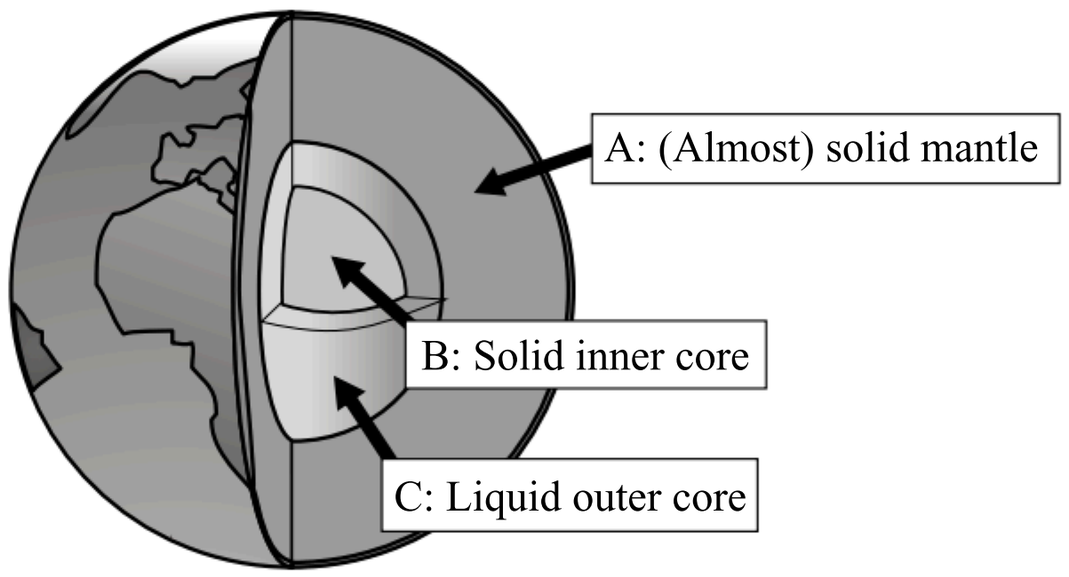
[1 mark]

2.3 What will happen to the ultrasound wave when it reaches point A?

[1 mark]

3 **Figure 3** shows the internal structure of the Earth. GRADE 6-7 GRADE

Figure 3



3.1 State which part(s) of the Earth, A, B or C, S-waves would be found in after an earthquake.
Explain your answer.

[3 marks]

3.2 Both P- and S-waves generally follow curved paths through the Earth.
At some points their paths abruptly change direction.
Explain what these observations tell us about the Earth's inner structure.

[4 marks]

4 Betelgeuse and Bellatrix are both stars in the constellation of Orion.
Betelgeuse is reddish while Bellatrix is bluish.
Red light has a wavelength range of approximately 630-700 nm.
Blue light has a wavelength range of approximately 450-490 nm. GRADE 7-9 GRADE

4.1 Which star you would expect to be hotter? Explain your answer.

[3 marks]

4.2 Betelgeuse is nearing the end of its life and will explode one day.
State whether people on Earth would expect to hear the explosion.
Explain your answer.

[2 marks]

Revision Summary Test for Topic 6b

Sadly, it's time to wave goodbye to that topic... At least there are a few great ways to see it off:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

Lenses (p.106-108) ☑

1) Describe the differences between convex and concave lenses.
2) What do these lens terms mean? a) principal focus b) axis c) focal length
3) Give the three rules for refraction of an incident ray in: a) a convex lens, b) a concave lens.
4) What is meant by a 'real image' and a 'virtual image'?
5) a) Draw a ray diagram for a convex lens. b) Now do one for — you guessed it — a concave lens.
6) State the size, nature and position of the image formed by a convex lens when the distance from the object to the lens is: a) greater than 2F, b) exactly 2F, c) between F and 2F, d) less than F.
7) Describe the image formed by a concave lens. How does this change as the distance between the object and the lens is increased?
8) What formula could you use to calculate the magnification of an object by a lens? What does it mean if the magnification of an object is greater than 1?

Visible Light and Black Body Radiation (p.109-112) ☑

9) What is visible light?
10) Name the primary colours of light.
11) Explain how different wavelengths of light behave when they hit an opaque object that is: a) black, b) white, c) a primary colour, d) a non-primary colour.
12) Describe how colour filters work.
13) Describe how the temperature of an object changes if it is: a) emitting less IR than it's absorbing, b) emitting more IR than it's absorbing, c) emitting the same amount of IR as it's absorbing.
14) What is a Leslie cube? How could you use one to investigate IR emitted by different surfaces? What results would you expect to get?
15) What is a "perfect black body"?
16) How does the temperature of a black body affect the intensity and wavelengths of radiation that it emits?
17) Explain how absorption, reflection and emission of radiation affect the Earth's temperature.

Sound Waves and Seismic Waves (p.113-115) ☑

18) How do sound waves travel through a medium? How does the speed of sound compare between solids, liquids and gases?
19) What is the typical range of human hearing? How does the structure of the ear determine that?
20) Explain how the wavelength, frequency and velocity of a sound wave change when it is refracted.
21) Explain how ultrasound can be used to detect the locations of boundaries in materials.
22) Describe these uses of ultrasound: a) medical imaging b) industrial imaging c) echo sounding
23) What are seismic waves? (And don't just say they're measured by seismologists with seismometers...)
24) What are the differences between S-waves and P-waves?
25) Explain how seismic waves can be used to explore the structure of the Earth's core.

Permanent and Induced Magnets

Quick Quiz

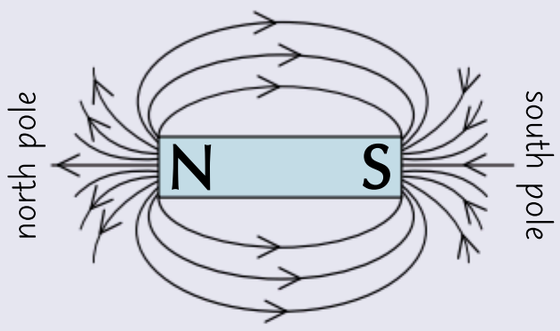
*I think magnetism is an **attractive** subject, but don't get **repelled** by the exam — **revise**.*

Magnets Produce **Magnetic Fields**

- All magnets have two poles — north (or north seeking) and south (or south seeking).
- All magnets produce a magnetic field — a region where other magnets or magnetic materials (e.g. iron, steel, nickel and cobalt) experience a magnetic force.

The magnetic force is a non-contact force — similar to the force on charges in an electric field, like you saw on page 38.

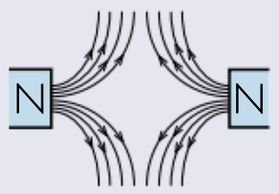
- You can show a magnetic field by drawing magnetic field lines:



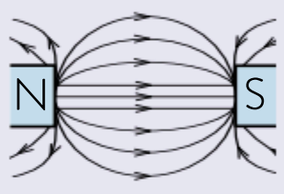
1) The lines always go from north to south and they show which way a force would act on a north pole if it was put at that point in the field.
2) The closer together the lines are, the stronger the magnetic field.
3) The further away from a magnet you get, the weaker the field is.
4) The magnetic field is strongest at the poles of a magnet. This means that the magnetic forces are also strongest at the poles.

- If two poles are put near each other, they will each exert a force on each other.

Two like poles will repel each other.

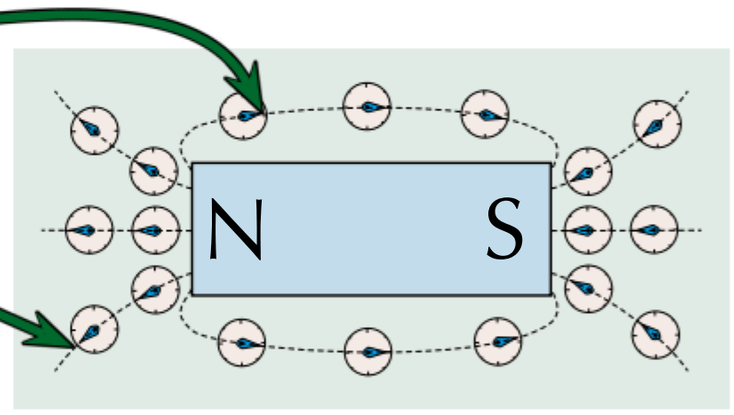


Two unlike poles will attract each other.



Compasses Show the **Directions** of Magnetic Fields

1) Inside a compass is a tiny bar magnet. The north pole of this magnet is attracted to the south pole and repelled by the north pole of any nearby magnet. So the compass points in the direction of the field it is in.
2) You can move a compass around a magnet and mark the direction it points in on some paper to build up a picture of what the magnetic field looks like.



3) When they're not near a magnet, compasses always point north. This is because the Earth generates its own magnetic field, which the compass lines up with — the inside (core) of the Earth is magnetic.

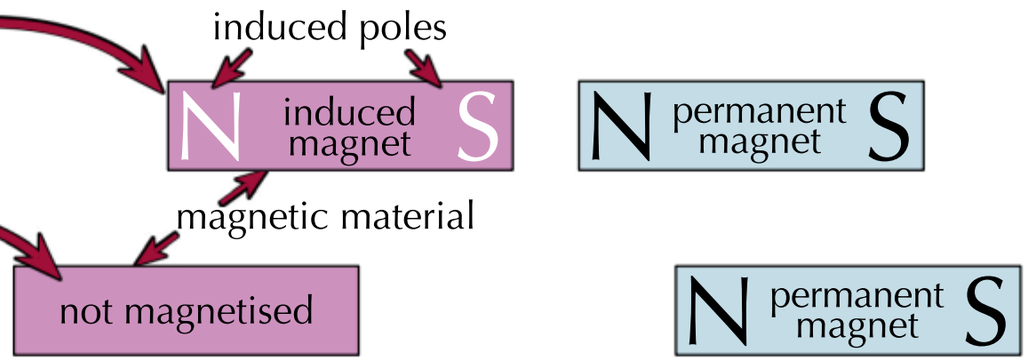
Magnets Can be **Permanent** or **Induced**

There are two types of magnet.

PERMANENT	Produce their own magnetic field.
INDUCED	Magnetic materials that turn into a magnet when they're put into a magnetic field.

For induced magnets:

- A magnetic material becomes magnetised when it is brought near a permanent magnet. It has its own poles and magnetic field.
- When you take away the magnetic field, induced magnets quickly lose their magnetism (or most of it) and stop producing a magnetic field.
- The force between permanent and induced magnets is always attractive.



A bar magnet's magnetic field lines go from north to south...

...no matter which direction the magnet is pointing. You can see the shape of a magnetic field using compasses or iron filings. Iron filings will give you a pretty pattern — but they won't show you the direction of the magnetic field. That's where compasses really shine. They'll always point to the south seeking pole. They're also a lot easier to clear up.

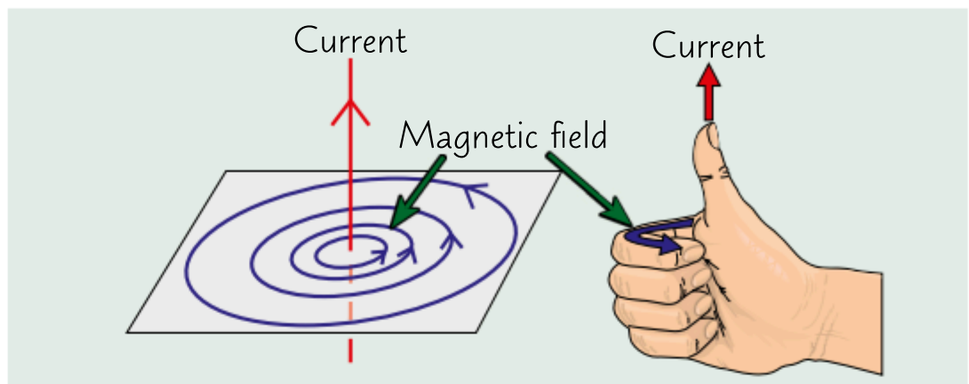
Quick Quiz

Electromagnetism

*On this page you'll see that a **magnetic field** is also found around a **wire** that has a **current** passing through it.*

A **Moving Charge** Creates a Magnetic Field

1) When a current flows through a wire, a magnetic field is created around the wire.
2) The field is made up of concentric circles perpendicular to the wire, with the wire in the centre.
3) You can see this by placing a compass near a wire that is carrying a current. As you move the compass, it will trace the direction of the magnetic field.
4) Changing the direction of the current changes the direction of the magnetic field — use the right-hand thumb rule to work out which way it goes.
5) The strength of the magnetic field produced changes with the current and the distance from the wire. The larger the current through the wire, or the closer to the wire you are, the stronger the field is.

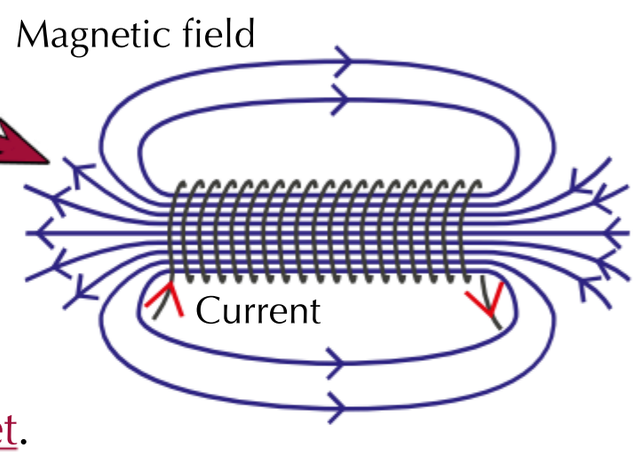


The Right-Hand Thumb Rule

Using your right hand, point your thumb in the direction of current and curl your fingers. The direction of your fingers is the direction of the field.

A **Solenoid** is a Coil of Wire

- You can increase the strength of the magnetic field that a wire produces by wrapping the wire into a coil called a solenoid.
- The field lines around each loop of wire in the solenoid line up with each other. They point in the same direction and are very close together, showing there is a strong magnetic field.
- The magnetic field inside a solenoid is strong and uniform.
- Outside the coil, the magnetic field is just like the one round a bar magnet.



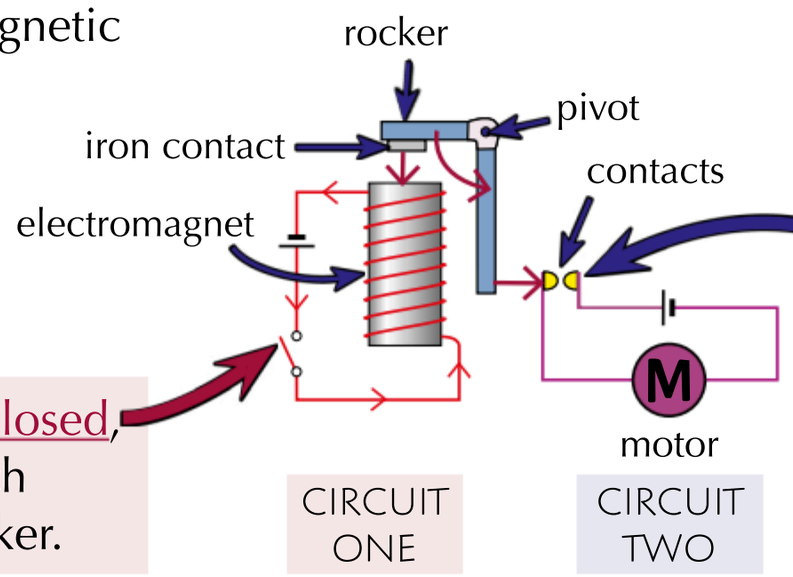
The field strength of a solenoid can be increased by putting a block of iron in the centre of the coil. This iron core becomes an induced magnet whenever current is flowing.

If you stop the current, the magnetic field disappears. A solenoid with an iron core (a magnet whose magnetic field can be turned on and off) is called an ELECTROMAGNET.

Electromagnets Have **Lots of Uses**

- Electromagnets are usually used because they're quick to turn on and off or because they can create a varying force (like in loudspeakers, p.123).
- Electromagnets are used in some cranes to attract, pick up, and drop things made from magnetic materials, e.g. in scrap yards.
- Electromagnets can also be used as switches (e.g. in the electric starters of motors) to link two separate circuits together, like this:

Video



1) When the switch in circuit one is closed, it turns on the electromagnet, which attracts the iron contact on the rocker.

2) The rocker pivots and closes the contacts, completing circuit two, and turning on the motor.

Just point your thumb in the direction of the current...

Electromagnets aren't all the same strength — the strength depends on stuff like the number of turns on the solenoid and the size of the current going through the wire. Different strength electromagnets can be used for different purposes.

Warm-Up & Exam Questions

It's time for another page of questions to check your knowledge retention. If you can do the warm-up questions without breaking into a sweat, then see how you get on with the exam questions below.

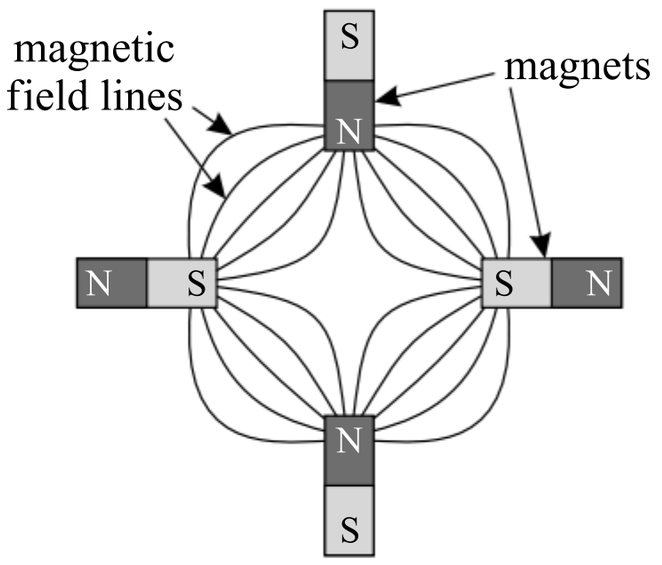
Warm-Up Questions

1) Draw a diagram to show the magnetic field around a single bar magnet.
2) Describe the magnetic field around a current-carrying wire.
3) Give one use of an electromagnet.

Exam Questions

1 A student draws the magnetic field lines between four bar magnets, as shown in **Figure 1**. GRADE 6-7

Figure 1



1.1 Describe an experiment that the student could have done to show this magnetic field pattern.

[2 marks]

The student arranges two of the magnets as shown in **Figure 2**.

Figure 2

1.2 Describe the magnetic field lines in the shaded region between the dotted lines.

[1 mark]

1.3 State whether there will be a force of attraction, repulsion, or no force between the two magnets. Explain your answer.

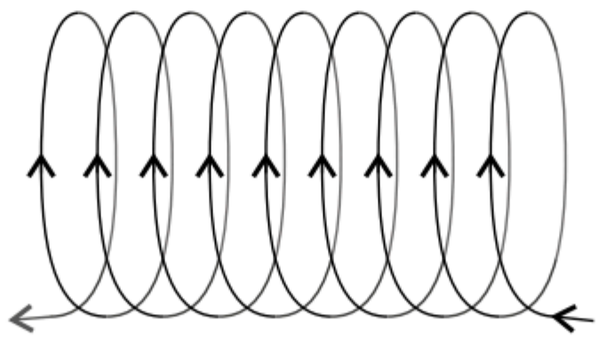
[2 marks]

2 Arnold is making an electromagnet using a current-carrying solenoid and a core. GRADE 6-7

2.1 Complete **Figure 3** of the solenoid to show the magnetic field inside and around it.

[2 marks]

Figure 3

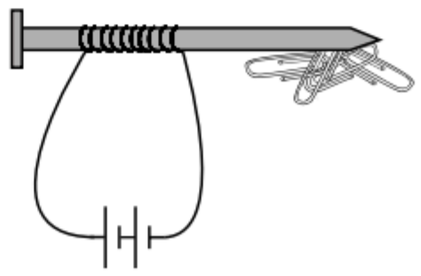


2.2 Suggest a material that would be suitable for the core.

[1 mark]

Arnold uses his electromagnet to pick up some paper clips, as shown in **Figure 4**.

Figure 4



2.3 State what happens when the current is turned off, and explain why this happens.

[2 marks]

The Motor Effect

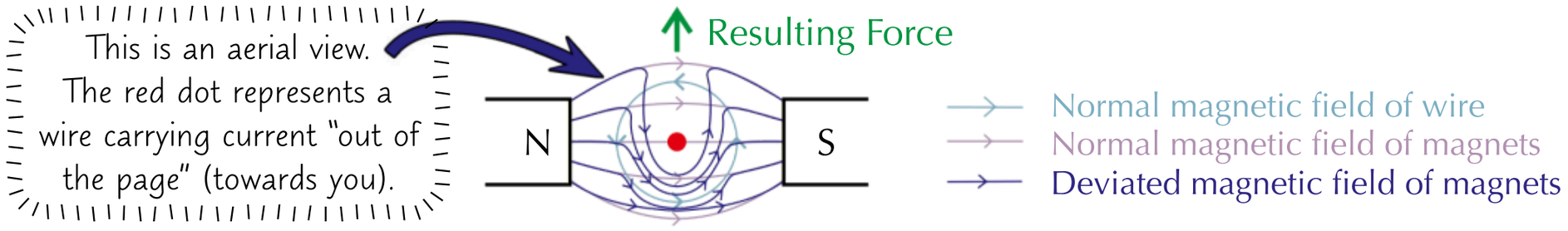
Quick Quiz

*The **motor effect** can happen when you put a **current-carrying wire** in a **magnetic field**. It's really useful in stuff like... well... electric motors. If you want to know exactly what it is, you'll have to **keep reading**.*

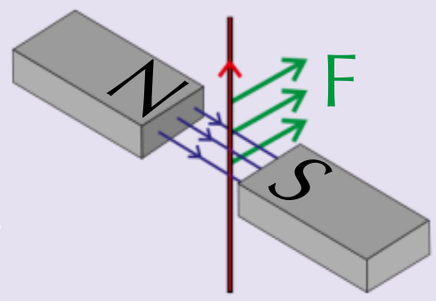
A **Current** in a Magnetic Field Experiences a **Force**

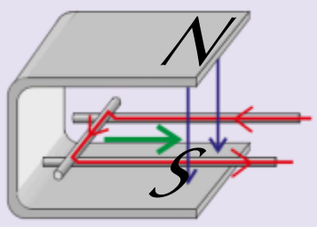
Video

When a wire (or any other conductor) carrying a current is put between magnetic poles, the magnetic field around the wire interacts with the magnetic field between the poles, causing them to exert a force on each other. This is called the motor effect and can cause the wire to move.



- To experience the full force, the wire has to be at 90° to the magnetic field. If the wire runs parallel to the magnetic field, it won't experience any force at all. At angles in between, it'll feel some force.
- The force always acts at right angles to the magnetic field of the magnets and the direction of the current in the wire.





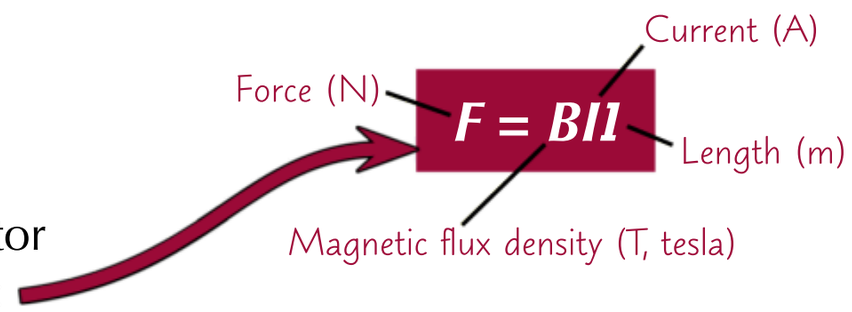
- A good way of showing the direction of the force is to apply a current to a set of rails inside a horseshoe magnet (shown to the left). A bar is placed on the rails, which completes the circuit. This generates a force that rolls the bar along the rails.

You Can Find the **Size** of the **Force**...

The force acting on a conductor in a magnetic field depends on three things:

1) The magnetic flux density — how many field (flux) lines there are in a region. This shows the strength of the magnetic field (p.119).
2) The size of the current through the conductor.
3) The length of the conductor that's in the magnetic field.

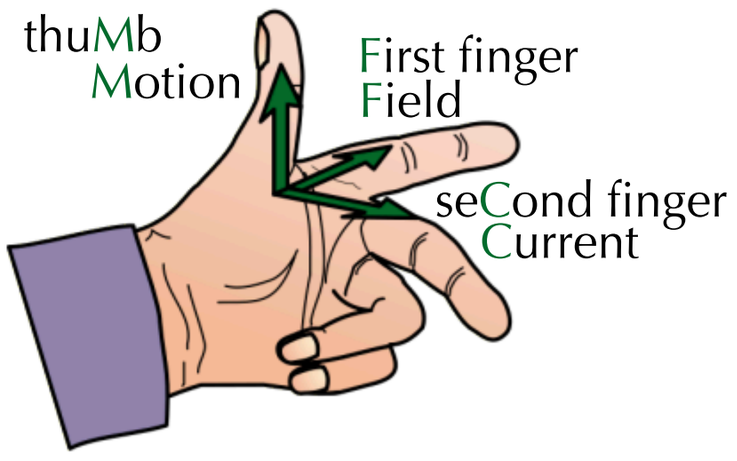
When the current is at 90° to the magnetic field the conductor is in, the force acting on it can be found using this equation:



... and **Which Way** it's Acting

You can find the direction of the force with Fleming's left-hand rule.

	POINTS IN THE DIRECTION OF...
FIRST FINGER	...the magnetic **F**ield between the poles.
SE**C**OND FINGER	...the direction of **C**urrent through the wire.
THU**M**B	...the force acting on the wire (**M**otion).



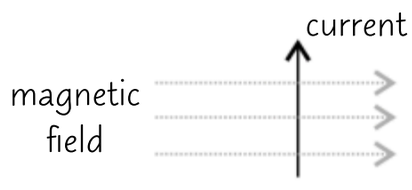
Fleming's left-hand rule shows that if either the current or the magnetic field is reversed, then the direction of the force will also be reversed. This can be used for all sorts of things — like motors on the next page.

Fleming's left-hand rule can really come in handy...

Q1 Video Solution

Use the left-hand rule in the exam. You might look a bit silly, but it makes getting the marks much easier.

Q1 A section of a current-carrying wire is in a magnetic field, as shown in the diagram. The wire is at 90° to the magnetic field. Find the direction of the force acting on the wire.



[1 mark]

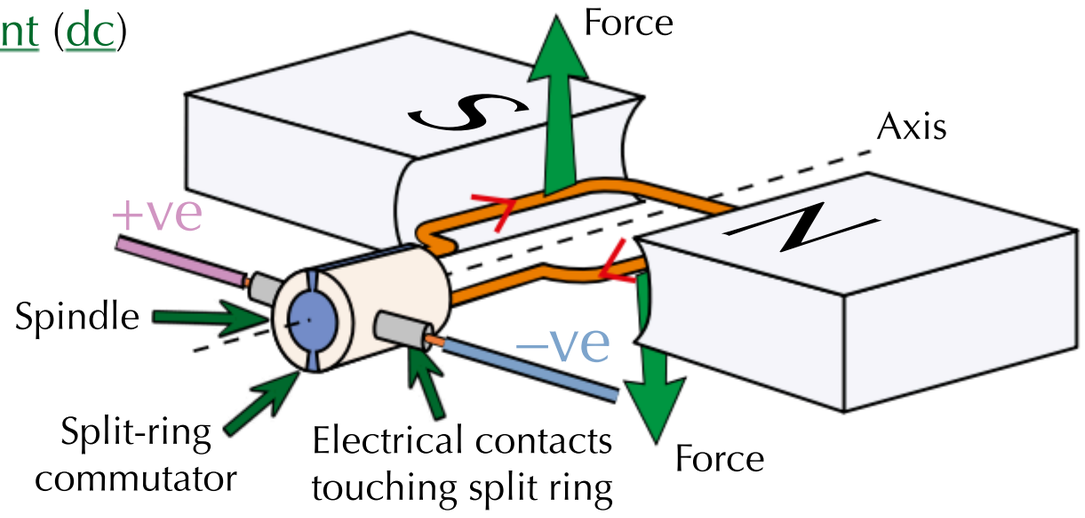
Electric Motors and Loudspeakers

Quick Quiz

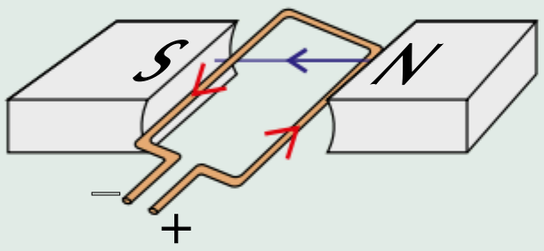
Electric motors *use the* ***motor effect*** *to get them* ***moving****, and* ***loudspeakers*** *use the effect to make* ***sound****.*
This is one of the favourite exam topics of all time. Read it. Understand it. Learn it. Lecture over.

A Current-Carrying **Coil** of Wire **Rotates** in a Magnetic Field

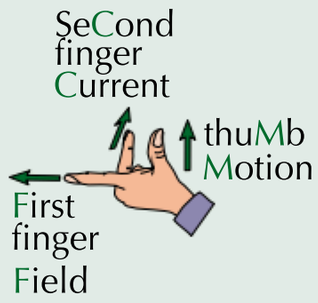
- The diagram on the right shows a basic direct current (dc) motor. Forces act on the two side arms of a coil of wire that's carrying a current.
- Because the coil is on a spindle and the forces act one up and one down, it rotates.
- The split-ring commutator has gaps in it that swap the electrical contacts every half turn to keep the motor rotating in the same direction.
- The direction of the motor can be reversed either by reversing the current of the dc supply or swapping the magnetic poles over (reversing the field).
- You can use Fleming's left-hand rule to work out which way the coil will turn:



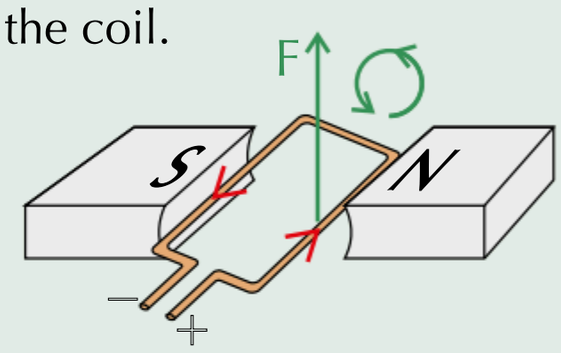
1) Draw in the direction of the magnetic field lines and the current. Current always flows from +ve to –ve.



2) Use Fleming's LHR on one side of the coil (here we've used the right-hand side).



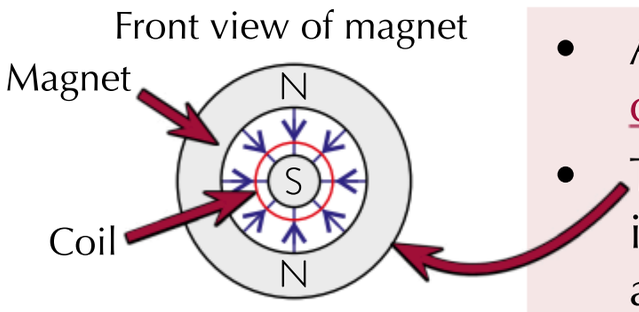
3) Draw in the direction of the force (motion) for this side of the coil.



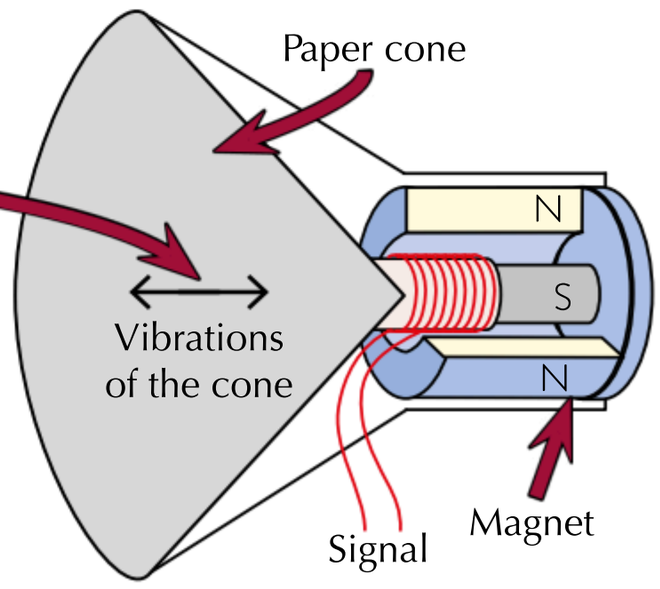
So — the coil is turning anticlockwise.

Loudspeakers Work Because of the **Motor Effect**

Loudspeakers and headphones (which are just tiny loudspeakers) both use electromagnets:



- An alternating current (ac) is sent through a coil of wire attached to a paper cone.
- The coil surrounds one pole of a permanent magnet, and itself is surrounded by the other pole, so the current causes a force on the coil (which causes the cone to move).
- When the current reverses, the force acts on the coil in the opposite direction, causing the cone that's attached to it to do the same.
- So variations in the current make the cone vibrate, which makes the air around the cone vibrate. This creates the variations in pressure that cause a sound wave (p.113).
- The frequency of the sound wave is the same as the frequency of the ac, so by controlling the frequency of the ac you can alter the pitch of the sound wave.



The motor effect has a lot of important applications...

Electric motors are important components in a lot of everyday items. Food mixers, washing machines, and anything that has a fan (hair dryers, laptops, vacuum cleaners, etc.) use electric motors to keep things turning.

Warm-Up & Exam Questions

Time to test your knowledge — as usual, check you can do the basics, then get stuck into some lovely exam questions. Don't forget to go back and check up on any niggling bits you can't do.

Warm-Up Questions

1) What is the motor effect?
2) In Fleming's left-hand rule, what's represented by the first finger, the second finger and the thumb?
3) Give two changes that can be made to make a dc motor run in reverse.

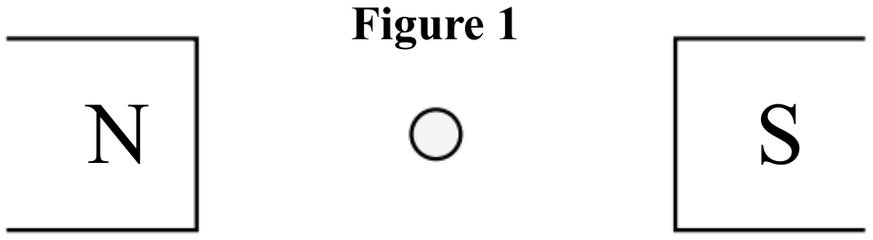
Exam Questions

For question 2, you can use the formula sheet on page 216.

1 **Figure 1** shows an aerial view of a current-carrying wire in a magnetic field. The circle represents the wire carrying current out of the page, towards you.

GRADE 4-6 GRADE

Figure 1



1.1 On **Figure 1**, draw an arrow to show the direction of the force acting on the current-carrying wire.

[1 mark]

1.2 Describe what would happen to the force acting on the current-carrying wire if the direction of the current was reversed.

[1 mark]

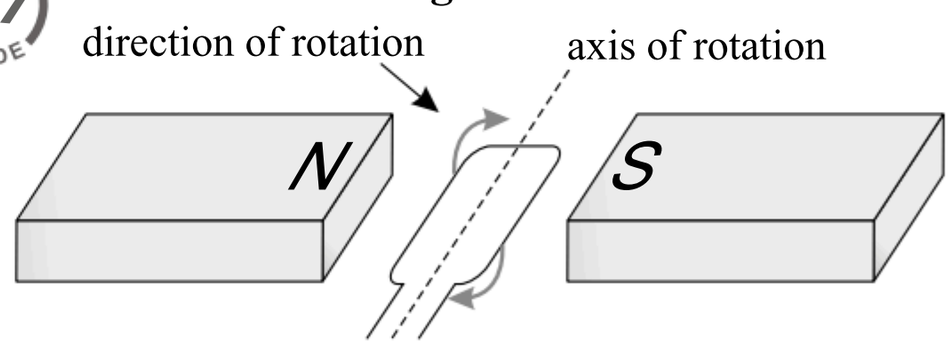
1.3 Describe how the size of the force acting on the wire would change if the wire was at 30° to the magnetic field.

[1 mark]

2 A student is building a simple dc motor. She starts by putting a loop of current-carrying wire that is free to rotate about an axis in a magnetic field, as shown in **Figure 2**. The magnetic field between the poles has a magnetic flux density of 0.2 T.

GRADE 6-7 GRADE

Figure 2



2.1 Add an arrow to **Figure 2** to show the direction of the current in the wire.

[1 mark]

2.2 The starting position of the loop is shown in **Figure 2**. The current in the wire is 15 A, and the length of each of the two longest sides is 10 cm. Calculate the force on one of the two sides of the coil as it starts to turn.

[2 marks]

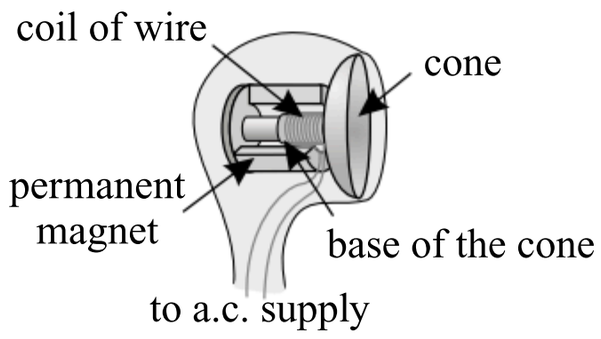
2.3 The motor will stop rotating in the same direction after 90° of rotation from its start position. Suggest and explain how the student could get the motor to keep rotating in the same direction.

[2 marks]

3 **Figure 3** shows the parts inside an earphone. Sound waves are caused by mechanical vibrations. Explain how the earphone uses an ac supply to produce sound waves.

GRADE 7-9 GRADE

Figure 3



[4 marks]

The Generator Effect

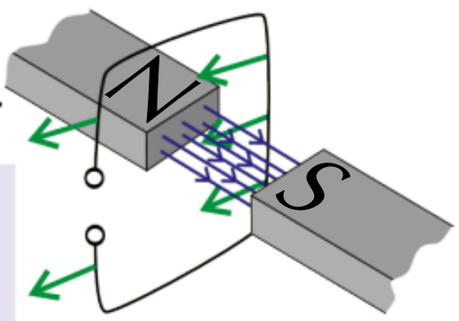
Quick Quiz

*Electricity is generated using the **generator effect**, also known as **electromagnetic induction**.*

Cutting Field Lines **Induces** a Potential Difference

THE GENERATOR EFFECT: The induction of a potential difference (and current if there's a complete circuit) in a wire which is moving relative to a magnetic field, or experiencing a change in magnetic field.

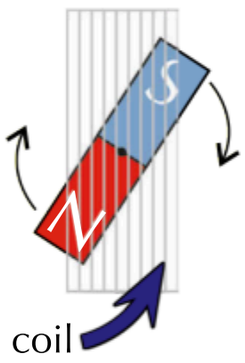
The effect can be created by moving a magnet in a coil of wire OR moving a conductor (wire) in a magnetic field ("cutting" magnetic field lines).



- Shifting a magnet sideways in a coil creates a little "blip" of current if the coil is part of a complete circuit.

- If you move the magnet in the opposite direction, the potential difference (pd) / current will be reversed. Likewise if the polarity of the magnet is reversed, that will also reverse the pd / current.
- If you continuously move the magnet (or the coil) backwards and forwards, you produce a pd / current that keeps swapping direction — an alternating current.

You can create the same effect by rotating a coil inside a magnetic field (this is how generators work — see next page). You can also do this by turning a magnet end-to-end in a coil:



1) As you turn the magnet, the magnetic field through the coil changes. This induces a potential difference, which can make a current flow in the wire.
2) After the magnet has gone through half a turn, the direction of the magnetic field through the coil will have reversed. As the magnet turns, the potential difference reverses, so the current flows in the opposite direction.
3) If you keep turning the magnet in the same direction — always clockwise, say — then the potential difference will keep on reversing every half turn and you'll get an alternating current.

You Can Change the **Size** of the **Induced Potential Difference**

Induced potential difference (and induced current) can be increased by either:

1) Increasing the speed of the movement — cutting more magnetic field lines in a given time.
2) Increasing the strength of the magnetic field (so there are more field lines available to be cut).

Induced Current **Opposes** the Change that Made It

- As you saw on page 120, when a current flows through a wire, a magnetic field is created around it. (That's a second magnetic field — different to the one whose field lines were being cut in the first place.)
- The magnetic field created by an induced current always acts against the change that made it (whether that's the movement of a wire or a change in the field it's in). Basically, it's trying to return things to the way they were.

The generator effect — works whether the coil or the field is moving...

Whether it's the magnet or the coil that's moving, the generator effect will work as long as field lines are being 'cut'. Remember, the current that's induced in the wire will always 'oppose' the change that generated it.

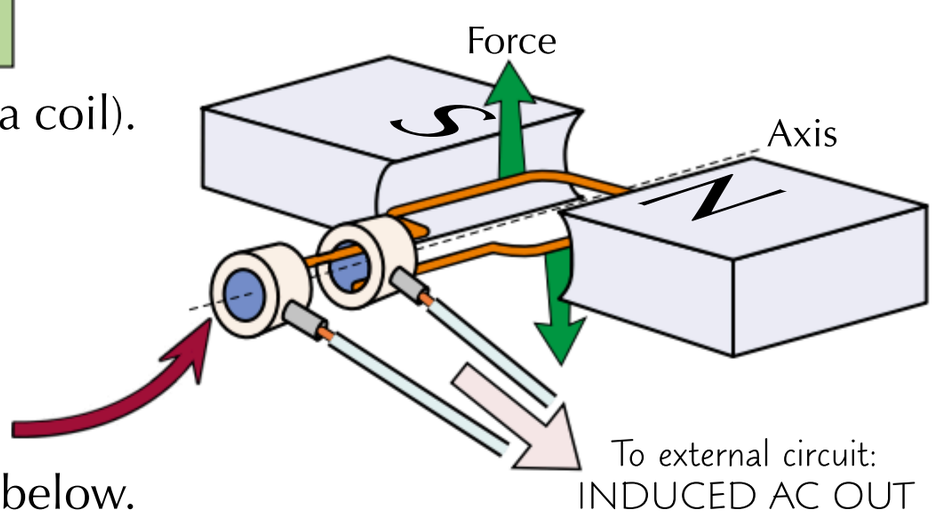
Generators and Microphones

Quick Quiz

Generators *make use of the* ***generator effect*** *to induce a current. Whether this current is* ***alternating*** *or* ***direct*** *depends on how the coil is connected to the circuit. Don't get the types of connection* ***mixed up***.

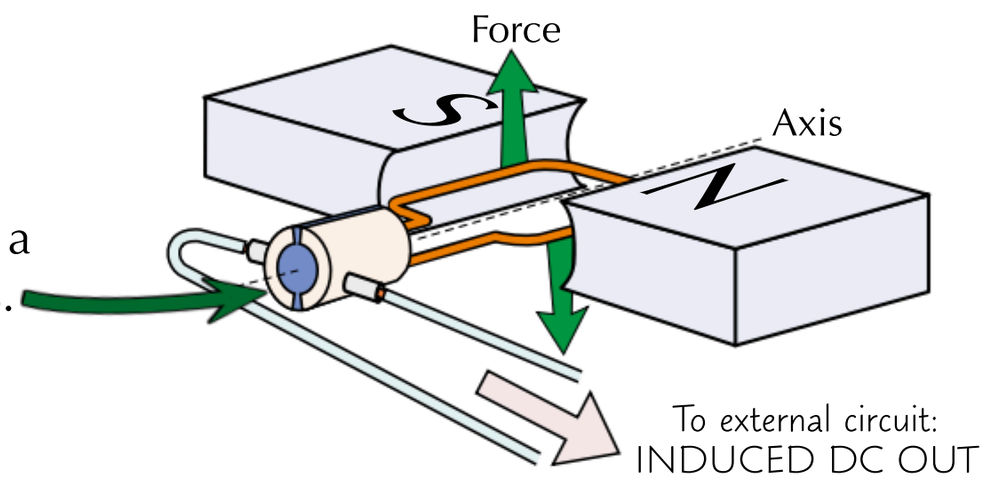
Alternators Generate **Alternating** Current

- Alternators rotate a coil in a magnetic field (or a magnet in a coil).
- As the coil (or magnet) spins, a current is induced in the coil. This current changes direction every half turn.
- Alternators have slip rings that rotate with the coil and brushes (electrical contacts that touch the rings but aren't attached to them). The contacts don't swap every half turn.
- This means they produce an alternating pd — more on this below.



Dynamos Generate **Direct** Current

- Dynamos work in the same way as alternators, apart from one important difference — they have a split-ring commutator (p.123) instead of slip rings.
- This swaps the connection every half turn to keep the current flowing in the same direction.

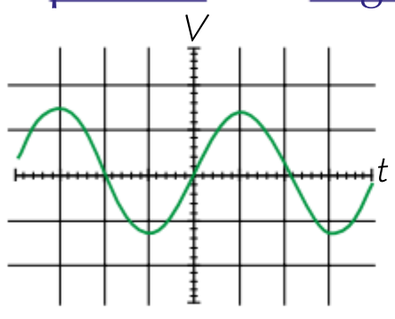


You Can Use an **Oscilloscope** To See the Generated pd

- Oscilloscopes show how the potential difference generated in the coil changes over time.
- The height of the line at a given point is the generated potential difference at that time.

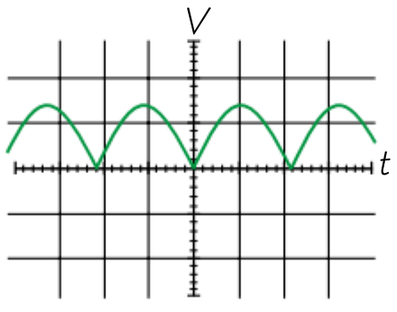
AC TRACE

For an alternator, the line crosses the horizontal axis, switching between positive and negative pd.



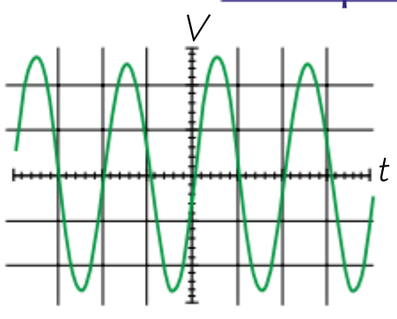
DC TRACE

For a dynamo, the line never crosses the axis so it's direct current.



INCREASING REVOLUTIONS

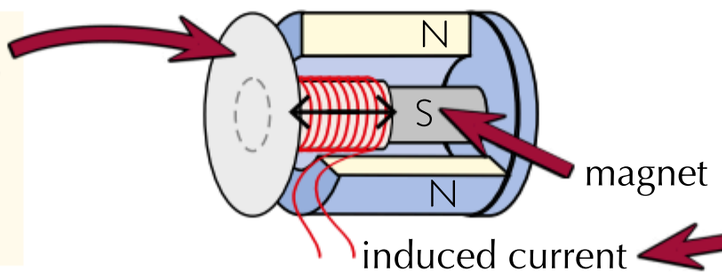
Increasing the frequency of revolutions increases the overall pd. It also creates more peaks too.



Microphones Generate Current From **Sound Waves**

Microphones are basically loudspeakers (p.123) in reverse.

A flexible diaphragm is attached to a coil of wire, wrapped around a magnet.



Sound waves hitting the diaphragm cause it to vibrate, moving the coil backwards and forwards in the magnetic field. This induces an ac current.

The induced current depends on the sound wave's properties, e.g. the frequency of the ac matches the frequency (pitch) of the sound and louder sounds produce a bigger current.

Loudspeakers and microphones have a lot of similarities...

...one is basically the other in reverse. Remember, loudspeakers (p.123) use an alternating current to create sound waves with the motor effect, while microphones use the generator effect to turn sound waves into alternating current.

Transformers

Quick Quiz

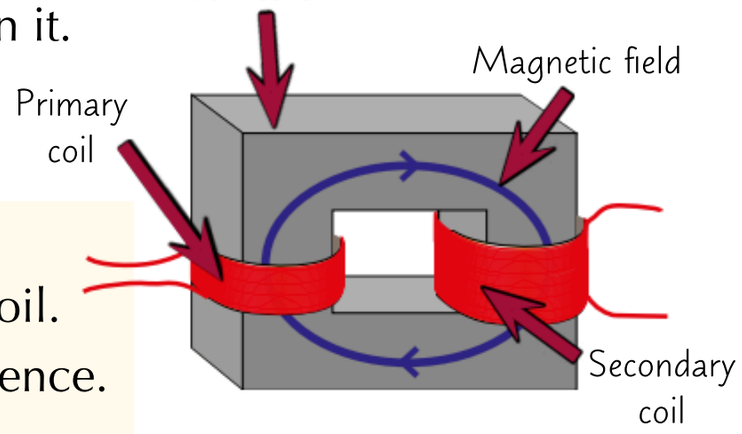
Transformers *change the size of a potential difference (pd). They will* ***only*** *work with* ***alternating current****.*

Transformers Change the pd — but Only for **Alternating** Current

- All transformers have two separate coils of wire, the primary and the secondary, joined with an iron core.
- When an alternating pd is applied across the primary coil, the iron core magnetises and demagnetises quickly. This changing magnetic field induces an alternating pd in the secondary coil (p.125).
- If the second coil is part of a complete circuit, a current is induced in it.
- The number of turns on each coil affects the pd induced in the secondary coil. There are two kinds of transformers:

1) STEP-UP TRANSFORMERS — increase the potential difference. They have more turns on the secondary coil than the primary coil.
2) STEP-DOWN TRANSFORMERS — decrease the potential difference. They have more turns on the primary coil than the secondary.

Iron is used because it's easily magnetised.



There are **Two** Really Useful **Equations** for Transformers

1) The ratio between the primary and secondary potential differences in a transformer is the same as the ratio between the number of turns on the primary and secondary coils.
2) As long as you know the pd across the primary coil and the number of turns on each coil, you can calculate the pd across the secondary coil from a transformer using the transformer equation:

Potential difference across primary coil (V) — V_p

Potential difference across secondary coil (V) — V_s

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

Number of turns on primary coil — n_p

Number of turns on secondary coil — n_s

So for a step-up transformer, $V_s > V_p$ and for a step-down transformer, $V_s < V_p$.

In an exam, you might have to rearrange this equation to find an answer (see page 144).

3) Transformers are almost 100% efficient, so the input power is equal to the output power. Since $P = VI$, this gives the equation:

Pd across primary coil (V) — V_p

Current through primary coil (A) — I_p

$$V_p I_p = V_s I_s$$

Current through secondary coil (A) — I_s

Pd across secondary coil (V) — V_s

$V_p \times I_p$ is the power input at the primary coil. $V_s \times I_s$ is the power output of the secondary coil.

4) You need to be able to relate both of these equations to power transmission in the National Grid, to explain how and why the National Grid transmits at very high pds:

- The equation in the red box above can be used to work out the number of turns needed to increase the pd (and decrease the current) to the right levels.
- A low current means that less energy is wasted heating the wires and the surroundings, making the National Grid an efficient way of transmitting power (see page 35). The equation in the green box shows why, for a given power, a high pd is needed for a low current.

Transformers increase or decrease the potential difference...

Q1 a) A transformer has 16 turns on its primary coil, 4 turns on its secondary coil and an output potential difference of 20 V. Calculate the potential difference across the primary coil. [3 marks]

b) Calculate the input current needed to produce an output power of 320 W. You can assume the transformer is 100% efficient. [3 marks]

Q1 Video Solution

Warm-Up & Exam Questions

There were lots of new ideas in that section, not to mention those equations on page 127. Better have a go at these questions so you can really see what's gone in and what you might need to go over again.

Warm-Up Questions

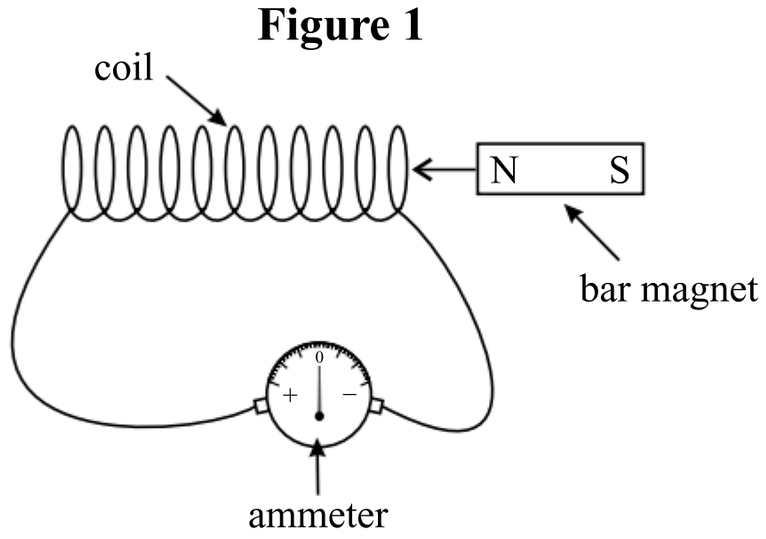
1) What kind of generator uses slip rings and brushes to transfer the induced current to a circuit?
2) What type of current does a dynamo generate?
3) What sort of current do transformers need to work?
4) Do step-up transformers have more turns on their primary or secondary coil?

Exam Questions

For question 2, you can use the formula sheet on page 216.

1 **Figure 1** shows a coil of wire connected to an ammeter. Ravi moves a bar magnet into the coil as shown. The pointer on the ammeter moves to the left. GRADE 4-6

Figure 1



1.1 Explain why the pointer moves.

[1 mark]

1.2 Suggest how Ravi could get the ammeter's pointer to move to the right.

[1 mark]

1.3 Suggest how Ravi could get a larger reading on the ammeter.

[1 mark]

1.4 What reading will the ammeter show if Ravi holds the magnet still inside the coil?

[1 mark]

2 A student is trying to test a transformer using a dc power supply. GRADE 6-7

2.1 Explain why the voltmeter connected to the secondary coil reads 0 V.

[3 marks]

The student finds an ac power supply and reconnects the transformer.
She finds that: $V_P = 12$ V, $I_P = 2.5$ A, $V_S = 4$ V (where V_P is the pd across the primary coil, etc.).

2.2 Calculate the power input to the transformer.

[2 marks]

2.3 Calculate the current in the secondary coil, I_S.

[4 marks]

2.4 The primary coil has 15 turns. How many turns must be on the secondary coil?

[3 marks]

Revision Summary Test for Topic 7

Just like opposite poles, I knew you'd be attracted towards these questions. Dive right in:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

Magnetism and Basic Electromagnetism (p.119-120)

1) What is a magnetic field? Give four examples of magnetic materials.
2) Explain what direction magnetic field lines point in. How do they show how strong a field is?
3) Are the magnetic forces between these pairs attractive or repulsive? a) like poles, b) unlike poles.
4) Describe how you could use a compass to build up a picture of a bar magnet's magnetic field lines.
5) What direction does a compass that isn't near a magnet point? What does this tell us about Earth?
6) What is the difference between a permanent magnet and an induced magnet?
   What happens to an induced magnet when it is moved far away from a permanent magnet?
7) Describe the magnetic field around a current-carrying wire.
   What factors affect the strength of the field?
8) Describe the magnetic field inside and outside a solenoid when a current flows through it.
9) What is an electromagnet? Describe how electromagnets can be used in electric starters of motors.
   Give one other practical use of electromagnets.

The Motor Effect (p.122-123)

10) Explain why a current-carrying wire in a magnetic field experiences a force.
    How does the angle of the wire affect the force?
11) In words, what is the formula used to calculate the magnetic force on a current-carrying conductor which is at right angles to a magnetic field?
12) What does each finger represent in Fleming's left-hand rule?
13) Describe the setup of a basic dc motor. How does it work?
14) Explain how loudspeakers use the motor effect to produce sound.

The Generator Effect (p.125-126)

15) What is the generator effect? Describe two ways that the generator effect can induce an alternating potential difference.
16) Explain two ways that an induced potential difference can be increased.
17) In which direction does the magnetic field created by an induced current always act?
18) What are the similarities and differences between how alternators and dynamos are set up?
19) Explain the difference between the oscilloscope trace from a dynamo and one from an alternator.
20) How do microphones work?

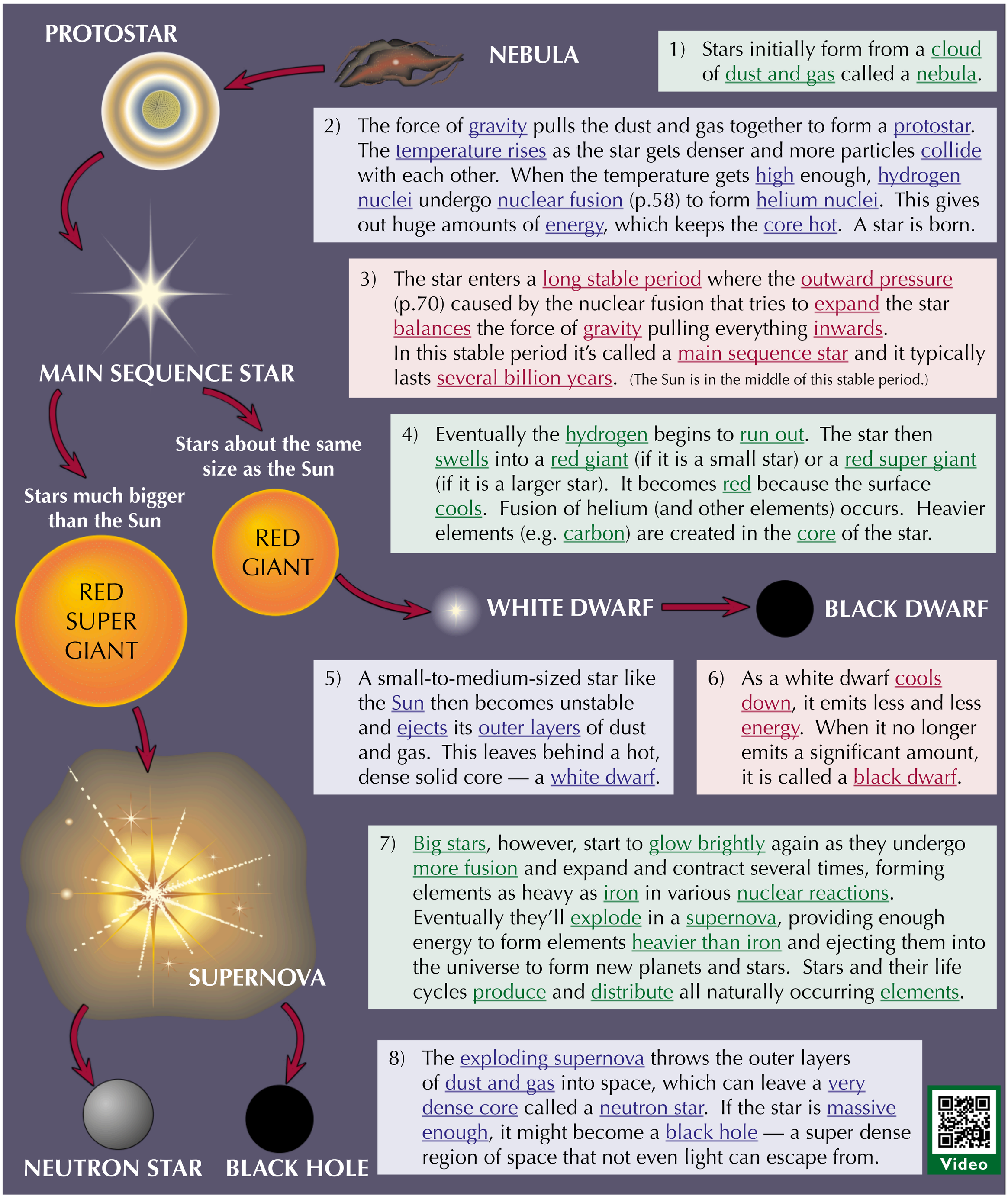
Transformers (p.127)

21) What is a transformer and how does it work?
22) What is the difference between a step-up transformer and a step-down transformer?
23) a) What is the equation that links the pds across the primary and secondary coils of a transformer with the number of turns on each coil?
    b) For a transformer to increase pd, does it need more turns on the primary or secondary coil?
24) Write the equation that relates the power input of a transformer to its power output.
    How does this equation explain why the National Grid transmits power at high pds?

The Life Cycle of Stars

Quick Quiz

*Stars go through **many traumatic stages** in their lives — just like teenagers.*



1) Stars initially form from a cloud of dust and gas called a nebula.

2) The force of gravity pulls the dust and gas together to form a protostar. The temperature rises as the star gets denser and more particles collide with each other. When the temperature gets high enough, hydrogen nuclei undergo nuclear fusion (p.58) to form helium nuclei. This gives out huge amounts of energy, which keeps the core hot. A star is born.

3) The star enters a long stable period where the outward pressure (p.70) caused by the nuclear fusion that tries to expand the star balances the force of gravity pulling everything inwards. In this stable period it's called a main sequence star and it typically lasts several billion years. (The Sun is in the middle of this stable period.)

4) Eventually the hydrogen begins to run out. The star then swells into a red giant (if it is a small star) or a red super giant (if it is a larger star). It becomes red because the surface cools. Fusion of helium (and other elements) occurs. Heavier elements (e.g. carbon) are created in the core of the star.

5) A small-to-medium-sized star like the Sun then becomes unstable and ejects its outer layers of dust and gas. This leaves behind a hot, dense solid core — a white dwarf.

6) As a white dwarf cools down, it emits less and less energy. When it no longer emits a significant amount, it is called a black dwarf.

7) Big stars, however, start to glow brightly again as they undergo more fusion and expand and contract several times, forming elements as heavy as iron in various nuclear reactions. Eventually they'll explode in a supernova, providing enough energy to form elements heavier than iron and ejecting them into the universe to form new planets and stars. Stars and their life cycles produce and distribute all naturally occurring elements.

8) The exploding supernova throws the outer layers of dust and gas into space, which can leave a very dense core called a neutron star. If the star is massive enough, it might become a black hole — a super dense region of space that not even light can escape from.

Video

Not all stars end their life cycle in the same way...

So stars that are a similar size to our Sun become red giants, and eventually black dwarves. It is only the most massive stars that enter the supernova stage of the life cycle, and end up becoming a neutron star or black hole.

The Solar System and Orbits

Quick Quiz

*The **Sun** is the centre of our **solar system**. It's **orbited** by **eight planets**, along with a bunch of other objects.*

Our Solar System has **One** Star — The **Sun**

A satellite is an object that orbits a second, more massive object.

The solar system is the Sun and all the stuff that orbits it. This includes:

PLANETS	Large objects that orbit a star. They have to be large enough to have "cleared their neighbourhoods", which means their gravity is strong enough to have pulled in any nearby objects other than their natural satellites.
DWARF PLANETS	Planet-like objects that orbit stars, but don't meet all of the rules for being a planet (like our pal, Pluto).
MOONS	Natural satellites (i.e. not man-made) that orbit planets.
ARTIFICIAL SATELLITES	Satellites built by humans — they generally orbit the Earth.

The planets of our solar system are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.

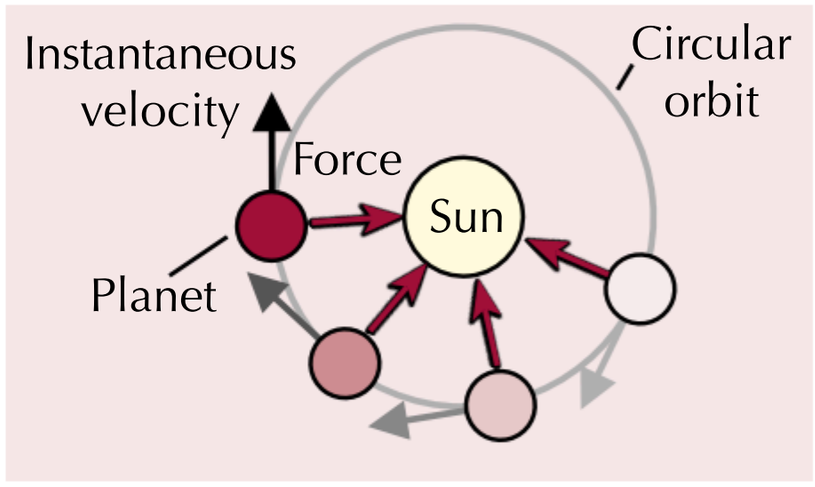
Our solar system is a tiny part of the Milky Way galaxy.
This is a massive collection of billions of stars that are all held together by gravity.

You are here.

Gravity Provides the **Force** That Creates Orbits

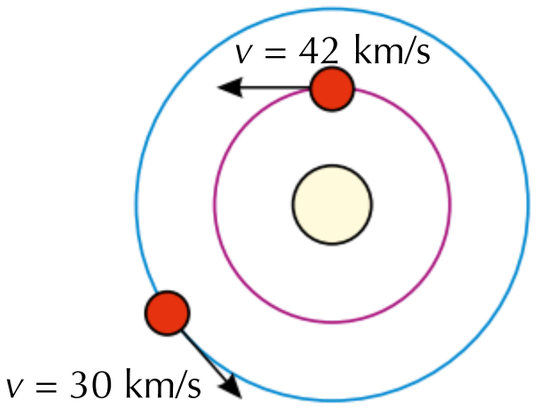
1) The planets move around the Sun in almost circular orbits (as does the Moon orbiting the Earth).
2) If an object is travelling in a circle, it's constantly changing direction, so its velocity is constantly changing. This means it's constantly accelerating.
3) For an object to accelerate, there must be a force acting on it (p.79). This force acts towards the centre of the circular orbit.
4) This force would cause the object to just fall towards whatever it was orbiting, but as the object is already moving, it just causes it to change direction.
5) The object keeps accelerating towards what it's orbiting but the instantaneous velocity (which is at a right angle to the acceleration) keeps it travelling in a circle.
6) The force that makes this happen is provided by gravity between the Sun and the planet, or between the planet and its satellite.

Be careful here — the object's velocity is constantly changing, but its speed isn't (see p.74).



The **Size** of the Orbit Depends on the Object's **Speed**

- The closer you get to a star or planet, the stronger the gravitational force, so the faster the orbiting object needs to travel to remain in orbit (to not crash into the object that it's orbiting).
- For an object in a stable orbit, if the speed of the object changes, the radius of its orbit must do so too. Faster moving objects that move in a stable, circular orbit have a smaller radius than slower moving ones.



Orbits — when things keep going round in (almost) circles...

Make sure you know what orbits what and how to tell a moon from a planet. Then try these questions.

Q1 Give one difference between natural and artificial satellites. [1 mark]

Q2 If the distance between the Moon and the Earth was smaller, how would the orbital speed of the Moon compare to its current orbital speed? Explain your answer. [3 marks]

Q2 Video Solution

Red-shift and the Big Bang

Quick Quiz

*'How it all began' is a tricky question that we just can't answer. Our **best guess** at the minute is the **Big Bang**.*

The Universe Seems to Be **Expanding**

The galaxies in the universe seem to be moving away from each other. There's good evidence for this...

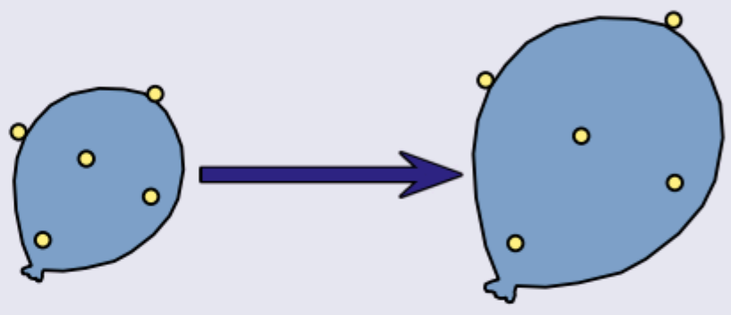
1) When scientists look at light travelling from most distant galaxies, they find that its wavelength has increased.
2) The wavelengths are all longer than they should be — they're shifted towards the red end of the spectrum. This is called red-shift.

You can think of the light wave 'stretching out' as the source moves away from us.

3) This suggests the source of the light is moving away from us.
4) The red-shift measured from more distant galaxies is greater than the red-shift from nearer ones — and it's the same result whichever direction you look in.
5) This means that more distant galaxies are moving away faster than nearer ones.
6) The inescapable conclusion appears to be that the whole universe (space itself) is expanding.

Imagine a balloon covered with pompoms.
As you blow into the balloon, it stretches.
The pompoms move further away from each other.

The balloon represents the universe and each pompom is a galaxy. As time goes on, space stretches and expands, moving the galaxies away from each other.



This Evidence Suggests the Universe **Started** with a **Bang**

So all the galaxies are moving away from each other at great speed — suggesting something must have got them going. That 'something' was probably a big explosion — the Big Bang. Here's the theory...

- Initially, all the matter we can see in the universe occupied a very small space. This tiny space was very dense and therefore very hot.
- Then it 'exploded' — space started expanding, and the expansion is still going on.

New **Evidence** Might **Change** Our Theories

1) The Big Bang theory is the best guess we have so far. When scientists discover new opposing evidence, they have to either make a new theory or change the current one to explain what they've observed.
2) There is still lots we don't know about the universe. Observations of supernovae from 1998 to the present day appear to show that the speed at which distant galaxies are moving away from us has been increasing over time.
3) Currently scientists think the universe is mostly made up of dark matter and dark energy:

DARK MATTER	An unknown substance which holds galaxies together. Unlike normal matter, it doesn't emit any electromagnetic radiation.
DARK ENERGY	Thought to be responsible for the accelerated expansion of the universe.

4) No one really knows what these things are, so there are lots of different theories about them. These theories get tested over time and are eventually either accepted or rejected.

Red-shift shows us that the universe is expanding at a rapid rate...

Dark matter and dark energy are really mysterious things. If scientists discover what they actually are, it could change our view of the universe and how it began. But for now, all we can assume is it all started with a Big Bang...

Warm-Up & Exam Questions

You know the drill — some warm-up questions to get you thinking about what you've just read, and some exam questions to see how well you can apply your knowledge. Get cracking.

Warm-Up Questions

1) Immediately after the end of its main sequence phase, what does a star much larger than the Sun become?
2) Name two types of object in the solar system that orbit the Sun.
3) What is the force responsible for keeping planets in orbit?
4) What is the name given to the observed increase in the wavelength of light from distant stars?
5) State two possible components of the universe which cannot currently be explained.

Exam Questions

1 Use the correct words from the box below to complete the following sentences. GRADE 4-6 GRADE

artificial satellites	**planets**	**the Sun**	**Jupiter**
the Earth	**natural satellites**	**stars**	

Moons are ..... . They orbit ..... .

..... are man-made, and most of them orbit ..... .

[4 marks]

2 Stars go through many stages in their lives. GRADE 6-7 GRADE

2.1 Describe how a star is formed.

[3 marks]

2.2 The stable period of a main sequence star can last billions of years. Explain why main sequence stars undergo a stable period.

[2 marks]

When main sequence stars begin to run out of hydrogen in their core, they swell and become either a red giant or a red super giant depending on their size.

2.3 Describe what happens to stars after their red giant phase until the end of their life cycle.

[3 marks]

2.4 Describe what happens to stars after their red super giant phase until the end of their life cycle.

[3 marks]

3* Describe red-shift and explain how it supports the Big Bang theory as an explanation for how the universe began. GRADE 7-9 GRADE

[6 marks]

Revision Summary Test for Topic 8

Space physics — the final ~~frontier~~ topic. Test your knowledge by answering these questions:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Light

Summary Test

The Life Cycle of Stars (p.130) ☑

1) What is a nebula?
2) How is a protostar formed?
3) What causes a protostar to turn into a main sequence star?
4) How long is a main sequence star typically stable for? What makes it stable?
5) What happens when the hydrogen in a main sequence star starts to run out?
6) What kind of size do main sequence stars have to be to turn into:
   a) red giants, b) red super giants.
7) Describe what happens to a red giant once it has run out of hydrogen. What does it turn into?
8) How does a black dwarf form from a white dwarf?
9) Will our Sun eventually turn into a black hole or a black dwarf? Explain your choice.
10) How are elements heavier than iron produced in massive stars?
    How do these elements get distributed throughout the universe?
11) What two things can be left behind by a star after a supernova?

The Solar System and Orbits (p.131) ☑

12) How many stars are there in our solar system? And how many planets?
13) What do planets and dwarf planets orbit? What is the difference between them?
14) What do moons orbit?
15) What is the difference between an artificial satellite and a natural satellite?
16) Which galaxy is our solar system part of?
17) Explain why the velocity of something with a circular orbit changes. What about the speed?
18) Why is a satellite in orbit constantly accelerating?
19) a) Name the force that causes an object to have a circular orbit around the Earth.
    b) In what direction does the force act?
    c) State the direction of the orbiting object's instantaneous velocity in relation to this force.
20) How does the strength of an object's gravitational force change as you get closer to its surface?
21) Planets A and B both have stable, circular orbits around a star.
    Planet A travels faster than Planet B. Which planet's orbit has a smaller radius?

Red-shift and the Big Bang (p.132) ☑

22) What is red-shift?
23) How can red-shift tell us how fast galaxies are moving?
    How does it tell us the universe is expanding?
24) Describe the ideas that make up the Big Bang theory.
25) How does red-shift support the Big Bang theory?
26) What have scientists observed about the movement of the galaxies from 1998 to the present day?
27) What is meant by dark matter and dark energy?

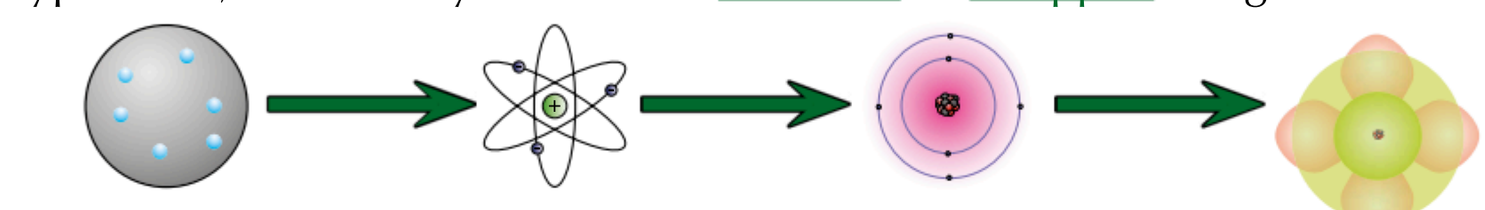
The Scientific Method

*This section **isn't** about how to 'do' science — but it does show you the way **most scientists** work.*

Scientists Come Up With **Hypotheses** — Then **Test** Them

1) Scientists try to explain things. They start by observing something they don't understand.
2) They then come up with a hypothesis — a possible explanation for what they've observed.
3) Next they make a prediction based on the hypothesis. They test the prediction by gathering evidence (i.e. data) from investigations. If the evidence backs up their prediction, then they're a step closer to figuring out if the hypothesis is true.
4) They share their findings at conferences, or in peer-reviewed journals.
5) Once other scientists know about a hypothesis, they'll start basing their own predictions on it and carry out their own experiments. They'll also try to reproduce the original experiments to check the results — and if all the experiments in the world back up the hypothesis, then scientists start to think it is true.
6) However, if results from lots of other experiments don't fit with the hypothesis, then it may need to be modified or scrapped altogether.

Peer-review is where other scientists check that methods, results and scientific explanations seem sensible before they're published. It helps to detect errors and false claims, but it doesn't mean that findings are correct — just that they're not wrong in any obvious way.



About 100 years ago, scientists hypothesised that atoms looked like this.

New evidence from further experiments meant they changed their hypothesis to this.

Once even more evidence was gathered, scientists changed their hypothesis again.

Now we think it's more like this.

If **All** the **Evidence** Supports a Hypothesis, It's **Accepted** — For Now

Accepted hypotheses often become the basis of theories. Our currently accepted theories are the ones that have survived many rounds of testing over the years. However, theories never become totally indisputable fact. If new evidence comes along that can't be explained using the existing theory, then the hypothesising and testing is likely to start all over again.

Theories Can Involve **Different Types** of **Models**

Scientists use models to explain observations and make predictions, for example:

1) Representational models — simplified descriptions or pictures of what's going on in real life. E.g. the Bohr model of an atom is a simplified way of showing the arrangement of electrons in an atom (see p.49). It can be used to explain electron excitations in atoms.

Scientists test models by carrying out experiments to check that the predictions made by the model happen as expected.

2) Computational models — computers are used to make simulations of complex real-life processes (e.g. climate change). Computational models are used when there's lots of different variables to consider, and because you can easily change their design to take into account new data.

All models have limitations on what they can explain or predict. E.g. the Big Bang model (p.132) can be used to explain why everything in the Universe is moving away from us. A limitation of it is that it doesn't explain the moments before the Big Bang.

Scientific models are constantly being refined...

The scientific method has developed over time. First, Aristotle (a Greek philosopher) realised that theories needed to be based on observations. Then, Muslim scholars introduced the ideas of testing hypotheses and repeating work to check results. And Roger Bacon took things further by writing down everything he did, so other scientists could reproduce his work.

Communication & Issues Created by Science

*Scientific developments can be great, but they can sometimes **raise more questions** than they answer...*

It's Important to **Communicate** Scientific Discoveries to the **Public**

Some scientific discoveries show that people should change their habits, or they might provide ideas that could be developed into new technology. So scientists need to tell the world about their discoveries.

Radioactive materials are used widely in medicine for imaging and treatment (see p.57). Information about these materials needs to be communicated to doctors (so they can make use of them) and to patients (so they can make informed decisions about their treatment).

Scientific **Evidence** can be **Presented** in a **Biased Way**

1) Media reports of scientific discoveries aren't peer-reviewed. So, even though news stories are often based on data that has been peer-reviewed, the data might be presented in a way that is over-simplified or inaccurate, making it open to misinterpretation.
2) People who want to make a point can sometimes present data in a biased way (sometimes without meaning to). For example, a scientist might overemphasise a relationship in some of their data, or a newspaper article might report data supporting an idea without giving any evidence against it.

Scientific Developments are **Great**, but they can **Raise Issues**

Economic issues	Society can't always afford to do what scientists recommend (e.g. investing in alternative energy sources) without cutting back elsewhere.
Social issues	Decisions based on scientific evidence affect people — e.g. should fossil fuels be taxed more highly? These decisions can have negative effects on some people's lifestyles, and some groups of people might be affected more than others.
Personal issues	Some decisions will affect individuals. For example, someone might support renewable energy, but object if a wind farm is built next to their house.
Environmental issues	Human activity can affect the natural environment. For example, building a hydroelectric dam might destroy the local habitats of some species. But it also reduces our need for fossil fuels, helping to reduce climate change.

Science **Can't Answer Every Question** — Especially **Ethical** Ones

- To answer scientific questions, scientists need data to provide evidence for their hypotheses.
- Some questions can't be answered yet because there's not enough data to support a theory, or because the data can't currently be collected. As more evidence is collected, more questions will be answered (like what the impact of global warming on sea levels will be).

But there will always be questions that experiments can't help us to answer...

Question: Should new drugs that boost 'brain power' be available to the public?

PRO
- Could improve your concentration or memory.

CONS
- Could give people an unfair advantage in exams.
- People might be pressured into taking them so that they could work more effectively, and for longer hours.

There are often issues with new scientific developments...

The trouble is, there's often no clear right answer where these issues are concerned — that's for society to sort out. Different people have different views, depending on their needs and priorities. These issues are full of grey areas.

Risk

Scientific discoveries *are often great, but they can prove* ***risky****. With dangers all around, you've got to be aware of hazards — this includes* ***how likely*** *they are to* ***cause harm*** *and* ***how serious*** *the effects may be.*

Every **Hazard** has a **Risk** Attached to it

HAZARD
Something that could potentially cause harm.

RISK
The chance that a particular hazard will cause harm.

You can estimate the size of a risk based on how many times something happens in a big sample over a given period. For example, you could assess the risk of a driver crashing by recording how many people in a group of 100 000 drivers crashed their cars over a year.

- There are risks that seem pretty obvious, or we've known about for a while (e.g. the risk of having a car accident while in a moving car).
- There are also new risks that scientists are unsure of that arise from scientific advances (e.g. whether nanoparticles used in cosmetics and sun cream might be harming the cells in our bodies).

People Make Their **Own Decisions** About Risk

Something's generally considered high risk if there is a high chance of it happening and if its consequences are serious. But there are lots of factors that can affect how people feel about taking different risks:

Consequences

Most people are happier to accept a higher probability of an accident if the consequences are short-lived and fairly minor. E.g. you're more likely to cut your finger when using a knife than you are to die when scuba-diving — cutting your finger is fairly minor compared to death so people are more willing to take that risk.

Familiarity

How risky people think things are isn't always accurate. They tend to view familiar activities as low-risk and unfamiliar activities as high-risk — even if that's not the case (e.g. cycling on roads is often high-risk, but some people are happy to do it because it's a familiar activity).

Visibility

People may underestimate the risk of things with long-term or invisible effects, e.g. using tanning beds.

Choice

People tend to be more willing to accept a risk if they choose to do something, compared to having the risk imposed on them (e.g. having a nuclear power station built next door).

Investigations Can be **Hazardous**

Hazards in experiments might include:

- Lasers — a laser directed into someone's eye can cause blindness.
- Gamma radiation — gamma-emitting radioactive sources can cause cancer.
- Fire — an unattended Bunsen burner is a fire hazard.
- Electricity — faulty electrical equipment could give you a shock.

You can find out about potential hazards by looking in textbooks, doing some internet research, or asking your teacher.

Part of planning an investigation is making sure that it's safe. You should always identify all the hazards that you might encounter, then think of ways of reducing the risks from these hazards. For example:

If you're working with springs, always wear safety goggles. This reduces the risk of the spring hitting your eye if it snaps.

If you're using a Bunsen burner, stand it on a heat proof mat. This will reduce the risk of starting a fire.

The pros and cons of new technology must be weighed up...

The world can be a dangerous place, and it's impossible to rule out the chance of an accident altogether. But if you can recognise hazards and decide how to reduce their risks, you're more likely to keep yourself, and others, safe.

Designing Investigations

*Dig out your lab coat and dust down your badly-scratched safety goggles... it's **investigation time**.*

Investigations **Produce Evidence** to **Support** or **Disprove** a **Hypothesis**

1) Like all great scientists, you'll need to be able to come up with a hypothesis to test an observation (see p.135).
2) To determine whether a hypothesis is right, you need to do an investigation to gather evidence. To do this, you'll need to use your hypothesis to make a prediction — something you think will happen that you can test.
3) Investigations are used to see if there are patterns or relationships between two variables, e.g. to see if there's a relationship between the variables 'number of spots' and 'size of feet'.

1. Observation: People have big feet and spots.
2. Hypothesis: Having big feet causes spots.
3. Prediction: People who have bigger feet will have more spots.

Evidence Needs to be **Repeatable, Reproducible** and **Valid**

REPEATABLE — if the same person does the same experiment again, they'll get similar results.

REPRODUCIBLE — if someone else does the experiment, or a different method or piece of equipment is used, the results will still be similar.

VALID RESULTS — results that are both repeatable and reproducible AND they answer the original question. They come from experiments that were designed to be a fair test.

If data is repeatable and reproducible, it's reliable and scientists are more likely to have confidence in it.

To Make an Investigation a **Fair Test** You Have to **Control the Variables**

1) In a lab experiment you usually change one variable and measure how it affects another.
2) To make it a fair test, everything else that could affect the results should stay the same — otherwise you can't tell if the thing you're changing is causing the results or not.
3) Because you can't always control all the variables, you often need to use a control experiment — an experiment that's kept under the same conditions as the rest of the investigation, but doesn't have anything done to it. This is so that you can see what happens when you don't change anything at all.

The different kinds of variables are:

Independent	The variable that you change
Dependent	The variable that you measure
Control	The variables you keep the same

E.g. you can find how changing the current through a component affects the measured potential difference. In this case, the independent variable is the current. The dependent variable is the potential difference. The control variables would include the temperature of the component, the p.d. of the power supply, etc.

The **Bigger** the **Sample Size** the **Better**

- A sample should represent the whole population — a small sample can't do that as well. It's also harder to spot anomalies if your sample size is too small.
- Scientists have to be realistic when choosing sample sizes, though. For example, if you were studying the effects of living near a nuclear power plant, it'd be great to study everyone who lived near a nuclear power plant (a huge sample), but it'd take ages and cost a tonne. It'd be more realistic to study a thousand people, with a range of ages, genders, and races to represent the population as a whole.

Designing an investigation is an involved process...

Not only do you need to be able to plan your own investigations, you should also be able to look at someone else's plan for their investigation and decide whether or not it needs improving. Those examiners aren't half demanding...

Collecting Data

You've designed the perfect investigation — now it's time to get out in the field and ***collect some data****.*

Your Data Should be **Repeatable**, **Reproducible**, **Accurate** and **Precise**

1) To check repeatability you need to repeat the readings and check that the results are similar. You need to repeat each reading at least three times.
2) To make sure your results are reproducible you can cross check them by taking a second set of readings with another instrument (or a different observer).
3) Your data also needs to be precise and accurate:

PRECISE RESULTS — where the data is all really close to the mean (average) of your repeated results (i.e. not spread out).

ACCURATE RESULTS — results that are really close to the true answer.

Repeat	Data set 1	Data set 2
1	12	11
2	14	17
3	13	14
Mean	13	14

Data set 1 is more precise than data set 2.

The accuracy of your results usually depends on your method. Estimating the volume of an irregularly shaped solid by measuring the sides isn't very accurate because this won't take into account any gaps in the object. It's more accurate to measure the volume using a eureka can (see p.148).

Your **Equipment** has to be **Right for the Job**

Sensitivity

The measuring equipment you use has to be sensitive enough to measure the changes you're looking for. E.g. if you need to measure changes of 1 cm^3 you need to use a measuring cylinder or burette that can measure in steps of 1 cm^3.

Resolution

Resolution is the smallest change a measuring instrument can detect. E.g. some mass balances have a resolution of 1 g, some have a resolution of 0.1 g, and some are even more sensitive.

Calibration

Equipment also needs to be calibrated by measuring a known value. If there's a difference between the measured and known value, you can use this to correct the inaccuracy of the equipment.

Look out for **Errors** and **Anomalous Results**

Type of error	Definition	Example	How to correct
Random	Unpredictable differences in results caused by things like human errors in measuring.	When taking a reading from a ruler, you have to estimate or round a reading when it's between two marks.	By taking repeat readings and finding the mean.
Systematic	When measurements are wrong by a consistent amount every time.	Background radiation will affect all readings when investigating the activity of a radioactive source.	By accounting for them, e.g. taking background radiation off readings.

A systematic error caused by using equipment that isn't zeroed properly is called a zero error. E.g. if a mass balance always reads 1 gram with nothing on it, all your measurements will be 1 gram too heavy. You can correct this error by subtracting 1 gram from all your results for mass.

If there's no systematic error, then doing repeats and calculating a mean could make your results more accurate.

An anomalous result is one that doesn't fit in with the rest of your data at all. You should investigate it and try to work out what happened. If you can work out what happened (e.g. you measured something totally wrong) you can ignore it when processing your results.

Many different things can cause errors in your data...

Weirdly, data can be really precise but not very accurate. For example, a fancy piece of lab equipment might give results that are really precise, but if it's not been calibrated properly those results won't be accurate.

Processing Data

*Read on to see how you can go about **processing** your data to make it **more useful**...*

Data Needs to be **Organised**

- Tables are very useful for organising data.
- When you draw a table, use a ruler and make sure each column has a heading (including the units).
- Here's a table about the time a student spent doing different activities:

Activity	Time spent (hours per week)				
	Week 1	Week 2	Week 3	Week 4	Week 5
Rock climbing	4.5	4.7	5.2	5.0	4.3
Table tennis	1.0	1.6	2.4	1.8	2.1
Chess	2.3	3.5	2.9	3.2	3.0

The green row is the header row

Categories of activity go down the left side

Units next to the header

Repeats go in the header row under a main header

You input your results in these rows underneath the header

You Might Have to **Process Your Data**

There are different ways you can calculate the average of a set of results.
They each tell you something different about your data:

- **MEAN** — calculated by adding all the values together and dividing by the total number of values used. When people say to calculate 'the average', they are usually talking about the mean.
- **MODE** — the value that occurs the most in a set of results.
- **MEDIAN** — the middle number when you write your results in numerical order.

EXAMPLE

The results of an experiment show the extension of two springs when a force is applied to both of them. Calculate the mean, mode and median of the extension for both springs.

MATHS SKILLS

Spring	Repeat (cm)					Mean (cm)	Mode (cm)	Median (cm)
	1	2	3	4	5			
A	18	26	22	26	28	(18 + 22 + 26 + 26 + 28) ÷ 5 = 24	26	26
B	11	14	20	15	20	(11 + 14 + 15 + 20 + 20) ÷ 5 = 16	20	15

Round to the **Lowest Number** of **Significant Figures**

The first significant figure of a number is the first digit that's not zero. The second and third significant figures come straight after (even if they're zeros). You should be aware of significant figures in calculations.

1) In any calculation, you should round to the lowest number of significant figures (s.f.) given.
2) Remember to write down how many significant figures you've rounded to after your answer.
3) If your calculation has multiple steps, only round the final answer, or it won't be as accurate.

EXAMPLE

The mass of a solid is 0.24 g and its volume is 0.715 cm^3. Calculate the density of the solid.

MATHS SKILLS

Density = mass ÷ volume

= 0.24 *g* ÷ 0.715 cm^3 = 0.33566... = 0.34 *g*/cm^3 (2 s.f.)

2 s.f. — 3 s.f. — Final answer should be rounded to 2 s.f.

Examiners might ask you to process some data in a certain way...

Always keep your eye on significant figures in your answers, and don't forget to write down your working.

Presenting Data

Once you've processed data, you can present your results in a nice ***chart*** *or* ***graph*** *to help you* ***spot any patterns*** *in your data. The type of data you record will affect how you should* ***present*** *it.*

If Your Data Comes in **Categories**, Present it in a **Bar Chart**

You should represent your data with a bar chart if your independent variable is one of the following types:

CATEGORIC
Data that comes in distinct categories (e.g. solid, liquid, gas).

DISCRETE
Data that can be counted in chunks, where there's no in-between value (e.g. number of protons — you can't have half a proton).

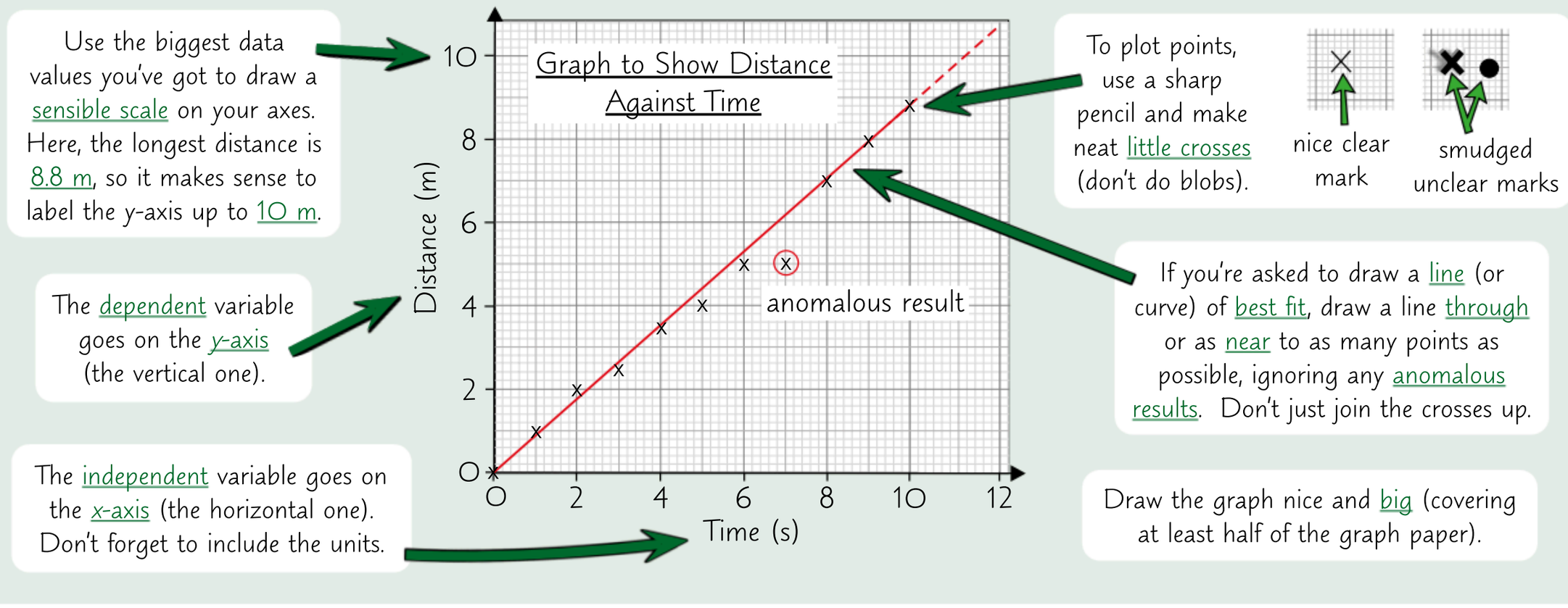
There are some golden rules you need to follow for drawing bar charts:

If Your Data is **Continuous**, Plot a **Graph**

If both variables are continuous, use a graph to display the data.

CONTINUOUS
Numerical data that can have any value within a range (e.g. length, volume, temperature).

Here's how to plot points on a graph:



Pick the right way to present your data...

If you're working with categoric or discrete data, draw a bar chart. If your data's continuous, plot points on a graph. Don't get them muddled up, or you'll be spending precious exam time on a graph that won't get you those marks.

Interpreting Graphs

*So you've plotted some graphs — what now? It's time to uncover some **trends** in your data...*

Graphs Can Give You a Lot of **Information** About Your **Data**

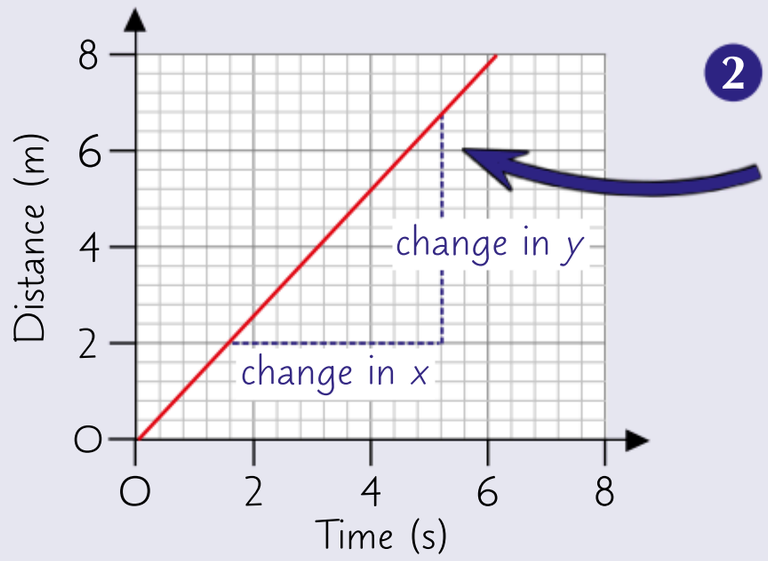
The gradient (slope) of a graph tells you how quickly the dependent variable changes if you change the independent variable.

$$\text{gradient} = \frac{\text{change in } y}{\text{change in } x}$$

This graph shows the distance walked by a person against time. The graph is linear (it's a straight line graph), so you can calculate the gradient of the line to find out the speed of the person.

1 To calculate the gradient, pick two points on the line that are easy to read and a good distance apart.

You can use this method to calculate other rates from a graph, so long as you always make *x* time.



2 Draw a line down from the top point and a line across from the bottom point to make a triangle. The line drawn down the side of the triangle is the change in *y* and the line across the bottom is the change in *x*.

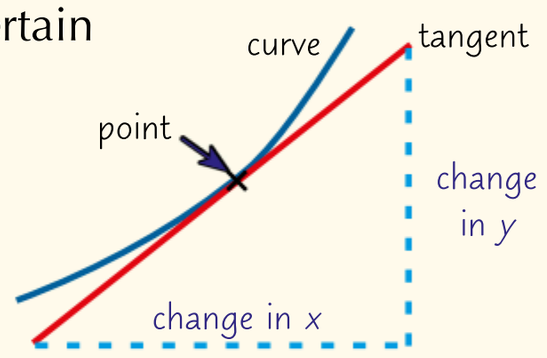
$$\text{Rate} = \text{gradient} = \frac{\text{change in } y}{\text{change in } x} = \frac{6.8 - 2}{5.2 - 1.6} = \frac{4.8\text{ m}}{3.6\text{ s}} = 1.3\text{ m/s}$$

The units of the gradient are (units of *y*) ÷ (units of *x*).

The intercept of a graph is where the line of best fit crosses one of the axes.

- The *x*-intercept is where the line of best fit crosses the *x*-axis...
- ...and the *y*-intercept is where it crosses the *y*-axis.

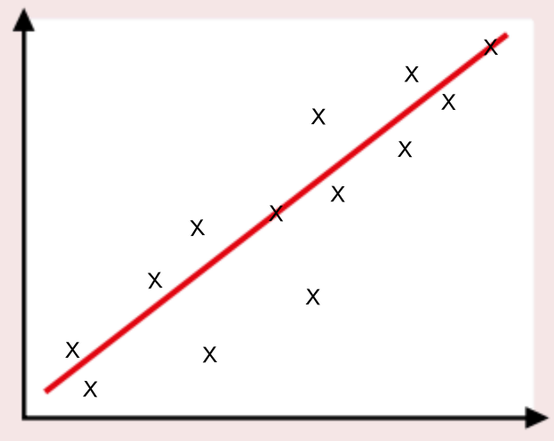
To find the gradient of a curve at a certain point, draw a tangent to the curve at that point. This is a straight line that touches the curve at that point only. Then just find the gradient of the tangent in the same way as above.



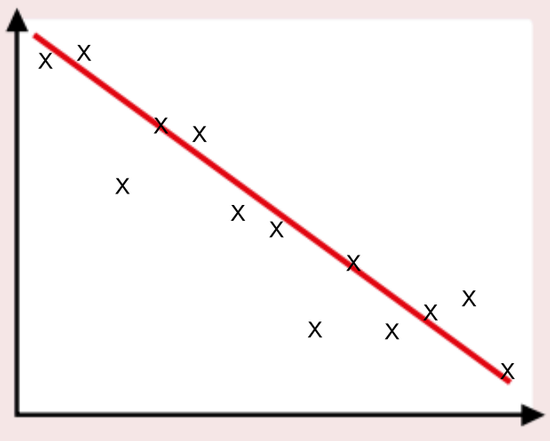
Graphs Show the **Relationship** Between Two Variables

You can get three types of correlation (relationship) between variables:

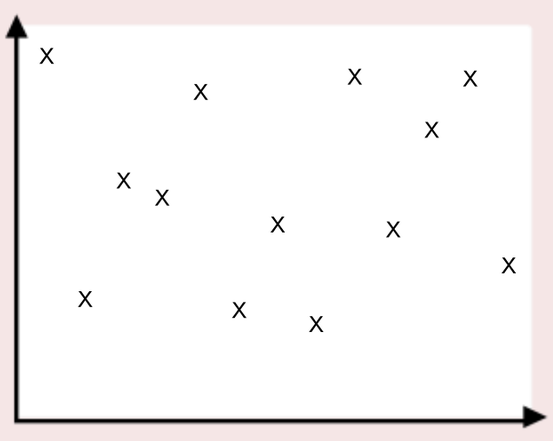
Positive correlation: as one variable increases, the other increases.



Inverse (negative) correlation: as one variable increases, the other decreases.



No correlation: there's no relationship between the two variables.



Just because there's correlation, it doesn't mean the change in one variable is causing the change in the other — there might be other factors involved (see page 145).

The gradient is change in y ÷ change in x...

You can tell whether a graph shows a positive or negative correlation from the gradient of the line of best fit. Positive gradient = uphill line = positive correlation. Negative gradient = downhill line = negative correlation.

Units

*You can be a human calculator, but numbers don't mean much if you don't get the **units** right.*

S.I. Units Are Used All Round the World

- Scientists use a set of standard units, called S.I. units, to measure their data. If everyone uses the same units, it's much easier to compare results as you don't need to do loads of conversions.
- For example, if you measured the speed of one car in m/s and another one in km/h, it'd be hard to tell straight away which car was going faster.

Here are some S.I. units you'll see in physics:

Quantity	S.I. Base Unit
mass	kilogram, kg
length	metre, m
time	second, s
temperature	kelvin, K

Scaling Prefixes Can Be Used for Large and Small Quantities

1) Quantities come in a huge range of sizes. For example, the volume of a swimming pool might be around 2 000 000 000 cm^3, while the volume of a cup is around 250 cm^3.
2) To make the size of numbers more manageable, larger or smaller units are used. These are the S.I. base unit (e.g. metres) with a prefix in front:

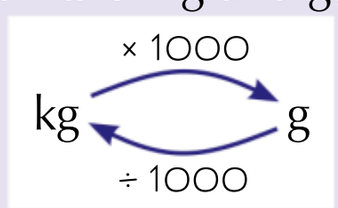
prefix	tera (T)	giga (G)	mega (M)	kilo (k)	deci (d)	centi (c)	milli (m)	micro (μ)	nano (n)
multiple of unit	10^{12}	10^{9}	1 000 000 (10^{6})	1000	0.1	0.01	0.001	0.000001 (10^{-6})	10^{-9}

3) These prefixes tell you how much bigger or smaller a unit is than the base unit. So one kilometre is one thousand metres.

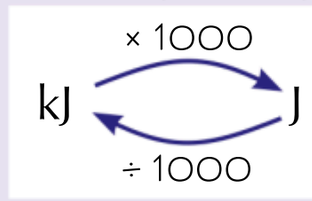
The conversion factor is the number of times the smaller unit goes into the larger unit.

4) To convert between units, you need to know the conversion factor:
   - To go from a bigger unit (like m) to a smaller unit (like cm), you multiply by the conversion factor.
   - To go from a smaller unit (like g) to a bigger unit (like kg), you divide by the conversion factor.
5) Here are some conversions that'll be useful for GCSE physics:

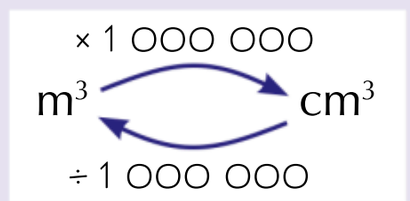
Mass can have units of kg and g.



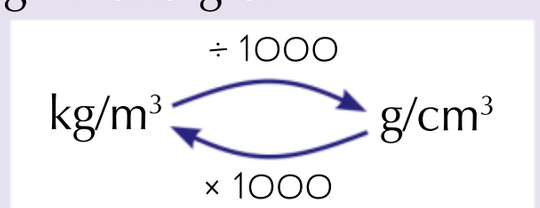
Energy can have units of J and kJ.



Volume can have units of m^3 and cm^3.



Density can have units of kg/m^3 and g/cm^3.



Always Check The Values Used in Equations Have the Right Units

Always make sure the values you put into an equation or formula have the right units. In some cases you'll have to convert a value into different units before you can put it into the equation.

To make sure your units are correct, it can help to write them down on each line of your calculation.

EXAMPLE

In an experiment to find the average speed of a trolley, the trolley travels 30 cm in 3 seconds. Calculate the average speed of the trolley in m/s.

MATHS SKILLS

1) Convert the distance measurement from cm into m by dividing by the correct conversion factor.
2) Find the speed using speed = distance ÷ time.

30 cm —(÷ 100)→ 0.3 m

$v = s \div t = 0.3 \div 3 = 0.1$ m/s

Multiply or divide by the conversion factor...

It's easy to get in a muddle when converting between units, but there's a handy way to check you've done it right. If you're moving from a smaller unit to a larger unit (e.g. g to kg) the number should get smaller, and vice versa.

Maths Skills

No, you've not opened up the wrong book — this is indeed a page on ***maths****. Maths comes up* ***all the time*** *in physics, so you'd better be* ***comfortable*** *using it. Read on for some of the* ***key*** *maths skills you need...*

Standard Form Is Often Used For **Really Big** or **Really Small** Numbers

You need to know how to deal with numbers in standard form. They always look like this:

A is a number between 1 and 10, but not including 10.

$$A \times 10^{n}$$

n is the number of places the decimal point would move if you wrote the number out fully. It's negative for numbers less than 1, and positive for numbers greater than 1.

So 200 000 Pa would be written as 2×10^{5} Pa in standard form. And 0.0000038 m is 3.8×10^{-6} m.

Do the **Same Thing** to **Both Sides** When **Rearranging Formulas**

Often you need to rearrange a formula to be able to solve for the variable you want. You just need to do the same thing to each side until you get that variable on its own.

EXAMPLE

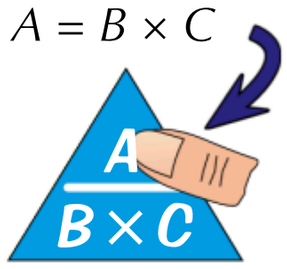
Rearrange $E_e = \frac{1}{2}ke^2$ to solve for k.

MATHS SKILLS

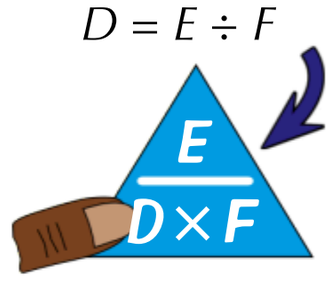
1) You're trying to get k on its own. $E_e = \frac{1}{2}ke^2$
2) Multiply both sides by 2 to get rid of the fraction. $2 \times E_e = 2 \times \frac{1}{2}ke^2$
3) Simplify to tidy things up. $2E_e = ke^2$
4) Divide both sides by e^2. $\frac{2E_e}{e^2} = \frac{ke^2}{e^2}$
5) Simplify again, and flip things around so k is on the left. $k = \frac{2E_e}{e^2}$

You can put simple formulas into formula triangles if you find them helpful:

$A = B \times C$



$D = E \div F$



To use them, cover up what you want to find and write down what's left showing.

Make Sure You Know How to Use Your **Calculator** Correctly

Here are a few things to remember to make sure you don't lose marks for silly calculator mistakes:

- You need to make sure that what you put in your calculator is the same calculation as is on the page, even if it looks a bit different.
- For example, to put $\frac{11.5 + 6.8}{3}$ into any standard calculator, you type (11.5 + 6.8) ÷ 3. Don't forget the brackets.
  You could also use the fraction button on a scientific calculator.
- Use exact values from previous calculations in your calculator rather than rounding (p.140). Make sure you know how to find these values on your calculator — e.g. by using the 'Ans' button or scrolling up.

Your calculator will follow the order of operations for what you've typed. If you forget the brackets, it won't do the calculation you want and you'll get 13.766... instead of 6.1.

Proportionality is a Way of Describing How Two Variables are **Related**

There are two types of proportionality:

This symbol means 'is proportional to'.

Direct	As one variable increases, the other increases in the same ratio.	$A \propto B$	E.g. force and acceleration
Inverse	As one variable increases, the other decreases in the same ratio.	$A \propto \frac{1}{B}$	E.g. acceleration and mass

Rearranging formulas is a really important skill...

Some people love formula triangles, some people not so much. You can use them as a tool if you want to, but you need to know how to rearrange formulas properly, not least because complicated formulas don't work in triangles.

Drawing Conclusions

Once you've designed your experiment, carried it out, processed and presented your data, it's finally time to sit down and work out exactly what your data tells you. Time to make some ***conclusions****...*

You Can **Only Conclude** What the Data Shows and **NO MORE**

1) Drawing conclusions might seem pretty straightforward — you just look at your data and say what pattern or relationship you see between the dependent and independent variables...

This table shows the current through a light bulb for three different potential differences across the bulb:

Potential difference (V)	Current (A)
6	4
9	10
12	13

Conclusion: when the pd across the bulb is higher, the current through the bulb is higher.

2) ...but you've got to be really careful that your conclusion matches the data you've got and doesn't go any further.

You can't conclude that the current through any circuit component will be higher for any larger potential difference — the results might be completely different for different components.

3) You also need to be able to use your results to justify your conclusion (i.e. back up your conclusion with some specific data).

E.g. the current through the bulb was 9 A higher with a potential difference of 12 V compared to the current at 6 V.

4) When writing a conclusion you need to refer back to the original hypothesis and say whether the data supports it or not:

If the hypothesis for this experiment was that the current through the bulb is higher when the potential difference across the bulb is higher, then the data supports this.

Correlation DOES NOT Mean **Cause**

If two things are correlated (i.e. there's a relationship between them) it doesn't necessarily mean a change in one variable is causing the change in the other — this is really important, so don't forget it. There are three possible reasons for a correlation:

1. Chance — it might seem strange, but two things can show a correlation purely due to chance.

   E.g. a study might find a correlation between people's hair colour and their frisbee skills. But, other scientists who investigate it don't get a correlation — the results of the first study are just a fluke.

2. Linked by a third variable — a lot of the time it may look as if a change in one variable is causing a change in the other, but it isn't. A third variable links the two things.

   E.g. there's a correlation between a higher water temperature and the amount of shark attacks. This isn't because warmer water makes sharks go crazy. The attacks are linked by a third variable — the number of people swimming (i.e. more people are swimming when the water's hotter, meaning more chance of a shark attack).

3. Cause — sometimes a change in one variable does cause a change in the other. You can only make this conclusion when you've controlled all variables that could be affecting the result.

   E.g. there is a correlation between smoking and lung cancer. This is because chemicals in tobacco smoke cause lung cancer. This conclusion was only made once other variables (such as age and exposure to other things that cause cancer) had been controlled. Once they were controlled, the effect of smoking was isolated and shown to increase the risk of someone getting lung cancer.

You should be able to justify your conclusion with your data...

You should always be able to explain how your data supports your conclusion. It's easy to go too far with conclusions and start making bold claims that your data simply can't back up. It's also important that you refer back to your initial hypothesis, the one you made right back at the start of the investigation, to see whether your data supports it or not.

Uncertainties and Evaluations

Uncertainty *measures how confident you are in your final answer. And an* ***evaluation*** *is the final step of any investigation — where you say how well things went, what you'd do differently next time... all that fun stuff.*

Uncertainty is the Amount of **Error** Your Measurements Might Have

1) The measurements you make will have some uncertainty in them, for example, due to limits in the resolution of the equipment you use (see page 139).
2) When you repeat a measurement, you often get a slightly different figure each time you do it due to random error. So random errors are another source of uncertainty — they can cause uncertainty in the mean of a set of results.
3) You can calculate the uncertainty of a mean result using the equation:

The range is the largest value minus the smallest value.

$$\text{uncertainty} = \frac{\text{range}}{2}$$

4) The larger the range, the less precise your results are and the more uncertainty there will be in your results. Uncertainties are shown using the '±' symbol.

EXAMPLE

The table below shows the results of an experiment to determine the speed of a trolley as it moves along a horizontal surface. Calculate the uncertainty of the mean.

Repeat	1	2	3	mean
Speed (m/s)	2.02	1.98	2.00	2.00

MATHS SKILLS

1) First work out the range:
Range = largest value − smallest value
= 2.02 − 1.98 = 0.04 m/s

2) Use the range to find the uncertainty:
Uncertainty = range ÷ 2 = 0.04 ÷ 2 = 0.02 m/s
So the uncertainty of the mean = 2.00 ± 0.02 m/s

5) Measuring a greater amount of something helps to reduce uncertainty. E.g. for the example above, measuring speed over a longer period will reduce the percentage uncertainty in measurements of speed.

Evaluations — Describe **How** it Could be **Improved**

1) An evaluation is a critical analysis of the whole investigation:

- You should comment on the method — was it valid? Did you control all the other variables to make it a fair test?
- Comment on the quality of the results — was there enough evidence to reach a valid conclusion? Were the results repeatable, reproducible, accurate and precise?
- Were there anomalous results? If there were, try to explain them — were they caused by errors in measurement? Were there other variables that could have affected the results? You should comment on the level of uncertainty in your results too.

2) Answering these questions will allow you to say how confident you are that your conclusion is right.
3) Then you can suggest any changes to the method that would improve the quality of the results and give you more confidence in your conclusion. E.g. you might increase the number of measurements you took. Taking more measurements at narrower intervals could give you a more accurate result:

Springs have a limit of proportionality (p.66). You could use several identical springs to find their limit of proportionality. If you apply forces of 1-5 N in increments of 1 N to one spring, and find that the limit is somewhere between 4 N and 5 N, you could repeat the experiment with one of the other springs, taking more measurements between 4 N and 5 N to get a more accurate value.

4) You could also make more predictions based on your conclusion, then further experiments could be carried out to test them.

When suggesting improvements to the investigation, always make sure that you say why you think this would make the results better.

Always look for ways to improve your investigations...

So there you have it — Working Scientifically. Make sure you know this stuff like the back of your hand. It's not just in the lab that you'll need to know how to work scientifically — you can be asked about it in the exams as well.

Apparatus and Techniques

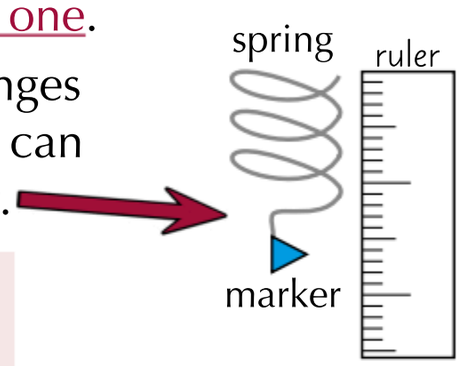
*Get your lab coat on, it's time to find out about the skills you'll need in **experiments**.*

Mass Should Be Measured Using a **Balance**

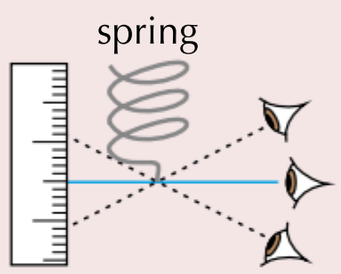
- For a solid object, set the balance to zero, put the object on the scale and read off the mass.
- For a liquid, put an empty container onto the balance then reset the balance to zero. Pour your liquid into the container and record the mass displayed. Easy peasy.

You Can Measure **Most Lengths** with a **Ruler**

- In most cases a bog-standard centimetre ruler can be used to measure length, but metre rulers are handy for large distances, while micrometers are used for measuring tiny things like the diameter of a wire.
- If you're using a ruler, it should always be parallel to what you want to measure.
- For something where it's tricky to measure just one accurately (e.g. water ripples, p.93), you can measure the length of ten of them and then divide to find the length of one.
- If you're taking multiple measurements of the same object (e.g. to measure changes in length) make sure you always measure from the same point on the object. It can help to draw or stick small markers onto the object to line up your ruler against.

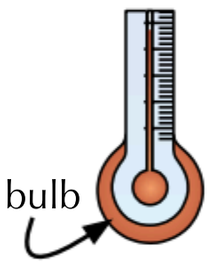


Make sure the ruler and the object are always at eye level. This reduces the effect of parallax on your results. Parallax is where a measurement appears to change based on the angle you're looking from. The blue line shows the correct length of the spring — a measurement taken at eye level.



When you're reading off a scale, use the value of the nearest mark on the scale.

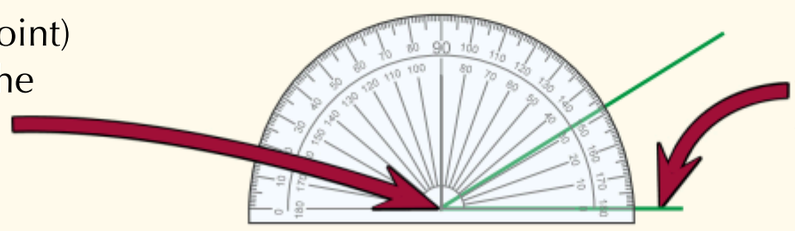
Measure **Temperature** Accurately with a **Thermometer**



1) Submerge the bulb of the thermometer completely in whatever it is you're measuring.
2) Wait for the temperature to stabilise before you take your initial reading.
3) Again, read your measurement off the scale on a thermometer at eye level.

Use a **Protractor** to Find **Angles**

Align the vertex (point) of the angle with the mark in the centre of the protractor.



Line up the base line of the protractor with one line that forms the angle and then measure the angle of the other line using the scale on the protractor.

1) If the lines are very thick, you should measure the angle from the centre of the lines. Using a sharp pencil to trace light rays or draw diagrams helps to reduce errors when measuring angles.
2) If the lines are too short to measure easily, you may have to extend them. Again, make sure you use a sharp pencil and a ruler to do this.

You May Have to Measure the **Time Taken** for a Change

- You should use a stopwatch to time most experiments — they're more accurate than regular watches. Always make sure you start and stop the stopwatch at exactly the right time.
- You might be able to use a light gate instead (p.149). This will reduce the errors in your experiment.

Make sure to take measurements at eye level to avoid parallax...

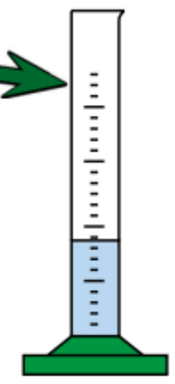
Some of this might feel a bit obvious, but it's really important that you know the right kit to use, and how to use it.

More Apparatus and Techniques

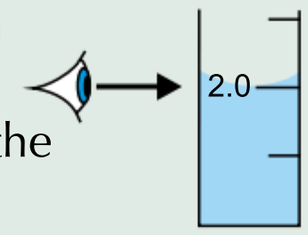
*More equipment on this page, all for measuring **volume**. Then a bit on how to **work safely** in the lab.*

Measuring Cylinders Measure Liquid Volume

- Measuring cylinders come in all different sizes. Make sure you choose one that's the right size for the measurement you want to make.
- It's no good using a huge 1 litre cylinder to measure out 2 cm^3 of a liquid — the graduations (markings for scale) will be too big and you'll end up with massive errors. It'd be much better to use one that measures up to 10 cm^3.



Always read the volume from the bottom of the meniscus (the curved upper surface of the liquid) when it's at eye level.



You might also use a pipette in the lab to suck up and transfer liquid between containers, or to measure small amounts of liquid accurately.

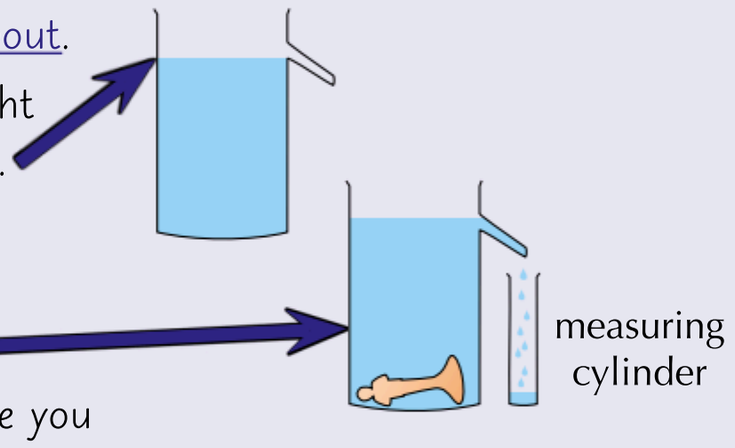
Eureka Cans Measure the Volume of Irregular Solids

A eureka can is essentially a beaker with a spout. You can use one (along with a measuring cylinder) to find the volume of a solid object.

Check out p.41 for more on using a eureka can in experiments.

How to use a eureka can

1) Fill the eureka can with water so the water level is above the spout.
2) Let the excess drain out, leaving the water level just at the height of the spout (this will help make sure your results are accurate).
3) Position a measuring cylinder under the end of the spout. When you place an object in the beaker, it causes the water level to rise and water to flow out of the spout.
4) Make sure you wait until the spout has stopped dripping before you measure the volume of the water in the measuring cylinder. And eureka! You know the object's volume.



Be Careful When You Do Experiments

There are always hazards in any experiment, so before you start an experiment you should read and follow any safety precautions to do with your method or the apparatus you're using:

MASSES AND CLAMP STANDS	Use masses that aren't too heavy, so they don't break the equipment. Use pulleys of a sensible length so that any hanging masses won't hit the floor during the experiment. Stop masses and equipment falling by using clamp stands.
HEATING EQUIPMENT	Let hot materials cool before moving them, or wear insulated gloves while handling them. If you're using an immersion heater to heat liquids, you should always let it dry out in air, just in case any liquid has leaked inside the heater.
LASERS	Always wear laser safety goggles and never look directly into the laser or shine it towards another person. Turn the laser off if it's not needed to avoid accidents.
ELECTRONICS	Use a low enough voltage and current to prevent wires overheating (and potentially melting) and avoid damage to components, like blowing a filament bulb.

You also need to be aware of general safety in the lab — handle glassware carefully so it doesn't break, don't stick your fingers in sockets and avoid touching frayed wires. That kind of thing.

BEWARE — hazardous physics experiments...

Before you carry out an experiment, it's important to consider all of the hazards. Hazards can be anything from lasers to weights. Whatever the hazards, make sure you know all the safety precautions you should follow to keep yourself safe.

Working with Electronics

Electrical devices *are used in a bunch of* ***experiments****, so make sure you know how to use them.*

You Have to Interpret **Circuit Diagrams**

Before you get cracking on an experiment involving any kind of electrical devices, you have to plan and build your circuit using a circuit diagram. Make sure you know all of the circuit symbols on page 21 so you're not stumped before you've even started.

There Are a Couple of Ways to Measure **Potential Difference** and **Current**

Voltmeters Measure Potential Difference

1) If you're using a voltmeter, choose one with the most appropriate unit (e.g. V or mV). If you're using a digital voltmeter, you might be able to switch between units.
2) Connect the voltmeter in parallel (p.22) across the component you want to test. The wires that come with a voltmeter are usually red (positive) and black (negative). These go into the red and black coloured ports on the voltmeter. Funnily enough.
3) Then simply read the potential difference from the scale (or from the screen if it's digital).

Ammeters Measure Current

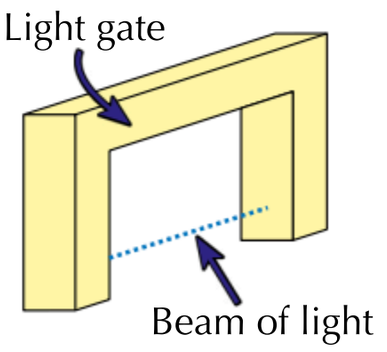
1) Just like with voltmeters, choose the ammeter with the most appropriate unit.
2) Connect the ammeter in series (p.22) with the component you want to test, making sure they're both on the same branch. Again, they usually have red and black ports.
3) Read off the current shown on the scale or screen.

Turn your circuit off between readings to prevent wires heating up and affecting your results (p.23).

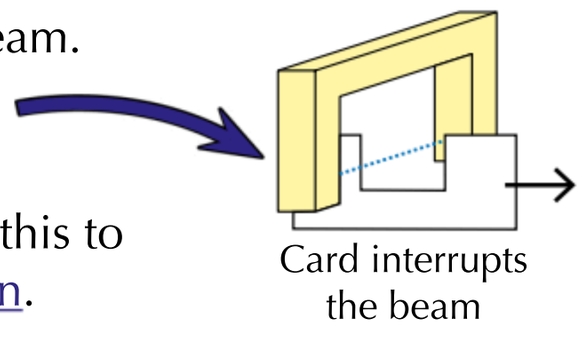
Multimeters Measure Both

- A multimeter is a single device that can measure pd, current and (usually) resistance.
- If you want to find potential difference, connect the multimeter in parallel and make sure the red wire is plugged into the port that has a 'V' (for volts).
- To find current, connect the multimeter in series and use the port labelled 'A' or 'mA'.
- The dial on the multimeter should then be turned to the relevant section, e.g. to 'A' to measure current in amps. The scale or screen will display the value you're measuring.

Light Gates Measure **Speed** and **Acceleration**



- A light gate (p.81) sends a beam of light from one side of the gate to a detector on the other side. When something passes through the gate, the beam of light is interrupted. The light gate then measures how long the beam was undetected.
- To find the speed of an object, connect the light gate to a computer. Measure the length of the object and input this using the software. It will then automatically calculate the speed of the object as it passes through the beam.
- To measure acceleration, use an object that interrupts the signal twice in a short period of time, e.g. a piece of card with a gap cut into the middle.
- The light gate measures the speed for each section of the object and uses this to calculate its acceleration. This can then be read from the computer screen.



Don't get your wires in a tangle when you're using circuits...

When you're dealing with voltmeters, ammeters and multimeters, make sure that you wire them into your circuit correctly. Just remember, the red wires should go into the red ports and the black wires should go into the black ports.

Practice Exams

Once you've been through all the questions in this book, you should feel pretty confident about the exams. As final preparation, here is a set of **practice exams** to really get you set for the real thing. The time allowed for each paper is 1 hour 45 minutes. These papers are designed to give you the best possible preparation for your exams.

CGP Practice Exam Paper GCSE Physics

GCSE Physics

Paper 1

Higher Tier

In addition to this paper you should have:
- A ruler.
- A calculator.
- The Physics Equation Sheet (on page 216).

Centre name					
Centre number					
Candidate number					

Time allowed:
- 1 hour 45 minutes

Surname
Other names
Candidate signature

Instructions to candidates
- Write your name and other details in the spaces provided above.
- Answer **all** questions in the spaces provided.
- Do all rough work on the paper.
- Cross out any work you do not want to be marked.

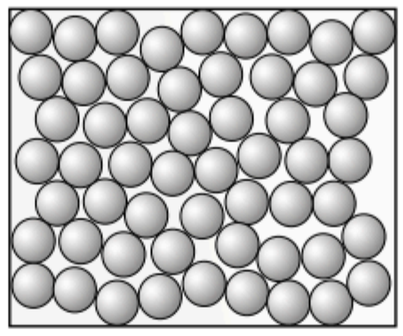
Information for candidates
- The marks available are given in brackets at the end of each question.
- There are 100 marks available for this paper.
- You are allowed to use a calculator.
- You should use good English and present your answers in a clear and organised way.

Advice to candidates
- In calculations show clearly how you worked out your answers.

For examiner's use							
Q	Attempt Nº			Q	Attempt Nº		
	1	2	3		1	2	3
1				8			
2				9			
3				10			
4				11			
5				12			
6				13			
7							
Total							

1 A representation of the particles of a substance is shown in **Figure 1**.

Figure 1



1.1 Name the state of matter of the substance in **Figure 1**.

.....

[1 mark]

1.2 The substance in **Figure 1** is heated.
Describe what happens to the internal energy of the substance when it is heated.
Explain why this occurs.

.....

.....

.....

[2 marks]

A student is carrying out an experiment into changes of state.

1.3 The student heats a sample of solid wax until it becomes a liquid.
Name this change of state.

.....

[1 mark]

The student wishes to find the density of the solid wax.
A sample of solid wax has a mass of 0.36 kg, and a volume of 4.0×10^{-4} m^3.

Use the Physics Equations Sheet on page 216 to answer questions **1.4** and **1.5**.

1.4 Write down the equation that links mass, volume and density.

.....

[1 mark]

1.5 Calculate the density of the wax.

.....

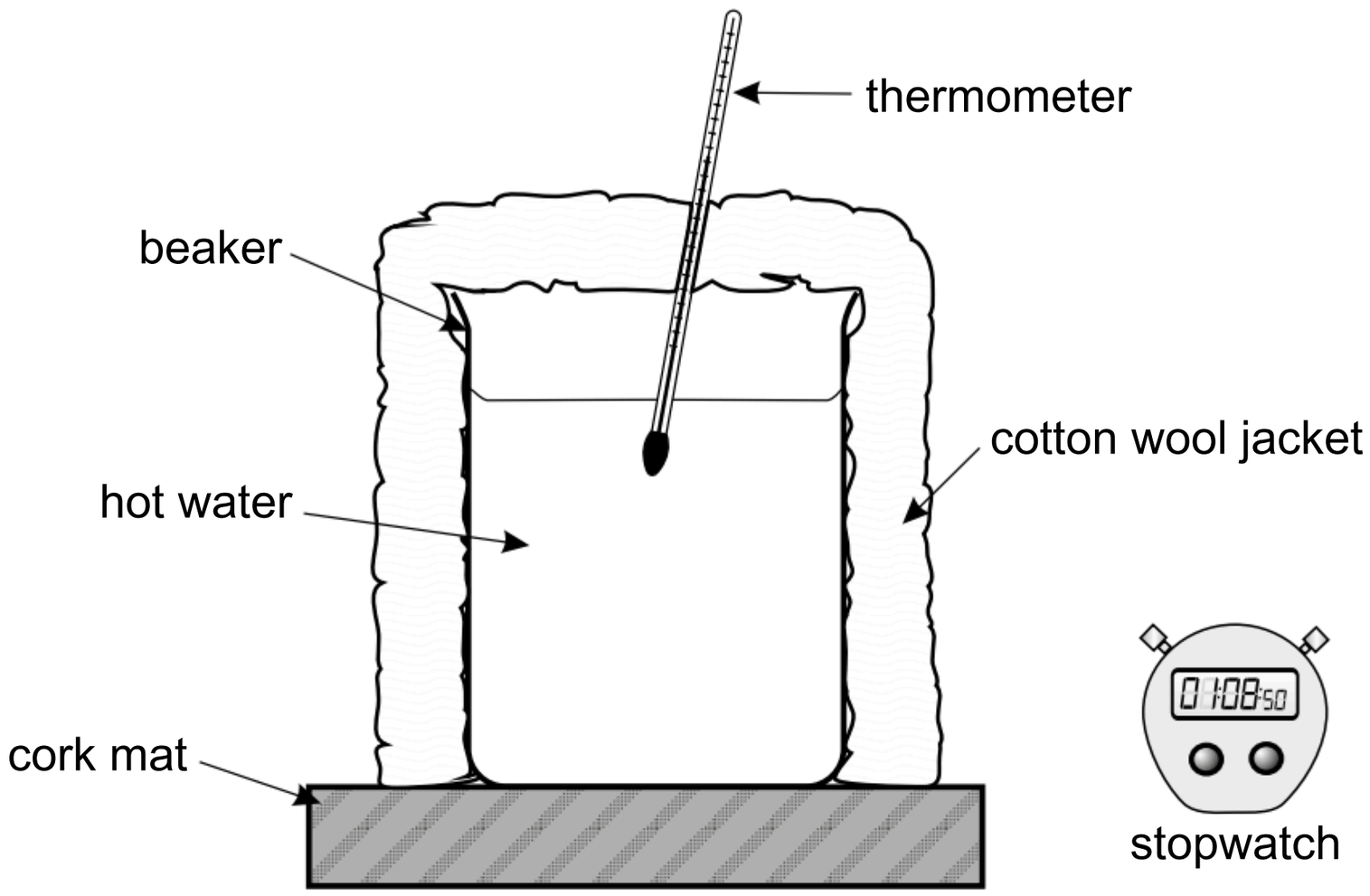
Density = ... kg/m^3

[2 marks]

Turn over for the next question

2 A student wanted to model how the thickness of an insulating layer affects how quickly the contents of a hot water tank cools. To model this system in the lab, the student carried out an investigation to test how the thickness of a cotton wool jacket affected the rate of cooling of a beaker of hot water.
The apparatus the student used is shown in **Figure 2**.

Figure 2



2.1 Give **one** strength and **one** limitation of using the apparatus shown in **Figure 2** to model a hot water tank.

Strength = ...

.....

Limitation = ...

.....

[2 marks]

2.2 State the **independent variable** in this investigation.

.....

[1 mark]

2.3 State **one** control variable in this investigation.

.....

.....

[1 mark]

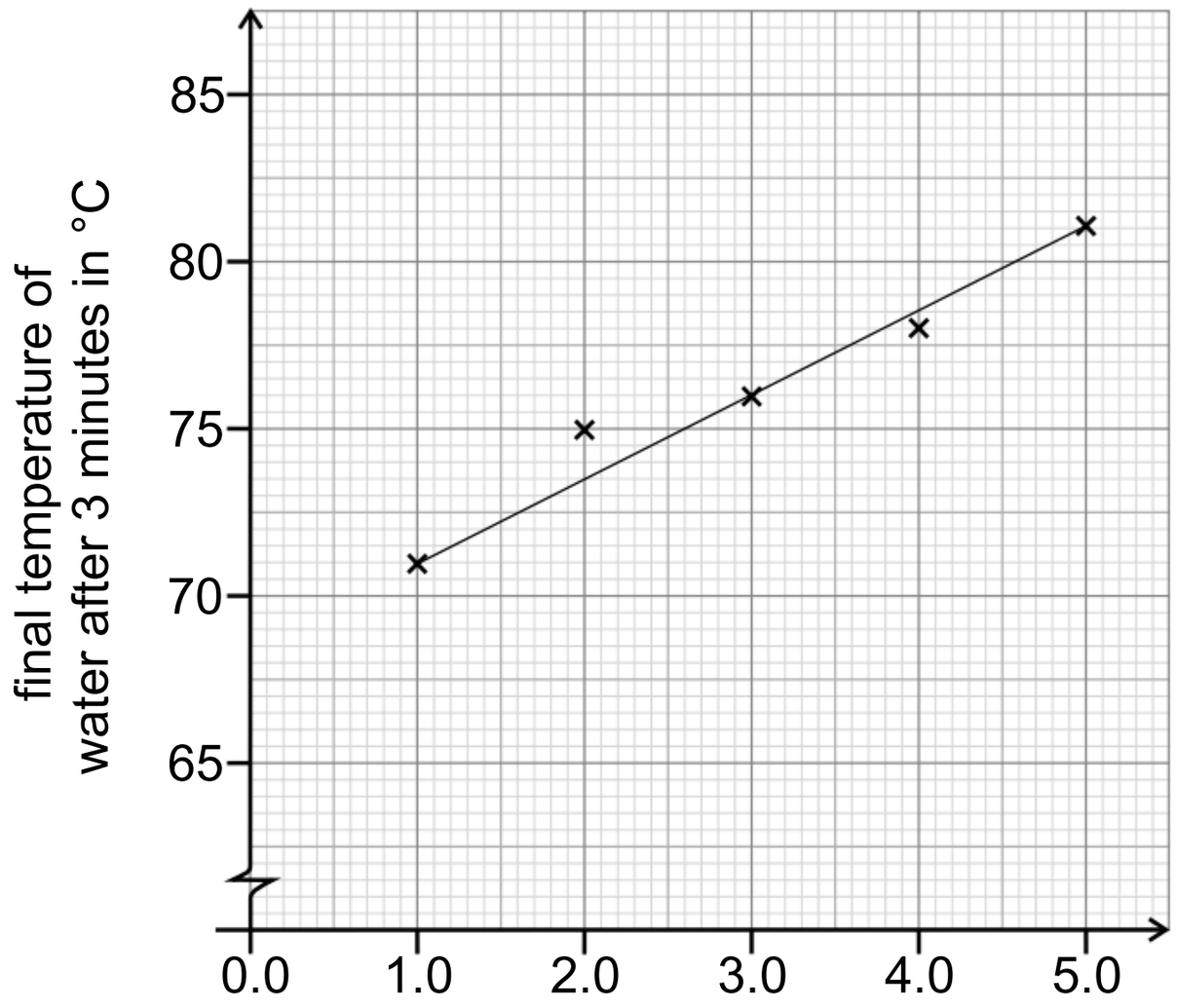
2.4 The student did the investigation again and got very similar results.
Which **one** of the following describes the student's results?
Tick **one** box.

- ☐ repeatable
- ☐ systematic
- ☐ anomalous

[1 mark]

The student's results are shown on the graph in **Figure 3**.

Figure 3



2.5 What conclusion can be made from the results in **Figure 3**?

.....

.....

[1 mark]

Turn over for the next question

3 The model of the atom has developed over time.

Figure 4 shows an early model of the atom, Model X, and a currently used model of the atom, Model Y.

Model X Model Y

Explain how experiments and scientific discoveries caused our understanding of the atom to develop from Model X to Model Y.
Your answer should include descriptions of the models of the atom shown in **Figure 4**.

.....

.....

.....

.....

.....

.....

.....

.....

.....

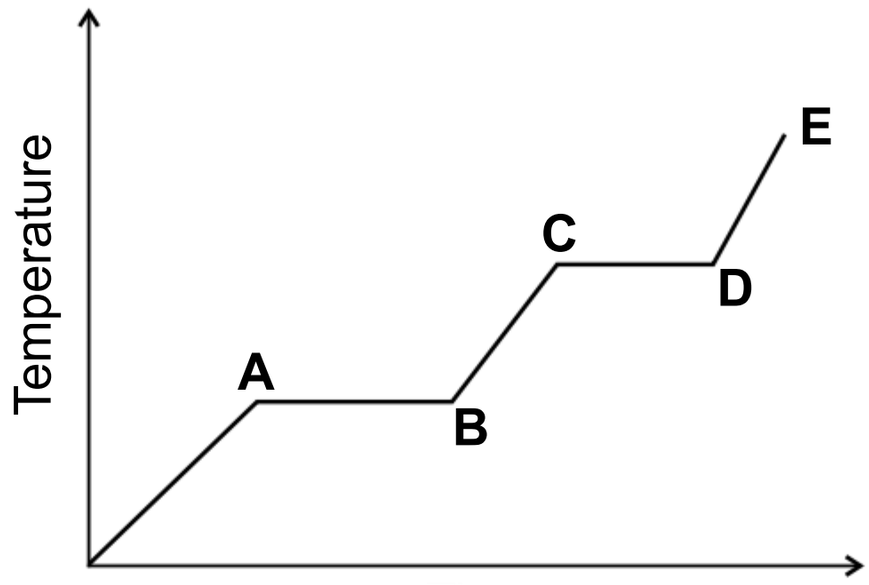
.....

.....

[6 marks]

4 **Figure 5** shows a graph of temperature against time for a substance as it is being continually heated.

Figure 5



4.1 Identify the state of matter of the substance at point **E**.

.....

[1 mark]

4.2 Which point represents the **boiling point**?
Tick **one** box.

- ☐ **A**
- ☐ **B**
- ☐ **C**

[1 mark]

4.3 Explain the shape of the line in **Figure 5** between points **A** and **B**.

.....

.....

.....

.....

[2 marks]

4.4 69 000 J of energy is transferred to 1.5 kg of the substance to completely melt it without changing its temperature.
Calculate the specific latent heat of fusion of the substance.
Use the correct equation from the Physics Equation Sheet on page 216.

.....

.....

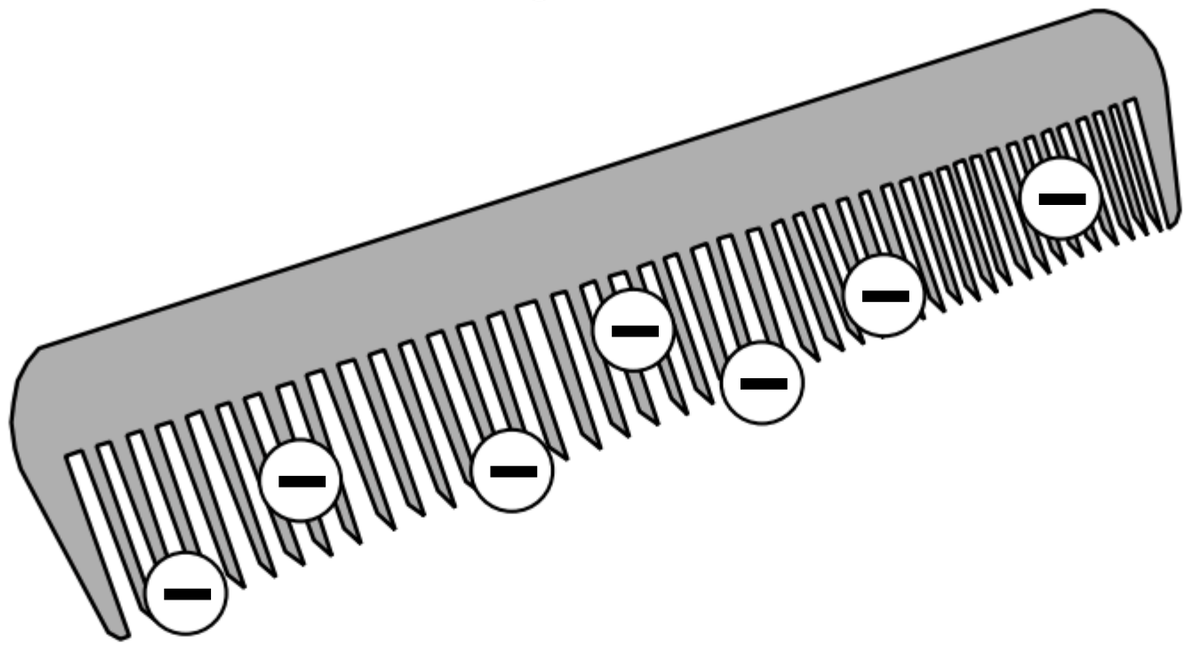
.....

Specific latent heat of fusion = ... J/kg
[3 marks]

Turn over for the next question

5 A scientist is experimenting with static electricity. The scientist's hair is combed fifty times with the same comb. Afterwards, some of the hairs on the scientist's head stand on end away from each other. The comb becomes negatively charged with a charge of –0.2 nC.

Figure 6



5.1 Explain why the comb has become negatively charged.

.....

.....

[1 mark]

5.2 Explain why some of the scientist's hairs stand on end away from each other.

.....

.....

.....

[2 marks]

5.3 Give the total charge on the scientist's hair.

Charge = ... nC

[1 mark]

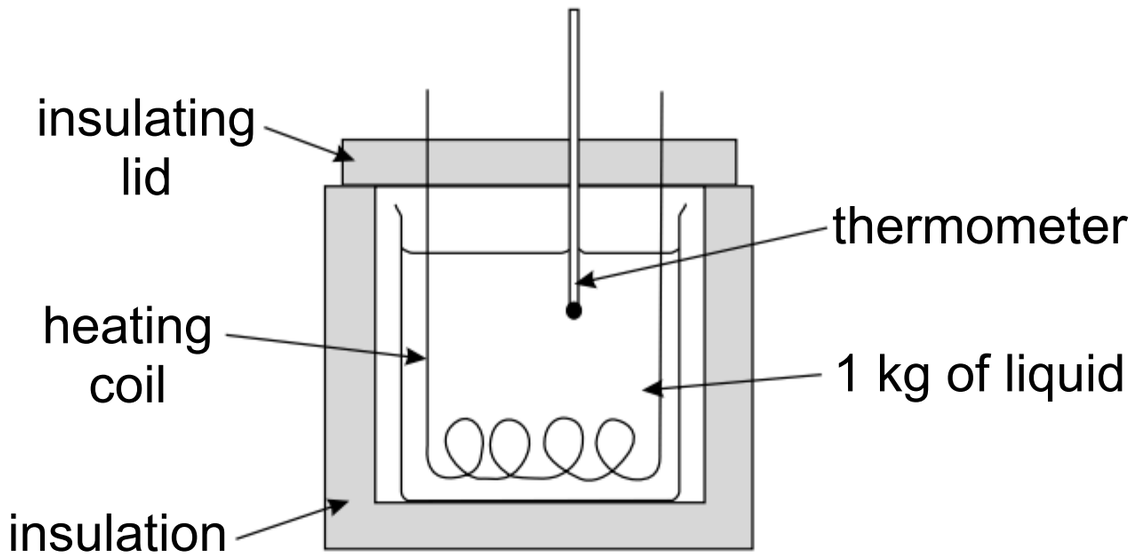
5.4 What will happen if the comb is brought close to the scientist's hair?
Tick **one** box.

- ☐ It will attract the hair.
- ☐ It will repel the hair.
- ☐ No interaction will be observed.

[1 mark]

6 A student carried out an investigation into the specific heat capacity of liquids using the apparatus shown in **Figure 7**. The student used identical electric heating coils to heat a beaker of water and a beaker of oil, and used exactly 1 kg of each liquid.

Figure 7



The student recorded the temperature of both the liquids before heating, and then again after ten minutes of heating. The student's results are shown in **Table 1**.

Table 1

	Water	**Oil**
Initial temperature in °C	18	18
Final temperature in °C	48	93

6.1 State what is meant by 'specific heat capacity'.

.....

.....

[1 mark]

6.2 During the experiment, the heating coil transferred 126 kJ of energy to each liquid.
Use the data from the experiment to calculate the specific heat capacity of the oil.
Use the correct equation from the Physics Equation Sheet on page 216.

.....

.....

.....

Specific heat capacity of oil = ... J/kg°C
[4 marks]

Question 6 continues on the next page

6.3 Both oil and water can be used in heating systems.
With reference to the specific heat capacities of oil and water, explain why most heating systems use water rather than oil.

.....

.....

.....

[2 marks]

7 **Table 2** gives details of some isotopes.

Table 2

Isotope	Symbol	Type of decay
Radium-226	$^{226}_{88}Ra$	alpha
Radon-222	$^{222}_{86}Rn$	alpha
Radon-224	$^{224}_{86}Rn$	beta
Bismuth-210	$^{210}_{83}Bi$	alpha, beta
Bismuth-214	$^{214}_{83}Bi$	alpha, beta
Lead-210	$^{210}_{82}Pb$	beta

7.1 Calculate the number of neutrons in a bismuth-214 nucleus.

.....

[1 mark]

7.2 Using data from **Table 2**, complete the equations in **Figure 8** to show how the following isotopes decay.

Figure 8

$^{226}_{88}Ra \longrightarrow$ [..... .....] + [$^{4}_{2}He$]

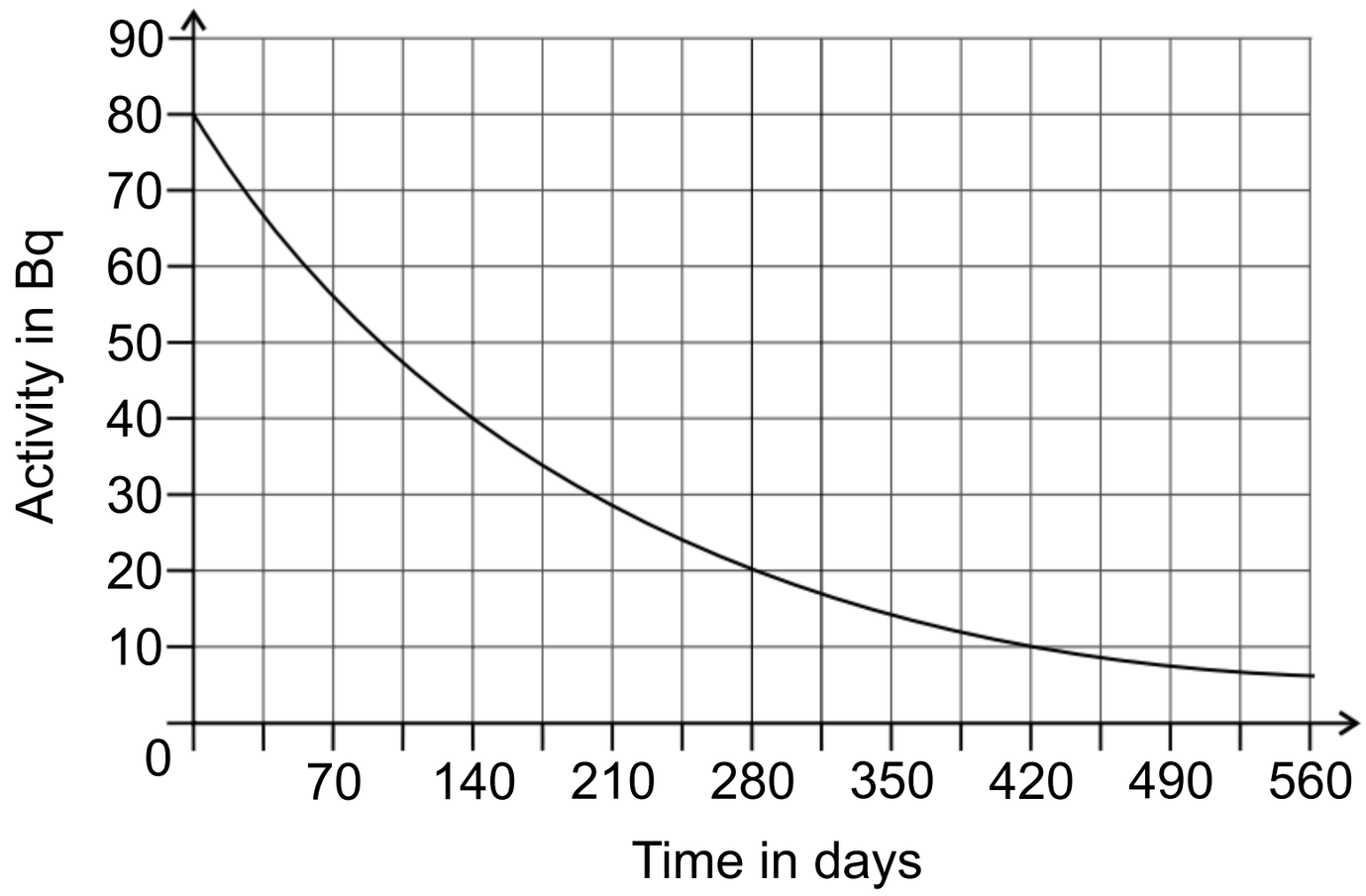
$^{210}_{82}Pb \longrightarrow$ [..... .....] + [$^{0}_{-1}e$]

[2 marks]

Question 7 continues on the next page

Figure 9 shows the activity-time graph of a sample of polonium-210.

Figure 9



7.3 Using the graph in **Figure 9**, determine the time it takes for the activity of the sample to drop from 80 Bq to 10 Bq.

Time taken = ... days

[1 mark]

7.4 Determine the half-life of polonium-210.

Half-life = ... days

[1 mark]

7.5 Polonium-210 emits alpha radiation.
Ingesting any amount of an alpha source can be very harmful.
Explain why sources of alpha radiation are much more dangerous inside the body than outside the body.

.....

.....

.....

.....

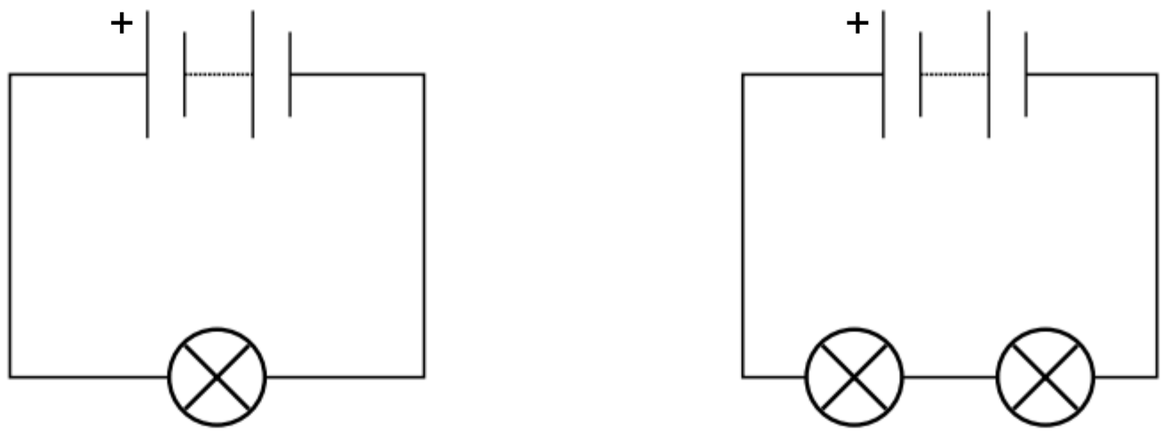
.....

.....

[4 marks]

8 A student is investigating the two electrical circuits shown in **Figure 10**.
All the lamps and batteries used are identical.

Figure 10



Circuit **A** Circuit **B**

8.1 Compare the current and total resistance in circuits **A** and **B**.

.....

.....

.....

[2 marks]

The student adds an ammeter and a voltmeter to circuit **A**.
They show readings of 0.30 A and 11 V respectively.

Use the Physics Equations Sheet on page 216 to answer questions **8.2** and **8.3**.

8.2 State the equation that links power, current and potential difference.

.....

[1 mark]

8.3 Calculate the power of the lamp.

.....

.....

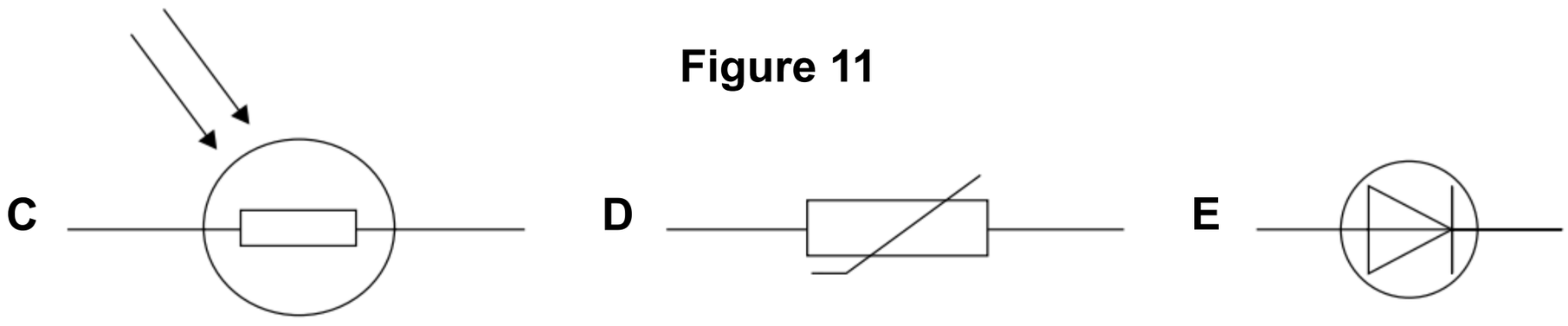
.....

Power = ... W
[2 marks]

Question 8 continues on the next page

The student considers adding one of the components in **Figure 11** to circuit **A**.

Figure 11



8.4 Describe what happens to the resistance of component **C** as the intensity of light that falls on it increases.

.....

.....

[1 mark]

8.5 Give an application for component **D**.

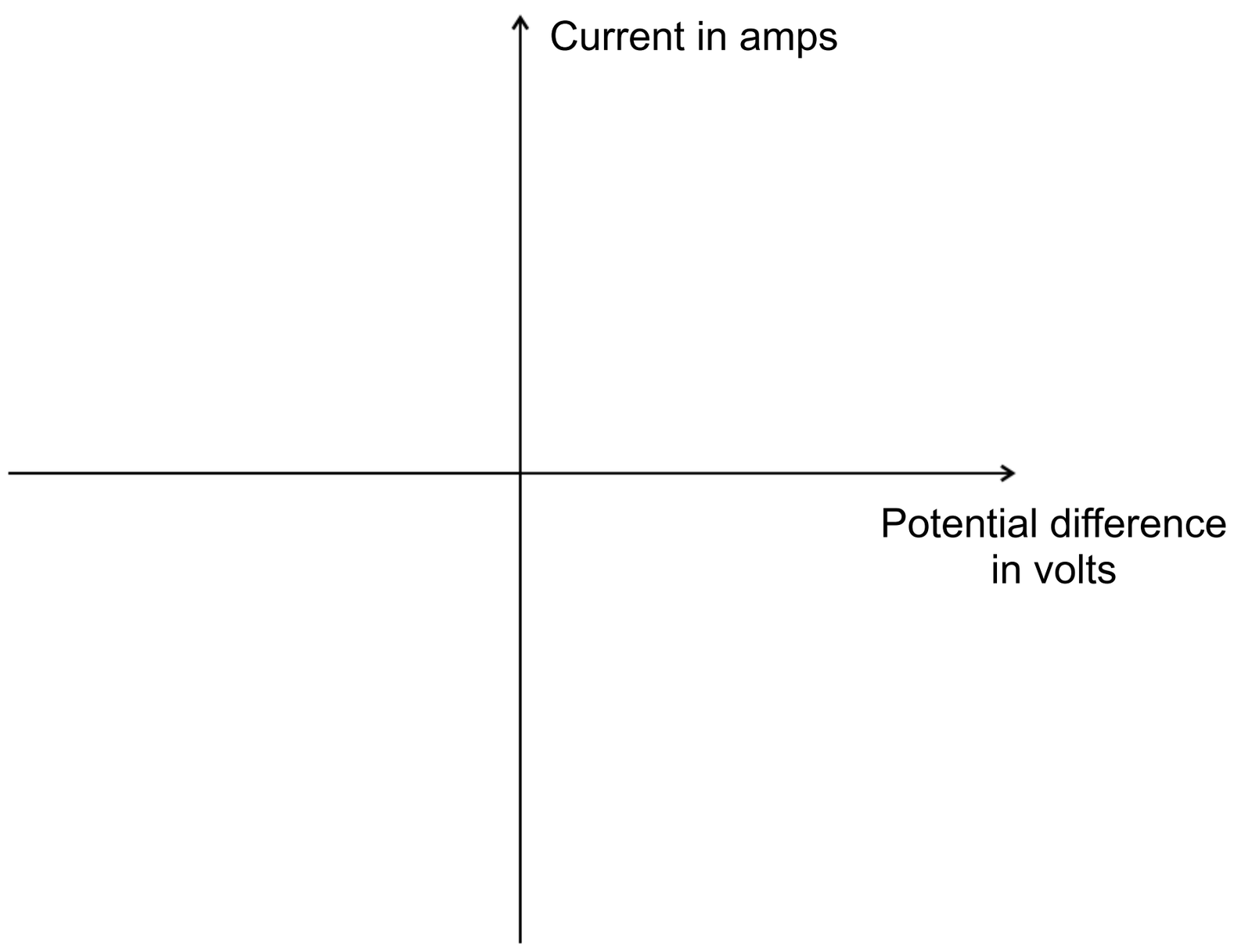
.....

.....

[1 mark]

8.6 On the axes in **Figure 12**, sketch the *I*-*V* characteristic of component **E**.

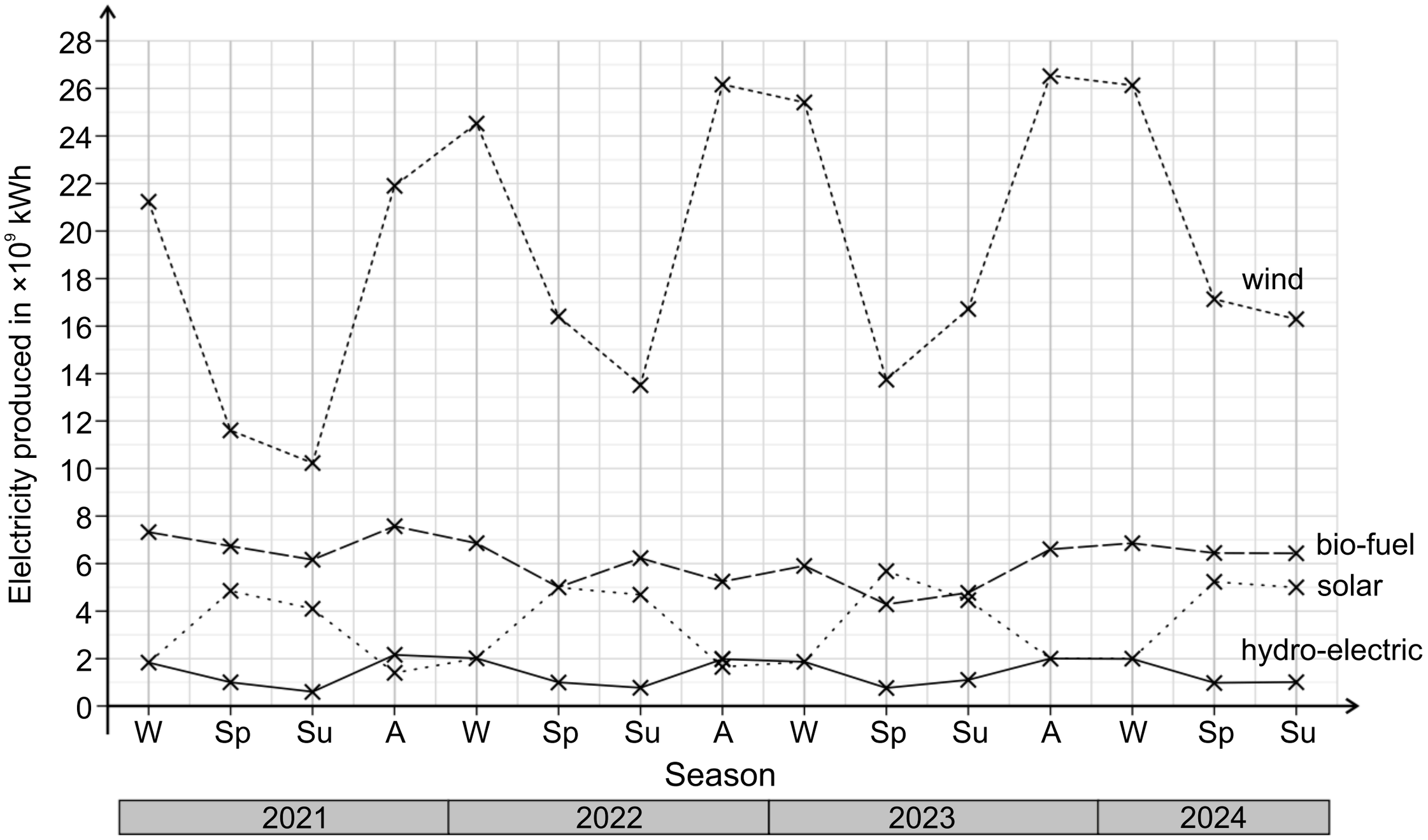
Figure 12



[1 mark]

9 **Figure 13** shows the amount of electricity generated by different renewable energy resources in the UK each season between 2021 and 2024.
A kilowatt hour (kWh) is a unit of energy equal to 3.6×10^6 J.

Figure 13



Key:			
W — Winter	Sp — Spring	Su — Summer	A — Autumn

9.1 Using **Figure 13**, determine the amount of electricity generated by solar power and by hydro-electric power in summer 2024.

Solar = ... kWh

Hydro-electric = ... kWh

[2 marks]

9.2 Using **Figure 13**, suggest which renewable energy resource usually provides the largest amount of electricity to the UK.

.....

[1 mark]

Question 9 continues on the next page

9.3 **Figure 13** shows that the amount of electricity generated from solar power during summer is always larger than the amount generated during winter of the same year.
Suggest a reason for this.

.....

.....

[1 mark]

9.4 Some electricity in the UK is generated from non-renewable energy resources.
Give **one** advantage and **one** disadvantage of using non-renewable energy resources to generate electricity.

Advantage = ...

.....

Disadvantage = ...

.....

[2 marks]

9.5 Give **two** limitations to increasing the amount of electricity produced from renewable energy resources in the UK.

.....

.....

.....

[2 marks]

9.6 Another renewable energy resource is geothermal power.
Suggest why geothermal power is **not** widely used in the UK.

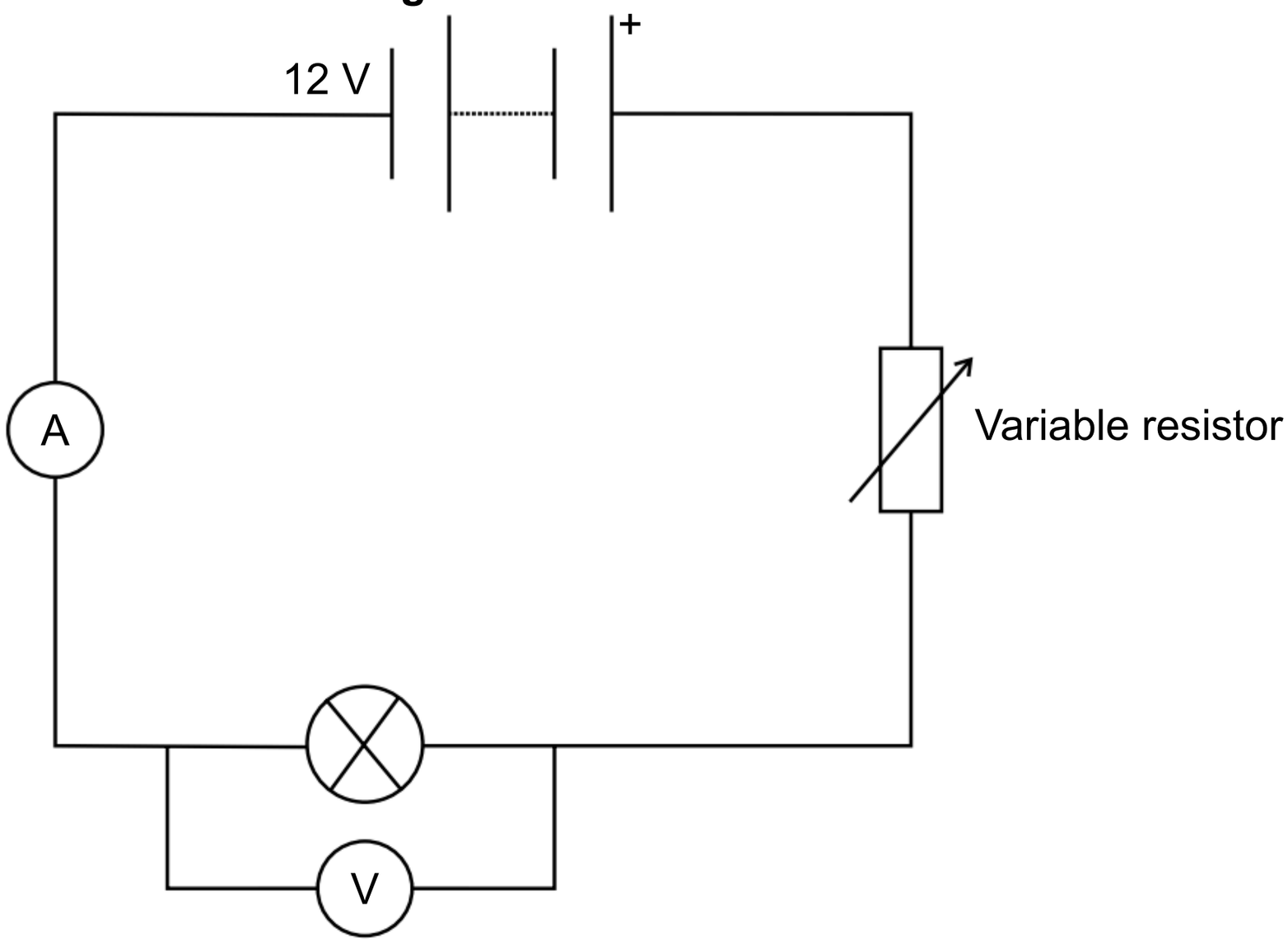
.....

.....

[1 mark]

10 A student wanted to know how the current flowing through a filament lamp changes with the potential difference across it. The student set up the circuit shown in **Figure 14**.

Figure 14



The student used the variable resistor to change the potential difference across the lamp. For each setting of the variable resistor, the student recorded the readings of the voltmeter and the ammeter. **Table 3** shows the student's results.

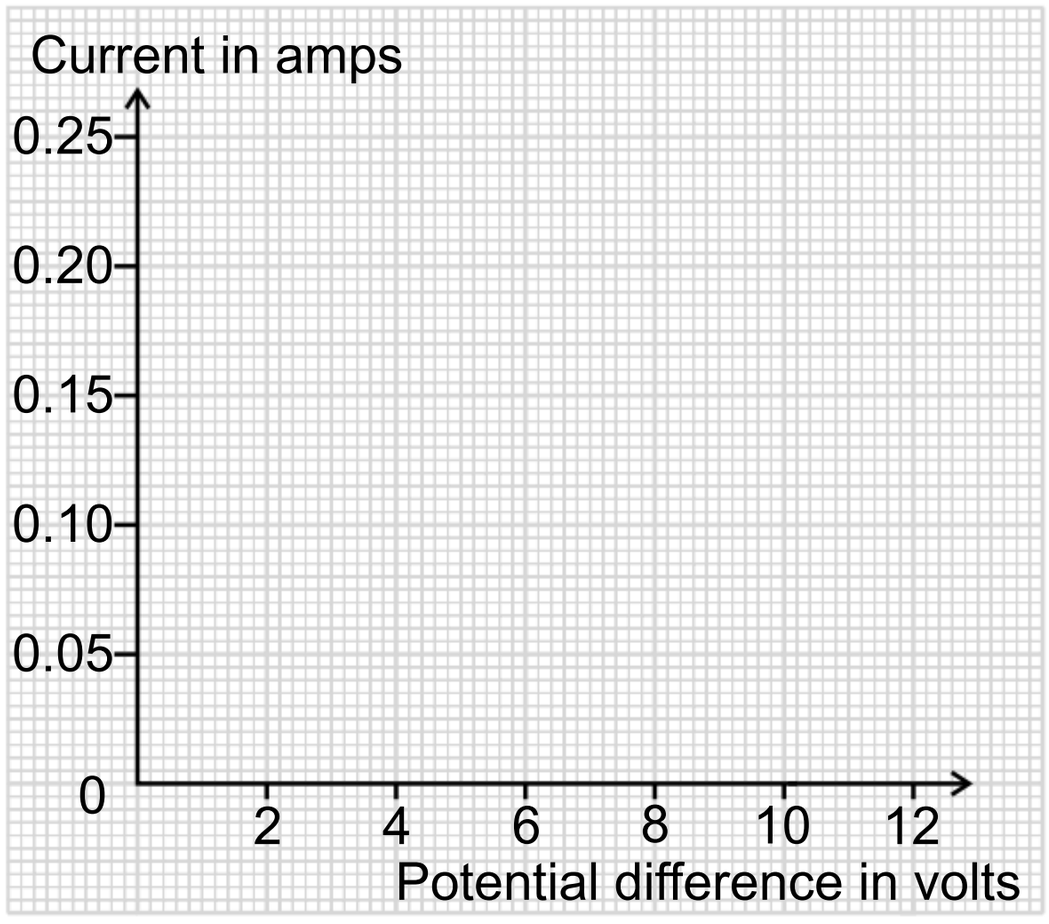
Table 3

Voltmeter reading in volts	Ammeter reading in amps
0.0	0.00
3.0	0.10
5.0	0.14
7.0	0.17
9.0	0.19
11.0	0.21

Question 10 continues on the next page

10.1 Plot the results displayed in **Table 3** on the grid in **Figure 15**.

Figure 15



[2 marks]

10.2 Draw a curve of best fit on your graph on the grid in **Figure 15**.

[1 mark]

10.3 Explain, with reference to energy transfers, what happens to the resistance of the lamp as the current through it increases.

.....

.....

.....

.....

.....

.....

[4 marks]

10.4 The lamp is disconnected from the test circuit, and is connected to a 15 V power supply.
At this potential difference the lamp has a resistance of 60 Ω.
The lamp operates at this potential difference for 180 s.
Calculate the amount of charge which passes through the lamp in this time.
Use the correct equation from the Physics Equation Sheet on page 216.

.....

.....

.....

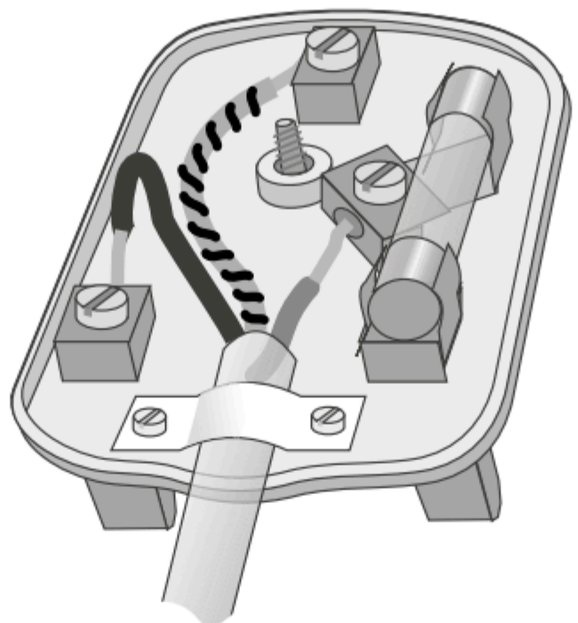
.....

Charge = ... C

[5 marks]

11 **Figure 16** shows the inside of the three-pin plug for an electric kettle.

Figure 16



11.1 Give the colours of the earth wire and the live wire.

Earth wire = ...

Live wire = ...

[2 marks]

11.2 Describe the purpose of the earth wire.

.....

.....

[1 mark]

11.3 Explain how the live wire can be dangerous even when the kettle is switched off.

.....

.....

.....

[2 marks]

11.4 The electric kettle is plugged into an electricity supply and is switched on for 180 s.
While switched on, a current of 10.0 A passes through the kettle.
The resistance of the kettle is 23.0 Ω. Calculate the energy transferred by the kettle.
Use the correct equation from the Physics Equation Sheet on page 216.

.....

.....

.....

Energy transferred = ... J

[4 marks]

Turn over for the next question

12 The National Grid is a network of pylons and cables that are used to transmit large amounts of electrical power throughout the UK.
The National Grid is considered to be an efficient way of transmitting electricity, through its use of devices such as transformers.

Explain how the National Grid can transmit high power electricity efficiently, and provide their consumers with useful electric power.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

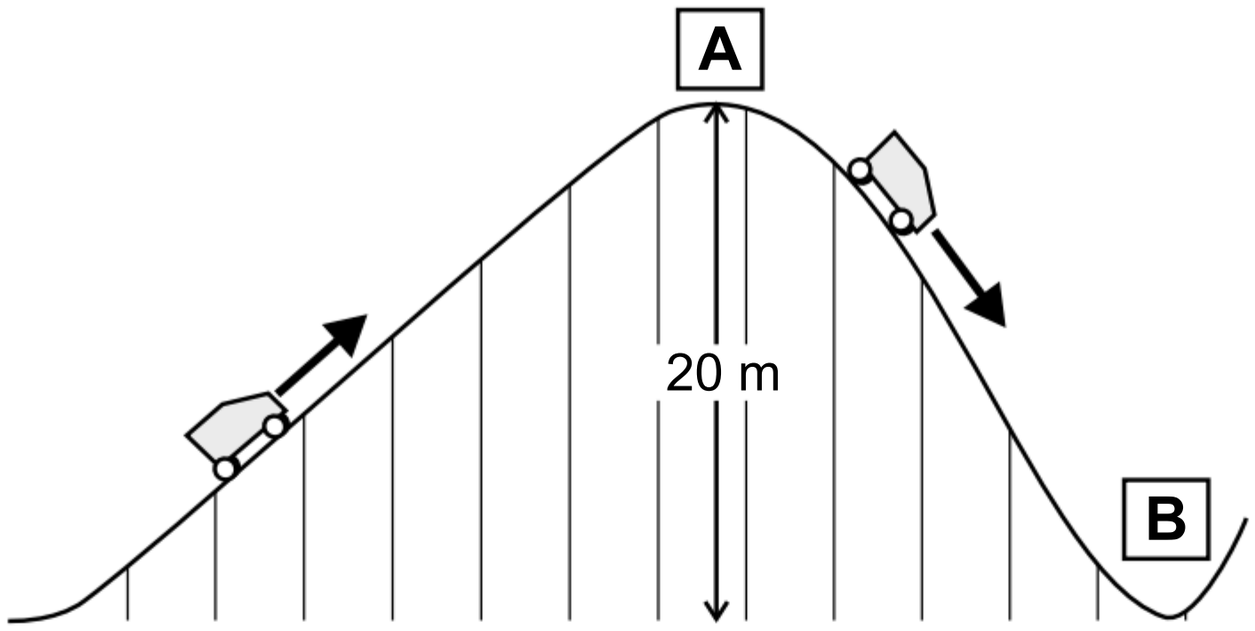
.....

.....

[6 marks]

13 At the start of a roller coaster ride, a carriage is raised through a vertical height of 20 m to point **A**, as shown in **Figure 17**.

Figure 17



The mass of the empty carriage is 600 kg.
The Earth's gravitational field strength is 9.8 N/kg.

Use the Physics Equations Sheet on page 216 to answer questions **13.1**, **13.2** and **13.4**.

13.1 Write down the equation which links gravitational potential energy, mass, the gravitational field strength and height.

.....

[1 mark]

13.2 Calculate the energy transferred (in kJ) to the gravitational potential energy store of the empty carriage as it is lifted to point **A**.
Give your answer to 2 significant figures.

.....

.....

.....

.....

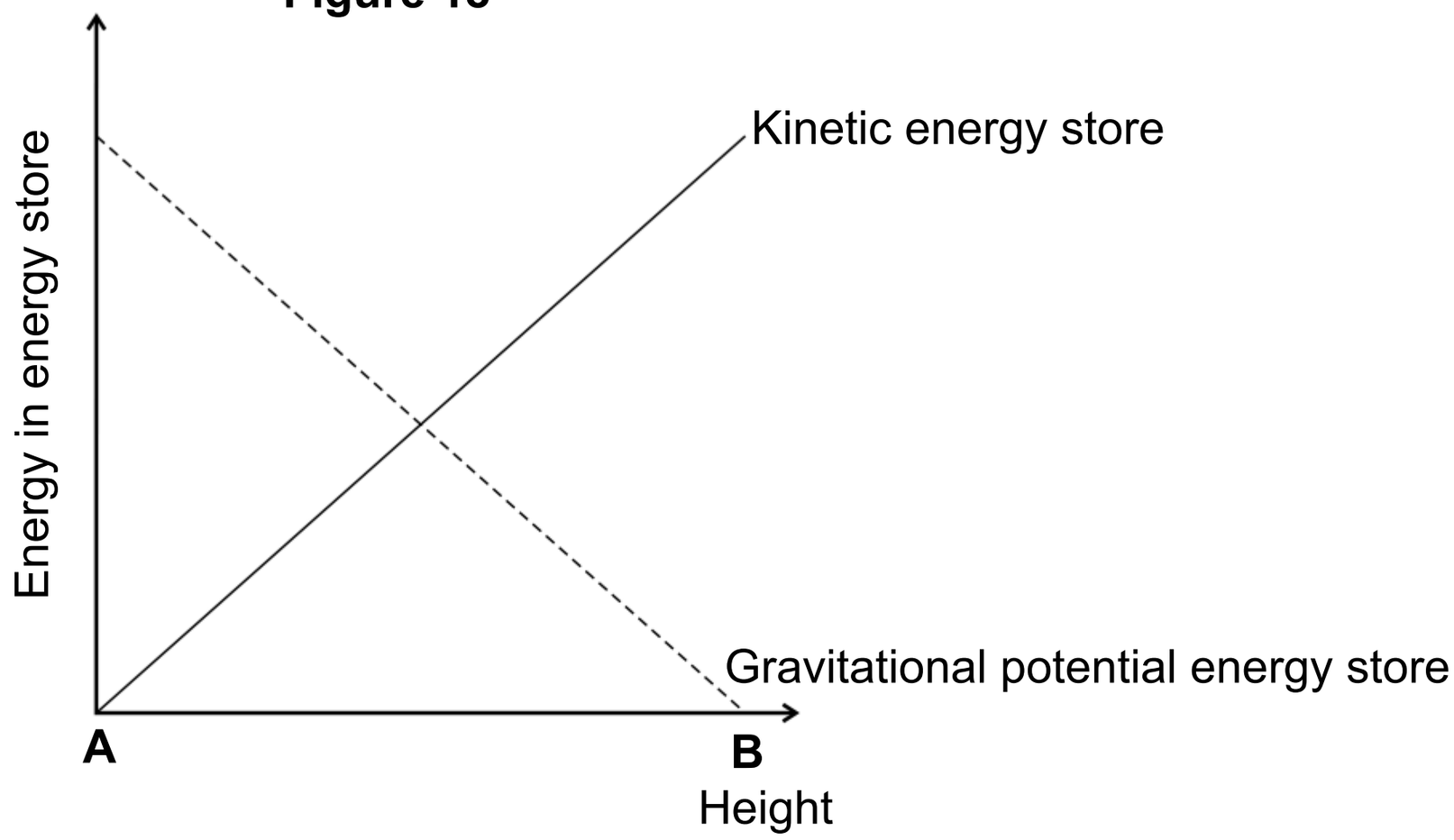
Energy transferred to gravitational potential energy store = ... kJ

[2 marks]

Question 13 continues on the next page

During the ride the carriage drops from point **A** to point **B**, as shown in **Figure 17**.
Figure 18 shows a graph of the energy in the carriage's gravitational potential energy store and kinetic energy store as it travels from point **A** to point **B**, assuming there is no friction or air resistance acting on the carriage.

Figure 18



13.3 Explain the shape of the lines in **Figure 18** in terms of energy transfer and conservation of energy.

.....

.....

.....

.....

[2 marks]

13.4 Point **A** is 20 m above point **B**.
Calculate the speed of the empty carriage as it reaches point **B**, assuming there is no friction or air resistance acting on the carriage.
Use your unrounded answer from **13.2**.
Give your final answer to 2 significant figures.

.....

.....

.....

.....

.....

Speed of carriage at point B = ... m/s

[4 marks]

END OF QUESTIONS

GCSE Physics

Paper 2

Higher Tier

In addition to this paper you should have:
- A ruler.
- A calculator.
- A protractor.
- The Physics Equation Sheet (on page 216).

Centre name					
Centre number					
Candidate number					

Time allowed:
- 1 hour 45 minutes

Surname
Other names
Candidate signature

Instructions to candidates
- Write your name and other details in the spaces provided above.
- Answer **all** questions in the spaces provided.
- Do all rough work on the paper.
- Cross out any work you do not want to be marked.

Information for candidates
- The marks available are given in brackets at the end of each question.
- There are 100 marks available for this paper.
- You are allowed to use a calculator.
- You should use good English and present your answers in a clear and organised way.

Advice to candidates
- In calculations show clearly how you worked out your answers.

For examiner's use							
Q	Attempt Nº			Q	Attempt Nº		
	1	2	3		1	2	3
1				7			
2				8			
3				9			
4				10			
5				11			
6				12			
Total							

1 X-rays can be used in hospitals for medical treatments and diagnoses.
Figure 1 shows X-ray images used for diagnoses.
X-rays are directed at a body part being examined.
A detector is placed behind the body part to detect the X-rays that reach it.

Figure 1

1.1 Explain how X-ray images, like those shown in **Figure 1**, are formed.

.....

.....

[2 marks]

1.2 Explain why precautions must be taken when using X-rays.

.....

.....

[1 mark]

1.3 When radiographers perform X-ray scans in hospitals, they often stand behind a lead screen while the X-rays are being used.
Suggest why the screen is made of lead.

.....

.....

[1 mark]

1.4 Give **two** medical conditions that X-rays can be used to either treat or diagnose.

.....

.....

[2 marks]

2 A remote control uses infrared radiation with a frequency of 2.5×10^{14} Hz to transmit signals to a television.

2.1 State what is meant by the 'frequency' of radiation.

.....

[1 mark]

Use the Physics Equations Sheet on page 216 to answer questions **2.2** and **2.3**.

The speed of infrared radiation travelling through air is approximately 3.0×10^8 m/s.

2.2 Write down the equation which links wave speed, frequency and wavelength.

.....

[1 mark]

2.3 Calculate the wavelength of the radiation.

.....

.....

.....

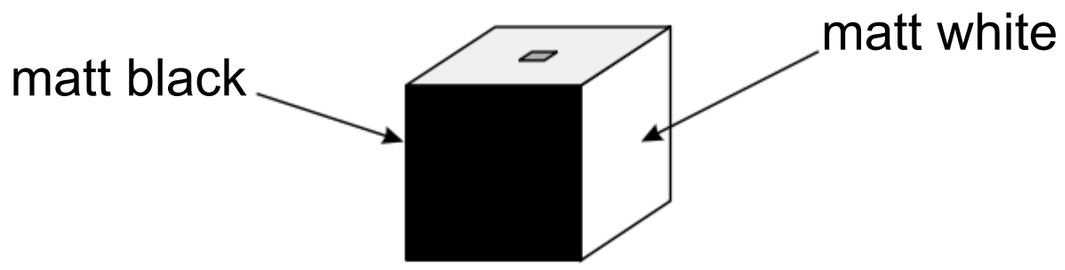
Wavelength = ... m

[3 marks]

Question 2 continues on the next page

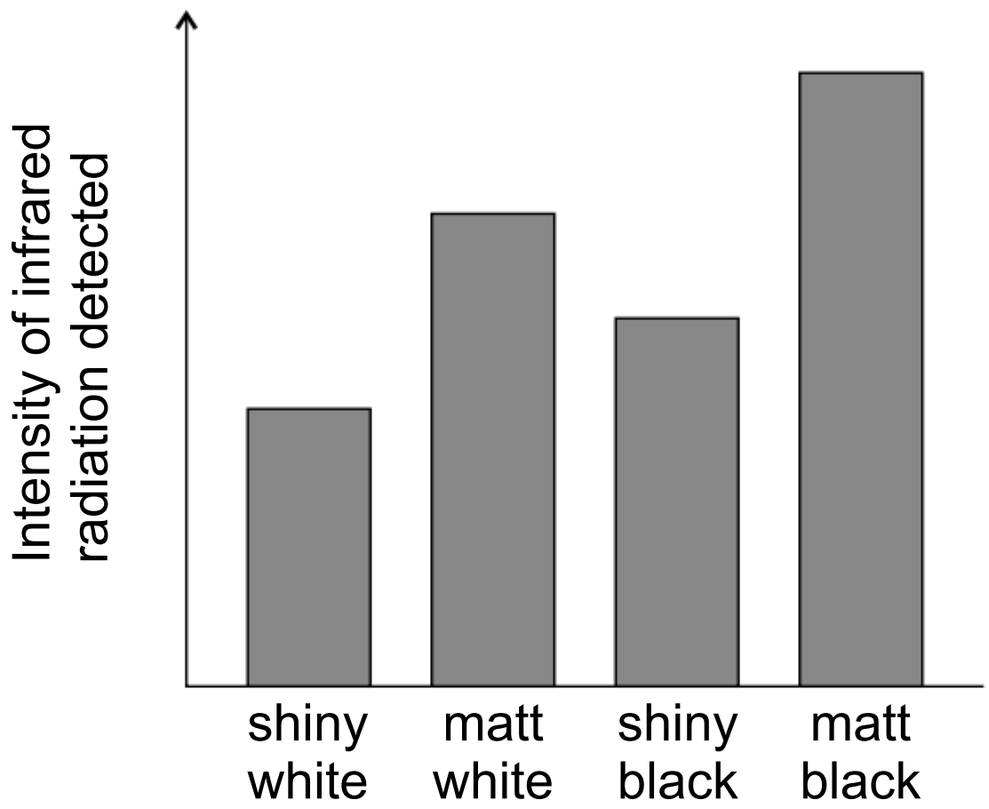
A student investigated the infrared radiation emitted by hot objects. **Figure 2** shows a Leslie cube, which was filled with hot water. The Leslie cube has four coated faces: shiny black, shiny white, matt white and matt black.

Figure 2



The student placed an infrared detector 10 cm from each face, one at a time, and recorded the intensity of infrared radiation detected. The results are shown on the bar chart in **Figure 3**.

Figure 3



2.4 Infrared emission is affected by both the texture and colour of a surface.
Write down a conclusion from the data in **Figure 3** about how each of these features affect infrared emission.

Texture: .....

.....

Colour: .....

.....

[2 marks]

2.5 The student repeated the experiment three times and used the repeat readings to calculate an average for the value of the intensity of infrared radiation detected at each face. Give **one** reason why it is important to do this.

.....

[1 mark]

3 The Sun is a main sequence star.

3.1 Stars are held together by gravitational attraction.
Explain how the Sun remains stable and does not collapse under gravity.

.....

.....

.....

[2 marks]

3.2 The Sun will eventually become a red giant star.
State what the Sun will become immediately after its red giant stage.

.....

[1 mark]

The Earth orbits the Sun. The planet Venus is sometimes called Earth's twin.
It has a very similar mass and diameter to the Earth, but is much hotter.
Table 1 shows some data about Venus and Earth.

Table 1

	Mass in kg	Diameter in km	Orbital radius in km	Orbital speed in km/s	Gravitational field strength in N/kg
Venus	5×10^{24}	12 000	1.1×10^{8}	.....	9
Earth	6×10^{24}	13 000	1.5×10^{8}	30	10

3.3 Using the data from **Table 1**, what is the orbital speed of Venus?
Tick **one** box.

☐ 35 km/s ☐ 30 km/s ☐ 25 km/s

[1 mark]

3.4 Using the data in **Table 1**, compare the weight of the same object when on Venus and Earth. Explain your answer.

.....

.....

.....

.....

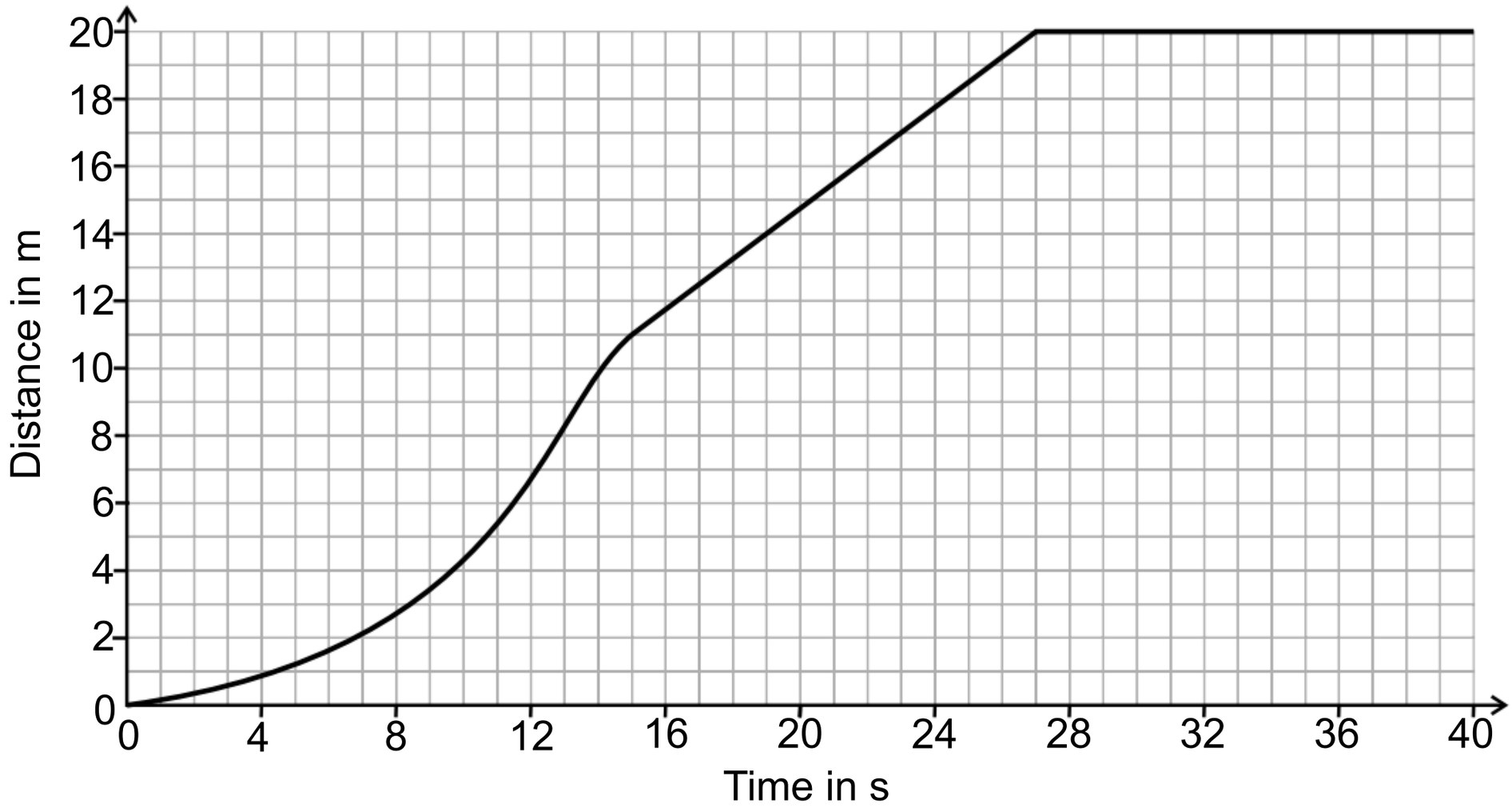
[2 marks]

Turn over for the next question

Turn over ▶

4 A swimmer swims a length of a 20 m swimming pool in a straight line.
The distance-time graph in **Figure 4** shows the swimmer's motion.

Figure 4



4.1 Calculate the time it takes for the swimmer to complete the length.

Time = ... s
[1 mark]

4.2 For part of the swim, the swimmer is travelling at a constant speed.
Work out the time the swimmer spends travelling at a constant speed.

.....

Time = ... s
[1 mark]

4.3 Determine the resultant force on the swimmer when travelling at a constant speed.

Force = ... N
[1 mark]

4.4 Between which of the following distances was the swimmer travelling fastest?
Tick **one** box.

- ☐ Between 14 m and 15 m.
- ☐ Between 9 m and 10 m.
- ☐ Between 0 m and 1 m.

[1 mark]

A camera travels along the length of the pool to film the swimmer's length.
It travels at a constant speed, and reaches the end of the pool in 25 s.

Use the Physics Equations Sheet on page 216 to answer questions **4.5** and **4.6**.

4.5 Write down the equation which links speed, distance and time.

.....

[1 mark]

4.6 Calculate the speed of the camera.

.....

.....

.....

Speed = ... m/s

[3 marks]

4.7 On **Figure 4**, draw a distance-time graph to represent the motion of the camera.

[2 marks]

4.8 The camera cannot film the swim if it is behind the swimmer.
Using **Figure 4**, explain whether the camera will be able to film the swimmer for the whole length.

.....

.....

.....

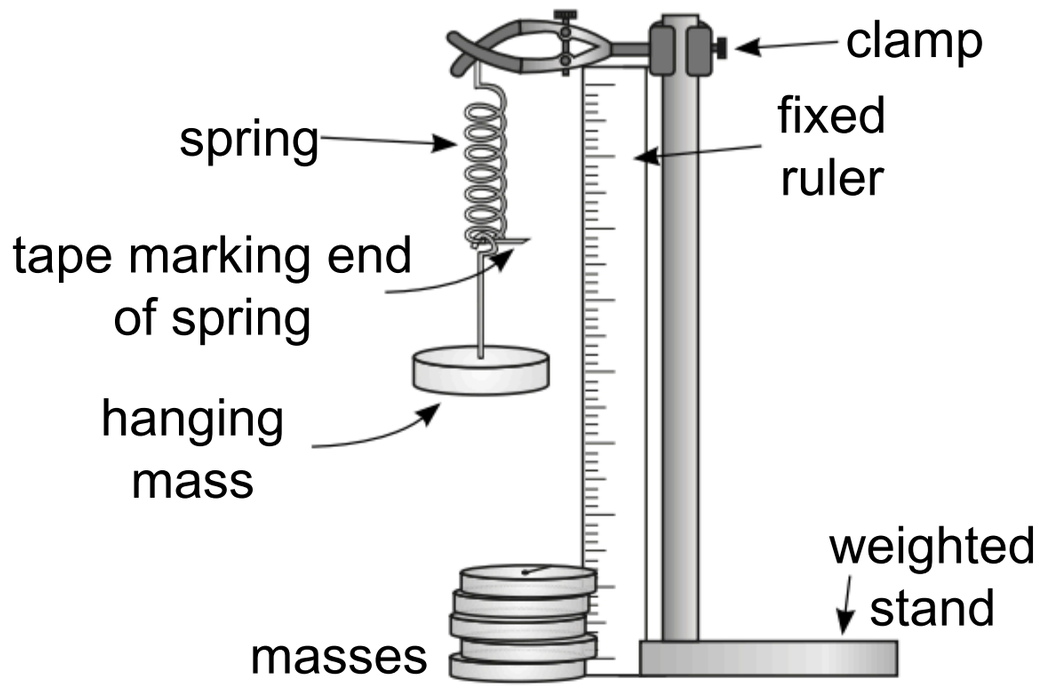
.....

[2 marks]

Turn over for the next question

5 A student is given a set of apparatus, set up as shown in **Figure 5**.

Figure 5



5.1 Name the type of error which may be reduced by the use of the tape marker at the end of the spring.

.....

[1 mark]

5.2 Describe how the student could use the experimental set-up in **Figure 5** to plot a force-extension graph for the spring and find the spring constant.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

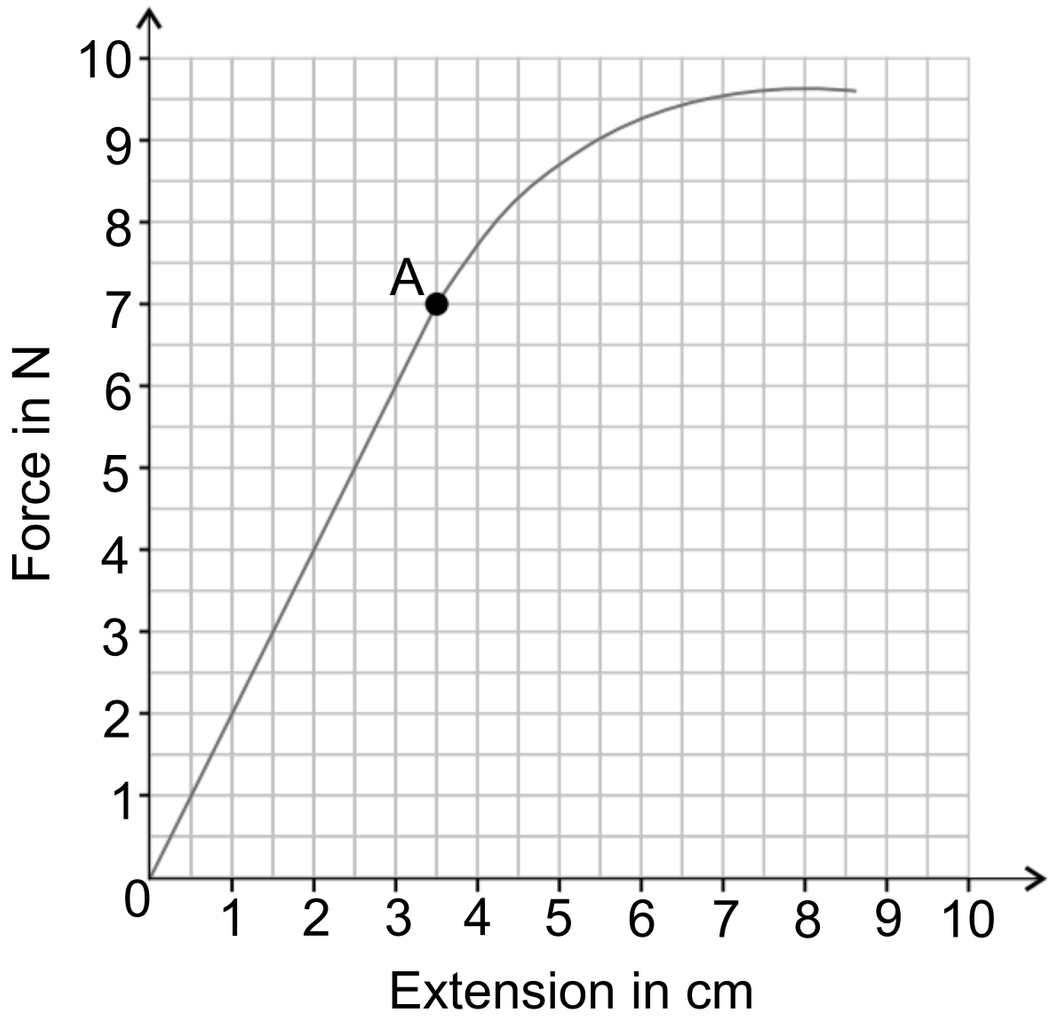
.....

.....

[6 marks]

The student used the apparatus in **Figure 5** to produce the graph shown in **Figure 6**.

Figure 6



5.3 The student marks point **A** on the graph where the shape of the graph changes.
Name point **A**.

.....

[1 mark]

5.4 Using **Figure 6**, calculate the spring constant of the spring used in the experiment.
Use the correct equation from the Physics Equations Sheet on page 216.

.....

.....

.....

.....

.....

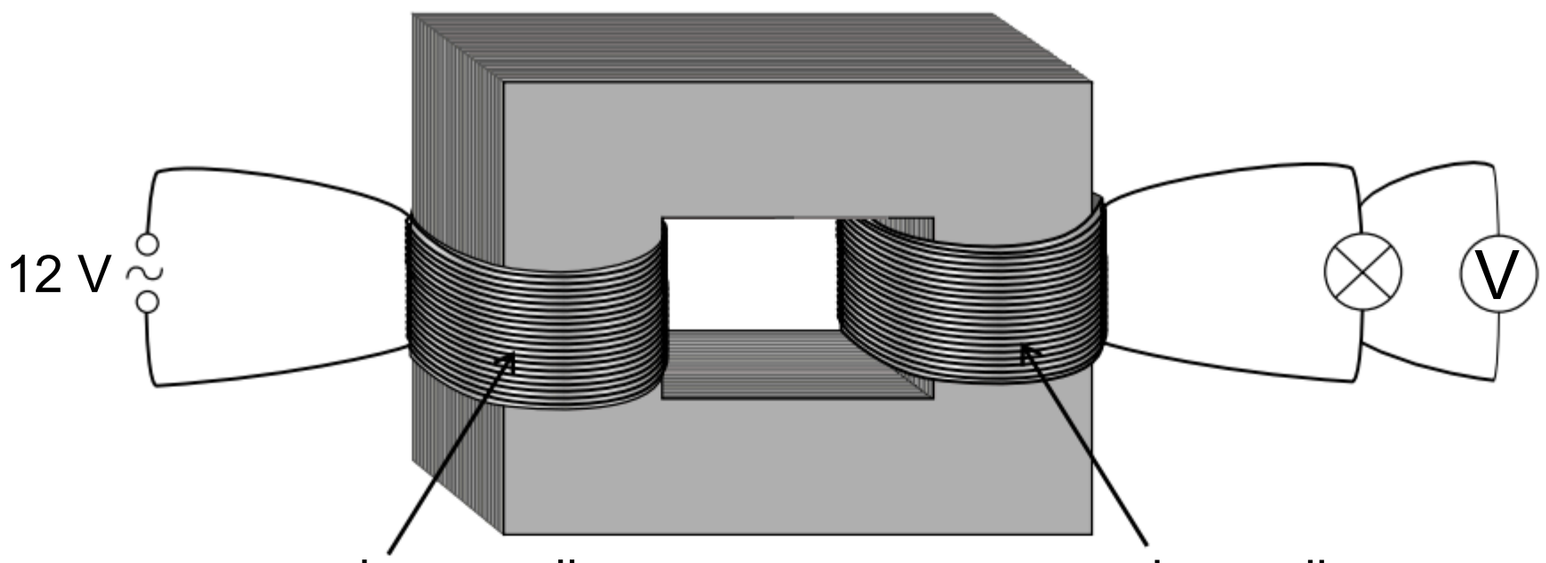
Spring constant = ... N/m

[3 marks]

Turn over for the next question

6 A student made a simple transformer from an iron core and two lengths of wire, as shown in **Figure 7**. A 12 V alternating power supply is connected to one of the coils and a lamp and voltmeter to the other coil.

Figure 7



6.1 Explain how a potential difference is generated in the secondary coil.

.....

.....

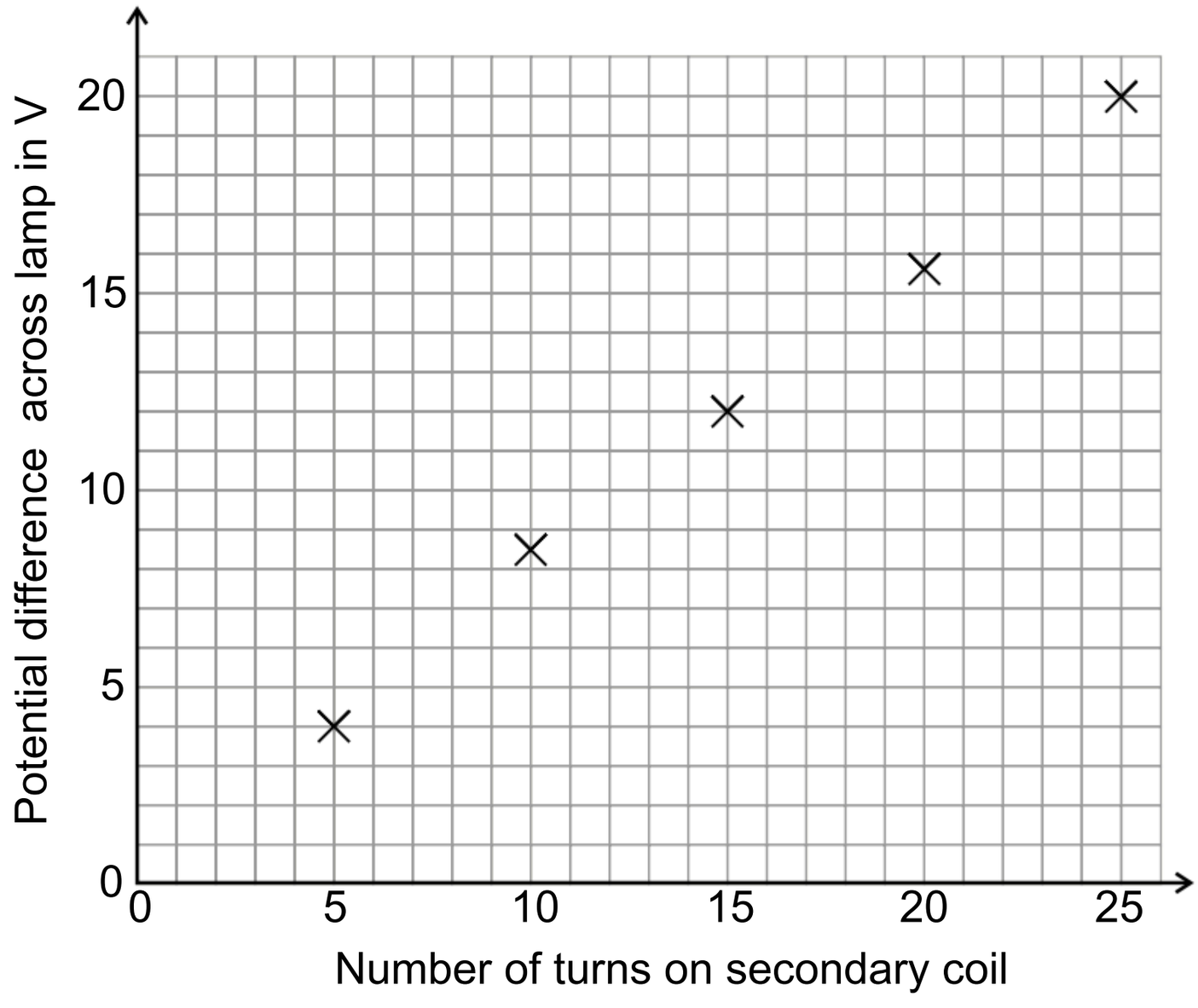
.....

.....

[2 marks]

The student experimented with the transformer by changing the number of turns on the secondary coil and measuring the potential difference across the lamp each time.
The student's results are plotted on **Figure 8**.

Figure 8

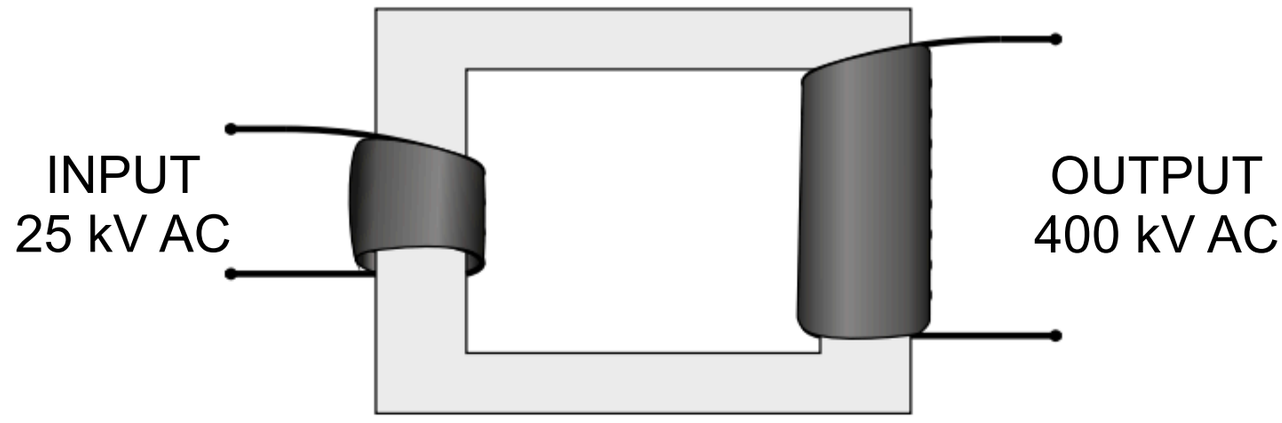


6.2 Draw a line of best fit on **Figure 8**.

[1 mark]

The generator in a power station produces an alternating voltage of 25 kV. This is changed to 400 kV by the transformer shown in **Figure 9**.

Figure 9



6.3 Name the type of transformer shown in **Figure 9**.

.....

[1 mark]

Use the Physics Equations Sheet on page 216 to answer questions **6.4** and **6.5**.

6.4 The primary coil has 5000 turns.
Calculate the how many turns must there be on the secondary coil.

.....

.....

.....

Number of turns = ...

[3 marks]

6.5 The output current is 250 A.
Calculate the input current. Assume the transformer is 100% efficient.

.....

.....

.....

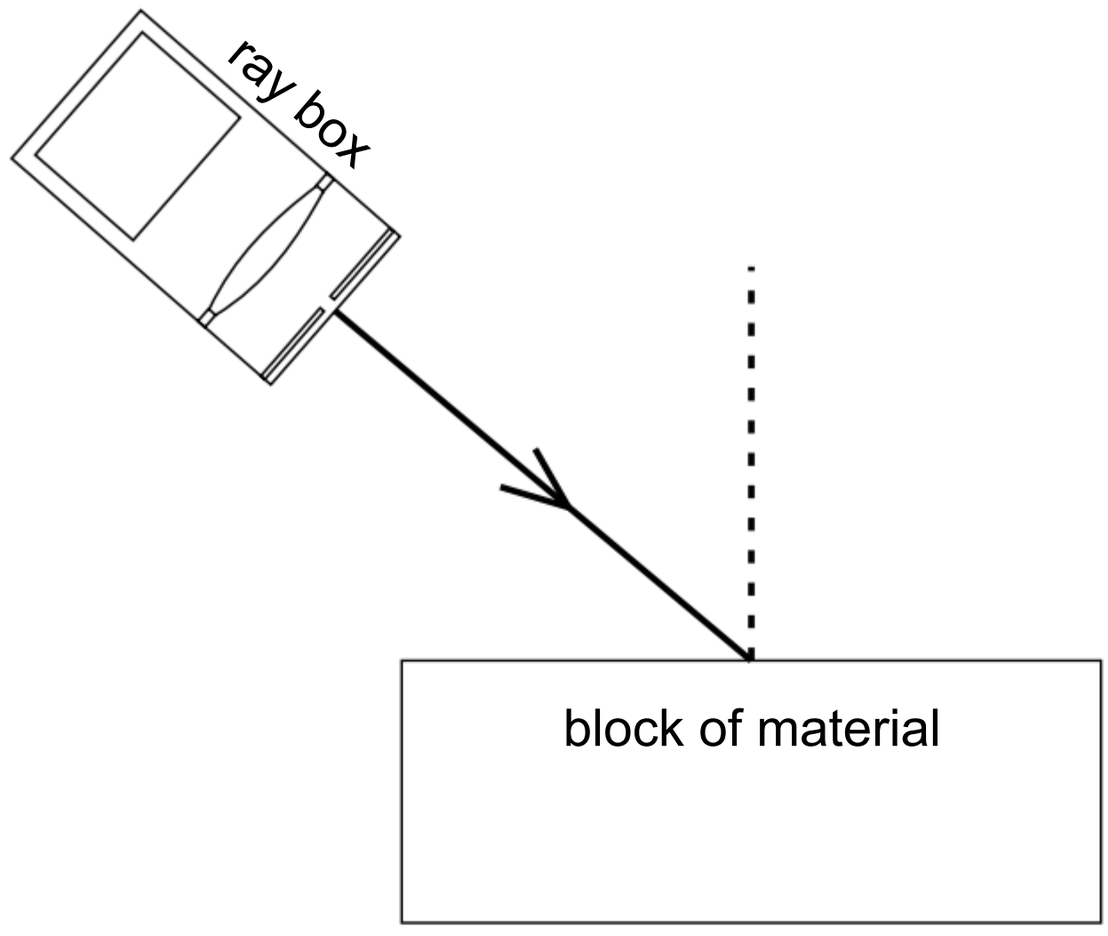
Input current = ... A

[3 marks]

Turn over for the next question

7 A student wants to investigate the reflection of light from different surfaces. A ray box is used to shine a ray of white light onto a block of material, as shown in **Figure 10**.

Figure 10



7.1 Measure the angle of incidence of the light ray in **Figure 10**.

Angle of incidence = ... °

[1 mark]

7.2 The student finds that when the light hits the block in **Figure 10**, it undergoes specular reflection. Complete the diagram in **Figure 10** by drawing in the reflected ray.

[2 marks]

The student replaces the block of material with another material. This block of material is transparent. When the ray hits this material, it is transmitted into the block.

7.3 When the light passes into the block, it is bent towards the normal.
Describe how the speed of the light in the block differs from its speed in air.

.....

.....

[1 mark]

8 A driving instructor is looking at the Highway Code. The instructor finds the data shown in **Table 3** about stopping distances for a well-maintained car travelling on dry roads at various speeds.

Table 3

Speed (km/h)	Thinking distance (m)	Braking distance (m)	Stopping distance (m)
32	6	6	12
48	9	14	23
64	12	24	36
80	15	38	53
96	18	55	73
112	21	75	96

8.1 The data in **Table 3** was obtained by observing a large number of drivers.
Explain why it was sensible to use a large sample of people.

.....

.....

.....

[2 marks]

8.2 Give **one** factor, other than speed, which affects the braking distance of a car.

.....

.....

[1 mark]

8.3 A car is travelling at 25 m/s and makes an emergency stop to avoid hitting a hazard.
The driver applies the brakes 150 m away from the hazard.
Calculate the minimum deceleration required for the car to stop before hitting the hazard.
Use the correct equation from the Physics Equation Sheet on page 216.

.....

.....

.....

.....

Deceleration = ... m/s^2

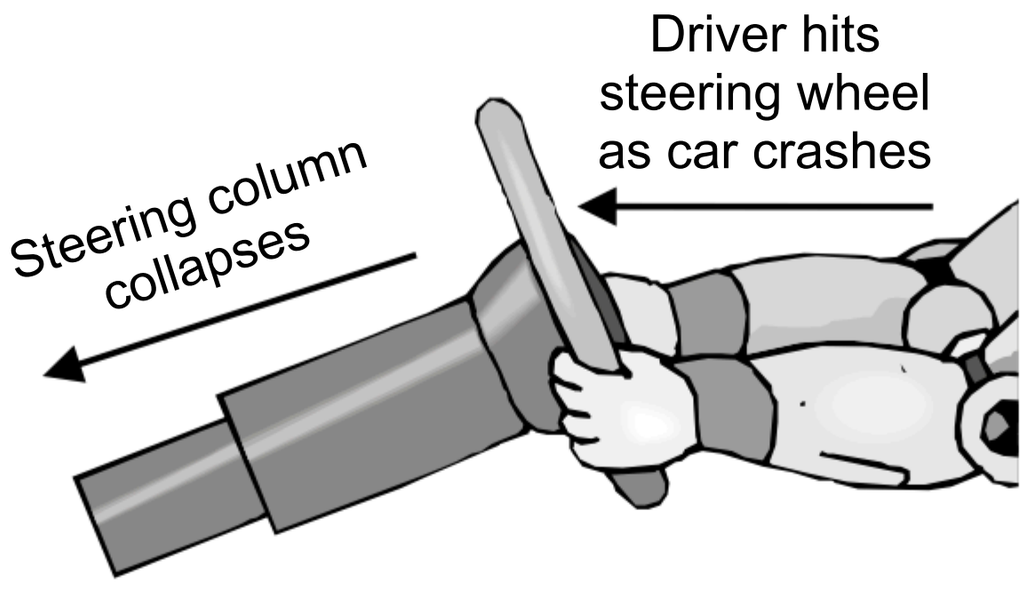
[3 marks]

Question 8 continues on the next page

Turn over ▶

Modern cars contain many safety features. **Figure 11** shows how a collapsible steering column works. When a crash occurs, the driver hits the steering wheel and the steering column collapses.

Figure 11



8.4 By considering rate of change of momentum, suggest how a collapsible steering column helps protect the driver from injury in the event of a car crash.

.....

.....

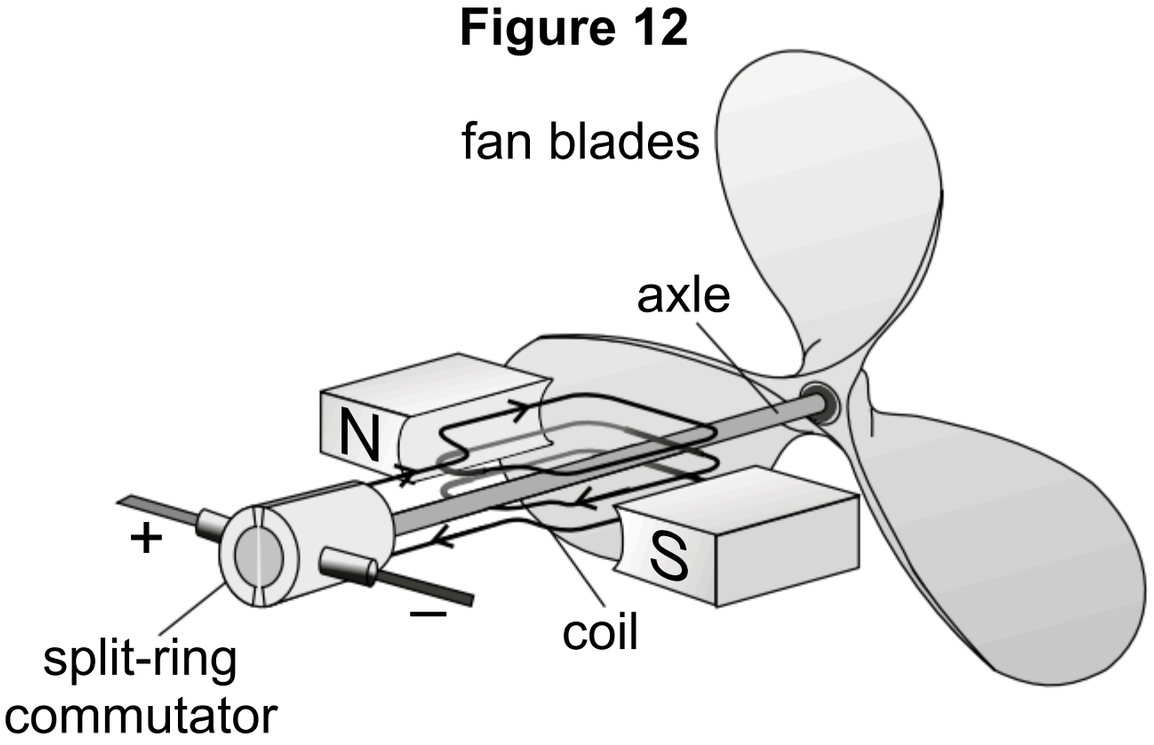
.....

.....

[3 marks]

9 **Figure 12** shows how a fan can be powered by an electric motor. When current flows in the wire coil, the fan blades rotate.

Figure 12



9.1 Name the effect used to rotate the fan.

.....

[1 mark]

9.2 Explain how the effect named in **9.1** causes the fan blades to rotate when current flows in the wire coil.

.....

.....

.....

.....

.....

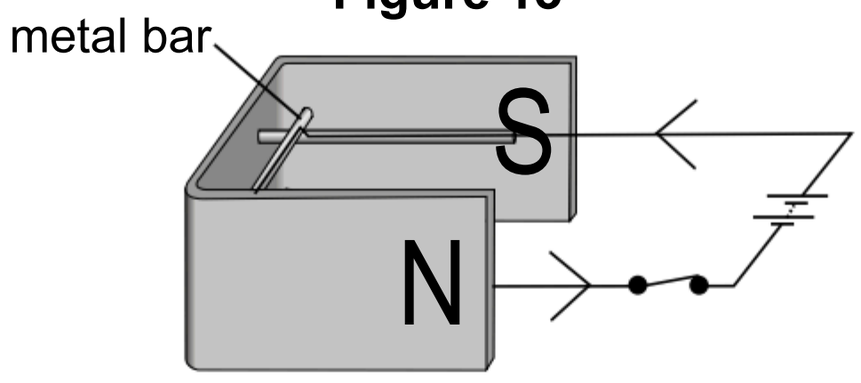
.....

[2 marks]

Question 9 continues on the next page

The apparatus shown in **Figure 13** can be used to show the force acting on a current-carrying bar in a magnetic field. When the switch is closed, current flows through the metal bar. The metal bar is free to move.

Figure 13



9.3 Explain whether or not a force will act on the bar when the switch is closed.
Give a reason for your answer.

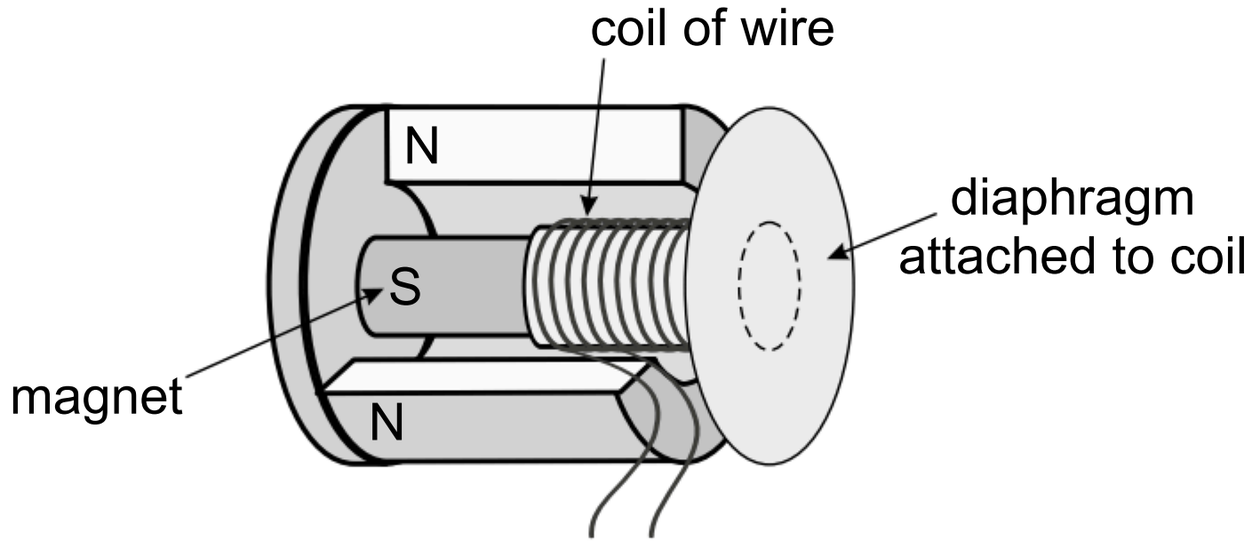
.....

.....

[2 marks]

Microphones also use magnets to work. **Figure 14** shows a diagram of the internal workings of a microphone.

Figure 14



9.4 Describe how a microphone converts a sound wave into an electrical signal.

.....

.....

.....

.....

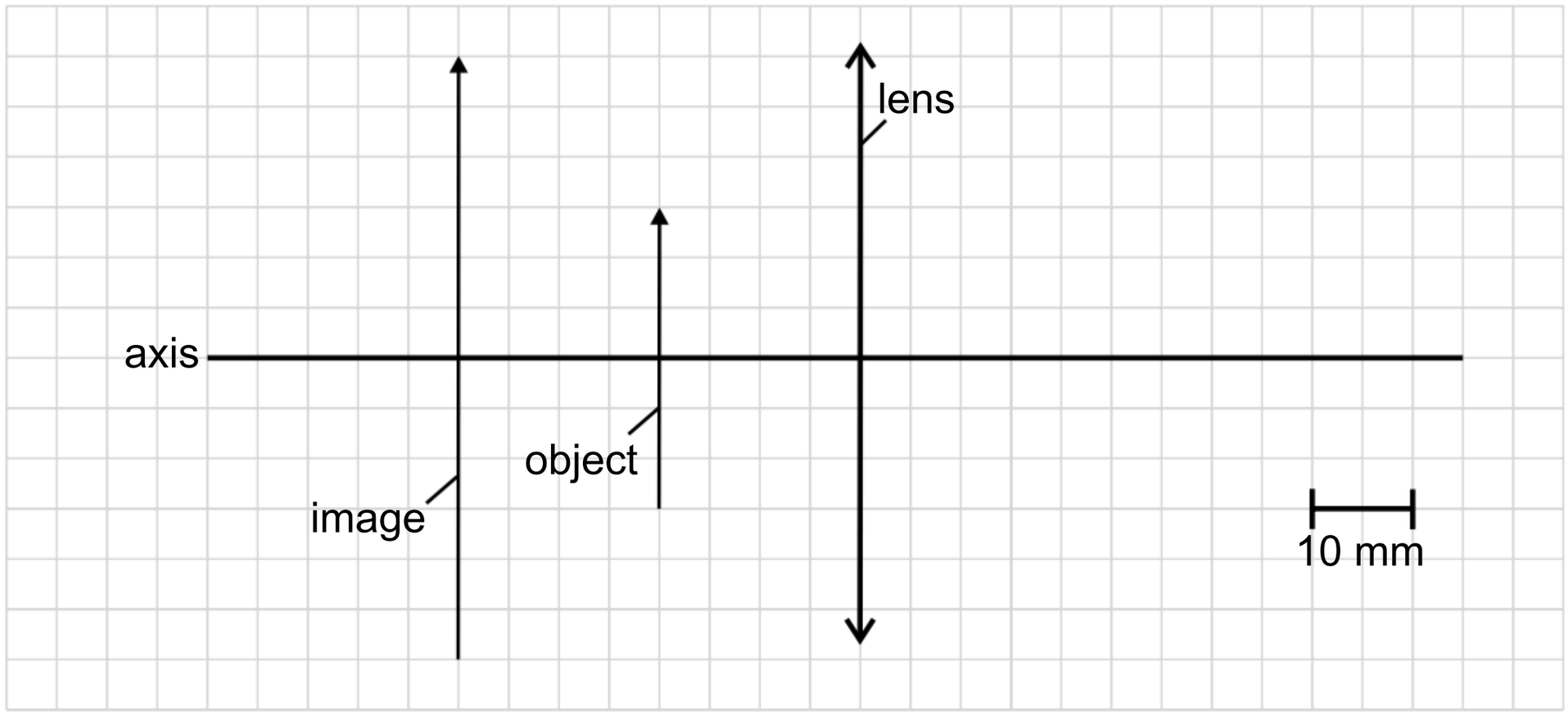
.....

.....

[4 marks]

10 A student is looking at an object through a lens.
A diagram of the object, image and lens is shown in **Figure 15**.

Figure 15



10.1 Name the type of lens the student is using in this experiment.

.....

[1 mark]

10.2 Complete the ray diagram in **Figure 15** to show how the image was formed.

[3 marks]

10.3 Use **Figure 15** to find the focal length of the lens.

.....

Focal length = ... mm

[1 mark]

10.4 Calculate the magnification of the lens using the heights of the object and image in **Figure 15**.
Use the correct equation from the Physics Equation Sheet on page 216.

.....

.....

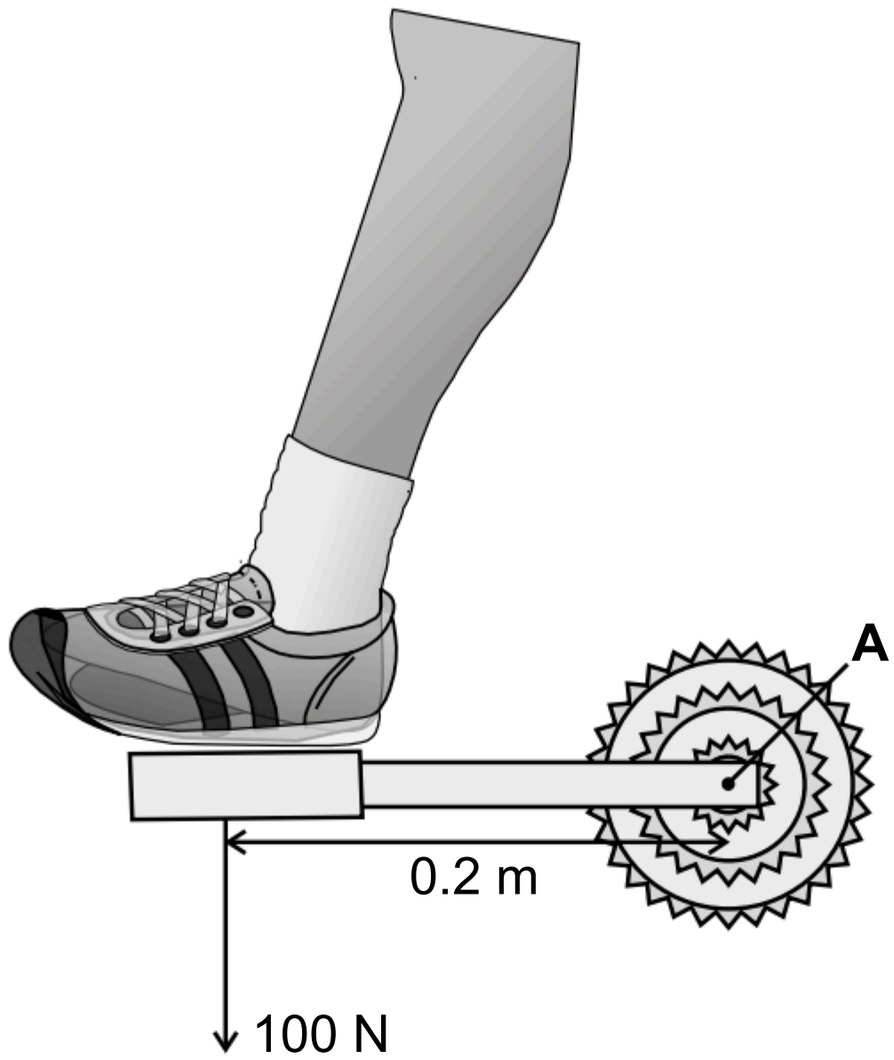
.....

Magnification = ...

[2 marks]

Turn over for the next question

11 A child is learning to ride a bicycle. The child pushes down on the pedal with a force of 100 N, as shown in **Figure 17**.



Use the Physics Equations Sheet on page 216 to answer questions **11.1** and **11.2**.

11.1 Write down the equation that links the moment, force, and the distance from the pivot.

.....

[1 mark]

11.2 Calculate the moment, about point **A**, that the child exerts, and give the unit.

.....

.....

.....

Moment = ... Units = ...

[3 marks]

11.3 Explain what will happen to the moment about point **A** if the child continues to push on the pedal with the same force while the pedal moves downwards.

.....

.....

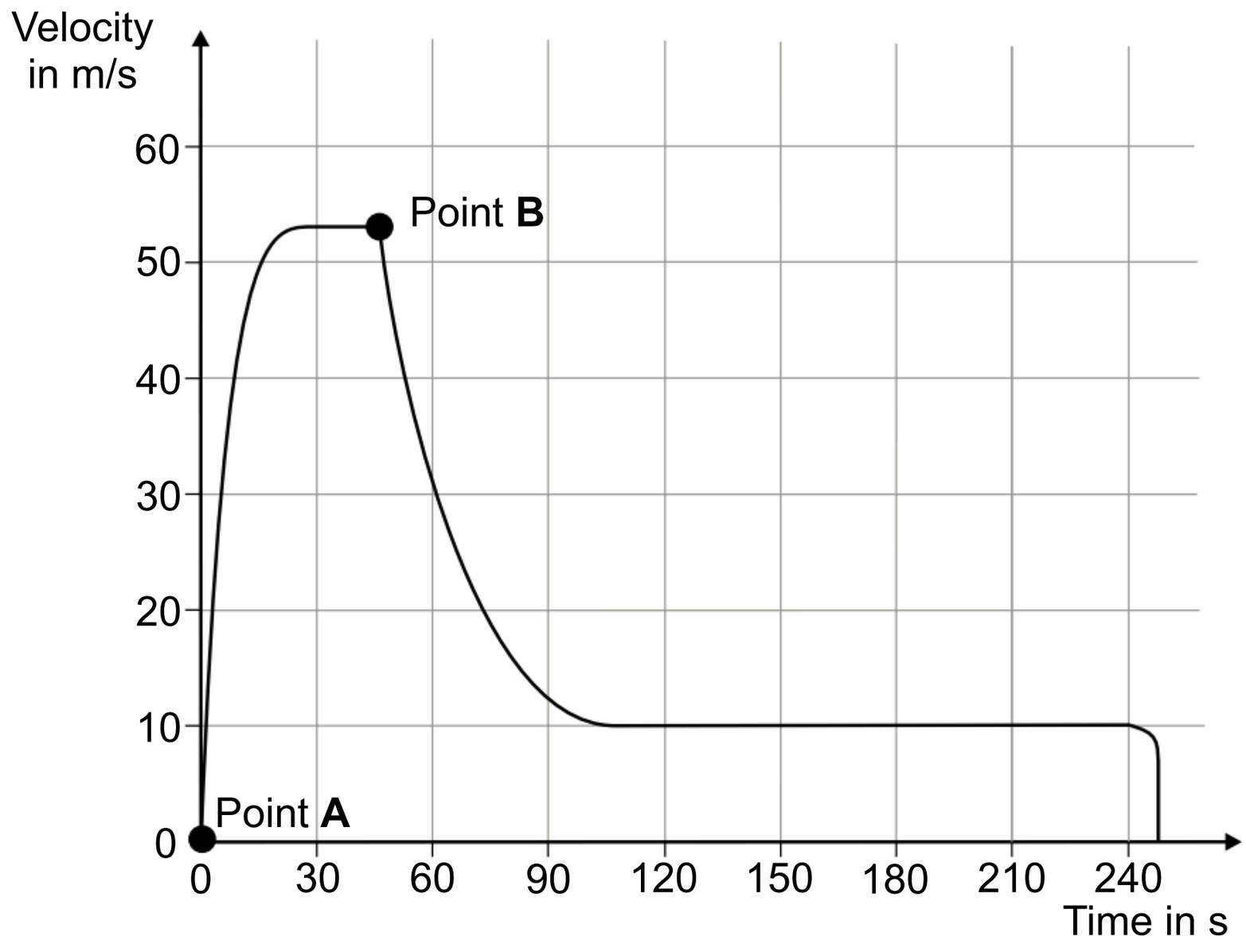
.....

.....

[2 marks]

12 The motion of a skydiver jumping from an aeroplane is recorded.
Figure 16 shows the velocity-time graph of the skydiver's fall.

Figure 16



Use the Physics Equations Sheet on page 216 to answer questions **12.1** and **12.2**.

12.1 Write down the equation that links weight, mass and the gravitational field strength.

.....

[1 mark]

12.2 The skydiver has a mass of 80.0 kg. Calculate the skydiver's weight.
Use gravitational field strength = 9.8 N/kg.

.....

.....

Weight = ... N

[2 marks]

Question 12 continues on the next page

12.3 Explain the shape of the graph in **Figure 16** from point **A** to point **B**, in terms of the motion of, and the forces acting on, the skydiver.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[6 marks]

12.4 Using **Figure 16**, estimate the total distance travelled by the skydiver.

.....

.....

.....

.....

Distance = ... m

[3 marks]

END OF QUESTIONS

Answers

Topic 1a — Energy Stores and Transfers

Page 3
Energy Stores and Systems

Q1 Energy is transferred mechanically ***[1 mark]*** from the kinetic energy store of the wind ***[1 mark]*** to the kinetic energy store of the windmill ***[1 mark]***.

Page 4
Kinetic and Potential Energy Stores

Q1 The change in height is 5.0 m. So the energy transferred from the gravitational potential energy store is:
$E_p = mgh = 2.0 \times 9.8 \times 5.0 = 98$ J ***[1 mark]***
This is transferred to the kinetic energy store of the object, so $E_k = 98$ J ***[1 mark]***
$E_k = \frac{1}{2}mv^2$ so
$v^2 = 2E_k \div m$ ***[1 mark]***
$= (2 \times 98) \div 2.0$ ***[1 mark]***
$= 98$ m^2/s^2
$v = \sqrt{98} = 9.899... =$ **9.9 m/s (to 2 s.f.)** ***[1 mark]***

Page 5
Specific Heat Capacity

Q1 $\Delta E = mc\Delta\theta$ so
$\Delta\theta = \Delta E \div (m \times c)$ ***[1 mark]***
$= 50\,000 \div (5 \times 4200)$
$= 2.380...$ °C ***[1 mark]***
So the new temperature $= 5 + 2.380...$
$= 7.380... =$ **7 °C (to 1 s.f.)** ***[1 mark]***

Page 6
Warm-Up Questions

1) Any two from: e.g. mechanically (by a force doing work) / electrically (work done by when a current flows) / by heating / by radiation.
2) Energy is transferred from the chemical energy store of the person's arm to the kinetic energy stores of the arm and the ball (and the gravitational energy store of the ball). Once the ball is released, energy is transferred from the kinetic energy store of the ball to its gravitational potential energy store.
3) kinetic energy = ½ × mass × (speed)^2 / $E_k = \frac{1}{2}mv^2$
4) A lorry travelling at 60 miles per hour.

Exam Questions

1.1 gravitational potential energy = mass × gravitational field strength × height / $E_p = mgh$ ***[1 mark]***
So, $h = E_p \div (m \times g)$ ***[1 mark]*** $= 137.2 \div (20 \times 9.8)$ ***[1 mark]***
$=$ **0.7 m** ***[1 mark]***

1.2 Energy is transferred from the gravitational potential energy store ***[1 mark]*** to the kinetic energy store of the load ***[1 mark]***.

1.3 Some of the energy would also be transferred to thermal energy store of the air (and the thermal energy store of the load) ***[1 mark]***.

2.1 change in thermal energy = mass × specific heat capacity × temperature change / $\Delta E = mc\Delta\theta$
So, $c = \Delta E \div (m \times \Delta\theta)$ ***[1 mark]***
$\Delta\theta = 100 - 20 = 80$ °C ***[1 mark]***
$c = 36\,000 \div (0.5 \times 80)$ ***[1 mark]*** $=$ **900 J/kg °C** ***[1 mark]***

2.2 Concrete has a higher specific heat capacity ***[1 mark]*** and so will be able to store a lot more energy in its thermal energy store ***[1 mark]***.

Even if you got the answer to 2.1 wrong, if your conclusion is correct for your answer to 2.1, you'd get the marks for this question.

3.1 Reading values from the start and end of the linear point on the graph:
E.g. $\Delta E = 3 - 1 = 2$ kJ $= 2000$ J
$\Delta\theta = 2.4 - 0.4 = 2$ °C
[1 mark for any values accurately calculated from a pair of points on the linear part of the graph]
change in thermal energy = mass × specific heat capacity × temperature change / $\Delta E = mc\Delta\theta$,
So, $c = \Delta E \div (m \times \Delta\theta)$ ***[1 mark]***
$= 2000 \div (1 \times 2)$ ***[1 mark]***
$=$ **1000 J/kg °C** ***[1 mark]***

3.2 Lower ***[1 mark]***. In the investigation, some of the energy transferred by the heater would have been transferred to the thermal energy stores of the surroundings rather than the block ***[1 mark]***. For the same temperature change to have occurred for a smaller amount of energy transferred, the specific heat capacity must be smaller ***[1 mark]***.

It's important to be able to spot reasons why the results of an investigation aren't perfectly accurate.

Page 7
Conservation of Energy and Power

Q1 $P = E \div t$
$t = 2 \times 60 = 120$ s ***[1 mark]***
$P = 4800 \div 120$ ***[1 mark]*** $=$ **40 W** ***[1 mark]***

Page 10
Lubrication and Efficiency

Q1 efficiency
= useful output energy transfer ÷ total input energy transfer
$= 225 \div 300$ ***[1 mark]*** $=$ **0.75** ***[1 mark]***

Q2 efficiency = useful power output ÷ total power input
$= 900 \div 1200 = 0.75$ ***[1 mark]***
useful output energy transfer
= efficiency × total input energy transfer ***[1 mark]***
$= 0.75 \times 72\,000$ ***[1 mark]***
$=$ **54 000 J** ***[1 mark]***

Pages 11-12
Warm-Up Questions

1) true
2) watts (W)
3) conduction, convection
4) The higher the thermal conductivity, the greater the rate of the energy transfer (i.e. the faster energy is transferred) through it.
5) E.g. lubrication
6) Some energy is always dissipated, so less than 100% of the input energy transfer is transferred usefully.

Exam Questions

1.1 power = work done ÷ time / $P = W \div t$ ***[1 mark]***
$P = 1000 \div 20$ ***[1 mark]*** $=$ **50 W** ***[1 mark]***

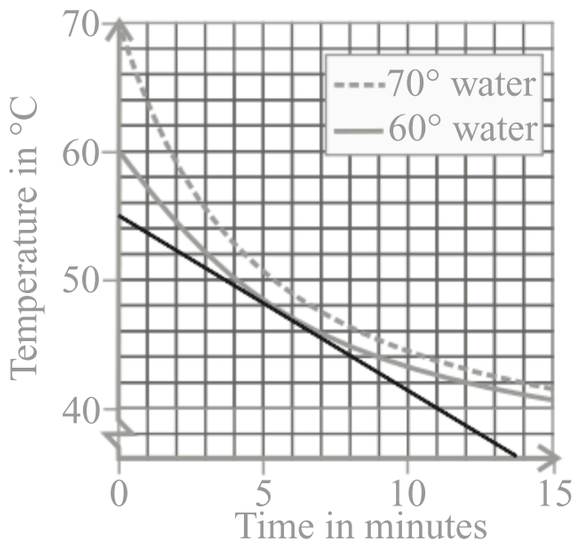
1.2 Less energy is transferred to the thermal energy store of the motor's parts and the surroundings ***[1 mark]***, so more is transferred to the scooter's kinetic energy store ***[1 mark]***.

1.3 It will be faster / complete the course in less time ***[1 mark]*** because the motor transfers the same amount of energy, but over a shorter time ***[1 mark]***.

2.1 Any two from: e.g. use identical flasks / same temperature gel in each box / same mass of gel in each box / same mass of water in each flask ***[1 mark for each correct control variable]***.

2.2 The water at 70 °C ***[1 mark]***. The initial gradient of this graph is steeper than the graph for the 60 °C water / there is a greater temperature drop for the 70 °C water over the first minute than for the 60 °C water ***[1 mark]***.

2.3



[1 mark for correct tangent drawn at 5 minutes]

E.g. Temperature change = change in y = 55 – 40 = 15°C
Change in time = change in x = 11 – 0 = 11 mins
Rate of temperature change = 15 ÷ 11
= 1.36... = **1.4 °C/min (2 s.f.)**
[2 marks for a rate between 1.1 and 1.7 °C/min, otherwise 1 mark for an attempt to calculate gradient of a tangent to curve at 5 minutes]

3.1 1000 – 280 – 200 = **520 J** ***[1 mark]***

3.2 efficiency = useful output energy transfer ÷ useful input energy transfer
useful energy = thermal and kinetic energy
= 200 + 520 = 720 W ***[1 mark]***
efficiency = 720 ÷ 1000 ***[1 mark]***
= **0.72 (or 72%)** ***[1 mark]***

3.3 1 minute = 60 seconds, so 4 minutes = 60 × 4 = 240 s
power = energy transferred ÷ time / $P = E \div t$
So, $E = P \div t$ ***[1 mark]***
E = 1000 × 240 ***[1 mark]*** = **240 000 J** ***[1 mark]***

4.1 efficiency = useful output energy transfer ÷ useful input energy transfer ***[1 mark]***

4.2 efficiency = useful output energy transfer ÷ useful input energy transfer
= 480 ÷ 1200 ***[1 mark]***
= **0.4 (or 40%)** ***[1 mark]***

4.3 power = energy transferred ÷ time / $P = E \div t$
1 minute = 60 seconds
So, output power = 600 ÷ 60 ***[1 mark]***
= **10 W** ***[1 mark]***

4.4 efficiency = useful output power ÷ total output power
So, total input power = useful output power ÷ efficiency ***[1 mark]***
total input power = 10 ÷ 0.55 ***[1 mark]***
= 18.181... = **18 W (to 2 s.f.)** ***[1 mark]***

4.5 Disagree ***[1 mark]***. Torch B has a lower input energy transfer than torch A, i.e. it transfers less energy per minute than torch A (as 18 × 60 = 1080, and 1080 < 1200) ***[1 mark]***.

Even if you got the answer to 4.4 wrong, if your conclusion is correct for your answer to 4.5, you'd get the marks for this question. You could also answer this question by comparing the input powers of the torches.

5.1 E.g. use cavity wall insulation ***[1 mark]***.

5.2 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief explanation of an experiment to measure how effective the films are as insulation using the equipment shown. The answer lacks coherency. ***[1 to 2 marks]***

Level 2: There is an explanation of an experiment to measure how effective the films are as insulation using the equipment shown. There is some discussion of how to compare the effectiveness of the two films. The answer has some structure. ***[3 to 4 marks]***

Level 3: There is a clear and detailed explanation of an experiment to measure how effective the films are as insulation using the equipment shown. There is a clear explanation of how to compare the effectiveness of the two films. The answer is well structured. ***[5 to 6 marks]***

Here are some points your answer may include:
Wrap one of the beakers in one type of film and another other in the second type of film.
Leave the third unwrapped. This is the control.
Pour boiling water from a kettle into all three beakers.
Cover the beakers with lids to minimise energy loss from the top between temperature readings.
Insert a thermometer through the lid of each beaker.
Record the starting temperature of the water in each beaker.
Take the temperature of the water after regular intervals of time, e.g. every minute, for ten minutes.
Compare the temperatures of the water in the wrapped beakers with the temperature of the water in the control beaker at each reading.
If the films prevent energy transfer, the temperature of the film-covered beakers should be higher than the control at most of the recorded times.
Compare the water temperatures of the water in the wrapped beakers to each other.
If one film is a better insulator than the other, the temperature of the water in that beaker should be higher than the other for most of the recorded times.

5.3 E.g. see if someone else gets the same result using different equipment ***[1 mark]***.

Topic 1b — Energy Resources

Page 19

Warm-Up Questions

1) An energy resource that can be renewed as it's being used, and so will never run out.

2) E.g. bio-fuels can be used to run vehicles / electricity generated using renewable resources can be used to power vehicles.

3) E.g. there's no pollution or environmental damage once the panels have been made / they are very reliable in sunny countries / running and maintenance costs are almost zero.

4) Any two from: e.g. it releases greenhouse gases and contributes to global warming / it causes acid rain / coal mining damages the landscape.

5) Any two from: e.g. renewable resources don't currently provide enough energy / energy from renewables cannot be relied upon currently / it's expensive to build new renewable power plants / it's expensive to switch to cars running on renewable energy.

Exam Questions

1.1 Any two from: e.g. wave / tidal / geothermal / bio-fuels ***[1 mark for each correct renewable energy resource]***.

1.2 Any one from: e.g. flooding a valley for a dam destroys animal habitats / carbon dioxide is released by rotting vegetation in flooded valley ***[1 mark]***.

2.1 From the gravitational potential energy store of the water ***[1 mark]*** to its kinetic energy store ***[1 mark]***.

2.2 E.g. they cause no pollution / they use a renewable energy source ***[1 mark for each correct advantage, up to a maximum of two]***.

3.1 power = energy transferred ÷ time / $P = E \div t$
So, $E = P \times t$ ***[1 mark]***
seconds in 5 hours = 5 × 60 × 60 = 18 000 s
Energy provided by 1 m^2 solar panel in 5 hours
= 200 × 18 000 ***[1 mark]*** = 3 600 000 J ***[1 mark]***
Number of panels needed = energy needed ÷ energy provided
= 32 500 000 ÷ 3 600 000 ***[1 mark]***
= 9.027... = **10 panels (to next whole number)** ***[1 mark]***

Remember, because you have to have a set number of whole panels, if you get a decimal answer, you need to round up to the next whole number to be able to provide the right amount of energy.

3.2 Ten 1 m^2 solar panels are needed, and they have 10 m^2 of space on their roof (10 × 1 m^2 = 10 m^2), so the family can install sufficient solar panels ***[1 mark]***.

3.3 E.g. Solar panels are less reliable than coal-fired power stations ***[1 mark]***. The energy output of the solar panels will vary based on the number of hours of good sunlight, and may not be able to provide enough energy on a given day ***[1 mark]***. The energy output of coal-fired power stations is not influenced by environmental factors like weather, and energy output can be increased to meet demand ***[1 mark]***.

Topic 2a — Circuits

Page 21

Current and Circuit Symbols

Q1 $Q = It$ so $t = Q \div I$ ***[1 mark]*** = 28 800 ÷ 8 ***[1 mark]***
= 3600 s ***[1 mark]***
t = 3600 ÷ 60 = **60 minutes** ***[1 mark]***

Page 22

Resistance and $V = IR$

Q1 $V = IR$
so $R = V \div I$ ***[1 mark]*** = 230 ÷ 5.0 ***[1 mark]*** = **46 Ω** ***[1 mark]***

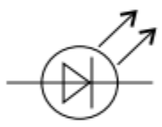
Page 23

Resistance and *I-V* Characteristics

Q1 As the current through the lamp increases, the temperature of its filament increases ***[1 mark]*** causing its resistance to increase ***[1 mark]***. A larger resistance means less current can flow per unit potential difference, and so the graph gets shallower ***[1 mark]***.

Page 25

Warm-Up Questions

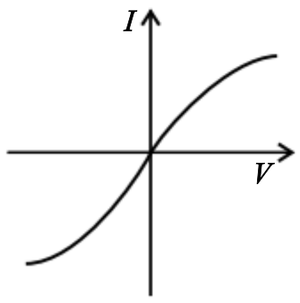
1) Ohms / Ω
2) The greater the resistance, the smaller the current / the smaller the resistance, the greater the current.
3) 
4) In parallel.
5) Any one from: e.g. a wire / a fixed resistor

Exam Questions

1.1 potential difference = current × resistance / $V = IR$
So, $R = V \div I$ ***[1 mark]*** = 1.5 ÷ 0.30 ***[1 mark]***
= **5.0 Ω (to 2 s.f.)** ***[1 mark]***

1.2 charge = current × time / $Q = It$
= 0.30 × 35 ***[1 mark]***
= 10.5 = **11 C (to 2 s.f.)** ***[1 mark]***

1.3 The amount of current flowing through the circuit will decrease ***[1 mark]***.

1.4 

[1 mark]

1.5 The resistance of the filament lamp increases as the temperature increases ***[1 mark]***.

2.1 Potential difference (across the component) ***[1 mark]***.

2.2 Diodes only allow current to flow in one direction ***[1 mark]***.

2.3 At point A, V = 6 V, I = 3 A ***[1 mark]***
potential difference = current × resistance / $V = IR$
$R = V \div I$ ***[1 mark]***
= 6 ÷ 3 ***[1 mark]***
= **2 Ω** ***[1 mark]***

Page 27

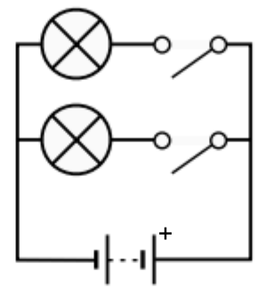
Series Circuits

Q1 R_{total} = 4 + 5 + 6 = 15 Ω ***[1 mark]***
$V = I \times R$ = 0.6 × 15 ***[1 mark]*** = **9 V** ***[1 mark]***

Page 28

Parallel Circuits

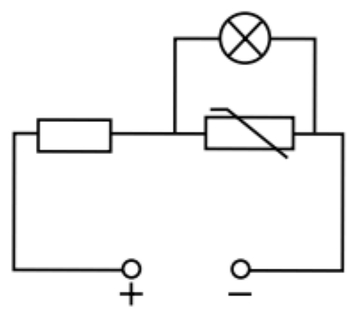
Q1 The total current through the circuit decreases ***[1 mark]*** as there are fewer paths for the current to take ***[1 mark]***. The total resistance of the circuit increases ***[1 mark]*** as, using $V = IR$, a decrease in the total current means an increase in the total resistance ***[1 mark]***.

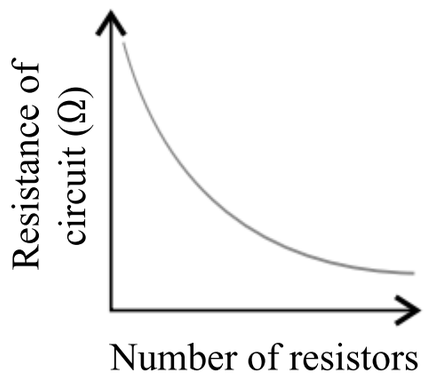
Q2 E.g. 

[1 mark for the correct circuit symbols, 1 mark for two bulbs connected in parallel, 1 mark for both switches being on the same branches as the lamps]

Page 30

Warm-Up Questions

1) Any one from: e.g. in automatic night lights / outdoor lighting / burglar detectors.
2) The resistance decreases.
3) E.g. 

4) Any one from: e.g. if you remove or disconnect one component, then the whole circuit is broken / you can't switch components on or off independently.
5) The total resistance is the sum of all the resistances.
6) Two resistors connected in series.
7) E.g. 

Exam Questions

1.1 total resistance = $R_1 + R_2 + R_3$ = 2 + 3 + 5 ***[1 mark]***
= **10 Ω** ***[1 mark]***

1.2 The current will be 0.4 A ***[1 mark]*** because in a series circuit, the same current flows through all parts of the circuit ***[1 mark]***.

1.3 V = source pd
$V_3 = V - V_1 - V_2$ = 4 – 0.8 – 1.2 ***[1 mark]*** = **2 V** ***[1 mark]***

2.1 15 V ***[1 mark]***
Potential difference is the same across each branch in a parallel circuit.

2.2 potential difference = current × resistance / $V = IR$
$I = V \div R$ ***[1 mark]***
= 15 ÷ 3 ***[1 mark]***
= **5 A** ***[1 mark]***

2.3 I_2 = 5 + 3.75 ***[1 mark]***
= **8.75 A** ***[1 mark]***
Even if you got the answer to 2.2 wrong, award yourself the marks for 2.3 if you did the sum above correctly.

Topic 2b — Everyday Electricity

Page 33

Power of Electrical Appliances

Q1 $P = E \div t = 6000 \div 30$ ***[1 mark]***
$= \mathbf{200\ W}$ ***[1 mark]***

Q2 $E = P \times t$ ***[1 mark]***
$= 250 \times (2 \times 60 \times 60)$
$= 1\ 800\ 000$ J ***[1 mark]***
$E = 375 \times (2 \times 60 \times 60)$
$= 2\ 700\ 000$ J ***[1 mark]***
So change in energy is
$2\ 700\ 000 - 1\ 800\ 000 =$ **900 000 J** ***[1 mark]***

Page 34

More on Energy Transfers and Power

Q1 $E = Q \times V = 10\ 000 \times 200$ ***[1 mark]*** = **2 000 000 J** ***[1 mark]***

Q2 $P = V \times I = 12 \times 4.0$ ***[1 mark]*** = **48 W** ***[1 mark]***

Q3 $R = P \div I^2$ ***[1 mark]*** $= 2300 \div 10.0^2$ ***[1 mark]*** = **23 Ω** ***[1 mark]***

Page 36

Warm-Up Questions

1) The live wire, neutral wire, and earth wire.
2) Energy is transferred electrically to the thermal energy store of the heating element.
3) power = (current)^2 × resistance / $P = I^2R$
4) The network of cables and transformers that distributes electricity across the country.

Exam Questions

1 The live wire is at a potential difference of around 230 V ***[1 mark]***. Touching the wire forms a low-resistance path from the wire to the earth through your body ***[1 mark]***, causing a large current to flow through you, which is an electric shock ***[1 mark]***.

2.1 power = potential difference × current / $P = V \times I$ ***[1 mark]***

2.2 $I = P \div V$ ***[1 mark]*** $= (2.8 \times 1000) \div 230$ ***[1 mark]***
$= 12.17...$ A
$=$ **12 (to 2 s.f.)** ***[1 mark]*** **A** ***[1 mark, allow 'amps']***

2.3 She should choose kettle B because it has the higher power rating ***[1 mark]***. This means that it transfers more energy usefully / to heat the water per unit time, so it will boil the water faster ***[1 mark]***.

3 Step-up transformers increase the potential difference ***[1 mark]*** of the electricity supply, allowing the electricity to be transmitted at a high potential difference, and so a low current ***[1 mark]***. This reduces the energy lost through heating in the cables ***[1 mark]***.

4.1 power = potential difference × current / $P = IV$
$P = 0.5 \times 3.0$ ***[1 mark]*** = **1.5 W** ***[1 mark]***
energy transferred = power × time / $E = Pt$
$t = 0.5$ hours $= (30 \times 60)$ s $= 1800$ s
$E = 1.5 \times 1800$ ***[1 mark]*** = **2700 J** ***[1 mark]***

4.2 energy transferred = charge flow × potential difference / $E = QV$, so
$Q = E \div V$ ***[1 mark]*** $= 2700 \div 3.0$ ***[1 mark]*** = **900 C** ***[1 mark]***

In the exam, you'd probably get all the marks for 4.2 if you did the sum above correctly, even if you got the answer to 4.1 wrong.

Page 39

Warm-Up Questions

1) electrons
2) It becomes positively charged.
3) It decreases.

Exam Questions

1.1 The charge on the cloth is +1.6 nC ***[1 mark]***.

1.2 Electrons are transferred from the cloth and onto the sphere ***[1 mark]***.

1.3 B ***[1 mark]***

1.4

[1 mark for arrow drawn at answer to 1.3, pointing away from sphere]

1.5 A strong electric field can ionise particles in the air ***[1 mark]***. This makes the air conductive and allows charge to flow to the earth in the form of a spark ***[1 mark]***.

2 A ***[1 mark]***. The rod is negatively charged, so it would repel the negative charges in the balloon ***[1 mark]***, causing them to move away from the rod ***[1 mark]***.

Topic 3 — Particle Model of Matter

Page 41

Density of Materials

Q1 Gemstone's mass $= 0.019$ kg $= 0.019 \times 1000 = 19$ g ***[1 mark]***
Gemstone's volume = volume of water pushed out of eureka can $= 7.0$ cm^3
$\rho = m \div V = 19 \div 7.0$ ***[1 mark]***
$= 2.714...$ = **2.7 g/cm^3 (to 2 s.f.)** ***[1 mark]***

Page 43

Specific Latent Heat

Q1 $E = m \times L = 0.25 \times 120\ 000$ ***[1 mark]*** = **30 000 J** ***[1 mark]***

Pages 46-47

Warm-Up Questions

1) Density is a measure of the amount of mass in a given volume / compactness.
2) In a liquid, the particles are held close together, but can move past each other. They form irregular arrangements. The particles move in random directions at low speeds.
3) Cooling a system decreases its internal energy.
4) The specific latent heat of vaporisation of a substance is the amount of energy needed to change 1 kg of that substance from a liquid to a gas.
5) J/kg

Exam Questions

1.1 Particles are held close together in a fixed, regular pattern ***[1 mark]***. They vibrate about fixed positions ***[1 mark]***.

1.2 melting ***[1 mark]***

1.3 evaporation ***[1 mark]***

2.1 The densities of each of the toy soldiers are the same, but their masses may vary ***[1 mark]***.

2.2 The volume of the toy soldier / the volume of water displaced by the toy soldier ***[1 mark]***.
The mass of the toy soldier ***[1 mark]***.

2.3 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief explanation of an experiment to measure the density of the toy soldier.
The answer lacks coherency. ***[1 to 2 marks]***

Level 2: There is an explanation of an experiment to measure the density of the toy soldier, with reference to the equipment needed. The answer has some structure. ***[3 to 4 marks]***

Level 3: There is a clear and detailed explanation of an experiment to measure the density of the toy soldier. The method includes details of the equipment needed and how to process the results to work out the density of the toy soldier.
The answer is well structured. ***[5 to 6 marks]***

Here are some points your answer may include:
Measure and record the mass of the toy soldier using the mass balance.
Fill a eureka can with water.
Place an empty measuring cylinder beneath the spout of the eureka can.
Submerge the toy soldier in the eureka can.
Measure the volume of water displaced from the eureka can using the measuring cylinder.
The volume of water displaced is equal to the volume of the soldier.
Use the equation 'density = mass ÷ volume / $\rho = m \div V$' to calculate the density of the toy soldier.

With questions where you have to describe a method, make sure your description is clear and detailed.

3.1 specific latent heat of fusion ***[1 mark]***

3.2 thermal energy for a change of state = mass × specific latent heat / $E = mL$
So, $L = E \div m$ ***[1 mark]***
40.8 g = (40.8 ÷ 1000) kg = 0.0408 kg
L = 47 100 ÷ 0.0408 ***[1 mark]***
= **$1.154... \times 10^6$ J/kg**
= **1.15×10^6 J/kg (to 3 s.f.)** ***[1 mark]***

3.3 The internal energy of the system increases as it's heated ***[1 mark]***. This is because heating a system transfers energy from thermal energy stores to the kinetic stores of the particles in the system ***[1 mark]***.

3.4 When a system is heated, energy is transferred to the particles in the system, causing them to gain energy in their kinetic stores/move faster ***[1 mark]***. If the substance is heated enough, the particles will have enough energy in their kinetic stores to break the bonds holding them together, so they change state ***[1 mark]***.

4.1 The particles in a gas have high energies ***[1 mark]***, move in random directions at high speeds ***[1 mark]*** and are not arranged in any pattern ***[1 mark]***.

4.2 In a sealed container, the gas particles collide with the container walls ***[1 mark]*** and exert a force on the walls, (creating an outward pressure) ***[1 mark]***.

4.3 density = mass ÷ volume / $\rho = m \div V$ ***[1 mark]***

4.4 ρ = 8.2 ÷ 6.69 ***[1 mark]***
= **1.2 g/cm^3 (to 2 s.f.)** ***[3 marks for correct answer, including units. Deduct 1 mark if reported to an incorrect number of significant figures and deduct 1 mark if incorrect units given]***

4.5 The pressure of the gas within the container increases as the container is heated ***[1 mark]***. This is because the increase in temperature causes the particles to move faster / increases the energy in the kinetic energy stores of the particles, meaning more force is exerted on the walls of the container when the particles collide with it ***[1 mark]***. The increased speed / average energy in kinetic stores of the particles also means there are more collisions, which also increases the total force exerted on the container ***[1 mark]***.

5 density = mass ÷ volume / $\rho = m \div V$
So, $m = \rho \times V$ ***[1 mark]***
volume of cube = 1.5 × 1.5 × 1.5 = 3.375 cm^3 ***[1 mark]***
The cube's density is 3500 kg/m^3.
1 g/cm^3 = 1000 kg/m^3, so this is
3500 ÷ 1000 = 3.5 g/cm^3 ***[1 mark]***
m = 3.5 × 3.375 ***[1 mark]***
= 11.8125 = **12 g (to 2 s.f.)** ***[1 mark]***

6 pV = constant
So, when V = 0.034 m^3 and p = 98 kPa,
pV = 0.034 × 98 = 3.332 ***[1 mark]***
When V = 0.031 m^3, $p \times 0.031 = 3.332$
So p = 3.332 ÷ 0.031 ***[1 mark]***
= **110 kPa (to 2 s.f.)** ***[1 mark]***

Topic 4 — Atomic Structure

Page 51

Isotopes and Nuclear Radiation

Q1 E.g. Alpha radiation would not be suitable because it is stopped by a few cm of air or a sheet of paper ***[1 mark]***. It would not be able to pass through the packaging to sterilise the equipment ***[1 mark]***.

Page 52

Nuclear Equations

Q1 Beta particles ***[1 mark]***

Q2 $^{219}_{86}\text{Rn} \rightarrow {}^{215}_{84}\text{Po} + {}^{4}_{2}\text{He}$
[1 mark for correct layout, 1 mark for correct symbol for an alpha particle, 1 mark for total atomic and mass numbers being equal on both sides]

Page 53

Half-life

Q1 After one half-life the count-rate will be 40 ÷ 2 = 20 cps ***[1 mark]***
After a second half-life: 20 ÷ 2 = 10 cps
After a third half-life: 10 ÷ 2 = 5 cps ***[1 mark]***
So the ratio is 5:40 = 1:8 ***[1 mark]***

Pages 54-55

Warm-Up Questions

1) electrons
2) Isotopes of an element are atoms with the same number of protons / the same atomic number but a different number of neutrons / a different mass number.
3) beta decay
4) gamma rays
5) alpha particles
6) Beta emitters are not immediately absorbed, like alpha radiation, and do not penetrate as far as gamma rays. Therefore variations in the thickness of the sheet significantly affect the amount of radiation passing through.
7) Substances with a short half-life decay very quickly so emit high amounts of radiation initially.

Exam Questions

1.1 Beta (particles) ***[1 mark]***, because the radiation passes through the paper, but not the aluminium, so it is moderately penetrating in comparison to the other two ***[1 mark]***.

1.2 E.g. a Geiger-Muller tube/counter ***[1 mark]***

2.1 The time taken for the number of radioactive nuclei in a sample to halve / the time taken for the count-rate or activity to fall to half of its initial level ***[1 mark]***.

2.2 4 minutes is equivalent to 4 ÷ 2 = 2 half-lives ***[1 mark]***.
After 1 half-life, there will be ½ of the unstable nuclei left.
So, after 2 half-lives, there will be ½ ÷ 2 = **¼ / one quarter** of the unstable nuclei left ***[1 mark]***.

3.1 2 × 60 = 120 seconds
120 ÷ 40 = 3 half-lives ***[1 mark]***
8000 ÷ 2 = 4000, 4000 ÷ 2 = 2000,
2000 ÷ 2 = **1000 Bq** ***[1 mark]***

3.2 8000 ÷ 2 = 4000, 4000 ÷ 2 = 2000, 2000 ÷ 2 = 1000,
1000 ÷ 2 = 500, 500 ÷ 2 = 250.
So it takes **5 half-lives** to drop to 250 Bq ***[2 marks for correct answer, otherwise 1 mark for attempting to halve values to find number of half-lives]***.

3.3 (100 ÷ 8000) × 100 ***[1 mark]*** = **1.25%** ***[1 mark]***

4.1 Protons ***[1 mark]*** and neutrons ***[1 mark]***.

4.2 The total number of protons and neutrons in the nucleus/atom ***[1 mark]***.

4.3 Atom A and atom B ***[1 mark]*** because isotopes of the same element have the same atomic number, but different mass numbers ***[1 mark]***.

4.4 They have opposite charges ***[1 mark]***.

You have to use some of your knowledge from Topic 2 here — an electric field will cause a force on a charged particle. The direction of the force will be opposite on opposite charges, so the alpha and beta particles will be deflected in opposite directions.

5.1 E.g. ${}^{0}_{-1}e$ / ${}^{0}_{-1}\beta$ ***[1 mark]***

5.2 The atomic number increases by 1 ***[1 mark]*** and the mass number stays the same ***[1 mark]***.

5.3 The atomic number doesn't change ***[1 mark]*** and neither does the mass number ***[1 mark]***.

5.4 ${}^{209}_{84}Po \rightarrow {}^{205}_{82}Pb + {}^{4}_{2}He$

[1 mark for both the mass number and atomic number of He, 1 mark for the mass number of Po, 1 mark for the atomic number of Pb]

Page 59

Warm-Up Questions

1) Radioactive contamination is caused when radioactive atoms get into or onto an object. Radioactive irradiation occurs when objects are exposed to radiation from a radioactive source.

2) Any two from: e.g. always handle a source with tongs / never allow the source to touch the skin / never have the source out of its lead-lined box for longer than necessary / wear gloves when handling the source.

3) E.g. because it is ionising and can damage cells.

4) Alpha radiation can't penetrate the skin and is easily blocked by a small air gap, so cannot damage tissue inside the body through irradiation. If a person is contaminated with alpha radiation, however, it is very dangerous as alpha sources are highly ionising and do lots of damage to nearby tissue.

Exam Questions

1.1 The cells will be killed ***[1 mark]***.

1.2 Any one from: e.g. so other scientists can check to see if the data is reproducible / so other scientists can check to see if the data is repeatable ***[1 mark]***.

2.1 A uranium-235 nucleus absorbs a neutron ***[1 mark]*** and splits into two smaller nuclei and releases 2 or 3 neutrons ***[1 mark]***. Some of these neutrons go on to start other fissions, and so on, creating a chain reaction ***[1 mark]***.

2.2 There could be an explosion ***[1 mark]***.

3.1 Iodine-123 is absorbed by the thyroid gland just like normal iodine-127, but it is radioactive / gives out radiation ***[1 mark]***. This radiation can be detected outside of the patient ***[1 mark]***. The radiation produced can be monitored to see if the thyroid gland is working correctly ***[1 mark]***.

3.2 Technetium-99m ***[1 mark]*** because it's got a short half-life, which means it won't be very radioactive inside the patient for long ***[1 mark]***. It's also a gamma source, which means the radiation will pass out of the patient without causing much damage/ionisation ***[1 mark]***.

Topic 5a — Forces, Moments & Pressure

Page 62

Weight, Mass and Gravity

Q1 a) $W = mg = 5 \times 9.8$ ***[1 mark]*** = **49 N** ***[1 mark]***

b) $W = 5 \times 1.6$ ***[1 mark]*** = **8 N** ***[1 mark]***

Page 63

Resultant Forces and Work Done

Q1 20 cm = 0.2 m ***[1 mark]***

$W = Fs = 20 \times 0.2$ ***[1 mark]***

= **4 J** ***[1 mark]***

Page 65

Warm-Up Questions

1) Vector quantity: e.g. velocity / momentum / force
Scalar quantity: e.g. mass / speed / volume

2) Contact force: e.g. air resistance
Non-contact force: e.g. gravitational attraction

3) a) N/kg
b) kg
c) N

4) If all force arrows placed tip-to-tail form a closed loop, the forces are balanced.

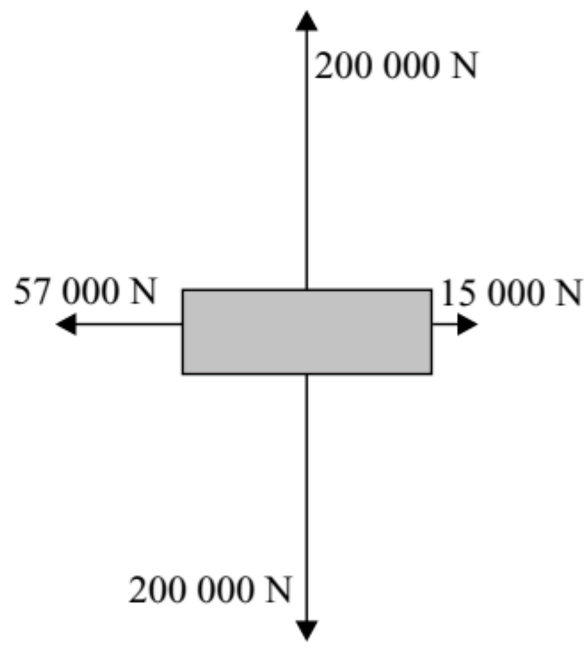
Exam Questions

1.1 Total force to the right = 1700 + 300 = 2000 N
Total force to the left = 2000 N
Total horizontal force = 2000 – 2000 = 0 N ***[1 mark]***
Resultant force = downwards force – upwards force
= 800 – 300
= **500 N** ***[1 mark]*** **downwards** ***[1 mark]***

1.2 Total vertical force = 0 N
so, y = **400 N** ***[1 mark]***
Total horizontal force = 0 N
so, x + 500 N = 2000 N
x = 2000 – 500 = **1500 N** ***[1 mark]***

2.1 work done = force × distance moved along the line of action of the force / $W = Fs$
$W = 42\,000 \times 700$ ***[1 mark]***
= 29 400 000 J ***[1 mark]***
= **29 400 kJ** ***[1 mark]***

2.2



[1 mark for four arrows in directions shown, 1 mark for upwards arrow same length as downwards arrow, 1 mark for driving force arrow bigger than resistive force arrow, 1 mark for all arrows labelled correctly]

You'd still get the marks if you've drawn the driving force to the right, and the resistive force to the left, as the direction that the train was travelling in was not specified in the question.

3.1 The spring would stretch less on Mars because the gravitational field strength on Mars is less than that on Earth ***[1 mark]***, so the ball would weigh less ***[1 mark]*** and the force on the spring would be lower ***[1 mark]***.

3.2 weight = mass × gravitational field strength / $W = mg$
so $g = W \div m$ ***[1 mark]*** = 0.37 ÷ 0.10 ***[1 mark]***
= **3.7 N/kg** ***[1 mark]***

Page 66

Forces and Elasticity

Q1 2 cm = 0.02 m ***[1 mark]***
$F = ke$ so $k = F \div e$ ***[1 mark]*** = 1 ÷ 0.02 ***[1 mark]***
= **50 N/m** ***[1 mark]***

Page 68

Moments

Q1 For forces to balance,
anticlockwise moment = clockwise moment ***[1 mark]***
Let your distance = y
So 300 × 2 = 600 × y ***[1 mark]***
y = 600 ÷ 600 = **1 m** ***[1 mark]***

Page 69

Warm-Up Questions

1) Force = spring constant × extension / $F = ke$
2) The limit of proportionality is the point at which the spring stops behaving according to $F = ke$ / F is no longer proportional to e.
3) moment = force × perpendicular distance from pivot / $M = Fd$
4) Nm (newton metres)

Exam Questions

1.1 The mass on the bottom of the spring / the force applied to the bottom of the spring ***[1 mark]***.
1.2 Any one from: e.g. the spring used throughout the experiment / the temperature the experiment is carried out at ***[1 mark]***.
1.3 extension = 2.5 cm = 0.025 m ***[1 mark]***
force = spring constant × extension / $F = ke$
so $k = F \div e$ ***[1 mark]*** = 4 ÷ 0.025 ***[1 mark]***
= **160 N/m** ***[1 mark]***

Remember to convert the measurement of extension from cm into m before you do your calculation.

1.4 The spring has been inelastically deformed ***[1 mark]***.
2.1 distance = 3 cm = 0.03 m ***[1 mark]***
moment = force × distance / $M = Fd$
M = 15 × 0.03 ***[1 mark]*** = **0.45 Nm** ***[1 mark]***
2.2 distance = 12 cm = 0.12 m ***[1 mark]***
$M = Fd$ = 15 × 0.12 ***[1 mark]*** = **1.8 Nm** ***[1 mark]***
2.3 The B end should be put into the bolt ***[1 mark]*** because it produces the greater moment, making the bolt easier to turn when a given force is applied ***[1 mark]***.

Page 70

Fluid Pressure

Q1 $p = F \div A$ so $F = p \times A$ ***[1 mark]*** = 200 000 × 10 ***[1 mark]***
= **2 000 000 N** ***[1 mark]***

Q2 $p = h\rho g$ so $\rho = p \div hg$ ***[1 mark]***
ρ = 450 ÷ (0.050 × 9.8) ***[1 mark]***
= 918.36... = **920 kg/m^3 (to 2 s.f.)** ***[1 mark]***

Page 72

Warm-Up Questions

1) Pressure is the force on a surface per unit area.
2) Height of the column, and the gravitational field strength.
3) 3.5 N
4) As altitude increases, atmospheric pressure decreases.

Exam Questions

1.1 It is more dense than water ***[1 mark]***.
1.2 pressure = height of column × density × gravitational field strength / $p = h\rho g$
h = (50 ÷ 100) – (8 ÷ 100) = 0.42 m ***[1 mark]***
p = 0.42 × 1000 × 9.8 ***[1 mark]*** = 4116 Pa ***[1 mark]***
= **4100 Pa (to 2 s.f.)** ***[1 mark]***

2.1 pressure = force (normal to surface) ÷ area / $p = F \div A$
p = 175 ÷ 0.25 ***[1 mark]*** = **700 Pa** ***[1 mark]***
2.2 Since liquid cannot be compressed, pressure on piston 2 is equal to the pressure applied by piston 1.
pressure = force (normal to surface) ÷ area / $p = F \div A$
So, $F = p \times A$ ***[1 mark]*** = 700 × 1.3 ***[1 mark]***
= **910 N** ***[1 mark]***

Topic 5b — Forces and Motion

Page 74

Distance, Displacement, Speed and Velocity

Q1 $s = vt$ so $v = s \div t$ ***[1 mark]*** = 200 ÷ 25 ***[1 mark]***
= **8 m/s** ***[1 mark]***

Q2 a) Distance travelled = 1500 m ***[1 mark]***
b) Marie's journey ends at the same position as it started, so the displacement is 0 m ***[1 mark]***.

Page 75

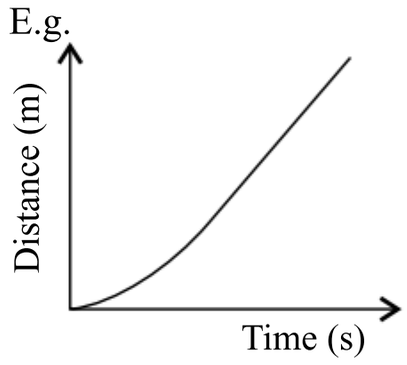
Acceleration

Q1 u = 0 m/s, v = 7.0 m/s, $a = g$ = 9.8 m/s^2,
$s = (v^2 - u^2) \div 2a$ ***[1 mark]*** = (49 – 0) ÷ (2 × 9.8) ***[1 mark]***
= **2.5 m** ***[1 mark]***

Page 76

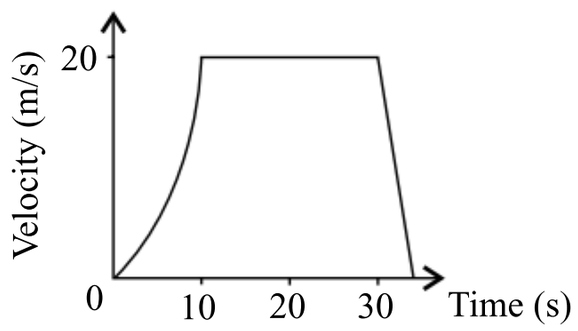
Distance-Time and Velocity-Time Graphs

Q1 E.g.



[1 mark for a curved line with an increasing positive gradient, 1 mark for the line becoming a straight line with a positive gradient]

Q2



[1 mark for an upwards curved acceleration line to 20 m/s, 1 mark for a straight line representing steady speed, 1 mark for a straight line representing deceleration]

Page 78

Warm-Up Questions

1) Speed is scalar, velocity is a vector / velocity has a direction, speed does not.
2) a) E.g. 3 m/s
b) E.g. 30 m/s
c) E.g. 250 m/s

Your answers may be slightly different to these, but as long as they're about the same size, you should be fine to use them in the exam.

3) An upwards curved line / a curve with increasing gradient.
4) A straight, horizontal line.
5) As the speed of the car increases, air resistance on the car increases.
6) Her new terminal velocity is lower than her original terminal velocity.

Exam Questions

1.1 The cyclist travels at a constant speed (of 3 m/s) between 5 s and 8 s ***[1 mark]***, then decelerates between 8 s and 10 s ***[1 mark]***.

1.2 Area of triangle = 0.5 × width × height
Width = 5 – 2 = 3 s
Height = 3 m/s
Distance = 0.5 × 3 × 3 = **4.5 m**
[2 marks, otherwise 1 mark for an attempt to calculate the area under the graph between 2 and 5 seconds]

1.3 Acceleration is given by the gradient of a velocity-time graph.
change in y = 3 – 0 = 3 m/s
change in x = 5 – 2 = 3 s
acceleration = 3 ÷ 3 = **1 m/s²**
[2 marks, otherwise 1 mark for an attempt to calculate the gradient of the line between 2 and 5 seconds]

You could also have used $a = \Delta v \div \Delta t$ here.

1.4 average acceleration = change in velocity ÷ change in time / $a = \Delta v \div \Delta t$
velocity at 8 s = 3 m/s; velocity at 10 s = 2 m/s
so Δv = 2 – 3 = –1 m/s ***[1 mark]***
a = –1 ÷ 2 ***[1 mark]***
= –0.5 m/s²
So the deceleration is **0.5 m/s² *[1 mark]***

Your answer should be positive since the question asks for deceleration, rather than acceleration.

2.1 0 N ***[1 mark]***

2.2 Parachutist A is travelling at a constant velocity / their terminal velocity ***[1 mark]***.

2.3 700 N ***[1 mark]*** because in free fall, at terminal velocity the air resistance is equal to the parachutist's weight ***[1 mark]***.

2.4 Parachutist A will have a higher terminal velocity ***[1 mark]***, since he weighs more ***[1 mark]***. Air resistance increases with speed ***[1 mark]*** so the parachutist will accelerate to a higher speed before air resistance can equal their weight and they reach terminal velocity ***[1 mark]***.

2.5 When the parachute opens, the surface area of the parachutist increases, so the air resistance increases ***[1 mark]*** and they decelerate until the forces rebalance and they reach a new, smaller terminal velocity ***[1 mark]***.

Page 80

Inertia and Newton's Third Law

Q1 Any one from: e.g the gravitational force of the Earth attracts the car and the gravitational force of the car attracts the Earth ***[1 mark]*** / the car exerts a normal contact force down against the ground and the normal contact force from the ground pushes up against the car ***[1 mark]*** / the car (tyres) pushes the road backwards and the road pushes the car (tyres) forwards ***[1 mark]***.

Page 82

Warm-Up Questions

1) 0 N

2) boulder B

Boulder B needs a greater force to accelerate it by the same amount as boulder A.

3) true

This is Newton's Third Law.

4) Masses should be added to the trolley.

Exam Questions

1.1 The ball exerts a force of –500 N on the bat ***[1 mark]***, because, due to Newton's Third Law, if the bat exerts a force on the ball, the ball exerts an equal force on the bat in the opposite direction ***[1 mark]***.

1.2 The acceleration of the ball is greater ***[1 mark]*** because it has a smaller mass, but is acted on by the same size force (and $F = ma$) ***[1 mark]***.

2.1 force = mass × acceleration / $F = ma$
So, $a = F \div m$ ***[1 mark]***
Set direction of van's motion to be positive, so F = –200 N
a = –200 ÷ 2500 ***[1 mark]***
= –0.08 m/s²
So, deceleration = **0.08 m/s² *[1 mark]***

The question asked for deceleration, so you should really quote your answer without the minus sign. However, you should get the marks either way.

2.2 force = mass × acceleration / $F = ma$
F = 10.0 × 29.0 ***[1 mark]***
= **290 N *[1 mark]***

2.3 By Newton's Third Law, force on van in collision is –290 N ***[1 mark]***.
force = mass × acceleration / $F = ma$
So, $a = F \div m$
a = –290 ÷ 2500 ***[1 mark]***
= –0.116 m/s²
So deceleration = **0.12 m/s² (to 2 s.f.) *[1 mark]***

You'd still get the marks here, even if you got 2.2 wrong, as long as your method's correct.

3* How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: A simple experiment to investigate force and acceleration which can be performed with the given equipment is partly outlined.
The answer lacks coherency. ***[1 to 2 marks]***

Level 2: An experiment to investigate force and acceleration which can be performed with the given equipment is outlined in some detail.
The answer has some structure. ***[3 to 4 marks]***

Level 3: An experiment to investigate force and acceleration which can be performed with the given equipment is fully described in detail.
The answer is well structured. ***[5 to 6 marks]***

Here are some points your answer may include:
Place all of the masses in the trolley.
Calculate and record the weight of the trolley.
Place the trolley on the starting line.
Release the trolley, so that it moves through the light gate, and record the acceleration measured.
Take one of the masses from the trolley, and attach it to the hook.
Calculate and record the new total weight of the hook and the new total weight of the trolley.
Reset the position of the trolley on the starting line.
Release the trolley again so that it moves through the light gate, and record the acceleration measured.
Repeat these steps until all the masses from the trolley have been moved to the hook.
Plot your results on a graph of acceleration against weight, and draw a line of best fit.

Topic 5c — Stopping Distances & Momentum

Page 85

More on Stopping Distances

Q1 Thinking distance increases linearly with speed, so thinking distance = 3 × 6 = 18 m ***[1 mark]***
Braking distance increases with speed by 3^2 times.
So braking distance = 3^2 × 6 ***[1 mark]*** = 54 m ***[1 mark]***
Stopping distance = 18 + 54 = **72 m *[1 mark]***

Page 86

Reaction Times

Q1 a) $v^2 - u^2 = 2as$

$v^2 = 2 \times 9.8 \times 0.162 + 0$ ***[1 mark]*** $= 3.1752$ m^2/s^2

$v = \sqrt{3.1752} = 1.781...$ m/s ***[1 mark]***

$a = \Delta v \div t$ so $t = \Delta v \div a$ ***[1 mark]***

$= 1.781... \div 9.8$ ***[1 mark]***

$= 0.181...$ s = **0.18 s (to 2 s.f.)** ***[1 mark]***

b) His reaction time is longer in the evening ***[1 mark]*** so whilst driving, he may take longer to react to a hazard, meaning his thinking distance would be longer ***[1 mark]***.

Page 87

Warm-Up Questions

1) The braking distance.

2) The thinking distance is the distance travelled during your reaction time (the time between seeing a hazard, and applying the brakes).

3) Any one from: e.g. poor grip on the roads increases braking distance / poor visibility delays when you see the hazard / distraction by the weather delays when you see the hazard.

4) Energy is transferred from the kinetic energy stores of the wheels to the thermal energy stores of the brakes.

5) Get the individual to sit with their arm resting on the edge of a table. Hold a ruler end-down so that the 0 cm mark hangs between their thumb and forefinger. Drop the ruler without warning. The individual must grab the ruler between their thumb and forefinger as quickly as possible. Measure the distance at which they have caught the ruler. Use $v^2 - u^2 = 2as$, $a = 9.8$ m/s^2 and $a = \Delta v \div t$ to calculate the time taken for the ruler to fall that distance. This is their reaction time.

Exam Question

1.1 stopping distance = braking distance + thinking distance,
So, braking distance = stopping distance – thinking distance ***[1 mark]***

At 40 mph,
stopping distance = 35 m (accept between 34 m and 36 m)
thinking distance = 13 m (accept between 12 m and 14 m) ***[1 mark]***

braking distance = 35 – 13 ***[1 mark]***

= **22 m** (Accept correct for above readings) ***[1 mark]***

1.2 braking distance ***[1 mark]***

The stopping distance is over twice as high as the thinking distance at 50 mph, so the braking distance must be bigger than the thinking distance.

1.3 Stopping distance is not directly proportional to speed ***[1 mark]***. If stopping distance and speed were directly proportional, the relationship between them would be shown by a straight line / would be linear ***[1 mark]***.

1.4 If the road were icy, the thinking distance graph would not change ***[1 mark]*** but the stopping distance graph would get steeper (as the braking distance would increase) ***[1 mark]***.

The thinking distance graph doesn't change, because the icy road won't change your reaction time. But it will decrease the friction between the car and the road, so the braking distance increases.

Page 88

Momentum

Q1 $p = mv = 60 \times 3$ ***[1 mark]***

= **180 kg m/s** ***[1 mark]***

Q2 Before the gun fires the bullet, the total momentum is zero (neither the gun nor the bullet are moving) ***[1 mark]***. When the bullet leaves the gun, it has momentum in one direction ***[1 mark]***. The gun moves backwards so it has momentum in the opposite direction ***[1 mark]***. This means that the total momentum after the bullet has been fired is zero. Momentum has been conserved ***[1 mark]***.

Page 89

Changes in Momentum

Q1 $p_{before} = (10 \times 6) + (20 \times 0)$

$= 60$ kg m/s ***[1 mark]***

$p_{after} = (10 + 20) \times v = 30v$ ***[1 mark]***

$p_{before} = p_{after}$

$60 = 30v$

so $v = 60 \div 30 =$ **2 m/s** ***[1 mark]***

Page 90

Warm-Up Questions

1) $p = mv = 2.5 \times 10 =$ **25 kg m/s**

2) The momentum is zero.

3) It decreases the force.

Exam Questions

1.1 momentum = mass × velocity / $p = mv$ ***[1 mark]***

1.2 $p = 60 \times 5.0$ ***[1 mark]***

= **300 kg m/s** ***[1 mark]***

1.3 force = change in momentum ÷ time taken / $F = m\Delta v \div \Delta t$

Gymnast comes to a stop, so $m\Delta v = 300$ kg m/s

$F = 300 \div 1.2$ ***[1 mark]***

= **250 N** ***[1 mark]***

2.1 momentum = mass × velocity / $p = mv$

$p = 650 \times 15.0$ ***[1 mark]***

= **9750 kg m/s** ***[1 mark]***

2.2 momentum before = momentum after

Set direction of first car to be positive.

momentum of second car = 750 × (–10.0)

= –7500 kg m/s ***[1 mark]***

Total momentum before = 9750 + (–7500)

= 2250 kg m/s ***[1 mark]***

Total momentum after = (mass of car 1 + mass of car 2) × v

$2250 = (650 + 750) \times v$

so, $v = 2250 \div (650 + 750)$ ***[1 mark]***

$= 2250 \div 1400$

$= 1.607142...$ ***[1 mark]***

= **1.61 m/s (to 3 s.f.)** ***[1 mark]***

2.3 The crumple zone increases the time taken by the car to stop / change its velocity ***[1 mark]***. The time over which momentum changes is inversely proportional to the force acting, so this reduces the force ***[1 mark]***.

3 momentum before = momentum after

momentum of neutron before = 1 × 14 000

= 14 000

momentum of uranium before = 235 × 0

= 0 ***[1 mark]***

momentum of neutron after = 1 × –13 000

= –13 000

momentum of uranium after = $235v$ ***[1 mark]***

So, $14\,000 = -13\,000 + 235v$ ***[1 mark]***

so, $v = (14\,000 + 13\,000) \div 235$

$= 114.8936...$ ***[1 mark]*** = **115 km/s (to 3 s.f.)** ***[1 mark]***

Don't worry too much about the units in this question. The masses given are relative masses, with no units, so we couldn't use the standard units for momentum. As you're only looking for the velocity though, you can just do the calculation as normal, and make sure that the units on your final answer match the units for velocity given in the question.

Topic 6a — Wave Basics & EM Waves

Page 92

Transverse and Longitudinal Waves

Q1 $7.5 \div 100 = 0.075$ m ***[1 mark]***

wave speed = frequency × wavelength,

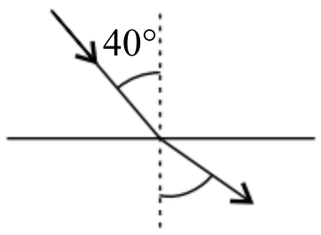
so frequency = wave speed ÷ wavelength ***[1 mark]***

= 0.15 ÷ 0.075 ***[1 mark]*** = **2 Hz** ***[1 mark]***

Page 95

Refraction

Q1



[1 mark for a correct diagram showing rays and the normal, 1 mark for an angle of incidence of 40°, 1 mark for an angle of refraction greater than 40°]

Pages 97-98

Warm-Up Questions

1) In a longitudinal wave, the vibrations are parallel to the direction of travel/energy transfer.

2) wave speed = frequency × wavelength / $v = f\lambda$

3) E.g. Set up and turn on a ripple tank. Then, dim the lights and turn on the lamp so a wave pattern can be seen on the screen below the ripple rank. From the screen, measure the distance between the shadow lines that are a certain number of wavelengths apart, e.g. ten wavelengths. Divide the distance by this number of wavelengths to find the average wavelength. Use the equation wave speed = frequency × wavelength / $v = f\lambda$ to find the speed of the waves.

4) true

5) E.g. Put one of the blocks on a piece of paper and trace round it. Shine a ray of light from a ray box onto one side of the block, at an angle to the block. Trace the incident ray and mark where it emerges from the block. Remove the block and draw a straight line (using a ruler) to connect the incident ray and the emerging ray. Draw the normal to the point of incidence and measure the angles of incidence and refraction. Repeat for the second block, making sure the angle of incidence is the same as for the first block. Compare the angles of refraction for the two blocks.

Exam Questions

1.1 transverse ***[1 mark]***

1.2 5 cm ***[1 mark]***

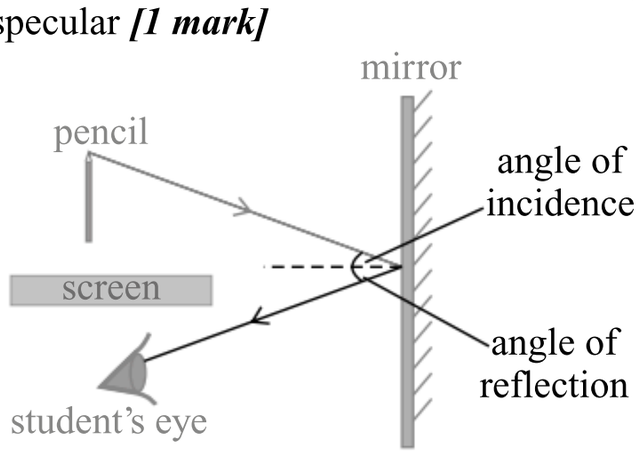
1.3 2 m ***[1 mark]***

1.4 It will halve ***[1 mark]***.

$v = f\lambda$, so if f doubles, then λ must halve, so that v stays the same.

2.1 specular ***[1 mark]***

2.2



[1 mark for line representing reflected ray, 1 mark for the angle of incidence and the angle of reflection being equal, 1 mark for labelling the angle of incidence and the angle of reflection]

3.1 0.005 × 4 ***[1 mark]*** = **0.02 s** ***[1 mark]***

3.2 period ***[1 mark]***

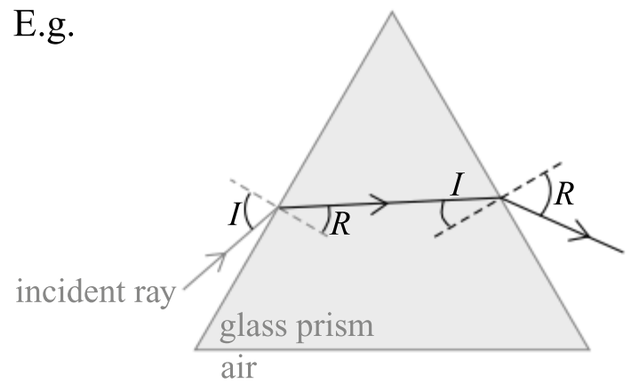
The period of a wave is the time taken for a full wave to pass a point.

3.3 One period is 8 divisions long,
so $T = 0.005 \times 8$ ***[1 mark]*** = 0.04 s ***[1 mark]***
$f = \frac{1}{T} = \frac{1}{0.04}$ ***[1 mark]*** = **25 Hz** ***[1 mark]***

4.1 The distance measured is 1 wavelength ***[1 mark]***. This can be used, together with the frequency of the signal, in the formula for wave speed, wave speed = frequency × wavelength / $v = f\lambda$ ***[1 mark]***.

4.2 wave speed = frequency × wavelength / $v = f\lambda$
So $v = 50 \times 6.8$ ***[1 mark]*** = **340** ***[1 mark]*** **m/s** ***[1 mark]***

5.1 E.g.



[1 mark for refracting the ray towards the normal upon entering the prism, 1 mark for refracting the ray away from the normal as it leaves the prism and 1 mark for correctly labelling all the angles of incidence and refraction]

5.2 E.g. Place the prism on a piece of paper and shine a ray of light at the prism. Trace the incident and emergent rays and the boundaries of the prism on the piece of paper ***[1 mark]***. Remove the prism and draw in the refracted ray through the prism by joining the ends of the other two rays with a straight line ***[1 mark]***. Draw in the normals using a protractor and ruler ***[1 mark]*** and use the protractor to measure I and R at both boundaries ***[1 mark]***.

Page 104

Warm-Up Questions

1) gamma rays

2) Radio waves absorbed by a receiver cause electrons in the receiver to oscillate back and forth. These oscillating electrons form an alternating current.

3) The microwaves penetrate a few centimetres into the food before being absorbed by water molecules. The energy from the absorbed microwaves causes the food to heat up.

4) E.g. to carry data over long distances.

5) It has enough energy to knock electrons off atoms — this can cause gene mutations, cell destruction and cancer.

Exam Questions

1.1 Any one of: e.g. fluorescent light bulbs / tanning lamps

1.2 Any one of: e.g. sunburn / premature ageing of skin / blindness / increased risk of skin cancer ***[1 mark]***.

1.3 A gamma ray emitter is injected into/swallowed by the patient ***[1 mark]*** and the gamma rays emitted are detected by an external detector ***[1 mark]***. By seeing where the gamma rays come from, they can track the progress of the tracer around the body and check it is functioning correctly ***[1 mark]***.

1.4 Gamma rays can pass out of the body without being absorbed ***[1 mark]***.

2.1 Radio waves can bend around / pass through objects so the signal can reach the inside of the house ***[1 mark]***. Light waves would be blocked by the mountain / walls, and so would not be able to reach the receiver inside the house ***[1 mark]***.

2.2 Microwave radiation ***[1 mark]***. It passes through Earth's watery atmosphere without being absorbed, so can reach the satellites ***[1 mark]***.

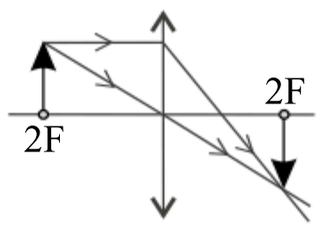
2.3 They are transmitted through the atmosphere into space, where they are picked up by a satellite receiver orbiting Earth ***[1 mark]***. The satellite transmits the signal back to Earth in a different direction, where it is received by a satellite dish connected to the house ***[1 mark]***.

Topic 6b — Light, Radiation & Other Waves

Page 107

Images and Convex Lenses

Q1



[1 mark for an image at a distance 2F in front of the lens, 1 mark for an inverted image, 1 mark for two correct light rays]

Page 108

Concave Lenses and Magnification

Q1 magnification = image height ÷ object height
= 6 ÷ 12 ***[1 mark]*** = **0.5** ***[1 mark]***

Q2 magnification = image height ÷ object height
so image height = magnification × object height ***[1 mark]***
= 2.5 × 10 ***[1 mark]*** = **25 cm** ***[1 mark]***

Page 110

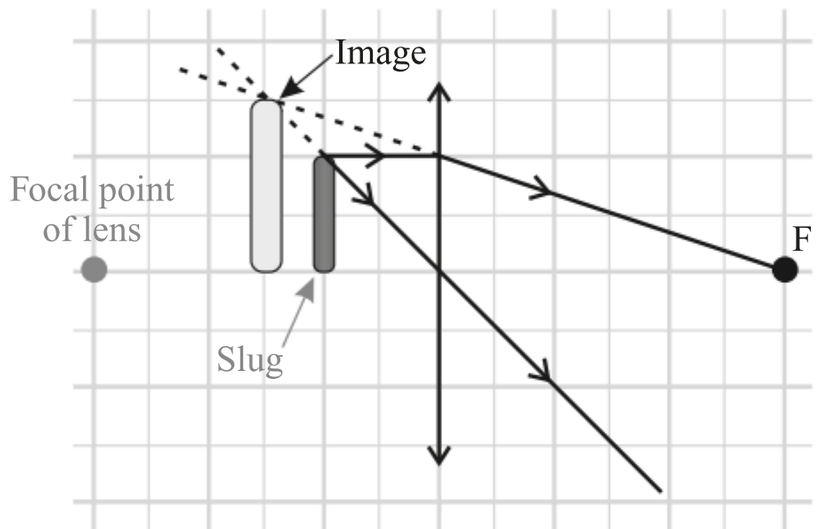
Warm-Up Questions

1) Concave/diverging and convex/converging.
2) Concave lenses always create virtual images. Convex lenses can create real or virtual images.
3) It most strongly reflects the wavelengths of light corresponding to the red part of the visible spectrum (all other wavelengths are absorbed).
4) The wavelengths reflected from the object (those corresponding to the green part of the visible spectrum) are absorbed by the red filter. The lack of any visible light transmitted is perceived as black, so the object appears black.

Exam Questions

1.1 A concave lens causes parallel rays of light to diverge (spread out) rather than converge (come together) ***[1 mark]***. This means that Sunita's lens cannot focus the sunlight to start a fire ***[1 mark]***.

1.2



[1 mark for showing a ray going to a correctly positioned focal point, 1 mark for showing a ray going through the centre of the lens, 1 mark for showing both rays extended backwards (dotted) and image drawn where they cross]

You sometimes need to draw another focal point on the opposite side of lens, the same distance away from the centre line of the lens.

1.3 magnification = image height ÷ object height
= 1.5 ÷ 1 ***[1 mark]***
= **1.5** ***[1 mark — accept any answer between 1.4 and 1.6]***

With this question, it's easiest to use the grid squares as units and just count them. Alternatively, you could measure the lengths of the object and the image with a ruler.

2.1 Daylight consists of an even distribution of all the wavelengths ***[1 mark]***. Light from a flat screen device consists of mostly wavelengths below 500 nm ***[1 mark]***. It emits far fewer wavelengths greater than 500 nm ***[1 mark]***.

2.2 E.g. use a colour (e.g. an orange) filter between the screen and the users eyes ***[1 mark]*** to absorb the blue light ***[1 mark]***.

Page 112

Black Body Radiation

Q1 The second star is cooler ***[1 mark]*** because it has the longer peak wavelength, and the colder an object is, the longer its peak wavelength ***[1 mark]***.

Page 114

Ultrasound

Q1 $s = vt$
Time taken for sound to reach the seabed:
$t = s \div v$ ***[1 mark]*** = 2500 ÷ 1520 = 1.64... ***[1 mark]***
Time taken for the sound to return:
1.64... × 2 = 3.28... = **3.3 s (to 2 s.f.)** ***[1 mark]***

Pages 116-117

Warm-Up Questions

1) Leslie cube
2) In a given place, more radiation is being emitted than is being absorbed.
3) longitudinal
4) Partial reflection is when some of the wave travels into the new medium (and is possibly refracted), whilst some is reflected.
5) E.g. earthquakes
6) P waves
7) They can't pass through the liquid outer core.

Exam Questions

1.1 Any one from: e.g. place the thermometers at equal distances away from the cube / place the thermometers at the same height as each other / make sure no thermometers are in direct sunlight/a draught ***[1 mark]***.

1.2 C, A, B, D ***[1 mark]***

Thermometer C will heat up fastest, since it is the one closest to the face which is most like a perfect black body and so the best infrared emitter.

1.3 The times recorded would be longer ***[1 mark]***. Cooler objects emit infrared radiation at a lower rate ***[1 mark]***.

1.4 E.g. the resolution of the digital thermometer is higher, so there will be less uncertainty in the results (the results will be more accurate) / the student is less likely to misread the temperature (human error is less likely) ***[1 mark for each correct reason, up to a maximum of 2]***.

2.1 20 kHz / 20 000 Hz ***[1 mark]***

2.2

[1 mark]

2.3 It will be partially reflected / some of it will be reflected and some will be transmitted into media 2 ***[1 mark]***.

3.1 A ***[1 mark]***. They only travel through solids ***[1 mark]***, and so are stopped by the liquid outer core ***[1 mark]***.

3.2 When the waves are following curved paths, it shows that the density of the material they are moving through is changing gradually ***[1 mark]***, suggesting there are areas of the Earth's inner structure made up of a similar material of a changing density ***[1 mark]***. When the waves abruptly change direction, it shows that the density at that point changes suddenly by a larger amount ***[1 mark]***, suggesting they have entered a layer of a new material in the Earth's structure ***[1 mark]***.

4.1 Bellatrix ***[1 mark]***. It emits blue light which has a lower wavelength than red light ***[1 mark]***. The peak wavelength of a hotter object is lower than the peak wavelength of a cooler object ***[1 mark]***.

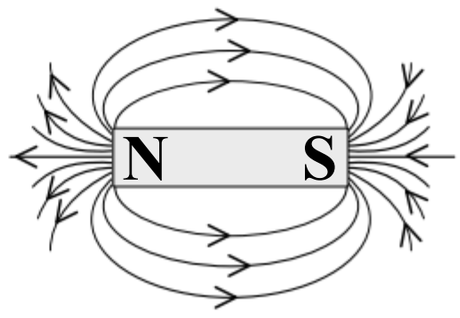
4.2 People on Earth wouldn't hear the explosion ***[1 mark]*** as sound can't travel through the vacuum of space ***[1 mark]***.

Topic 7 — Magnetism & Electromagnetism

Page 121

Warm-Up Questions

1)



2) It's made up of concentric circles around the wire (with the wire at the centre).
3) E.g. cranes use them to pick up iron and steel in scrap yards.

Exam Questions

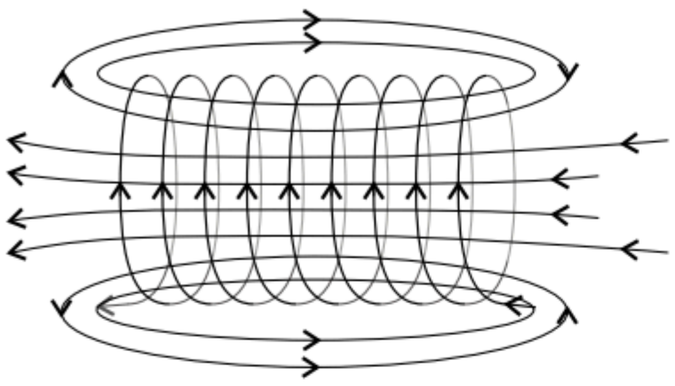
1.1 E.g. Put the magnets on a piece of paper and place many compasses in different places between the magnets to show the magnetic field at those points ***[1 mark]***. The compass needles will line up with the magnetic field lines ***[1 mark]***.

They could also use iron filings to show the pattern.

1.2 The field lines point straight across from the north pole towards the south pole ***[1 mark]***.

1.3 Attraction ***[1 mark]***, as opposite poles are facing each other and opposite poles attract ***[1 mark]***.

2.1



[1 mark for correct field shape, 1 mark for direction]

2.2 E.g. iron ***[1 mark]***

2.3 The paperclips will fall ***[1 mark]***, because an electromagnet only has a magnetic field if a current is flowing ***[1 mark]***.

Page 122

The Motor Effect

Q1 Into the page ***[1 mark]***.

Page 124

Warm-Up Questions

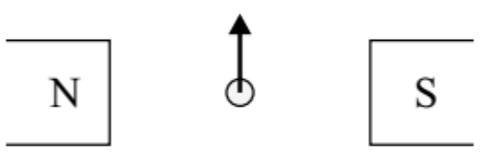
1) When a current-carrying wire in a magnetic field experiences a force.

2) First finger — magnetic field
Second finger — current
Thumb — force (motion)

3) Swapping the polarity of the dc supply (reversing the current), or swapping the magnetic poles over (reversing the field).

Exam Questions

1.1

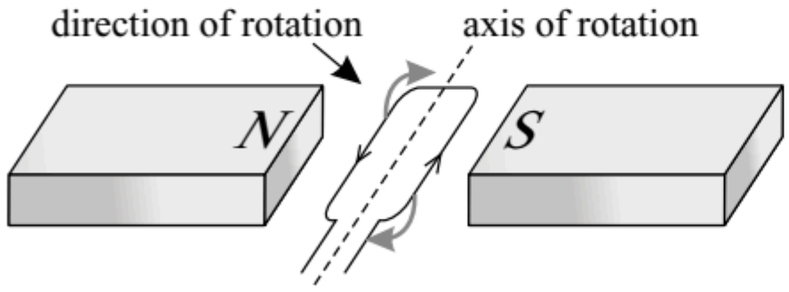


[1 mark]

1.2 The direction of the force would be reversed too ***[1 mark]***.

1.3 The force would be lower ***[1 mark]***.

2.1 E.g.



[1 mark for any indication that the current goes anticlockwise]

2.2 force = magnetic flux density × current × length of conductor inside field / $F = BIl$
$F = 0.2 \times 15 \times 0.1$ ***[1 mark]*** = **0.3 N** ***[1 mark]***

2.3 By swapping the direction of the current/contacts every half turn (using a split-ring commutator) ***[1 mark]*** so the forces on the loop always act in a way that keeps the loop rotating ***[1 mark]***.

3 When the ac current flows through the coil of wire in the magnetic field of the permanent magnet, the coil of wire experiences a force ***[1 mark]***. The force causes the coil, and so the cone, to move ***[1 mark]***. The alternating current is constantly changing direction and so the direction of the force on the coil is constantly changing and the cone vibrates back and forth ***[1 mark]***. The vibrations cause pressure variations in the air that cause sound waves ***[1 mark]***.

Page 127

Transformers

Q1 a) $V_p \div V_s = n_p \div n_s$
$V_p = (n_p \div n_s) \times V_s$ ***[1 mark]*** $= (16 \div 4) \times 20$ ***[1 mark]***
$=$ **80 V** ***[1 mark]***

b) output power = input power
$P = VI$ so $V_s \times I_s = V_p \times I_p$
$320 = 80 \times I_p$ ***[1 mark]***
$I_p = 320 \div 80$ ***[1 mark]*** = **4 A** ***[1 mark]***

Page 128

Warm-Up Questions

1) an alternator

2) direct current

3) alternating current

4) More turns on the secondary coil.

Exam Questions

1.1 Because electrical current is generated ***[1 mark]***.

1.2 Any one from: e.g. move the magnet out of the coil / move the coil away from the magnet / insert the south pole of the magnet into the same end of the coil / insert the north pole of the magnet into the other end of the coil ***[1 mark]***.

1.3 Any one from: e.g. push the magnet into the coil more quickly / use a stronger magnet ***[1 mark]***.

1.4 Zero / no reading ***[1 mark]***.

Movement is needed to generate a current.

2.1 A potential difference will not be induced in the secondary coil when using the dc supply ***[1 mark]*** because the magnetic field generated in the iron core is not changing ***[1 mark]***. A pd is only induced if the secondary coil experiences a change in magnetic field ***[1 mark]***.

2.2 power = potential difference × current / $P = VI$
$= 12 \times 2.5$ ***[1 mark]*** = **30 W** ***[1 mark]***

2.3 Assuming that the transformer is 100% efficient, the output power is 30 W ***[1 mark]***.
power = potential difference × current / $P = VI$
So, $I = P \div V$ ***[1 mark]***
$= 30 \div 4$ ***[1 mark]***
$=$ **7.5 A** ***[1 mark]***

2.4 $V_s \div V_p = n_s \div n_p$
So, $n_s = (V_s \div V_p) \times n_p$ ***[1 mark]***
$n_s = (4 \div 12) \times 15$ ***[1 mark]***
$=$ **5 turns** ***[1 mark]***

Topic 8 — Space Physics

Page 131

The Solar System and Orbits

Q1 E.g. artificial satellites are made by humans, whilst natural satellites are not ***[1 mark]***.

Q2 Its orbital speed would be faster ***[1 mark]***. The force due to gravity increases the closer you get to the Earth's surface, so if the radius of the Moon's orbit was smaller, the force acting on it would be larger ***[1 mark]***.
So to remain in a stable orbit, the Moon's instantaneous velocity would need to be larger ***[1 mark]***.

Page 133

Warm-Up Questions

1) A red super giant.

2) E.g. planets and dwarf planets.

3) gravity

4) red-shift

5) E.g. dark matter and dark energy.

Exam Questions

1 Moons are **natural satellites** ***[1 mark]***. They orbit **planets** ***[1 mark]***. **Artificial satellites** ***[1 mark]*** are man-made, and most of them orbit **the Earth** ***[1 mark]***.

2.1 Stars form from clouds of dust and gas which are pulled together by gravitational attraction ***[1 mark]***. As the density increases, the temperature rises ***[1 mark]***. When the temperature gets high enough, nuclear fusion happens and a huge amount of energy is emitted ***[1 mark]***.

2.2 The forces acting on a main sequence star are balanced, so it doesn't collapse or explode ***[1 mark]***. The energy released by nuclear fusion provides an outward force to balance the force of gravity pulling everything inwards ***[1 mark]***.

2.3 They become unstable and eject their outer layer of dust and gases ***[1 mark]*** which leaves a hot, dense, solid core known as a white dwarf ***[1 mark]***. White dwarfs then cool to become black dwarfs ***[1 mark]***.

2.4 They start to glow brightly again, undergo more fusion, and expand and contract several times ***[1 mark]***. Heavier elements are formed and the star eventually explodes in a supernova ***[1 mark]***. The supernova leaves behind a neutron star or a black hole ***[1 mark]***.

3 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief description of red-shift and the Big Bang theory. The answer lacks coherency. ***[1 to 2 marks]***

Level 2: There is a description of red-shift and the Big Bang theory, and some explanation of how red-shift provides evidence for the Big Bang theory. The answer has some structure. ***[3 to 4 marks]***

Level 3: There is a description of red-shift and the Big Bang theory, and a clear and detailed explanation of how red-shift provides evidence of the expansion of the universe, and so evidence for the Big Bang theory. The answer is well structured. ***[5 to 6 marks]***

Here are some points your answer may include:
Red-shift is when the wavelengths of observed light from a source are longer than that of the light emitted by the source.
This occurs when the source of light is moving away from the observer.
The light from most distant galaxies is observed to be red-shifted.
The light from galaxies that are further away from us is red-shifted more than the light from nearer galaxies.
This indicates that all of these galaxies are moving away from us and each other.
This suggests that the universe is expanding.
The Big Bang theory is the theory that the universe began as a small region of space that was very hot and dense, which exploded and has been expanding ever since.
Therefore, the observations of red-shift support the idea of the origin of the universe suggested by the Big Bang theory.

Practice Paper 1

Pages 150-170

1.1 liquid ***[1 mark]***

1.2 The internal energy of the substance increases ***[1 mark]*** as energy is transferred by heating to the (kinetic and potential) energy stores of the particles ***[1 mark]***.

1.3 melting ***[1 mark]***

1.4 density = mass ÷ volume / $\rho = m \div V$ ***[1 mark]***

1.5 $\rho = 0.36 \div (4.0 \times 10^{-4})$ ***[1 mark]*** = **900 kg/m^3** ***[1 mark]***

2.1 Strength: Any one from: e.g. the hot water in the beaker cools by conduction and convection, as in a hot water tank / the water can be raised to a similar temperature as in a hot water tank / both the tank and the model involve heat loss from water ***[1 mark]***.
Weakness: Any one from: e.g. the mass of water used is much less than in a hot water tank / the insulation of a hot water tank wouldn't be cotton wool ***[1 mark]***.

2.2 The thickness of the cotton wool jacket ***[1 mark]***.

2.3 E.g. starting temperature of the water / length of time water is left to cool / volume of water used ***[1 mark]***.

2.4 repeatable ***[1 mark]***

2.5 The thicker the cotton wool jacket, the smaller the temperature change / the higher the final temperature of the water (and hence the less energy transferred from the water's thermal energy store) ***[1 mark]***.

3 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief description of both models of the atom. The answer lacks coherency. ***[1 to 2 marks]***

Level 2: There is a description of both models of the atom, and some description of the scientific discoveries that led to the development of the nuclear model. The answer has some structure. ***[3 to 4 marks]***

Level 3: There is a clear and detailed description of both models of the atom, and of the scientific discoveries and experiments which led to the development of the nuclear model. The answer is well structured. ***[5 to 6 marks]***

Here are some points your answer may include:
Model X is the plum pudding model of the atom.
The plum pudding model describes the atom as a sphere of positive charge, with negatively charged electrons within it.
Model Y is the nuclear model of the atom.
The nuclear model of the atom describes the atom as a nucleus, made up of positively charged protons and uncharged neutrons, orbited by electrons.
In the early 20th century (1909), Rutherford performed the alpha scattering experiment.
They fired a beam of positively charged alpha particles at a thin gold foil.
Based on the plum pudding model, they expected all the alpha particles to pass through the foil, with some deflection.
However, they found that most alpha particles passed through the foil without deflecting, while a small few were deflected back towards the emitter.
This suggested that most of the atom is empty space, since so many of the alpha particles passed through without deflecting.
It also suggested there was a small, positively charged 'nucleus' in the centre of the atom, which caused the backwards deflection of the alpha particles.
This formed the basis of the nuclear model of the atom.
Later, Niels Bohr adapted the nuclear model, when he discovered electrons orbited the nucleus at specific distances called energy levels.
In 1932, James Chadwick discovered the neutron, an uncharged particle with the same mass as a proton.
This led to the idea that the nucleus is made of protons and neutrons, explaining the imbalance between the atomic number and mass number of an atom.

4.1 gas ***[1 mark]***

4.2 C ***[1 mark]***

4.3 The line is flat because a change of state is occurring ***[1 mark]*** and all energy transferred to the substance is being used to break intermolecular bonds and change the state of the substance, and not increase its temperature ***[1 mark]***.

4.4 thermal energy for a change of state = mass × specific latent heat / $E = mL$
So, $L = E \div m$ ***[1 mark]***
$L = 69\,000 \div 1.5$ ***[1 mark]***
= **46 000 J/kg** ***[1 mark]***

5.1 Electrons have been transferred from the scientist's hair to the comb ***[1 mark]***.

5.2 The hairs all have the same (positive) charge ***[1 mark]***. Like charges repel, so the hairs repel each other and so stand on end ***[1 mark]***.

5.3 (+) 0.2 nC ***[1 mark]***

The hair will have an equal and opposite charge to the comb.

5.4 It will attract the hair ***[1 mark]***.

6.1 The specific heat capacity of a substance is the energy required to raise the temperature of 1 kg of that substance by 1 °C ***[1 mark]***.

6.2 change in thermal energy = mass × specific heat capacity × temperature change / $\Delta E = mc\Delta\theta$
So, $c = \Delta E \div m\Delta\theta$ ***[1 mark]***
$\Delta\theta = 93 - 18 = 75$ ***[1 mark]***
$c = 126\,000 \div (1 \times 75)$ ***[1 mark]***
= 1680 = **1700 J/kg°C (to 2 s.f.)** ***[1 mark]***

6.3 Water has a higher specific heat capacity than oil ***[1 mark]*** so it transfers more energy to the surroundings compared to oil, for the same decrease in temperature ***[1 mark]***.

7.1 214 – 83 = **131 neutrons** ***[1 mark]***

7.2 $^{222}_{86}Rn$ ***[1 mark]***

$^{210}_{83}Bi$ ***[1 mark]***

7.3 420 days ***[1 mark]***

7.4 Half-life when activity drops to half original total.
Activity after 1 half-life = 80 ÷ 2 = 40 Bq
Using Figure 9, when activity = 40 Bq, time = 140 days.
Therefore, half-life = **140 days** ***[1 mark]***

7.5 Alpha radiation is strongly ionising ***[1 mark]***, and when it enters living cells it can easily kill them, or damage them and cause cancer ***[1 mark]***. Alpha radiation is easily absorbed by thin barriers (e.g. skin) or the air, so is unlikely to reach the body's delicate organs if the source is outside the body ***[1 mark]***. However it is much more dangerous inside the body, where it is almost certain to be absorbed by living cells and cause damage ***[1 mark]***.

8.1 The current is higher in circuit A ***[1 mark]***.
The total resistance is higher in circuit B ***[1 mark]***.

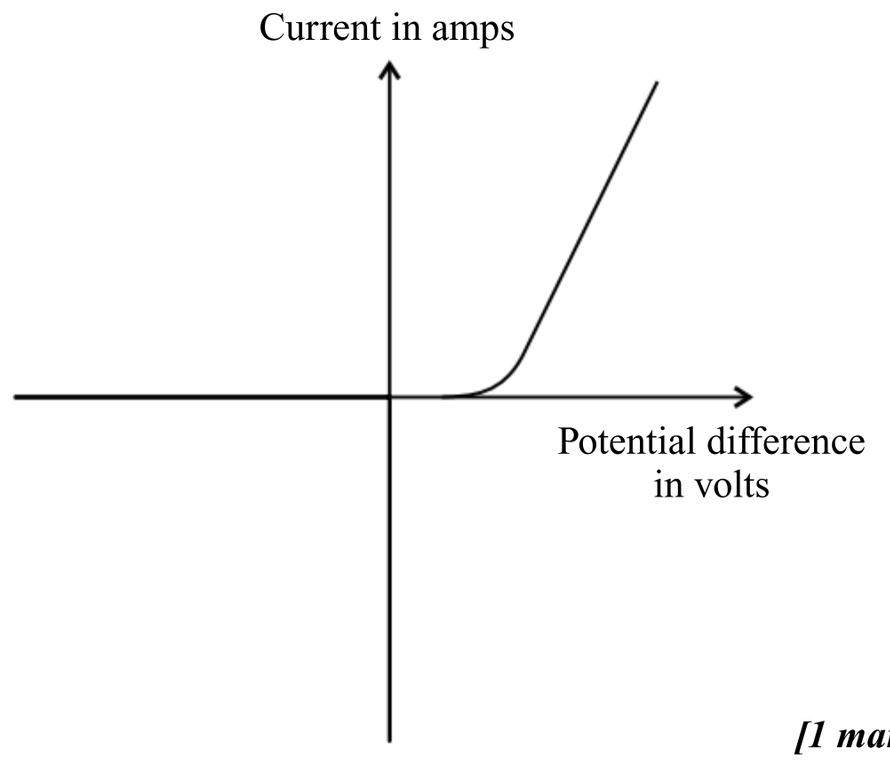
8.2 power = potential difference × current / $P = VI$ ***[1 mark]***

8.3 $P = 11 \times 0.30$ ***[1 mark]***
= **3.3 W** ***[1 mark]***

8.4 As the intensity of light increases, the resistance of component C decreases ***[1 mark]***.

8.5 E.g. a car engine temperature sensor ***[1 mark]***.

8.6



[1 mark]

9.1 Solar = 5×10^9 kWh ***[1 mark]***
Hydro-electric = 1×10^9 kWh ***[1 mark]***
[Allow 1 mark only if the answers are 5 kWh and 1 kWh and × 10^9 is omitted]

9.2 Wind power ***[1 mark]***

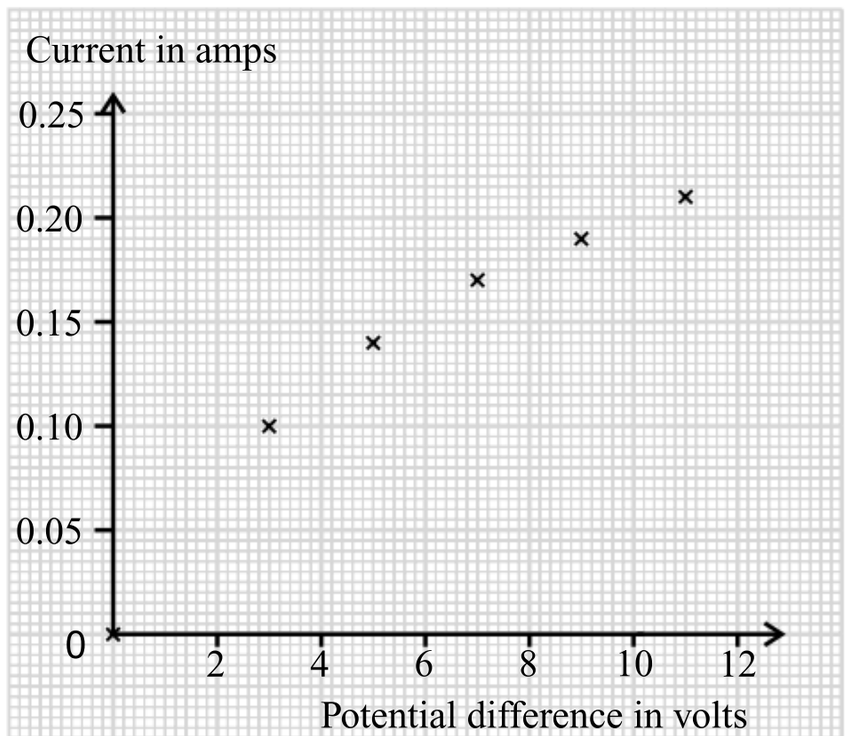
9.3 Any one from: e.g. there are more hours of daylight in summer than in winter, so more electricity can be generated from solar power / there are generally more clear days during summer so more electricity can be generated from solar power ***[1 mark]***.

9.4 Advantage: Any one from: e.g. non-renewables are reliable / we can easily alter energy output to meet demand ***[1 mark]***.
Disadvantage: Any one from: e.g. non-renewables will eventually run out / burning some non-renewables, e.g. fossil fuels, produces pollutants / burning some non-renewables, e.g. fossil fuels can cause global warming ***[1 mark]***.

9.5 Any two from: e.g. they're not as reliable as non-renewables / they can have large initial setup costs / a number of power stations that use renewable resources can only be built in certain locations / a lot of renewable resources cannot easily have their power output increased to meet demand ***[1 mark for each correct limitation]***.

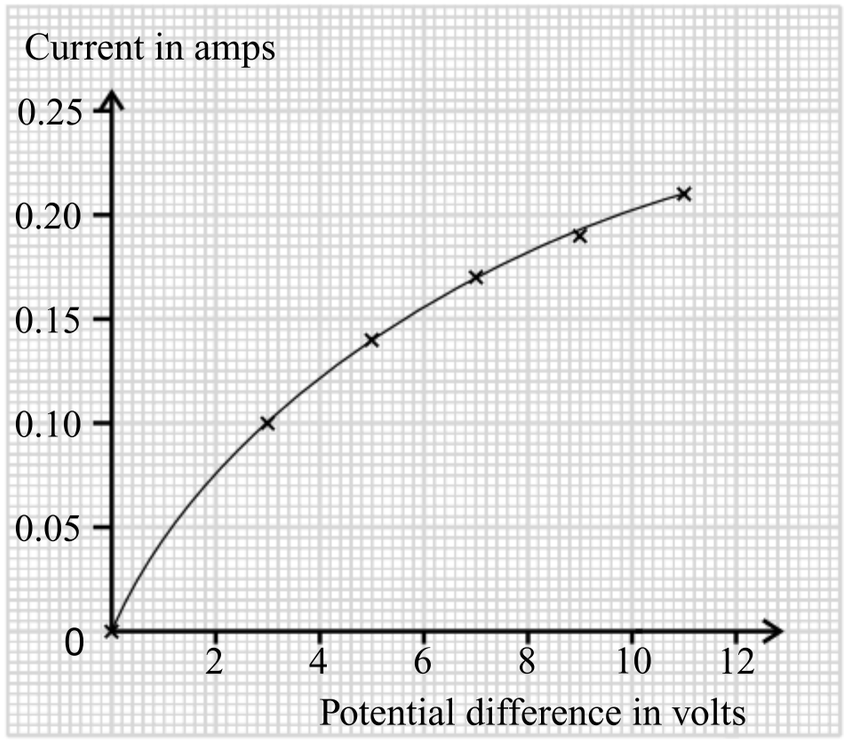
9.6 Geothermal power requires a suitably volcanically active site to work, and there are few viable volcanic sites in the UK ***[1 mark]***.

10.1



[2 marks for all points plotted correctly, otherwise 1 mark for any 3 plotted correctly]

10.2



[1 mark for curved line close to all points and through the origin]

10.3 As an electrical charge flows through the lamp, energy is transferred to its thermal energy store ***[1 mark]***. As the current increases, more energy is transferred electrically to the thermal energy store of the component, increasing the temperature ***[1 mark]***. Resistance increases with temperature ***[1 mark]***, and so resistance increases with increasing current ***[1 mark]***.

10.4 potential difference = current × resistance / $V = IR$
So, $I = V \div R$ ***[1 mark]*** $= 15 \div 60$ ***[1 mark]*** $= 0.25$ A ***[1 mark]***
charge flow = current × time / $Q = It$
$Q = 0.25 \times 180$ ***[1 mark]*** = **45 C** ***[1 mark]***

11.1 Earth wire: yellow and green (stripes) ***[1 mark]***
Live wire: brown ***[1 mark]***

11.2 The earth wire stops exposed metal parts / the casing of the device from becoming live in the event of a fault ***[1 mark]***.

11.3 A live wire may still have a potential difference in it even when this circuit is broken / current is now flowing ***[1 mark]***. If you touch the live wire, your body forms a link between the wire and the earth, and current flows through you, and can give you an electric shock ***[1 mark]***.

11.4 power = (current)^2 × resistance / $P = I^2R$
$P = 10.0^2 \times 23.0$ ***[1 mark]*** $= 2300$ W ***[1 mark]***
energy transferred = power × time / $E = Pt$
$E = 2300 \times 180$ ***[1 mark]*** = **414 000 J** ***[1 mark]***

12 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief description of devices used in the National Grid. The answer lacks coherency. ***[1 to 2 marks]***

Level 2: There is a description of devices used in the National Grid, and some explanation of how they are used to efficiently transmit electricity. The answer has some structure. ***[3 to 4 marks]***

Level 3: There is a clear and detailed description of how devices such as transformers are used in the National Grid to transmit electricity efficiently. There is also some mention of how the electricity is made suitable for home use after transmission. The answer is well structured. ***[5 to 6 marks]***

Here are some points your answer may include:
The National Grid needs to transmit high power electricity from power stations to consumers.
Because the power is high, and $P = IV$, the electricity must be transmitted with either high potential difference (V) or high current (I).
When a current flows through a conductor, work is done against its resistance and energy is dissipated to the thermal energy store of the conductor.
The higher the current, the more energy is transferred to thermal energy stores and wasted.
Transformers are a device which can change the potential difference of electricity without changing the power.
In the National Grid, step-up transformers are used to increase the potential difference of the electricity before transmission.
Because the power is constant, increasing the potential difference of the electricity decreases the current.
So step-up transformers allow the electricity to be transmitted at very high potential difference and very low currents.
As it is transmitted at a very low current, very little energy is dissipated to thermal energy stores.
So transmission at low current is more efficient.
The transmission potential difference is too large to be used by consumers.
Before reaching the consumer, step-down transformers are used.
Step-down transformers decrease the potential difference back to a usable level for consumer appliances.

13.1 gravitational potential energy = mass × gravitational field strength (g) × height / $E_p = mgh$ ***[1 mark]***

13.2 $E_p = 600 \times 9.8 \times 20$ ***[1 mark]***
$= 117\,600$ J = **120 kJ (to 2 s.f.)** ***[1 mark]***

13.3 As the carriage drops, the energy is transferred from its gravitation potential energy store to its kinetic energy store, so the kinetic energy store graph slopes upwards as the g.p.e store graph slopes downwards ***[1 mark]***. Due to conservation of energy, as there are no resistive forces, the sum of the energy in the gravitational potential and kinetic energy stores at any given point between A and B is constant (so the lines have the same, but opposite, gradient) ***[1 mark]***.

13.4 kinetic energy = 0.5 × mass × (speed)^2 / $E_k = \frac{1}{2}mv^2$
Conservation of energy:
total energy before = total energy after
No resistive forces, so by conservation of energy, all energy transferred from g.p.e. store goes to the kinetic energy store.
So, $E_p = E_k = 117\,600$ J ***[1 mark]***

So, $v = \sqrt{\frac{2E_k}{m}}$ ***[1 mark]*** $= \sqrt{\frac{2 \times 117\,600}{600}}$ ***[1 mark]***
$= 19.79... =$ **20 m/s (to 2 s.f.)** ***[1 mark]***

Practice Paper 2

Pages 171-190

1.1 X-rays are less easily absorbed by tissues, but are absorbed more by denser materials like bones and metal ***[1 mark]***. The detector detects the X-rays transmitted at each point and creates a picture showing areas of tissue and bone ***[1 mark]***.

1.2 X-rays are ionising radiation which can kill living cells or cause mutations and cancer ***[1 mark]***.

1.3 Lead absorbs X-rays and prevents them reaching the radiographer ***[1 mark]***.

1.4 Any two from: e.g. X-rays can be used to treat cancers / X-ray photographs can be used to diagnose bone fractures / X-ray photographs can be used to diagnose dental problems (problems with your teeth) ***[1 mark for each correct answer]***.

2.1 E.g. frequency is the number of complete waves passing a certain point per second / the number of waves produced by a source each second ***[1 mark]***.

2.2 wave speed = frequency × wavelength / $v = f\lambda$ ***[1 mark]***

2.3 So, $\lambda = v \div f$ ***[1 mark]***
$\lambda = 3.0 \times 10^8 \div 2.5 \times 10^{14}$ ***[1 mark]*** = **1.2×10^{-6} m** ***[1 mark]***

2.4 Texture: Matt surfaces are better infrared emitters then shiny surfaces ***[1 mark]***.
Colour: Black surfaces are better infrared emitters than white surfaces ***[1 mark]***.

2.5 Any one from: e.g. to reduce the effect of random errors / to make the results more accurate ***[1 mark]***.

3.1 The outward pressure/expansion due to nuclear fusion ***[1 mark]*** balances the inwards force due to gravitational attraction ***[1 mark]***.

3.2 A white dwarf ***[1 mark]***.

3.3 35 km/s ***[1 mark]***

The smaller the orbit, the faster the orbital speed.

3.4 The object will weigh less on Venus than on Earth ***[1 mark]*** as the gravitational field strength of Venus is lower than Earth's (and weight is proportional to gravitational field strength) ***[1 mark]***.

4.1 27 s ***[1 mark]***

4.2 Graph is linear between 15 s and 27 s.
27 – 15 = **12 s** ***[1 mark]***

4.3 0 N ***[1 mark]***

When an object is travelling at a constant speed in a fixed direction, all the forces are balanced and so the resultant force will be zero.

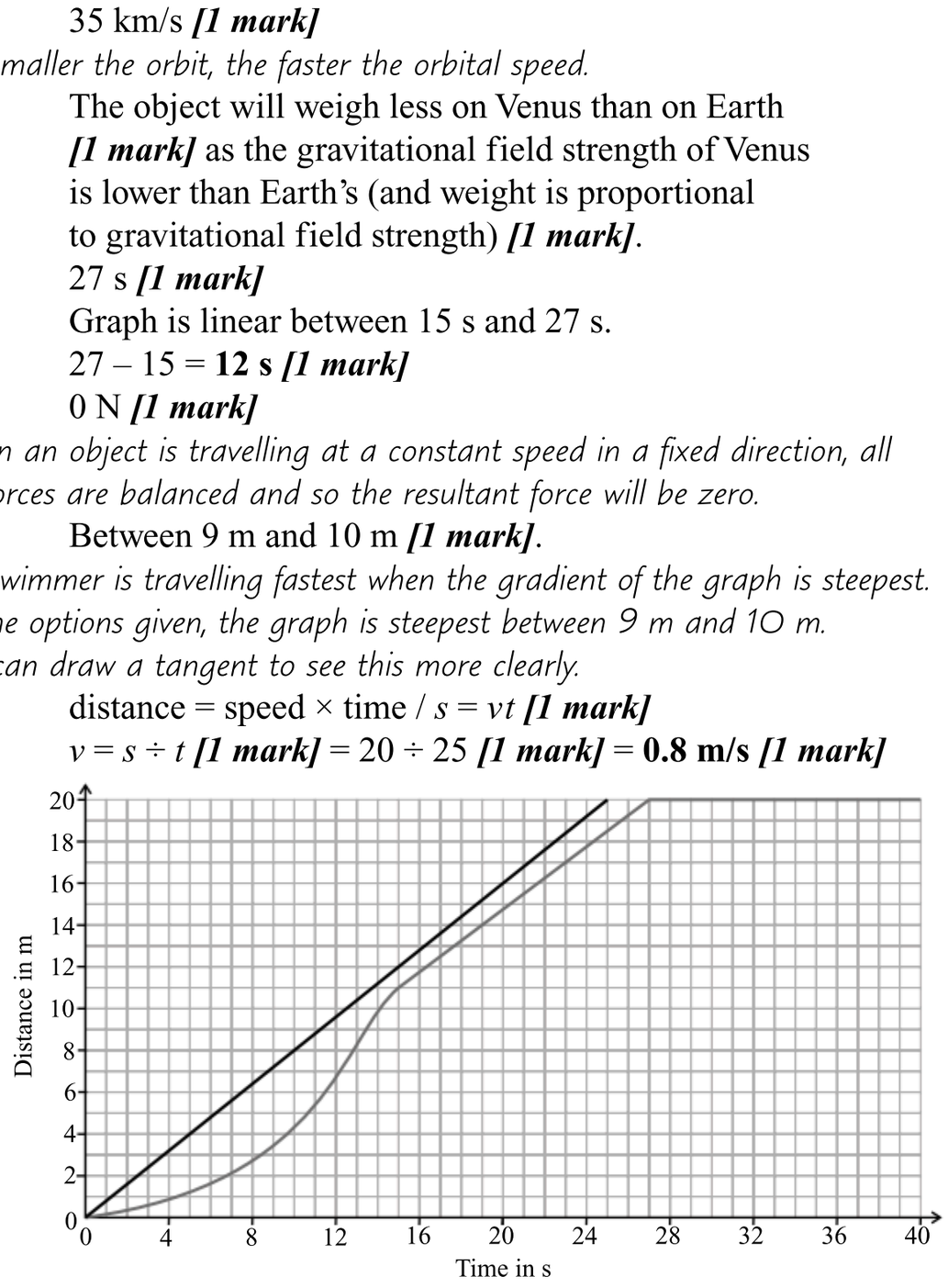
4.4 Between 9 m and 10 m ***[1 mark]***.

The swimmer is travelling fastest when the gradient of the graph is steepest. Of the options given, the graph is steepest between 9 m and 10 m. You can draw a tangent to see this more clearly.

4.5 distance = speed × time / $s = vt$ ***[1 mark]***

4.6 $v = s \div t$ ***[1 mark]*** = 20 ÷ 25 ***[1 mark]*** = **0.8 m/s** ***[1 mark]***

4.7



[1 mark for straight line, 1 mark for correct start and end point]

4.8 Yes, the camera will be able to film the swimmer for the whole length because its distance-time graph is always above the swimmer's ***[1 mark]***, and so the camera is always ahead of the swimmer (or level at $t = 0$) ***[1 mark]***.

5.1 Random errors ***[1 mark]***

5.2* How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief description of an experiment using the equipment shown. The answer lacks coherency. ***[1 to 2 marks]***

Level 2: There is a description of an experiment which can be performed with the equipment shown, and some description of how the results should be processed to calculate the spring constant. The answer has some structure. ***[3 to 4 marks]***

Level 3: There is a clear and detailed description of an experiment which can be performed using the equipment shown, and of how to calculate the spring constant from the resulting force-extension graph. The answer is well structured. ***[5 to 6 marks]***

Here are some points your answer may include:
Measure the mass of each of the masses using a mass balance.
Calculate the weight of each of the masses using $W = mg$.
Using the ruler, measure the length of the spring when it has no masses hanging from it (the unstretched length).
Hang a mass from the spring, and record the force applied by the mass (the weight of the mass) and the new length of the spring.
Calculate the extension of the spring by subtracting the unstretched length from the new length.
Repeat these steps, increasing the mass hanging from the spring, recording the new weight and calculating the extension each time.
After you have a suitable number of points, plot your results on a force-extension graph, with force on the y-axis, and tension on the x-axis.
Draw a line of best fit on your results.
Identify the linear part of your graph.
Since $F = ke$, $k = F \div e$, the gradient of the linear part of the graph is equal to the spring constant.
Calculate the spring constant by calculating the gradient of the linear part of the graph.

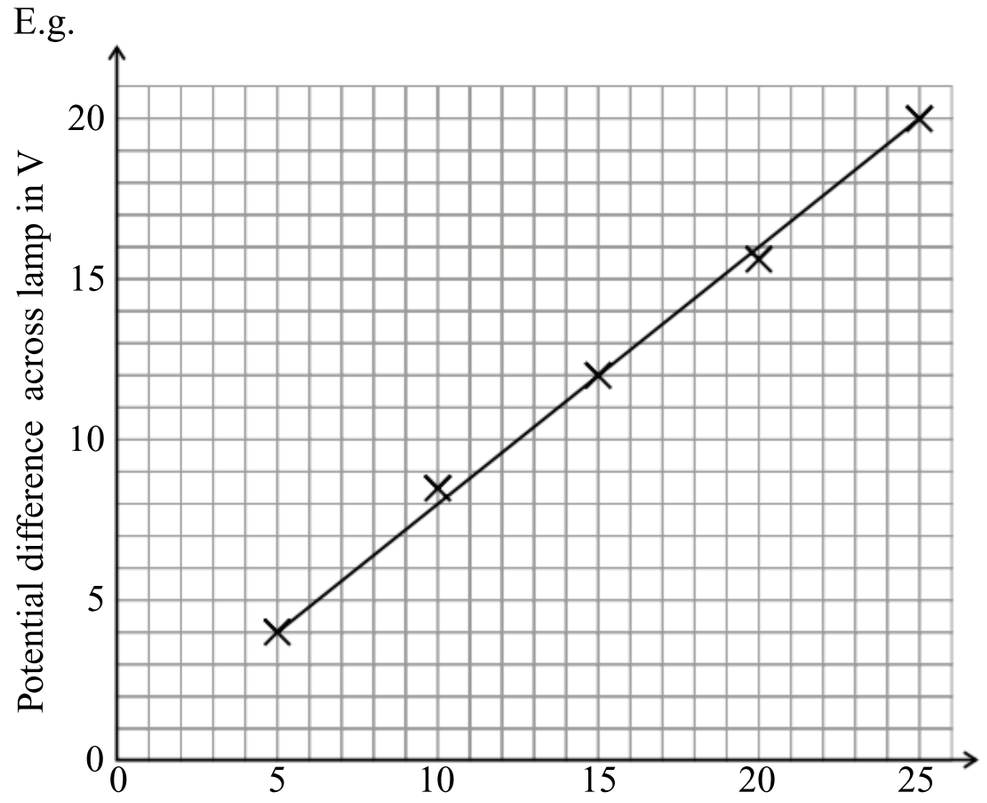
5.3 The limit of proportionality ***[1 mark]***.

5.4 Below the limit of proportionality (where the graph is linear) the spring obeys $F = ke$.
So, k = gradient = change in y ÷ change in x
change in y = 7.0 – 0 = 7.0 N
change in x = 3.5 – 0 = 3.5 cm = 0.035 m
k = 7.0 ÷ 0.035 = **200 N/m**
[3 marks, otherwise 1 mark for an attempt to calculate the gradient of the linear part of the graph and 1 mark for the correct units]

You'd get the marks here for using any values for the change in y and x, as long as they're from the linear part of the graph.

6.1 The alternating current in the primary coil causes a changing magnetic field in the iron core and so in the secondary coil ***[1 mark]***. The changing magnetic field induces an alternating potential difference in the secondary coil (the generator effect) ***[1 mark]***.

6.2 E.g.



[1 mark for appropriate line of best fit]

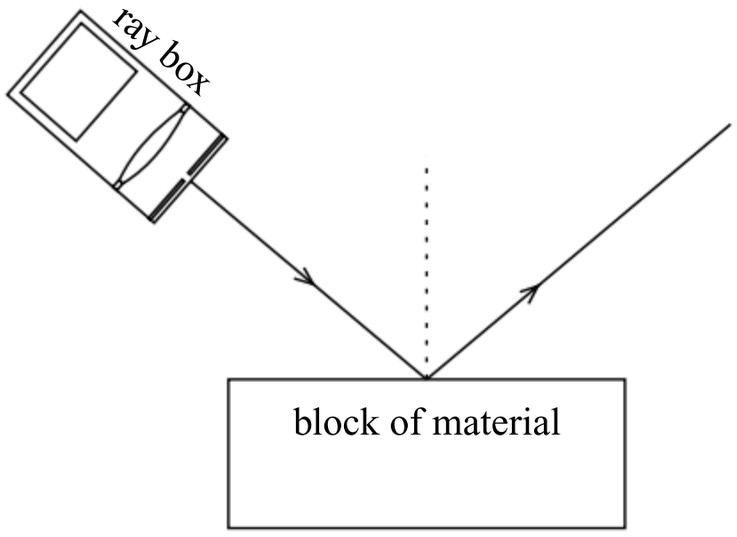
6.3 A step-up transformer ***[1 mark]***

6.4 $V_p \div V_s = n_p \div n_s$
So, $n_s = (V_p \div V_s) \times n_p$ ***[1 mark]***
n_s = (400 000 ÷ 25 000) × 5000 ***[1 mark]***
= **80 000** ***[1 mark]***

6.5 $V_s I_s = V_p I_p$
So, $I_p = (V_s \times I_s) \div V_p$ ***[1 mark]***
I_p = (400 000 × 250) ÷ 25 000 ***[1 mark]***
= **4000 A** ***[1 mark]***

7.1 50° (Allow values between 48° and 52°) ***[1 mark]***

7.2



[1 mark for straight line drawn from point of incidence, 1 mark for angle with the normal equal to the angle stated in 7.1]

7.3 The speed of light is slower in the block than in air ***[1 mark]***.

8.1 E.g. Reaction times vary from person to person ***[1 mark]***. A sample needs to be large to be representative of the population ***[1 mark]***.

To get a single representative value, it's a good idea to take an average from a large number of results.

8.2 Any one from: e.g. the condition of the tyres / the weather / the condition of the road surface / the condition of the brakes ***[1 mark]***.

8.3 $v^2 - u^2 = 2as$

So, $a = (v^2 - u^2) \div 2s$ ***[1 mark]***

$= (0 - 25^2) \div (2 \times 150)$ ***[1 mark]*** $= -2.0833...$ m/s^2

So, deceleration = **2.1 m/s^2 (to 2 s.f.)** ***[1 mark]***

Since the question asked for deceleration, you should ignore the minus sign when you quote your answer.

8.4 The steering column collapsing slows the driver down more gradually. This increases the time over which the change of momentum happens ***[1 mark]*** which reduces the forces acting on the driver ***[1 mark]*** since force is equal to the rate of change of momentum ***[1 mark]***.

9.1 The motor effect ***[1 mark]***

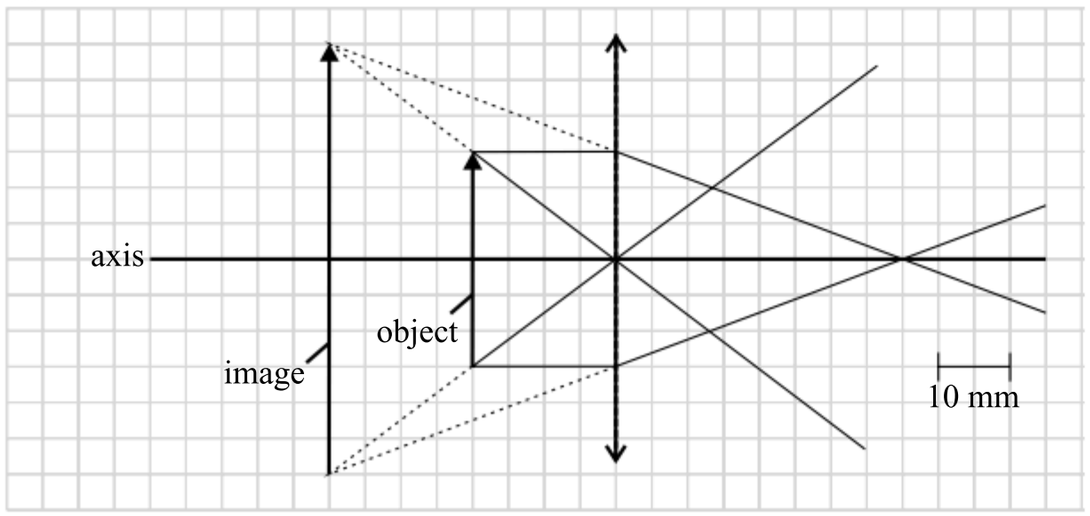
9.2 The motor effect causes a current-carrying wire in a magnetic field to experience a force, so when current flows in the coil, the wire on each side feels a force ***[1 mark]***. The force on one side of the coil acts upwards, and the force on the other side acts downwards, so the coil rotates, and turns the attached fan blades ***[1 mark]***.

9.3 The metal bar won't feel any force ***[1 mark]*** because the current through it is parallel to the magnetic field ***[1 mark]***.

9.4 Sound waves hit the diaphragm that is attached to a coil of wire ***[1 mark]***. This causes the coil of wire to move back and forward in the magnetic field ***[1 mark]***, which generates an alternating current in the coil ***[1 mark]***. The generated current depends on the properties of the sound wave, so variations in sound are converted into variations in current ***[1 mark]***.

10.1 Convex lens ***[1 mark]***

10.2



[3 marks for complete ray diagram, otherwise 1 mark for a ray from the object travelling parallel to the axis and refracting so that it travels through the axis and can be traced back to the image, and 1 mark for a ray passing through the centre of the lens without bending that can be traced back to the image.]

10.3 focal length = 8 squares,
2 squares = 10 mm
so focal length = $(8 \times 10) \div 2$
= **40 mm** ***[1 mark]***

10.4 object height = 6 squares = 30 mm
image height = 12 squares = 60 mm
magnification = image height ÷ object height
= 60 ÷ 30 ***[1 mark]***
= **2** ***[1 mark]***

You could also have used the heights in squares instead. As long as the image and object heights are in the same units, you'll get the same answer.

11.1 moment = force × (perpendicular) distance from the pivot / $M = Fd$ ***[1 mark]***

11.2 $M = 100 \times 0.2$ ***[1 mark]***
= **20** ***[1 mark]*** **Nm** ***[1 mark]***

11.3 The moment will decrease ***[1 mark]*** because the perpendicular distance between the line of action of the force and the axis of rotation/pivot/point A will decrease ***[1 mark]***.

12.1 weight = mass × gravitational field strength / $W = mg$ ***[1 mark]***

12.2 $W = 80.0 \times 9.8$ ***[1 mark]***
= 784 N = **780 N (to 2 s.f.)** ***[1 mark]***

12.3* How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief description of the skydiver's motion. The answer lacks coherency. ***[1 to 2 marks]***

Level 2: There is a description of the skydiver's motion, with brief reference to the forces acting on the skydiver. The answer has some structure. ***[3 to 4 marks]***

Level 3: There is a clear and detailed description of the skydiver's motion, including details of the forces acting on the skydiver. The answer is well structured. ***[5 to 6 marks]***

Here are some points your answer may include:
At first, the gradient of the graph is steep, showing that the skydiver is accelerating quickly.
This is because the force of gravity acting on the skydiver is much more than the air resistance slowing the skydiver down.
The graph then starts to level off, showing that acceleration is decreasing.
This is because the air resistance increases with speed, so the resultant force acting on the skydiver decreases.
Eventually, the air resistance becomes equal to the accelerating force of weight.
At this point, the skydiver travels at a constant (terminal) velocity.
This is represented by the straight horizontal line on the graph.

12.4 Area under the graph = ~16 squares
(Allow between 15 and 17) ***[1 mark]***
Each square = 30 × 10 = 300 m
Distance fallen = 16 × 300 ***[1 mark]***
= **4800 m**
[1 mark, allow between 4500 and 5100]

Glossary

A

Acceleration A change in velocity in a certain amount of time.

Accurate result A result that is very close to the true answer.

Activity (radioactive) The number of nuclei of a sample that decay per second.

Air resistance The frictional force caused by air on a moving object.

Alpha decay A type of radioactive decay in which an alpha particle is given out from a decaying nucleus.

Alpha particle A positively-charged particle made up of two protons and two neutrons (a helium nucleus).

Alpha particle scattering experiment An experiment in which alpha particles were fired at gold foil to see if they were deflected. It led to the plum pudding model being abandoned in favour of the nuclear model of the atom.

Alternating current (ac) Current that is constantly changing direction.

Alternator A device which generates an ac supply using the generator effect.

Ammeter A component used to measure the current through a component. It is always connected in series with the component.

Amplitude The maximum displacement of a point on a wave from its undisturbed position.

Angle of incidence The angle the incoming ray makes with the normal at a boundary.

Angle of reflection The angle a reflected ray makes with the normal at a boundary.

Angle of refraction The angle a refracted ray makes with the normal when a wave refracts at a boundary.

Anomalous result A result that doesn't seem to fit with the rest of the data.

Artificial satellite A man-made satellite (normally orbiting the Earth).

Atmosphere A relatively thin layer of air that surrounds the Earth.

Atmospheric pressure The pressure felt by any surface within the atmosphere, due to air molecules colliding with the surface.

Atom A particle that makes up matter.

Atomic number The number of protons in the nucleus of an atom.

Axis (of a lens) A line passing through the middle of a lens, perpendicular to the lens.

B

Background radiation The radiation which surrounds us at all times, arising from both natural and man-made sources.

Beta decay A type of radioactive decay in which a beta particle is given out from a decaying nucleus.

Beta particle A high-speed electron emitted in beta decay.

Bias Unfairness in the way data is presented, possibly because the presenter is trying to make a particular point (sometimes without knowing they're doing it).

Big Bang theory The idea that the universe began from a small, very hot and dense region of space, which exploded and has been expanding ever since.

Bio-fuel A renewable energy resource made from plant products or animal dung.

Black body An object that absorbs all the electromagnetic radiation that hits it. A black body is also the best possible emitter of radiation.

Black dwarf The remains of a star that are left behind when a white dwarf cools.

Black hole A super dense point in space that light cannot escape from.

Braking distance The braking distance is the distance a vehicle travels after the brakes are applied until it comes to a complete stop, as a result of the braking force.

C

Calibrate Measure something with a known quantity to make sure the instrument being used to measure that quantity gives the correct value.

Categoric data Data that comes in distinct categories, e.g. metals (copper, zinc, etc.).

Chain reaction A reaction which keeps going (without any outside input) because the products of the reaction cause further reactions (e.g. nuclear fission).

Closed system A system where neither matter nor energy can enter or leave. The net change in total energy in a closed system is always zero.

Concave lens A lens that curves inwards and causes rays of light parallel to the axis to diverge (spread out) so they appear to have come from the principal focus.

Conduction A method of energy transfer by heating where vibrating particles transfer energy through a material by colliding with neighbouring particles and transferring energy between their kinetic energy stores.

Conservation of energy principle Energy can be transferred usefully from one energy store to another, stored or dissipated — but it can never be created or destroyed.

Conservation of momentum In a closed system, the total momentum before an event is the same as the total momentum after the event.

Contamination (radioactive) The presence of unwanted radioactive atoms on or inside an object.

Continuous data Numerical data that can have any value within a range (e.g. length, volume or temperature).

Control experiment An experiment that's kept under the same conditions as the rest of the investigation, but where the independent variable isn't altered.

Control variable A variable in an experiment that is kept the same.

Convection A method of energy transfer by heating in liquids and gases in which energetic particles move away from hotter regions to cooler regions.

Conversion factor A number which you must multiply or divide a unit by to convert it to a different unit.

Convex lens A lens that bulges outwards and causes rays of light parallel to the axis to converge (come together) at the principal focus.

Correlation A relationship between two variables.

Cosmic ray Radiation from space.

Current The flow of electric charge. The size of the current is the rate of flow of charge. Measured in amperes (A).

Glossary

D

Density A substance's mass per unit volume.

Dependent variable The variable in an experiment that is measured.

Diffuse reflection When parallel waves are reflected by a rough surface (e.g. a piece of paper) and the reflected rays are scattered in lots of different directions.

Diode A circuit component that only allows current to flow through it in one direction. It has a very high resistance in the other direction.

Direct current (dc) Current that always flows in the same direction.

Discrete data Numerical data that can only take a certain value, with no in-between value (e.g. number of protons).

Displacement The straight-line distance and direction from an object's starting position to its finishing position.

Distance-time graph A graph showing how the distance travelled by an object changes over a period of time.

Drag The frictional force caused by any fluid (a liquid or gas) on a moving object.

Dwarf planet A planet-like object in space that orbits a star but which doesn't match all of the rules for being a planet.

Dynamo A device which generates a dc supply using the generator effect.

E

Ear drum The part of the ear which vibrates when sound waves enter the ear. It passes on these vibrations to other parts of the ear, which convert them to electrical signals that cause the sensation of hearing.

Earth wire The green and yellow wire in an electrical cable that only carries current when there's a fault. It stops exposed metal parts of an appliance from becoming live.

Efficiency The proportion of input energy transfer which is usefully transferred. Also the proportion of input power which is usefully output.

Elastic deformation An object undergoing elastic deformation will return to its original shape and length once any forces being applied to it are removed.

Elastic object An object which can be elastically deformed.

Elastic potential energy store Anything that has been stretched or compressed, e.g. a spring, has energy in its elastic potential energy store.

Electric field A region in which an electrically charged object experiences an electrostatic force.

Electromagnet A solenoid with an iron core.

Electromagnetic (EM) spectrum A continuous spectrum of all the possible wavelengths of electromagnetic waves.

Electron A subatomic particle with a relative charge of –1.

Electrostatic attraction/repulsion The non-contact force which acts to bring together opposite charges (attraction) / push apart like charges (repulsion).

Energy store A means by which an object stores energy. There are 8 different types of energy store: thermal, kinetic, gravitational potential, elastic potential, chemical, magnetic, electrostatic and nuclear.

Equilibrium A state in which all the forces acting on an object are balanced, so the resultant force is zero.

F

Fair test A controlled experiment where the only thing being changed is the independent variable.

Fleming's left-hand rule The rule used to work out the direction of the force produced by the motor effect. Your first finger points in the direction of the magnetic field, your second finger points in the direction of the current and your thumb points in the direction of the force (or motion).

Fluid A substance that can flow — either a liquid or a gas.

Focal length (of a lens) The distance from the centre of a lens to the principal focus.

Force A push or a pull on an object caused by it interacting with something.

Fossil fuel The fossil fuels are coal, oil and natural gas. They're non-renewable energy resources that we burn to generate electricity.

Free body diagram A diagram that shows all the forces acting on an isolated object, the direction in which the forces are acting and their (relative) magnitudes.

Frequency The number of complete waves passing a certain point per second. Measured in hertz (Hz).

Friction A force that opposes an object's motion. It acts in the opposite direction to motion.

G

Gamma decay A type of radioactive decay in which a gamma ray is given out from a decaying nucleus.

Gamma ray A high-frequency, short-wavelength electromagnetic wave.

Gear A circular disc with teeth round its edge. It can be used to transmit the rotational effect of a force.

Geiger-Müller tube A particle detector that is used with a counter to measure count rate.

Generator effect The induction of a potential difference across a conductor which is experiencing a change in external magnetic field. If the conductor is part of a complete circuit, this will cause a current to flow.

Geothermal power A renewable energy resource where energy is transferred from the thermal energy stores of hot rocks underground and is used to generate electricity or to heat buildings.

Gradient The slope of a line graph. It shows how quickly the variable on the *y*-axis changes with the variable on the *x*-axis.

Gravitational potential energy (g.p.e) store Anything that has mass and is in a gravitational field has energy in its gravitational potential energy store.

H

Half-life The time it takes for the number of nuclei of a radioactive isotope in a sample to halve, or the time it takes for the count rate (or activity) of a radioactive sample to fall to half its initial level.

Hazard Something that has the potential to cause harm (e.g. fire, electricity, etc.).

Hypothesis A possible explanation for a scientific observation.

Glossary

I

Independent variable The variable in an experiment that is changed.

Induced magnet A magnetic material that turns into a magnet when it is placed inside another magnetic field.

Inelastic deformation An object undergoing inelastic deformation will not return to its original shape and length once the forces being applied to it are removed.

Inertia The tendency of an object to remain stationary or continue travelling at a constant velocity.

Inertial mass The ratio between the resultant force acting on an accelerating object and its acceleration.

Infrared (IR) radiation A type of electromagnetic wave continually emitted and absorbed by all objects.

Intensity The power per unit area, i.e. how much energy is transferred to a given area in a certain amount of time.

Internal energy The total energy that a system's particles have in their kinetic and potential energy stores.

Ionising radiation Radiation that has enough energy to knock electrons off atoms.

Irradiation Exposure to radiation.

Isotopes Different forms of the same element, which have atoms with the same number of protons (atomic number), but different numbers of neutrons (and so different mass numbers).

***I-V* characteristic** A graph of current against potential difference for a component.

K

Kinetic energy store Anything that's moving has energy in its kinetic energy store.

L

Law of reflection The angle of reflection of a reflected ray is always equal to the angle of incidence.

Lever A device that increases the distance between an applied force and a pivot, making it easier to do work.

Light-dependent resistor (LDR) A resistor whose resistance decreases as light intensity increases.

Limit of proportionality The point beyond which the force applied to an elastic object is no longer directly proportional to the extension of the object.

Live wire The brown wire in an electrical cable that carries an alternating potential difference from the mains.

Longitudinal wave A wave in which the oscillations are parallel to the direction of energy transfer.

Lubricant A substance (usually a liquid) that can flow easily between two objects. Used to reduce friction between surfaces.

M

Magnetic field A region where magnetic materials (like iron and steel) and current-carrying wires experience a force.

Magnetic flux density The number of magnetic field lines per unit area. Its symbol is *B* and it is measured in tesla, T.

Magnetic material A material (such as iron, steel, cobalt or nickel) which can become an induced magnet while it's inside another magnetic field.

Magnification The ratio of image size to object size.

Main sequence star A star in a stable period of its life, where the outward pressure from fusion balances the inward pull of gravity.

Mass number The number of neutrons and protons in the nucleus of an atom.

Mean (average) A measure of average found by adding up all the data and dividing by the number of values there are.

Median (average) A measure of average found by selecting the middle value from a data set arranged in ascending order.

Medical tracer A radioactive isotope that can be injected into or swallowed by people. Their progress around the body can be followed using an external detector.

Microwave A type of electromagnetic wave that can be used for cooking and satellite communications.

Mode (average) A measure of average found by selecting the most frequent value from a data set.

Model Used to describe or display how an object or system behaves in reality.

Moment The turning effect of a force.

Momentum A property of a moving object that is the product of its mass and velocity.

Moon A natural satellite which orbits a planet.

Motor effect When a current-carrying wire in a magnetic field experiences a force.

N

National Grid The network of transformers and cables that distributes electrical power from power stations to consumers.

Nebula A cloud of dust and gas in space.

Neutral wire The blue wire in an electrical cable that current in an appliance normally flows through. It is around 0 V.

Neutron A subatomic particle with a relative charge of 0.

Neutron star The very dense core of a star that is left behind when a red super giant explodes in a supernova.

Newton's First Law An object will remain at rest or travelling at a constant velocity unless it is acted on by a resultant force.

Newton's Second Law The acceleration of an object is directly proportional to the resultant force acting on it, and inversely proportional to its mass.

Newton's Third Law When two objects interact, they exert equal and opposite forces on each other.

Non-contact force A force that can act between objects that are not touching.

Non-renewable energy resource An energy resource that will run out, as it cannot be replaced as quickly as it is being used.

Normal (at a boundary) A line that's perpendicular (at 90°) to a surface at the point of incidence (where a wave hits the surface).

Nuclear fission When an atomic nucleus splits up to form two smaller nuclei.

Nuclear fusion When two nuclei join to create a heavier nucleus.

Nuclear model A model of the atom that says that the atom has a small, central, positively-charged nucleus with negatively-charged electrons moving around the nucleus, and that most of the atom is empty space. The nucleus is made up of protons and neutrons.

Nucleus (atom) The centre of an atom, containing protons and neutrons.

Glossary

O

Ohmic conductor A conductor with resistance that is constant at a constant temperature. It has a linear *I*-*V* characteristic.

P

Parallel circuit A circuit in which every component is connected separately to the positive and negative ends of the battery.

Partial reflection When waves are incident on a boundary, some are reflected and some are transmitted.

Peer-review The process in which other scientists check the results and explanations of an investigation before they are published.

Period (of a wave) The time taken for one full cycle of a wave to be completed.

Permanent magnet A magnetic material that always has its own magnetic field around it.

Physical change A change where you don't end up with a new substance — it's the same substance as before, just in a different form. (A change of state is a physical change.)

Planet A natural object in space which orbits a star.

Potential difference The driving force that pushes electric charge around a circuit, measured in volts (V). Also known as pd or voltage.

Power The rate of transferring energy (or doing work). Normally measured in watts (W).

Precise result When all the data is close to the mean.

Prediction A statement that can be tested and is based on a hypothesis.

Pressure The force per unit area exerted on a surface.

Principal focus of a concave lens The point where rays hitting the lens parallel to the axis appear to have come from.

Principal focus of a convex lens The point where rays hitting the lens parallel to the axis all meet.

Proton A subatomic particle with a relative charge of +1.

Protostar The earliest stage in the life cycle of a star. Protostars are formed when the force of gravity causes clouds of dust and gas to pull together.

R

Radiation dose A measure of the risk of harm to your body due to exposure to radiation.

Radioactive decay The random process of a radioactive substance giving out radiation from the nuclei of its atoms.

Radioactive substance A substance that spontaneously gives out radiation from the nuclei of its atoms.

Radiotherapy A treatment of cancer that uses ionising radiation (such as gamma rays and X-rays) to kill cancer cells.

Radio wave A type of electromagnetic wave mainly used for radio and TV signals.

Random error A difference in the results of an experiment caused by unpredictable events, e.g. human error in measuring.

Range The difference between the smallest and largest values in a set of data.

Range of human hearing The range of frequencies of sound waves that humans can hear. It's 20 Hz to 20 kHz.

Ray A straight line showing the path along which a wave moves.

Reaction time The time taken for a person to react after an event (e.g. seeing a hazard).

Real image An image formed when light rays from a point on an object come together at another point — the light rays actually pass through that point.

Red giant A type of star that is formed when a star around the same size as the Sun expands as it starts to run out of hydrogen.

Red-shift The shift in observed wavelength of light from a source moving away from a stationary observer. The wavelength is shifted towards the red end of the electromagnetic spectrum.

Red super giant A type of star that is formed when a large star (much bigger than the Sun) expands as it starts to run out of hydrogen.

Reflection When a wave bounces back as it meets a boundary between two materials.

Refraction When a wave changes direction as it passes across the boundary between two materials at an angle to the normal.

Reliable result A result that is repeatable and reproducible.

Renewable energy resource An energy resource that is renewable is one that is being, or can be, made at the same rate (or faster) than it's being used, and so will never run out.

Repeatable result A result that will come out the same if the experiment is repeated by the same person using the same method and equipment.

Reproducible result A result that will come out the same if someone different does the experiment, or a slightly different method or piece of equipment is used.

Resistance Anything in a circuit that reduces the flow of current. Measured in ohms (Ω).

Resolution The smallest change a measuring instrument can detect.

Resultant force A single force that can replace all the forces acting on an object to give the same effect as the original forces acting altogether.

Right-hand thumb rule The rule to work out the direction of the magnetic field around a current-carrying wire. Your thumb points in the direction of the current, and your fingers curl in the direction of the magnetic field.

Risk The chance that a hazard will cause harm.

S

Scalar A quantity that has magnitude but no direction.

Scaling prefix A word or symbol which goes before a unit to indicate a multiplying factor (e.g. 1 km = 1000 m).

Seismic wave A wave which travels through (or over the surface of) the Earth when an earthquake occurs. Two important types are P-waves and S-waves.

Series circuit A circuit in which every component is connected in a line, end to end.

Significant figure The first significant figure of a number is the first non-zero digit. The second, third and fourth significant figures follow on immediately after it.

S.I. unit A unit recognised as standard by scientists all over the world.

Glossary

Solar cell A device that generates electricity directly from the Sun's radiation.

Solenoid A coil of wire often used in the construction of electromagnets.

Sound wave A longitudinal wave caused by vibrating particles. When they reach the ears, they can cause vibrations that are converted into electrical signals, which cause the sensation of hearing.

Spark The passage of electrons across a (usually) small gap between a static charge and an earthed conductor.

Specific heat capacity The amount of energy (in joules) needed to raise the temperature of 1 kg of a material by 1°C.

Specific latent heat (SLH) The amount of energy needed to change 1 kg of a substance from one state to another without changing its temperature. (For cooling, it is the energy released by a change in state.)

Specific latent heat of fusion The specific latent heat for changing between a solid and a liquid (melting or freezing).

Specific latent heat of vaporisation The specific latent heat for changing between a liquid and a gas (evaporating, boiling or condensing).

Specular reflection When parallel waves are reflected in a single direction by a smooth surface.

Split-ring commutator A ring with gaps in it that swaps the electrical contacts of a device every half-turn.

Standard form A number written in the form $A \times 10^n$, where A is a number between 1 and 10.

State of matter The form which a substance can take — e.g. solid, liquid or gas.

Static charge An electric charge that cannot move. It often forms on electrical insulators, where charge cannot flow freely.

Stopping distance The distance covered by a vehicle in the time between the driver spotting a hazard and the vehicle coming to a complete stop. It's the sum of the thinking distance and the braking distance.

Supernova The explosion of a red super giant.

System The object, or group of objects, that you're considering.

Systematic error An error that is consistently made throughout an experiment.

T

Tangent A straight line that touches a curve at a point but doesn't cross it.

Terminal velocity The maximum velocity a falling object can reach without any added driving forces. It's the velocity at which the resistive forces (drag) acting on the object match the force due to gravity (weight).

Theory An explanation or system of ideas that has been accepted by the scientific community because there is good evidence to back it up.

Thermal conductivity A measure of how quickly an object transfers energy by heating through conduction.

Thermal insulator A material with a low thermal conductivity.

Thermistor A resistor whose resistance is dependent on the temperature. The resistance decreases as temperature increases.

Thinking distance The distance a vehicle travels during the driver's reaction time (before the brakes have been applied).

Three-core cable An electrical cable containing a live wire, a neutral wire and an earth wire.

Transformer A device which can change the potential difference of an ac supply.

Transverse wave A wave in which the oscillations are perpendicular (at 90°) to the direction of energy transfer.

U

Ultrasound Sound with a frequency higher than the range of human hearing (i.e. greater than 20 000 Hz).

Ultraviolet (UV) radiation A type of electromagnetic wave, the main source of which is sunlight.

Uncertainty The amount by which a given result may differ from the true value.

Upthrust The resultant force acting upwards on an object submerged in a liquid, due to the pressure of the liquid being greater at the bottom of the object than at the top.

V

Valid result A result that is repeatable, reproducible and answers the original question.

Vector A quantity which has both magnitude (size) and a direction.

Velocity The speed and direction of an object.

Velocity-time graph A graph showing how the velocity of an object changes over a period of time.

Virtual image An image that is formed when light rays appear to have come from one point, but have actually come from another — the light rays don't actually pass through that point.

Visible light The part of the electromagnetic spectrum that we can see with our eyes.

Voltmeter A component used to measure the potential difference across a component. Always connected in parallel with the component.

W

Wave An oscillation that transfers energy without transferring any matter.

Wavelength The length of a full cycle of a wave, e.g. from a crest to the next crest.

Weight The force acting on an object due to gravity.

White dwarf The hot, dense core left behind when a red giant becomes unstable and ejects its outer layer of dust and gas.

Work done The energy transferred when a force moves an object through a distance, or by a moving charge.

X

X-ray A high-frequency, short-wavelength electromagnetic wave. It is mainly used in medical imaging and treatment.

Z

Zero error A type of systematic error caused by using a piece of equipment that isn't zeroed properly.

Index

A

acceleration 75, 76, 79, 81
  in orbits 131
accuracy 139
activity (radioactivity) 53
air bags 89
air resistance 77
alpha radiation 49, 51, 52, 56
alpha scattering 49
alternating currents (ac) 32, 125, 126
alternators 126
ammeters 22, 149
amplitude 92
angle of incidence 94-96
angle of reflection 94, 96
angle of refraction 95, 96
anomalous results 139
area under a graph 76
artificial satellites 131
atmospheric pressure 45, 71
atomic models 49, 50
atomic number 51, 52
average speed 74
averages 140

B

background radiation 56
beta radiation 51, 52, 56, 57
Big Bang theory 132
bio-fuels 14, 17
black body radiation 112
black dwarfs 130
black holes 130
boiling 42, 43
brakes 84, 85
braking distance 84, 85

C

cancer 57
cavity wall insulation 9
centre of mass 62
chain reactions 58
changes of state 42, 43
charge
  in circuits 21
  ions 50
  of a nucleus 51, 52
  of an atom 50
  relative charge of particles 50
  static 37, 38
circuit symbols 21
circuits 21-24, 26-29, 32-34
  parallel circuits 28, 29
  series circuits 27, 29
circular motion 131
closed systems 3, 7
coal 14, 17
collecting data 139
colour filters 109
colours 109
compasses 119
components of a force 64
compressions (in waves) 92, 113
concave lenses 106, 108
conclusions 145
condensing 42, 43
conduction (heat) 8
conservation of energy 7
conservation of momentum 88, 89
contact forces 61
contamination (radioactivity) 56
control rods 58
control variables 138
convection 8
convex lenses 106-108
correlations 142, 145
cosmic rays 56
count-rate 53
crumple zones 89
current (electrical) 21, 34
  alternating 32, 125, 126
  direct 32, 126
  I-V characteristics 23, 24
  magnetic field around a current-carrying wire 120, 122

D

dangers of ionising radiation 56, 57, 103
dark energy 132
dark matter 132
deceleration 75
density 41, 70, 71
dependent variables 138
designing investigations 138
diffuse reflection 94, 96
diodes 21, 23, 24
direct currents (dc) 32, 126
displacement 74
dissipated energy 7, 9
distance 74
distance-time graphs 76
double-glazing 9
drag 77
draught excluders 9
dwarf planets 131
dynamos 126

E

ear drum 113
earth wires 32
earthing 37
earthquakes 115
Earth's structure 115
Earth's temperature 112
echo sounding 114
efficiency 10
elastic deformation 66
elastic objects 66
elastic potential energy stores 4, 66, 67
electric charge 21
electric fields 38
electric motors 123
electric shocks 32
electrical (energy transfers) 3
electricity 21-24, 26-29, 32-35, 37, 38
  supply and demand 35
electromagnetic induction 125-127
electromagnetic spectrum 99
electromagnetic waves 99, 109
  black body radiation 112
  dangers 103
  uses 100-102
electromagnetism 120
  uses 120, 123
electrons 37, 49-52
electrostatic attraction 37, 38
electrostatic repulsion 37, 38
elements 52
energy 3-5, 7-10, 14-18
  conservation of 7
  internal 42-44
energy efficient light bulbs 102
energy levels in atoms 49, 50, 99
energy resources 14-18
  non-renewables 14, 17
  renewables 14-17
  transport 14, 18
  trends in use 18
energy stores 3, 42, 44
  elastic potential 4, 66, 67
  gravitational potential 3, 4
  kinetic 3, 4, 8, 42, 44
  thermal 3, 5, 8-10
energy transfers 3, 63
  by heating 3, 5, 8, 43
  by waves 92
  efficiency 10
  electrical 3, 33, 34
  mechanical 3
  rate of 7
  reducing 9
  work done 3, 44, 63
equilibrium (forces) 64, 80
ethics 136
eureka cans 41, 148
evaluations 146
evaporation 42, 43
expansion of the universe 132
experimental safety 148
extension (of objects) 66, 67

F

fair tests 138
field lines 38, 119
fields
  electric 38
  gravitational 4, 62
  magnetic 119, 120, 122, 123, 125-127
filament lamps 21, 23, 24
filters (colour) 109
fission 58
Fleming's left-hand rule 122, 123
floating 71
fluid pressure 70, 71
fluids 70, 71
  drag 77
focal length 106
force-extension graphs 66, 67
forces 61
  atmospheric pressure 45, 71
  change in momentum 89
  contact 61
  electrostatic 37, 38
  frictional 10, 63, 77, 84
  gravitational 3, 62, 130, 131
  interaction pair 61, 80
  magnetic 119, 120, 122, 123
  moments 68
  Newton's laws 79, 80
  non-contact 61
  normal contact 61, 80
  pressure 70, 71
  resolving 64
  resultant 63, 64, 79, 89
  upthrust 71
  weight 62
formation of stars 130
formulas 144, 216
fossil fuels 14, 17, 18
free body diagrams 63
freezing 42, 43
frequency 92
  human hearing range 113
  of EM waves 99, 103
  of mains supply 32
friction 10, 63, 77, 84
fusion 58, 130

Index

G

galaxies 131, 132
gamma rays 51, 52, 99
  dangers 56, 57, 103
  uses 57, 102
gases
  gas pressure 44, 45
  natural gas 14, 17
  particle motion 41, 44, 45
  states of matter 41-43
  work done on a gas 44
gears 68
Geiger-Muller tube 53
gene mutation 103
generator effect 125, 126
generators (electrical) 126
geothermal power 15
global warming 17
graphs 76, 141, 142
  black body radiation 112
  distance-time 76
  force-extension 66, 67
  heating and cooling 43
  radioactive decay 53
  velocity-time 76
gravitational field strength 4, 62
gravitational force 62
  orbits 131
  star formation 130
gravitational potential energy stores 3, 4

H

half-life 53
headphones 123
heating 3, 5, 42, 43
  conduction 8
  convection 8
human hearing 113
hydroelectric power 14, 16
hypotheses 135

I

images (types of) 107
induced current 125-127
induced magnets 119
induced potential difference 125-127
industrial imaging 114
inelastic deformation 66
inertia 80
inertial mass 80
infrared cameras 101
infrared radiation 99, 101, 111
insulation 9
intensity 112
interaction pairs 61, 80
internal energy 42-44
ionisation 38, 51
ionising power 51
ionising radiation 51-53, 56, 57, 103
  alpha 51, 52, 56
  beta 51, 52, 56, 57
  dangers 56, 57, 103
  gamma 51, 52, 56-58
  uses 57, 102
ions 49, 51
irradiation 56
isotopes 51
I-V characteristics 23, 24

K

kinetic energy stores 3, 8

L

laws of motion 79, 80
LDRs 26
lenses 106-108
  concave 106, 108
  convex 106-108
Leslie cube 111
levers 68
life cycle of stars 130
light gates 149
limit of proportionality 66, 67
linear components 23
liquids 41-43
live wires 32
loft insulation 9
longitudinal waves 92
loudspeakers 123
lubricants 10

M

magnetic fields 119, 120, 122, 123, 125-127
  field lines 119
  field strength 119, 120, 122
  flux density 122
  of the Earth 119
magnetic forces 119, 122, 123
magnetic materials 119
magnification formula 108
magnifying glasses 108
main sequence stars 130
mass 62
  inertial mass 79-81
  in fusion reactions 58
  mass balance 147
  particle model 41, 42
mass number 51, 52
mean 140
measuring cylinders 148
mechanical (energy transfers) 3
median 140
medical imaging 114
medical tracers 57
melting 42, 43
micrometers 147
microphones 126
microwaves 99, 101
Milky Way galaxy 131
models 135
mode 140
moments 68
momentum 88, 89
moons 131
motor effect 122, 123
motors 123

N

National Grid 35, 127
natural gas 14, 17
natural satellites 131
nebulae 130
neutral wires 32
neutron stars 130
neutrons 49-52, 58
newton meters 62
Newton's First Law 79
Newton's Second Law 79, 81
Newton's Third Law 61, 80
non-contact forces 61
non-linear components 23, 26
non-renewables 14, 17
north poles 119
nuclear equations 52
nuclear fission 58
nuclear fuels 14, 17, 58
nuclear fusion 58, 130
nuclear model 49, 50
nuclear power 17
nuclear waste 17
nuclear weapons 56, 58
nuclei 49-51, 58

O

ohmic conductors 23, 24
oil (energy resource) 14, 17, 18
opaque objects 109
optical density 95, 96
optical fibres 102
orbits 131
oscilloscopes (generated potential difference) 126

P

parallax 147
parallel circuits 28, 29
particle model of matter 41-45
perfect black bodies 112
period 92
permanent magnets 119
physical changes 42
planets 131
plugs 32
plum pudding model 49
plutonium 14, 58
potential difference 21, 22, 125-127
  energy transferred 34
  induced 125-127
  in parallel circuits 28
  in series circuits 27
  I-V characteristics 23, 24
  measuring 149
  National Grid 35
  transformers 127
power 7, 10
  electrical 33, 34
power ratings 33
precision 139
pressure 70, 71
  atmospheric 71
  in fluids 70, 71
  in gases 44, 45
principal focus 106
proportionality 144
protons 49-52
protostars 130
P-waves 115

R

radiation (energy transfers) 3
radiation dose 56, 103
radiation sickness 57
radio waves 99, 100
radioactive decay 51-53, 56, 57
  alpha 49, 51, 52, 56
  background 53, 56
  beta 51, 52, 56, 57
  gamma 51, 57, 99
radiotherapy 57, 102
random errors 139, 146
rarefactions 92, 113
ray boxes 96
ray diagrams
  concave lenses 106, 108
  convex lenses 106-108
  reflection 94, 96
  refraction 95, 96
reaction times 84-86
real images 107

Index

red giants 130
red super giants 130
red-shift 132
reducing energy transfers 9
reflection 94, 96, 109
  sound 113
  ultrasound 114
refraction 95, 96, 106-108
  seismic waves 115
  sound 113
renewable energy resources 14-17
repeatability 138, 139
reproducibility 138, 139
resistance (air) 77
resistance (electrical) 21-24, 26-29, 33
  in parallel circuits 28, 29
  in series circuits 27, 29
resolving (a force) 64
resultant force 63, 64, 79, 89
right-hand thumb rule 120
ripple tanks 93
risk 137
  and radiation 56, 57, 102, 103
ruler drop test 86

S

S.I. units 143
safety
  during experiments 148
  features in cars 89
  handling radioactive sources 56
satellites 101, 131
scalar quantities 61
scale drawings 64
scaling prefixes 143
scientific method 135
seat belts 89
seismic waves 115
sensing circuits 26
series circuits 27, 29
shock waves 115
significant figures 140
sinking 71
skin cancer 103
smoke detectors 51
solar power 15
solar system 131
solenoids 120
solids 41-43
sound waves 92, 113, 114
south poles 119
space 130-132
sparks 37, 38
specific heat capacity 5
specific latent heat 43
specular reflection 94, 96
speed 74-77
  braking distances 84, 85
  of sound in air 74
  typical speeds 74
  wave speed 92, 93, 95, 99
spring constant 66
standard form 144
stars 130, 131
  life cycle 130
states of matter 41-43
static electricity 37, 38
step-down transformers 35, 127
step-up transformers 35, 127
stopping distances 84, 85
stopwatches 147
streamlining 77
sublimation 42
Sun 130, 131
sunburn 103
supernovae 130, 132
S-waves 115
systematic errors 139
systems 3

T

taking measurements 147-149
temperature
  Earth's 112
  internal energy 42, 44, 45
  monitoring with IR 101
  specific heat capacity 5
  specific latent heat 43
terminal velocity 77
thermal conductivity 8, 9
thermal energy stores 3, 5, 8
thermal insulation 9
thermistors 26
thinking distance 84, 85
three-core cables 32
tidal barrages 16
timing 147
tracers 57
transformers 35, 127
  equation 127
translucent objects 109
transparent objects 109
transverse waves 92, 99
trends in electricity use 35
trends in energy resources 18
turning effects 68
TV signals 100
typical speeds 74

U

UK mains supply 32
ultrasound 114
ultraviolet (UV) 99, 102, 103
uncertainties 146
uniform acceleration 75
uniform magnetic fields 120
Universe 132
upthrust 71
uranium 14, 15, 58

V

validity 138
variables 138
vectors 61, 64, 74
velocity 74-76
  momentum 88, 89
  orbits 131
  terminal 77
velocity-time graphs 76
virtual images 107, 108
visible light 99, 109
  refraction and reflection 96, 102
voltage 21
voltmeters 22, 149
volume
  measurement 148
  of a gas 45

W

wasted energy 7, 10
wave equation 92
wave front diagrams 95
wave power 16
wavelength 92
  black body radiation 112
  measuring 93
  of EM waves 99
  red-shift 132
waves 92-96, 99-103, 106-109, 111-115
  electromagnetic 99-103, 109, 111, 112
  longitudinal 92
  reflection 94
  refraction 95
  seismic 115
  sound 113, 114
  speed 92, 93, 95, 99
  transverse 92
  wavelength 92, 93, 95, 99
weight 62
white dwarfs 130
white light 109
wind power 15
work done 3, 7, 33, 63
  on a gas 44

X

X-rays 99, 102, 103

Z

zero errors 139

Formulas

In your GCSE Physics exams, you'll be given an equations sheet listing the formulas you might need. That means you don't have to learn them all (hurrah), but you do need to be able to pick out the right formulas to use and be really confident using them. Here are the formulas that you'll be given:

kinetic energy = 0.5 × mass × (speed)^2 $E_k = \frac{1}{2}mv^2$

elastic potential energy = 0.5 × spring constant × (extension)^2 $E_e = \frac{1}{2}ke^2$

gravitational potential energy = mass × gravitational field strength × height $E_p = mgh$

change in thermal energy = mass × specific heat capacity × temperature change $\Delta E = mc\Delta\theta$

efficiency = $\frac{\text{useful output energy transfer}}{\text{total input energy transfer}}$

efficiency = $\frac{\text{useful power output}}{\text{total power input}}$

power = $\frac{\text{energy transferred}}{\text{time}}$ $P = \frac{E}{t}$

power = $\frac{\text{work done}}{\text{time}}$ $P = \frac{W}{t}$

charge flow = current × time $Q = It$

potential difference (pd) = current × resistance $V = IR$

power = pd × current $P = VI$

energy transferred = power × time $E = Pt$

energy transferred = charge flow × pd $E = QV$

power = (current)^2 × resistance $P = I^2R$

weight = mass × gravitational field strength $W = mg$

force = spring constant × extension $F = ke$

pressure = $\frac{\text{force normal to surface}}{\text{area of surface}}$ $p = \frac{F}{A}$

work done = force × distance (along the line of action of the force) $W = Fs$

moment of a force = force × distance (normal to direction of force) $M = Fd$

pressure (due to column of liquid) = height (of column) × density (of liquid) × gravitational field strength $p = h\rho g$

distance = speed × time $s = vt$

force = mass × acceleration $F = ma$

momentum = mass × velocity $p = mv$

acceleration = $\frac{\text{change in velocity}}{\text{time taken}}$ $a = \frac{\Delta v}{t}$

force = $\frac{\text{change in momentum}}{\text{time taken}}$ $F = \frac{m\Delta v}{\Delta t}$

(final velocity)^2 – (initial velocity)^2 = 2 × acceleration × distance $v^2 - u^2 = 2as$

density = $\frac{\text{mass}}{\text{volume}}$ $\rho = \frac{m}{V}$

thermal energy for a change of state = mass × specific latent heat $E = mL$

pressure (of a gas) × volume (of a gas) = constant $pV = \text{constant}$

period = $\frac{1}{\text{frequency}}$ $T = \frac{1}{f}$

magnification = $\frac{\text{image height}}{\text{object height}}$

wave speed = frequency × wavelength $v = f\lambda$

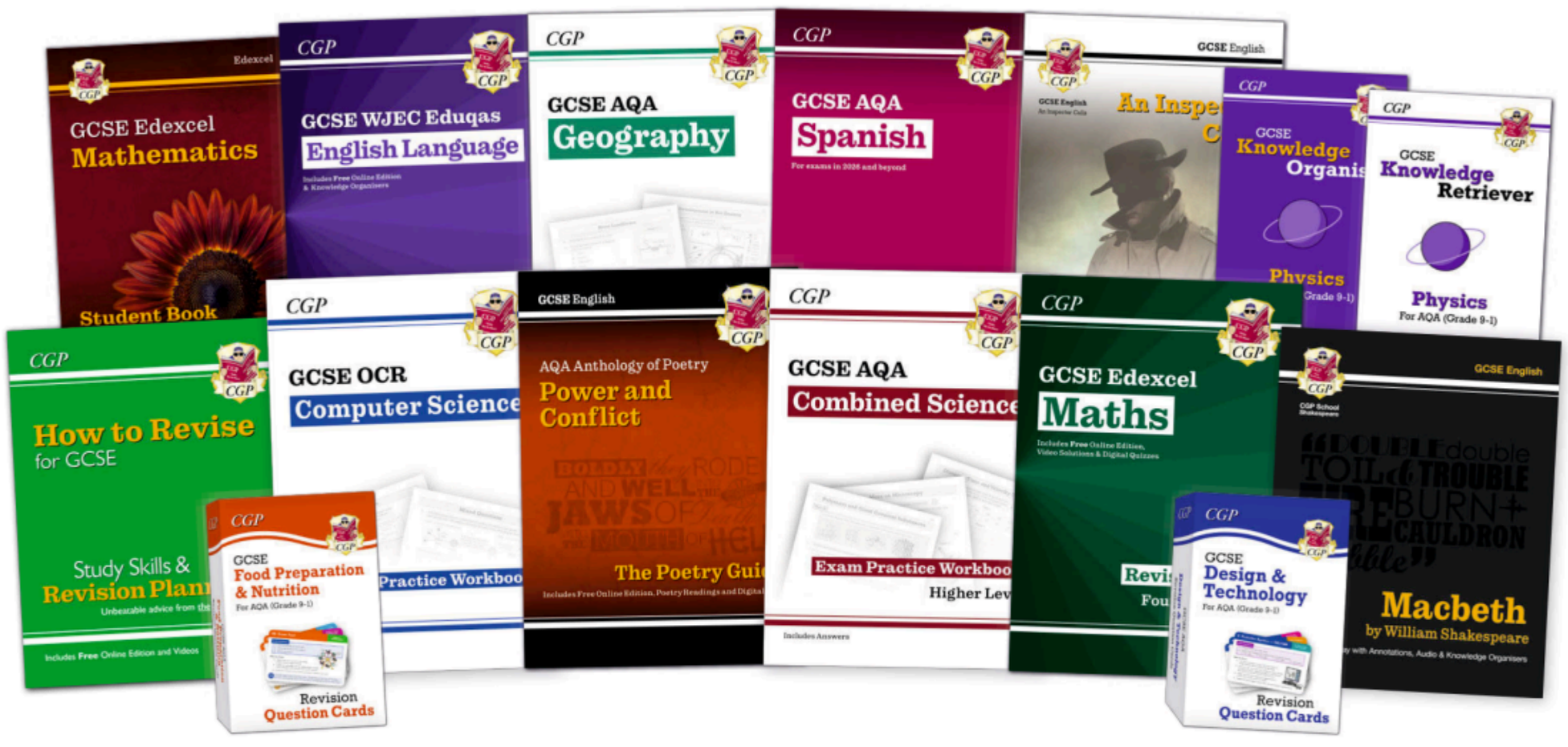
force (on a current-carrying conductor at right angles to a magnetic field) = magnetic flux density × current × length $F = BIl$

$\frac{\text{pd across primary coil}}{\text{pd across secondary coil}} = \frac{\text{number of turns in primary coil}}{\text{number of turns in secondary coil}}$ $\frac{V_p}{V_s} = \frac{n_p}{n_s}$

pd across primary coil × current in primary coil = pd across secondary coil × current in secondary coil $V_p I_p = V_s I_s$

PAS49

CGP books — they might just save your life...



OK, maybe not your *life*. But when it comes to exam revision, CGP are the undisputed champions.
You can order any of our books and cards (with next-day delivery!) from us, online or by phone:

cgpbooks.co.uk • 0800 1712 712

CGP Books — The Choice of Champions!

CGP

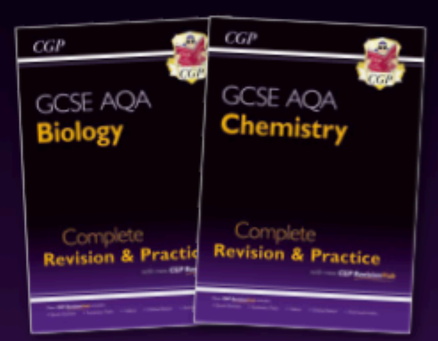
Nobody beats CGP for Physics revision!

You'll be totally prepared for the GCSE exams with this fantastic CGP book...

- **Notes for everything on the course...**
  All the topics, all the required practicals — the lot!
- **A huge serving of exam-style questions...**
  With all the answers included
- **Plus full practice papers...**
  To make sure you're confident for the real thing
- **With the new *CGP RevisionHub*...**
  Oh, yes — all the online revision you'll need!

You *could* choose someone else's dreary book...
...but really — why, why would you do that? ☻

P.S. CGP also makes Complete Revision & Practice books for GCSE **Biology** and **Chemistry**!



CGP Shakespeare: Is this a digger I see before me?

www.cgpbooks.co.uk